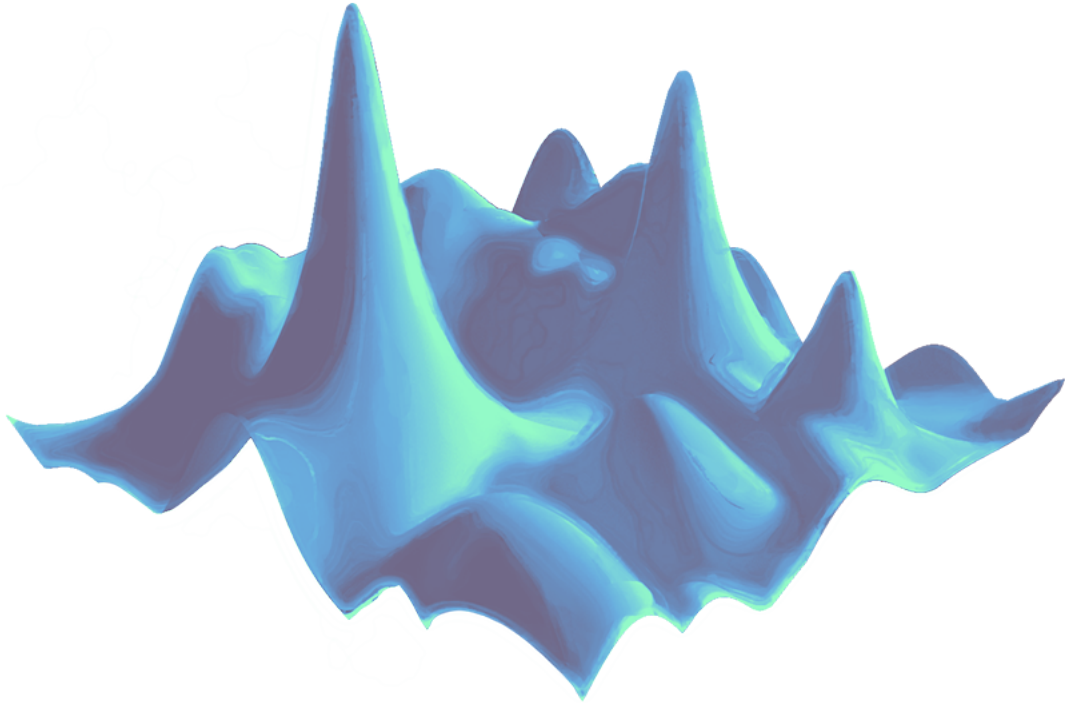


Quantum Fields in Curved Spacetime



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Part I

Quantum Fields and Horizons

It is often said that quantum mechanics and general relativity are fundamentally incompatible, and that a new theory of quantum gravity is needed to describe situations where both are important. This is an overstatement. While it is true that general relativity is a non-renormalizable theory that breaks down near the Planck scale $L_P \equiv \sqrt{G\hbar/c^3} \sim 10^{-33}$ cm, at long distances, $L \ll L_P$ it is a perfectly well-defined effective field theory (EFT) that can be treated quantum mechanically like any conventional quantum field theory (QFT) [1]. In particular, the large quantum fluctuations at $L \sim L_P$ “decouple” from the physics at long distances (low energies). Of course, this decoupling is essential for physics to be predictive.

In this part of the book, we will study the effects of *classical* gravity on other *quantum* fields. To focus on the essential physics, we take the matter fields to be scalar fields and don’t include their self-interactions (already the free theory is remarkably rich in a curved spacetime). For a single scalar field ϕ , we will therefore consider the action

$$S = \int d^4x \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} m^2 \phi^2 \right],$$

which includes a coupling to the metric $g_{\mu\nu}$. We will then consider the quantum theory of the field ϕ on a fixed spacetime background whose curvature is nonzero, but small in Planck units.

We will encounter interesting subtleties, especially in spacetimes with horizons. For example, the notion of “particles” is difficult to make precise in a curved spacetime and even the concept of a “vacuum” is observer-dependent. These difficulties don’t arise in flat space and are therefore an important novelty of QFT in curved spacetime.

After reviewing the conceptual foundations of quantum field theory, both in flat space (see Chapter 1) and in curved spacetime (see Chapter 2), we will study a number of important applications:

- In Chapter 3, we will discuss the physics of an accelerated observer in flat space and the associated *Unruh effect*. Such an observer experiences an effective horizon and will detect a thermal spectrum of particles, with a temperature [2]

$$k_B T = \frac{\hbar}{c} \frac{a}{2\pi}.$$

where a is the acceleration. The appearance of Planck’s constant $\hbar$ makes manifest that this is a quantum effect.

- In Chapter 4, we will explore the behavior of quantum fields in the vicinity of black holes and the associated *Hawking radiation*. We will show that a distant observer will measure a thermal spectrum of particles, with a temperature [3, 4]

$$k_B T = \frac{\hbar c^3}{8\pi G M}.$$

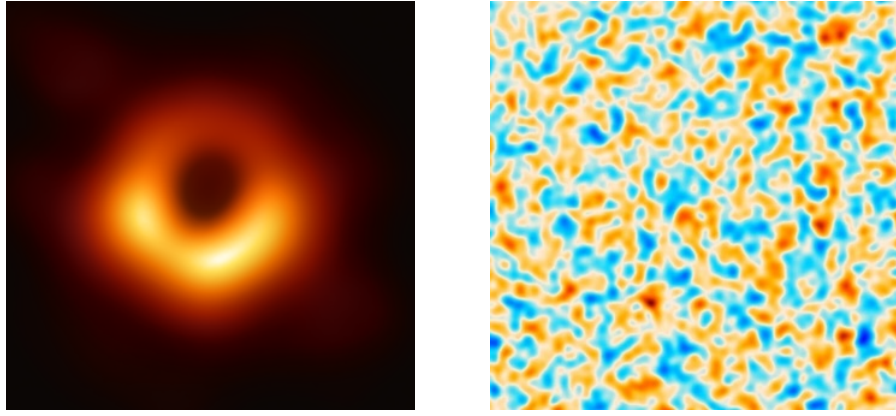


Figure 1: Two key applications of QFT in curved spacetime are: 1) Hawking radiation and the thermodynamics of black holes (*left*) and 2) the quantum origin of structure including the anisotropies of the cosmic microwave background radiation (*right*).

where M is the black hole’s mass. For ordinary astrophysical black holes this is a tiny effect, e.g. the temperature of a Solar mass black hole is about 60 nK, which is much too small to be observable. In fact, this temperature is much smaller than the temperature of the cosmic microwave background (2.7 K), so astrophysical black holes will actually absorb radiation rather than emit it. Nevertheless, the conceptual consequences of Hawking radiation are enormous. It proves that black holes are thermodynamic objects with a temperature and an entropy. We will show that the entropy is proportional to the area of the event horizon

$$S = \frac{Ac^3}{4\hbar G} = \frac{A}{4L_P^2}.$$

For ordinary astrophysical black holes, this entropy is enormous. For example, a Solar mass black hole has an entropy $S \sim 10^{77}$, which implies a gigantic number of microstates $N \sim e^S$. The microscopic origin of this entropy remains mysterious.

- In Chapter 5, we will study the analog of Hawking radiation in de Sitter space, where it leads to a thermal spectrum with a temperature [5]

$$k_B T = \frac{\hbar}{2\pi L},$$

where L is the de Sitter radius. This is again a small quantum effect, but this time with enormous phenomenological implications. There is intriguing observational evidence that the early universe experienced a burst of very rapid expansion called “cosmic inflation”, during which the spacetime geometry was approximately that of de Sitter space. Quantum fluctuations during this inflationary era got stretched to large scales and became density fluctuations at the origin of the hot Big Bang. Eventually, these primordial density fluctuations grew into the large-scale structure of the Universe.

- Finally, in Chapter 6, we will present an alternative approach—called “Euclidean quantum gravity”—to derive the thermodynamics of black holes and de Sitter space. The basic idea

is to complexify the metric by introducing “imaginary time” $\tau = it$ and then define the “gravitational path integral” as

$$Z(\beta) = \int \mathcal{D}g e^{-I_E[g]},$$

where $I_E[g]$ is the “Euclidean action” of the metric $g_{\mu\nu}$. If the path integral is over solutions that are periodic in imaginary time, with periodicity $\tau_\beta = \beta\hbar$, then (0.1) is the partition function of a field theory at temperature $T = (k_B\beta)^{-1}$ and the entropy of the system is

$$S = (1 - \beta\partial_\beta) \log Z.$$

We will show that this gives the correct results for black holes and de Sitter space.

QFT in curved spacetime is only an approximation to a more complete theory of quantum gravity. Nevertheless, it predicts important phenomena that have both observational and conceptual value. Moreover, although it is an old subject, it continues to be a topic of active research. In particular, perturbative computations of QFTs in curved spacetimes continue to provide important “theoretical data” in the quest for a nonperturbative theory of quantum gravity. We hope that these notes will provide the necessary background to follow recent developments in the field.

Recommended Resources

The following books and lecture notes have been useful in the preparation of these lectures:

- Carroll, *Spacetime and Geometry* (Chapter 9)
- Mukhanov and Winitzki, *Introduction to Quantum Effect in Gravity*
- Birrell and Davies, *Quantum Fields in Curved Spacetime*
- Toms and Parker, *Quantum Field Theory in Curved Spacetime*
- Preskill, *Quantum Field Theory in Curved Spacetime*
- Jacobson, *Introduction to Quantum Fields in Curved Spacetime*
- Marino, *QFT in Curved Spacetime*
- Bomans, *QFT in Curved Spacetime*
- Hartman, *Quantum Gravity and Black Holes*
- Reall, *Black Holes*
- Townsend, *Black Holes*
- Spradlin, Strominger and Volovich, *Les Houches Lectures on De Sitter Space*
- Witten, *Black Hole Thermodynamics*

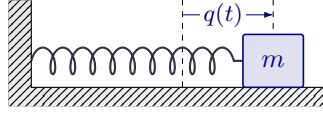
1 QFT in Flat Spacetime

In this chapter, we will give a quick refresher of the canonical quantization of free (non-interacting) fields in Minkowski space. We will present this familiar material in a way that will prepare us for the generalization to curved spacetime.

1.1 Quantum Oscillators

Through the magic of the Fourier expansion, a quantum field can be expressed as an infinite number of coupled harmonic oscillators. As a warmup for quantum field theory, we therefore begin by reviewing the harmonic oscillator in quantum mechanics.

Classical dynamics Consider a mass on a spring moving in one dimension:



Let $q(t)$ be the deviation from the equilibrium point defined by a quadratic potential $V(q) = \frac{1}{2}\omega^2 q^2$, where ω is a constant parameter. The Lagrangian is

$$L = \frac{1}{2}\dot{q}^2 - \frac{1}{2}\omega^2 q^2, \quad (1.1)$$

where we have set the particle's mass to unity. The equation of motion then becomes

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) = \frac{\partial L}{\partial q} \quad \Rightarrow \quad \ddot{q} + \omega^2 q = 0. \quad (1.2)$$

The most general solution of this equation is

$$\begin{aligned} q(t) &= A \cos(\omega t) + B \sin(\omega t) \\ &= \frac{A + iB}{2} e^{-i\omega t} + \frac{A - iB}{2} e^{i\omega t}, \end{aligned} \quad (1.3)$$

where A and B are constants that are fixed by the initial conditions $q(0) = A$ and $\dot{q}(0) = \omega B$.

In quantum mechanics, we typically work with the Hamiltonian of a system, which is related to the Lagrangian by a Legendre transform

$$H = p\dot{q} - L, \quad (1.4)$$

where $p \equiv \partial L / \partial \dot{q} = m\dot{q}$ is the conjugate momentum. For the harmonic oscillator in (1.1), we get

$$H = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 q^2. \quad (1.5)$$

As expected, the Hamiltonian is the sum of kinetic and potential energies. To “solve the theory” means finding the spectrum of energy eigenstates of this Hamiltonian.

Canonical quantization First, we promote the classical variables q, p to quantum operators $\hat{q}, \hat{p}$ and impose the canonical commutation relation

$$[\hat{q}(t), \hat{p}(t)] = i\hbar, \quad (1.6)$$

where $\hbar = 1.05 \times 10^{-34}$ Js is the reduced Planck constant. Note that we work in the Heisenberg picture where the operators are time-dependent. In particular, the position operator $\hat{q}(t)$ satisfies the equation of motion (10.29). Using the equation of motion it can also be shown that the commutator (1.6) is time-independent. Hence, if it holds at some initial time, it will hold at all times.

The most general solution for the operator $\hat{q}(t)$ can be written as

$$\hat{q}(t) = \sqrt{\frac{\hbar}{2\omega}} \left[\hat{a} e^{-i\omega t} + \hat{a}^\dagger e^{i\omega t} \right], \quad (1.7)$$

where $\hat{a}$ and $\hat{a}^\dagger$ are time-independent non-Hermitian operators, and the overall factor was chosen for future convenience. The conjugate momentum $\hat{p}(t) = \partial_t \hat{q}$ then is

$$\hat{p}(t) = -i\sqrt{\frac{\hbar\omega}{2}} \left[\hat{a} e^{-i\omega t} - \hat{a}^\dagger e^{i\omega t} \right]. \quad (1.8)$$

Using (1.7) and (1.8), it is easy to show that the commutation relation (1.6) is satisfied if

$$[\hat{a}, \hat{a}^\dagger] = 1. \quad (1.9)$$

You may recognize this as the standard commutation relation of the raising and lowering operators for the quantum oscillator.

The Hamiltonian (1.5) can then be written as

$$\begin{aligned} \hat{H} &= \frac{1}{2}\hat{p}^2 + \frac{1}{2}\omega^2\hat{q}^2 \\ &= \frac{1}{2}\hbar\omega(\hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a}) = \hbar\omega\left(\hat{N} + \frac{1}{2}\right), \end{aligned} \quad (1.10)$$

where $\hat{N} \equiv \hat{a}^\dagger\hat{a}$ is the number operator. The ground state of the oscillator is defined by

$$\hat{a}|0\rangle = 0, \quad (1.11)$$

and has energy $E_0 \equiv \langle 0|\hat{H}|0\rangle = \frac{1}{2}\hbar\omega$. We see that, unlike in classical physics, the ground state energy doesn't vanish in quantum mechanics. These zero-point fluctuations of the quantum vacuum play an important role in many areas of physics. Excited states are constructed by repeated application of the creation operator

$$|n\rangle = \frac{1}{\sqrt{n!}}(\hat{a}^\dagger)^n|0\rangle, \quad (1.12)$$

and have energies $E_n \equiv \langle n|\hat{H}|n\rangle = \hbar\omega(n + \frac{1}{2})$, where $\hat{N}|n\rangle = n|n\rangle$.

Time-dependent oscillator As we will see later, in a curved background, the harmonic oscillators that make up a quantum field can receive time-dependent frequencies due to their couplings to the time-dependent spacetime curvature. To prepare for this possibility, we now consider the simple harmonic oscillator with a time-dependent frequency $\omega(t)$. We will follow the treatment in [6].

The solution for the operator $\hat{q}(t)$ now is

$$\hat{q}(t) = f(t) \hat{a} + f^*(t) \hat{a}^\dagger, \quad (1.13)$$

where $f(t)$ is a complex function solving the classical equation of motion, $\ddot{f} + \omega^2(t)f = 0$. Substituting (1.13) into (1.6), with $\hat{p} = \partial_t \hat{q}$, we find

$$\langle f, f \rangle [\hat{a}, \hat{a}^\dagger] = 1, \quad \text{where} \quad \langle f, g \rangle \equiv \frac{i}{\hbar} (f^* \partial_t g - g \partial_t f^*). \quad (1.14)$$

Note that the bracket $\langle f, g \rangle$ is time-independent if f and g satisfy the equation of motion. We assume that the function f is chosen so that the real number $\langle f, f \rangle$ is positive. By a rescaling of the amplitude, we can set

$$\langle f, f \rangle \equiv 1, \quad (1.15)$$

so that (1.14) becomes

$$[\hat{a}, \hat{a}^\dagger] = 1. \quad (1.16)$$

We have therefore found the same raising and lowering operators as before, and we can use them to construct the Hilbert space of the theory.

So far, the solution $f(t)$ is still not completely fixed. This implies that the definition of the operator $\hat{a}$ and the associated state $|0\rangle$ is not yet unique. In the special case of a constant frequency $\omega(t) = \omega$, a unique mode function $f(t)$ is selected by requiring the state $|0\rangle$ to be the ground state of the Hamiltonian. To see this, we first compute the Hamiltonian for a general $f(t)$:

$$\begin{aligned} \hat{H} &= \frac{1}{2} \hat{p}^2 + \frac{1}{2} \omega^2 \hat{q}^2 \\ &= \frac{1}{2} \left[(\dot{f}^2 + \omega^2 f^2) \hat{a} \hat{a} + (\dot{f}^2 + \omega^2 f^2)^* \hat{a}^\dagger \hat{a}^\dagger + (|\dot{f}|^2 + \omega^2 |f|^2) (\hat{a} \hat{a}^\dagger + \hat{a}^\dagger \hat{a}) \right]. \end{aligned} \quad (1.17)$$

This implies

$$\hat{H}|0\rangle = \frac{1}{2} (\dot{f}^2 + \omega^2 f^2)^* \hat{a}^\dagger \hat{a}^\dagger |0\rangle + \frac{1}{2} (|\dot{f}|^2 + \omega^2 |f|^2) |0\rangle. \quad (1.18)$$

In order for the state $|0\rangle$ to be an eigenstate of the Hamiltonian, the first term on the right-hand side must vanish, which requires that

$$\dot{f} = \pm i\omega f. \quad (1.19)$$

The bracket in (1.14) then is

$$\langle f, f \rangle = \mp \frac{2\omega}{\hbar} |f|^2, \quad (1.20)$$

which is positive if we pick the minus sign in (1.19). This selects the *positive-frequency solution*:

$$f(t) = \sqrt{\frac{\hbar}{2\omega}} e^{-i\omega t}, \quad (1.21)$$

where the normalization is fixed by $\langle f, f \rangle \equiv 1$. Using (1.21), the Hamiltonian (1.18) becomes

$$\hat{H} = \frac{1}{2} \hbar \omega (\hat{a} \hat{a}^\dagger + \hat{a}^\dagger \hat{a}) = \hbar \omega \left(\hat{N} + \frac{1}{2} \right), \quad (1.22)$$

as before. Since the eigenvalues of the number operator are non-negative integers, the state $|0\rangle$ is the minimum-energy state. If any function other than (1.21) had been chosen to expand the position operator, this would not be the case.

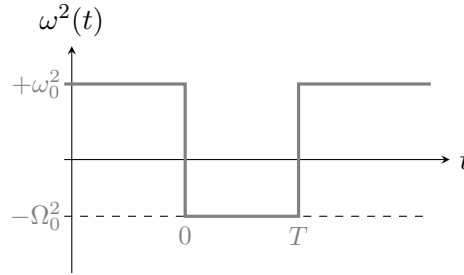
Exercise 1.1 The most general solution for a simple harmonic oscillator (with constant frequency ω) is

$$f(t) = c_1 \frac{1}{\sqrt{2\omega}} e^{-i\omega t} + c_2 \frac{1}{\sqrt{2\omega}} e^{i\omega t}, \quad (1.23)$$

where c_1 and c_2 are constants. Setting $\hbar \equiv 1$, the normalization is $\langle f, f \rangle \equiv i(f^* \dot{f} - \dot{f}^* f) = 1$, so that $[\hat{a}, \hat{a}^\dagger] = 1$. By considering the vacuum expectation value of the Hamiltonian, $\langle 0 | \hat{H} | 0 \rangle$, show that the vacuum $|0\rangle$ is the minimum energy state iff $c_1 = 1$ and $c_2 = 0$. $\rightarrow$

Particle creation An important feature of having a time-dependent frequency is that the oscillator can get excited. In QFT in curved spacetime, a similar phenomenon will lead to the quantum creation of particles during the period of changing frequency.

As a concrete example, we will assume the following evolution of the frequency:



The asymptotic solutions can be written as

$$t < 0 : \quad q_{\text{in}}(t) = A_{\text{in}} \sin(\omega_0 t), \quad (1.24)$$

$$t > T : \quad q_{\text{out}}(t) = A_{\text{out}} \sin(\omega_0 t + \alpha), \quad (1.25)$$

where A_{in} , A_{out} and α are constants. We wish to determine how the change in the frequency creates a boost in the amplitude of the oscillator, $A_{\text{out}} > A_{\text{in}}$.

We first write the solution in the transition region as

$$0 < t < T : \quad q(t) = A \cosh(\Omega_0 t) + B \sinh(\Omega_0 t). \quad (1.26)$$

At $t = 0$, we match this solution (and its first derivative) to the solution (1.24):

$$q(0) = q_{\text{in}}(0) \quad \Rightarrow \quad A = 0, \quad (1.27)$$

$$\dot{q}(0) = \dot{q}_{\text{in}}(0) \quad \Rightarrow \quad B = \frac{\omega_0}{\Omega_0} A_{\text{in}}. \quad (1.28)$$

Having fixed the coefficients in (1.26), we then match this solution (and its first derivative) to the solution (1.25) at $t = T$:

$$q(T) = q_{\text{out}}(T) \quad \Rightarrow \quad A_{\text{out}} \sin(\omega_0 T + \alpha) = B \sinh(\Omega_0 T), \quad (1.29)$$

$$\dot{q}(T) = \dot{q}_{\text{out}}(T) \quad \Rightarrow \quad A_{\text{out}} \cos(\omega_0 T + \alpha) = \frac{\Omega_0}{\omega_0} B \cosh(\Omega_0 T). \quad (1.30)$$

This gives

$$\begin{aligned} A_{\text{out}}^2 &= B^2 \left[\sinh^2(\Omega_0 T) + \frac{\Omega_0^2}{\omega_0^2} \cosh^2(\Omega_0 T) \right] \\ &= B^2 \frac{\Omega_0^2}{\omega_0^2} \left[\cosh^2(\Omega_0 T) + \frac{\omega_0^2}{\Omega_0^2} \sinh^2(\Omega_0 T) \right] \\ &= A_{\text{in}}^2 \left[1 + \left(1 + \frac{\omega_0^2}{\Omega_0^2} \right) \sinh^2(\Omega_0 T) \right]. \end{aligned} \quad (1.31)$$

For $\Omega_0 T \gg 1$, we then have

$$A_{\text{out}} \approx \frac{A_{\text{in}}}{2} e^{\Omega_0 T} \sqrt{1 + \frac{\omega_0^2}{\Omega_0^2}} \gg A_{\text{in}}. \quad (1.32)$$

The final state of the oscillator therefore has a larger amplitude (and hence a larger energy) than its initial state.

1.2 From Oscillators to Fields

Having reviewed the physics of a single harmonic oscillator, we now generalize this to a system of many coupled oscillators. We will show that the continuum limit of an infinite number of these oscillators is described by a scalar field $\phi(x^\mu)$, whose amplitude is a function of the spacetime position $x^\mu = (t, \mathbf{x})$.

The Lagrangian describing N coupled simple harmonic oscillators is

$$L = \frac{1}{2} \left[\sum_{i=1}^N \dot{q}_i^2 - \sum_{i,j=1}^N M_{ij} q_i q_j \right], \quad (1.33)$$

where M_{ij} is a symmetric positive-definite matrix. By defining the normal coordinates $\tilde{q}_\alpha = \sum_i C_{\alpha i} q_i$, the matrix can be diagonalized, $M_{ij} \rightarrow M_{\alpha\beta} = \omega_\alpha^2 \delta_{\alpha\beta}$, so that (1.33) becomes

$$L = \frac{1}{2} \sum_{\alpha=1}^N \left[\dot{\tilde{q}}_\alpha^2 - \omega_\alpha^2 \tilde{q}_\alpha^2 \right]. \quad (1.34)$$

The oscillators have “decoupled” and we can quantize each of them separately as before.

Next, we consider a field $\phi(t, \mathbf{x})$ which has a value at every time t and spatial point $\mathbf{x}$. We can think of a field as an infinite set of oscillators attached to each point $\mathbf{x}$, i.e. $\phi_{\mathbf{x}}(t) \Leftrightarrow q_i(t)$. We see that the spatial coordinate $\mathbf{x}$ plays the role of the index i labelling the oscillators q_i . The continuum analog of (1.33) therefore is

$$L = \frac{1}{2} \left[\int d^3x \dot{\phi}^2(t, \mathbf{x}) - \int d^3x d^3y M(\mathbf{x}, \mathbf{y}) \phi(t, \mathbf{x}) \phi(t, \mathbf{y}) \right], \quad (1.35)$$

where the function $M(\mathbf{x}, \mathbf{y})$ is still to be determined. Imposing Lorentz symmetry requires that $M(\mathbf{x}, \mathbf{y}) = [-\nabla^2 + m^2]\delta(\mathbf{x} - \mathbf{y})$, and hence the Lagrangian becomes

$$L = \frac{1}{2} \int d^3x \left[\dot{\phi}^2 - (\nabla\phi)^2 - m^2\phi^2 \right], \quad (1.36)$$

where $\dot{\phi}^2 - (\nabla\phi)^2 = -\eta^{\mu\nu}\partial_\mu\phi\partial_\nu\phi$ is the Lorentz-invariant “kinetic term”. To decouple the oscillators like in (1.34), we then write the field in terms of its Fourier expansion

$$\phi(t, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \phi_{\mathbf{k}}(t) e^{i\mathbf{k}\cdot\mathbf{x}}, \quad (1.37)$$

where $\phi_{\mathbf{k}}(t)$ are the complex “Fourier modes”. Note that for a real scalar field $\phi(t, \mathbf{x})$, we have $(\phi_{\mathbf{k}})^* = \phi_{-\mathbf{k}}$. Substituting (1.37) into (1.36), we find

$$L = \frac{1}{2} \int d^3k \left[\dot{\phi}_{\mathbf{k}}\dot{\phi}_{-\mathbf{k}} - \omega^2 \phi_{\mathbf{k}}\phi_{-\mathbf{k}} \right], \quad (1.38)$$

where $\omega^2 \equiv \mathbf{k}^2 + m^2$ is the frequency of each Fourier mode. Equation (1.38) is the continuum analog of (1.34).

1.3 Classical Solutions

We have found that the Lagrangian density of a free relativistic scalar field is

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2. \quad (1.39)$$

We will now discuss the canonical quantization of this theory, following the same procedure as for the harmonic oscillator described in Section 1.1.

First, we consider the classical dynamics of the field. From the Euler–Lagrange equation, we derive the Klein–Gordon equation for the field:

$$\frac{d}{dx^\mu} \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \right) = \frac{\partial\mathcal{L}}{\partial\phi} \quad \Rightarrow \quad (\partial_\mu\partial^\mu + m^2)\phi(x) = 0. \quad (1.40)$$

A complex solution of this equation can be written as

$$\phi(x^\mu) = \phi_0 e^{ik_\mu x^\mu} = \phi_0 e^{-i\omega t + i\mathbf{k}\cdot\mathbf{x}}, \quad (1.41)$$

where $k^\mu = (\omega, \mathbf{k})$, with $\omega^2 = \mathbf{k}^2 + m^2$. Note that ω can be positive or negative. We define the most general solution by constructing a complete, orthonormal set of modes. This requires us to first introduce an inner product on the space of solutions of the Klein–Gordon equation:

$$\langle\phi_1, \phi_2\rangle \equiv i \int_{\Sigma_t} d^3x (\phi_1^* \partial_t \phi_2 - \phi_2 \partial_t \phi_1^*), \quad (1.42)$$

where Σ_t is a constant-time hypersurface. It can be shown (using Stoke’s theorem and the Klein–Gordon equation) that this inner product is independent of the choice of hypersurface Σ_t . It is easy to see from its definition that the inner product satisfies $\langle\phi_1, \phi_2\rangle^* = \langle\phi_2, \phi_1\rangle$ and

$\langle \phi_1^*, \phi_2^* \rangle = -\langle \phi_2, \phi_1 \rangle$. The latter implies that $\langle \phi, \phi \rangle = -\langle \phi^*, \phi^* \rangle$, so that the inner product is *not* positive definite.

Applying this inner product to two solutions with different wave vectors gives

$$\begin{aligned} \langle e^{ik_1, \mu x^\mu}, e^{ik_2, \mu x^\mu} \rangle &= i \int_{\Sigma_t} d^3x \left(e^{i\omega_1 t - i\mathbf{k}_1 \cdot \mathbf{x}} \partial_t e^{-i\omega_2 t + i\mathbf{k}_2 \cdot \mathbf{x}} - e^{-i\omega_2 t + i\mathbf{k}_2 \cdot \mathbf{x}} \partial_t e^{i\omega_1 t - i\mathbf{k}_1 \cdot \mathbf{x}} \right) \\ &= (\omega_2 + \omega_1) e^{i(\omega_1 - \omega_2)t} \int_{\Sigma_t} d^3x e^{-i(\mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{x}} \\ &= (\omega_2 + \omega_1) e^{i(\omega_1 - \omega_2)t} (2\pi)^3 \delta(\mathbf{k}_1 - \mathbf{k}_2). \end{aligned} \quad (1.43)$$

We see that the inner product vanishes unless the two spatial wave vectors (and hence the frequencies) are equal for both modes. An orthonormal set of mode functions therefore is

$$f_{\mathbf{k}}(x^\mu) = \frac{e^{ik_\mu x^\mu}}{\sqrt{(2\pi)^3 2\omega}}, \quad (1.44)$$

so that

$$\langle f_{\mathbf{k}_1}, f_{\mathbf{k}_2} \rangle = \delta(\mathbf{k}_1 - \mathbf{k}_2). \quad (1.45)$$

Recall that $\mathbf{k}$ determines ω only up to a sign. Our convention will be to take $\omega \equiv \sqrt{\mathbf{k}^2 + m^2} > 0$, and create a complete set of mode functions by including the complex conjugate solutions $f_{\mathbf{k}}^*(x^\mu)$. The complex conjugate modes are orthogonal to the original modes and have negative norm:

$$\begin{aligned} \langle f_{\mathbf{k}_1}, f_{\mathbf{k}_2}^* \rangle &= 0, \\ \langle f_{\mathbf{k}_1}^*, f_{\mathbf{k}_2}^* \rangle &= -\delta(\mathbf{k}_1 - \mathbf{k}_2). \end{aligned} \quad (1.46)$$

The modes $f_{\mathbf{k}}$ are called *positive-frequency* solutions satisfying

$$\partial_t f_{\mathbf{k}} = -i\omega f_{\mathbf{k}}, \quad \omega > 0, \quad (1.47)$$

while $f_{\mathbf{k}}^*$ are *negative-frequency* solutions, with

$$\partial_t f_{\mathbf{k}}^* = +i\omega f_{\mathbf{k}}^*, \quad \omega > 0. \quad (1.48)$$

Together, $f_{\mathbf{k}}$ and $f_{\mathbf{k}}^*$ form a complete set, and any solution can be written as

$$\phi(t, \mathbf{x}) = \int d^3k \left[a_{\mathbf{k}} f_{\mathbf{k}}(t, \mathbf{x}) + a_{\mathbf{k}}^* f_{\mathbf{k}}^*(t, \mathbf{x}) \right], \quad (1.49)$$

where $a_{\mathbf{k}}$ is a complex constant (indexed by the wave vector $\mathbf{k}$).

The wave vector $\mathbf{k}$ is a continuous parameter. As we will see below, this can lead to “volume divergences”. These divergences can be avoided by putting theory in a box of side length L and volume $V = L^3$. Imposing periodic boundary conditions, the wave vectors then become quantized, $\mathbf{k} = (2\pi/L)\mathbf{n}$, where $\mathbf{n}$ is a vector of integer entries. We will use a discrete index to label these wave vectors. The mode functions are then written as f_i and the integral in (1.49) becomes an infinite sum:

$$\phi(t, \mathbf{x}) = \sum_i \left[a_i f_i(t, \mathbf{x}) + a_i^* f_i^*(t, \mathbf{x}) \right], \quad (1.50)$$

where

$$\begin{aligned}\langle f_i, f_j \rangle &= \delta_{ij}, \\ \langle f_i^*, f_j^* \rangle &= -\delta_{ij}, \\ \langle f_i, f_j^* \rangle &= 0.\end{aligned}\tag{1.51}$$

Below we will freely move between the discrete and continuous notation.

1.4 Canonical Quantization

To quantize the theory, we promote the field ϕ and its conjugate momentum $\pi = \partial\mathcal{L}/\partial\dot{\phi} = \dot{\phi}$ to operators $\hat{\phi}, \hat{\pi}$, and impose canonical commutation relations on equal-time hypersurfaces

$$\begin{aligned}[\hat{\phi}(t, \mathbf{x}), \hat{\phi}(t, \mathbf{x}')] &= 0, \\ [\hat{\pi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{x}')] &= 0, \\ [\hat{\phi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{x}')] &= i\delta(\mathbf{x} - \mathbf{x}').\end{aligned}\tag{1.52}$$

Just like the classical solution (1.49), the quantum operator $\hat{\phi}$ can be expanded in positive- and negative-frequency modes

$$\hat{\phi}(t, \mathbf{x}) = \int d^3k \left[\hat{a}_{\mathbf{k}} f_{\mathbf{k}}(t, \mathbf{x}) + \hat{a}_{\mathbf{k}}^\dagger f_{\mathbf{k}}^*(t, \mathbf{x}) \right],\tag{1.53}$$

where the operators $\hat{a}_{\mathbf{k}}$ and $\hat{a}_{\mathbf{k}}^\dagger$ must obey

$$\begin{aligned}[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{k}'}] &= 0, \\ [\hat{a}_{\mathbf{k}}^\dagger, \hat{a}_{\mathbf{k}'}^\dagger] &= 0, \\ [\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{k}'}^\dagger] &= \delta(\mathbf{k} - \mathbf{k}').\end{aligned}\tag{1.54}$$

These commutation relations imply that $\hat{a}_{\mathbf{k}}^\dagger$ and $\hat{a}_{\mathbf{k}}$ play the role of creation and annihilation operators.

Exercise 1.2 Prove the commutation relations (1.54) for the creation and annihilation operators, starting from (1.52). ➔

As in the case of the quantum oscillator in Section 1.1, we would like to express the Hamiltonian in terms of these operators. The classical Hamiltonian is

$$\begin{aligned}H &= \int d^3x (\dot{\phi}\pi - \mathcal{L}) \\ &= \int d^3x \left[\frac{1}{2}\dot{\phi}^2 + \frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}m^2\phi^2 \right].\end{aligned}\tag{1.55}$$

Promoting this to an operator, and substituting (1.53), one finds

$$\hat{H} = \frac{1}{2} \int d^3k \left[\hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} + \hat{a}_{\mathbf{k}} \hat{a}_{\mathbf{k}}^\dagger \right] \omega = \int d^3k \left[\hat{N}_{\mathbf{k}} + \frac{1}{2}\delta(0) \right] \omega,\tag{1.56}$$

where $\hat{N}_{\mathbf{k}} \equiv \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}}$.

Exercise 1.3 Derive the result (1.56).



We note that second term in (1.56) contains two types of divergences: First, the delta function is evaluated at zero where it is infinite. Second, the integral over $\omega = \sqrt{\mathbf{k}^2 + m^2}$ diverges at large $\mathbf{k}$. The delta-function divergence arises because we are integrating over an infinite space. To see this, we put the theory in a box of side length L and volume $V = L^3$. Taking the limit $L \rightarrow \infty$, we then get

$$(2\pi)^3 \delta(0) = \lim_{L \rightarrow \infty} \int_{-L/2}^{L/2} d^3x e^{i\mathbf{k} \cdot \mathbf{x}} \Big|_{\mathbf{k}=0} = \lim_{L \rightarrow \infty} \int_{-L/2}^{L/2} d^3x = V. \quad (1.57)$$

We can remove the *infrared divergence* associated with $\delta(0)$ by computing the energy *density*, rather than the total energy, i.e.

$$\rho_0 \equiv \frac{E_0}{V} = \int \frac{d^3k}{(2\pi)^3} \frac{\omega}{2}, \quad (1.58)$$

which is the sum of the ground state energies of each harmonic oscillator. The integral in (1.58) is still infinite because it diverges at high momenta. This *ultraviolet divergence* arises because we are extending the theory to infinite energies. This extrapolation is clearly naive. In reality, we expect the theory to break down at a finite energy, so the integral in (1.58) should contain a finite momentum cutoff. A more detailed description of this *renormalization* can be found in any QFT textbook.

As in the case of the harmonic oscillator in quantum mechanics, we use the creation and annihilation operators to construct a *Hilbert space* of states. The vacuum state is defined as

$$\hat{a}_{\mathbf{k}}|0\rangle = 0, \quad \text{for all } \mathbf{k}, \quad (1.59)$$

and a state with $n_{\mathbf{k}}$ excitations of identical momenta $\mathbf{k}$ is

$$|n_{\mathbf{k}}\rangle = \frac{1}{\sqrt{n_{\mathbf{k}}!}} (\hat{a}_{\mathbf{k}}^\dagger)^{n_{\mathbf{k}}} |0\rangle. \quad (1.60)$$

Similarly, a state with n_i excitations of distinct momenta $\mathbf{k}_i$ is

$$|n_1, n_2, \dots, n_N\rangle = \frac{1}{\sqrt{n_1! n_2! \dots n_N!}} (\hat{a}_{\mathbf{k}_1}^\dagger)^{n_1} (\hat{a}_{\mathbf{k}_2}^\dagger)^{n_2} \dots (\hat{a}_{\mathbf{k}_N}^\dagger)^{n_N} |0\rangle. \quad (1.61)$$

These states are eigenstates of the number operators $\hat{N}_{\mathbf{k}_i} = \hat{a}_{\mathbf{k}_i}^\dagger \hat{a}_{\mathbf{k}_i}$ and hence the Hamiltonian operator (1.56). It therefore makes sense to identify these states with “particles” of a fixed energy (and momentum).

An important feature of QFT in flat spacetime is that all inertial observers agree on the choice of vacuum and the associated particle excitations. To illustrate this, consider a Lorentz boost with velocity $\mathbf{v} = d\mathbf{x}/dt$. The new coordinates are

$$t' = \gamma t - \gamma \mathbf{v} \cdot \mathbf{x}, \quad \mathbf{x}' = \gamma \mathbf{x} - \gamma \mathbf{v} t, \quad (1.62)$$

where $\gamma \equiv \sqrt{1 - v^2}$. We will also need that the inverse transformation is

$$t = \gamma t' + \gamma \mathbf{v} \cdot \mathbf{x}', \quad \mathbf{x} = \gamma \mathbf{x}' + \gamma \mathbf{v} t'. \quad (1.63)$$

The time derivative of the mode function in the boosted frame then is

$$\begin{aligned} \partial_{t'} f_{\mathbf{k}} &= \frac{\partial x^\mu}{\partial t'} \partial_\mu f_{\mathbf{k}} \\ &= \gamma(-i\omega) f_{\mathbf{k}} + \gamma \mathbf{v} \cdot (i\mathbf{k}) f_{\mathbf{k}} \\ &\equiv -i\omega' f_{\mathbf{k}}, \end{aligned} \quad (1.64)$$

where we have defined the boosted frequency

$$\omega' \equiv \gamma\omega - \gamma \mathbf{v} \cdot \mathbf{k}. \quad (1.65)$$

We see that the functions are eigenfunctions of the time evolution operators ∂_t and $\partial_{t'}$. All inertial observers will therefore agree on the separation into positive- and negative-frequency modes. This also means that these observers will agree on the definition of creation and annihilation operators, and hence on a unique vacuum state. A state containing a collection of particles with certain momenta will simply be boosted into a state with same particles, but with boosted momenta. The number of particles in a given state is an invariant. This will *not* be true in a curved spacetime.

2 QFT in Curved Spacetime

We are now ready to attack the problem of quantum fields living on a curved spacetime. Although much of it will be a straightforward generalization of the canonical quantization of fields in flat spacetime, we will also encounter some conceptual subtleties. Most importantly, in a non-stationary spacetime, there is no unique separation into positive- and negative-frequency modes. This means that different observers might not agree on the choice of vacuum and the associated particle excitations. What one observer might declare to be the vacuum state, another might find to be a state full of particles. This effect becomes especially interesting in spacetimes with horizons.

2.1 From Flat to Curved

There is a straightforward procedure to take a theory defined in flat spacetime and generalize it to curved spacetime. We will illustrate this for the case of a free relativistic scalar field, which in flat space has the action

$$S = \int d^4x \left[-\frac{1}{2} \eta^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} m^2 \phi^2 \right]. \quad (2.1)$$

We then make the following substitutions: 1) The Minkowski metric $\eta_{\mu\nu}$ is replaced by the metric $g_{\mu\nu}$ of the curved spacetime; 2) The partial derivatives ∂_μ are replaced by covariant derivatives ∇_μ ; 3) The integral measure d^4x is replaced by $\sqrt{-g} d^4x$, where $g = \det(g_{\mu\nu})$ is the determinant of the metric. The action (2.1) then becomes

$$S = \int d^4x \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{2} m^2 \phi^2 \right]. \quad (2.2)$$

By construction, this action reduces to the flat-space action in a local inertial frame, where $g_{\mu\nu} \rightarrow \eta_{\mu\nu}$ and $\nabla_\mu \rightarrow \partial_\mu$. The Klein–Gordon equation for the field is

$$(\square + m^2) \phi = 0, \quad (2.3)$$

where $\square$ is the d'Alembertian on the curved spacetime:

$$\square \phi \equiv g^{\mu\nu} \nabla_\mu \nabla_\nu \phi = \frac{1}{\sqrt{-g}} \partial_\mu (\sqrt{-g} g^{\mu\nu} \partial_\nu \phi). \quad (2.4)$$

Equation (2.3) can be obtained either by varying the action (2.2) or by performing the above substitutions directly at the level of the equation of motion (1.40).

There is one more term that we could add to the action (2.2), which wouldn't change the flat-space limit, namely

$$\delta S = \int d^4x \sqrt{-g} \left(-\frac{1}{2} \xi R \phi^2 \right), \quad (2.5)$$

where ξ is a dimensionless constant and R is the Ricci scalar. This term is still quadratic in the field ϕ and vanishes in flat space, where $R = 0$. The choice $\xi = 0$ is called *minimal coupling*. Another special value is $\xi = 1/6$, where the theory exhibits *conformal invariance* for $m^2 = 0$, i.e. the action is invariant under the transformation $g_{\mu\nu} \rightarrow \Omega^2(x) g_{\mu\nu}$ and $\phi \rightarrow \Omega^{-1}(x) \phi$.

Exercise 2.1 Consider a free massless scalar field in D dimensions. A conformal transformation is

$$g_{\mu\nu} \rightarrow \bar{g}_{\mu\nu} = \Omega^2(x) g_{\mu\nu} \quad \text{and} \quad \phi \rightarrow \bar{\phi} = \Omega^{-\Delta}(x) \phi, \quad (2.6)$$

where $\Omega(x)$ is an arbitrary function and Δ is a constant parameter (called the scaling dimension of the field). Using that

$$R \rightarrow \bar{R} = \Omega^{-2} R - 2(D-1) \Omega^{-3} g^{\mu\nu} \nabla_\mu \nabla_\nu \Omega - (D-1)(D-4) \Omega^{-4} g^{\mu\nu} \nabla_\mu \Omega \nabla_\nu \Omega, \quad (2.7)$$

determine the values of ξ and Δ for which the action is invariant under this transformation. $\rightarrow$

2.2 Classical Solutions

The solutions to the Klein–Gordon equation (2.3) will depend on the form of the spacetime metric $g_{\mu\nu}(t, \mathbf{x})$. We will study explicit examples below. Given two solutions ϕ_1 and ϕ_2 , we define their inner product as

$$\langle \phi_1, \phi_2 \rangle \equiv i \int_\Sigma d\Sigma^\mu \left(\phi_1^* \nabla_\mu \phi_2 - \phi_2 \nabla_\mu \phi_1^* \right) \equiv \int_\Sigma d\Sigma^\mu J_\mu, \quad (2.8)$$

where Σ is a spacelike surface. Since $\phi_{1,2}$ satisfy the Klein–Gordon equation, the integrand is a conserved current:

$$\begin{aligned} \nabla^\mu J_\mu &= i \phi_1^* (\nabla^\mu \nabla_\mu \phi_2) - i \phi_2 (\nabla^\mu \nabla_\mu \phi_1^*) \\ &= i \phi_1^* (-m^2 \phi_2) - i \phi_2 (-m^2 \phi_1^*) = 0. \end{aligned} \quad (2.9)$$

This also implies that inner product (ϕ_1, ϕ_2) does not depend on the choice of the hypersurface Σ , if the fields decay sufficiently fast at spatial infinity. To see this, consider another spacelike hypersurface Σ' . We then have

$$\int_\Sigma d\Sigma^\mu J_\mu - \int_{\Sigma'} d\Sigma^\mu J_\mu = \int_M d^4x \sqrt{-g} \nabla^\mu J_\mu = 0, \quad (2.10)$$

where M is the spacetime region bounded by Σ and Σ' .

In the Hamiltonian description of the theory, we introduce a preferred time coordinate t and write the spacetime coordinates as $x^\mu = (t, x^i)$. The inner product (2.8) can then be written as

$$\langle \phi_1, \phi_2 \rangle = i \int_\Sigma d^3x \sqrt{\gamma} n^\mu \left(\phi_1^* \nabla_\mu \phi_2 - \phi_2 \nabla_\mu \phi_1^* \right), \quad (2.11)$$

where $\gamma = \det(\gamma_{ij})$ is the determinant of the induced metric of the spacelike hypersurface and $n^\mu = g^{\mu 0} / \sqrt{g^{00}}$ is the normal vector.

Using the inner product, we can define a set of orthonormal mode functions:

$$\begin{aligned} \langle f_i, f_j \rangle &= \delta_{ij}, \\ \langle f_i^*, f_j^* \rangle &= -\delta_{ij}, \\ \langle f_i, f_j^* \rangle &= 0, \end{aligned} \quad (2.12)$$

where $f_i(x^\mu)$ is a complex solution of the Klein–Gordon equation (2.3). The index i may be continuous or discrete. For simplicity, we will write expressions in discrete form. A general solution can then be written as

$$\phi(x^\mu) = \sum_i \left[a_i f_i(x^\mu) + a_i^* f_i^*(x^\mu) \right]. \quad (2.13)$$

The problem is that the normalization in (2.12) doesn't fix the solutions $f_i(x^\mu)$ uniquely. In flat space, we therefore imposed the extra condition that $f_{\mathbf{k}}$ are positive-frequency modes satisfying $\partial_t f_{\mathbf{k}} = -i\omega f_{\mathbf{k}}$. We also showed that this condition is preserved under Lorentz boosts. In a non-stationary spacetime, however, we don't have a timelike Killing vector and therefore cannot classify modes as positive- or negative-frequency. This means that instead of $f_i(x^\mu)$ we could have chosen an alternative set of orthonormal functions $g_i(x^\mu)$ and expanded the field as

$$\phi(x^\mu) = \sum_i \left[b_i g_i(x^\mu) + b_i^* g_i^*(x^\mu) \right]. \quad (2.14)$$

In the quantum theory, this ambiguity will lead to an ambiguity of the vacuum state.

2.3 Canonical Quantization

To quantize the theory, we proceed as before: We promote the field ϕ and its conjugate momentum $\pi = \partial\mathcal{L}/\partial(\partial_0\phi) = \sqrt{-g}g^{0\mu}(\partial_\mu\phi)$ to operators $\hat{\phi}, \hat{\pi}$, and impose canonical commutation relations:

$$\begin{aligned} [\hat{\phi}(t, \mathbf{x}), \hat{\phi}(t, \mathbf{x}')] &= 0, \\ [\hat{\pi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{x}')] &= 0, \\ [\hat{\phi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{x}')] &= i\delta(\mathbf{x} - \mathbf{x}'), \end{aligned} \quad (2.15)$$

where now the δ -function is normalized with respect to the volume form on the hypersurface.

Given a complete set of orthonormal mode functions f_i , the quantum operator $\hat{\phi}$ can be expanded as

$$\hat{\phi}(x^\mu) = \sum_i \left[\hat{a}_i f_i(x^\mu) + \hat{a}_i^\dagger f_i^*(x^\mu) \right], \quad (2.16)$$

where the operators $\hat{a}_i$ and $\hat{a}_i^\dagger$ must obey

$$\begin{aligned} [\hat{a}_i, \hat{a}_j] &= 0, \\ [\hat{a}_i^\dagger, \hat{a}_j^\dagger] &= 0, \\ [\hat{a}_i, \hat{a}_j^\dagger] &= \delta_{ij}. \end{aligned} \quad (2.17)$$

These commutation relations imply that $\hat{a}_i^\dagger$ and $\hat{a}_i$ play the role of creation and annihilation operators. The state annihilated by the operators $\hat{a}_i$ is the “ a -vacuum”:

$$\hat{a}_i |0_a\rangle = 0, \quad \text{for all } i, \quad (2.18)$$

where role of the subscript on $|0_a\rangle$ will become clear in a moment.

As we discussed before, in a non-stationary spacetime, there is no unique way to define the mode expansion. Using an alternative set of mode functions $g_i(x^\mu)$, we can write

$$\hat{\phi}(x^\mu) = \sum_i \left[\hat{b}_i g_i(x^\mu) + \hat{b}_i^\dagger g_i^*(x^\mu) \right], \quad (2.19)$$

where

$$\begin{aligned} [\hat{b}_i, \hat{b}_j] &= 0, \\ [\hat{b}_i^\dagger, \hat{b}_j^\dagger] &= 0, \\ [\hat{b}_i, \hat{b}_j^\dagger] &= \delta_{ij}. \end{aligned} \quad (2.20)$$

We now have an alternative set of creation and annihilation operators $\hat{b}_i^\dagger$ and $\hat{b}_i$. The state annihilated by the operators $\hat{b}_i$ is the “ b -vacuum”:

$$\hat{b}_i |0_b\rangle = 0, \quad \text{for all } i, \quad (2.21)$$

where the subscript on $|0_b\rangle$ is used to distinguish it from the “ a -vacuum” $|0_a\rangle$ defined in (2.18).

What we have lost in the transition from flat to curved spacetime is any reason to prefer one set of modes over any other. In flat spacetime, we defined positive-frequency modes with respect to the time coordinate. Although the time coordinate is not unique, since it changes under Lorentz transformations, we have seen that the vacuum state and the number operator are invariant under such transformations. Every inertial observer therefore agrees on the vacuum state and the number of particles in an excited state. This is *not* the case in a curved spacetime.

2.4 Bogolubov Transformations

Since f_i are a complete set of functions, we can write the alternative mode functions g_i as

$$g_i \equiv \sum_j \left(\alpha_{ij} f_j + \beta_{ij} f_j^* \right). \quad (2.22)$$

This transformation from one set of mode functions to another is called a *Bogolubov transformation*, and the matrices α_{ij} and β_{ij} are Bogolubov coefficients. Using (2.12), we have

$$\begin{aligned} \langle f_i, g_j \rangle &= \langle f_i, \sum_k [\alpha_{jk} f_k + \beta_{jk} f_k^*] \rangle \\ &= \sum_k [\alpha_{jk} \langle f_i, f_k \rangle + \beta_{jk} \langle f_i, f_k^* \rangle] \\ &= \sum_k [\alpha_{jk} \delta_{ik} + \beta_{jk} \times 0] = \alpha_{ji}, \end{aligned} \quad (2.23)$$

$$\begin{aligned} \langle f_i^*, g_j \rangle &= \langle f_i^*, \sum_k [\alpha_{jk} f_k + \beta_{jk} f_k^*] \rangle \\ &= \sum_k [\alpha_{jk} \langle f_i^*, f_k \rangle + \beta_{jk} \langle f_i^*, f_k^* \rangle] \\ &= \sum_k [\alpha_{jk} \times 0 - \beta_{jk} \delta_{ik}] = -\beta_{ji}, \end{aligned} \quad (2.24)$$

and hence the Bogolubov coefficients are

$$\begin{aligned}\alpha_{ij} &= \langle f_j, g_i \rangle, \\ \beta_{ij} &= -\langle f_j^*, g_i \rangle.\end{aligned}\tag{2.25}$$

Conversely, we may also write

$$f_i \equiv \sum_j \left(\lambda_{ij} g_j + \mu_{ij} g_j^* \right),\tag{2.26}$$

where

$$\begin{aligned}\lambda_{ij} &= \langle g_j, f_i \rangle = \langle f_i, g_j \rangle^* = \alpha_{ji}^*, \\ \mu_{ij} &= -\langle g_j^*, f_i \rangle = \langle f_i^*, g_j \rangle = -\beta_{ji},\end{aligned}\tag{2.27}$$

so that

$$f_i = \sum_j \left(\alpha_{ji}^* g_j - \beta_{ji} g_j^* \right).\tag{2.28}$$

Substituting (2.28) into (2.16), we find

$$\begin{aligned}\hat{\phi}(x^\mu) &= \sum_i \left[\hat{a}_i f_i(x^\mu) + \hat{a}_i^\dagger f_i^*(x^\mu) \right] \\ &= \sum_i \left[\hat{a}_i \sum_j \left(\alpha_{ji}^* g_j - \beta_{ji} g_j^* \right) + \hat{a}_i^\dagger \sum_j \left(\alpha_{ji} g_j^* - \beta_{ji}^* g_j \right) \right] \\ &= \sum_j \left[\sum_i \left(\alpha_{ji}^* \hat{a}_i - \beta_{ji}^* \hat{a}_i^\dagger \right) g_j + \sum_i \left(\alpha_{ji} \hat{a}_i^\dagger - \beta_{ji} \hat{a}_i \right) g_j^* \right],\end{aligned}\tag{2.29}$$

and comparing this to (2.19), we infer that

$$\hat{b}_i = \sum_j \left(\alpha_{ij}^* \hat{a}_j - \beta_{ij}^* \hat{a}_j^\dagger \right).\tag{2.30}$$

Similarly, substituting (2.22) into (2.19) and comparing the result to (2.16), we find

$$\hat{a}_i = \sum_j \left(\alpha_{ji} \hat{b}_j - \beta_{ji}^* \hat{b}_j^\dagger \right).\tag{2.31}$$

We see that the Bogolubov transformation mixes annihilation and creation operators. The vacuum with respect to one set of oscillators is therefore a populated state with respect to the other set (if $\beta_{ij} \neq 0$).

Exercise 2.2 Show that the commutation relation $[\hat{b}_i, \hat{b}_j^\dagger] = \delta_{ij}$ implies the constraint

$$\delta_{ij} = \sum_k (\alpha_{ik}^* \alpha_{jk} - \beta_{ik}^* \beta_{jk}).\tag{2.32}$$

Additional constraints can be derived from $[\hat{a}_i, \hat{a}_j^\dagger] = \delta_{ij}$ and $[\hat{a}_i, \hat{a}_j] = [\hat{b}_i, \hat{b}_j] = 0$. ➔

What is the vacuum?

Consider the a -vacuum $|0_a\rangle$. How many b -particles does it contain? To answer this, we compute the expectation value of the number operator $\hat{N}_{b,i}$ in the a -vacuum:

$$\begin{aligned}
\langle 0_a | \hat{N}_{b,i} | 0_a \rangle &= \langle 0_a | \hat{b}_i^\dagger \hat{b}_i | 0_a \rangle \\
&= \langle 0_a | \sum_{j,k} \left(\alpha_{ij} \hat{a}_j^\dagger - \beta_{ij} \hat{a}_j \right) \left(\alpha_{ik}^* \hat{a}_k - \beta_{ik}^* \hat{a}_k^\dagger \right) | 0_a \rangle \\
&= \sum_{j,k} (-\beta_{ij}) (-\beta_{ik}^*) \langle 0_a | \hat{a}_j \hat{a}_k^\dagger | 0_a \rangle \\
&= \sum_{j,k} \beta_{ij} \beta_{ik}^* \langle 0_a | \left(\hat{a}_k^\dagger \hat{a}_j + \delta_{jk} \right) | 0_a \rangle \\
&= \sum_j |\beta_{ij}|^2.
\end{aligned} \tag{2.33}$$

We see that this will not be zero if $\beta_{ij} \neq 0$, for any j . Looking at (2.30), this is no surprise: A nonzero β_{ij} implies that the annihilation operators $\hat{b}_i$ contain an admixture of the creation operators $\hat{a}_i$.

2.5 Cosmological Particle Production

We will now study particle production in a cosmological spacetime. The metric is

$$ds^2 = -dt^2 + a^2(t) \gamma_{ij} dx^i dx^j, \tag{2.34}$$

where the scale factor $a(t)$ describes the cosmological expansion and γ_{ij} is the metric on the homogeneous and isotropic spatial slices. For simplicity, we will assume the spatial geometry to be flat, so that $\gamma_{ij} = \delta_{ij}$ (in Cartesian coordinates). In terms of the conformal time, $\eta = \int dt/a(t)$, the metric then becomes

$$ds^2 = a^2(\eta) [-d\eta^2 + \delta_{ij} dx^i dx^j]. \tag{2.35}$$

This form of the metric is convenient when thinking about the causal structure of the spacetime, since null geodesics are simply straight lines (as in Minkowski space).

Scalar field dynamics We will consider the evolution of a massive scalar field in this spacetime. Evaluating the action (2.2) in the spacetime (2.35), we get

$$S = \frac{1}{2} \int d\eta d^3x a^2 \left[(\phi')^2 - (\nabla\phi)^2 - m^2 a^2 \phi^2 \right], \tag{2.36}$$

where the prime denotes a derivative with respect to the conformal time η . It is convenient to define $f \equiv a(\eta)\phi$, so that

$$\begin{aligned}
a^2(\phi')^2 &= a^2[(f/a)']^2 \\
&= a^2[f'/a - (a'/a^2)f]^2 \\
&= (f')^2 - 2(a'/a)f'f + (a'/a)^2 f^2 \\
&= (f')^2 - (a'/a)(f^2)' + (a'/a)^2 f^2 \\
&= (f')^2 - [(a'/a)f^2]' + (a'/a)'f^2 + (a'/a)^2 f^2 \\
&= (f')^2 + (a''/a)f^2 - [(a'/a)f^2]'.
\end{aligned} \tag{2.37}$$

Dropping the total-derivative term, the action (2.36) becomes

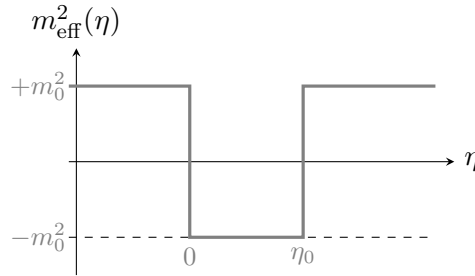
$$S = \frac{1}{2} \int d\eta d^3x \left[(f')^2 - (\nabla f)^2 - \left(m^2 a^2 - \frac{a''}{a} \right) f^2 \right], \quad (2.38)$$

and the equation of motion for the field is

$$f'' - \nabla^2 f + \left(m^2 a^2 - \frac{a''}{a} \right) f = 0. \quad (2.39)$$

This is the same equation as for a field in Minkowski space, but with a time-dependent effective mass, $m_{\text{eff}}^2(\eta) \equiv m^2 a^2 - a''/a$. It is this time dependence that can lead to spontaneous particle production in a cosmological spacetime.

Particle creation To illustrate this effect, we will use a simple toy model where the effective mass in (2.39) evolves as:



This connects two asymptotic Minkowski regions where the positive-frequency solutions are

$$f_k^{(\text{in})}(\eta) = \frac{1}{\sqrt{2\omega_k}} e^{-i\omega_k \eta}, \quad \eta < 0, \quad (2.40)$$

$$f_k^{(\text{out})}(\eta) = \frac{1}{\sqrt{2\omega_k}} e^{-i\omega_k(\eta-\eta_0)}, \quad \eta > \eta_0, \quad (2.41)$$

with $\omega_k \equiv \sqrt{k^2 + m_0^2}$. Due to the time dependence of the effective mass, a positive-frequency solution in the “in” region will evolve into a sum of positive- and negative-frequency solutions in the “out” region:

$$f_k^{(\text{out})}(\eta) = \frac{\alpha_k}{\sqrt{2\omega_k}} e^{-i\omega_k(\eta-\eta_0)} + \frac{\beta_k}{\sqrt{2\omega_k}} e^{i\omega_k(\eta-\eta_0)}, \quad \eta > \eta_0. \quad (2.42)$$

To determine the coefficients of this Bogolubov transformation, we write the general solution in the transition region $0 < \eta < \eta_0$ as

$$f_k(\eta) = \frac{A_k}{\sqrt{2\Omega_k}} e^{-i\Omega_k \eta} + \frac{B_k}{\sqrt{2\Omega_k}} e^{i\Omega_k \eta}, \quad 0 < \eta < \eta_0, \quad (2.43)$$

where $\Omega_k \equiv \sqrt{k^2 - m_0^2}$. We first match this solution (and its first derivative) to the positive-frequency solution (2.40) at $\eta = 0$. This gives

$$f_k(0) = f_k^{(\text{in})}(0) \Rightarrow A_k + B_k = \sqrt{\Omega_k/\omega_k}, \quad (2.44)$$

$$\dot{f}_k(0) = \dot{f}_k^{(\text{in})}(0) \Rightarrow A_k - B_k = \sqrt{\omega_k/\Omega_k}, \quad (2.45)$$

and hence

$$A_k = \frac{1}{2} \left(\sqrt{\frac{\Omega_k}{\omega_k}} + \sqrt{\frac{\omega_k}{\Omega_k}} \right), \quad (2.46)$$

$$B_k = \frac{1}{2} \left(\sqrt{\frac{\Omega_k}{\omega_k}} - \sqrt{\frac{\omega_k}{\Omega_k}} \right). \quad (2.47)$$

With these coefficients fixed, we then match the solutions (2.43) and (2.42) at $\eta = \eta_0$. This gives

$$f_k(\eta_0) = f_k^{(\text{out})}(\eta_0) \Rightarrow \alpha_k + \beta_k = \sqrt{\frac{\omega_k}{\Omega_k}} \left(A_k e^{-i\Omega_k \eta_0} + B_k e^{i\Omega_k \eta_0} \right), \quad (2.48)$$

$$\dot{f}_k(\eta_0) = \dot{f}_k^{(\text{out})}(\eta_0) \Rightarrow \alpha_k - \beta_k = \sqrt{\frac{\Omega_k}{\omega_k}} \left(A_k e^{-i\Omega_k \eta_0} - B_k e^{i\Omega_k \eta_0} \right), \quad (2.49)$$

and hence

$$\alpha_k = \frac{A_k}{2} \left(\sqrt{\frac{\omega_k}{\Omega_k}} + \sqrt{\frac{\Omega_k}{\omega_k}} \right) e^{-i\Omega_k \eta_0} + \frac{B_k}{2} \left(\sqrt{\frac{\omega_k}{\Omega_k}} - \sqrt{\frac{\Omega_k}{\omega_k}} \right) e^{i\Omega_k \eta_0}, \quad (2.50)$$

$$\beta_k = \frac{A_k}{2} \left(\sqrt{\frac{\omega_k}{\Omega_k}} - \sqrt{\frac{\Omega_k}{\omega_k}} \right) e^{-i\Omega_k \eta_0} + \frac{B_k}{2} \left(\sqrt{\frac{\omega_k}{\Omega_k}} + \sqrt{\frac{\Omega_k}{\omega_k}} \right) e^{i\Omega_k \eta_0}. \quad (2.51)$$

Substituting (2.46) and (2.47), this becomes

$$\alpha_k = \frac{1}{4} \left(\sqrt{\frac{\omega_k}{\Omega_k}} + \sqrt{\frac{\Omega_k}{\omega_k}} \right)^2 e^{-i\Omega_k \eta_0} - \frac{1}{4} \left(\sqrt{\frac{\omega_k}{\Omega_k}} - \sqrt{\frac{\Omega_k}{\omega_k}} \right)^2 e^{i\Omega_k \eta_0}, \quad (2.52)$$

$$\beta_k = \frac{1}{2} \left(\frac{\omega_k}{\Omega_k} - \frac{\Omega_k}{\omega_k} \right) \sin(\Omega_k \eta_0). \quad (2.53)$$

In the out-region, the in-vacuum $|0\rangle_{\text{in}}$ contains particles, with the density of a mode $\mathbf{k}$ given by

$$n_k \equiv |\beta_k|^2 = \frac{m_0^4}{|k^4 - m_0^4|} \left| \sin \left(\sqrt{k^2 - m_0^2} \eta_0 \right) \right|^2. \quad (2.54)$$

It is interesting to consider two limits of this result:

- For $k \gg m_0$, the particles are ultra-relativistic and their density is highly suppressed:

$$k \gg m_0 : \quad n_k \approx \frac{m_0^4}{k^4} \sin^2(k\eta_0) \ll 1. \quad (2.55)$$

We conclude that very few relativistic particles are created.

- For $k \ll m_0$, the particles are non-relativistic and the density becomes

$$k \ll m_0 : \quad n_k \approx \sinh^2(m_0 \eta_0). \quad (2.56)$$

If $m_0 \eta_0 \gg 1$, then the density of produced particles is exponentially large.

Although these facts were derived for a simple toy model, these qualitative features of cosmological particle production remain valid more generally.

Exercise 2.3 Consider a two-dimensional FRW spacetime, $ds^2 = a^2(\eta)(-d\eta^2 + dx^2)$, with scale factor [7]

$$a(\eta) = \sqrt{A + B \tanh(\rho\eta)}, \quad (2.57)$$

where A , B and ρ are constants. This scale factor interpolates between two Minkowski regions in the asymptotic past and future (see Fig. 2).

1. The mode function of massive scalar field, $\phi_k(\eta, x) \equiv f_k(\eta)e^{ikx}/\sqrt{2\pi}$, satisfies the equation of motion $f_k'' + (k^2 + m^2 a^2(\eta))f_k = 0$. For the scale factor (2.57), this equation can be solved in terms of a hypergeometric function:

$$f_k^{(\text{in})}(\eta) = \frac{g(\omega_{\pm}, \eta)}{\sqrt{2\omega_{\text{in}}}} {}_2F_1 \left[\begin{matrix} 1 + \frac{i\omega_-}{\rho}, \frac{i\omega_-}{\rho} \\ 1 - \frac{i\omega_{\text{in}}}{\rho} \end{matrix} \middle| \frac{1}{2}(1 + \tanh(\rho\eta)) \right], \quad (2.58)$$

$$f_k^{(\text{out})}(\eta) = \frac{g(\omega_{\pm}, \eta)}{\sqrt{2\omega_{\text{out}}}} {}_2F_1 \left[\begin{matrix} 1 + \frac{i\omega_-}{\rho}, \frac{i\omega_-}{\rho} \\ 1 - \frac{i\omega_{\text{out}}}{\rho} \end{matrix} \middle| \frac{1}{2}(1 - \tanh(\rho\eta)) \right], \quad (2.59)$$

where

$$g(\omega_{\pm}, \eta) \equiv \exp \left[-i\omega_+ \eta - \frac{i\omega_-}{\rho} \log(2 \cosh(\rho\eta)) \right], \quad (2.60)$$

with

$$\begin{aligned} \omega_{\text{in}} &\equiv \sqrt{k^2 + m^2(A - B)}, \\ \omega_{\text{out}} &\equiv \sqrt{k^2 + m^2(A + B)}, \\ \omega_{\pm} &\equiv \frac{1}{2}(\omega_{\text{out}} \pm \omega_{\text{in}}). \end{aligned} \quad (2.61)$$

Show that these solutions have the correct limits as $\eta \rightarrow \pm\infty$.

2. The positive-frequency mode functions in the “in” and “out” regions are different. Writing $f_k^{(\text{in})} = \alpha_k f_k^{(\text{out})} + \beta_k f_k^{(\text{out})*}$, show that

$$|\alpha_k|^2 = \frac{\sinh^2(\pi\omega_+/\rho)}{\sinh(\pi\omega_{\text{in}}/\rho) \sinh(\pi\omega_{\text{out}}/\rho)}, \quad (2.62)$$

$$|\beta_k|^2 = \frac{\sinh^2(\pi\omega_-/\rho)}{\sinh(\pi\omega_{\text{in}}/\rho) \sinh(\pi\omega_{\text{out}}/\rho)}. \quad (2.63)$$

Hint: You may use the identity

$$\begin{aligned} {}_2F_1 \left[\begin{matrix} a, b \\ c \end{matrix} \middle| z \right] &= \frac{\Gamma(c)\Gamma(c-a-b)}{\Gamma(c-a)\Gamma(c-b)} {}_2F_1 \left[\begin{matrix} a, b \\ 1+a+b-c \end{matrix} \middle| 1-z \right] \\ &+ \frac{\Gamma(c)\Gamma(a+b-c)}{\Gamma(a)\Gamma(b)} (1-z)^{c-a-b} z^{1-c} {}_2F_1 \left[\begin{matrix} 1-b, 1-a \\ 1+c-a-b \end{matrix} \middle| 1-z \right]. \end{aligned} \quad (2.64)$$

3. Comment on the number density of produced particles, $n_k \equiv |\beta_k|^2$, for $m = 0$.
4. Show that the density of particles is suppressed for large k and large m . ➔

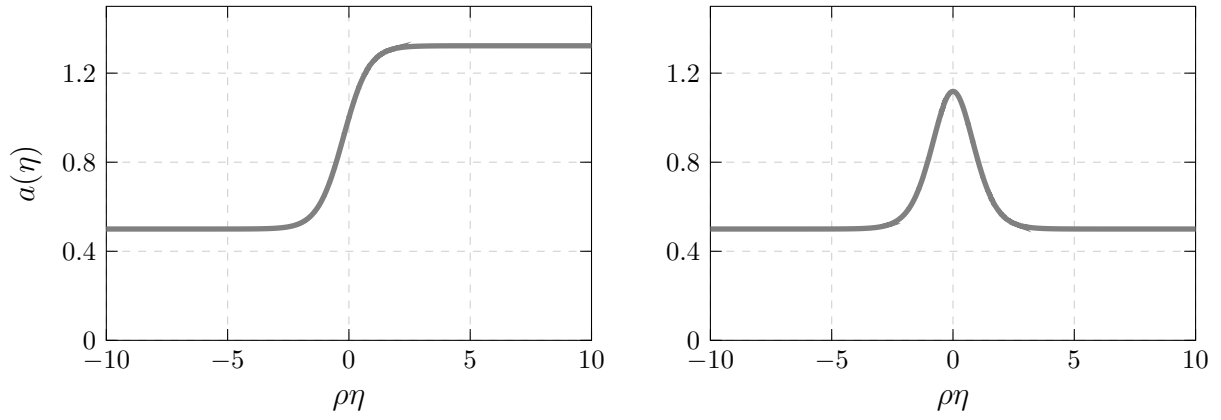


Figure 2: Sketch of the scale factors in the models in Exercise 2.3 (*left*) and in Exercise 2.4 (*right*).

Exercise 2.4 Perform the same analysis as in Exercise 2.3, but with the scale factor [8]

$$a(\eta) = \sqrt{A + B \operatorname{sech}^2(\rho\eta)}, \quad (2.65)$$

where A , B and ρ are constants. A sketch of this scale factor can be found in Fig. 2.

Show that the number density of produced particles with momentum k is

$$|\beta_k|^2 = \frac{\cos^2(\pi\kappa)}{\sinh^2(\pi\omega_{\text{in}}/\rho)}, \quad (2.66)$$

where κ and ω_{in} are to be determined. Comment on the values of κ for which this density vanishes. →

3 Rindler Space and Unruh Effect

Having introduced the general formalism of QFT in curved spacetime, we are now ready to study some applications. We will begin with the so-called “Unruh effect”, which refers to the phenomenon that an accelerating observer in Minkowski space sees particles with a thermal spectrum. For simplicity, we will compute this effect for massless particles in two spacetime dimensions. We will see, in the next chapter, that this also provides a useful toy model for Hawking radiation.

3.1 Accelerating Observer

We begin by deriving the trajectory of a uniformly accelerating observer in two-dimensional Minkowski spacetime:

$$ds^2 = -dt^2 + dx^2 = \eta_{\mu\nu} dx^\mu dx^\nu. \quad (3.1)$$

The observer’s trajectory is described by the function $x^\mu(\tau)$, where τ is proper time. The two-velocity of the observer is $u^\mu \equiv dx^\mu/d\tau$ and its acceleration is $a^\mu \equiv du^\mu/d\tau$. The timelike velocity satisfies the constraint $u^\mu u_\mu = -1$ and constant acceleration implies $a^\mu a_\mu = \alpha^2$, where α is a constant parameter. Taking a time derivative of $u^\mu u_\mu = -1$, we also find that $a^\mu u_\mu = 0$. Note that in the instantaneous rest-frame, we have $u^\mu = (1, 0)$ and $a^\mu = (0, \alpha)$.

We will now derive the function $x^\mu(\tau)$ for the case of constant acceleration. It is convenient to work in the lightcone coordinates of the inertial frame:

$$u \equiv t - x, \quad v \equiv t + x. \quad (3.2)$$

The metric (3.1) in these coordinates becomes

$$ds^2 = -du dv \equiv g_{\mu\nu} dx^\mu dx^\nu, \quad g_{\mu\nu} \equiv \begin{pmatrix} 0 & -1/2 \\ -1/2 & 0 \end{pmatrix}, \quad (3.3)$$

where $x^0 \equiv u$ and $x^1 \equiv v$. The trajectory of the observer is $x^\mu(\tau) = (u(\tau), v(\tau))$ and the constraints $g_{\mu\nu} u^\mu u^\nu = -1$ and $g_{\mu\nu} a^\mu a^\nu = \alpha^2$ become

$$\dot{u} \dot{v} = 1, \quad (3.4)$$

$$\ddot{u} \ddot{v} = -\alpha^2. \quad (3.5)$$

Using $\dot{u} = 1/\dot{v}$, we have $\ddot{u} = -\ddot{v}/\dot{v}^2$ and (3.5) becomes

$$\left(\frac{\ddot{v}}{\dot{v}}\right)^2 = \alpha^2 \quad \implies \quad \frac{d \log \dot{v}}{d\tau} = \alpha \quad \implies \quad v(\tau) = \frac{A}{\alpha} e^{\alpha\tau} + B, \quad (3.6)$$

where A and B are integration constants. Since $\dot{u} = 1/\dot{v}$, we also have

$$u(\tau) = -\frac{1}{A\alpha} e^{-\alpha\tau} + C, \quad (3.7)$$

where C is another integration constant. Note that $u \rightarrow \lambda u$ and $v \rightarrow v/\lambda$, with λ a constant, is a Lorentz transformation that leaves the metric (3.3) invariant. Using this freedom, we can

set $A \equiv 1$. Moreover, by shifting the origin of the inertial frame, we can achieve that $B = C \equiv 0$. We then have

$$u(\tau) = -\frac{1}{\alpha}e^{-\alpha\tau}, \quad v(\tau) = \frac{1}{\alpha}e^{\alpha\tau}. \quad (3.8)$$

Going back to the original coordinates $t = (u + v)/2$ and $x = (v - u)/2$, we get

$$\begin{aligned} t(\tau) &= \frac{1}{\alpha} \sinh(\alpha\tau), \\ x(\tau) &= \frac{1}{\alpha} \cosh(\alpha\tau). \end{aligned} \quad (3.9)$$

The components of the two-acceleration then are

$$\begin{aligned} a^0 &= \ddot{t} = \alpha \sinh(\alpha\tau), \\ a^1 &= \ddot{x} = \alpha \cosh(\alpha\tau), \end{aligned} \quad (3.10)$$

and its magnitude is

$$\sqrt{a^\mu a_\mu} = \sqrt{-\alpha^2 \sinh^2(\alpha\tau) + \alpha^2 \cosh^2(\alpha\tau)} = \alpha, \quad (3.11)$$

as required.

We see from (3.9) that the trajectory of the accelerated observer obeys

$$x^2(\tau) - t^2(\tau) = \frac{1}{\alpha^2}, \quad (3.12)$$

which describes a hyperbola in the x - t plane (see Fig. 3). For large $|t|$, the trajectory asymptotes to the lightcone. Signals emitted outside the shaded region in Fig. 3 can't reach the observer.

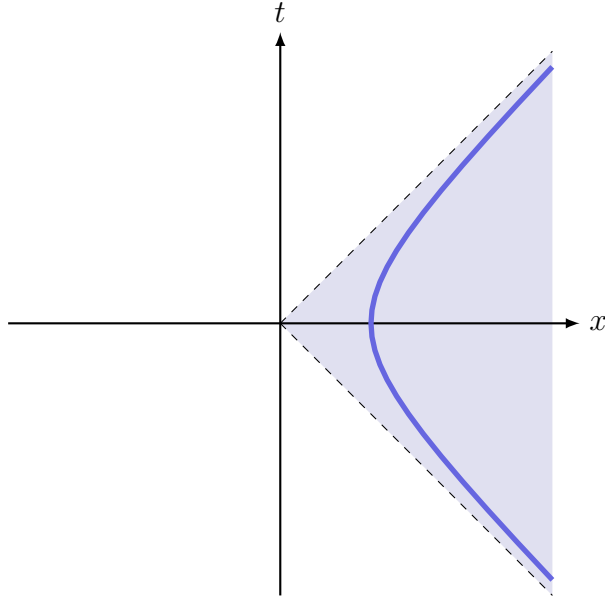


Figure 3: Trajectory of an accelerated observer in two-dimensional Minkowski space.

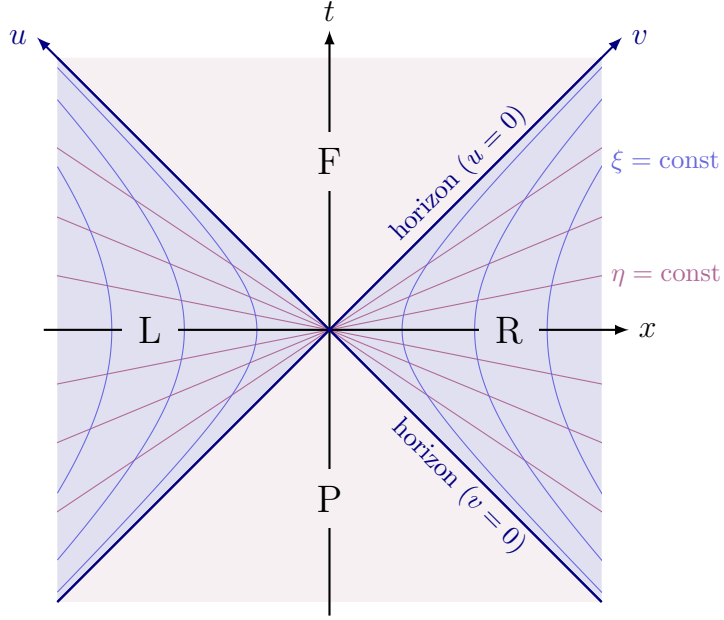


Figure 4: Illustration of the Rindler coordinates in two-dimensional Minkowski space. The coordinates in (3.13) and (3.19) cover the two causally disconnected regions R and L, respectively.

3.2 Rindler Space

We will now introduce a set of coordinates (η, ξ) that is adapted to the motion of an accelerated observer. Consider the transformation

$$\begin{aligned} t &= \frac{1}{a} e^{a\xi} \sinh(a\eta), \\ x &= \frac{1}{a} e^{a\xi} \cosh(a\eta), \quad (x > |t|). \end{aligned} \tag{3.13}$$

The range of the new coordinates is $-\infty < \eta, \xi < +\infty$ and they cover the wedge $x > |t|$, labeled as region R in Fig. 4. We will also find it convenient to work with the lightcone coordinates

$$U \equiv \eta - \xi, \quad V \equiv \eta + \xi, \tag{3.14}$$

which are related to the lightcone coordinate (3.2) in the inertial frame by

$$\begin{aligned} -aU &= \log(-au), \\ aV &= \log(av). \end{aligned} \tag{3.15}$$

This relation will be important when we compare the mode functions in the two frames.

The trajectory of the accelerated observer in (3.9) maps to

$$\begin{aligned} \eta(\tau) &= \frac{\alpha}{a} \tau, \\ \xi(\tau) &= \frac{1}{a} \log(a/\alpha). \end{aligned} \tag{3.16}$$

We see that the coordinate η is proportional to the proper time τ and ξ is a constant. For an observer with $\alpha = a$, we have $\eta = \tau$ and $\xi = 0$. More generally, lines of constant ξ in the Rindler metric describe uniformly accelerated observers with acceleration $\alpha = ae^{-a\xi}$ and proper time η . Near the horizon ($\xi \rightarrow -\infty$), we have $\alpha \rightarrow \infty$, i.e. the observers feel an infinite acceleration.

Using

$$\begin{aligned} dt &= e^{a\xi} (\sinh(a\eta) d\xi + \cosh(a\eta) d\eta), \\ dx &= e^{a\xi} (\cosh(a\eta) d\xi + \sinh(a\eta) d\eta), \end{aligned} \quad (3.17)$$

we find that the metric in the new coordinates is

$$ds^2 = e^{2a\xi} (-d\eta^2 + d\xi^2), \quad (3.18)$$

which is conformally equivalent to Minkowski space. Since the metric is independent of η , it has a timelike Killing vector ∂_η .

We can also define coordinates that cover the wedge $x < |t|$, labeled as region L in Fig. 4. Changing the overall signs in (3.13), we have

$$\begin{aligned} t &= -\frac{1}{a} e^{a\xi} \sinh(a\eta), \\ x &= -\frac{1}{a} e^{a\xi} \cosh(a\eta), \quad (x < |t|). \end{aligned} \quad (3.19)$$

Note that the two regions R and L are causally disconnected. The metric in region L will be same as in region R; cf. (3.18). Because of the change of sign in (3.19), however, the future-directed Killing vector will be $\partial_{(-\eta)} = -\partial_\eta$ rather than ∂_η . This will be relevant when we define positive-frequency modes in the left Rindler wedge.

3.3 Field Quantization

We now consider quantization of a massless scalar field in two-dimensional Minkowski spacetime. In the right Rindler wedge, the equation of motion becomes $\square\phi = e^{-2a\xi}(-\partial_\eta^2 + \partial_\xi^2)\phi = 0$, and the positive-frequency solutions are

$$g_k^R(x^\mu) = \frac{1}{\sqrt{4\pi\omega}} e^{-i\omega\eta + ik\xi}, \quad (3.20)$$

with $\omega = |k|$. These modes are *not* defined on a Cauchy surface for the whole Minkowski space. To define global modes, we write

$$g_k^R(x^\mu) = \begin{cases} \frac{1}{\sqrt{4\pi\omega}} e^{-i\omega\eta + ik\xi} & (\text{R}), \\ 0 & (\text{L}). \end{cases} \quad (3.21)$$

However, this is not a complete set of modes. To achieve this, we define the following modes on the left Rindler wedge:

$$g_k^L(x^\mu) = \begin{cases} 0 & (\text{R}), \\ \frac{1}{\sqrt{4\pi\omega}} e^{+i\omega\eta + ik\xi} & (\text{L}). \end{cases} \quad (3.22)$$

These are positive-frequency solutions with respect to the Killing vector $-\partial_\eta$, i.e. we have $(-\partial_\eta)g_k^L = -i\omega g_k^L$, with $\omega > 0$.

The modes in (3.21) and (3.22) form a complete basis, so the operator $\hat{\phi}$ can be written as

$$\hat{\phi} = \int dk \left(\hat{b}_k^R g_k^R + \hat{b}_k^L g_k^L + \text{h.c.} \right), \quad (3.23)$$

where $\hat{b}_k^R$ and $\hat{b}_k^L$ are annihilation operators for the right and left Rindler modes, respectively. This is to be compared to the standard expansion of the operator in the Minkowski basis

$$\hat{\phi} = \int dk \left(\hat{a}_k f_k + \text{h.c.} \right), \quad (3.24)$$

where $f_k(x^\mu) = e^{-i\omega t + ikx} / \sqrt{4\pi\omega}$. We have two different vacuum states: The *Minkowski vacuum* defined by $\hat{a}_k|0_M\rangle = 0$ and the *Rindler vacuum* which satisfies $\hat{b}_k^R|0_R\rangle = \hat{b}_k^L|0_R\rangle = 0$. As we will now show, the Rindler observer will detect particles in the Minkowski vacuum.

3.4 Unruh Effect

We would like to compute the number of particles that an accelerating observer in the right Rindler wedge will associate to the Minkowski vacuum:

$$\langle \hat{N}_k^R \rangle \equiv \langle 0_M | \hat{b}_k^{R\dagger} \hat{b}_k^R | 0_M \rangle. \quad (3.25)$$

Using (2.30), the Bogolubov transformation relating the Rindler and Minkowski operators is

$$\hat{b}_k^R = \int dk' \left(\alpha_{k,k'}^{R*} \hat{a}_{k'} - \beta_{k,k'}^{R*} \hat{a}_{k'}^\dagger \right), \quad (3.26)$$

and substituting this into (3.27), we get

$$\begin{aligned} \langle \hat{N}_k^R \rangle &= \langle 0_M | \int dk' \left(\alpha_{k,k'}^R \hat{a}_{k'}^\dagger - \beta_{k,k'}^R \hat{a}_{k'} \right) \int dk'' \left(\alpha_{k,k''}^{R*} \hat{a}_{k''} - \beta_{k,k''}^{R*} \hat{a}_{k''}^\dagger \right) | 0_M \rangle \\ &= \int dk' dk'' \beta_{k,k'}^R \beta_{k,k''}^{R*} \langle 0_M | \hat{a}_{k'} \hat{a}_{k''}^\dagger | 0_M \rangle \\ &= \int dk' dk'' \beta_{k,k'}^R \beta_{k,k''}^{R*} \langle 0_M | [\hat{a}_{k'}, \hat{a}_{k''}^\dagger] | 0_M \rangle \\ &= \int dk' |\beta_{k,k'}^R|^2. \end{aligned} \quad (3.27)$$

To compute $\langle \hat{N}_k^R \rangle$, we therefore have to compute the Bogolubov coefficient $\beta_{k,k'}^R = -\langle f_k^*, g_{k'}^R \rangle$.

Recall that the positive-frequency Minkowski mode functions are

$$f_k(x^\mu) = \frac{1}{\sqrt{4\pi\omega}} e^{-i\omega t + ikx}, \quad (3.28)$$

with $\omega = |k| > 0$. It is useful to separate this into “right-moving” modes (with $k > 0$ and $\omega = k$) and “left-moving” modes (with $k < 0$ and $\omega = -k$). These different modes don’t mix under a

Bogolubov transformation, so we can treat them separately. For concreteness, we will focus on the right-moving modes:

$$f_\omega(u) = \frac{1}{\sqrt{4\pi\omega}} e^{-i\omega u}, \quad (3.29)$$

where $u = t - x$. Similarly, the solution for the right-moving modes in the right Rindler wedge is

$$g_\omega^R(U) = \frac{1}{\sqrt{4\pi\omega}} e^{-i\omega U} = \frac{1}{\sqrt{4\pi\omega}} e^{i(\omega/a) \log(-au)}, \quad (3.30)$$

where $U \equiv \eta - \xi$ and we have used the relation $-aU = \log(-au)$, which in the second equality.

Using the Bogolubov transformation (2.22), we can write

$$\begin{aligned} g_\omega^R(u) &= \int d\omega' \left(\alpha_{\omega,\omega'}^R f_{\omega'}(u) + \beta_{\omega,\omega'}^R f_{\omega'}^*(u) \right) \\ &= \int_0^\infty \frac{d\omega'}{2\pi} \left(\alpha_{\omega,\omega'}^R \sqrt{\frac{\pi}{\omega'}} e^{-i\omega' u} + \beta_{\omega,\omega'}^R \sqrt{\frac{\pi}{\omega'}} e^{i\omega' u} \right). \end{aligned} \quad (3.31)$$

It is instructive to look at the (inverse) Fourier transform

$$\begin{aligned} g_\omega^R(u) &= \int_{-\infty}^\infty \frac{d\omega'}{2\pi} e^{-i\omega' u} \tilde{g}_\omega(\omega') \\ &= \int_0^\infty \frac{d\omega'}{2\pi} e^{-i\omega' u} \tilde{g}_\omega(\omega') + \int_0^\infty \frac{d\omega'}{2\pi} e^{i\omega' u} \tilde{g}_\omega(-\omega'). \end{aligned} \quad (3.32)$$

Comparing this to (3.31), we see that

$$\begin{aligned} \alpha_{\omega,\omega'}^R &= \sqrt{\frac{\omega'}{\pi}} \tilde{g}_\omega(\omega'), \\ \beta_{\omega,\omega'}^R &= \sqrt{\frac{\omega'}{\pi}} \tilde{g}_\omega(-\omega'). \end{aligned} \quad (3.33)$$

Our task therefore is to compute the Fourier transform of the function $g_\omega^R(u)$ in (3.30). An explicit computation leads to

$$\alpha_{\omega,\omega'}^R = \frac{1}{2\pi a} \sqrt{\frac{\omega}{\omega'}} \left(\frac{\omega'}{a} \right)^{i\omega/a} \Gamma(-i\omega/a) e^{\pi\omega/2a}, \quad (3.34)$$

$$\beta_{\omega,\omega'}^R = -\frac{1}{2\pi a} \sqrt{\frac{\omega}{\omega'}} \left(\frac{\omega'}{a} \right)^{i\omega/a} \Gamma(-i\omega/a) e^{-\pi\omega/2a}, \quad (3.35)$$

so that

$$|\alpha_{\omega,\omega'}^R|^2 = e^{2\pi\omega/a} |\beta_{\omega,\omega'}^R|^2. \quad (3.36)$$

This relation can also be obtained without computing the integrals explicitly.

Exercise 3.1 Without explicitly computing the Fourier transform, show that

$$\tilde{g}_\omega(-\omega') = -e^{-\pi\omega/a} \tilde{g}_\omega(\omega') \implies |\alpha_{\omega,\omega'}^R|^2 = e^{2\pi\omega/a} |\beta_{\omega,\omega'}^R|^2. \quad (3.37)$$

This will suffice to compute (3.27). →

From the commutation relation of $\hat{b}_\omega^R$ and $\hat{b}_\omega^{R\dagger}$, we get

$$\begin{aligned} \delta(\omega - \omega') &= [\hat{b}_\omega^R, \hat{b}_{\omega'}^{R\dagger}] \\ &= \left[\int d\lambda \left(\alpha_{\omega,\lambda}^{R*} \hat{a}_\lambda - \beta_{\omega,\lambda}^{R*} \hat{a}_\lambda^\dagger \right), \int d\lambda' \left(\alpha_{\omega',\lambda'}^R \hat{a}_{\lambda'}^\dagger - \beta_{\omega',\lambda'}^R \hat{a}_{\lambda'} \right) \right] \\ &= \int d\lambda d\lambda' \left(\alpha_{\omega,\lambda}^{R*} \alpha_{\omega',\lambda'}^R - \beta_{\omega,\lambda}^{R*} \beta_{\omega',\lambda'}^R \right) \delta(\lambda - \lambda') \\ &= \int d\lambda \left(\alpha_{\omega,\lambda}^{R*} \alpha_{\omega',\lambda}^R - \beta_{\omega,\lambda}^{R*} \beta_{\omega',\lambda}^R \right). \end{aligned} \quad (3.38)$$

Setting $\omega = \omega'$, this implies

$$\int d\lambda \left(|\alpha_{\omega,\lambda}^R|^2 - |\beta_{\omega,\lambda}^R|^2 \right) = \delta(0). \quad (3.39)$$

Using (3.37), this then leads to

$$\langle \hat{N}_\omega \rangle = \int d\lambda |\beta_{\omega,\lambda}^R|^2 = \frac{\delta(0)}{e^{2\pi\omega/a} - 1}. \quad (3.40)$$

As shown in (1.57), the divergent factor $\delta(0)$ comes from working with an infinite volume. For a finite volume, the number density is

$$n_\omega \equiv \frac{\langle \hat{N}_\omega \rangle}{V} = \frac{1}{e^{2\pi\omega/a} - 1}. \quad (3.41)$$

We see that this is a thermal spectrum with temperature¹

$$T = \frac{a}{2\pi}. \quad (3.42)$$

This is the main result of this section: A uniformly accelerated observer moving through the Minkowski vacuum observes a thermal spectrum of particles with a temperature that is proportional to the acceleration.

Exercise 3.2 A glass of water is moving with a constant acceleration. Determine the acceleration that makes the water boil due to the Unruh effect. →

¹This is the temperature that would be measured by an observer moving along the path $\xi = 0$, who feels the acceleration $\alpha = a$. An observer with $\xi \neq 0$ feels an acceleration $\alpha = ae^{-a\xi}$ and will thus measure thermal radiation with the temperature $T = \alpha/(2\pi)$. As $\xi \rightarrow \infty$, the temperature vanishes, which is consistent with the fact that in this limit the Rindler observer is nearly inertial.

3.5 Particle Detectors

We have seen that the notion of a particle is observer-dependent. To clarify this further, we now introduce a simple model of a particle detector and use it to analyze the Unruh effect.

A detector is a device with some internal energy states E_i that can interact with the scalar field by exchanging energy, i.e. by emitting or absorbing scalar particles. We model this by coupling the scalar field $\phi(x(\tau))$ along the world-line $x(\tau)$ of the observer to some operator $m(\tau)$ acting on the internal detector states:

$$g \int d\tau \hat{m}(\tau) \hat{\phi}(x(\tau)), \quad (3.43)$$

where g is the coupling constant. The detector has a Hamiltonian H_0 , with energy eigenstates $|E_i\rangle$, so that $\hat{H}_0|E_i\rangle = E_i|E_i\rangle$. We are interested in the transition amplitude from a state $|0\rangle \otimes |E_i\rangle$ to a state $|\psi\rangle \otimes |E_j\rangle$, where $|0\rangle$ and $|\psi\rangle$ are states of the scalar field. For small coupling, this amplitude is

$$\mathcal{A}(E_i \rightarrow E_j) = g \int_{-\infty}^{\infty} d\tau \langle E_j | \otimes \langle \psi | \hat{m}(\tau) \hat{\phi}(x(\tau)) | 0 \rangle \otimes | E_i \rangle. \quad (3.44)$$

Using $\hat{m}(\tau) = e^{i\hat{H}_0\tau} \hat{m}(0) e^{-i\hat{H}_0\tau}$, this becomes

$$\mathcal{A}(E_i \rightarrow E_j) = g m_{ji} \int_{-\infty}^{\infty} d\tau e^{i(E_j - E_i)\tau} \langle \psi | \hat{\phi}(x(\tau)) | 0 \rangle, \quad (3.45)$$

where we have defined the matrix element $m_{ij} \equiv \langle E_i | \hat{m}(0) | E_j \rangle$. To obtain the probability for the detector to make the transition from E_i to E_j , we square the amplitude and sum over the final state $|\psi\rangle$ of the scalar field (which is not measured). Using $\sum_{\psi} |\psi\rangle \langle \psi| = 1$, this gives

$$P(E_i \rightarrow E_j) = g^2 |m_{ij}|^2 \int_{-\infty}^{\infty} d\tau d\tau' e^{-i(E_j - E_i)(\tau' - \tau)} G(x(\tau), x(\tau')), \quad (3.46)$$

where we have introduced the *Wightman function*

$$G(x(\tau), x(\tau')) \equiv \langle 0 | \hat{\phi}(x(\tau)) \hat{\phi}(x(\tau')) | 0 \rangle. \quad (3.47)$$

Dropping the prefactor in (3.46), which depends on the details of the detector, we obtain the detector's *response function*:

$$\mathcal{F}(E) \equiv \int_{-\infty}^{\infty} d\tau d\tau' e^{-iE(\tau' - \tau)} G(x(\tau), x(\tau')). \quad (3.48)$$

Assuming that the Wightman function only depends on $\Delta\tau \equiv \tau' - \tau$, we can change variables to $\Delta\tau$ and $\bar{\tau} \equiv \frac{1}{2}(\tau' + \tau)$ to factorize the integrals. Dropping the (divergent) integral over $\bar{\tau}$, this leads to

$$f(E) = \int_{-\infty}^{\infty} d\Delta\tau e^{-iE\Delta\tau} G(\Delta\tau). \quad (3.49)$$

For a given theory, we compute the Wightman function G and then perform this integral to determine the response of the detector. We will do this in a few examples.

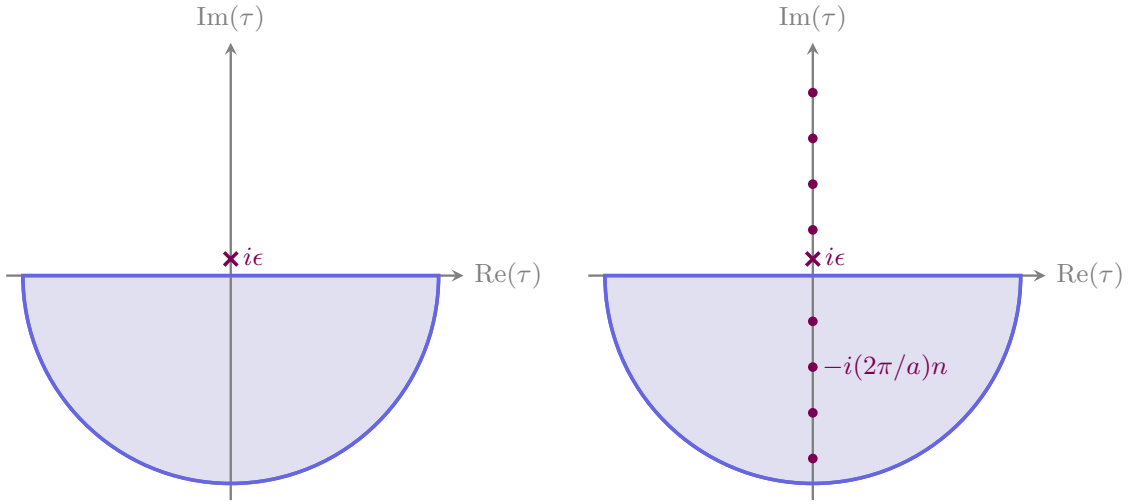


Figure 5: Integration contours for the integrals defining the detector's response function of an inertial observer (*left*) and an accelerating observer (*right*).

Exercise 3.3 Show that the Wightman function of a massless scalar field in four-dimensional Minkowski space is

$$G(x, x') = -\frac{1}{4\pi^2} \frac{1}{(t - t' - i\epsilon)^2 - (\mathbf{x} - \mathbf{x}')^2}, \quad (3.50)$$

where $\epsilon > 0$ was introduced to regulate a divergence on the lightcone. ➔

Using (3.50), the Wightman function in Minkowski space can be written as

$$G(\Delta\tau) = \frac{1}{4\pi^2} \frac{1}{(\Delta\tau - i\epsilon)^2}. \quad (3.51)$$

To determine the detector's response, we substitute this into (3.49) and evaluate the integral by the method of residues. For $E > 0$, we close the integration contour for $\Delta\tau$ in the lower half-plane where $e^{-iE\Delta\tau}$ leads to an exponential suppression for large $|\Delta\tau|$; see Fig. 24. However, since (3.51) is analytic in the lower half-plane, we conclude that

$$f(E) = 0. \quad (3.52)$$

As expected, there is no particle detection in the Minkowski vacuum.

We would now like to repeat the exercise for an accelerating Rindler observer. In that case, we can write

$$\begin{aligned} t &= a^{-1} \sinh(a\tau), \\ x &= a^{-1} \cosh(a\tau), \quad y = z = 0, \end{aligned} \quad (3.53)$$

for a detector undergoing uniform acceleration a . From (3.50), we then find

$$\begin{aligned}
G(x(\tau), x(0)) &= -\frac{1}{4\pi^2} \frac{a^2}{(\sinh(a\tau) - ia\epsilon)^2 - (\cosh(a\tau) - 1)^2} \\
&= -\frac{1}{8\pi^2} \frac{a^2}{\cosh(a\tau) - 1 - ia\epsilon \sinh(a\tau)} \\
&= -\frac{1}{8\pi^2} \frac{a^2}{\cosh[a(\tau - i\epsilon)] - 1} \\
&= -\frac{1}{16\pi^2} \frac{a^2}{\sinh^2[\frac{1}{2}a(\tau - i\epsilon)]}.
\end{aligned} \tag{3.54}$$

To determine the detector's response, we therefore have to compute the following integral:

$$f(E) = -\frac{a^2}{16\pi^2} \int_{-\infty}^{\infty} d\tau \frac{e^{-iE\tau}}{\sinh^2[\frac{1}{2}a(\tau - i\epsilon)]}. \tag{3.55}$$

Using $\sinh(iy) = i \sin y$, and the identity

$$\frac{\pi^2}{\sin^2(\pi y)} = \sum_{n=-\infty}^{\infty} (y - n)^{-2}, \tag{3.56}$$

this can be written as

$$f(E) = -\frac{1}{4\pi^2} \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} d\tau \frac{e^{-iE\tau}}{[\tau - i\epsilon + i(2\pi/a)n]^2}. \tag{3.57}$$

We see that the integrand now has double poles at $\tau_n = -i(2\pi/a)n$, with $n = 1, 2, \dots$, in the lower half-plane; see Fig. 24. Computing (3.49) using the method of residues then leads to

$$\begin{aligned}
f(E) &= -\frac{1}{4\pi^2} \times (-2\pi i) \times \left[(-iE) \times \sum_{n=1}^{\infty} e^{-iE\tau_n} \right] \\
&= \frac{E}{2\pi} \sum_{n=1}^{\infty} e^{-2\pi En/a} \\
&= \frac{E}{2\pi} \frac{1}{e^{2\pi E/a} - 1},
\end{aligned} \tag{3.58}$$

as expected for a detector immersed in a thermal bath at the Unruh temperature $T = a/2\pi$.

4 Black Holes and Hawking Radiation

One of the most famous applications of QFT in curved spacetimes is to black holes. In 1974, Stephen Hawking discovered (to everyone’s surprise) that black holes radiate and therefore have a temperature and an entropy. This addressed the concern that black holes may violate the second law of thermodynamics by reducing the overall entropy of the Universe when an object falls into a black hole. By carrying an entropy, black holes can balance the apparent loss of entropy, thereby saving the second law.

In this chapter, we will compute the thermal spectrum of Hawking radiation. We will begin with a review of the geometry and causal structure of Schwarzschild black holes, but readers who are familiar with this material may jump directly to the quantization of fields in the Schwarzschild geometry in Section 4.2. We will provide two derivations of Hawking radiation—one heuristic and one more technical—and discuss some of its theoretical implications.

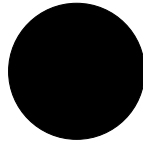
4.1 Schwarzschild Black Holes

The metric around a spherically symmetric object of mass M is

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2\theta d\varphi^2), \quad (4.1)$$

where we have used units with $G \equiv 1$. This spacetime has two singularities: a physical singularity at $r = 0$ and a coordinate singularity at $r = 2M$. The latter can be removed by a change of coordinates.

An object compressed to a size smaller than its Schwarzschild radius $R_S \equiv 2GM/c^2$ will form a black hole. For an object of the mass of the Earth, $M_\oplus = 6 \times 10^{24}$ kg, the corresponding Schwarzschild radius is $R_{S,\oplus} = 2GM_\oplus/c^2 = 8.9$ mm. Such a black hole can therefore be drawn to scale:



Of course, this is much smaller than the actual size of the Earth, $R_\oplus \approx 6400$ km, which is why the Earth is *not* a black hole. Similarly, taking the mass of the Sun, $M_\odot = 2 \times 10^{30}$ kg gives $R_{S,\odot} \approx 3$ km compared to $R_\odot \approx 7 \times 10^5$ km for the radius of the Sun.

For ordinary planets or stars, we have $R_S \ll R$, so that the would-be event horizon is not part of the spacetime. In order for a black hole to form, the mass must be compressed into an incredibly small region of space. This can happen when a star with a mass above the Tolman–Oppenheimer–Volkoff limit, $M > 4 M_\odot$, runs out of fuel and collapses. (Stars with smaller masses will become white dwarfs or neutron stars.) We also believe that there are supermassive black holes, with masses up to $M \sim 10^{10} M_\odot$, at the centers of most galaxies. In 2019, the EHT collaboration released the first image of the black hole at the center of M87. The inferred mass of the central black hole is 6.5 billion times the mass of our Sun. At the center of our galaxy lives a black hole of 4.3 million Solar masses. The existence of this black hole was first inferred

indirectly, by observing the orbits of stars around it. (Reinhard Genzel and Andrea Ghez received the 2020 Nobel Prize for these observations.)

Exercise 4.1 Consider the metric (4.1) of the Schwarzschild black hole. Let $r = 2M + \rho$. Expanding the metric for small $\rho \ll 2M$, show that the near-horizon geometry is the product of a two-sphere and two-dimensional Rindler space; cf. (3.18). →

Kruskal coordinates

To study null geodesics in the spacetime, it is useful to make a change of coordinates. First, we note that the metric (4.1) can be written as

$$ds^2 = \left(1 - \frac{2M}{r}\right) \left[-dt^2 + \left(\frac{r dr}{r - 2M}\right)^2\right] + r^2 d\Omega^2. \quad (4.2)$$

This suggests that it will be convenient to introduce the so-called *tortoise coordinate*:

$$dr^* \equiv \frac{r dr}{r - 2M} \quad \implies \quad r^* = r + 2M \log \left(\frac{r - 2M}{2M}\right), \quad (4.3)$$

so that the metric becomes

$$ds^2 = \left(1 - \frac{2M}{r}\right) (-dt^2 + dr^{*2}) + r^2 d\Omega^2, \quad (4.4)$$

where r should be thought of as a function of r^* . Note that the tortoise coordinate r^* is only defined outside the black hole. The event horizon, at $r = 2M$, has been pushed to $r^* = -\infty$. We will also work with the *tortoise lightcone coordinates*:

$$U \equiv t - r^*, \quad V \equiv t + r^*. \quad (4.5)$$

We have used capital letters since these coordinates will play the same role as the Rindler coordinates in Section 3 where we capitalized the lightcone coordinates.

We mentioned that the singularity at $r = 2M$ is just a coordinate singularities. To make this manifest, we use the *Kruskal coordinates*:

$$\kappa u = -e^{-\kappa U}, \quad \kappa v \equiv e^{\kappa V}, \quad (4.6)$$

where $\kappa \equiv (4M)^{-1}$. Notice the similarity with (3.19). The exterior of the black hole corresponds to $u < 0$ and $v > 0$. Outside the horizon, we have

$$\kappa^2 uv = -e^{2\kappa r^*} = -\left(\frac{r}{2M} - 1\right) e^{r/2M}, \quad (4.7)$$

$$\frac{u}{v} = -e^{-2\kappa t}. \quad (4.8)$$

The metric (4.4) then becomes

$$\begin{aligned}
ds^2 &= - \left(1 - \frac{2M}{r}\right) dU dV + r^2 d\Omega^2 \\
&= - \left(1 - \frac{2M}{r}\right) \frac{(4M)^2}{-uv} dudv + r^2 d\Omega^2 \\
&= - \left(1 - \frac{2M}{r}\right) \left(\frac{r}{2M} - 1\right)^{-1} e^{-r/2M} dudv + r^2 d\Omega^2 \\
&= - \frac{2M}{r} e^{-r/2M} dudv + r^2 d\Omega^2.
\end{aligned} \tag{4.9}$$

Although the original Schwarzschild coordinates only cover the region of the spacetime with $u < 0$ and $v > 0$, nothing stops us now from extending this to $u, v \in \mathbb{R}$. The metric is manifestly smooth and non-degenerate at $r = 2M$.

The Kruskal coordinates in (4.6) can also be written as

$$u \equiv T - X, \quad v \equiv T + X, \tag{4.10}$$

where

$$\begin{aligned}
T &= \frac{1}{2}(v + u) = 4M \left(\frac{r}{2M} - 1\right)^{1/2} e^{r/4M} \sinh\left(\frac{t}{4M}\right), \\
X &= \frac{1}{2}(v - u) = 4M \left(\frac{r}{2M} - 1\right)^{1/2} e^{r/4M} \cosh\left(\frac{t}{4M}\right).
\end{aligned} \tag{4.11}$$

Note that

$$T^2 - X^2 = (4M)^2 \left(1 - \frac{r}{2M}\right) e^{r/2M}, \tag{4.12}$$

so that hypersurfaces of constant r are hyperbolae in the X - T plane. Unlike in the tortoise coordinates (t, r^*) , the horizon $r = 2M$ is not infinitely far away, but maps to

$$T = \pm X. \tag{4.13}$$

Surfaces of constant t are given by

$$\frac{T}{X} = \tanh\left(\frac{t}{4M}\right), \tag{4.14}$$

i.e. straight lines with slope $\tanh(t/4M)$. Note that as $t \rightarrow \pm\infty$ the curves given by (4.14) become the same as (4.13); therefore $t = \pm\infty$ represents the same surface as $r = 2M$. All of this is very similar to what we found in Rindler space.

Kruskal diagram

Figure 6 shows the Schwarzschild spacetime in the Kruskal coordinates. Shown are both the X - T coordinates and the rotated u - v coordinates. As we have seen in (4.13), the horizon $r = 2M$ corresponds to the two null surfaces:

$$r = 2M \quad \Rightarrow \quad T = \pm X \quad (uv = 0). \tag{4.15}$$

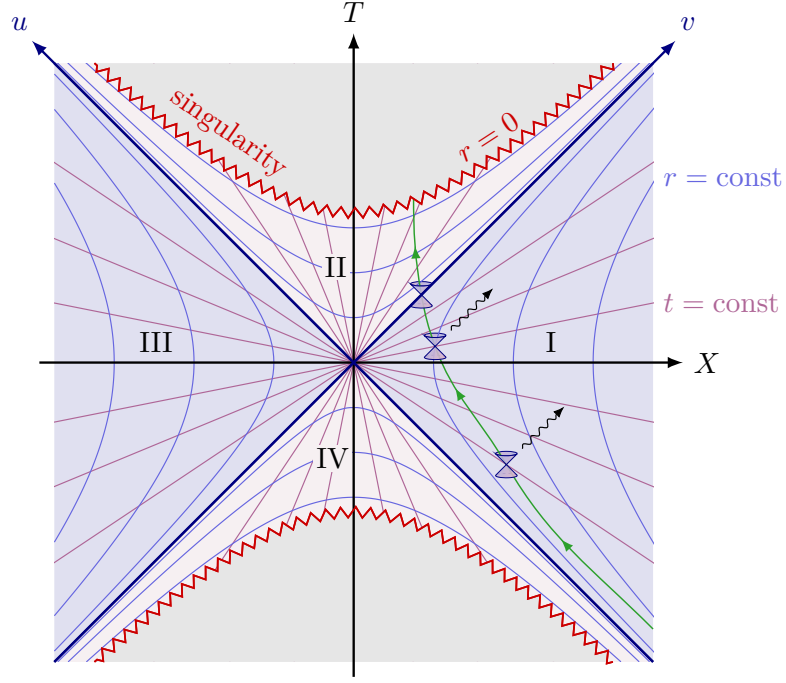


Figure 6: Kruskal diagram of the Schwarzschild solution (figure by Izaak Neutelings). Region I corresponds to the outside of the black hole. Region II is the inside of the black hole, while region IV is the inside of the white hole. Region III is the mirror image of region I. Regions I and III are connected by a wormhole.

The null surface $T = X$ (or $u = 0$) is the horizon of the black hole (the *future horizon*), while the null surface $T = -X$ (or $v = 0$) is the horizon of the white hole (the *past horizon*). Region I is the spacetime outside of the black hole (white hole). This is similar to the Rindler geometry shown in Fig. 4. Regions II and III and the inside of the black hole and the white hole, respectively.

The *singularity* is mapped to two spacelike surfaces:

$$r = 0 \quad \Rightarrow \quad T = \pm \sqrt{X^2 + (4M)^2} \quad (\kappa^2 uv = 1). \quad (4.16)$$

In Fig. 6, these are shown as two disconnected hyperbolae. The surface $T = \sqrt{X^2 + (4M)^2}$ (or $u, v > 0$) is the singularity of the black hole, while $T = -\sqrt{X^2 + (4M)^2}$ (or $u, v < 0$) is the singularity of the white hole. You may have thought that the singularity of a black hole was a point that traces out a timelike worldline (like a massive particle). The diagram shows that this is not the case. David Tong describes this very clearly [9]: “Once you pass through the horizon, the singularity isn’t something that sits to your left or to your right: It is something that lies in your future. This makes it clear why you cannot avoid the singularity when inside a black hole. It is your fate. Similarly, the singularity of the white hole lies in the past. It is similar to the singularity of the Big Bang.”

What is region III in the Kruskal diagram? It is another mirror copy of the black hole, now covered by $u > 0$ and $v < 0$. To see this, we can write the Kruskal coordinates as in (4.6), but

with different signs,

$$\begin{aligned}\kappa u &= +e^{-2\kappa U}, \\ \kappa v &= -e^{2\kappa V}.\end{aligned}\tag{4.17}$$

Doing all the coordinate transformations in reverse then shows that region III is again described by the Schwarzschild metric. Note that regions I and III are spacelike separated, so that an observer in I cannot send a signal to III. The regions are *causally disconnected*. Nevertheless, it is still rather freaky. The full spacetime has *two* copies of the black hole exterior. The two regions are connected by a wormhole (or *Einstein-Rosen bridge*). Because the regions are spacelike separated, however, it is not like the science fiction wormholes that you could travel through. Moreover, as we will discuss below, we don't really expect regions III and IV to exist for real black holes in the Universe (which form by collapsing matter), but instead they are an artifact of assuming an *eternal* Schwarzschild geometry.

Penrose diagram

A black hole is defined as the region of space from which light cannot escape to infinity. The boundary of that region is the event horizon. In the Kruskal diagram, infinity is still a large distance away. A more precise way to capture the black geometry maps the points at infinity to a finite distance. This leads to the famous *Penrose diagram* which allows us to draw the entire spacetime on a sheet of paper. For the Schwarzschild black hole, the Penrose diagram is very similar to the Kruskal diagram; we just have to straighten out a few lines. Penrose diagrams play an important role in exhibiting the causal structure of the spacetime, so it is worth learning what they are all about.

Two-dimensional Minkowski.—Let us start with a simple example: two-dimensional Minkowski space, with metric

$$ds^2 = -dt^2 + dx^2.\tag{4.18}$$

We first introduce lightcone coordinates,

$$\begin{aligned}u &= t - x, \\ v &= t + x,\end{aligned}\tag{4.19}$$

so that the metric becomes

$$ds^2 = -dudv.\tag{4.20}$$

The range of the coordinates is the entire real line, $u, v \in (-\infty, \infty)$. We would like to map this to a finite range. One choice of such a mapping is

$$\begin{aligned}u &= \tan \tilde{u}, \\ v &= \tan \tilde{v},\end{aligned}\tag{4.21}$$

so that $\tilde{u}, \tilde{v} \in (-\pi/2, +\pi/2)$. In the new coordinates, the metric becomes

$$ds^2 = -\frac{1}{\cos^2 \tilde{u} \cos^2 \tilde{v}} d\tilde{u} d\tilde{v}.\tag{4.22}$$

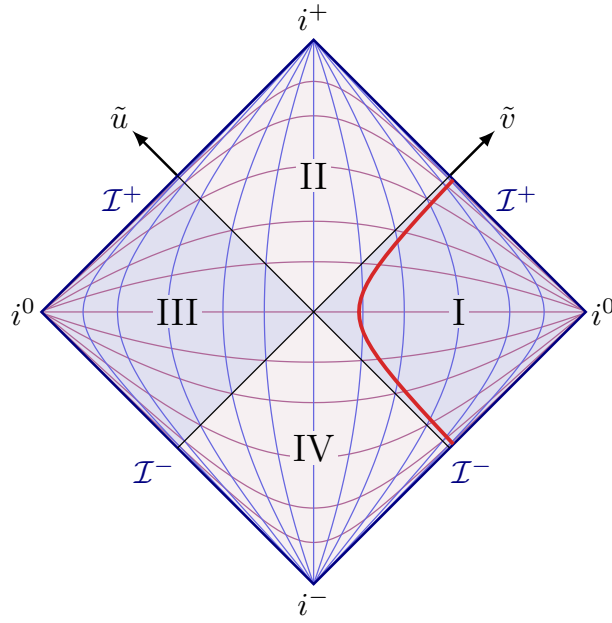


Figure 7: Penrose diagram of two-dimensional Minkowski space. An accelerated Rindler observer can only communicate with points within region I of the spacetime.

The crucial point is that the overall factor does *not* change the causal structure since it doesn't affect null geodesics which obey $ds^2 = 0$. We therefore define a new metric

$$d\tilde{s}^2 = (\cos^2 \tilde{u} \cos^2 \tilde{v}) ds^2 = -d\tilde{u} d\tilde{v}. \quad (4.23)$$

The two line elements $d\tilde{s}^2$ and ds^2 are related by a *conformal transformation* and have the same causal structure. The Penrose diagram is the graphical representation of the spacetime in the compactified coordinates $\tilde{u}$ and $\tilde{v}$.

We draw the lightcone coordinates $\tilde{u}$ and $\tilde{v}$ at 45 degrees, so that light rays travel at 45 degrees. Figure 7 shows the resulting Penrose diagram. The boundaries of the diagram are different types of infinity:

- $i^\pm$: All timelike geodesics start at i^- , with $(\tilde{u}, \tilde{v}) = (-\pi/2, -\pi/2)$ and end at i^+ , with $(\tilde{u}, \tilde{v}) = (+\pi/2, +\pi/2)$. These points are called *past* and *future timelike infinity*, respectively.
- i^0 : All spacelike geodesics start and end at the two points labelled i^0 , either $(\tilde{u}, \tilde{v}) = (-\pi/2, +\pi/2)$ or $(\tilde{u}, \tilde{v}) = (+\pi/2, -\pi/2)$. These points are called *spacelike infinity*.
- $\mathcal{I}^\pm$: All null geodesics start on $\mathcal{I}^-$ (sometimes called “scri-minus”), with $\tilde{u} = -\pi/2$ or $\tilde{v} = -\pi/2$, and end on $\mathcal{I}^+$ (“scri-plus”), with $\tilde{u} = +\pi/2$ or $\tilde{v} = +\pi/2$. These boundaries are called *past* and *future null infinity*, respectively.

Figure 7 also shows the trajectory of an accelerated Rindler observer in two-dimensional Minkowski space. Notice that this observer stays confined to region I in the Penrose diagram. Light from regions II and III cannot reach the Rindler observer who is separated from those regions by an

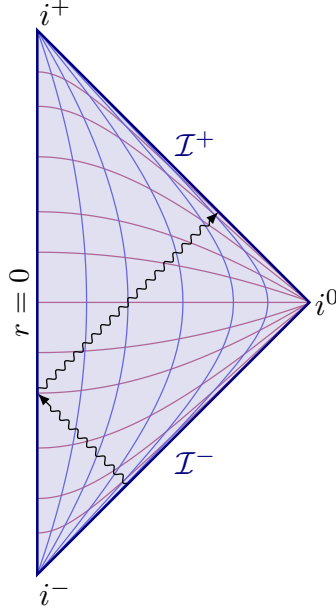


Figure 8: Penrose diagram of four-dimensional Minkowski space. Shown is also a null geodesic starting at $\mathcal{I}^-$ and ending at $\mathcal{I}^+$.

effective horizon. The Rindler observer can receive signals from region IV, but it cannot send any signals to that region, while it can send signals to region II. By virtue of his acceleration, the Rindler observer has cut himself off from part of the spacetime. A non-accelerating observer, in contrast, can communicate with the entire spacetime. This is the equivalence principle in action: an accelerating observer in Minkowski space experiences the same kind of horizons as a free-falling observer in the Schwarzschild geometry.

Four-dimensional Minkowski.—Let us repeat this exercise for four-dimensional Minkowski space:

$$ds^2 = -dt^2 + dr^2 + r^2 d\Omega^2. \quad (4.24)$$

Going to light cone coordinates,

$$\begin{aligned} u &= t - r, \\ v &= t + r, \end{aligned} \quad (4.25)$$

the metric becomes

$$ds^2 = -du dv + \frac{1}{4}(u - v)^2 d\Omega^2, \quad (4.26)$$

and, using the same mapping as in (4.21), we get

$$ds^2 = \frac{1}{4 \cos^2 \tilde{u} \cos^2 \tilde{v}} (-4 d\tilde{u} d\tilde{v} + \sin^2(\tilde{u} - \tilde{v}) d\Omega^2). \quad (4.27)$$

To study the causal structure of the spacetime, it again suffices to use the new metric

$$d\tilde{s}^2 = -4 d\tilde{u} d\tilde{v} + \sin^2(\tilde{u} - \tilde{v}) d\Omega^2. \quad (4.28)$$

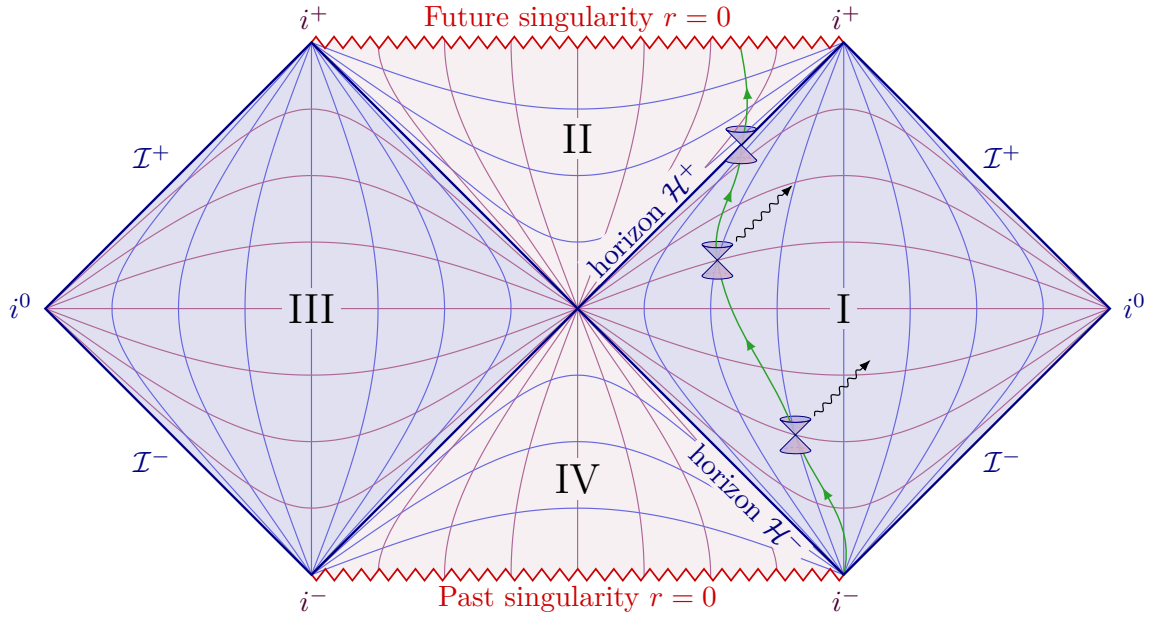


Figure 9: Penrose diagram for the extended Schwarzschild black hole. (Figure by Izaak Neutelings.)

One difference compared to the 2D case is that $v \geq u$ because $r \geq 0$. This means that the compactified coordinates obey

$$-\frac{\pi}{2} \leq \tilde{u} \leq \tilde{v} \leq \frac{\pi}{2}. \quad (4.29)$$

To draw a two-dimensional diagram, we suppress the angular coordinates. Each point in the the Penrose diagram shown in Fig. 8 therefore has a two-sphere attached to it. The vertical line corresponds to the point $r = 0$ and is *not* a boundary of the spacetime. A null geodesic that starts on $\mathcal{I}^-$ will simply be reflected at the vertical line and end up at $\mathcal{I}^+$.

Back to Schwarzschild.—After this digression, we are ready to return to the Schwarzschild geometry. The metric in the light cone Kruskal coordinates is

$$ds^2 = -\frac{2M}{r} e^{-r/2M} du dv + r^2 d\Omega^2. \quad (4.30)$$

As in (4.21), we define

$$\begin{aligned} \kappa u &= \tan \tilde{u}, \\ \kappa v &= \tan \tilde{v}, \end{aligned} \quad (4.31)$$

so that $\tilde{u}, \tilde{v} \in (-\pi/2, +\pi/2)$. The metric then becomes

$$ds^2 = \frac{1}{\cos^2 \tilde{u} \cos^2 \tilde{v}} \left[-\frac{32M^3}{r} e^{-r/2M} d\tilde{u} d\tilde{v} + r^2 \cos^2 \tilde{u} \cos^2 \tilde{v} d\Omega^2 \right]. \quad (4.32)$$

Dropping the conformal factor, we define

$$d\tilde{s}^2 = -\frac{32M^3}{r} e^{-r/2M} d\tilde{u} d\tilde{v} + r^2 \cos^2 \tilde{u} \cos^2 \tilde{v} d\Omega^2. \quad (4.33)$$

To draw a Penrose diagram, we again ignore the angular coordinates. Each point in the diagram therefore corresponds to a two-sphere whose size depends on the radial position r .

The singularity at $r = 0$ (or $\kappa^2 uv = 1$) is at

$$\begin{aligned} \tan \tilde{u} \tan \tilde{v} = 1 &\Rightarrow \sin \tilde{u} \sin \tilde{v} - \cos \tilde{u} \cos \tilde{v} = 0 \\ \cos(\tilde{u} + \tilde{v}) = 0 &\Rightarrow \tilde{u} + \tilde{v} = \pm\pi/2. \end{aligned} \quad (4.34)$$

The singularities are therefore straight, horizontal lines in the Penrose diagram. In the absence of these singularities, the Penrose diagram would be diamond-shaped, like that of 2D Minkowski. The singularities cut off the top and bottom and the Penrose diagram of the Schwarzschild geometry is that shown in Fig. 9.

We don't think that the regions III and IV of the Kruskal diagram can arise in a physical situation such as a black hole forming from a collapsing star. Figure 10 shows the alternative Penrose diagram for matter collapsing into a black hole. The diagram is a hybrid of the Penrose diagram for the Schwarzschild geometry (see Fig. 9) and that of four-dimensional Minkowski space (see Fig. 8). We see that the spacetime of a realistic black hole shares the singularity and the future event horizon with the maximally extended Schwarzschild solution, without any white hole, past horizon, or separate asymptotic region.

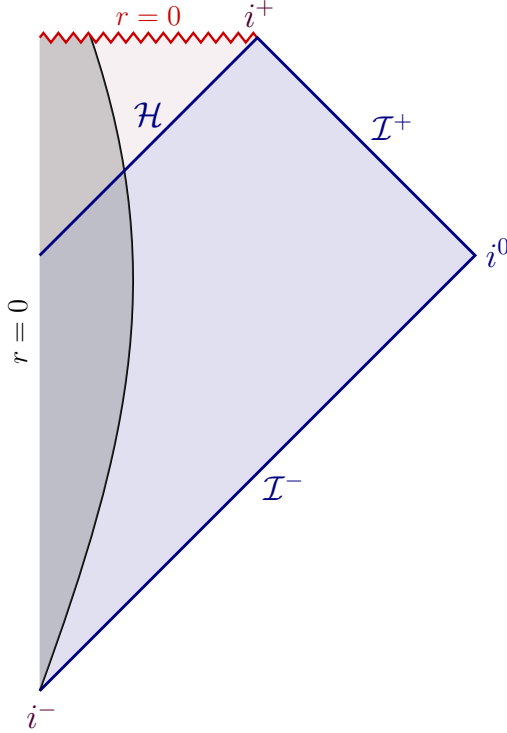


Figure 10: Penrose diagram for a real black hole formed from a collapsing cloud of dust. In interior of the cloud (gray region) is nonvacuum and therefore is not described by the Schwarzschild metric. Once the cloud enters the horizon $\mathcal{H}$, a black hole is formed.

4.2 Derivation of Hawking Radiation

We are now ready to consider the quantization of fields in the Schwarzschild geometry. We will first do this by ignoring the angular dependence of the metric and study a simple model for a “black hole” in two dimensions, where the analysis becomes the same as that for the Unruh effect. After that, we will describe Hawking’s original derivation which is much more involved, but leads to the same conclusion.

Toy model

Consider the metric (4.4), but drop its angular part, so that

$$\begin{aligned} ds^2 &= \left(1 - \frac{2M}{r}\right) (-dt^2 + dr^{*2}), \\ &= -\left(1 - \frac{2M}{r}\right) dU dV, \\ &= -\frac{2M}{r} e^{-r/2M} du dv. \end{aligned} \tag{4.35}$$

The analysis is then greatly simplified because a massless scalar in two dimensions is conformally invariant. Since the metric is conformally flat, solutions of the equation of motion $\square\phi = 0$ will be the same as those in flat space. We will write those solutions both in the lightcone tortoise coordinates (U, V) and the lightcone Kruskal coordinates (u, v) . The latter will play the role of the Minkowski coordinates of Section 3, while the former correspond to the Rindler coordinates in the analysis of the Unruh effect.

As before, the solution can be separated into right-moving (outgoing) modes (which are functions of U or u) and left-moving (incoming) modes (which are functions of V or v). Focusing on the outgoing modes in region I, the mode functions are

$$\begin{aligned} f_\omega(u) &= \frac{1}{\sqrt{4\pi\omega}} e^{-i\omega u}, \\ g_\omega^{(I)}(U) &= \frac{1}{\sqrt{4\pi\omega}} e^{-i\omega U} = \frac{1}{\sqrt{4\pi\omega}} e^{i(\omega/\kappa) \log(-\kappa u)}, \end{aligned} \tag{4.36}$$

where we have used (4.6) to write U in terms of u . The field operator can then be written in two different ways:

$$\begin{aligned} \hat{\phi} &= \int d\omega \left(\hat{a}_\omega f_\omega(u) + \text{h.c.} \right) \\ &= \int d\omega \left(\hat{b}_\omega g_\omega^{(I)}(U) + \text{h.c.} \right), \end{aligned} \tag{4.37}$$

and we again have two different vacuum states. The *Boulware vacuum* is

$$\hat{b}_\omega |0_B\rangle = 0. \tag{4.38}$$

It describes the state with no particles from the point of view of a remote observer. It is analogous to the Rindler vacuum in our treatment of the Unruh effect. The *Kruskal vacuum* is

$$\hat{a}_\omega |0_K\rangle = 0. \tag{4.39}$$

It describes the state that contains particles from the point of view of a remote observer, and is therefore the analog of the Minkowski vacuum in the Unruh effect.

Note that functions in (4.36) are of the same form as the mode functions in the analysis of the Unruh effect; cf. (3.29) and (3.30). In fact, comparing (3.19) to (4.6), we see that the two situations are identical if we identify the acceleration with the *surface gravity*

$$\kappa = \frac{1}{4M}. \quad (4.40)$$

Without doing any further calculation, we therefore infer that a remote observer will see particles in the Kruskal vacuum with a thermal spectrum

$$n_\omega \equiv \frac{\langle \hat{N}_\omega \rangle}{V} = \frac{1}{e^{2\pi\omega/\kappa} - 1}, \quad (4.41)$$

so that the effective temperature of the black hole is

$$T_{\text{BH}} = \frac{\kappa}{2\pi} = \frac{1}{8\pi M}. \quad (4.42)$$

We see that the temperature scales inversely with the mass, so large astrophysical black holes will be very cold.

Exercise 4.2 Determine the Hawking temperature of black holes of mass $M_\odot \approx 2 \times 10^{30} \text{ kg}$ (Solar mass), $M_\oplus \approx 6 \times 10^{24} \text{ kg}$ (Earth mass) and $M_{\text{Pl}} \approx 2 \times 10^{-8} \text{ kg}$ (Planck mass). →

Black hole evaporation

Hawking radiation leads to a slow evaporation of black holes. The rate of mass loss can be estimated by using the Stefan–Boltzmann law:

$$\frac{dM}{dt} = -\frac{dE}{dt} \approx -\sigma T_{\text{BH}}^4 A, \quad (4.43)$$

where A is the surface area of the black hole and $\sigma = \pi^2/60$ is the Stefan–Boltzmann constant (in Planck units). We have dropped an order-one constant related to the number of degrees of freedom that are emitted as Hawking radiation. Substituting $A = 4\pi R_S^2 = 16\pi M^2$ and $T_{\text{BH}} = (8\pi M)^{-1}$ in (4.43), we find

$$\frac{dM}{dt} = -\frac{1}{15360\pi M^2}, \quad (4.44)$$

which can be integrated to give

$$M(t) = M_0 \left(1 - \frac{t}{t_{\text{BH}}}\right)^{1/3}, \quad t_{\text{BH}} \equiv 5120\pi M_0^3, \quad (4.45)$$

where M_0 is the initial mass of the black hole and t_{BH} is its lifetime. Towards the end of the evaporation, the size of the black hole reaches the Planck scale and the effects of quantum gravity cannot be ignored. What happens then isn't known.

Exercise 4.3 Determine the lifetimes of black holes of mass $M_{\odot} \approx 2 \times 10^{30}$ kg (Solar mass), $M_{\oplus} \approx 6 \times 10^{24}$ kg (Earth mass) and $M_{\text{Pl}} \approx 2 \times 10^{-8}$ kg (Planck mass). →

Graybody factor

Our derivation of Hawking radiation was greatly simplified by ignoring the angular part of the metric (4.1) and focusing on an effective two-dimensional spacetime. Due to conformal invariance, the mode functions of the massless scalar were then the same as in flat space. Solving the real problem is significantly more complicated because the mode functions are more complicated. Here, we just sketch how the result is modified. In the next section, we will give more details of the exact computation.

We first write the field in a harmonic expansion

$$\phi(t, r, \theta, \varphi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{r} \phi_{lm}(t, r) Y_{lm}(\theta, \varphi), \quad (4.46)$$

where $\phi_{lm}(t, r)$ are the multipole moments. Using the tortoise coordinate (4.3), the equation of a massless field, $\square\phi = 0$, then becomes

$$\left[\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial r^{*2}} + V_l(r) \right] \phi_{lm}(t, r) = 0, \quad (4.47)$$

where we have introduced the effective potential

$$V_l(r) \equiv \left(1 - \frac{2M}{r} \right) \left(\frac{l(l+1)}{r^2} + \frac{2M}{r^3} \right). \quad (4.48)$$

Equation (4.47) can therefore be interpreted as the equation for waves propagating in the two-dimensional spacetime (4.35) with an additional potential term. Figure 11 shows the shape of the potential (for $l = 1$). We see that the potential vanishes exponentially for $r^* \rightarrow -\infty$ (near the horizon $\mathcal{H}^+$) and as a power law for $r^* \rightarrow \infty$ (near future null infinity $\mathcal{I}^+$).

Since the potential is localized away from the black hole, the main features of our simplified derivation of Hawking radiation survive. However, part of the waves escaping from the black hole will be reflected by the potential and part will be transmitted, with reflection coefficient $R(\omega)$ and transmission coefficient $T(\omega)$, respectively. A more detailed calculation shows that an observer near $\mathcal{I}^+$ will measure the following spectrum of particles:

$$n_{\omega} = \frac{\Gamma(\omega)}{e^{\omega/T_{\text{BH}}} - 1}, \quad (4.49)$$

where $\Gamma(\omega) = |T(\omega)|^2$ is called the *graybody factor*. For high frequencies, $\omega \gg 1/M$, the attenuation due to the potential is negligible, while for low frequencies, $\omega \ll 1/M$, it is significant. In those limits, the graybody factor becomes

$$\begin{aligned} \Gamma(\omega) &\approx 1, & \omega &\gg 1/M, \\ \Gamma(\omega) &\approx \frac{A}{4\pi} \omega^2, & \omega &\ll 1/M, \end{aligned} \quad (4.50)$$

where A is the area of the event horizon.

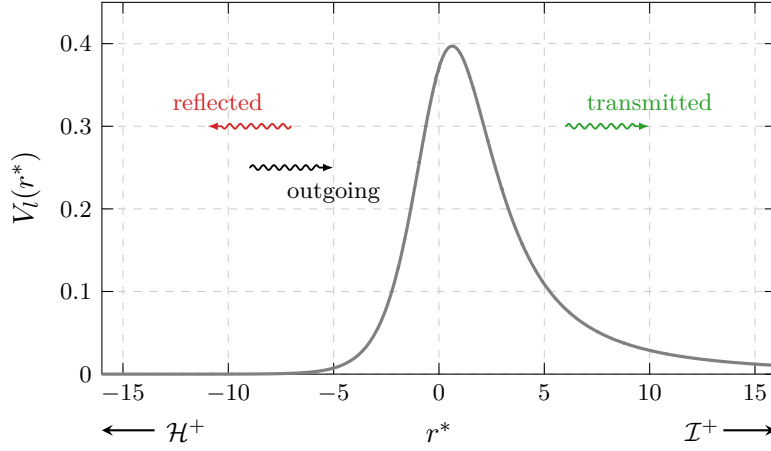


Figure 11: Sketch of the effective potential in the wave equation (4.47) of a massless scalar field in the Schwarzschild geometry.

Hawking's computation*

So far, we have only considered the case of an eternal Schwarzschild black hole. However, realistic black holes form by the collapse of matter; see Fig. 10. As we will now describe, this complication was included in Hawking's original derivation [4]. Our discussion will be very schematic. The technical details can be found in Hawking's paper or Reall's lecture notes [10].

As before, we consider the quantization of a massless scalar field. Near $\mathcal{I}^\pm$, the spacetime is stationary, so we can consider modes with definite (positive) frequency $\omega > 0$. The multiple moments in (4.46) can then be written as $\phi_{lm}(t, r) = e^{-i\omega t} R_{\omega lm}(r)$, and the differential equation (4.47) becomes

$$\left[-\frac{d^2}{dr^{*2}} + V_l(r^*) \right] R_{\omega lm} = \omega^2 R_{\omega lm}, \quad (4.51)$$

which has the form of a Schrödinger equation. Since the potential vanishes for $|r^*| \rightarrow \infty$, the asymptotic solution is $R_{\omega lm} \sim e^{\pm i\omega r^*}$ and hence

$$\phi_{\omega lm} \propto e^{-i\omega(t \mp r^*)} = \begin{cases} e^{-i\omega u} & \text{outgoing,} \\ e^{-i\omega v} & \text{ingoing.} \end{cases} \quad (4.52)$$

We will choose ingoing modes on $\mathcal{I}^-$ and outgoing modes on $\mathcal{I}^+$.²

Let $f_\omega(v)$ be a basis of positive-frequency “in” modes on $\mathcal{I}^-$, where we suppress the quantum numbers l and m . In terms of these modes, the field operator is

$$\hat{\phi} = \int d\omega (\hat{a}_\omega f_\omega(v) + \text{h.c.}). \quad (4.53)$$

The vacuum at early times is then defined by $\hat{a}_\omega|0\rangle = 0$. This will play the role of the Kruskal vacuum in our previous analysis.

²The plane-wave solutions in (4.52) are delocalized on $\mathcal{I}^\pm$ which leads to divergences in the computation. These divergences are regulated by considering localized “wavepackets” made of a superposition of plane waves. In the following, we will always implicitly assume a treatment in terms of such wavepackets.

The outgoing modes must include both modes escaping to future null infinity $\mathcal{I}^+$ and those entering the event horizon $\mathcal{H}$. Since the spacetime is static near $\mathcal{I}^+$, it contains a natural notion of positive-frequency modes on $\mathcal{I}^+$. We denote a basis of such “out” modes by $g_\omega(u)$. In addition, we have a second set of modes on the horizon $\mathcal{H}$. Since the geometry is not static everywhere on $\mathcal{H}$, there is no natural notion of positive-frequency modes. We will pick an arbitrary basis $h_\omega(u)$ for these modes. Since an observer at $\mathcal{I}^+$ will trace over the modes at the horizon, this choice will not affect the result. In terms of these outgoing modes, the field operator is

$$\hat{\phi} = \int d\omega (\hat{b}_\omega g_\omega(u) + \hat{c}_\omega h_\omega(u) + \text{h.c.}), \quad (4.54)$$

where $\hat{b}_\omega$ and $\hat{c}_\omega$ are the annihilation operators for outgoing particles at late times. The vacuum at $\mathcal{I}^+$ is defined by $\hat{b}_\omega|0\rangle_B = 0$. It plays the role of the Boulware vacuum.

To compute the density of emitted particles, we must derive the Bogolubov coefficients

$$\begin{aligned} \alpha_{\omega,\omega'} &= \langle f_\omega, g_{\omega'} \rangle, \\ \beta_{\omega,\omega'} &= -\langle f_\omega^*, g_{\omega'} \rangle, \end{aligned} \quad (4.55)$$

and then evaluate

$$\langle \hat{N}_\omega \rangle \equiv \langle 0 | \hat{b}_\omega^\dagger \hat{b}_\omega | 0 \rangle = \int d\omega' |\beta_{\omega,\omega'}|^2. \quad (4.56)$$

Although this computation is conceptually straightforward, it is technically rather complicated. In the following, we will sketch the basic logic of Hawking’s paper [4]. More details can be found in the lecture notes by Reall [10].

To find the expansion of the out-modes g_ω on $\mathcal{I}^+$ in terms of the in-modes f_ω on $\mathcal{I}^-$, Hawking traced the solution backwards in time. When the wave interacts with the potential, it splits into a “reflected” part and a “transmitted” part:

$$g_\omega = g_\omega^{(r)} + g_\omega^{(t)}. \quad (4.57)$$

Since $g_\omega^{(r)}$ is scattered by the static Schwarzschild field outside the collapsing matter, it will end up on $\mathcal{I}^-$ with the same frequency ω . It therefore does *not* contribute to the particle creation captured by the Bogolubov coefficient β . In contrast, the transmitted part $g_\omega^{(t)}$ enters the collapsing matter. Since it travels through a time-dependent geometry, the resulting solution on $\mathcal{I}^-$ will be a mixture of positive and negative frequency modes. Hence, $g_\omega^{(t)}$ determines the Bogolubov coefficient β and hence the particle creation on $\mathcal{I}^+$.

To compute $\beta_{\omega,\omega'}$, we must determine $g_\omega^{(t)}$ on $\mathcal{I}^-$. One might worry that this depends on the complicated dynamics of the gravitational collapse and is therefore highly model-dependent. We will now explain that this is fortunately not the case.

On $\mathcal{I}^+$, we consider a wavepacket localized around some retarded time (outgoing null coordinate) u_0 (see Fig. 12). When the waves making up the wavepacket are propagated backwards in time, there will be infinitely many oscillations between the line $u = u_0$ and the event horizon at $u = \infty$. Due to the gravitational blueshift, the wavepacket will be peaked at a very high frequency when it enters the collapsing matter. This allows us to use the “geometric optics”

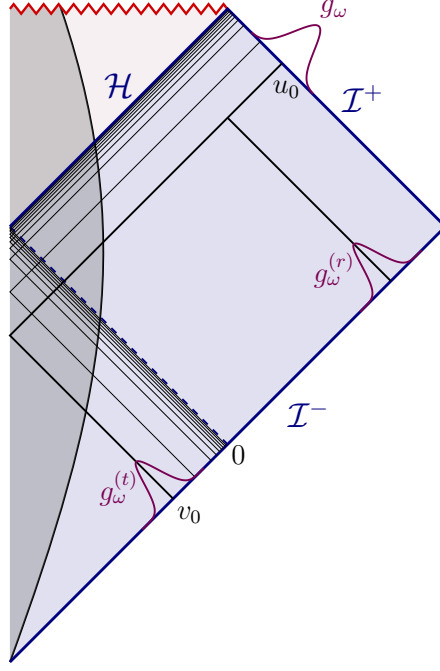


Figure 12: Backwards propagation of an out mode in the collapsing spacetime. The mode splits into a reflected part outside the collapsing matter and a transmitted part entering the collapsing matter.

approximation: The mode g_ω takes the form $A(x)e^{iS(x)}$, where $A(x)$ is a slowly-varying function. Substituting this into the Klein–Gordon equation gives $\nabla_\mu S \nabla^\mu S = 0$, which means that surfaces of constant phase are null. We can therefore trace a wave’s surfaces of constant phase back in time by following null geodesics. Using this fact, it can be shown that the transmitted part $g_\omega^{(t)}$ on $\mathcal{I}^-$ takes the form [4, 10]:

$$g_\omega^{(t)}(v) \approx \begin{cases} A(v) \exp[i(\omega/\kappa) \log(-\kappa v)] & v < 0, \\ 0 & v > 0, \end{cases} \quad (4.58)$$

where v is the advanced time (incoming null coordinate) on $\mathcal{I}^-$ and we absorbed an overall phase in the function $A(v)$.

The Bogolubov coefficients can then be written as

$$\alpha_{\omega, \omega'} = C \int_{-\infty}^0 dv \left(\frac{\omega}{\omega'} \right)^{1/2} e^{i\omega v} \exp[i(\omega'/\kappa) \log(-\kappa v)], \quad (4.59)$$

$$\beta_{\omega, \omega'} = C \int_{-\infty}^0 dv \left(\frac{\omega}{\omega'} \right)^{1/2} e^{-i\omega v} \exp[i(\omega'/\kappa) \log(-\kappa v)]. \quad (4.60)$$

where C is a constant. These are the same integrals as we found in our discussion of the Unruh effect in Section 3.1. Performing the same analysis as in Exercise 3.1 therefore leads to

$$|\alpha_{\omega, \omega'}|^2 = e^{2\pi\omega/\kappa} |\beta_{\omega, \omega'}|^2. \quad (4.61)$$

The normalization of the transmitted modes is

$$\begin{aligned}\Gamma(\omega) \delta(\omega - \omega') &\equiv \langle g_{\omega}^{(t)}, g_{\omega'}^{(t)} \rangle \\ &= \int d\lambda (\alpha_{\omega, \lambda}^* \alpha_{\omega', \lambda} - \beta_{\omega, \lambda}^* \beta_{\omega', \lambda}) = \left(e^{2\pi\omega/\kappa} - 1 \right) \int d\lambda \beta_{\omega, \lambda}^* \beta_{\omega', \lambda},\end{aligned}\tag{4.62}$$

and the density of particles observed on $\mathcal{I}^+$ becomes

$$n_{\omega} \equiv \frac{\langle \hat{N}_{\omega} \rangle}{V} = \frac{\langle 0 | \hat{b}_{\omega}^{\dagger} \hat{b}_{\omega} | 0 \rangle}{V} = \frac{\Gamma(\omega)}{e^{\omega/T_{\text{BH}}} - 1},\tag{4.63}$$

where $\Gamma(\omega)$ is the graybody factor and $T_{\text{BH}} = \kappa/(2\pi)$ is the Hawking temperature.

4.3 Black Hole Thermodynamics

In 1972, Jacob Bekenstein—then a graduate student at Princeton—worried what black holes would do to the second law of thermodynamics [11]. Prompted by a question of his advisor John Wheeler, Bekenstein wondered what would happen when an object with entropy S falls into a black hole. If black holes don't carry any entropy, this seemed to violate the second law by decreasing the total entropy of the Universe. To save the second law, Bekenstein therefore suggested that black holes must have an entropy, so that the total energy of the Universe—including the black hole entropy—never decreases.

Not long before, Hawking had proven that in classical general relativity the horizon area of a black hole always increases [12]. This area theorem states, for example, that a rotating black hole can lose mass and angular momentum by the Penrose process, but its horizon area nevertheless increases. It was therefore natural for Bekenstein to associate the entropy of a black hole with its entropy. Hawking's discovery of black hole radiation further showed that this association was more than an analogy, but that black holes actually were thermodynamic objects whose entropy was proportional to their area.

Black hole entropy Consider the case of a Schwarzschild black hole. The area of the black hole's event horizon is $A = 4\pi R_S^2 = 16\pi M^2$. Taking a differential of this expression gives

$$dA = 32\pi M dM \quad \implies \quad dM = \frac{1}{8\pi M} d\left(\frac{A}{4}\right).\tag{4.64}$$

Comparing this to the first law of thermodynamics, $dE = TdS$, and using $T = (8\pi M)^{-1}$ from (4.42), we identify the black hole entropy as

$$S_{\text{BH}} = \frac{A}{4}.\tag{4.65}$$

The result $S_{\text{BH}} \propto A$ was argued by Bekenstein on general grounds, but Hawking's calculation of the black hole's temperature fixed the precise relation including the constant of proportionality.

How large is this entropy for a typical astrophysical black hole? Restoring the fundamental constants, we have

$$S_{\text{BH}} = \frac{Ac^3}{4\hbar G} = \frac{A}{4\ell_P^2},\tag{4.66}$$

where $\ell_P \equiv (\hbar G/c^3)^{1/2}$ is the Planck length. Since we measure temperature in units of energy, with Boltzmann's constant k_B set to 1, the entropy is dimensionless. For ordinary astrophysical black holes, the entropy is enormous. For example, a black hole with the mass of the Sun has an entropy $S \sim 10^{77}$, which implies a gigantic number of microstates $N \sim e^S$. The microscopic origin of this entropy remains mysterious (but see [13]).

Laws of black hole mechanics Even before the discovery of Hawking radiation, it was noticed that the classical Einstein equations lead to laws of black hole mechanics that are “analogous” to the laws of thermodynamics [14]. In the following, we will briefly describe these observations. The discovery of Hawking radiation then showed that these results are not merely an analogy, but a deep property of the quantum nature of black holes.

- *Zeroth law* In thermodynamics, the zeroth law states that if two systems are in thermal equilibrium with a third system, then they are in thermal equilibrium with each other. This implies that the temperature T of a body in equilibrium is a constant. Similarly, for black holes the surface gravity κ is constant on the future event horizon of a stationary black hole spacetime obeying the dominant energy condition.³ This suggested that surface gravity was “analogous” to temperature.
- *First law* In thermodynamics, the first law is the conservation of energy. The change in energy dE must equal the sum of heat transfer $d\tilde{Q}$ and the work done dW . For simplicity, we will set $dW = 0$. For reversible changes between equilibrium states, we have $d\tilde{Q} = TdS$, so that $dE = TdS$. In the presence of an electric potential Φ and an angular potential Ω , this becomes

$$TdS = dE - \Phi dQ - \Omega dJ, \quad (4.67)$$

where Q is the electric charge and J is the angular momentum. Similarly, using the Einstein equations, one can prove that the change of the horizon area of a black hole (possibly charged and rotating) satisfies

$$\frac{\kappa}{8\pi} dA = dM - \Phi dQ - \Omega dJ. \quad (4.68)$$

The similarity with (4.67) is obvious.

- *Second law* The entropy of an isolated system never decreases, $dS \geq 0$. Similarly, for black holes, Hawking proved that the area of its horizon never decreases, $dA \geq 0$ [12]. For example, although a rotating black hole can lose mass through the Penrose process, its area still increases. This *area theorem* further motivates the correspondence between area and entropy of black holes.

The theorem even applies to multiple black holes. Consider the merger of two non-rotating black holes with masses M_1 and M_2 . The system will emit gravitational wave and then settle down to a black hole with mass $M_3 < M_1 + M_2$. Although the total mass has decreased in the process, the total area, $A \propto M^2$, has not, as long as $(M_1 + M_2)^2 > M_1^2 + M_2^2$.

³The proof of this statement is relatively simple, but will not be given here.

- *Third law* In thermodynamics, the third law states that the temperature of a system can never be reduced to absolute zero (in a finite number of steps). The corresponding statement for black holes would be that the surface gravity cannot be reduced to zero. This is believed to be true, but has not been proven. In fact, there are claimed counterexamples.

Exercise 4.4 Consider the merger of two non-rotating black holes. Show that at most 29% of the initial rest energy can be emitted in gravitational waves. →

4.4 Holography and Information

The consequences of Hawking’s discovery continue to inspire important work in quantum gravity. In this section, we will describe some of the more recent developments (with pointers to the relevant literature).

Holography The most striking feature of Hawking’s formula for the thermodynamic entropy of a black hole is that it is proportional to the area of the black hole’s event horizon, rather than its volume. This is unusual because entropy is typically an extensive quantity that scales with the volume of a system. Somehow for black holes, all information about the interior is encoded on the surface. Inspired by this result, ‘t Hooft and Susskind suggested the “holographic principle” [15, 16] which posits that in any theory of quantum gravity the information contained within a volume of space can be fully described by information on its boundary, just like a hologram encodes a three-dimensional image in a two-dimensional surface. This idea was made very precise when Juan Maldacena discovered that string theory in anti-de Sitter (AdS) spacetime is fully described by a conformal field theory (CFT) on its boundary [17]; see Fig. 13. Remarkably, this AdS/CFT correspondence relates a theory with gravity in the bulk to a lower-dimensional theory without gravity on the boundary.

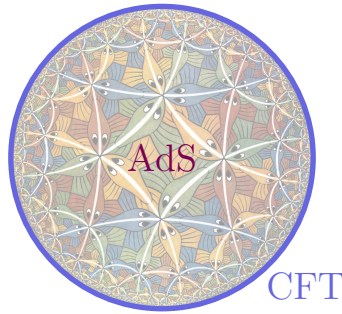


Figure 13: Two-dimensional representation of anti-de Sitter spacetime.

Microstates What is the microscopic origin of the black hole entropy? Above we have seen that the entropy of typical astrophysical black holes is exponentially large and therefore black holes must be described by an enormous number of microstates $N \sim e^S$. Naively, this seems to be in tension with the “no-hair theorem”, which states that, in classical general relativity, black holes are completely described by only a few macroscopic parameters: mass, charge, and angular

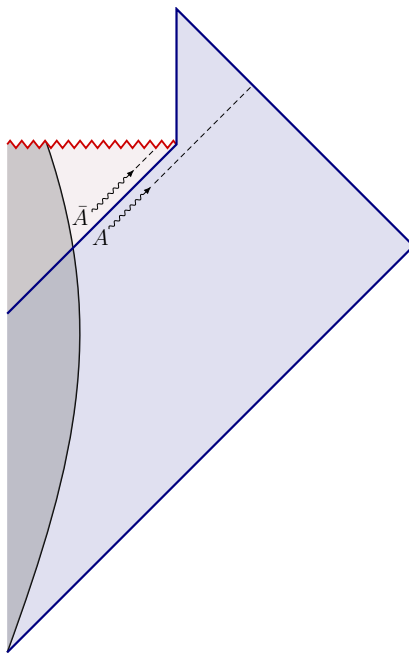


Figure 14: Penrose diagram for an evaporating black hole. The Hawking radiation is described by two entangled quanta A and $\bar{A}$, one outside the black hole and one inside. The outside radiation evolves into a mixed state.

momentum. The resolution must come from quantum gravity. In fact, for certain black holes in string theory, Strominger and Vafa were able to reproduce the Bekenstein–Hawking entropy by counting the microstates associated to configurations of D-branes [13].

Information paradox We have seen that in classical general relativity the horizon area of black holes (and hence the associated entropy) always increases in accordance with the second law of thermodynamics. However, we also showed that quantum mechanically black holes shrink by Hawking evaporation. How is this consistent with the second law? Of course, an easy resolution is that the missing entropy is carried away by the Hawking radiation. Thinking about it more carefully, however, a new puzzle arises: the “black hole information paradox” [18].

To describe the problem, we return to the no-hair theorem. According to this theorem, two black holes with the same mass, charge, and angular momentum are indistinguishable from one another, regardless of the specific details of the matter that formed them. The information about the initial state of the system seems to be lost. One may think that the information is simply hidden behind the black hole horizon, however, the black hole evaporates and eventually disappears. Moreover, Hawking’s computation showed that the outgoing radiation is purely thermal and therefore doesn’t carry away the missing information. (This is different from the smoke of a fire which *in principle* contains all information of the objects being burnt.)

The problem sharpens when one looks more closely at the details of black hole evaporation. We can think of the Hawking process as the pair creation of entangled particles near the horizon, with one particle escaping as Hawking radiation to infinity and the other falling toward the singularity (see Fig. 14). This entanglement is crucial for a precise formulation of the information

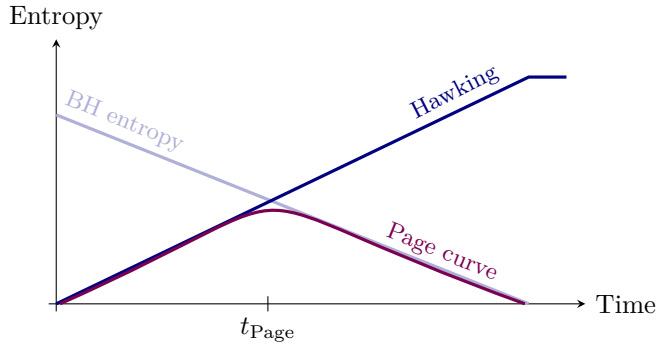


Figure 15: Schematic evolution of the entropy of the outgoing radiation. Shown are the result of Hawking’s calculation and the Page curve (predicted by unitarity). At the Page time, t_{Page} , the entropy of the radiation exceeds the thermodynamic entropy of the black hole and the information paradox arises.

paradox. Initially, the system described by the black hole and its emitted radiation would be in a pure state. However, as the black hole evaporates, the entanglement evolves, leading to a mixed state for the emitted radiation. At first, this isn’t a problem since the black hole degrees of freedom together with the radiation are still in a pure state. However, eventually the black hole has fully evaporated and only the thermal radiation remains. The system has evolved from a pure to a mixed state in violation of the principle of *unitarity* in quantum mechanics. In fact, the problem arise even earlier when the entropy of radiation has become bigger than the thermodynamic entropy of the black hole.

To quantify the issue, we consider the *von Neumann entropy* of the radiation:

$$S_R = -\text{Tr}[\rho \log \rho], \quad (4.69)$$

where ρ is the density matrix of the radiation. The “fine-grained” entropy vanishes for a pure state, so it is a useful diagnostic for the information problem. According to Hawking’s calculation [18], the entropy of the radiation grows monotonically as the black hole evaporates, while the “coarse-grained” Bekenstein–Hawking entropy of the black hole decreases to zero. This leads to a contradiction: Unitarity requires that von Neumann entropy of an isolated system stays constant. If the initial state of the black hole is pure then the final state of the radiation must also be pure and $S_R \rightarrow 0$ at late times. Moreover, general properties of entangled quantum systems require that $S_R \leq S_{\text{BH}}$.

In Figure 15, we have sketched the *Page curve* [19] for the evolution the entropy of the outgoing Hawking radiation as predicted by unitarity. We see that Hawking’s calculation is in conflict with this prediction. In particular, after the Page time (roughly half the evaporation time), Hawking’s computation gives a radiation entropy that exceeds the thermodynamic entropy of the black hole. From that point on, the entangled black hole–radiation system cannot be in a pure state anymore and the information paradox arises. Note that the problem appears when the black hole is still very large, so we cannot solve the problem by appealing to the breakdown of the semi-classical computation during the late stages of the evaporation.

Recently, the paradox was revisited using new insights from holography. In two breakthrough

papers [20, 21], it was shown that a “quantum extremal surface” develops just inside the black hole horizon. After crossing this surface, the inner particles of the entangled pairs are no longer part of the black hole and don’t count towards the von Neumann entropy of the Hawking radiation. As the black hole evaporates, the entropy therefore drops to zero and the evolution follows the Page curve. Although these calculations prove that information is indeed preserved during black hole evaporation, they don’t show exactly how the information is encrypted in the outgoing radiation. Comprehensive reviews of these recent developments can be found in [22–24].

4.5 Selected Problems

The following are a few longer problems related to the material covered in this chapter.

Exercise 4.5 The Reissner–Nordström metric of a charged non-rotating black hole is

$$ds^2 = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2 d\Omega^2, \quad f(r) \equiv 1 - \frac{2M}{r} + \frac{Q^2}{r^2}, \quad (4.70)$$

where M is the mass and Q is the (electric) charge (in units where $G = 4\pi\epsilon_0 \equiv 1$). The four-vector potential of the solution is $A_\mu = (-Q/r, 0, 0, 0)$, so that non-zero component of the field strength is $F_{tr} = Q/r^2$ (i.e. a radial electric field). This exercise is adapted from [25].

1. Determine the two event horizons r_+ and r_- .
2. Assuming $S = A/4$, where $A = 4\pi r_+^2$ is the area of the outer horizon, show that

$$T dS = dM - \Phi dQ, \quad (4.71)$$

where T and Φ should be related to M and Q .

3. Imagine now that a particle of energy (per unit mass) $e \ll M$ and charge (per unit mass) $q \ll Q$ falls into a Reissner–Nordström black hole along a radial trajectory. The mass and charge of the black hole changes by $dM = e$ and $dQ = q$, so that the associated change in its entropy is

$$dS = \frac{1}{T}(e - \Phi q). \quad (4.72)$$

You will now show that this change must be positive.

The conserved energy of the particle is

$$e = -(p_\mu + qA_\mu)K^\mu, \quad (4.73)$$

where $p^\mu = dx^\mu/d\tau$ is the four-momentum of the particle (with τ being the proper time along the trajectory $x^\mu(\tau)$) and $K^\mu = (1, 0, 0, 0)$ is a Killing vector of the metric (4.70).

- (a) Find e in terms of $f(r)$ and the components of p^μ .

- (b) Assume $Q > 0$. For which sign of the charge q can the energy e be negative? If we drop such a negative-energy particle into a black hole, the mass of the black hole will decrease.
- (c) Use the normalization $p_\mu p^\mu = -1$, to write e in terms of p^r .
- (d) Since e is conserved, we can evaluate it at the horizon. Show that the minimal value of the energy is

$$e_{\min} = q\Phi(r_+), \quad (4.74)$$

and that the second law therefore isn't violated.

In this exercise, you have shown that although you can decrease the energy of a charged black hole, its entropy still increases, so that the second law is satisfied. ➔

Exercise 4.6 The Kerr metric of a rotating black hole of mass M and angular momentum J is

$$\begin{aligned} ds^2 = & - \left(1 - \frac{2Mr}{\rho^2}\right) dt^2 - \frac{4Mr a}{\rho^2} \sin^2 \theta d\phi dt + \frac{\rho^2}{\Delta} dr^2 + \rho^2 d\theta^2 \\ & + \left(r^2 + a^2 + \frac{2Mr a^2 \sin^2 \theta}{\rho^2}\right) \sin^2 \theta d\phi^2, \end{aligned} \quad (4.75)$$

where $a \equiv J/M$, $\rho^2 \equiv r^2 + a^2 \cos^2 \theta$ and $\Delta \equiv r^2 - 2Mr + a^2$.

1. For $a < M$, derive the two event horizons r_+ and r_- .
2. Show that the area of the outer horizon is $A = 8\pi M r_+$.
3. Assuming $S = A/4$, show that

$$T dS = dM - \Omega dJ, \quad (4.76)$$

where T and Ω should be related to M and a .

4. The metric (4.75) has the Killing vectors $K = \partial_t$ and $R = \partial_\phi$.
 - (a) Determine the surface on which K is null. The region between the outer event horizon and this surface is the “ergoregion”. Is K timelike or spacelike inside the ergoregion?
 - (b) A particle with four-momentum $p^\mu = dx^\mu/d\tau$ has the conserved energy $E = -p_\mu K^\mu$. Argue why it is possible to have an object with negative energy inside the ergoregion.
 - (c) Consider now the following combination of the two Killing vectors:

$$\xi^\mu \equiv K^\mu + \Omega R^\mu, \quad \Omega \in \mathbb{R}. \quad (3)$$

Determine the function $\Omega(r, \theta)$ such that ξ^μ is a null vector, i.e. $\xi_\mu \xi^\mu = 0$. Write the result in terms of g_{tt} , $g_{t\phi}$, $g_{\phi\phi}$. Show that

$$\Omega_H \equiv \Omega(r_+, \theta) = \frac{a}{2Mr_+} . \quad (4)$$

5. Imagine now that the black hole absorbs a particle with energy $E = -p_\mu K^\mu$ and angular momentum $L = p_\mu R^\mu$. Note that $\xi_\mu p^\mu \leq 0$ for any timelike p^μ . By evaluating $\xi_\mu p^\mu$ at the outer horizon, find a relation between E and L .

After the absorption, the black hole's mass changes by $\delta M = E \ll M$ and its angular momentum by $\delta J = L \ll J$. Compute the change of the horizon area δA and show that it is always positive. This shows that this process cannot be used to shrink the event horizon area of the black hole to zero.

In this exercise, you proved a special case of the area theorem [12].



Exercise 4.7 Consider a black hole of mass M in a finite reservoir of radiation. This question concerns the stability of the system as function of the volume.

A finite reservoir of radiation of volume V at temperature T has an energy $E_R = \sigma VT^4$ and entropy $S_R = \frac{4}{3}\sigma VT^3$, where σ is a constant. A Schwarzschild black hole of mass M is placed in the reservoir.

1. Assuming that the black hole has entropy $S_{\text{BH}} = 4\pi M^2$, show that the total entropy $S = S_{\text{BH}} + S_R$ is extremized for fixed total energy $E = M + E_R$, when

$$T = T_{\text{BH}} , \quad (4.77)$$

where $T_{\text{BH}} = (8\pi M)^{-1}$ is the black hole temperature.

2. Show that the extremum is a maximum if and only if $V < V_c$, where the critical value is

$$V_c = \frac{2^{20}\pi^4}{5^5} \frac{E^5}{\sigma} . \quad (4.78)$$

What happens when $V > V_c$?



Exercise 4.8 Recall that the metric of a charged black hole is

$$ds^2 = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2 d\Omega^2 , \quad f(r) \equiv 1 - \frac{2M}{r} + \frac{Q^2}{r^2} . \quad (4.79)$$

In Chapter 6, we will show that the black hole's temperature is $T_{\text{BH}} \equiv f'(r_+)/4\pi$, where r_+ is

the (outer) horizon. At fixed charge Q , the specific heat of the black hole is

$$C \equiv T_{\text{BH}} \left. \frac{\partial S_{\text{BH}}}{\partial T_{\text{BH}}} \right|_Q, \quad (4.80)$$

where S_{BH} is its entropy.

1. Assuming that the black hole's entropy is $S_{\text{BH}} = A/4$, where A is the area of the event horizon, show that its specific heat is

$$C = 2S_{\text{BH}} \frac{\sqrt{M^2 - Q^2}}{M - 2\sqrt{M^2 - Q^2}}. \quad (4.81)$$

Hence, show that C^{-1} changes sign when M passes through $2|Q|/\sqrt{3}$.

2. Repeat Exercise 4.7 for this charged black hole. Specifically, show that the critical reservoir volume, V_c , is infinite for $|Q| \leq M \leq 2|Q|/\sqrt{3}$. Why is this result to be expected from your previous result for C ? 

5 Quantum Fields in De Sitter Space

It came as a great shock when supernova observations discovered that the current expansion of the Universe is accelerating [26, 27]. In the future, this expansion will dilute all matter and radiation and only a form of dark energy—possibly a cosmological constant—will remain. We will be causally disconnected from the rest of the Universe by a cosmic horizon, some 17 billion light years in size. It will be a cold and lonely world.

Observations of the cosmic microwave background have also given us tantalizing evidence that the early universe experienced another phase of accelerated expansion called “cosmic inflation” [28–31]. During this inflationary period, small quantum fluctuations were stretched to cosmological distances creating the primordial density fluctuations that seeded the large-scale structure of the Universe. The energy density during inflation was approximately constant, mimicking for a brief period the effects of a cosmological constant.

We therefore have observational evidence for two periods of accelerated expansion, in the far past and future of our Universe. In both eras, the spacetime geometry is well approximated by *de Sitter space*, which is a maximally symmetric spacetime of positive curvature:

$$R = \frac{D(D-1)}{L^2}, \quad (5.1)$$

where L is the de Sitter radius. It is a solution to the vacuum Einstein equations with positive cosmological constant $\Lambda > 0$, where $L = \sqrt{(D-1)(D-2)/(2\Lambda)}$. The measured value of the dark energy density in our Universe today implies $L \sim 10^{26}$ m, which sets the scale of the cosmic horizon. Given the importance of de Sitter space in the history of our Universe, we will now study the behavior of quantum fields in this geometry. We will follow closely the treatment in [32, 33].

5.1 Geometry of De Sitter

We will begin with some mathematical preliminaries on the geometry of de Sitter space. We will present results in D spacetime dimensions, although we are ultimately interested in the real world with $D = 4$.

D -dimensional de Sitter space can be viewed as a hyperboloid embedded in $(D+1)$ -dimensional Minkowski space

$$-X_0^2 + \sum_{i=1}^D X_i^2 = L^2, \quad (5.2)$$

The embedding makes the $SO(D, 1)$ symmetry of de Sitter space manifest. The de Sitter metric can be derived by projecting the metric of the embedding space, $ds^2 = -dX_0^2 + \sum_{i=1}^D dX_i^2$, onto the hyperboloid. We will present this metric in several different coordinate systems.

Global coordinates The solution to the embedding constraint (5.2) can be written as

$$\begin{aligned} X_0 &= L \sinh(t/L), \\ X_i &= L \cosh(t/L) \omega_i, \end{aligned} \quad (5.3)$$

where $-\infty < t < \infty$ and $\omega^i \omega_i = 1$. The vector ω_i defines the points on the unit $(D-1)$ -sphere S_{D-1} . The coordinates (t, ω_i) cover the entire hyperboloid and are therefore called the

global coordinates of de Sitter space.⁴ For $D = 4$, we can use the following parameterization in terms of polar angles

$$\begin{aligned}\omega_1 &= \cos \psi, \\ \omega_2 &= \sin \psi \cos \theta, \\ \omega_3 &= \sin \psi \sin \theta \cos \phi, \\ \omega_4 &= \sin \psi \sin \theta \sin \phi,\end{aligned}\tag{5.4}$$

where $0 \leq \psi < \pi$, $0 \leq \theta < \pi$ and $0 \leq \phi < 2\pi$. The metric on the three-sphere S^3 is

$$\begin{aligned}d\Omega_3^2 &= d\psi^2 + \sin^2 \psi (d\theta^2 + \sin^2 \theta d\phi^2) \\ &\equiv d\psi^2 + \sin^2 \psi d\Omega^2.\end{aligned}\tag{5.5}$$

In general spacetime dimensions, we similarly foliate the $(D-1)$ -sphere by $(D-2)$ -spheres, and write the metric as

$$d\Omega_{D-1}^2 = d\psi^2 + \sin^2 \psi d\Omega_{D-2}^2,\tag{5.6}$$

where $0 \leq \psi < \pi$.

Using (5.3), the induced metric on dS_D is

$$ds^2 = -dt^2 + L^2 \cosh^2(t/L) d\Omega_{D-1}^2.\tag{5.7}$$

The global coordinates are also called the *closed slicing* of de Sitter, because the metric takes the form of a closed FRW universe. The scale factor $a(t) = L \cosh(t/L)$ first decreases exponentially, then reaches the minimum $a(0) = L$ at $t = 0$, and finally expands exponentially into the future.

Penrose diagram To draw de Sitter space on a finite sheet of paper, we introduce the conformal transformation $\cosh(t/L) = (\cos T)^{-1}$, where $-\pi/2 \leq T < \pi/2$. The metric (5.7) then takes the form

$$ds^2 = \frac{L^2}{\cos^2 T} (-dT^2 + d\Omega_{D-1}^2),\tag{5.8}$$

which is conformally related to the Minkowski metric $d\tilde{s}^2 = -dT^2 + d\Omega_{D-1}^2$. Null geodesics on ds^2 are also null with respect to $d\tilde{s}^2$. Using (5.5), we have

$$ds^2 = \frac{L^2}{\cos^2 T} (-dT^2 + d\psi^2 + \sin^2 \psi d\Omega_{D-2}^2).\tag{5.9}$$

Suppressing the sphere S_{D-2} , this is conformally related to two-dimensional Minkowski space, $d\tilde{s}^2 = -dT^2 + d\psi^2$. Since the coordinates T and ψ both have range π , the corresponding *Penrose diagram* is a square (see Fig. 16). The fixed T slices of the Penrose diagram are $(D-1)$ -spheres. Each point on the inside of the diagram is a $(D-2)$ -sphere. The North and South poles (at $\psi = 0$ and $\psi = \pi$, respectively) are points where the size of the suppressed $(D-2)$ -spheres shrink to zero. The left and right “edges” of the diagram therefore aren’t real boundaries, but are just the poles of the $(D-1)$ -spheres at each time slice. Remembering that light rays travel at 45° in a Penrose diagram, it is easy to determine the causal relations of points in the spacetime. For

⁴Calling these coordinates “global” is a slight exaggeration since it is impossible to cover the spheres S_{D-1} with a single coordinate patch. In particular, the standard polar coordinates have isolated singularities at the poles.

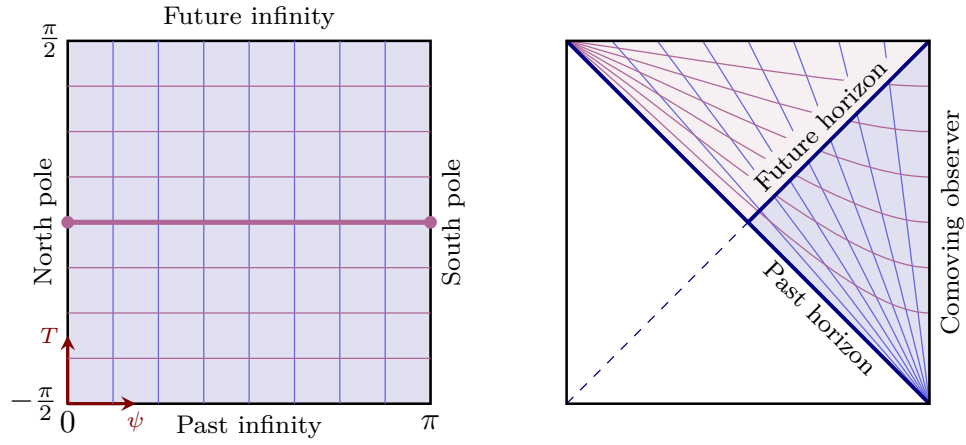


Figure 16: Penrose diagram of D -dimensional de Sitter space (for $D \neq 2$). Each time slice is a $(D - 1)$ -sphere and each point is a $(D - 2)$ -sphere. The left and right “edges” aren’t boundaries, but are the North and South poles of the S_{D-1} . The bottom and top are the past and future boundaries.

example, a comoving observer at the South pole has both a “past horizon” (limiting the region that this observer can influence) and a “future horizon” (limiting the region from which this observer can be influenced). The observer has causal contact only with the “causal patch”, a region that is bounded by both horizons.

Planar coordinates Another solution to the embedding constraint (5.2) is

$$X_0 = L \sinh(\tilde{t}/L) + \frac{\tilde{r}^2}{2L} e^{\tilde{t}/L}, \quad X_1 = L \cosh(\tilde{t}/L) - \frac{\tilde{r}^2}{2L} e^{\tilde{t}/L}, \quad X_i = e^{\tilde{t}/L} \tilde{x}_i, \quad (5.10)$$

where $i = 2, \dots, D$ and $\tilde{r}^2 = \tilde{x}^i \tilde{x}_i$. These *planar coordinates* are used to describe the inflationary spacetime. Note that these coordinates only cover $X_0 + X_1 = L e^{\tilde{t}/L} > 0$, i.e. half of the spacetime (see Fig. 16).

Exercise 5.1 Comparing (5.3) and (5.10), show that the global and planar coordinates are related by

$$e^{\tilde{t}/L} = \cos \psi \cosh(t/L) + \sinh(t/L), \quad \frac{L}{\tilde{r}} e^{\tilde{t}/L} = \sin \psi \cosh(t/L). \quad (5.11)$$

Note that the time coordinates becomes the same at late times.

Show that the induced metric is

$$ds^2 = -d\tilde{t}^2 + e^{2\tilde{t}/L} \delta_{ij} d\tilde{x}^i d\tilde{x}^j, \quad (5.12)$$

These coordinates are also called the *flat slicing* of de Sitter, because the metric takes the form of a flat FRW universe. The scale factor $a(\tilde{t}) = e^{\tilde{t}/L}$ grows exponentially. ➔

Both global and planar coordinates, of course, describe the same de Sitter spacetime. There is no physics in the choice of coordinates. Beyond the idealized de Sitter approximation, however, an FRW universe contains additional matter contributions (like the inflaton field in single-field slow-

roll inflation). The average rest frame of the matter selects a preferred frame and breaks the time translation symmetry of de Sitter spacetime. The closed and flat slicings then become inequivalent FRW universes. (The spacetimes are approximately identical, but the initial conditions for the matter are different.) Inflation picks out the flat slicing as the preferred foliation of the spacetime.

Conformal coordinates It is often useful to work with the *conformal time* defined by

$$\eta \equiv \int_{-\infty}^{\tilde{t}} \frac{dt}{a(t)} = -Le^{-\tilde{t}/L}. \quad (5.13)$$

Note that $\eta = -\infty$ corresponds to the infinity past, while $\eta = 0$ is the infinite future. In these coordinates, the embedding constraint (5.2) is solved by

$$X_0 = \frac{1}{2\eta}(\eta^2 - r^2 - L^2), \quad X_1 = \frac{1}{2\eta}(\eta^2 - r^2 + L^2), \quad X_i = -\frac{L}{\eta}x_i, \quad (5.14)$$

and the induced metric is

$$ds^2 = \frac{L^2}{\eta^2} \left(-d\eta^2 + \delta_{ij}dx^i dx^j \right). \quad (5.15)$$

We see that this is conformally flat, which is one of the benefits of working with conformal time.

Static coordinates De Sitter space has a time-translation symmetry (i.e. all physical quantities are time-independent constants), but this isn't manifest in the global and planar coordinates. Consider therefore the following alternative solution of the embedding constraint (5.2):

$$\begin{aligned} X_0 &= \sqrt{L^2 - r^2} \sinh(t/L), \\ X_1 &= \sqrt{L^2 - r^2} \cosh(t/L), \\ X_i &= r\omega_i, \end{aligned} \quad (5.16)$$

where $i = 2, \dots, D$ and $\omega^i \omega_i = 1$. In these coordinates, the induced metric on dS_D is

$$ds^2 = -\left(1 - \frac{r^2}{L^2}\right) dt^2 + \left(1 - \frac{r^2}{L^2}\right)^{-1} dr^2 + r^2 d\Omega_{D-2}, \quad (5.17)$$

with $0 < r < L$. These “static coordinates” cover the *causal patch* of an observer sitting at the South pole (see Fig. 16). They are the analog of the “Schwarzschild coordinates” for a black hole. The metric is now manifestly static, since its components are independent of t . The horizon is at $r = L$.

Exercise 5.2 Show that (5.17) is a solution of the vacuum Einstein equation with a positive cosmological constant. You may specialize to $D = 4$.

Hint: By Birkhoff's theorem, any spherically symmetric solution of the vacuum field equations must be static. You may therefore use the ansatz

$$ds^2 = -e^{2\alpha(r)} dt^2 + e^{2\beta(r)} dr^2 + r^2 d\Omega^2, \quad (5.18)$$

for which the Ricci tensor becomes

$$\begin{aligned}
R_{tt} &= e^{2(\alpha-\beta)} \left[\partial_r^2 \alpha + (\partial_r \alpha)^2 - \partial_r \alpha \partial_r \beta + \frac{2}{r} \partial_r \alpha \right] \\
R_{rr} &= -\partial_r^2 \alpha - (\partial_r \alpha)^2 + \partial_r \alpha \partial_r \beta + \frac{2}{r} \partial_r \beta \\
R_{\theta\theta} &= e^{-2\beta} \left[r(\partial_r \beta - \partial_r \alpha) - 1 \right] + 1 \\
R_{\phi\phi} &= \sin^2 \theta R_{\theta\theta},
\end{aligned} \tag{5.19}$$

Show that the Einstein equation reads $R_{\mu\nu} = \Lambda g_{\mu\nu}$, and determine $\alpha(r)$ and $\beta(r)$. →

Exercise 5.3 Let $(\tilde{t}, \tilde{r})$ be the coordinates in the flat slicing. Comparing (5.10) and (5.16), show that the coordinates on the static slices are related by

$$r = \tilde{r} e^{\tilde{t}/L}, \quad e^{-2t/L} = e^{-2\tilde{t}/L} - \frac{\tilde{r}^2}{L^2}. \tag{5.20}$$

We see that the horizon at $r = L$ corresponds to $\tilde{r} = L e^{-\tilde{t}/L}$ in the flat slicing. →

De Sitter as a sphere In Euclidean signature, D -dimensional de Sitter space dS_D becomes a D -dimensional sphere S_D . This is easily seen from the embedding condition (5.2). Setting $X_0 = -iX_{0,E}$, we get

$$X_{0,E}^2 + \sum_{i=1}^D X_i^2 = L^2, \tag{5.21}$$

which describes a four-sphere of radius L . Equivalently, we can define the following analytic continuation of the time coordinate $t = -i\tau$ in (5.7) and write

$$\Psi \equiv \frac{\tau}{L} + \frac{\pi}{2}, \tag{5.22}$$

where the shift by $\pi/2$ is just for convenience. The metric (5.7) then becomes

$$ds^2 = L^2 (d\Psi^2 + \sin^2 \Psi d\Omega_{D-1}^2) \equiv L^2 d\Omega_D^2, \tag{5.23}$$

which we recognize as the metric of S_D .

Geodesics An important quantity is the *geodesic distance*. To derive this, it is useful to first define the distance between two points in the embedding space and then constrain these points to lie on the hyperboloid.

Let us first illustrate this for a D -dimensional sphere of radius L . The geodesic distance between two points on the sphere is $d = L\theta$. However, we can also think of the sphere as embedded in $(D+1)$ -dimensional Euclidean space, with the embedding constraint $\delta_{AB} X^A X^B = L^2$, with $A, B = 1, \dots, D+1$. It is then useful to define the quantity

$$P(X, X') \equiv \frac{\delta_{AB} X^A X'^B}{L^2} = \cos \theta, \tag{5.24}$$

and write the geodesic distances as $d(x, x') = L \cos^{-1}(P)$. We can do something similar for dS_D although it is harder to visualize.

Consider the auxiliary quantity

$$P(X, X') \equiv \frac{\eta_{AB} X^A X'^B}{L^2}, \quad \eta_{AB} \equiv (-1, 1, \dots, 1). \quad (5.25)$$

Notice that, if $X' = X$ then $P = 1$, while if X and X' are antipodal, i.e. $X' = -X$, then $P = -1$. Substituting the parameterization (B.843), we get

$$P(X, X') = 1 + \frac{(\eta - \eta')^2 - (\mathbf{x} - \mathbf{x}')^2}{2\eta\eta'}. \quad (5.26)$$

An important property of the function $P(X, X')$ is that it is $SO(1, D)$ invariant on de Sitter space, since it is constructed out of the Lorentz product in $\mathbb{R}^{1, D}$. We will exploit this in Section 5.3.

Exercise 5.4 Show that the geodesic distance between two points in de Sitter space is

$$d(x, x') = \begin{cases} L \cosh^{-1}(P) & \text{timelike,} \\ L \cos^{-1}(P) & \text{spacelike,} \\ 0 & \text{lightlike.} \end{cases} \quad (5.27)$$

Hint: Use the Lagrangian

$$\mathcal{L} = -\frac{1}{2}\eta_{AB}\dot{X}_A\dot{X}^B + \lambda(\eta_{AB}X^AX^B - L^2), \quad (5.28)$$

where λ is a Lagrange multiplier enforcing the embedding constraint and the derivative is with respect to proper time. ➔

5.2 Canonical Quantization

Next, we consider the quantization of a massive scalar field in a fixed de Sitter background. We consider the action

$$S = \int d^D x \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{2} m^2 \phi^2 - \frac{1}{2} \xi R \phi^2 \right], \quad (5.29)$$

where we have allowed for a curvature coupling. We will often specialize to a massless field with a conformal coupling. In Exercise 2.1, we showed that this implies

$$m^2 = 0, \quad \xi = \frac{D-2}{4(D-1)} \equiv \bar{\xi}(D). \quad (5.30)$$

We now go through the standard quantization procedure.

Equation of motion To derive the equation of motion for the field, we evaluate the action (5.29) in the spacetime (5.15). We will use that

$$a^2 R = 2(D-1) \left[\frac{a''}{a} + \frac{D-4}{2} \left(\frac{a'}{a} \right)^2 \right] = \frac{D(D-1)}{\eta^2}, \quad (5.31)$$

where the final result can also be obtained directly from (5.1), with $a(\eta) = -L/\eta$. After some straightforward algebra, the equation of motion for the canonically-normalized field $f \equiv a^{(D-2)/2}(\eta)\phi$ is

$$f'' - \nabla^2 f + \frac{m^2 L^2 + D(D-1)(\xi - \bar{\xi})}{\eta^2} f = 0. \quad (5.32)$$

We see that for a massless field with a conformal coupling this reduces to $f'' - \nabla^2 f = 0$, which is simply the equation of motion of a massless field in Minkowski space.

Exercise 5.5 Derive the equation of motion (5.32).



Mode functions In Fourier space, the mode function satisfies

$$f_k'' + \left[k^2 + \frac{m^2 L^2 + D(D-1)(\xi - \bar{\xi})}{\eta^2} \right] f_k = 0. \quad (5.33)$$

To solve this, we define $f_k(\eta) \equiv \sqrt{s} h(s)$, with $s \equiv -k\eta$. Straightforward algebra then shows that $h(s)$ satisfies the *Bessel equation* (see Appendix A):

$$s^2 h'' + s h' + (s^2 - \nu^2) h = 0, \quad (5.34)$$

where

$$\nu^2 \equiv \frac{1}{4} - m^2 L^2 - D(D-1)(\xi - \bar{\xi}) \xrightarrow{D=4} \frac{9}{4} - m^2 L^2 - 12\xi. \quad (5.35)$$

Before presenting the general solution for the mode function, let us discuss its late-time behavior. Taking the limit $s \rightarrow 0$ of the Bessel equation, we find that the mode function asymptotically behaves as

$$f_k(\eta) \propto (-k\eta)^{1/2 \pm \nu}. \quad (5.36)$$

Notice that for small m and ξ , the parameter ν is real and the function either blows up or vanishes asymptotically, while for large masses and/or ξ it becomes oscillatory.

The general solution to (5.33) can then be written as

$$f_k(\eta) = \sqrt{k|\eta|} \left[A_k J_\nu(k|\eta|) + B_k Y_\nu(k|\eta|) \right], \quad (5.37)$$

where J_ν and Y_ν are *Bessel functions* of the first and second kind. The solution for a single mode of the original field ϕ is

$$\phi_{\mathbf{k}}(\eta, \mathbf{x}) = \frac{e^{i\mathbf{k} \cdot \mathbf{x}}}{\sqrt{(2\pi)^{D-1}}} a^{(D-2)/2}(\eta) f_k(\eta), \quad (5.38)$$

where we have used spatial translation invariance to extract the factor of $e^{i\mathbf{k} \cdot \mathbf{x}}$. As you will show in the following exercise, imposing the normalization $\langle \phi_{\mathbf{k}}, \phi_{\mathbf{k}'} \rangle = \delta(\mathbf{k} - \mathbf{k}')$ then leads to

$$A_k B_k^* - A_k^* B_k = -\frac{i\pi}{2k}. \quad (5.39)$$

This will fix the normalization of our mode functions.

Exercise 5.6 Derive the result (5.39).

Hint: You may use that the Wronskian of the Bessel functions is

$$W[J_\nu(s), Y_\nu(s)] = \frac{2}{\pi s}, \quad (5.40)$$

where $W[f, g] \equiv fg' - f'g$. ➔

Vacuum choice To choose the vacuum state, we consider the behavior of the modes in the infinite past, $k|\eta| \ll 1$. In this limit, the solution (5.37) becomes (see Appendix A)

$$f_k \rightarrow \sqrt{\frac{2}{\pi}} \left[\frac{A_k - iB_k}{2} e^{-i(k\eta + \Phi)} + \frac{A_k + iB_k}{2} e^{i(k\eta + \Phi)} \right], \quad \Phi \equiv \frac{\nu\pi}{2} + \frac{\pi}{4}. \quad (5.41)$$

To select the positive frequency mode, $e^{-ik\eta}$, we set $A_k + iB_k \equiv 0$. The normalization condition (5.39) becomes

$$|A_k|^2 = \frac{\pi}{4k}. \quad (5.42)$$

Up to an irrelevant phase, the solution (5.37) then reads

$$f_k(\eta) = \frac{1}{2} \sqrt{\pi|\eta|} \left[J_\nu(k|\eta|) + iY_\nu(k|\eta|) \right] \equiv \frac{1}{2} \sqrt{\pi|\eta|} H_\nu^{(1)}(k|\eta|), \quad (5.43)$$

where $H_\nu^{(1)}$ is the *Hankel function* of the first kind (see Appendix A). The vacuum associated to this choice of mode function is the *Bunch–Davies vacuum*.

Let us check the case of a massless, conformally coupled scalar. Setting $m = 0$ and $\xi = \bar{\xi}(D)$, we get $\nu = 1/2$. The Hankel function reduces to

$$H_{1/2}^{(1)}(k|\eta|) = -i \left(\frac{2}{\pi k|\eta|} \right)^{1/2} e^{-ik\eta}, \quad (5.44)$$

and the mode function (5.43) becomes

$$f_k(\eta) = -\frac{i}{\sqrt{2k}} e^{-ik\eta}. \quad (5.45)$$

As expected, this is the flat-space mode function.

5.3 Wightman Function

In Section 3.5, we showed how the temperature measured by an accelerating observer can be derived using a simple model of a particle detector. We want to use the same method to derive the de Sitter temperature. Recall that the derivation required computing the Wightman function

$$G(x, x') \equiv \langle 0 | \hat{\phi}(x) \hat{\phi}(x') | 0 \rangle = \int d^{D-1}k a^{-(D-2)} f_k(x) f_k^*(x'). \quad (5.46)$$

To prepare for this computation, we therefore now derive the Wightman function in de Sitter space. For a massive field in a D -dimensional spacetime with arbitrary curvature coupling parameter ξ , this will be a bit of a special function slugfest.

In principle, we could just substitute (5.43) into (5.46) and perform the integration. In fact, this has been done in [34], but the computation is rather involved. Here, we will pursue a more elegant approach that exploits the symmetry of the problem.

If the Wightman function is computed in a state that is invariant under the de Sitter isometry group, it should only depend on the $SO(1, D)$ -invariant distance between two points. This is the geodesic distance and we have

$$G(x, x') = G(P(x, x')), \quad (5.47)$$

where $P(x, x')$ was defined in (5.25). We can then compute the Wightman function by solving the homogeneous Klein–Gordon equation

$$(\square_x + m^2 + \xi R) G(P(x, x')) = 0. \quad (5.48)$$

This is easier than integrating products of Hankel functions in (5.46).

Exercise 5.7 Show that

$$\nabla^\mu P \nabla_\mu P = \frac{1 - P^2}{L^2}, \quad \nabla_\mu \nabla_\nu P = -g_{\mu\nu} \frac{P}{L^2}. \quad (5.49)$$

Hint: Since these are tensorial expansions, they hold in any coordinate system. The computation is particularly simple in conformal coordinates $(\eta, \mathbf{x})$, where the only nonzero Christoffel symbols are $\Gamma_{\eta\eta}^\eta = a'/a$, $\Gamma_{j\eta}^i = (a'/a)\delta_j^i$ and $\Gamma_{ij}^\eta = (a'/a)\delta_{ij}$, with $a'/a = -1/\eta$. $\rightarrow$

Using the identities in (5.49), we get

$$\begin{aligned} \square_x G(P) &= -g^{\mu\nu} \nabla_\mu \nabla_\nu G \\ &= -g^{\mu\nu} \nabla_\mu (G'(P) \nabla_\nu P) \\ &= -G''(P) \nabla^\mu P \nabla_\mu P - G'(P) g^{\mu\nu} \nabla_\mu \nabla_\nu P \\ &= \frac{P^2 - 1}{L^2} G''(P) + \frac{DP}{L^2} G'(P), \end{aligned} \quad (5.50)$$

and the Wightman function satisfies

$$(P^2 - 1) \partial_P^2 G + DP \partial_P G + (m^2 L^2 + D(D - 1)\xi) G = 0. \quad (5.51)$$

Defining the new variable $z \equiv \frac{1}{2}(1 + P)$, this becomes a *hypergeometric equation* (see Appendix A):

$$z(1 - z) \partial_z^2 G + \left(\frac{D}{2} - Dz \right) \partial_z G - (m^2 L^2 + D(D - 1)\xi) G = 0. \quad (5.52)$$

Comparison with (A.29) shows that the solution of this equation is

$$G(P) = c_{m,D} {}_2F_1 \left[\begin{matrix} h_+, h_- \\ D/2 \end{matrix} \middle| \frac{1 + P}{2} \right], \quad (5.53)$$

where ${}_2F_1$ is the Gauss hypergeometric function and we have defined the parameters

$$h_\pm \equiv \frac{D - 1}{2} \pm \sqrt{\frac{(D - 1)^2}{4} - (m^2 L^2 + D(D - 1)\xi)}. \quad (5.54)$$

The hypergeometric function in (5.53) has a pole at $z = 1$ (or $P = 1$) and a branch cut for $1 < P < \infty$. The pole occurs when the points x and x' are separated by a null geodesic. At short distances, the scalar field is insensitive to the fact that it lives in a de Sitter space and the form of the singularity should be the same as that of the propagator in flat Minkowski space. We can use this fact to fix the normalisation constant $c_{m,D}$.

Exercise 5.8 Show that

$$c_{m,D} = \frac{L^{2-D}}{(4\pi)^{D/2}} \frac{\Gamma(h_+)\Gamma(h_-)}{\Gamma(D/2)}. \quad (5.55)$$

Hint: You may use that the short-distance limit of the Wightman function in flat space is

$$G(x, x') \approx (-1)^{(D-2)/2} \frac{\Gamma((D-2)/2)}{4\pi^{D/2}} \frac{1}{[(x-x')^2]^{(D-2)/2}}, \quad (5.56)$$

where Γ is the Euler gamma function (see Appendix A). →

As a sanity check, we consider the case of a massless, conformally coupled scalar. Setting $m = 0$ and $\xi = \bar{\xi}(D)$, we get $h_+ = D/2$ and $h_- = D/2 - 1$. The hypergeometric function then reduces to an elementary function

$${}_2F_1\left[\begin{matrix} D/2, D/2-1 \\ D/2 \end{matrix} \middle| z\right] = (1-z)^{1-D/2}, \quad (5.57)$$

and the Wightman function becomes

$$\begin{aligned} G(x, x') &= (-1)^{(D-2)/2} \frac{\Gamma((D-2)/2)}{4\pi^{D/2} L^{D-2}} \left(\frac{\eta\eta'}{(\Delta\eta)^2 - (\Delta\mathbf{x})^2} \right)^{(D-2)/2} \\ &= a(\eta)^{-(D-2)/2} G_M(x, x') a(\eta')^{-(D-2)/2}, \end{aligned} \quad (5.58)$$

which is the correct conformal rescaling of the Wightman function in Minkowski.

5.4 De Sitter Temperature

In Section 3.5, we showed that the response function of a particle detector is

$$f(E) = \int_{-\infty}^{\infty} d\tau e^{-iE\tau} G(\tau), \quad (5.59)$$

where $G(x(\tau), x(0))$ is the Wightman function evaluated along the worldline of the observer. We would now like to apply this to a comoving observer in de Sitter space. For simplicity, we will consider the special case of a massless, conformally coupled scalar field in $D = 4$ dimensions. The Wightman function is then

$$G(x, x') = -\frac{1}{4\pi^2 L^2} \frac{\eta\eta'}{(\Delta\eta - i\epsilon)^2 - (\Delta\mathbf{x})^2}. \quad (5.60)$$

To use this in (5.59), we need to write the result in terms of the proper time τ . In Exercise 5.4, we showed that the geodesic distance between two timelike-separated points is

$$d(x, x') = \Delta\tau = L \cosh^{-1}(P), \quad \text{with} \quad P = 1 + \frac{(\Delta\eta)^2 - (\Delta\mathbf{x})^2}{2\eta\eta'}. \quad (5.61)$$

We therefore have

$$\frac{2\eta\eta'}{(\Delta\eta)^2 - (\Delta\mathbf{x})^2} = \left[\cosh\left(\frac{\Delta\tau}{L}\right) - 1 \right]^{-1} = \left[2 \sinh^2\left(\frac{\Delta\tau}{2L}\right) \right]^{-1}, \quad (5.62)$$

and the detector's response function becomes

$$f(E) = -\frac{1}{16\pi^2 L^2} \int_{-\infty}^{\infty} d\tau \frac{e^{-iE\tau}}{\sinh^2[\frac{1}{2L}(\tau - i\epsilon)]}. \quad (5.63)$$

Notice that this is exactly of the same form as the result (3.55) for the Unruh effect if we make the identification $a \leftrightarrow 1/L$. Without doing any more work, we can therefore infer that the *de Sitter temperature* is

$$T_{\text{dS}} = \frac{1}{2\pi L}. \quad (5.64)$$

This is the temperature an inertial observer will measure in de Sitter space. Using the measure value of the de Sitter length scale for our Universe, $L \sim 10^{26}$ m, we conclude that the de Sitter temperature of our Universe is $T_{\text{dS}} \sim 10^{-30}$ K. In the next chapter, we will see that the associated entropy is proportional to the area of the cosmic horizon, $S_{\text{dS}} = \pi L^2/G$.

Although the physics of the de Sitter horizon and the associated radiation is similar to the physics of black holes, there are also some important differences [35]. First of all, the cosmic horizon is personal and moves with the observer who can therefore never approach it. Unlike the black hole horizon, the cosmic horizon is therefore not an object localized in space. Moreover, while black holes in flat space decay through Hawking emission, the radiation of the de Sitter horizon is reabsorbed by the horizon itself, so that the spacetime doesn't shrink (at least not through this perturbative mechanism).

6 Euclidean Quantum Gravity

In this chapter, we will describe an alternative approach to derive the thermodynamics of black holes and de Sitter space. The central object will be the gravitational path integral⁵

$$Z(\beta) = \int \mathcal{D}g e^{-I_E[g]}, \quad (6.1)$$

where $I_E[g]$ is the “Euclidean action” of the metric $g_{\mu\nu}$ defined in terms of the “imaginary time” $\tau = it$. If the path integral is over solutions that are periodic in imaginary time, with periodicity $\tau_\beta = \beta\hbar$, then (6.1) is the partition function of a field theory at temperature $T = (k_B\beta)^{-1}$. This magical fact provides a deep connection between thermodynamics and geometry. In particular, we can use the standard thermodynamic relation, $S = (1 - \beta\partial_\beta) \log Z$, to derive the entropy of the system by computing the gravitational path integral. We will illustrate this for black holes and de Sitter space, and derive the famous relation between entropy and horizon area:

$$S = \frac{k_B c^3}{4G\hbar} A. \quad (6.2)$$

Our treatment of Euclidean quantum gravity will be rather minimal, but more details can be found in [36].

6.1 Statistical Field Theory

We begin by motivating the relation between quantum field theory in imaginary time and statistical field theory at finite temperature.

Quantum mechanics To avoid unnecessary distractions, we start with a simple example in quantum mechanics. Consider a non-relativistic particle in one dimension with a potential $V(q)$, where q is the position of the particle. The corresponding Hamiltonian is

$$H = \frac{p^2}{2m} + V(q). \quad (6.3)$$

In the Heisenberg picture, states $|q\rangle$ evolve according to the unitary evolution operator:

$$\hat{U}(t) = e^{-i\hat{H}t/\hbar}. \quad (6.4)$$

The transition amplitude between an initial state $|q_1\rangle$ at time t_1 and a final state $|q_2\rangle$ at time t_2 is then defined as

$$K(q_2, t_2; q_1, t_1) \equiv \langle q_2 | \hat{U}(t_2 - t_1) | q_1 \rangle = \int_{q(t_1)=q_1}^{q(t_2)=q_2} \mathcal{D}q e^{iI[q]/\hbar}, \quad (6.5)$$

where I is the classical action

$$I[q] = \int_{t_1}^{t_2} dt \left(\frac{m}{2} (\partial_t q)^2 - V(q) \right). \quad (6.6)$$

⁵In this section, we will use I for the action to avoid confusion with S for entropy.

The final equality in (6.5) expresses the transition amplitude as an integral over all possible paths $q(t)$, with fixed endpoints $q(t_1) = q_1$ and $q(t_2) = q_2$. Each path is weighted by a factor of $e^{iI[q]/\hbar}$ and the classical path q_{cl} is the one for which the phase becomes stationary. A derivation of this path integral can be found in any quantum mechanics textbook. This textbook will also describe the meaning of the integral measure $\mathcal{D}q$. We will never actually have to compute a path integral explicitly, so we can afford to stay a bit vague in our treatment.

Next, we put the particle into equilibrium with a thermal bath. The particle is then in a thermal state described by the density matrix, $\hat{\rho} = e^{-\beta\hat{H}}/Z$, where $\beta = (k_B T)^{-1}$ is the inverse temperature and Z is the partition function

$$Z = \text{Tr}(e^{-\beta\hat{H}}) = \int dq \langle q | e^{-\beta\hat{H}} | q \rangle. \quad (6.7)$$

The factor of $e^{-\beta\hat{H}}$ in these expressions looks like the evolution operator $e^{-i\hat{H}t/\hbar}$ in “imaginary time” $\tau = it$, if we identify $\tau = \hbar\beta$. Using

$$\hat{U}_{\text{E}}(\tau) \equiv \hat{U}(-i\tau) = e^{-\beta\hat{H}}, \quad \text{with} \quad \beta = \frac{\tau}{\hbar}, \quad (6.8)$$

we can therefore write the transition amplitude (6.5) as

$$K(q_2, \tau_2; q_1, \tau_1) = \langle q_2 | \hat{U}_{\text{E}}(\tau_2 - \tau_1) | q_1 \rangle = \int_{q(\tau_1)=q_1}^{q(\tau_2)=q_2} \mathcal{D}q e^{-I_{\text{E}}[q]/\hbar}, \quad (6.9)$$

where I_{E} is the “Euclidean action”⁶

$$I_{\text{E}}[q] = \int_{\tau_1}^{\tau_2} d\tau \left(\frac{m}{2} (\partial_{\tau} q)^2 + V(q) \right). \quad (6.10)$$

We now associate a Boltzmann factor $e^{-I_{\text{E}}[q]/\hbar}$ to each trajectory $q(\tau)$ and the path integral computes the statistical average over all trajectories.

If we set $|q_1\rangle = |q_2\rangle$ and sum over all position eigenstates, then the transition amplitude becomes the partition function:

$$Z = \int_{q(0)}^{q(\beta\hbar)=q(0)} \mathcal{D}q e^{-I_{\text{E}}[q]/\hbar}, \quad (6.11)$$

where the path integral is over all trajectories with periodic boundary conditions, $q(\beta\hbar) = q(0)$. The periodic boundary conditions of the path integral correspond to the trace of the thermal sum, and the period corresponds to the inverse temperature.

⁶This is called Euclidean because the Minkowski metric in imaginary time becomes, $ds^2 = d\tau^2 + d\mathbf{x}^2$.

Quantum field theory The observations we just made are easily generalized to quantum field theory. Most introductory treatments of QFT only deal with systems at zero temperature where all correlation functions are evaluate in the vacuum state $|0\rangle$. For example, the Feynman propagator is defined as

$$G_F(x-y) = \langle 0|T(\hat{\phi}(x)\hat{\phi}(y))|0\rangle. \quad (6.12)$$

At finite temperature, we would instead have

$$G_F^{T>0}(x-y) = \frac{1}{Z} \sum_n \langle n|T(\hat{\phi}(x)\hat{\phi}(y))|n\rangle e^{-\beta E_n}. \quad (6.13)$$

where $|n\rangle$ are the eigenstates of the Hamiltonian, with energy E_n , and $Z \equiv \sum_n e^{-\beta E_n}$ is the partition function. As before, the right-hand side of (6.13) can be written as a path integral using imaginary time and periodic boundary conditions. Here, we will show that the original definition of the Feynman propagator indeed has this periodicity:

$$\begin{aligned} G_F^{T>0}(x^0, \mathbf{x}; 0, \mathbf{y}) &= \frac{1}{Z} \text{Tr} \left[e^{-\beta \hat{H}} T \left(\hat{\phi}(x^0, \mathbf{x}) \hat{\phi}(0, \mathbf{y}) \right) \right] \\ &= \frac{1}{Z} \text{Tr} \left[e^{-\beta \hat{H}} \hat{\phi}(x^0, \mathbf{x}) \hat{\phi}(0, \mathbf{y}) \right] \quad (\text{assuming } x^0 > 0) \\ &= \frac{1}{Z} \text{Tr} \left[\hat{\phi}(0, \mathbf{y}) e^{-\beta \hat{H}} \hat{\phi}(x^0, \mathbf{x}) \right] \\ &= \frac{1}{Z} \text{Tr} \left[e^{-\beta \hat{H}} e^{\beta \hat{H}} \hat{\phi}(0, \mathbf{y}) e^{-\beta \hat{H}} \hat{\phi}(x^0, \mathbf{x}) \right] \\ &= \frac{1}{Z} \text{Tr} \left[e^{-\beta \hat{H}} T \left(\hat{\phi}(x^0, \mathbf{x}) \hat{\phi}(-i\hbar\beta, \mathbf{y}) \right) \right] \\ &= G_F^{T>0}(x^0, \mathbf{x}; -i\hbar\beta, \mathbf{y}). \end{aligned} \quad (6.14)$$

Next, we will use this feature of thermal field theory as an inspiration to define thermal properties in quantum gravity.

Quantum gravity In ordinary QFT, we fix the spacetime manifold M , and then perform a path integral over fields defined on M . In quantum gravity, however, we also have to integrate over the spacetime geometry:

$$Z = \int \mathcal{D}g \mathcal{D}\phi e^{-I_E[g, \phi]}. \quad (6.15)$$

If we impose periodic boundary conditions in imaginary time, with periodicity $\tau_\beta = \beta\hbar$, then (6.15) is the partition function of a thermal field theory. At leading order, the path integral is given by its saddle-point approximation:

$$Z \approx e^{-I_E[g_{\text{cl}}, \phi_{\text{cl}}]}, \quad (6.16)$$

where $(g_{\mu\nu})_{\text{cl}}$ and ϕ_{cl} are solutions to the classical equations of motion.

In general, the partition function of a thermal system at inverse temperature β is $Z = e^{S-\beta E}$, where S is the entropy and E is the energy. Alternatively, this can be written as $Z = e^{-\beta F}$,

where $F = E - TS$ is the free energy. Using the first law of thermodynamics, $dE = TdS$, one finds

$$E = -\frac{d}{d\beta} \log Z, \quad (6.17)$$

$$S = \left(1 - \beta \frac{d}{d\beta}\right) \log Z. \quad (6.18)$$

Combining this with (6.16) provides the geometry of the spacetime and its thermodynamic interpretation.

6.2 Gravitational Path Integral

To derive (6.16), we have to compute the on-shell Euclidean action. Our starting point is the Einstein–Hilbert action on a manifold M of Euclidean signature (i.e. after Wick rotating to imaginary time):⁷

$$I_{\text{EH}} = -\frac{1}{16\pi G} \int_M d^4x \sqrt{g} R. \quad (6.20)$$

It is often taught that variation of this action leads to the Einstein equation for the metric. However, this requires dropping a boundary term which will be important for us.

The integrand of (6.20) is schematically of the form $R \sim g^{-1} \partial^2 g$. It is the second derivative that leads to a subtlety. To illustrate the problem, let us consider the action for a free point particle, $I_1 = \frac{1}{2} \int dt \dot{q}^2$. Variation of this action leads to the equation of motion, $\ddot{q} = 0$. Naively, the same equation of motion can also be derived from the alternative action

$$I_2 = -\frac{1}{2} \int dt q \ddot{q}. \quad (6.21)$$

However, the second time-derivative in the action leads to a subtlety that is analogous of what we find for (6.20). Let us see what happens when we vary this action on a finite time interval $[t_1, t_2]$, such that $\delta q(t) = 0$, for $t = t_1, t_2$. We get

$$\delta I_2 = -\int_{t_1}^{t_2} dt \ddot{q} \delta q - \frac{1}{2} [q \delta \dot{q}]_{t_1}^{t_2}. \quad (6.22)$$

This only gives the correct equation of motion, if we ignore the boundary term (or enforce $q(t_1) = q(t_2) = 0$). The problem is circumvented if we define

$$I'_2 = I_2 + \frac{1}{2} [q \delta \dot{q}]_{t_1}^{t_2}, \quad (6.23)$$

i.e. we added a boundary term that precisely cancels the unwanted contribution in the variation (6.22). Indeed, after integration by parts, we have $I'_2 = I_1 = \frac{1}{2} \int dt \dot{q}^2$. In the following exercise, you will show that the variation of the Einstein–Hilbert action (6.20) leads to a similar boundary term as the variation of I_2 .

⁷The overall minus sign comes from carrying out the Wick rotation, $t = -i\tau$, of the Lorentzian action

$$I = \frac{1}{16\pi G} \int_M d^4x \sqrt{-g} R, \quad (6.19)$$

and imposing $I_E = -iI$.

Exercise 6.1 Show that the variation of the Einstein–Hilbert action gives

$$(16\pi G) \delta I_{\text{EH}} = - \int_M G_{\mu\nu} \delta g^{\mu\nu} \sqrt{g} d^4x + \int_{\partial M} h^{\alpha\beta} \delta g_{\alpha\beta,\mu} n^\mu \sqrt{h} d^3x, \quad (6.24)$$

where $h_{\mu\nu}$ is the induced metric on the boundary ∂M , n^μ is an outward directed unit normal vector and $\sqrt{h} d^3x$ is the volume form on ∂M .

Hint: You may use that

$$\delta \sqrt{g} = -\frac{1}{2} \sqrt{g} g_{\mu\nu} \delta g^{\mu\nu}, \quad (6.25)$$

$$\delta R_{\mu\nu} = \nabla_\alpha (\delta \Gamma_{\mu\nu}^\alpha) - \nabla_\nu (\delta \Gamma_{\mu\alpha}^\alpha). \quad (6.26)$$

You will also need some basic mathematical facts about the geometry of hypersurfaces. →

To address the issue of the boundary term in (6.24), we add a compensating “Gibbons–Hawking–York boundary term” to the Einstein–Hilbert action

$$I'_{\text{EH}} = I_{\text{EH}} + I_{\text{GHY}}, \quad \text{where} \quad I_{\text{GHY}} = -\frac{1}{8\pi G} \int_{\partial M} d^3x \sqrt{h} K. \quad (6.27)$$

This modified action is the analog of I'_2 in (6.23). The boundary term is an integral over the trace of the “extrinsic curvature” on ∂M , which is given by $K \equiv \nabla_\mu n^\mu$.

Exercise 6.2 Show that the variation of the Gibbons–Hawking–York boundary term gives

$$(16\pi G) \delta I_{\text{GHY}} = - \int_{\partial M} h^{\alpha\beta} \delta g_{\alpha\beta,\mu} n^\mu \sqrt{h} d^3x, \quad (6.28)$$

which precisely cancels the boundary term in (6.24). →

To compute the GHY action, it is convenient to introduce local coordinates x^i (with $i = 1, 2, 3$) on ∂M . Near ∂M , the coordinates on M are (y, x^i) , where $y = 0$ on ∂M and $y > 0$ in its neighborhood. The induced metric on ∂M is h_{ij} . The outward-directed unit vector normal to ∂M is n^μ . The extrinsic curvature is $K_{ij} \equiv \nabla_i n_j$ and its trace is $K \equiv h^{ij} K_{ij}$. In the local coordinates, these can be written as

$$K_{ij} = -\frac{1}{2} \partial_y h_{ij} \Big|_{y=0}, \quad (6.29)$$

$$K = -\partial_y \log \sqrt{h} \Big|_{y=0}, \quad (6.30)$$

where $h \equiv \det(h_{ij})$. Defining the volume of a hypersurface of fixed y by $V(y) \equiv \int_{\partial M} d^3x \sqrt{h}$, the Gibbons–Hawking–York term can be expressed as

$$I_{\text{GHY}} = -\frac{1}{8\pi G} \frac{d}{dy} V(y) \Big|_{y=0}. \quad (6.31)$$

We will use this formula to compute the Euclidean action (6.27).

6.3 Black Hole Entropy

Let's see that this magic really works. As a first example, we will re-derive the thermodynamics of black holes using the Euclidean formalism.

Euclidean black hole Consider the metric of the Schwarzschild black hole in Euclidean signature:

$$ds^2 = \left(1 - \frac{2GM}{r}\right) d\tau^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\Omega^2, \quad (6.32)$$

where $\tau = it$ is imaginary time and $d\Omega^2$ is the metric of a unit two-sphere. As emphasized by Gibbons and Hawking [37], this metric is singularity-free if we interpret τ as an angular variable, with periodicity $8\pi GM$, and the horizon $r = 2GM$ as the origin of polar coordinates. To see this, we consider the following change of variables

$$\rho = 4GM \left(1 - \frac{2GM}{r}\right)^{1/2}. \quad (6.33)$$

so that the metric (6.32) becomes

$$ds^2 = \rho^2 \frac{d\tau^2}{(4GM)^2} + \left(\frac{r}{2GM}\right)^4 d\rho^2 + r^2 d\Omega^2. \quad (6.34)$$

Near $r = 2GM$, this reduces to

$$ds^2 = \underbrace{d\rho^2 + \rho^2 \frac{d\tau^2}{(4GM)^2}}_{\mathbb{R}^2} + \underbrace{(2GM)^2 d\Omega^2}_{S^2}. \quad (6.35)$$

This looks like the product of two-dimensional flat space $\mathbb{R}^2$ in polar coordinates, with metric $d\rho^2 + \rho^2 d\phi^2$, and a two-sphere S^2 of radius $2GM$. The usual periodicity of the angular coordinate, $\phi \sim \phi + 2\pi$, then implies that τ should be periodic:

$$\tau \sim \tau + 8\pi GM. \quad (6.36)$$

Any other periodicity would imply a conical singularity at the origin and the metric would not satisfy the vacuum Einstein equations. Having a smooth metric satisfying the Einstein equation therefore requires that we impose (6.36). Our previous discussion implies that this period in imaginary time implies $\hbar\beta = 8\pi GM$, so that the black hole's temperature is

$$T_{\text{BH}} = \frac{\hbar c^3}{k_B} \frac{1}{8\pi GM}, \quad (6.37)$$

which agrees with the result (4.42) derived previously.

On-shell action To derive the entropy of the black hole, we have to compute its on-shell action. Since (6.32) is a solution to the vacuum Einstein equations, $R_{\mu\nu} = 0$, the bulk action (6.20) vanishes and on-shell action will only be non-zero because of the GHY boundary term in (6.27).

To compute this boundary term for the Euclidean Schwarzschild solution, we introduce a cutoff at large radius, $r = \bar{r}$. We denote the result of computing (6.27) on this surface by $I_{\text{GHY}}(\bar{r}, \text{BH})$. Unfortunately, the result diverges in the limit $\bar{r} \rightarrow \infty$. To get a finite result, Gibbons and Hawking subtracted to corresponding boundary term for the flat manifold $\mathbb{R}^3 \times S^1$, i.e. in the absence of the black hole. The periodicity of the S^1 has to be chosen to be the same as for the black hole solution, so that the boundary geometries are the same. We denote this “counterterm” by $I_{\text{GHY}}(\bar{r}, \text{flat})$ and the renormalized boundary term by

$$I_{\text{BH}} \equiv I_{\text{GHY}}(\bar{r}, \text{BH}) - I_{\text{GHY}}(\bar{r}, \text{flat}). \quad (6.38)$$

Computing this is now relatively straightforward.

By construction, the circumference of the circle at $r = \bar{r}$ for both the black hole solution and the flat $\mathbb{R}^3 \times S^1$ manifold is $\beta\sqrt{1 - 2GM/\bar{r}}$. The volume near the boundary then is

$$V(r) = 4\pi r^2 \times \beta \begin{cases} \sqrt{1 - \frac{2GM}{\bar{r}}} & \text{flat} \\ \sqrt{1 - \frac{2GM}{r}} & \text{BH} \end{cases} \quad (6.39)$$

Note that the square root factor contains $\bar{r}$ for the flat metric and r for the black hole spacetime. To evaluate the extrinsic curvature on the boundary, we use the normal derivative

$$\frac{d}{dy} = -\frac{d}{dr} \begin{cases} 1 & \text{flat} \\ \sqrt{1 - \frac{2GM}{\bar{r}}} & \text{BH} \end{cases} \quad (6.40)$$

Using (6.31), we then have

$$\begin{aligned} I_{\text{BH}} &= -\lim_{\bar{r} \rightarrow \infty} \frac{\beta}{8\pi G} \sqrt{1 - \frac{2GM}{\bar{r}}} \frac{d}{dr} \left(4\pi r^2 \sqrt{1 - \frac{2GM}{r}} - 4\pi r^2 \right) \Big|_{r=\bar{r}} = \frac{\beta M}{2} \\ &= \frac{\beta^2}{16\pi G}. \end{aligned} \quad (6.41)$$

Knowing the action I_{BH} as a function of β will then allow us to derive the black hole entropy.

Entropy Using (6.18), the black hole entropy is

$$S = -\left(1 - \beta \frac{d}{d\beta}\right) I_{\text{BH}} = \frac{\beta^2}{16\pi G} = \frac{(8\pi GM)^2}{16\pi G} = 4\pi GM^2. \quad (6.42)$$

Since $A = 4\pi(2GM)^2 = 16\pi G^2 M^2$, this corresponds to the correct Bekenstein–Hawking entropy:

$$S = \frac{A}{4G}. \quad (6.43)$$

Although this was derived for a Schwarzschild black hole, the formula holds more generally. For example, it also applies to rotating Kerr black holes and charged Reissner–Nordstrom black holes. The derivation did, however, assume that gravity is described by the Einstein–Hilbert action. A generalization of the black hole entropy to gravity theories with higher-derivative corrections was proposed by Wald [38].

Exercise 6.3 The metric of a Schwarzschild black hole in four-dimensional anti-de Sitter (AdS) spacetime with negative cosmological constant, $\Lambda < 0$, is

$$ds^2 = -f(r)dt^2 + \frac{1}{f(r)}dr^2 + r^2d\Omega^2, \quad f(r) \equiv 1 - \frac{2GM}{r} + \frac{r^2}{L^2}, \quad (6.44)$$

where $L \equiv \sqrt{-3/\Lambda}$.

1. Show that the Euclidean AdS-Schwarzschild metric is smooth if the imaginary time $\tau = -it$ is a periodic variable with period

$$\beta = \frac{4\pi r_+ L^2}{L^2 + 3r_+^2}, \quad (6.45)$$

where r_+ is the black hole's horizon.

Hint: Expand the Euclidean metric near r_+ , by writing $r = r_+ + \rho^2$, for $\rho^2 \ll r_+$.

2. To obtain a finite action for the black hole solution, we introduce a cutoff at $r = \bar{r}$ for some large $\bar{r}$. The regularized action is then the difference between the action of the Euclidean black hole and the action of empty AdS (with the same boundary metric). Assuming that the difference of the GHY boundary terms vanishes for $\bar{r} \rightarrow \infty$, show that the regularized Euclidean action is

$$I_E = \frac{\pi r_+^2 (L^2 - r_+^2)}{G(L^2 + 3r_+^2)}, \quad (6.46)$$

and hence that black hole entropy is $S = A/4G$.

3. Show that there is a minimum temperature $T_0 = \sqrt{3}/(2\pi L)$ and that there are two black hole solutions for $T > T_0$. Thermodynamic stability requires the specific heat $\partial M/\partial T$ to be positive. Use this to determine which of the two black hole solutions is stable.

More on the thermodynamics of black holes in AdS can be found in the paper by Hawking and Page [39]. ➔

6.4 De Sitter Entropy

An important feature of cosmological spacetimes is that they can also have *horizons*. As we will see, just like for black holes, we can associate an entropy, $S = A/4G$, to these horizons. An important special case is de Sitter space, which we will describe in this section.

Temperature To infer the temperature associated to the de Sitter horizon, we continue the static patch metric (5.17) to imaginary time $\tau = it$:

$$ds^2 = \left(1 - \frac{r^2}{L^2}\right) d\tau^2 + \left(1 - \frac{r^2}{L^2}\right)^{-1} dr^2 + r^2 d\Omega^2. \quad (6.47)$$

Near the horizon, we write $r = L - \rho$ and expand in the ratio $\rho/L \ll 1$. This gives

$$ds^2 = \frac{2\rho}{L} d\tau^2 + \left(\frac{2\rho}{L}\right)^{-1} d\rho^2 + L^2 d\Omega^2. \quad (6.48)$$

Defining $R \equiv \sqrt{2\rho L}$ and $\Theta \equiv \tau/L$, we get

$$ds^2 = \underbrace{dR^2 + R^2 d\Theta^2}_{\mathbb{R}^2} + L^2 d\Omega^2. \quad (6.49)$$

As before, we avoid a conical singularity at the origin, by imposing the periodicity $\tau \sim \tau + 2\pi L$. Identifying the period of imaginary time with the inverse temperature, we find

$$T_{\text{dS}} = \frac{1}{2\pi L}, \quad (6.50)$$

which is the temperature measured by an inertial observer in de Sitter. We are currently at the start of a new de Sitter epoch and the present-day de Sitter temperature is $T \sim 10^{-29}$ K.

Entropy Finally, we compute the de Sitter entropy using the Euclidean path integral. Recall that in asymptotically flat (or AdS) space, this path integral was interpreted as $Z = e^{-\beta F}$, where $F = E - TS$ is the free energy. Here, the energy E is the ADM energy measured at spatial infinity. Since de Sitter space is a closed universe, it doesn't have a spatial infinity where a conserved energy could be defined. The proposal of Gibbons and Hawking [5] was therefore to set $E = 0$ and interpret the partition function simply as $Z = e^S$. In the semi-classical limit, where $Z \approx e^{-I_E}$, this implies $S = -I_E$.

In the case of de Sitter, the GHY boundary term vanishes and result for the Euclidean action I_E comes purely from the bulk Einstein–Hilbert action

$$I_E = -\frac{1}{16\pi G} \int d^4x \sqrt{g} (R - 2\Lambda), \quad (6.51)$$

where we have added the cosmological constant Λ . Using that $R = 4\Lambda$ (from the Einstein equation), we have

$$I_E = -\frac{\Lambda}{8\pi G} V = -\frac{3}{8\pi G L^2} V, \quad (6.52)$$

where V is the volume of the spacetime. where V is the volume of the Euclidean solution. Since this is a four-sphere of radius L , we have $V = (8\pi^2/3)L^4$ and hence $I_E = -\pi L^2/G$. Using $A = 4\pi L^2$, we then have

$$S_{\text{dS}} = \frac{A}{4G}. \quad (6.53)$$

For our Universe, with its small cosmological constant, this de Sitter entropy is estimated to be $\sim 10^{120}$, which is vastly larger than the entropy of all CMB photons ($\sim 10^{89}$), all stars ($\sim 10^{81}$) and all supermassive black holes ($\sim 10^{104}$) [40].

Part II

Cosmological Correlations

One of the biggest questions in all of science is the origin of structure in the Universe. What created everything that we see around us? An important clue lies in the fact that the structures in the Universe aren't distributed randomly, but displays large-scale correlations. These correlations are a fossil record of the early universe, and by measuring them we hope to uncover how the cosmological perturbations formed and evolved.

Our earliest snapshot of the large-scale structure of the Universe comes from measurements of the cosmic microwave background (CMB), whose intensity varies as a function of position on the sky. These CMB anisotropies reflect variations in the matter density at the time when the first atoms formed, about 380 000 years after the Big Bang (see Fig. 17). Over time, the density fluctuations grew, until clouds of gas started to collapse into stars and galaxies. In principle, the distribution of the collapsed objects still contains correlations that carry information about the primordial universe. Extracting this information is a fundamental challenge of modern cosmology.

We understand the late-time evolution of the Universe well enough to trace it back until the beginning of the hot Big Bang. While doing this, cosmologists recently discovered a remarkable fact: the initial fluctuations which ultimately led to the large-scale structure of the Universe must have been created *before* the hot Big Bang. The conventional Big Bang was therefore *not* the beginning of time, but the end of an earlier high-energy period, and the primordial fluctuations were created in this pre-Big Bang phase. But, how exactly did this happen? The challenge is to

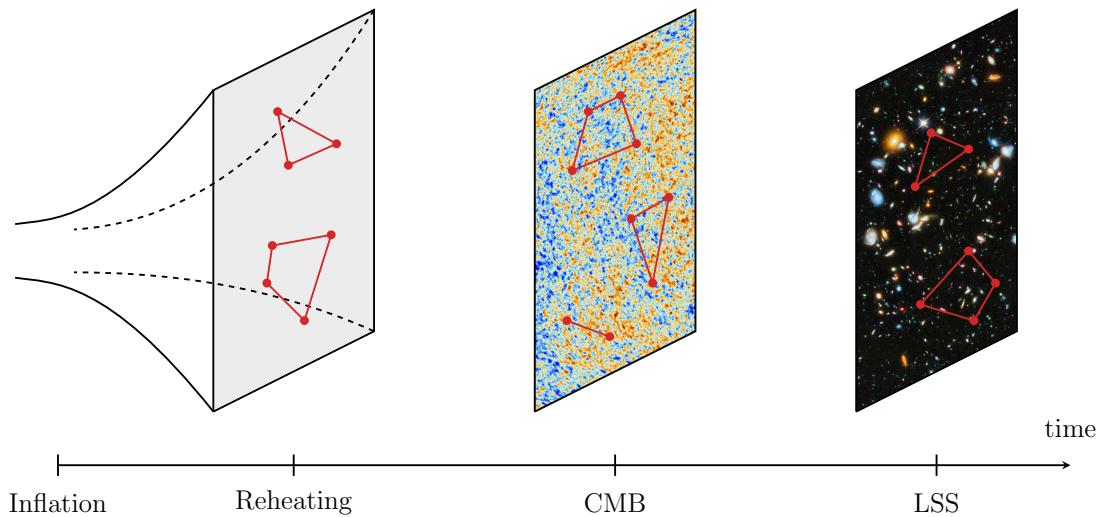


Figure 17: The correlations in late-time cosmological observables—like the distribution of the large-scale structure (LSS) or the anisotropies of the cosmic microwave background (CMB)—can be traced back in time to primordial correlations on the “reheating surface” (marking the initial surface of the hot Big Bang universe or the final surface at the end of inflation). Imprinted in these correlations is the physics of the inflationary era.

extract this information from the pattern of correlations in the late universe.

We have some evidence for a period of inflationary expansion before the hot Big Bang. The details of inflation are still uncertain, but it provides an elegant mechanism to convert quantum vacuum fluctuations into large-scale cosmological perturbations. The main observable predictions of inflation are spatial correlation functions and our goal in these lectures is to explain how these are computed.

- We begin, in Chapter 7, by describing how cosmological theories make predictions for the statistical properties of the Universe and introduce the concept of “cosmological correlators”. We describe that the late-time cosmological correlations depend both on the composition of the Universe and its initial conditions. We present four interesting facts about the cosmological initial conditions that we have learned from recent observations of the CMB.
- In Chapter 8, we derive the quantum initial conditions predicted by inflation. Using the standard quantization of fields in a curved spacetime, we derive the power spectrum of inflationary scalar and tensor fluctuations. We also discuss current constraints on primordial non-Gaussianity (higher-point correlators).

We will then describe three different approaches to compute higher-order cosmological correlation functions:

- In Chapter 9, we review the **in-in formalism** and use it to compute correlation functions both in flat space and in de Sitter space. We describe in detail Maldacena’s famous computation [41] of the bispectrum in slow-roll inflation, and derive the bispectrum in the EFT of inflation [42] where the signal can be enhanced by a small sound speed. Finally, we also show that the soft limit of the inflationary correlation functions can receive a contribution from new massive particles during inflation, allowing us to look for these particles in a kind of “cosmological collider physics” [43].
- In Chapter 10, we introduce the **wavefunction approach** and use it to compute a wide variety of correlators. We show in detail how the wavefunction computations reproduce the results from the in-in computations. We also highlight new insights into the analytic structure of the cosmological wavefunction.
- A drawback to the in-in and wavefunction methods is that they involve complex time integrals that often cannot be performed analytically. Since the time evolution during inflation is not actually observable, it stands to reason that the late-time correlators can be determined more directly without having to follow this evolution. In Chapter 11, we therefore introduce a new approach to the problem of cosmological correlations called the **cosmological bootstrap** [44, 45]. Instead of struggling with complex time integrals, the bootstrap method tries to infer the spatial correlators directly by imposing consistency requirements. The cosmological bootstrap is still an active area of research and we therefore only provide a limited pedagogical introduction to the bootstrapping of simple scalar correlators.

Our goal is to explain in detail how to perform computations in all three of these approaches.

7 Primordial Statistics

An important characteristic of cosmology is that we can only predict the statistical properties of the Universe. This means that we can't predict whether a galaxy will be found at a specific position $\mathbf{x}$. Instead, our theories can only tell us about the probability $P(\mathbf{x})$ of finding a galaxy at $\mathbf{x}$, or the conditional probability $P(\mathbf{x}_2, \mathbf{x}_1)$ that a second galaxy is at $\mathbf{x}_2$ given a galaxy at $\mathbf{x}_1$. Using these probabilities, we can derive spatial correlation functions, which are the main observables in cosmology and play a fundamental role in extracting information about the early universe from cosmological observations.

Another special feature of cosmology is that the cosmological “experiment”—the Big Bang—cannot be repeated. This leads to some conceptual subtleties in the statistical interpretation of cosmological observations. In a typical physical experiment, we can repeat the measurement and compute averages from the different realizations of the statistical ensemble. In cosmology, however, we should think of the Universe as a single realization of a statistical ensemble. Defining expectation values for observable quantities is then more subtle. In practice, one uses the so-called “ergodic hypothesis” which relates averages over the statistical ensemble (which are not observable) to spatial averages over one realization (which are observable).

In this chapter, we will review fundamental concepts of statistics used in the study of cosmological correlators. We begin with the idea of a Gaussian random field and its correlation functions.

7.1 Gaussian Random Fields

The large-scale structure of the Universe is characterized by “random fields”, such as the field $\rho(t, \mathbf{x})$ describing the density of matter at position $\mathbf{x}$ and time t . We are interested in the statistical properties of such fields. Our treatment will follow nice lecture notes by Roman Scoccimarro [46].

Random fields

Consider first the simpler concept of a random variable $r \in \mathbb{R}$, with probability density function (PDF) denoted by $P(r)$. The PDF is normalized, so that $\int P(r) dr = 1$. The *average* of the variable r is then defined as

$$\langle r \rangle \equiv \int r P(r) dr. \quad (7.1)$$

Since we have integrated over the PDF, the average is *not* a random variable. Similarly, we define higher moments of the distribution as

$$\langle r^n \rangle \equiv \int r^n P(r) dr, \quad (7.2)$$

where n is a positive integer. From the second moment, we can extract the variance $\sigma_r^2 \equiv \langle r^2 \rangle - \langle r \rangle^2$. Knowing all moments would allow us to reconstruct the PDF, but usually only a finite number of moments is measured. This nevertheless can give a good approximation to the underlying PDF. For a Gaussian PDF, all odd moments vanish and the even moments can be expressed in terms of the variance.

A random field $\Phi(\mathbf{x})$ is a random variable at each point in space $\mathbf{x}$. In principle, the field will also depend on time t , but in a deterministic system the statistics of the field at time t will be determined by its properties at $t' < t$. On large scales, cosmological perturbations are small and this evolution can be derived analytically, while on smaller scales one might need numerical simulations or phenomenological models.

Perturbation theory is defined by expanding the field around its mean value. Given $\Phi(t, \mathbf{x})$, we therefore write

$$\Phi(t, \mathbf{x}) \equiv \bar{\Phi}(t) + \delta\Phi(t, \mathbf{x}), \quad (7.3)$$

where we have assumed that the mean $\bar{\Phi}(t) \equiv \langle \Phi(t, \mathbf{x}) \rangle$ only depends on time. As before, the mean is a deterministic object and not a random field. The perturbative expansion is defined in terms of the fractional deviation from the mean, which we write as

$$\delta\Phi(t, \mathbf{x}) \equiv \frac{\Phi(t, \mathbf{x}) - \bar{\Phi}(t)}{\bar{\Phi}(t)}. \quad (7.4)$$

By definition, this has zero mean, $\langle \delta\Phi \rangle = 0$, and perturbation theory is valid as long as $|\delta\Phi| \ll 1$.

We now have to face the fact that, in cosmology, we only observe one Universe and cannot repeat the experiment. Since we only observe a single realization of a statistical ensemble, we cannot construct ensemble averages.

To derive expectation values from a single realization of the observable universe, we rely on the *ergodic hypothesis*: We assume that averages over the statistical ensemble are equal to spatial averages over one realization. For instance, the mean of a random field is then defined as

$$\langle \Phi(t, \mathbf{x}) \rangle = \int \Phi P(\Phi) d\Phi = \underset{\substack{\uparrow \\ \text{ergodicity}}}{\frac{1}{V} \int d^3x \Phi(t, \mathbf{x})} \equiv \bar{\Phi}(t). \quad (7.5)$$

It can formally be proven that the ergodic hypothesis holds for “statistically homogeneous Gaussian random fields with continuous power spectra”. (We will learn below what this means.) This is fortunate because cosmology as we know it would be impossible without the ergodic property.

Correlation functions

As we explained above, a random field can be thought of as an infinite number of random variables, one for each point in space. The statistical properties of a field are then encoded in the possible averages of these random variables, i.e. products of the field at different points. For example, the N -point correlation function is the average of the field at N points,

$$\langle \Phi(\mathbf{x}_1) \Phi(\mathbf{x}_2) \cdots \Phi(\mathbf{x}_N) \rangle = \left(\prod_i^N d\Phi(\mathbf{x}_i) \Phi(\mathbf{x}_i) \right) P(\Phi(\mathbf{x}_1), \dots, \Phi(\mathbf{x}_N)), \quad (7.6)$$

where we have assumed that all fields are evaluated at the same time (and dropped the time argument). When the points are taken to be equal, $\mathbf{x}_1 = \mathbf{x}_2 = \cdots = \mathbf{x}_N$, this reduces to the moments of a random variable defined in (7.2).

The functional form of cosmological correlators is constrained by the symmetries of the background spacetime. Since the Universe is homogeneous and isotropic, we will assume that the correlations are *statistically homogeneous* (invariant under spatial translations) and *statistically isotropic* (invariant under spatial rotations). Invariance under translations means that the correlators only depend on the separations between the points, $\mathbf{x}_i - \mathbf{x}_j$, while invariance under rotations implies that they only depend on the distances, $|\mathbf{x}_i - \mathbf{x}_j|$, and not the directions. This leads to a significant reduction in the complexity of these correlators. For example, defining $\mathbf{x}_{ij} \equiv \mathbf{x}_i - \mathbf{x}_j$ and $x_{ij} \equiv |\mathbf{x}_{ij}|$, the form of the two- and three-point functions is constrained to be

$$\langle \Phi(\mathbf{x}_1) \Phi(\mathbf{x}_2) \rangle = \xi_\Phi(\mathbf{x}_{12}) = \xi_\Phi(x_{12}), \quad (7.7)$$

$$\langle \Phi(\mathbf{x}_1) \Phi(\mathbf{x}_2) \Phi(\mathbf{x}_3) \rangle = B_\Phi(\mathbf{x}_{12}, \mathbf{x}_{13}) = B_\Phi(x_{12}, x_{13}, x_{23}). \quad (7.8)$$

$\uparrow \qquad \qquad \uparrow$
 homogeneity isotropy

We see that, for the two-point function, the number of independent variables is reduced from six (the six components of two three-dimensional vectors) to just one (the distance x_{12}). Similarly, for the three-point functions, we go from 9 variables to just 3.

Fourier space

In cosmology, we like to work in Fourier space. There are two reasons for this: 1) For small fluctuations, we can linearize the equations of motion and then each Fourier mode evolves independently. 2) Correlators “diagonalize” in Fourier space. More precisely, statistical homogeneity implies that all N -point functions are proportional to the Dirac delta function $\delta(\mathbf{k}_1 + \dots + \mathbf{k}_N)$, which constrains the sum of all wavevectors to vanish.

The Fourier expansion of the field is

$$\Phi(t, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} e^{i\mathbf{k} \cdot \mathbf{x}} \Phi_{\mathbf{k}}(t), \quad (7.9)$$

where the Fourier components $\Phi_{\mathbf{k}}(t)$ are complex. If the field $\Phi(t, \mathbf{x})$ is real, we must have $\Phi_{\mathbf{k}}^* = \Phi_{-\mathbf{k}}$. We can think of the Fourier modes $\Phi_{\mathbf{k}}(t)$ as a random field, but now with two components for each $\mathbf{k}$, because the field is complex. We can then construct correlators directly in Fourier space:

$$\langle \Phi_{\mathbf{k}_1} \Phi_{\mathbf{k}_2} \dots \Phi_{\mathbf{k}_N} \rangle, \quad (7.10)$$

where again all fields are evaluated at the same time (and we dropped the time argument). For example, consider the two-point function:

$$\langle \Phi_{\mathbf{k}_1} \Phi_{\mathbf{k}_2} \rangle = \int d^3x_1 d^3x_2 e^{-i\mathbf{k}_1 \cdot \mathbf{x}_1} e^{-i\mathbf{k}_2 \cdot \mathbf{x}_2} \langle \Phi(\mathbf{x}_1) \Phi(\mathbf{x}_2) \rangle. \quad (7.11)$$

Assuming statistical homogeneity and isotropy, we have $\langle \Phi(\mathbf{x}_1) \Phi(\mathbf{x}_2) \rangle = \xi_\Phi(|\mathbf{x}_1 - \mathbf{x}_2|)$, and hence get

$$\langle \Phi_{\mathbf{k}_1} \Phi_{\mathbf{k}_2} \rangle = \underbrace{\int d^3x_2 e^{-i(\mathbf{k}_1 + \mathbf{k}_2) \cdot \mathbf{x}_2}}_{(2\pi)^3 \delta(\mathbf{k}_1 + \mathbf{k}_2)} \underbrace{\int d^3x_{12} e^{-i\mathbf{k}_1 \cdot \mathbf{x}_{12}} \xi_\Phi(x_{12})}_{\equiv P_\Phi(k_1)}, \quad (7.12)$$

where $k_1 \equiv |\mathbf{k}_1|$. As promised, the result is proportional to $\delta(\mathbf{k}_1 + \mathbf{k}_2)$. This implies that the two-point function is “diagonal” in Fourier space, i.e. nonzero only when $\mathbf{k}_2 = -\mathbf{k}_1$. We call this diagonal because $\mathbf{k}_2$ then belongs to the same complex Fourier mode as $\mathbf{k}_1$ since $\Phi_{\mathbf{k}}^* = \Phi_{-\mathbf{k}}$.

We see from (7.12) that the *power spectrum* is the Fourier transform of the two-point correlations function:

$$P_{\Phi}(k) = \int d^3x e^{-i\mathbf{k}\cdot\mathbf{x}} \xi_{\Phi}(x). \quad (7.13)$$

Considering the inverse transform, we can read off the variance of the field as

$$\xi_{\Phi}(0) = \int \frac{d^3k}{(2\pi)^3} P_{\Phi}(k) = \int d \log k \frac{k^3}{2\pi^2} P_{\Phi}(k), \quad (7.14)$$

where the integrand is the *dimensionless power spectrum*:

$$\Delta_{\Phi}^2(k) \equiv \frac{k^3}{2\pi^2} P_{\Phi}(k). \quad (7.15)$$

The spectrum is called *scale-invariant* if $\Delta_{\Phi}^2(k)$ is a constant. The variance then receives equal contributions from every logarithmic interval in k .

Let us visualize how the shape of the power spectrum affects the correlations in position space. We first note that the two-point correlation function, in d spatial dimensions, is given by the Fourier transform of the power spectrum:

$$\langle \Phi(\mathbf{x}_1) \Phi(\mathbf{x}_2) \rangle = \int \frac{d^d k}{(2\pi)^d} e^{i\mathbf{k}\cdot(\mathbf{x}_1 - \mathbf{x}_2)} P_{\Phi}(k). \quad (7.16)$$

We then consider a power spectrum of the form

$$P_{\Phi}(k) \propto k^{-n}, \quad (7.17)$$

where n is an integer. Although eventually we are interested in $d = 3$, we now choose $d = 2$, so that we can plot the field in two dimensions (see Fig. 18). Let us choose different values for the spectral index n in (7.17):

- For $n = 0$, the Fourier transform in (7.17) leads to

$$\langle \Phi(\mathbf{x}_1) \Phi(\mathbf{x}_2) \rangle \propto \delta(\mathbf{x}_1 - \mathbf{x}_2). \quad (7.18)$$

This means that there is no correlation between the value of the field at separated points. A typical configuration of $\Phi(\mathbf{x})$ is shown in the left of Figure 18. A distribution like this, with no correlation between neighboring points, is known as *white noise*.

- For $n = 1$, the correlation between neighboring points becomes stronger. As we can see in the second panel in Fig. 18, the field takes a particular value at some point, there is now an increased likelihood that it takes similar values at neighboring points.

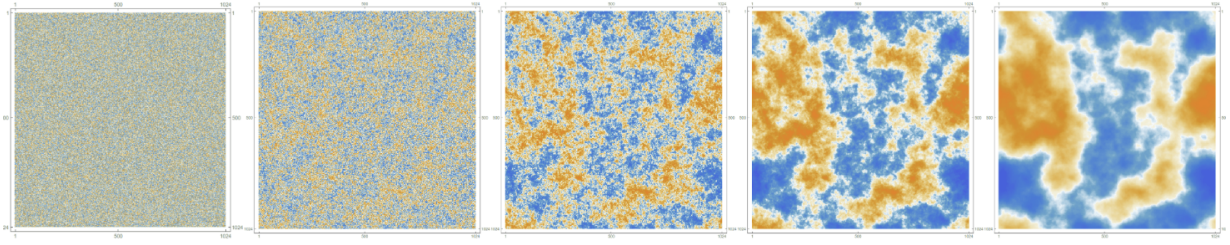


Figure 18: Realizations for $P(k) \propto k^{-n}$, with $n \in \{0, 1, 2, 3, 4\}$ from left to right. The case $n = 0$ corresponds to white noise, while $n = 2$ is scale invariant. The length scale over which fluctuations are correlated increases as n increases. (The plots were created using a Mathematica code by Garrett Goon.)

- For $n = 2$, the distribution is *scale invariant*, meaning that the correlation between any two points is independent of the distance between those points. This is illustrated in the middle panel of Fig. 18. To understand this scale invariance, we rescale the positions in (7.19):

$$\begin{aligned} \langle \Phi(\lambda \mathbf{x}_1) \Phi(\lambda \mathbf{x}_2) \rangle &\propto \int \frac{d^d k}{(2\pi)^d} \frac{e^{i\lambda \mathbf{k} \cdot (\mathbf{x}_1 - \mathbf{x}_2)}}{k^d} \\ &= \int \frac{d^d \tilde{\mathbf{k}}}{(2\pi)^d} \frac{e^{i\tilde{\mathbf{k}} \cdot (\mathbf{x}_1 - \mathbf{x}_2)}}{\tilde{k}^d} = \langle \Phi(\mathbf{x}_1) \Phi(\mathbf{x}_2) \rangle, \end{aligned} \quad (7.19)$$

where we have defined $\tilde{\mathbf{k}} = \lambda \mathbf{k}$ in the second equality.

- Increasing n further, increases the length scale over which the fluctuations are correlated. Two examples are shown in Fig. 18.

As we will see in Section 7.4, the initial conditions in our Universe have a near scale-invariant power spectrum which in three dimensions is $P_\Phi(k) \propto k^{-3}$.

Connected correlators

As we take two (or more) points far apart, we expect the correlations between those points to vanish. To make this manifest, we have to defined the so-called “connected correlators”.

As a simple example, consider a random field Φ , with mean $\langle \Phi \rangle \equiv \bar{\Phi}(t)$ and two-point function $\langle \Phi(\mathbf{x}_1) \Phi(\mathbf{x}_2) \rangle \equiv \langle \Phi_1 \Phi_2 \rangle$. A trivial two-point function would just be the product of the mean values at the two positions $\mathbf{x}_1$ and $\mathbf{x}_2$, which doesn’t vanishes in the limit $|\mathbf{x}_1 - \mathbf{x}_2| \rightarrow \infty$. We therefore define the “true correlation” as the connected two-point function:

$$\langle \Phi_1 \Phi_2 \rangle_c \equiv \langle \Phi_1 \Phi_2 \rangle - \langle \Phi_1 \rangle \langle \Phi_2 \rangle. \quad (7.20)$$

It may seem that we could also solve the problem by subtracting the mean and working with the field $\delta\Phi \equiv \Phi - \bar{\Phi}$. We then have

$$\langle \delta\Phi \rangle = 0, \quad (7.21)$$

$$\langle \delta\Phi_1 \delta\Phi_2 \rangle = \langle \delta\Phi_1 \delta\Phi_2 \rangle_c, \quad (7.22)$$

$$\langle \delta\Phi_1 \delta\Phi_2 \delta\Phi_3 \rangle = \langle \delta\Phi_1 \delta\Phi_2 \delta\Phi_3 \rangle_c. \quad (7.23)$$

However, the issue then still arises for the four-point function:

$$\begin{aligned} \langle \delta\Phi_1\delta\Phi_2\delta\Phi_3\delta\Phi_4 \rangle &= \langle \delta\Phi_1\delta\Phi_2 \rangle \langle \delta\Phi_3\delta\Phi_4 \rangle + \langle \delta\Phi_1\delta\Phi_3 \rangle \langle \delta\Phi_2\delta\Phi_4 \rangle + \langle \delta\Phi_1\delta\Phi_4 \rangle \langle \delta\Phi_2\delta\Phi_3 \rangle \\ &\quad + \langle \delta\Phi_1\delta\Phi_2\delta\Phi_3\delta\Phi_4 \rangle_c, \end{aligned} \quad (7.24)$$

which contains trivial disconnected contributions made from products of two-point functions. In the following, we will describe the relation between general N -point functions and their connected parts.

We start with the case of a random variable r , with moments $\langle r^n \rangle$ and connected moments (or “cumulants”) $\langle r^n \rangle_c$. The moment generating function is

$$M(t) \equiv \langle e^{tr} \rangle = \int e^{tr} P(r) dr = \sum_{n=0}^{\infty} \frac{r^n}{n!} \langle r^n \rangle, \quad (7.25)$$

so that

$$\langle r^n \rangle = \left(\frac{d^n M(t)}{dt^n} \right)_{t=0}. \quad (7.26)$$

Note that $M(it) = \langle e^{itr} \rangle$ can be interpreted as the Fourier transform of $P(r)$. The relation between moments and cumulants is then provided by defining the following cumulant generating function, $C(t) \equiv \log[M(t)]$, so that

$$\langle r^n \rangle_c = \left(\frac{d^n C(t)}{dt^n} \right)_{t=0}. \quad (7.27)$$

It is then straightforward to generate the expected relations between the moments $\langle r^n \rangle$ and the connected moments $\langle r^n \rangle_c$.

Exercise 7.1 Show that

$$\begin{aligned} \langle r \rangle_c &= \langle r \rangle, \\ \langle r^2 \rangle_c &= \langle r^2 \rangle - \langle r \rangle^2, \\ \langle r^3 \rangle_c &= \langle r^3 \rangle - 3\langle r \rangle \langle r^2 \rangle + 2\langle r \rangle^3, \\ \langle r^4 \rangle_c &= \langle r^4 \rangle - 4\langle r \rangle \langle r^3 \rangle - 3\langle r^2 \rangle^2 + 12\langle r^2 \rangle \langle r \rangle^2 - 6\langle r \rangle^4. \end{aligned} \quad (7.28)$$

If $\langle r \rangle = 0$, the last expression reduces to $\langle r^4 \rangle_c = \langle r^4 \rangle - 3\langle r^2 \rangle^2$, as expected from (7.24). ➔

It is straightforward to generalize this to the case of random fields. The generating function for N -point correlators is

$$M[t(\mathbf{x})] = \langle e^{\int t(\mathbf{x})\Phi(\mathbf{x}) d^3\mathbf{x}} \rangle, \quad (7.29)$$

so that

$$\langle \Phi(\mathbf{x}_1) \cdots \Phi(\mathbf{x}_N) \rangle = \left(\frac{\mathcal{D}^N M[t(\mathbf{x})]}{\mathcal{D}t(\mathbf{x}_1) \cdots \mathcal{D}t(\mathbf{x}_N)} \right)_{t(\mathbf{x})=0}. \quad (7.30)$$

As before, the relation to connected N -point correlators is then given by the generation function $C[t(\mathbf{x})] = \log[M[t(\mathbf{x})]]$ and its functional derivatives.

Gaussianity

Random fields (with zero mean) are called Gaussian if they are completely determined by their power spectrum (two-point function) because all “connected correlators” with $N > 2$ vanish.

For a random variable, a Gaussian PDF is

$$P_G(r) = \frac{1}{\sqrt{2\pi\sigma_r^2}} \exp\left[-\frac{1}{2} \frac{r^2}{\sigma_r^2}\right], \quad (7.31)$$

where we assumed $\langle r \rangle = 0$. This PDF depends only on the variance σ_r^2 and not on any higher moments.

For a field at N points, this generalizes to multivariate Gaussian:

$$P_G[\Phi_1, \dots, \Phi_N] = \frac{1}{(2\pi)^{N/2} \sqrt{\det \xi}} \exp\left[-\frac{1}{2} \Phi_1 \xi_{ij}^{-1} \Phi_j\right], \quad (7.32)$$

where we use the shorthand $\Phi_n \equiv \Phi(\mathbf{x}_n)$ and defined the two-point function $\xi_{ij} \equiv \langle \Phi_i \Phi_j \rangle$. Repeated indices are summed over. This PDF takes a particular nice form in Fourier space where the two-point function is diagonal:

$$P_G[\Phi_{\mathbf{k}}] = \prod_{\mathbf{k}} \frac{1}{\sqrt{2\pi P_{\Phi}(k)}} \exp\left[-\frac{1}{2} \frac{|\Phi(\mathbf{k})|^2}{P_{\Phi}(k)}\right]. \quad (7.33)$$

All higher-order correlators are then simply functions of the power spectrum $P_{\Phi}(k)$.

7.2 Cosmological Perturbations

We will be interested in the correlators of cosmological perturbations with initial conditions set by quantum fluctuations during the inflationary era. To prepare for this discussion, we will review the theory of cosmological perturbations. We will keep it brief, but further details and explanations can be found in textbooks like [47].

The basic idea is to write the metric and stress-energy tensor as

$$\begin{aligned} g_{\mu\nu}(t, \mathbf{x}) &= \bar{g}_{\mu\nu}(t) + \delta g_{\mu\nu}(t, \mathbf{x}), \\ T_{\mu\nu}(t, \mathbf{x}) &= \bar{T}_{\mu\nu}(t) + \delta T_{\mu\nu}(t, \mathbf{x}), \end{aligned} \quad (7.34)$$

and then expand $\nabla^\mu T_{\mu\nu} = 0$ and $G_{\mu\nu} = 8\pi G T_{\mu\nu}$ to linear order in the perturbations. This is conceptually straightforward to algebraically a bit involved. One important subtlety is that there is an ambiguity in what we call matter perturbations and what are metric perturbations—by a change of coordinates we can transform one into the other.

Metric perturbations

Assuming that the background is a flat FRW spacetime, we write the perturbed metric as

$$ds^2 = a^2(\eta) \left\{ -(1 + 2A) dt^2 + 2B_i dx^i d\eta + [\delta_{ij} + 2E_{ij}] dx^i dx^j \right\}, \quad (7.35)$$

where A , B_i and E_{ij} are functions of space and (conformal) time. The factors of 2 are introduced for later convenience. We will adopt the useful convention that the indices on the spatial vectors and tensors are raised and lowered with δ_{ij} .

It will be extremely useful to perform a scalar-vector-tensor (SVT) decomposition of the perturbations. For three-vectors, this should be familiar. It simply means that we can split any three-vector into the gradient of a scalar and a divergenceless vector

$$B_i = \underbrace{\partial_i B}_{\text{scalar}} + \underbrace{\hat{B}_i}_{\text{vector}}, \quad (7.36)$$

with $\partial^i \hat{B}_i = 0$. In Fourier space, the decomposition in (7.36) becomes $B_i = i\hat{k}_i B + \hat{B}_i$, where $\hat{\mathbf{k}} \equiv \mathbf{k}/|\mathbf{k}|$ is a unit vector. The magnitude of the wavevector, $k \equiv |\mathbf{k}|$, has been absorbed into $B(\eta, \mathbf{k})$, so that B and B_i have the same dimensions. We see that the vector has been split into a piece that is parallel to the direction of the Fourier mode $\mathbf{k}$ and a piece orthogonal to it. Similarly, any rank-2 symmetric tensor can be written

$$E_{ij} = \underbrace{C\delta_{ij} + \partial_{\langle i}\partial_{j\rangle}E}_{\text{scalar}} + \underbrace{\partial_{\langle i}\hat{E}_{j\rangle}}_{\text{vector}} + \underbrace{\hat{E}_{ij}}_{\text{tensor}}, \quad (7.37)$$

where

$$\partial_{\langle i}\partial_{j\rangle}E \equiv \left(\partial_i\partial_j - \frac{1}{3}\delta_{ij}\nabla^2\right)E, \quad (7.38)$$

$$\partial_{\langle i}\hat{E}_{j\rangle} \equiv \frac{1}{2}\left(\partial_i\hat{E}_j + \partial_j\hat{E}_i\right). \quad (7.39)$$

The first term in (7.37) contains the trace of the spatial perturbation, $E^i_i = C\delta^i_i = 3C$, while the remaining terms are traceless. As before, the hatted quantities have vanishing divergence, $\partial^i \hat{E}_i = \partial^i \hat{E}_{ij} = 0$, and so correspond to transverse vector and tensor perturbations. The tensor perturbations describe gravitational waves and we will also denote them by $h_{ij} \equiv 2\hat{E}_{ij}$. In Fourier space, we write

$$\partial_{\langle i}\partial_{j\rangle}E \equiv -\left(\hat{k}_i\hat{k}_j - \frac{1}{3}\delta_{ij}\right)E, \quad (7.40)$$

$$\partial_{\langle i}\hat{E}_{j\rangle} \equiv \frac{i}{2}\left(\hat{k}_i\hat{E}_j + \hat{k}_j\hat{E}_i\right), \quad (7.41)$$

where the magnitude k has been absorbed in E and $\hat{E}_i$.

The 10 degrees of freedom of the metric have thus been decomposed into 4 scalar modes (A, B, C, E), 4 vector modes ($\hat{B}_i, \hat{E}_i$) and 2 tensor modes ($\hat{E}_{ij}$). What makes the SVT decomposition so powerful is the fact that the Einstein equations for scalars, vectors and tensors don't mix at linear order and can therefore be treated separately. We will mostly be interested in scalar fluctuations and the associated density perturbations, and at linear order it is consistent to set vectors and tensors to zero. Vector perturbations aren't produced by inflation and even if they were, they would decay quickly with the expansion of the universe. Tensor perturbations are an important prediction of inflation and we will discuss them at various points in the text.

The metric perturbations in (7.35) aren't uniquely defined, but depend on our choice of coordinates or the *gauge choice*. In particular, when we wrote down the perturbed metric, we implicitly introduced a specific time slicing of the spacetime and defined specific spatial coordinates on these time slices. Making a different choice of coordinates can change the values of the perturbation

variables. It may even introduce “fictitious perturbations,” which are fake perturbations that can arise from an inconvenient choice of coordinates even if the background is perfectly homogeneous.

Consider the following change of coordinates

$$x^\mu \mapsto \tilde{x}^\mu \equiv x^\mu + \xi^\mu(x), \quad \text{where} \quad \begin{aligned} \xi^0 &\equiv T, \\ \xi^i &\equiv L^i = \partial^i L + \hat{L}^i. \end{aligned} \quad (7.42)$$

We assume that ξ^μ is small and can therefore be treated as a small perturbation. The function $T(\eta, \mathbf{x})$ defines the hypersurfaces of constant time in the new coordinates, while $L^i(\eta, \mathbf{x})$ determines the spatial coordinates on these hypersurfaces. We have split the spatial shift L^i into a scalar, L , and a divergenceless vector, $\hat{L}^i$.

To determine the transformation of the metric, we use that the spacetime interval is an invariant:

$$ds^2 = g_{\mu\nu}(x) dx^\mu dx^\nu = \tilde{g}_{\alpha\beta}(\tilde{x}) d\tilde{x}^\alpha d\tilde{x}^\beta, \quad (7.43)$$

where we have used a different set of dummy indices on both sides to make the next few lines clearer. Writing $d\tilde{x}^\alpha = (\partial\tilde{x}^\alpha/\partial x^\mu) dx^\mu$ (and similarly for $d\tilde{x}^\beta$), we find

$$g_{\mu\nu}(x) = \frac{\partial\tilde{x}^\alpha}{\partial x^\mu} \frac{\partial\tilde{x}^\beta}{\partial x^\nu} \tilde{g}_{\alpha\beta}(\tilde{x}), \quad (7.44)$$

which relates the metric in the old coordinates, $g_{\mu\nu}$, to the metric in the new coordinates, $\tilde{g}_{\alpha\beta}$. Substituting (7.35) on both sides of (7.44), and using (7.42), we find that the metric perturbations transform as

$$A \mapsto A - T' - \mathcal{H}T, \quad (7.45)$$

$$B \mapsto B + T - L', \quad \hat{B}_i \mapsto \hat{B}_i - \hat{L}'_i, \quad (7.46)$$

$$C \mapsto C - \mathcal{H}T - \frac{1}{3}\nabla^2 L, \quad (7.47)$$

$$E \mapsto E - L, \quad \hat{E}_i \mapsto \hat{E}_i - \hat{L}_i, \quad \hat{E}_{ij} \mapsto \hat{E}_{ij}, \quad (7.48)$$

where the primes denote derivatives with respect to conformal time η and $\mathcal{H} \equiv a'/a$ is the conformal Hubble parameter.

Exercise 7.2 Derive the transformations of the metric perturbations. →

Exercise 7.3 Show that the following quantities are invariant under these transformations

$$\begin{aligned} \Psi &\equiv A + \mathcal{H}(B - E') + (B - E')', \\ \Phi &\equiv -C + \frac{1}{3}\nabla^2 E - \mathcal{H}(B - E'). \end{aligned} \quad (7.49)$$

These Bardeen variables are the “real” spacetime perturbations since they cannot be removed by a gauge transformation. →

Often we choose to work in a fixed gauge where the coordinate freedom is used to put the metric into a specific form.

- *Newtonian gauge*: A popular choice is the Newtonian gauge, where $B = E = 0$ and the metric has only 2 scalar perturbations $A \equiv \Psi$ and $C \equiv -\Phi$, which play the role of the gravitational potentials. Newtonian gauge is the preferred gauge for studying structure formation, since the equations of motion have the closest resemblance to the equations in Newtonian gravity.
- *Spatially flat gauge*: A convenient gauge for computing inflationary perturbations is the spatially flat gauge, with $C = E = 0$. In this gauge, we will be able to focus most directly on the fluctuations in the inflaton field $\delta\phi$.

To describe the evolution of the perturbed spacetime, we must consider its coupling to matter perturbations via the Einstein equations. We turn to this next.

Matter perturbations

Let us discuss the perturbations of the energy-momentum tensor. We start with the energy-momentum tensor of a non-ideal fluid:

$$T_{\mu\nu} = (\rho + P)u_\mu u_\nu + P g_{\mu\nu} + \Pi_{\mu\nu}, \quad (7.50)$$

where ρ and P are the density and pressure, $\Pi_{\mu\nu}$ is the anisotropic stress and u^μ is the four-velocity of a fluid element. The anisotropic stress satisfies $\Pi^\mu{}_\mu = 0$ and $\Pi^\mu{}_\nu u^\nu = 0$. The four-velocity is defined as $u^\mu \equiv dx^\mu/d\tau$, where τ is the proper time. We can write this as

$$u^\mu = \frac{dx^\mu}{d\tau} = \frac{dx^\mu}{d\eta} \frac{d\eta}{d\tau}. \quad (7.51)$$

The first factor is $dx^\mu/d\tau = (1, v^i)$, where $v^i \equiv dx^i/d\tau$ is the three-velocity. We will assume that $|v^i| \ll 1$. To determine the second factor, we look at

$$\begin{aligned} \frac{d\tau}{d\eta} &= \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\eta} \frac{dx^\nu}{d\eta}} = \sqrt{-g_{00} - 2g_{0i}v^i - g_{ij}v^i v^j} \\ &\approx \sqrt{-g_{00}} \\ &= \sqrt{a^2(1 + 2A)} \\ &= a(1 + A). \end{aligned} \quad (7.52)$$

We therefore have

$$\frac{d\eta}{d\tau} = \left(\frac{d\tau}{d\eta}\right)^{-1} = \frac{1}{a(1 + A)} \approx \frac{1 - A}{a}. \quad (7.53)$$

The four-velocity then is

$$u^\mu = \frac{1}{a}(1 - A, v^i). \quad (7.54)$$

Note that this involves the metric perturbation A . We will also need the four-velocity with a lower index, $u_\mu = g_{\mu\nu}u^\nu$, which becomes

$$u_\mu = a(-(1 + A), v_i + B_i). \quad (7.55)$$

Note that this also depends on the metric perturbation B_i . At linear order in the perturbations, we have the correct normalization $u^\mu u_\mu = -1$.

The components of (7.50) then are

$$\begin{aligned} T^0_0 &= -(\bar{\rho} + \delta\rho), \\ T^i_0 &= -(\bar{\rho} + \bar{P})v^i, \\ T^0_i &= (\bar{\rho} + \bar{P})(v_i + B_i), \\ T^i_j &= (\bar{P} + \delta P)\delta^i_j + \Pi^i_j, \quad \Pi^i_i \equiv 0, \end{aligned} \tag{7.56}$$

where we have used mixed upper and lower indices to minimize the appearance of metric perturbations. We have split the density and the pressure into background values (denoted by an overbar) and perturbations. As for the metric, we can apply an SVT decomposition to the perturbations of the energy-momentum tensor: $\delta\rho$ and δP have scalar parts only, while v_i and Π_{ij} can be split into scalar, vector and tensor parts.

Under a coordinate transformation, $x^\mu \mapsto \tilde{x}^\mu$, the tensor $T^\mu{}_\nu$ transforms as

$$T^\mu{}_\nu(x) = \frac{\partial x^\mu}{\partial \tilde{x}^\alpha} \frac{\partial \tilde{x}^\beta}{\partial x^\nu} \tilde{T}^\alpha{}_\beta(\tilde{x}). \tag{7.57}$$

Using this, one can show that the perturbations of the energy-momentum tensor transform as

$$\delta\rho \mapsto \delta\rho - \bar{\rho}' T, \tag{7.58}$$

$$\delta P \mapsto \delta P - \bar{P}' T, \tag{7.59}$$

$$v_i \mapsto v_i + L'_i, \tag{7.60}$$

$$\Pi_{ij} \mapsto \Pi_{ij}. \tag{7.61}$$

Note that ρ and P are Lorentz scalars, so the transformation of their perturbations only arises because the time-dependent backgrounds are expressed in a different temporal gauge.

Exercise 7.4 Derive the above transformations of the perturbations of the energy-momentum tensor. ➔

We see that the scalar and vector parts of the perturbations can be changed by a change of coordinates. As before, we can define specific combinations of variables for which these transformations cancel. There are various gauge-invariant quantities that can be formed from the metric and matter variables. For our purposes, two important gauge-invariant variables are⁸

$$\zeta_d \equiv C - \frac{1}{3}\nabla^2 E - \mathcal{H} \frac{\delta\rho}{\bar{\rho}'}, \tag{7.62}$$

$$\zeta_c \equiv C - \frac{1}{3}\nabla^2 E + \mathcal{H}(v + B). \tag{7.63}$$

These are called *curvature perturbations* because they reduce to the intrinsic curvature of the spatial slices in the uniform density and comoving gauges, respectively.

⁸In [47], these are denoted by $\zeta = -\zeta_d$ and $\mathcal{R} = -\zeta_c$.

Exercise 7.5 Consider the induced metric on surfaces of constant time

$$\gamma_{ij} = a^2 \left[(1 + 2C) \delta_{ij} + 2E_{ij} \right]. \quad (7.64)$$

Show that the intrinsic curvature on the spatial slices is

$$\begin{aligned} a^2 R_{(3)}[\gamma_{ij}] &= -4\nabla^2 C + 2\partial_i \partial_j E^{ij} \\ &= 4\nabla^2 \left(-C + \frac{1}{3} \nabla^2 E \right), \end{aligned} \quad (7.65)$$

which explains why $-C + \frac{1}{3} \nabla^2 E$ is called the “curvature perturbation.” ➔

Above we have used the coordinate freedom to put the perturbed spacetime metric into a specific form. Alternatively, we can also use this freedom to remove some of the matter perturbations, which leads to alternative gauge choices.

- *Uniform density gauge*: We can use the freedom in the time slicing to set the total density perturbation to zero, $\delta\rho \equiv 0$. The main scalar perturbation in this gauge is the curvature perturbation, $\delta g_{ij} = a^2(1 + 2\zeta_d)\delta_{ij}$.
- *Comoving gauge*: Similarly, we can ask for the covariant velocity perturbation to vanish, $v + B \equiv 0$. The main scalar perturbation in this gauge is then the curvature perturbation, $\delta g_{ij} = a^2(1 + 2\zeta_c)\delta_{ij}$. This is another convenient gauge for the computation of the inflationary quantum fluctuations.

There are different versions of the uniform density and comoving gauges depending on which of the metric fluctuations is set to zero.

On large scales, the curvature perturbations ζ_d and ζ_c become equal and can therefore be treated interchangeably. In the rest of these lectures, we will exclusively be working with the comoving curvature perturbation ζ_c and drop the subscript. A critical feature of the curvature perturbations is that they stay constant on superhorizon scales if the matter perturbations are “adiabatic”, even when the equation of state changes. This is particularly important for the transition from inflation to the radiation-dominated universe. Since the equation of state during reheating is unknown, it is crucial that we can track a quantity whose evolution doesn’t depend on these details. This will allow us to match the predictions made by inflation to the fluctuations in the primordial plasma after inflation.

Exercise 7.6 Slow-roll inflation is modeled by a scalar field ϕ —called the inflaton—minimally coupled to gravity (see Section 8.1):

$$S = \int d^4x \sqrt{-g} \left(-\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) \right), \quad (7.66)$$

where $V(\phi)$ is the inflationary potential.

1. By varying this action with respect to the (inverse) metric, show that the energy-momentum tensor is

$$T_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi - g_{\mu\nu} \left(\frac{1}{2} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi + V(\phi) \right). \quad (7.67)$$

Hint: You may use that $\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}g_{\mu\nu}\delta g^{\mu\nu}$ and $\delta S \equiv -\frac{1}{2}\int d^4x \sqrt{-g} T_{\mu\nu} \delta g^{\mu\nu}$.

2. Find the energy density and pressure of the homogeneous background, with $\phi = \bar{\phi}(t)$.
3. Show that the comoving curvature perturbation is

$$\zeta_c = -\frac{\mathcal{H}}{\dot{\phi}} \delta\phi, \quad (7.68)$$

where $\delta\phi$ is the field fluctuation in the spatially flat gauge. ➔

7.3 Observed Correlations

We measure cosmological perturbations by observing different *tracers* of the distribution of structure in the Universe. For example, we observe the positions and redshifts of galaxies to determine the late-time distribution of matter, or the angular variations in the CMB temperature to infer the primordial density perturbations at the time of recombination. We will collectively denote the different cosmological observables by $\mathcal{O}(t, \mathbf{x})$. It is a crucial fact that the large-scale structure in the Universe isn't distributed randomly, but has interesting correlations between spatially separated points:

$$\langle \mathcal{O}(t, \mathbf{x}_1) \mathcal{O}(t, \mathbf{x}_2) \dots \mathcal{O}(t, \mathbf{x}_N) \rangle. \quad (7.69)$$

These N -point correlators are to the cosmologist what N -point scattering amplitudes are to the particle physicist.

Figure 19 shows the two-point correlation function of the CMB temperature anisotropies, $\langle \delta T(\boldsymbol{\theta}_1) \delta T(\boldsymbol{\theta}_2) \rangle$. The importance of this graph for cosmology cannot be overstated. The shape of the CMB power spectrum depends sensitively on the cosmological parameters and CMB measurements have therefore been used to determine the parameters of the Λ CDM cosmology to great precision [47].

- *Hubble constant* (H_0):

The value of the Hubble constant affects the size of the observable universe and therefore the apparent angular size of features in the CMB. A larger Hubble constant corresponds to a smaller universe, making the surface of last-scattering appear closer. As a result, the characteristic features in the CMB, such as the acoustic peaks, appear larger on the sky and are shifted to lower multipoles. However, in the Λ CDM model, H_0 is not an independent parameter—it is determined by the matter and dark energy densities (Ω_m and Ω_Λ) assuming a flat universe—so its effect is typically explored through changes in these parameters.

- *Curvature* (Ω_K):

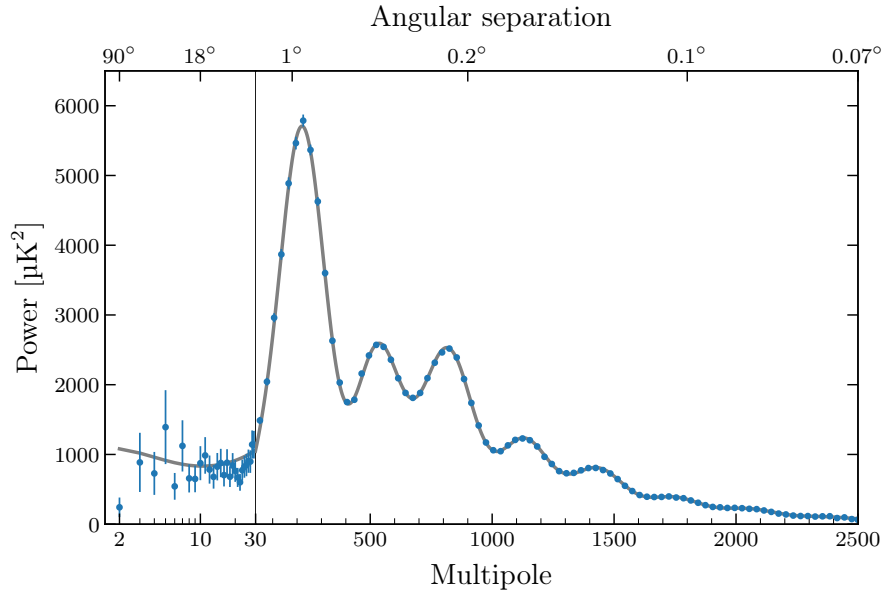


Figure 19: Power spectrum of CMB temperature anisotropies as measured by the Planck satellite [48].

Changing the spatial curvature of the universe alters how physical scales are projected onto the sky, shifting the positions of the peaks in the CMB power spectrum. In a closed universe (positive curvature), light paths curve inward, making features appear larger and moving the peaks to lower multipoles (larger angular scales). In an open universe (negative curvature), light paths diverge, making features appear smaller and shifting the peaks to higher multipoles (smaller angular scales). Observations favor a flat universe ($\Omega_K = 0$).

- *Dark energy* (Ω_Λ):

Changing the dark energy density affects the CMB spectrum mainly through its effect on the distance to the last-scattering surface. Increasing Ω_Λ (at fixed Ω_m) increases the Hubble rate and hence decreases the distance to last-scattering, so that the peaks move to larger angular scales (lower multipoles). To break the “geometric degeneracy” between changes in Ω_Λ and Ω_K requires external data sets or a measurement of CMB lensing.

- *Dark matter* (Ω_m):

The matter density affects how the universe expanded before the CMB was released. Increasing the matter density makes the universe expand faster, which causes the sound waves in the early plasma to travel shorter distances. This shifts the peaks in the CMB power spectrum to higher multipoles (smaller angular scales). Changing the matter density also affects the height of the peaks. This effect is largest for the first peak, which increases when reducing the matter density.

- *Baryon density* (Ω_b):

Changing the baryon density alters the relative heights of the peaks in the CMB spectrum. An increase in baryon density enhances the gravitational compression of the photon-baryon

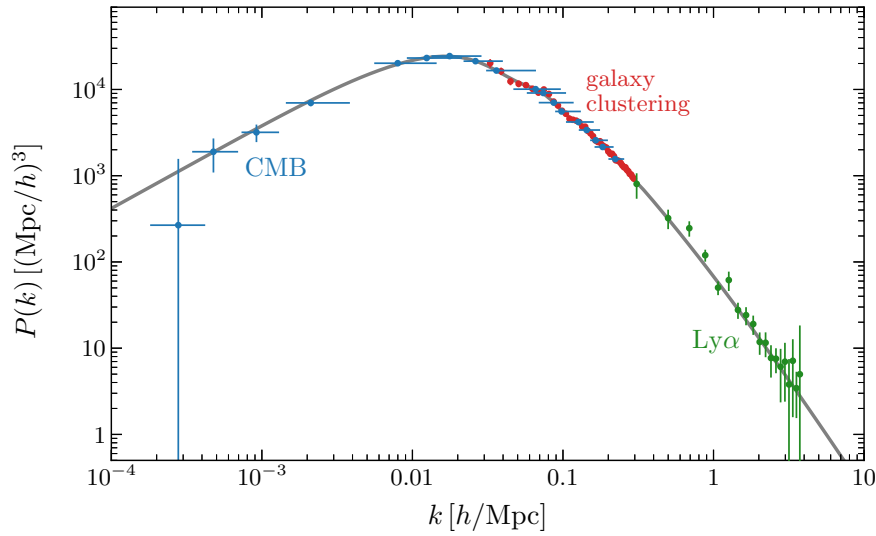


Figure 20: Measurements of the linear matter power spectrum (figure adapted from [49]). The “galaxy clustering” constraints are from the luminous red galaxy sample of the Sloan Digital Sky Survey (SDSS), the “CMB” constraints are derived from the Planck measurements of the CMB anisotropies, and “Ly α ” refers to the Lyman-alpha forest.

fluid, which amplifies the odd-numbered peaks (compression peaks) more than the even-numbered ones (rarefaction peaks). As a result, the relative heights of the peaks provide a sensitive probe of the universe’s baryon content.

- *Neutrinos* (Ω_ν):

Neutrinos suppress the CMB spectrum on small scales by free-streaming out of density fluctuations, smoothing them and reducing anisotropy at high multipoles.

- *Amplitude* ($A_s e^{-2\tau}$):

The overall amplitude of the CMB spectrum is determined by a combination of the amplitude of primordial density fluctuations (A_s) and the optical depth to reionization (τ). The latter suppresses the CMB anisotropies through the scattering by free electrons released during reionization. Additionally, it enhances the large-scale polarization signal due to these re-scatterings. By measuring CMB polarization, we can therefore break the degeneracy between A_s and τ .

- *Scalar tilt* (n_s):

As we will discuss below, the spectrum of primordial perturbations was close to scale-invariant, but not exactly. This is reflected in the CMB spectrum by a slight increase in power on large scales (low multipoles).

The CMB anisotropies predict additional correlations in CMB polarization and CMB lensing. These have also been observed.

Figure 20 displays measurements of the two-point function of matter density fluctuations, $\langle \delta\rho_m(\mathbf{x}_1)\delta\rho_m(\mathbf{x}_2) \rangle$, using different tracers of the matter perturbations. The shape of this two-

point function can be understood from the difference in the clustering of subhorizon modes during the radiation and matter eras. (Matter only starts to cluster significantly when the universe becomes matter dominated.) The dependence on the cosmological parameters is easy to understand.

- Matter density (Ω_m):

Increasing the matter density shifts the turnover scale of the power spectrum to smaller scales (higher wavenumbers) because matter-radiation equality occurs earlier, allowing structures to start growing sooner. This boosts power on small scales (which enter the horizon earlier), while reducing it on large scales. Additionally, a higher matter density increases the overall growth rate of structure, enhancing the amplitude of fluctuations after equality.

- Amplitude (σ_8):

The overall amplitude of the matter power spectrum is characterized by the parameter σ_8 , the amplitude of matter density fluctuations on scales of $8 h^{-1}\text{Mpc}$. It is related to the amplitude of primordial density fluctuations A_s .

In summary, we have learned a great deal about the geometry and composition of the Universe by measuring cosmological correlations. In the following, we will describe what these observations have taught us about the initial conditions of the Universe.

7.4 Initial Conditions

As long as the fluctuations are small (i.e. at early times or on large scales), their evolution equations can be linearized. It is convenient to describe this linear evolution in Fourier space, since the individual Fourier modes evolve independently. Each observable $\mathcal{O}$ is then related to the primordial perturbations $\zeta_{\mathbf{k}}(t_i)$ by a *transfer function*:

$$\mathcal{O}(\mathbf{x}, t) = \int \frac{d^3k}{(2\pi)^3} T_{\mathcal{O}}(k; t, t_i) \zeta_{\mathbf{k}}(t_i) e^{i\mathbf{k}\cdot\mathbf{x}}. \quad (7.70)$$

As indicated by the notation, there are different transfer functions $T_{\mathcal{O}}$ for each component of the Universe (photons, baryons, dark matter, etc.) At early times, photons and baryons were strongly coupled and acted as a single fluid. The large pressure of the radiation supported sound waves in the primordial plasma that left an imprint in the pattern of CMB anisotropies that we can still observe today. All of this interesting physics is captured by the CMB transfer function. Dark matter, on the other hand, is pressureless and therefore started to cluster earlier than the baryons. Only after recombination, did the baryons lose their pressure support and started collapsing into denser structure. It is important that the clustering of dark matter had a head start and then assisted the formation of baryonic structures (like stars and galaxies). Without the scaffolding provided by the dark matter structures wouldn't have formed fast enough to produce the Universe we see around us. All of this is encoded in the (dark) matter transfer function. More on the linear transfer functions for cosmological fluctuations can be found in any cosmology textbook [47, 50]. For the purpose of these lectures, it is only important to know that

the physics of the linear evolution is extremely well-understood, so that the fluctuations can be traced back to their primordial origins, captured by the initial curvature perturbation $\zeta_{\mathbf{k}}(t_i)$.⁹

There are *four important facts* that recent observations have revealed about the initial conditions of our Universe:

1. *Superhorizon*

The fluctuations spanned superhorizon scales at photon decoupling. Specifically, the detection of anti-correlation between CMB temperature fluctuations and E-mode polarization at angular scales larger than about one degree cannot be produced by causal processes occurring within the horizon after recombination. Causality then demands that these fluctuations must have been created *before* the hot Big Bang.

2. *Scale-invariant*

The primordial fluctuations were scale invariant. This is a natural prediction of inflation [28–31] which creates scale-invariant fluctuations from quantum fluctuations in a phase of nearly time translation-invariant dynamics.

3. *Adiabatic*

The primordial fluctuations were adiabatic, i.e. there were no relative fluctuations between the different matter species (dark matter, baryons, photons). All fluctuations are sourced by a single scalar mode:

$$\delta\rho_i(t, \mathbf{x}) = \rho_i(t + \pi(t, \mathbf{x})) - \rho_i(t) \quad \Rightarrow \quad \pi(t, \mathbf{x}) \approx \left(\frac{\delta\rho}{\dot{\rho}} \right)_i, \quad (7.71)$$

for $i = \text{dark matter, baryons, photons}$. An important consequence of adiabaticity is that all matter perturbations $\delta\rho_i$ can be removed by a choice of time slicing. The fluctuations in that gauge are then captured by the curvature perturbation $\zeta = -H\pi$. The initial conditions could also contain tensor fluctuations, $g_{ij} = a^2(t)(e^{2h(t, \mathbf{x})})_{ij}$, but we haven't seen them yet.

4. *Gaussian*

The primordial fluctuations were highly Gaussian. For example, the typical size of three-point correlations is

$$\frac{\langle \zeta \zeta \zeta \rangle}{\langle \zeta \zeta \rangle^{3/2}} < 0.1\%. \quad (7.72)$$

This Gaussianity is a natural prediction for quantum fluctuations of free fields. Having said that, a lot of the physics of inflation is encoded in small non-Gaussian correlations (see Section 8.5), although these have not been observed yet.

In the rest of these lectures, we will study different ways of computing the correlations predicted in the inflationary scenario.

⁹At some point—at late times and on small scales—the fluctuations become large and the linear approximation breaks down. The relation between the observables $\mathcal{O}$ and the initial conditions ζ then isn't described by linear transfer functions, but instead must be described by numerical simulations or a phenomenological model.

8 Inflationary Fluctuations

One of the most remarkable features of inflation is that it provides a natural mechanism for creating the primordial density fluctuations that seeded the structure in the Universe. In this chapter, we will derive the spectrum of quantum fluctuations produced during inflation.

An influential toy model for the inflationary dynamics is *slow-roll inflation*. In these models, a scalar field ϕ (the *inflaton*) dominates the energy density of the Universe and dictates the evolution of the background spacetime. The action of the field, minimally coupled to gravity, is

$$S = \int d^4x \sqrt{-g} \left(\frac{M_{\text{Pl}}^2}{2} R - \frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) \right). \quad (8.1)$$

If the potential $V(\phi)$ is sufficiently flat, it acts like a temporary cosmological constant sourcing accelerated expansion. Inflation ends when the potential steepens and the field picks up kinetic energy (see Fig. 21). If the field were perfectly homogeneous, then inflation would end at the same time everywhere in space. Quantum mechanics, however, creates spatial fluctuations of the field, $\delta\phi(t, \mathbf{x})$, which modulate the end of inflation throughout space. Although these fluctuations are a small quantum effect, they trigger large differences in the local expansion histories, which lead to significant differences in the local densities after inflation, $\delta\rho(t, \mathbf{x})$. Regions of space that end inflation first will start diluting and therefore end up having smaller densities, while regions that inflate for a longer time will have higher densities. It is worth emphasizing that the theory was not engineered to produce these fluctuations, but that their origin is instead a natural consequence of treating inflation quantum mechanically.

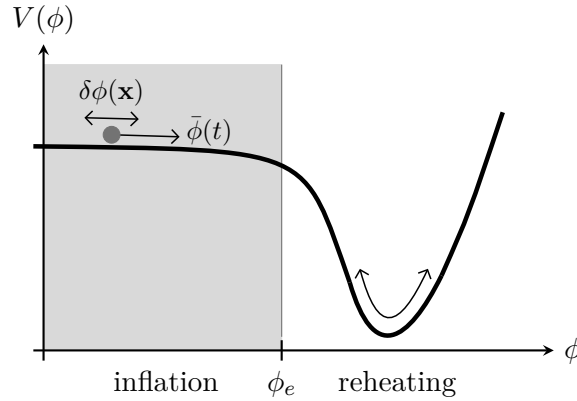


Figure 21: Quantum fluctuations around the classical background evolution affect the duration of inflation. Regions acquiring negative fluctuations remain potential-dominated longer, and hence inflate longer, than regions with positive fluctuations. Different parts of the Universe therefore undergo slightly different evolution, which induces variations in the density after inflation.

8.1 Basics of Inflation

We start with a lightning review of the physics of inflation. More details can be found in any modern cosmology textbook (e.g. Chapter 4 of [47]).

Varying the inflaton-part of the action (8.1) with respect to the metric gives the energy-momentum tensor of the scalar field (see Exercise 7.6):

$$T_{\mu\nu} = \partial_\mu\phi\partial_\nu\phi - g_{\mu\nu} \left(\frac{1}{2}g^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi + V(\phi) \right). \quad (8.2)$$

Consistency with the symmetries of the FRW spacetime requires that the background value of the inflaton only depends on time, $\phi = \phi(t)$. From the time-time component $T^0_0 = -\rho_\phi$, we infer that the energy density of the field is

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi). \quad (8.3)$$

We see that this is simply the sum of the kinetic energy density, $\frac{1}{2}\dot{\phi}^2$, and the potential energy density, $V(\phi)$. Feeding the density as a source in the Friedmann equation, we find

$$H^2 = \frac{1}{3M_{\text{Pl}}^2} \left(\frac{1}{2}\dot{\phi}^2 + V(\phi) \right), \quad (8.4)$$

where $M_{\text{Pl}} = (8\pi G)^{-1/2}$ is the reduced Planck mass and $H \equiv \dot{a}/a$ is the expansion rate. During inflation, the expansion rate is nearly constant, $H \approx \text{const.}$, and the scale factor grows exponentially $a(t) \approx e^{Ht}$. The metric during inflation is approximately that of *de Sitter space* (see Chapter 5).

Varying the action (8.1) with respect to the inflaton field gives its equation of motion

$$\ddot{\phi} + 3H\dot{\phi} = -\frac{dV}{d\phi}. \quad (8.5)$$

Sustained slow-roll inflation occurs when the Hubble friction term, $3H\dot{\phi}$, is large compared to the acceleration of the field, $\ddot{\phi}$. Many models of inflation have been constructed that satisfy these slow-roll conditions (see e.g. [51, 52] for more details).

Exercise 8.1 Show that the conservation of the energy-momentum tensor, $\nabla_\mu T^{\mu\nu} = 0$, implies the Klein–Gordon equation (8.5).

Hint: You may use that $\Gamma^0_{0\nu} = 0$ and $\Gamma^0_{ij} = Hg_{ij}$.



The accelerated expansion during inflation also implies that the comoving Hubble radius is shrinking,

$$\frac{d}{dt}(aH)^{-1} = \frac{d}{dt}(\dot{a})^{-1} = -\frac{\ddot{a}}{(\dot{a})^2} < 0, \quad (8.6)$$

which has important consequences for the evolution of perturbations during inflation. Quantum mechanics creates fluctuations on small scales, inside the Hubble radius (see Fig. 22). These fluctuations are then stretched by the inflationary expansion and freeze when they cross Hubble radius¹⁰ (“horizon”), $\lambda > (aH)^{-1}$. After inflation, the comoving Hubble radius increases and the fluctuations re-enter the horizon, becoming the primordial density fluctuations at the origin of the hot Big Bang.

¹⁰Although we call this “horizon crossing”, it is crucial that the Hubble radius is *not* equal to the particle horizon. In fact, it is precisely by making the particle horizon much larger than the Hubble radius that inflation is able to solve the horizon problem of the standard Big Bang cosmology [47].

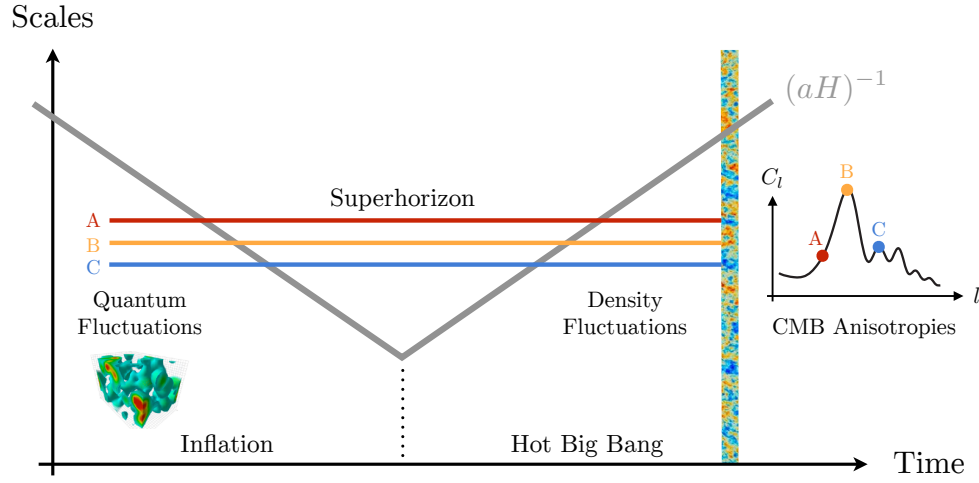


Figure 22: Evolution of fluctuations in inflationary cosmology. Perturbations are generated quantum mechanically on small scales and are then stretched by the cosmological expansion to become large-scale density fluctuations (like those seen in the CMB).

The evolution of the comoving Hubble radius can also be written as

$$\frac{d}{dt}(aH)^{-1} = -\frac{\dot{a}H + a\dot{H}}{(aH)^2} = -\frac{1}{a}(1 - \varepsilon), \quad (8.7)$$

where we have introduced the *slow-roll parameter*

$$\varepsilon \equiv -\frac{\dot{H}}{H^2}. \quad (8.8)$$

Inflation (i.e. a shrinking Hubble radius) is therefore also associated with $\varepsilon < 1$. Moreover, as we will see below, the near scale-invariance of the observed fluctuations in fact requires that $\varepsilon \ll 1$. In the limit $\varepsilon \rightarrow 0$, the dynamics become time-translation invariant, $H = \text{const}$, and the spacetime is de Sitter space:

$$ds^2 = -dt^2 + e^{2Ht}d\mathbf{x}^2. \quad (8.9)$$

Of course, inflation has to end, so the time-translation symmetry must be broken and the spacetime must deviate from a perfect de Sitter space. However, for small, but finite $\varepsilon \neq 0$, the line element (8.9) is still a good approximation to the inflationary background.

Exercise 8.2 Show that for single-field slow-roll inflation the parameter ε in (8.8) becomes

$$\varepsilon = \frac{\frac{3}{2}\dot{\phi}^2}{\frac{1}{2}\dot{\phi}^2 + V}. \quad (8.10)$$

When does inflation occur?



Exercise 8.3 In order for the slow-roll behavior to persist, the acceleration of the scalar field also has to be small. We quantify this by the parameter

$$\delta \equiv -\frac{\ddot{\phi}}{H\dot{\phi}}. \quad (8.11)$$

Use the slow-roll conditions, $\varepsilon \ll 1$ and $|\delta| \ll 1$, to simplify the equations of motion and show that

$$\varepsilon \approx \frac{M_{\text{Pl}}^2}{2} \left(\frac{V_{,\phi}}{V} \right)^2 \ll 1, \quad (8.12)$$

$$\delta + \varepsilon \approx M_{\text{Pl}}^2 \frac{V_{,\phi\phi}}{V} \ll 1, \quad (8.13)$$

which puts constraints on the inflationary potentials. ➔

8.2 Free Scalar in De Sitter

Before attacking the real problem of interest, namely the quantization of coupled inflaton-metric fluctuations during inflation, we will consider the simpler case of a free scalar field in de Sitter space. We will assume that the scalar field carries an insignificant amount of the total energy density and, hence, doesn't backreact on the de Sitter geometry. Such a field is sometimes called a *spectator field*.

Classical dynamics

In Section 2.5, we already discussed the evolution of a massive scalar field in a cosmological spacetime. Defining $\phi(\eta, \mathbf{x}) = u(\eta, \mathbf{x})/a(\eta)$, we found the following equation of motion

$$u'' - \nabla^2 u + \left(m^2 a^2 - \frac{a''}{a} \right) u = 0. \quad (8.14)$$

We will first study the case of a massless field, $m = 0$, while the case of a massive field is treated in Exercise 8.6. Using $a(\eta) = -(H\eta)^{-1}$, each Fourier mode then satisfies

$$u''_{\mathbf{k}} + \left(k^2 - \frac{2}{\eta^2} \right) u_{\mathbf{k}} = 0. \quad (8.15)$$

This is the equation of a harmonic oscillator, with a frequency given by the wavenumber k and a time-dependent (tachyonic) mass term.

Before solving (8.15), it is useful to determine the limiting behavior of the solutions:

- At *early times*, most modes of interest had wavelengths much smaller than the comoving Hubble scale, $k^{-1} \ll (aH)^{-1} = -\eta$. In this regime, the mass term can be dropped and equation (8.15) simplifies to

$$u''_{\mathbf{k}} + k^2 u_{\mathbf{k}} = 0, \quad (8.16)$$

which is exactly the same as the equation satisfied by a massless scalar field in Minkowski space.¹¹ The solutions to this equation are the standard solutions of the simple harmonic oscillator

$$u_{\mathbf{k}} \xrightarrow{k\eta \rightarrow -\infty} \frac{1}{\sqrt{2k}} e^{\pm i k \eta}, \quad (8.17)$$

where we have chosen the overall normalization, so that the solutions have Wronskian $\pm i$.

- At *late times*, the tachyonic mass term becomes relevant, and the fluctuations are amplified. In this limit, equation (8.15) becomes

$$u_{\mathbf{k}}'' - \frac{2}{\eta^2} u_{\mathbf{k}} = 0, \quad (8.18)$$

which is solved by

$$u_{\mathbf{k}} = A_{\mathbf{k}} \eta^{-1} + B_{\mathbf{k}} \eta^2 \xrightarrow{k\eta \rightarrow 0} A_{\mathbf{k}} \eta^{-1}. \quad (8.19)$$

The growing mode solution implies $\phi_{\mathbf{k}} = a^{-1} f_{\mathbf{k}} \propto \eta \times \eta^{-1} = \text{const}$, so that the original field fluctuations are *frozen*.

Of course, we can also solve (8.15) directly to find

$$u_{\mathbf{k}}(\eta) = C_{\mathbf{k}} \frac{1}{\sqrt{2k}} \left(1 - \frac{i}{k\eta}\right) e^{-ik\eta} + D_{\mathbf{k}} \frac{1}{\sqrt{2k}} \left(1 + \frac{i}{k\eta}\right) e^{ik\eta}, \quad (8.20)$$

where the free coefficients $C_{\mathbf{k}}, D_{\mathbf{k}}$ will be fixed by boundary conditions. From these solutions, we can explicitly see that there is a crossover in behavior when the wavelength of a mode becomes larger than the comoving Hubble radius, where the mode goes from being oscillatory to growing like a power law, reproducing the behaviors we saw above in the relevant regimes. Now that we have a handle on the classical dynamics, we turn to the quantization of this field.

Canonical quantization

We have seen that, at early times, the field fluctuations $u_{\mathbf{k}}$ satisfy the equation of a simple harmonic oscillator. Quantum zero-point fluctuations of this collection of oscillators produce a nonzero variance of the field. The expansion of the universe amplifies these fluctuations and stretches them to large scales. In the following, we will compute the spectrum of these quantum fluctuations.

We quantize the theory by promoting the classical fields u and $\pi \equiv u'$ to quantum operators $\hat{u}$ and $\hat{\pi}$, and impose the canonical commutation relation

$$[\hat{u}(\eta, \mathbf{x}), \hat{\pi}(\eta, \mathbf{x}')] = i\delta(\mathbf{x} - \mathbf{x}'), \quad (8.21)$$

in units where $\hbar \equiv 1$. We then decompose the field operator in terms of creation and annihilation operators

$$\hat{u}(\eta, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \left(\hat{a}_{\mathbf{k}} u_{\mathbf{k}}(\eta) e^{i\mathbf{k} \cdot \mathbf{x}} + \hat{a}_{\mathbf{k}}^\dagger u_{\mathbf{k}}^*(\eta) e^{-i\mathbf{k} \cdot \mathbf{x}} \right), \quad (8.22)$$

¹¹Physically, this is a manifestation of the equivalence principle, which states that at scales much smaller than the radius of curvature of de Sitter space, the effects of the geometry are invisible.

where the mode function $u_k(\eta)$ is a solution to the classical equation of motion (8.15), and $u_k^*(\eta)$ is its complex conjugate.¹² The commutator of the ladder operators is

$$[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{k}'}^\dagger] = (2\pi)^3 \delta(\mathbf{k} - \mathbf{k}'). \quad (8.23)$$

The field operator and its conjugate momentum operator then satisfy the commutation relation (8.21), as long as the mode functions obey the *Wronskian condition*:

$$u_k(\eta) \partial_\eta u_k^*(\eta) - u_k^*(\eta) \partial_\eta u_k(\eta) = i. \quad (8.24)$$

Up to an irrelevant phase, this fixes the normalization of the solution $u_k(\eta)$.

Exercise 8.4 Consider the alternative mode function

$$v_k(\eta) = \alpha_k u_k(\eta) + \beta_k u_k^*(\eta), \quad (8.25)$$

where α_k and β_k are time-independent complex coefficients. Show that $v_k(\eta)$ also obeys the Wronskian condition (8.24) as long as $|\alpha_k|^2 - |\beta_k|^2 = 1$. →

Bunch–Davies vacuum

At this point, we have not yet picked a unique solution for the mode function $u_k(\eta)$. In fact, there is a one-parameter family of solutions that satisfy the Wronskian condition (8.24), and our selection of $u_k(\eta)$ is related to choosing the vacuum state of the theory. As in the ordinary harmonic oscillator, the vacuum state is defined to be the state that satisfies

$$\hat{a}_{\mathbf{k}}|0\rangle = 0. \quad (8.26)$$

Effectively, different choices of u_k in (8.22) correspond to different choices of vacuum state. In flat space, there is a natural choice, which is to choose the vacuum to be the (unique) state of lowest energy (see Exercise 1.1).

The most natural choice of vacuum in a cosmological spacetime is to require that modes at large k (or, equivalently, in the infinite past) reduce to their Minkowski space counterparts:

$$\lim_{k\eta \rightarrow -\infty} u_k(\eta) = \frac{1}{\sqrt{2k}} e^{-ik\eta}, \quad (8.27)$$

which is known as the *Bunch–Davies* vacuum. This additional boundary condition fixes the solution (8.20) to be

$$u_k(\eta) = \frac{1}{\sqrt{2k}} \left(1 - \frac{i}{k\eta} \right) e^{-ik\eta}. \quad (8.28)$$

This specifies the ground state of the system and fixes the mode expansion (8.22), so that we can compute its variance.

¹²Because the evolution is isotropic, the mode functions depend only on the magnitude of the wavevector, $k \equiv |\mathbf{k}|$, and not its direction. The creation and annihilation operators, on the other hand, do depend on direction and only their statistical properties have to be isotropic.

Zero-point fluctuations

Finally, we can predict the quantum statistics of the operator

$$\hat{u}(\eta, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \left[u_k(\eta) \hat{a}_{\mathbf{k}} + u_k^*(\eta) \hat{a}_{-\mathbf{k}}^\dagger \right] e^{i\mathbf{k} \cdot \mathbf{x}}. \quad (8.29)$$

The expectation value of $\hat{u}$ vanishes, i.e. $\langle \hat{u} \rangle \equiv \langle 0 | \hat{u} | 0 \rangle = 0$. However, the variance of inflaton fluctuations receives non-zero quantum fluctuations:

$$\begin{aligned} \langle |\hat{u}|^2 \rangle &\equiv \langle 0 | \hat{u}(\eta, \mathbf{0}) \hat{u}(\eta, \mathbf{0}) | 0 \rangle \\ &= \int \frac{d^3k}{(2\pi)^3} \int \frac{d^3k'}{(2\pi)^3} \langle 0 | \left(u_k^*(\eta) \hat{a}_{-\mathbf{k}}^\dagger + u_k(\eta) \hat{a}_{\mathbf{k}} \right) \left(u_{k'}(\eta) \hat{a}_{\mathbf{k}'} + u_{k'}^*(\eta) \hat{a}_{-\mathbf{k}'}^\dagger \right) | 0 \rangle \\ &= \int \frac{d^3k}{(2\pi)^3} \int \frac{d^3k'}{(2\pi)^3} u_k(\eta) u_{k'}^*(\eta) \langle 0 | [\hat{a}_{\mathbf{k}}, \hat{a}_{-\mathbf{k}'}^\dagger] | 0 \rangle \\ &= \int \frac{d^3k}{(2\pi)^3} |u_k(\eta)|^2 \\ &= \int d \log k \frac{k^3}{2\pi^2} |u_k(\eta)|^2. \end{aligned} \quad (8.30)$$

We define the (dimensionless) *power spectrum* as

$$\Delta_u^2(k, \eta) \equiv \frac{k^3}{2\pi^2} |u_k(\eta)|^2. \quad (8.31)$$

We see that the square of the classical solution determines the variance of quantum fluctuations. Substituting the Bunch–Davies mode function (8.28) into (8.31), we find

$$\Delta_\phi^2(k, \eta) = \frac{\Delta_u^2(k, \eta)}{a^2(\eta)} = \left(\frac{H}{2\pi} \right)^2 \left[1 + (k\eta)^2 \right] \xrightarrow{k\eta \rightarrow 0} \left(\frac{H}{2\pi} \right)^2. \quad (8.32)$$

Note that in the superhorizon limit, $k\eta \rightarrow 0$, the dimensionless power spectrum Δ_ϕ^2 approaches the same constant for all momenta. This is the characteristic of a *scale-invariant* spectrum.

Exercise 8.5 Show that the scale invariance of the power spectrum for a massless field follows from the dilatation symmetry of the de Sitter geometry. Use the same symmetry argument, to derive the k -dependence of the spectrum of a scalar with a small mass, $m^2 \ll H^2$. ➔

Exercise 8.6 Repeat the canonical quantization procedure for a massive scalar field in de Sitter space, starting from the action

$$S = \int d^4x \sqrt{-g} \left(-\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - \frac{m^2}{2} \phi^2 \right). \quad (8.33)$$

Determine the Bunch–Davies mode function for the field, $\phi_k(\eta)$, and then derive the late-time power spectrum. What is the difference of the results for small masses and large masses? ➔

8.3 Curvature Perturbations

So far, we have only considered the quantization of a spectator field in a fixed de Sitter geometry. We now move to studying the fluctuations in the inflaton field during inflation. These fluctuations cannot be treated independently from fluctuations in the metric, since the two are coupled by the Einstein equations. This leads to some technical complications, but conceptually the quantization of the coupled inflaton-metric fluctuations is the same as before.

Metric fluctuations

We will treat the fluctuations of the metric in the so-called *ADM formalism* [53] (developed by Arnowitt, Deser and Misner). We start by writing the perturbed line element as

$$ds^2 = -N^2 dt^2 + h_{ij}(N^i dt + dx^i)(N^j dt + dx^j), \quad (8.34)$$

where $N \equiv N(t, \mathbf{x})$ is the lapse function, $N^i \equiv N^i(t, \mathbf{x})$ is the shift vector, and $h_{ij} \equiv h_{ij}(t, \mathbf{x})$ is the induced metric on three-dimensional hypersurfaces of constant time t . The metric and its inverse are

$$g_{\mu\nu} = \left(\frac{-N^2 + N_i N^i}{N_i} \middle| \frac{N_j}{h_{ij}} \right), \quad g^{\mu\nu} = \frac{1}{N^2} \left(\frac{-1}{N^i} \middle| \frac{N^j}{N^2 h^{ij} - N^i N^j} \right). \quad (8.35)$$

Note that $g^{ij} \neq h^{ij}$ and $\sqrt{-g} = N\sqrt{h}$. The geometry of the spatial slices is characterized by the intrinsic curvature, $R_{ij}^{(3)}$, i.e. the Ricci tensor of the induced metric h_{ij} , and by the extrinsic curvature (see Chapter 6)

$$K_{ij} \equiv \frac{1}{2N} \left(\dot{h}_{ij} - \nabla_i N_j - \nabla_j N_i \right) \equiv \frac{1}{N} E_{ij}, \quad (8.36)$$

where ∇_i is the three-dimensional covariant derivative, defined from h_{ij} . The four-dimensional Ricci scalar, R , can be written in terms of the three-dimensional Ricci scalar, $R^{(3)}$, and the extrinsic curvature tensor as

$$R = R^{(3)} + N^{-2} (E^{ij} E_{ij} - E^2) + \text{total derivative}, \quad (8.37)$$

where indices are raised with h^{ij} , and $E \equiv h^{ij} E_{ij}$.

The inflaton action can then be written as

$$\begin{aligned} S &= \frac{1}{2} \int d^4x \sqrt{-g} \left[R - g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - 2V \right] \\ &= \frac{1}{2} \int d^4x \sqrt{h} N \left[R^{(3)} - 2V + N^{-2} (E^{ij} E_{ij} - E^2) + N^{-2} (\dot{\phi} - N^i \partial_i \phi)^2 - h^{ij} \partial_i \phi \partial_j \phi \right], \end{aligned} \quad (8.38)$$

where for the moment we have set $M_{\text{Pl}} \equiv 1$. Note that N and N_i do not have time derivatives acting on them and are therefore non-dynamical fields that will be fixed by constraint equations. This is the beauty of the ADM formalism. Indeed, varying the action with respect to N and N^i , we find

$$R^{(3)} - 2V - h^{ij} \partial_i \phi \partial_j \phi + N^{-2} [E_{ij} E^{ij} - E^2 + (\dot{\phi} - N^i \partial_i \phi)^2] = 0, \quad (8.39)$$

$$\nabla_i [N^{-1} (E_j^i - E \delta_j^i)] = 0. \quad (8.40)$$

The basic idea is to solve these equations for N and N^i , and plug the solutions back into the action, which leaves ϕ and h_{ij} as the only dynamical variables. We will perform this procedure in perturbation theory. For the time being, we will focus on scalar perturbations. To fix time and space reparameterizations, we will have to choose a gauge for ϕ and h_{ij} .

Comoving gauge

We will work in the comoving gauge, where the inflaton field is taken to be unperturbed: $\phi = \bar{\phi}(t)$. All perturbations are then carried by the metric:

$$h_{ij} = a^2 e^{2\zeta(t, \mathbf{x})} \delta_{ij}, \quad N \equiv 1 + \alpha(t, \mathbf{x}), \quad N_i \equiv \partial_i \beta(t, \mathbf{x}), \quad (8.41)$$

where ζ is the comoving curvature perturbation. We first compute the perturbed intrinsic and extrinsic curvature in this gauge.

Exercise 8.7 Show that

$$R^{(3)} = -2a^{-2}e^{-2\zeta} [2\partial^2\zeta + (\partial\zeta)^2], \quad (8.42)$$

$$E_{ij} = a^2 e^{2\zeta} (H + \dot{\zeta}) \delta_{ij} - \partial_{(i} N_{j)} + 2N_{(i} \partial_{j)} \zeta - N_k \partial_k \zeta \delta_{ij}. \quad (8.43)$$

Hint: Resist the temptation to expand $e^{2\zeta} = (1 + 2\zeta + 2\zeta^2 + \dots)$. →

Exercise 8.8 Substituting the ansatz (8.41) into the constraint equations (8.39) and (8.40), show that

$$\alpha = \frac{\dot{\zeta}}{H}, \quad \partial^2 \beta = -\frac{\partial^2 \zeta}{H} + a^2 \frac{\dot{\phi}^2}{2H^2} \dot{\zeta}, \quad (8.44)$$

to linear order in the fluctuations. →

Plugging these results into the action (8.38), expanding to second order, performing integrations by parts and using the background equations of motion, we get a remarkably simple result

$$S_2 = \int dt d^3x a^3 \varepsilon \left(\dot{\zeta}^2 - \frac{1}{a^2} (\partial_i \zeta)^2 \right). \quad (8.45)$$

We note two important features of this result: First, the Lagrangian density is proportional to the slow-roll parameter ε and therefore vanishes in the de Sitter limit $\varepsilon \rightarrow 0$. This is consistent with the fact that the curvature perturbation ζ becomes a pure gauge mode in that limit. Second, the action doesn't have a mass term proportional to ζ^2 . An important consequence of this is that ζ becomes a constant on super-Hubble scales. This is a special case of the theorem that we cited in Section 7.2 that ζ is conserved on super-Hubble scales (for adiabatic perturbations).

Exercise 8.9 Derive the result (8.45). →

Switching to conformal time, the action (8.45) becomes

$$S_2 = \frac{1}{2} \int d\eta d^3x z^2 \left[(\zeta')^2 - (\partial_i \zeta)^2 \right], \quad (8.46)$$

where $z \equiv a\sqrt{2\varepsilon}$ (in units where $M_{\text{Pl}} \equiv 1$). Varying the action with respect to ζ gives the following equation of motion

$$\zeta'' - 2\frac{z'}{z}\zeta' - \nabla^2\zeta = 0, \quad (8.47)$$

where

$$\frac{z'}{z} = \mathcal{H} \left[1 + \frac{1}{2}\kappa \right], \quad \text{with} \quad \kappa \equiv \frac{\varepsilon'}{\varepsilon\mathcal{H}} \quad \text{and} \quad \mathcal{H} \equiv aH. \quad (8.48)$$

Before solving (8.47) explicitly, we will determine the limiting behavior of the solutions:

- At *early times*, the friction term is subdominant and a Fourier mode of the curvature perturbation satisfies

$$\zeta'' + k^2\zeta = 0. \quad (8.49)$$

The solution in this regime is $e^{\pm ik\eta}$.

- At *late times*, the friction term dominates and the equation becomes

$$\zeta'' = 2\frac{z'}{z}\zeta'. \quad (8.50)$$

The growing mode solution in this regime is $\zeta = \text{const}$, i.e. the curvature perturbation freezes.

To derive the general solution to (8.47) it is useful to define the variable $u \equiv z\zeta$. We then have

$$\zeta' = \frac{u'}{z} - \frac{uz'}{z^2}, \quad (8.51)$$

$$\zeta'' = \frac{u''}{z^2} - 2\frac{u'z'}{z^2} - \frac{u}{z} \left[\frac{z''}{z} - 2\left(\frac{z'}{z}\right)^2 \right]. \quad (8.52)$$

Substituting this into (8.47), we get

$$u'' - \nabla^2 u - \frac{z''}{z}u = 0, \quad (8.53)$$

and each Fourier mode therefore satisfies

$$u_k'' + \left(k^2 - \frac{z''}{z} \right) u_k = 0. \quad (8.54)$$

This is the *Mukhanov–Sasaki equation*.

Exercise 8.10 Show that

$$\frac{z''}{z} = \frac{1}{\eta^2} \left[2 + 3 \left(\varepsilon + \frac{1}{2}\kappa \right) \right], \quad (8.55)$$

where ε and κ are the two Hubble slow-variation parameters. ➔

Using the result of the previous exercise, equation (8.54) can be written as

$$u_k'' + \left(k^2 - \frac{\nu^2 - 1/4}{\eta^2} \right) u_k = 0, \quad \text{where} \quad \nu \equiv \frac{3}{2} + \varepsilon + \frac{1}{2}\kappa. \quad (8.56)$$

This equation was solved in Problem 8.6. It was found that the Bunch–Davies solution is

$$u_k(\eta) = \frac{\sqrt{\pi}}{2} \sqrt{-\eta} H_\nu^{(1)}(-k\eta), \quad (8.57)$$

where $H_\nu^{(1)}$ is a Hankel function of the first kind. The power spectrum of u is the square of this mode function.

Exercise 8.11 To compute the power spectrum of $\zeta = u/z$, we need to know the function $z(\eta)$. Show that, for slow-roll inflation, we have

$$z(\eta) = z_*(\eta/\eta_*)^{\frac{1}{2}-\nu}, \quad (8.58)$$

where η_* is some reference time. It will be convenient to take $\eta_* = -k_*^{-1}$, i.e. the moment of horizon crossing of a mode with wavenumber k_* . ➔

Using the result of the previous exercise, the dimensionless power spectrum of ζ is

$$\begin{aligned} \Delta_\zeta^2(k) &= \frac{k^3}{2\pi^2} \frac{|u_k(\eta)|^2}{z^2(\eta)} \\ &= \frac{k^3}{2\pi^2} \frac{1}{2\varepsilon_* a_*^2} (-k_*\eta)^{2\nu-1} \frac{\pi}{4} (-\eta) |H_\nu^{(1)}(-k\eta)|^2. \end{aligned} \quad (8.59)$$

Using $a_* = k_*/H_*$ and the late-time limit of the Hankel function,

$$\lim_{k\eta \rightarrow 0} |H_\nu^{(1)}(-k\eta)|^2 = \frac{2^{2\nu} \Gamma(\nu)^2}{\pi^2} (-k\eta)^{-2\nu} \approx \frac{2}{\pi} (-k\eta)^{-2\nu}, \quad (8.60)$$

we get

$$\Delta_\zeta^2(k) = \frac{1}{8\pi^2 \varepsilon_*} \frac{H_*^2}{M_{\text{Pl}}^2} (k/k_*)^{3-2\nu}, \quad (8.61)$$

where we have restored the scale M_{Pl} by dimensional analysis.

We see that the result has a slight k -dependence which characterizes a breaking of scale invariance. It is conventional to parameterize this in terms of the *spectral index*

$$n_s - 1 \equiv \frac{d \log \Delta_\zeta^2}{d \log k} = -2\varepsilon_* - \kappa_*, \quad (8.62)$$

where the quantities on the right-hand side are evaluated at the time when the reference scale k_* exited the horizon. The observed spectral index is $n_s = 0.965 \pm 0.004$ (for $k_* = 0.05 \text{ Mpc}^{-1}$) [48]. This percent-level deviation from the scale-invariant value, $n_s = 1$, is the first direct measurement of the time dependence during inflation.

8.4 Gravitational Waves

A key prediction of all models of inflation is a stochastic background of gravitational waves (GWs), or tensor metric perturbations:

$$ds^2 = a^2(\eta) [-d\eta^2 + (\delta_{ij} + h_{ij})dx^i dx^j]. \quad (8.63)$$

We will now derive the tensor power spectrum predicted by inflation, which we define in such a way that $\langle h_{ij}^2 \rangle = \int d \log k \Delta_h^2(k)$. Other definitions exist in the literature which differ from ours by factors of 2.

We first need the quadratic action for the tensor fluctuations h_{ij} , which we obtain from the Einstein–Hilbert action for the metric. Substituting (8.63) into this action and expanding to second order gives

$$\begin{aligned} S &= \frac{M_{\text{Pl}}^2}{2} \int d^4x \sqrt{-g} R \\ &= \frac{M_{\text{Pl}}^2}{8} \int d\eta d^3x a^2 [(h'_{ij})^2 - (\nabla h_{ij})^2] + \dots, \end{aligned} \quad (8.64)$$

where $M_{\text{Pl}} \equiv 1/\sqrt{8\pi G}$. Instead of computing this by brute force, we could first carry out the simpler calculation on a Minkowski background and then apply a conformal transformation to move into an inflationary background. This exploits the fact that the tensor perturbation h_{ij} is invariant under a conformal transformation of the form $g_{\mu\nu} \mapsto \Omega^2(x)g_{\mu\nu}$. Finally, we could also have guessed the final form of the action which is the same as the action for a massless scalar field. The only thing we gained from the explicit computation was the overall normalization $M_{\text{Pl}}^2/8$, which doesn't affect the equation of motion, but fixes the normalization of the conjugate momentum and is therefore essential in the quantization procedure.

Varying the action (8.64) with respect to h_{ij} , we find that tensor fluctuations in an FRW background satisfy the following wave equation:

$$h''_{ij} + 2\mathcal{H}h'_{ij} - \nabla^2 h_{ij} = 0. \quad (8.65)$$

This equation could also have been derived more simply from the linearized vacuum Einstein equation, $G_{\mu\nu} = 0$. The benefit of deriving it from an action is that we automatically obtain the correct normalization of the action.

It is convenient to use rotational symmetry to align the z -axis of the coordinate system with the momentum of the mode, i.e. $\mathbf{k} \equiv (0, 0, k)$, and write

$$\frac{M_{\text{Pl}}}{\sqrt{2}} a h_{ij} \equiv \begin{pmatrix} u_+ & u_\times & 0 \\ u_\times & -u_+ & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (8.66)$$

where u_+ and $u_\times$ describe the two polarization modes of the gravitational wave. Note that $h_{ij}^2 = (4/M_{\text{Pl}}^2)(u_+^2 + u_\times^2)/a^2$. The action (8.64) then becomes

$$S = \frac{1}{2} \sum_{\lambda=+,\times} \int d\eta d^3x \left[(u'_\lambda)^2 - (\nabla u_\lambda)^2 + \frac{a''}{a} u_\lambda^2 \right], \quad (8.67)$$

which is just two copies of the action of a massless scalar field. The equation of motion for each polarization mode is

$$u''_k + \left(k^2 - \frac{a''}{a} \right) u_k = 0, \quad (8.68)$$

where the effective mass can be written as

$$\frac{a''}{a} = \frac{\nu^2 - 1/4}{\eta^2}, \quad \text{with} \quad \nu \approx \frac{3}{2} + \varepsilon. \quad (8.69)$$

The Bunch–Davies mode function is again given by (8.57). The superhorizon limit of the power spectrum of tensor fluctuations then is

$$\Delta_h^2(k) = 2 \times \left(\frac{2}{aM_{\text{Pl}}} \right)^2 \lim_{k\eta \rightarrow 0} \frac{k^3}{2\pi^2} |u_k(\eta)|^2 = \frac{2}{\pi^2} \frac{H_*^2}{M_{\text{Pl}}^2} (k/k_*)^{3-2\nu}, \quad (8.70)$$

where H_* is the Hubble parameter evaluated at the time when the reference scale k_* exited the horizon. Observational constraints on the tensor amplitude are usually expressed in terms of the *tensor-to-scalar ratio*:

$$r \equiv \frac{\Delta_h^2(k_*)}{\Delta_\zeta^2(k_*)} = 16\varepsilon_*. \quad (8.71)$$

Since the amplitude of scalar fluctuations has been measured, $\Delta_\zeta^2(k_*) = (2.098 \pm 0.023) \times 10^{-9}$, the tensor-to-scalar ratio quantifies the size of the tensor fluctuations. The current upper bound on the tensor-to-scalar ratio is $r < 0.03$. The search for inflationary gravitational waves remains an important target of observational cosmology.

8.5 Non-Gaussianity

We have seen that inflation predicts that the initial correlations are highly Gaussian and therefore described well by the power spectrum alone. Nevertheless, in principle, a significant amount of information about the physics of inflation can be encoded in small levels of non-Gaussianity. In particular, the higher-order correlations associated with non-Gaussian initial conditions are sensitive to nonlinear interactions, while the power spectrum only probes the free theory. Measurements of primordial non-Gaussianity would therefore probe the particle content and the interactions during inflation.¹³

The leading signature of non-Gaussianity is a nonzero three-point correlation function, or its Fourier equivalent, the *bispectrum*:

$$\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle = (2\pi)^3 \delta(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3) \frac{(2\pi^2)^2}{(k_1 k_2 k_3)^2} B_\zeta(k_1, k_2, k_3), \quad (8.72)$$

where the delta function is again a consequence of the homogeneity of the background. The sum of the three wavevectors must therefore form a closed triangle and the strength of the signal will depend on the shape of the triangle. As indicated by the notation, the bispectrum is a function of the magnitudes of the wavevectors, $k_{n=1,2,3}$, as required by the isotropy of the background. The factor of $(k_1 k_2 k_3)^{-2}$ has been extracted in to make the function $B_\zeta(k_1, k_2, k_3)$ dimensionless. The

¹³An analogy with particle physics may be informative: The measurements of the power spectrum are the analog of measuring the propagators of particles. While these propagators are important features of long-lived particles, their form is completely fixed by Lorentz symmetry and therefore doesn't contain dynamical information. To learn about the dynamics of the theory, we collide particles and study the resulting interactions. Measurements of higher-order correlations (or non-Gaussianity) are the analog of measuring collisions in particle physics. The study of non-Gaussianity therefore also goes by the name of “cosmological collider physics” [43].

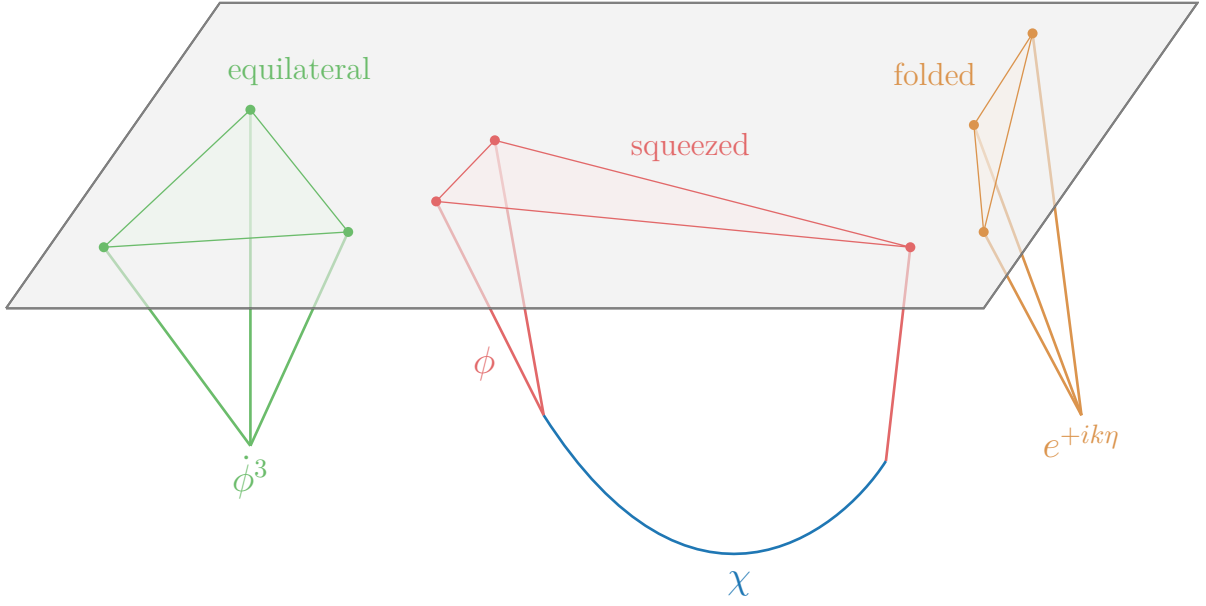


Figure 23: Illustration of the different types of non-Gaussianity described in the text: 1) Local interactions in the bulk produce the “equilateral” shape; 2) Excited initial states create an enhanced signal in the “folded” configuration; and 3) The production and decay of massive particles leave an imprint in the “squeezed” limit.

amplitude of the non-Gaussianity is then defined as the size of the bispectrum in the equilateral configuration:

$$f_{\text{NL}}(k) \equiv \frac{5}{18} \frac{B_{\zeta}(k, k, k)}{\Delta_{\zeta}^4(k)}, \quad (8.73)$$

where the numerical factor is a historical accident. In general, the parameter f_{NL} can depend on the overall wavenumber (or the size of the triangle), but for scale-invariant initial conditions it would be a constant. The bispectrum can then be written as

$$B_{\zeta}(k_1, k_2, k_3) = \frac{18}{5} f_{\text{NL}} \times S(x_2, x_3) \times \Delta_{\zeta}^4, \quad (8.74)$$

where $x_2 \equiv k_2/k_1$ and $x_3 \equiv k_3/k_1$. The shape function $S(x_2, x_3)$ is normalized so that $S(1, 1) \equiv 1$. As we will see below, the *shape* of the non-Gaussianity contains a lot of information about the microphysics of inflation (see Fig. 23). This is to be contrasted with the power spectrum, which is described by just two numbers, A_s and n_s , and not a whole function.

Equilateral In slow-roll inflation, the flatness of the inflationary potential constrains the size of the inflaton self-interactions. However, interesting models of inflation have been suggested in which higher-derivative corrections—such as $(\partial\phi)^4$ —play an important role during inflation. These interactions lead to cubic interactions of the inflaton perturbations—like $\dot{\phi}^3$ and $\dot{\phi}(\partial_i\phi)^2$ —and hence a nonzero bispectrum.¹⁴ Since the inflaton fluctuations interact locally at points in the

¹⁴A systematic way to classify these derivative interactions is in terms of an EFT for the inflationary fluctuations [42, 54].

bulk spacetime, this produces a bispectrum with an enhanced signal for “equilateral” triangles, with $k_1 \approx k_2 \approx k_3$.

Folded The Gaussianity of slow-roll inflation also relies on the fact that we evaluated the quantum fluctuations in the Bunch–Davies vacuum (corresponding to the ground state of the harmonic oscillator). In contrast, starting from an excited initial state would lead to non-Gaussianity. The detailed shape of this non-Gaussianity depends on the model for the excited initial state. A universal feature is that the correlations are enhanced for “folded” triangles where two of the wavevectors become colinear, so that $k_1 + k_2 \approx k_3$. The signal in the folded configuration also provides an interesting test of the quantum origin of the fluctuations [55]. While classical fluctuations would generically have non-vanishing correlations in the folded limit, quantum fluctuations in the Bunch–Davies vacuum are characterized by the absence of such a signal.

Squeezed Finally, non-Gaussianity is a sensitive detector of new particles during inflation. These particles are created by the rapid expansion of the spacetime and, through their decays, produce distinct correlations in the primordial fluctuations. The massive particles can decay at widely separated points in the bulk spacetime, leading to an imprint in the so-called “squeezed” limit, $k_3 \ll k_1 \approx k_2$. This “cosmological collider” phenomenon will be described in Sections 9.8, 10.9 and 11.5.

In the following chapters, we will describe three methods for computing cosmological correlators: 1) the in-in formalism, 2) the wavefunction approach, and 3) the cosmological bootstrap.

9 In-In Formalism

In particle physics, the main observables are scattering amplitudes which predict the transition probabilities between some initial (“in”) and final (“out”) states (defined in terms free particles in the far past and future). In cosmology, however, there are no well-defined asymptotic regions where particles behave freely, which prevents a clean definition of scattering amplitudes. Instead, the primary observables are correlation functions, defined as expectation values at a fixed time in a quantum state evolved from an initial “in” state.

In this chapter, we introduce the “in-in” formalism to compute cosmological correlators (also known as the Schwinger–Keldysh or closed-time-path formalism) [56–60], and apply it to a few illustrative examples. The use of the in-in formalism in cosmology was reinvigorated by Maldacena, who used it to compute the three-point function in slow-roll inflation [41], and then further systematized by Weinberg [61]. Nice reviews of the in-in formalism with cosmological applications in mind can be found in [62, 63].¹⁵

9.1 In-In Versus In-Out

In flat-space quantum field theory, the S -matrix is defined as a transition amplitude between in and out states:¹⁶

$$S_{\beta\alpha} = \langle \beta | \alpha \rangle, \quad (9.1)$$

which captures the probability for a state prepared in the far past with particles α to evolve into a state with particles β in the far future. To actually compute the S -matrix, we specify the initial state in the asymptotic past, $|\alpha_0\rangle \equiv |\alpha_0(t_i)\rangle$, where the theory is effectively free and evolve this state using the interaction Hamiltonian H_{int} :

$$|\alpha_0(t_f)\rangle = T \exp \left(-i \int_{t_i}^{t_f} dt \hat{H}_{\text{int}} \right) |\alpha_0(t_i)\rangle, \quad (9.2)$$

where T denotes time-ordering. We then measure the overlap of the evolved initial state $|\alpha_0(t_f)\rangle$ with some other multi-particle final state, $|\beta_0\rangle \equiv |\beta_0(t_f)\rangle$, specified again in the free theory. The S -matrix for this process then is

$$S_{\beta\alpha} = \langle \beta_0 | T e^{-i \int_{-\infty}^{\infty} dt \hat{H}_{\text{int}}} | \alpha_0 \rangle. \quad (9.3)$$

Importantly, to specify the S -matrix, we choose boundary conditions for our measurement *both* in the past and in the future.

In cosmology, the situation is different. Roughly speaking, none of the conditions we require to define the S -matrix hold in cosmological backgrounds. First of all, it is not possible to localize wave packets and separate them asymptotically in order to map interacting states to free states (at least not for light fields). Relatedly, the gravitational production of particles cannot be shut off in asymptotic regions, so in and out states with definite particle numbers are not well-defined. Moreover, in de Sitter space a single observer cannot access an entire Cauchy slice of

¹⁵Aside from cosmology, the in-in formalism is useful any time one is interested in a question where only the initial state can be prepared, and is frequently used in the study of thermal or out-of-equilibrium systems.

¹⁶See Chapter 3 of [64] for an authoritative discussion.

the spacetime, so they cannot even in principle set up a scattering experiment. Finally, as far as we know, the Universe did not start in some state specified by an experimentalist, so even if they could be defined, transition amplitudes are of questionable practical utility.

This motivates us to ask a different question. Given some initial state of the Universe, $|0\rangle$, which we evolve from an early time t_i to a late time t , what is the quantum-mechanical expectation value of an operator $\hat{\mathcal{O}}(t)$ in this (time-evolved) state? This expectation value can be written as

$$\langle \hat{\mathcal{O}}(t) \rangle = \langle 0 | \bar{T} e^{i \int_{-\infty}^t dt' \hat{H}_{\text{int}}} \hat{\mathcal{O}}(t) T e^{-i \int_{-\infty}^t dt' \hat{H}_{\text{int}}} | 0 \rangle, \quad (9.4)$$

where T again denotes time-ordering and $\bar{T}$ denotes anti time-ordering. While this formula bears some formal similarity to (9.3), the important difference is that we only specify the initial state of the system, and compute expectation values sandwiched between this in-state on both sides (hence the name “in-in” formalism).

In the following, we will first derive the formula (9.4), and then utilize it to compute expectation values in a number of situations.

9.2 Master Formula

We are interested in computing expectation values of quantum operators, $\hat{\mathcal{O}}(t)$, at some time t in a state $|\Omega\rangle$:

$$\langle \hat{\mathcal{O}}(t) \rangle \equiv \langle \Omega | \hat{\mathcal{O}}(t) | \Omega \rangle. \quad (9.5)$$

The operator $\hat{\mathcal{O}}$ can be either a single operator or a string of operators evaluated at the same time.¹⁷ Typically, we take the state $|\Omega\rangle$ to be the vacuum state of the full interacting theory. Much as in the construction of the S -matrix, we compute these expectation values by envisioning $|\Omega\rangle$ as the time-evolution of some state specified in the infinite past as a state of the corresponding free theory (typically the free vacuum).

In the Heisenberg picture, states are time-independent and operators evolve according to the full Hamiltonian H of the theory:

$$\hat{\mathcal{O}}(t) = \hat{U}^\dagger(t, t_i) \hat{\mathcal{O}}(t_i) \hat{U}(t, t_i), \quad (9.6)$$

where t_i is some initial time, and the unitary time-evolution operator $\hat{U}(t, t_i)$ satisfies

$$\frac{d\hat{U}}{dt} = -i\hat{H}\hat{U}, \quad \text{with} \quad \hat{U}(t_i, t_i) = 1. \quad (9.7)$$

The exact evolution of $\hat{\mathcal{O}}(t)$ is complicated by the fact that $\hat{H}$ contains interaction terms (when written in terms of the field perturbations), which leads to nonlinear equations of motion. We deal with this by working in the *interaction picture*, where operators evolve according to the free Hamiltonian $\hat{H}_0$ and states according to the interaction Hamiltonian $\hat{H}_{\text{int}} = \hat{H} - \hat{H}_0$. The interaction picture operator at time t is then given by

$$\hat{\mathcal{O}}_I(t) = \hat{U}_0^\dagger(t, t_i) \hat{\mathcal{O}}(t_i) \hat{U}_0(t, t_i), \quad (9.8)$$

¹⁷It is not difficult to generalize the formalism to compute expectation values involving operator insertions at different times. One just inserts time evolution operators between the various operator insertions. We will specialize to equal-time correlators because they are typically the observables of interest in cosmology.

where $\hat{U}_0(t, t_i)$ satisfies

$$\frac{d\hat{U}_0}{dt} = -i\hat{H}_0\hat{U}_0, \quad \text{with} \quad \hat{U}_0(t_i, t_i) = 1. \quad (9.9)$$

The main idea is that (9.8) encodes the leading evolution in terms of interaction picture fields and the effects of interactions will be treated perturbatively as a power series in $\hat{H}_{\text{int}}$.

We now write (9.5) in terms of interaction-picture operators as

$$\begin{aligned} \langle \Omega | \hat{\mathcal{O}}(t) | \Omega \rangle &= \langle \Omega | \hat{U}^\dagger(t, t_i) \hat{\mathcal{O}}(t_i) \hat{U}(t, t_i) | \Omega \rangle \\ &= \langle \Omega | \hat{F}^\dagger(t, t_i) \hat{U}_0^\dagger(t, t_i) \hat{\mathcal{O}}(t_i) \hat{U}_0(t, t_i) \hat{F}(t, t_i) | \Omega \rangle \\ &= \langle \Omega | \hat{F}^\dagger(t, t_i) \hat{\mathcal{O}}_I(t) \hat{F}(t, t_i) | \Omega \rangle, \end{aligned} \quad (9.10)$$

where we have defined

$$\hat{F}(t, t_i) \equiv \hat{U}_0^\dagger(t, t_i) \hat{U}(t, t_i). \quad (9.11)$$

Using (9.7) and (9.9), we find that

$$\begin{aligned} \frac{d\hat{F}}{dt} &= \frac{d}{dt}(\hat{U}_0^\dagger \hat{U}) = \frac{d\hat{U}_0^\dagger}{dt} \hat{U} + \hat{U}_0^\dagger \frac{d\hat{U}}{dt} = (i\hat{H}_0 \hat{U}_0^\dagger) \hat{U} + \hat{U}_0^\dagger (-i\hat{H} \hat{U}) \\ &= -i\hat{U}_0^\dagger (\hat{H} - \hat{H}_0) \hat{U}_0 \hat{U}_0^\dagger \hat{U} \\ &= -i(\hat{U}_0^\dagger \hat{H}_{\text{int}} \hat{U}_0) \hat{U}_0^\dagger \hat{U} \\ &= -i\hat{H}_{\text{int}}^I \hat{F}, \end{aligned} \quad (9.12)$$

where the superscript on $\hat{H}_{\text{int}}^I$ indicates that the operators inside $\hat{H}_{\text{int}}$ are evolved in the interaction picture (using the free Hamiltonian $\hat{H}_0$). We can think of $\hat{F}$ as evolving the state using the interaction Hamiltonian: $|\Omega_I(t)\rangle = \hat{F}(t, t_i)|\Omega_I(t_i)\rangle$, where $|\Omega_I(t_i)\rangle \equiv |\Omega\rangle$. Equation (9.10) then implies consistency of the expectation value in the Heisenberg and interaction pictures: $\langle \Omega | \hat{\mathcal{O}}(t) | \Omega \rangle = \langle \Omega_I(t) | \hat{\mathcal{O}}_I(t) | \Omega_I(t) \rangle$.

A formal solution of (9.12) is

$$\hat{F}(t, t_i) = T \exp \left(-i \int_{t_i}^t dt' \hat{H}_{\text{int}}^I(t') \right), \quad (9.13)$$

where T denotes the time-ordering operation. The expectation value (9.5) can then be written as

$$\langle \hat{\mathcal{O}}(t) \rangle = \langle \Omega | \bar{T} e^{i \int_{t_i}^t dt' \hat{H}_{\text{int}}^I(t')} \hat{\mathcal{O}}_I(t) T e^{-i \int_{t_i}^t dt' \hat{H}_{\text{int}}^I(t')} | \Omega \rangle, \quad (9.14)$$

where $\bar{T}$ denotes anti time-ordering.

We would like the state $|\Omega\rangle$ to be the “in” vacuum of the *interacting theory*. In the interaction picture, all states can be expressed in terms of the Fock states of the free Hamiltonian (which can be constructed by applying creation operators on the free vacuum $|0\rangle$). We would therefore like to express the interacting vacuum at early times in terms of this Fock (or *Bunch–Davies*) vacuum $|0\rangle$. We first write $|0\rangle$ in terms of a complete set of energy eigenstates of the *full* theory:

$$|0\rangle = \sum_N |N\rangle \langle N|0\rangle. \quad (9.15)$$

Near $t_i \rightarrow -\infty$, the evolution of the Fock vacuum with the full Hamiltonian $\hat{H}$ gives

$$\begin{aligned} e^{-i\hat{H}(t-t_i)}|0\rangle &= \sum_N e^{-iE_N(t-t_i)}|N\rangle\langle N|0\rangle \\ &= e^{-iE_\Omega(t-t_i)}|\Omega\rangle\langle\Omega|0\rangle + \sum_{N'\neq\Omega} e^{-iE_{N'}(t-t_i)}|N'\rangle\langle N'|0\rangle, \end{aligned} \quad (9.16)$$

where $|N'\rangle$ denotes all states excluding the vacuum. We can now employ the same trick as in standard in-out field theory: we rotate time slightly into the imaginary direction, so that

$$t \mapsto t(1 - i\epsilon) \equiv t^-, \quad (9.17)$$

where $\epsilon > 0$ is an infinitesimal constant. Near $t_i \rightarrow -\infty$, the $i\epsilon$ factor leads to factors of the form $e^{-\infty \times \epsilon E_N}$. Since E_Ω is the smallest eigenvalue of the Hamiltonian $\hat{H}$, only the state $|\Omega\rangle$ survives in this limit, and will project the interacting vacuum onto the free vacuum:

$$e^{-i\hat{H}(t^- - t_i^-)}|\Omega\rangle = \frac{e^{-iH(t^- - t_i^-)}|0\rangle}{\langle\Omega|0\rangle}. \quad (9.18)$$

Equation (9.14) can then be written as

$$\langle\Omega|\hat{\mathcal{O}}(t)|\Omega\rangle = \frac{1}{|\langle\Omega|0\rangle|^2} \langle 0 | \bar{T} e^{i \int_{-\infty+}^t dt' \hat{H}_{\text{int}}^I(t')} \hat{\mathcal{O}}_I(t) T e^{-i \int_{-\infty-}^t dt' \hat{H}_{\text{int}}^I(t')} | 0 \rangle, \quad (9.19)$$

where we have introduced the shorthand $-\infty^\pm \equiv -\infty(1 \pm i\epsilon)$. Note that the anti-time-ordering changes the sign of the $i\epsilon$ prescription. Taking $\hat{\mathcal{O}}(t)$ to be the identity operator, equation (9.19) implies

$$|\langle\Omega|0\rangle|^2 = \frac{\langle 0 | \hat{F}^\dagger \hat{F} | 0 \rangle}{\langle\Omega|\Omega\rangle} = \frac{\langle 0 | 0 \rangle}{\langle\Omega|\Omega\rangle} = 1. \quad (9.20)$$

We can therefore drop the prefactor in (9.19) and obtain our final *master formula*

$$\boxed{\langle\hat{\mathcal{O}}(t)\rangle = \langle 0 | \bar{T} e^{i \int_{-\infty+}^t dt' \hat{H}_{\text{int}}^I(t')} \hat{\mathcal{O}}(t) T e^{-i \int_{-\infty-}^t dt' \hat{H}_{\text{int}}^I(t')} | 0 \rangle}. \quad (9.21)$$

To avoid clutter, we have dropped the labels I on the right-hand side, but evolution in the interaction picture will always be implied. A graphical illustration of the in-in contour is given in Figure 24. Operators will be time-ordered on $\mathcal{C}_+$ and anti-time-ordered on $\mathcal{C}_-$.

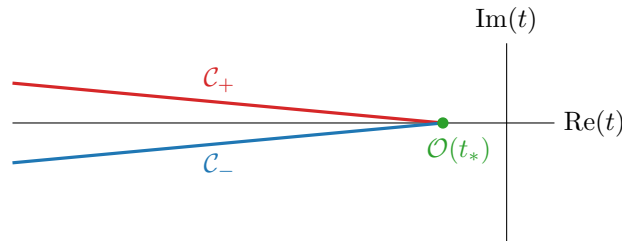


Figure 24: Graphical illustration of the in-in contour. The contour has small imaginary parts at early times to project the interacting vacuum onto the free vacuum.

9.3 Perturbative Expansion

Equation (9.21) is the in-in analogue of Dyson's formula. In practice, we use it to compute correlation functions in perturbation theory by expanding in powers of H_{int} and evaluating the resulting expressions by writing a mode expansion of the field operators in canonical quantization.

At lowest order, we find that the expectation value of $\hat{\mathcal{O}}$ is given by

$$\langle \hat{\mathcal{O}}(t) \rangle = 2 \text{Im} \left(\int_{-\infty}^t dt' \langle 0 | \hat{\mathcal{O}}(t) \hat{H}_{\text{int}}(t') | 0 \rangle \right) + \dots, \quad (9.22)$$

where we have assumed that $\langle 0 | \hat{\mathcal{O}}(t) | 0 \rangle = 0$. This expression captures the leading-order perturbative effect of interactions, which from a diagrammatic viewpoint correspond to contact interactions. To make this less abstract, we work out an explicit example in the box below.

Example To show how to use (9.22), let us consider the three-point function coming from a ϕ^3 interaction in an arbitrary FRW spacetime. We take the interaction Lagrangian (density) to be

$$\mathcal{L}_{\text{int}} = -\frac{1}{3!} \lambda \phi^3, \quad (9.23)$$

where λ is the coupling constant. In this case, the interaction Hamiltonian is then just minus the interaction Lagrangian

$$\begin{aligned} H_{\text{int}}(t) &= \frac{\lambda}{3!} \int d^3x a^3(t) \phi^3(\mathbf{x}, t) \\ &= \frac{\lambda}{3!} a^3(t) \int d^3x \int \frac{d^3q_1}{(2\pi)^3} \frac{d^3q_2}{(2\pi)^3} \frac{d^3q_3}{(2\pi)^3} \phi_{\mathbf{q}_1}(t) \phi_{\mathbf{q}_2}(t) \phi_{\mathbf{q}_3}(t) e^{-i(\mathbf{q}_1 + \mathbf{q}_2 + \mathbf{q}_3) \cdot \mathbf{x}} \\ &= \frac{\lambda}{3!} a^3(t) \int \frac{d^3q_1}{(2\pi)^3} \frac{d^3q_2}{(2\pi)^3} \frac{d^3q_3}{(2\pi)^3} \phi_{\mathbf{q}_1}(t) \phi_{\mathbf{q}_2}(t) \phi_{\mathbf{q}_3}(t) (2\pi)^3 \delta(\mathbf{q}_1 + \mathbf{q}_2 + \mathbf{q}_3). \end{aligned} \quad (9.24)$$

Equation (9.22) then implies

$$\langle \phi_{\mathbf{k}_1}(t) \phi_{\mathbf{k}_2}(t) \phi_{\mathbf{k}_3}(t) \rangle = 2 \text{Im} \left(\int_{-\infty}^t dt' \langle 0 | \hat{\phi}_{\mathbf{k}_1}(t) \hat{\phi}_{\mathbf{k}_2}(t) \hat{\phi}_{\mathbf{k}_3}(t) \hat{H}_{\text{int}}(t') | 0 \rangle \right), \quad (9.25)$$

and we therefore need to compute the following expectation value

$$\langle 0 | \hat{\phi}_{\mathbf{k}_1}(t) \hat{\phi}_{\mathbf{k}_2}(t) \hat{\phi}_{\mathbf{k}_3}(t) \hat{\phi}_{\mathbf{q}_1}(t') \hat{\phi}_{\mathbf{q}_2}(t') \hat{\phi}_{\mathbf{q}_3}(t') | 0 \rangle. \quad (9.26)$$

To evaluate this, we write each field operator in terms of annihilation and creation operators,

$$\hat{\phi}_{\mathbf{k}}(t) = f_k(t) \hat{a}_{\mathbf{k}} + f_k^*(t) \hat{a}_{-\mathbf{k}}^\dagger, \quad (9.27)$$

and then compute (9.26) by shuffling the $\hat{a}_{\mathbf{k}}$'s all the way to the right using the commutator (8.23), so that they can annihilate the vacuum. This somewhat tedious procedure can be streamlined using Wick's theorem. We define the *Wick contraction* between two operators as

$$\overline{\hat{\phi}_{\mathbf{k}}(t) \hat{\phi}_{\mathbf{q}}(t')} = f_k(t) f_q^*(t') \delta(\mathbf{k} + \mathbf{q}) \equiv W_k(t, t') \delta(\mathbf{k} + \mathbf{q}), \quad (9.28)$$

Feynman rules

Diagrammatic rules for the computation of correlation functions were developed in [61, 65]. We will state these rules without explicit derivation and then illustrate them in a few examples.

1. Draw the final time surface ($t = t_*$).
2. Draw all diagrams contributing to the relevant process, allowing vertices both above and below this surface.
3. Lines that cross or end on the time surface represent a *Wightman propagator*,

$$W_k(t, t') = f_k(t) f_k^*(t'), \quad (9.34)$$

where the left argument is given by the time of the uppermost vertex and the right argument is given by the lowermost vertex. Lines which lie entirely below the fixed time surface represent *Feynman propagators*,

$$G_F(k; t, t') = W_k(t, t') \theta(t - t') + W_k(t', t) \theta(t' - t), \quad (9.35)$$

while lines which lie entirely above the surface are replaced with time-reversed Feynman propagators, $G_F(k; t', t) = G_F^*(k; t, t')$.

4. Vertices below the time surface get vertex factors $-iV$, where V is obtained by functional differentiation of the interaction Hamiltonian.¹⁸ Vertices above the time surface get factors of $iV^\dagger$. In Fourier space, momentum is conserved at each vertex, and there is an overall momentum-conserving delta function.
5. All vertex insertions must be integrated over time, using the measure $Dt \equiv dt a^3(t)$. If the diagram involves loops, integrate over the loop momenta.
6. If there are symmetry factors in diagrams, divide by them. In the second step above, we can mod out by reflections across the final time surface, but then we have to take 2Re of the answer.

Example Let us apply these rules to the three-point example described above. We are instructed to consider two diagrams—one with the interaction vertex above and one with it below the final time surface. Using the Feynman rules presented above, the first diagram gives

$$I_+ \equiv \text{Diagram} = -iV \int_{-\infty}^{t_*} Dt W_{k_1}(t_*, t) W_{k_2}(t_*, t) W_{k_3}(t_*, t) \quad (9.36)$$

$$= -iV f_{k_1}(t_*) f_{k_2}(t_*) f_{k_3}(t_*) \int_{-\infty}^{t_*} Dt f_{k_1}^*(t) f_{k_2}^*(t) f_{k_3}^*(t), \quad (9.37)$$

where we are being a bit schematic about the meaning of the vertex factor V . In cases that involve derivatives, we have to include the relevant factors of spatial momenta and appropriately act with time derivatives on the mode functions inside the integral. Similarly, the second

¹⁸For example, a theory with $\mathcal{L}_{\text{int}} = -g\phi^n/n!$ would get a factor of $-iV = -ig$ at an n -point vertex.

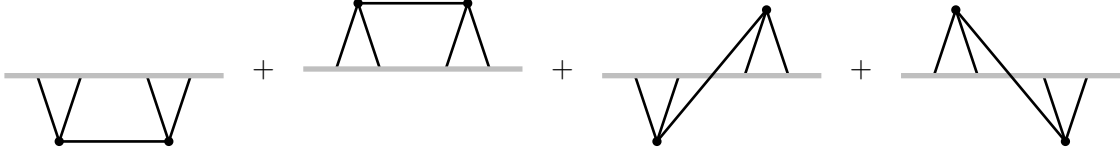
diagram leads to

$$I_- \equiv \text{[triangle diagram]} = +iV \int_{-\infty}^{t_*} Dt W_{k_1}(t, t_*) W_{k_2}(t, t_*) W_{k_3}(t, t_*) \quad (9.38)$$

$$= +iV f_{k_1}^*(t_*) f_{k_2}^*(t_*) f_{k_3}^*(t_*) \int_{-\infty}^{t_*} Dt f_{k_1}(t) f_{k_2}(t) f_{k_3}(t). \quad (9.39)$$

The sum of the two diagrams, $\langle \phi \phi \phi \rangle' = I_+ + I_-$, therefore indeed reproduces the result in (9.30).

Things become more interesting for exchange diagrams. For the tree-level four-point function in ϕ^3 theory, there are four different time orderings for the two vertices of the exchange diagram, corresponding to four distinct diagrams (in the s -channel):



Rather than writing complicated abstract formulas for each of the contributions, we will evaluate them in specific examples.

9.4 Examples in Flat Space

Though our primary interest is in cosmology, an instructive warmup is to consider correlators in flat space. In that case, the mode functions are simple enough that all time integrals can be evaluated explicitly. The computation in de Sitter space will then be conceptually exactly the same, but just algebraically more complicated.

We assume that the correlations are computed at a fixed time $t_* = 0$. This time slice is the analog of the future boundary in the de Sitter case.¹⁹ The interactions which create the late-time correlations live at $-\infty < t < 0$. For simplicity, we first consider a massless ϕ^3 theory, with the action

$$S = \int d^4x \left(-\frac{1}{2}(\partial\phi)^2 - \frac{\lambda}{3!}\phi^3 \right). \quad (9.40)$$

The flat-space mode function for this massless scalar is

$$f_k(t) = \frac{e^{-ikt}}{\sqrt{2k}}, \quad (9.41)$$

which implies that the Wightman and Feynman propagators are given by

$$W_k(t, t') = \frac{1}{2k} e^{-ik(t-t')}, \quad (9.42)$$

$$G_F(k; t, t') = \frac{1}{2k} \left(e^{-ik(t-t')} \theta(t-t') + e^{ik(t-t')} \theta(t'-t) \right). \quad (9.43)$$

The vertex factors are simple because $H_{\text{int}} = -L_{\text{int}}$, so that $V = \lambda$. We can now combine together these building blocks to compute some example correlators.

¹⁹Unlike the de Sitter case, there is no privileged time slice in Minkowski space. The choice of such a slice does break the time translation symmetry of the spacetime, but all such constant time slices are equivalent because they can be transformed into each other by time translations. It is therefore convenient to choose the slice $t_* = 0$.

Three-point function We first compute the three-point function $\langle\phi\phi\phi\rangle$. Substituting the flat-space mode function into the expressions (9.37) and (9.39), and taking the limit $t_* \rightarrow 0$, gives

$$I_+ = -\frac{i\lambda}{8k_1k_2k_3} \int_{-\infty}^0 dt e^{iKt} = -\frac{\lambda}{8k_1k_2k_3} \frac{1}{K}, \quad (9.44)$$

$$I_- = \frac{i\lambda}{8k_1k_2k_3} \int_{-\infty}^0 dt e^{-iKt} = -\frac{\lambda}{8k_1k_2k_3} \frac{1}{K}, \quad (9.45)$$

where we have defined the sum of energies $K \equiv k_1 + k_2 + k_3$. Notice that in order to get these oscillatory integrals to converge as $t \rightarrow -\infty$, we have implicitly implemented the $i\epsilon$ prescription of (9.21) and rotated the contour slightly. Summing these answers, we get the final correlator

$$\boxed{\langle\phi_{\mathbf{k}_1}\phi_{\mathbf{k}_2}\phi_{\mathbf{k}_3}\rangle' = -\frac{\lambda}{4k_1k_2k_3} \frac{1}{(k_1 + k_2 + k_3)}}. \quad (9.46)$$

Notice from (9.37) and (9.39) that I_+ and I_- in general are complex conjugates of each other. In this specific case, however, both contributions are purely real, so we just get two times the contribution of one of the graphs.

Four-point function As a more nontrivial example, we next compute the four-point function $\langle\phi\phi\phi\phi\rangle$, which involves the exchange of a particle.²⁰ In the s -channel, there are four relevant diagrams (corresponding to the four different time orderings of the interaction vertices), which we will now evaluate in turn.

We first consider the diagram where both vertices are below the final time surface. Using the Feynman rules, we find

$$\begin{aligned} I_{++} &\equiv \text{Diagram: A horizontal line at the top with two vertical lines extending downwards from it. These two vertical lines meet at two vertices, which are connected by a horizontal line.} \\ &= (-i\lambda)^2 \int_{-\infty}^0 dt dt' W_{k_1}(0, t) W_{k_2}(0, t) G_F(k_I; t, t') W_{k_3}(0, t') W_{k_4}(0, t') \\ &= -\frac{\lambda^2}{16k_1k_2k_3k_4} \int_{-\infty}^0 dt dt' e^{ik_{12}t} G_F(k_I; t, t') e^{ik_{34}t'}, \end{aligned} \quad (9.47)$$

where we have defined $k_{12} \equiv k_1 + k_2$, $k_{34} \equiv k_3 + k_4$ and $k_I \equiv |\mathbf{k}_1 + \mathbf{k}_2|$. (Due to momentum conservation at the bulk vertices, the momentum of the internal line is $\mathbf{k}_I = \mathbf{k}_1 + \mathbf{k}_2 = -\mathbf{k}_3 - \mathbf{k}_4$.)

²⁰There can, of course, also be contributions to the four-point function coming from contact interactions, which will involve a single insertion of the quartic interaction Hamiltonian. See Section 9.5, for an example.

Substituting the Feynman propagator (9.43), this becomes

$$\begin{aligned}
I_{++} &= -\frac{\lambda^2}{16k_1k_2k_3k_4} \frac{1}{2k_I} \int_{-\infty}^0 dt dt' e^{ik_{12}t} e^{-ik_I(t-t')} \theta(t-t') e^{ik_{34}t'} + (k_{12} \leftrightarrow k_{34}) \\
&= -\frac{\lambda^2}{16k_1k_2k_3k_4} \frac{1}{2k_I} \int_{-\infty}^0 dt e^{i(k_{12}-k_I)t} \int_{-\infty}^t dt' e^{i(k_{34}+k_I)t'} + (k_{12} \leftrightarrow k_{34}) \\
&= -\frac{\lambda^2}{16k_1k_2k_3k_4} \frac{1}{2k_I} \int_{-\infty}^0 dt \frac{e^{i(k_{12}+k_{34})t}}{i(k_{34}+k_I)} + (k_{12} \leftrightarrow k_{34}) \\
&= \frac{\lambda^2}{16k_1k_2k_3k_4} \frac{1}{2k_I} \frac{1}{k_{12}+k_{34}} \left(\frac{1}{k_{12}+k_I} + \frac{1}{k_{34}+k_I} \right) \\
&= \frac{\lambda^2}{16k_1k_2k_3k_4} \left(\frac{1}{EE_LE_R} + \frac{1}{2k_I E_L E_R} \right), \tag{9.48}
\end{aligned}$$

where we have defined $E \equiv k_{12} + k_{34}$, $E_L \equiv k_{12} + k_I$ and $E_R \equiv k_{34} + k_I$. While in more general situations in cosmology these nested time integrals are challenging to compute, in this flat-space example the final result is a relatively simple rational function of the energies involved in the process.

Similarly, when both vertices are above the final time surface, the relevant integral is

$$\begin{aligned}
I_{--} &\equiv \text{Diagram: A horizontal line with two vertices above it. Each vertex is connected to the line by two lines, forming two triangles.} \\
&= (i\lambda)^2 \int_{-\infty}^0 dt dt' W_{k_1}(t, 0) W_{k_2}(t, 0) G_F(k_I; t', t) W_{k_3}(t', 0) W_{k_4}(t', 0). \tag{9.49}
\end{aligned}$$

Note the difference in the orderings of the arguments of the propagators in (9.47) and (9.49). Using that $W_k(t', t) = W_k^*(t, t')$ and $G_F(k; t', t) = G_F^*(k; t, t')$, we see that $I_{--} = I_{++}^*$ and hence $I_{++} + I_{--} = 2 \text{Re}(I_{++})$. In this specific case, both contributions are purely real (so $I_{--} = I_{++}$) and the sum of the two diagrams is just twice the answer in (9.48):

$$I_{++} + I_{--} = \frac{\lambda^2}{8k_1k_2k_3k_4} \left(\frac{1}{EE_LE_R} + \frac{1}{2k_I E_L E_R} \right). \tag{9.50}$$

We have separated the answer into a “connected part” (with an $E = 0$ pole) and a “disconnected part” (without an $E = 0$ pole, but with a $k_I = 0$ pole).

Next, we consider the case where one vertex is below the final time surface and one is above:

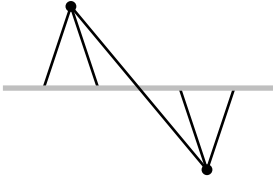
$$\begin{aligned}
I_{+-} &\equiv \text{Diagram: A horizontal line with one vertex below it and one above it. The bottom vertex is connected to the line by two lines, and the top vertex is connected to the line by two lines.} \\
&= (-i\lambda)(i\lambda) \int_{-\infty}^0 dt dt' W_{k_1}(0, t) W_{k_2}(0, t) W_{k_I}(t', t) W_{k_3}(t', 0) W_{k_4}(t', 0)
\end{aligned}$$

$$\begin{aligned}
&= \frac{\lambda^2}{16k_1k_2k_3k_4} \frac{1}{2k_I} \int_{-\infty}^0 dt dt' e^{ik_{12}t} e^{-ik_I(t'-t)} e^{-ik_{34}t'} \\
&= \frac{\lambda^2}{16k_1k_2k_3k_4} \frac{1}{2k_I} \int_{-\infty}^0 dt e^{i(k_{12}+k_I)t} \int_{-\infty}^0 dt' e^{-i(k_{34}+k_I)t'} \\
&= \frac{\lambda^2}{16k_1k_2k_3k_4} \frac{1}{2k_I} \frac{1}{(k_{12}+k_I)(k_{34}+k_I)} \\
&= \frac{\lambda^2}{16k_1k_2k_3k_4} \frac{1}{2k_I E_L E_R} .
\end{aligned} \tag{9.51}$$

Since all propagators are Wightman propagators, the two time integrations decoupled and could be performed separately. The answer therefore only includes a “disconnected part.” We will see later that it is useful to combine this with the disconnect part inside I_{++} .

Finally, we consider the contribution I_{-+} , with opposite time ordering of the interaction vertices:

$I_{-+} \equiv$



$$= i\lambda(-i\lambda) \int_{-\infty}^0 dt dt' W_{k_1}(t, 0) W_{k_2}(t, 0) W_{k_I}(t, t') W_{k_3}(0, t') W_{k_4}(0, t') . \tag{9.52}$$

It is straightforward to check that $I_{-+} = I_{+-}^*$, and since I_{+-} is purely real, it will just give us a factor of 2:

$$I_{+-} + I_{-+} = \frac{\lambda^2}{8k_1k_2k_3k_4} \frac{1}{2k_I E_L E_R} . \tag{9.53}$$

Notice that this contribution to the correlator only has the singularities when the energy flowing into a vertex is conserved, but does not have a singularity when the total energy is conserved.

Putting everything together, the final result for the s -channel exchange is then the sum of (9.50) and (9.53):

$$\boxed{\langle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \phi_{\mathbf{k}_4} \rangle' = \frac{\lambda^2}{8k_1k_2k_3k_4} \left[\frac{1}{E E_L E_R} + \frac{1}{k_I E_L E_R} \right]} . \tag{9.54}$$

One of the benefits of this simplified situation is that we can obtain completely explicit expressions for all the constituent pieces of the correlator, and get a sense of how they fit together in the final answer. When it comes time to examine the singularity structure of correlation functions, we will see that this is enormously useful.

One can proceed to compute higher-point correlation functions in this model, or consider more nontrivial types of interactions. One such example is given in the following exercise.

Exercise 9.2 Compute the four-point function arising from a cubic interaction involving derivatives

$$S = \int d^4x \left(-\frac{1}{2}(\partial\phi)^2 - \frac{\dot{\phi}^3}{6\Lambda^2} \right). \quad (9.55)$$

Note that the interaction involves $\dot{\phi}$, so be careful about the definition of H_{int} . →

9.5 Examples in De Sitter

Having gained some experience with the computation of correlators in flat space, we are now ready to attack the corresponding computations in de Sitter space. Conceptually, these computations are exactly the same, only algebraically do they become a bit more involved. In fact, in many cases, the resulting time integrals cannot be performed explicitly in de Sitter space, which will motivate us to develop a bootstrap method that avoids these integrals altogether (see Section 11).

For concreteness, we again consider Φ^3 theory, where the field Φ is a generic scalar field of mass m . As we showed in Exercise 8.6, the Bunch–Davies mode function of a massive field is

$$f_k(\eta) = H \sqrt{\frac{\pi}{4}} e^{\frac{i\pi}{4}(1+2\nu)} (-\eta)^{3/2} H_\nu^{(1)}(-k\eta), \quad \text{where } \nu = \sqrt{\frac{9}{4} - \frac{m^2}{H^2}}. \quad (9.56)$$

We will often specialize to *conformally coupled scalars*, with $m^2 = 2H^2$. In that case, the order of the Hankel function becomes $\nu = 1/2$ and the mode function is almost as simple as in flat space:

$$f_k(\eta) = H(-\eta) \frac{e^{-ik\eta}}{\sqrt{2k}}. \quad (9.57)$$

The in-in computations simplify drastically for conformally coupled scalars and can often be performed explicitly.²¹

Three-point function We first compute the three-point function $\langle \phi\phi\phi \rangle$. Equation (9.37) still applies, so we have

$$I_+ = I_-^* = -i\lambda f_{k_1}(\eta_*) f_{k_2}(\eta_*) f_{k_3}(\eta_*) \int_{-\infty}^{\eta_*} D\eta f_{k_1}^*(\eta) f_{k_2}^*(\eta) f_{k_3}^*(\eta), \quad (9.58)$$

where $D\eta \equiv d\eta a^4(\eta)$. One conceptual difference with the flat-space case is that there is a distinguished time slice in de Sitter space ($\eta_* = 0$), where it is most natural to compute correlators. This future boundary is particularly nice because it is mapped to itself under de Sitter transformations. In the following, we will always compute correlators in the $\eta_* \rightarrow 0$ limit for this reason.

For the totally generic case involving massive fields, this involves an integral over a product of three Hankel functions. This integral can be evaluated in terms of Appell’s F_4 function [67],

²¹Aside from being simple to compute, it is often possible to transform the correlation functions of conformally coupled scalar fields into those of fields with other mass values, corresponding to half-integer values of ν , by a “weight-shifting” procedure [66]. The correlators of conformal scalars therefore become seeds for more complicated correlators.

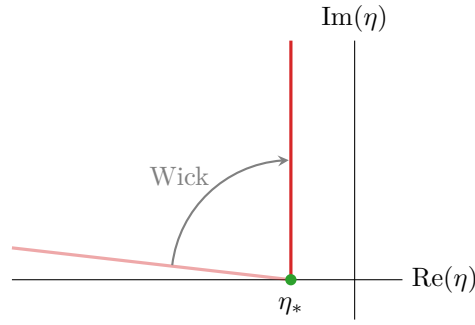


Figure 25: Graphical illustration of Wick rotation of the contour used in the integral in (9.59).

but unfortunately the properties of this special function are still somewhat mysterious, making this representation less useful than it could be. However, for special masses the integral simplifies and can be evaluated explicitly.

One such example is the three-point function of conformally coupled scalars. In this case, the relevant integral (in the limit $\eta_* \rightarrow 0$) is

$$I_+ = -\frac{i\lambda}{8} \frac{H^2 \eta_*^3}{k_1 k_2 k_3} \int_{-\infty}^{\eta_*} \frac{d\eta}{\eta} e^{iK\eta} = -\frac{ig}{8} \frac{H^2 \eta_*^3}{k_1 k_2 k_3} \log(-iK\eta_*), \quad (9.59)$$

where again $K \equiv k_1 + k_2 + k_3$. Two aspects of this expression merit discussion. First, this is our first encounter with a correlator that diverges as $\eta_* \rightarrow 0$. We regulate this IR divergence by cutting the integration off at finite $\eta = \eta_*$. Second, the answer has a factor of i in the argument of the log. This arises from the $i\epsilon$ prescription used to define the integral (see Fig. 25 for a graphical illustration of the Wick rotation of the integration contour).²²

Combining I_+ with its complex conjugate I_- , we obtain the full correlator

$$\langle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \rangle' = -\frac{\lambda}{8} \frac{H^2 \eta_*^3}{k_1 k_2 k_3} \left(i \log(-iK\eta_*) - i \log(iK\eta_*) \right) = \boxed{\frac{\pi}{8} \lambda \frac{H^2 \eta_*^3}{k_1 k_2 k_3}}. \quad (9.60)$$

Notice that the two branches of the in-in contour conspire to cancel the $\log \eta_*$ divergence, and to remove the singularity as $K \rightarrow 0$ that I_+ and I_- have individually.

Another particularly relevant mass value is $m^2 = 0$, where the field ϕ is related to the fluctuations of the inflaton field in the conformal limit, $\varepsilon \rightarrow 0$. This means that correlation functions in slow-roll inflation can often be obtained from those involving massless scalars.

Exercise 9.3 Consider a massless scalar field with a ϕ^3 interaction in de Sitter space. Show that the bispectrum, in the limit $\eta_* \rightarrow 0$, is [68]

$$\langle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \rangle' = -\frac{\lambda H^2}{12} \frac{(\gamma + \log(|K\eta_*|)) \sum_i k_i^3 - 3K \sum_{i<j} k_i k_j + K^3}{(k_1 k_2 k_3)^3}, \quad (9.61)$$

²²The $i\epsilon$ prescription for I_+ selects a contour that extends to infinity above the x -axis. Since the integrand does not have any singularities in the upper left quadrant of the complex plane, we can rotate it to run along the imaginary axis, $\eta \rightarrow -i\eta$.

where γ is the Euler–Mascheroni constant. What is the origin of the log divergence?

Hint: You may use that

$$\int_{-\infty}^{\eta_*} d\eta \frac{e^{iK\eta}}{\eta} \rightarrow \gamma + \log(|K\eta_*|), \quad (9.62)$$

in the limit $\eta_* \rightarrow 0$.



It is also possible to directly integrate (9.58) when two of the fields involved are conformally coupled and the third has an arbitrary mass, as you will see in the following exercise. This correlator is useful as a building block to construct more complicated processes, and gives us some intuition for how correlators involving massive fields behave.

Exercise 9.4 Derive an expression for the three-point function of two conformally coupled scalars and one scalar of generic mass. You should find

$$\begin{aligned} \langle \phi \phi \chi \rangle' = & -\frac{\lambda(H\eta_*)^2}{16\sqrt{\pi}} \frac{1}{k_1 k_2 k_3^2} \Gamma[\tfrac{1}{2} - \nu] \Gamma[\tfrac{1}{2} + \nu] P_{\nu-\frac{1}{2}}^0(u^{-1}) \\ & \times \left[\cos\left(\frac{\pi}{4}(1+2\nu)\right) \Gamma[\nu] \left(\frac{-k_3\eta_*}{2}\right)^{\frac{3}{2}-\nu} + \cos\left(\frac{\pi}{4}(1-2\nu)\right) \Gamma[-\nu] \left(\frac{-k_3\eta_*}{2}\right)^{\frac{3}{2}+\nu} \right], \end{aligned} \quad (9.63)$$

where $u \equiv k_3/(k_1 + k_2)$ and $P_\nu^\mu(x)$ is an associated Legendre function. Show that you recover (9.60) in the appropriate limit.

Hints: You may find the following integral representations of the Hankel function and the hypergeometric function useful

$$H_\nu^{(2)}(x) = \frac{i}{\sqrt{\pi}} \frac{2^{\nu+1} e^{i\nu\pi}}{\Gamma[\nu + \frac{1}{2}]} x^\nu e^{-ix} \int_0^\infty ds e^{-2ixs} s^{\nu-\frac{1}{2}} (1+s)^{\nu-\frac{1}{2}}, \quad (9.64)$$

$${}_2F_1\left[\begin{matrix} a, & b \\ c \end{matrix} \middle| z\right] = \frac{\Gamma[c]}{\Gamma[b]\Gamma[c-b]} \int_0^\infty dt t^{-b+c-1} (t+1)^{a-c} (t-z+1)^{-a}. \quad (9.65)$$

Moreover, you may use that

$$P_\nu^\mu(x) = \left(\frac{x+1}{x-1}\right)^{\mu/2} \frac{1}{\Gamma[1-\mu]} {}_2F_1\left[\begin{matrix} \nu+1, & -\nu \\ 1-\mu \end{matrix} \middle| \frac{1}{2} - \frac{x}{2}\right], \quad (9.66)$$

and assume that the Legendre function has no branch cut in the physical region.



Four-point function We next consider the four-point function $\langle \phi \phi \phi \phi \rangle$. The simplest such four-point functions arise from contact interactions, much like the three-point functions that we have studied so far. In terms of (9.21), they arise from a single insertion of the quartic interactions inside H_{int} , such as the bulk interaction ϕ^4 . As a concrete example, we can consider the four-point function of a conformally coupled scalar field. From the Feynman rules, we find that the

I_+ contribution is given by

$$\begin{aligned}
I_+ &= \text{Diagram: a vertex connected to four external lines} = -iV \int_{-\infty}^{\eta_*} d\eta W_{k_1}(\eta_*, \eta) W_{k_2}(\eta_*, \eta) W_{k_3}(\eta_*, \eta) W_{k_4}(\eta_*, \eta) \\
&= -i \frac{\lambda(H\eta_*)^4}{16 k_1 k_2 k_3 k_4} \int_{-\infty}^{\eta_*} d\eta e^{i(k_1+k_2+k_3+k_4)\eta} \\
&= -\frac{\lambda(H\eta_*)^4}{16 k_1 k_2 k_3 k_4 (k_1 + k_2 + k_3 + k_4)}. \tag{9.67}
\end{aligned}$$

This contribution is again purely real, so the full correlator is just twice the above answer:

$$\langle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \phi_{\mathbf{k}_4} \rangle' = -\frac{\lambda(H\eta_*)^4}{8 k_1 k_2 k_3 k_4} \frac{1}{E}, \tag{9.68}$$

where $E \equiv k_1 + k_2 + k_3 + k_4$. In this particular case, the flat-space version of the correlator is very simply related to the one that we have just calculated—one just removes the $(H\eta_*)^4 = a^{-4}(\eta_*)$ factor. This is special to four dimensions where the theory is conformal.²³ Notice that (9.67) decays as $\eta_* \rightarrow 0$, but the coefficient of this power-law decay is still physically meaningful, as we will see more explicitly in Section 11.

Exercise 9.5 Consider a conformally coupled scalar field in D -dimensional de Sitter space with a quartic interaction

$$S = \int d^D x \sqrt{-g} \left(-\frac{1}{2}(\partial\phi)^2 - \frac{D(D-2)H^2}{8} \phi^2 - \frac{\lambda}{4!} \phi^4 \right). \tag{9.69}$$

Compute the four-point function at tree level. Do you notice a difference between even and odd D ? ➔

We can also compute tree-level exchange contributions to four-point functions in the same way that we did in flat space. It is again useful to begin by considering a conformally coupled scalar. We can write the Wightman and Feynman propagators explicitly using (9.57):

$$W_k(\eta, \eta') = H^2 \eta \eta' \frac{1}{2k} e^{-ik(\eta - \eta')}, \tag{9.70}$$

$$G_F(k; \eta, \eta') = H^2 \eta \eta' \frac{1}{2k} \left(e^{-ik(\eta - \eta')} \theta(\eta - \eta') + e^{ik(\eta - \eta')} \theta(\eta' - \eta) \right). \tag{9.71}$$

Notice that these are exactly $H^2 \eta \eta'$ times their flat-space counterparts (9.42) and (9.43). (This is another manifestation of the conformal invariance of the conformally coupled scalar.) The I_{++}

²³Since a conformally coupled scalar with a $\lambda\phi^4$ interaction is conformally invariant, it can be related to the corresponding flat-space theory by a Weyl rescaling with $g_{\mu\nu}^{\text{dS}} = a^2(\eta)\eta_{\mu\nu}$ and $\phi_{\text{dS}} = a^{-1}(\eta)\phi_{\text{flat}}$.

contribution is

$$\begin{aligned}
I_{++} &\equiv \text{Diagram: A trapezoid with a horizontal top line, a horizontal bottom line, and two slanted sides. The bottom line has two black dots at its endpoints. The top line is slightly longer than the bottom line and is parallel to it. The slanted sides connect the endpoints of the top line to the endpoints of the bottom line.} \\
&= (-i\lambda)^2 \int_{-\infty}^{\eta_*} d\eta d\eta' W_{k_1}(\eta_*, \eta) W_{k_2}(\eta_*, \eta) G_F(k_I; \eta, \eta') W_{k_3}(\eta_*, \eta') W_{k_4}(\eta_*, \eta') \\
&= -\frac{\lambda^2 H^2 \eta_*^4}{16k_1 k_2 k_3 k_4} \int_{-\infty}^{\eta_*} \frac{d\eta}{\eta} \int_{-\infty}^{\eta_*} \frac{d\eta'}{\eta'} e^{ik_{12}\eta} G_F^{(\text{flat})}(k_I; \eta, \eta') e^{ik_{34}\eta'}, \tag{9.72}
\end{aligned}$$

where, in the second line, we have suggestively written the expression in terms of the flat-space Feynman propagator (9.43). This integral is a bit tricky to evaluate because it is divergent. As we will see, the divergences in I_{--} and I_{++} cancel against similar divergences in I_{-+} and I_{+-} . In order to isolate the divergent piece, it is therefore useful to add and subtract a contribution to the integral (9.72) as²⁴

$$I_{++} = -\frac{\lambda^2 H^2 \eta_*^4}{16k_1 k_2 k_3 k_4} \int_{-\infty}^{\eta_*} \frac{d\eta}{\eta} \int_{-\infty}^{\eta_*} \frac{d\eta'}{\eta'} e^{ik_{12}\eta} \left(G_B^{(\text{flat})}(k_I; \eta, \eta') + \frac{1}{2k_I} e^{ik_I(\eta+\eta')} \right) e^{ik_{34}\eta'}, \tag{9.73}$$

where we have defined²⁵

$$G_B^{(\text{flat})}(k_I; \eta, \eta') \equiv G_F^{(\text{flat})}(k_I; \eta, \eta') - \frac{1}{2k_I} e^{ik_I(\eta+\eta')}. \tag{9.74}$$

We can now perform the integral involving G_B via a trick. We note that

$$-i \int_a^\infty dx e^{ix\eta} = \frac{1}{\eta} e^{ia\eta}, \tag{9.75}$$

so that (9.73) can be written as

$$\begin{aligned}
I_{++} &= \frac{\lambda^2 H^2 \eta_*^4}{16k_1 k_2 k_3 k_4} \left[\int_{k_{12}}^\infty dx \int_{k_{34}}^\infty dx' \int_{-\infty}^{\eta_*} d\eta d\eta' e^{ix\eta} e^{ix'\eta'} G_B^{(\text{flat})}(k_I; \eta, \eta') \right. \\
&\quad \left. - \frac{1}{2k_I} \int_{-\infty}^{\eta_*} \frac{d\eta}{\eta} \int_{-\infty}^{\eta_*} \frac{d\eta'}{\eta'} e^{i(k_{12}+s)\eta} e^{i(k_{34}+s)\eta'} \right], \tag{9.76}
\end{aligned}$$

The time integrals in the first line now only involve flat-space quantities and can therefore easily be computed

$$I(x, x') \equiv \int_{-\infty}^0 d\eta d\eta' e^{ix\eta} e^{ix'\eta'} G_B^{(\text{flat})}(k_I; \eta, \eta') = -\frac{1}{(x+x')(x+k_I)(x'+k_I)}. \tag{9.77}$$

It is then useful to split I_{++} into two pieces: a “connected” component, $I_{++}^{(c)}$, which is the energy integral of $I(x, y)$, and a “disconnected” component, $I_{++}^{(d)}$, which involves un-ordered time integrals.

²⁴We have dropped a factor of $e^{-iE\eta_*}$ which becomes unity in the limit $\eta_* \rightarrow 0$.

²⁵In Section 10, we will encounter this object as the *bulk-to-bulk propagator*.

Performing the energy integrals, the connected piece becomes [43]

$$\begin{aligned}
I_{++}^{(c)} &\equiv \frac{\lambda^2 H^2 \eta_*^4}{16k_1 k_2 k_3 k_4} \int_{k_{12}}^{\infty} dx \int_{k_{34}}^{\infty} dx' I(x, x') \\
&= \frac{\lambda^2 H^2 \eta_*^4}{32k_1 k_2 k_3 k_4 k_I} \left[\text{Li}_2 \left(\frac{E - E_L}{E} \right) + \text{Li}_2 \left(\frac{E - E_R}{E} \right) + \log \left(\frac{E_L}{E} \right) \log \left(\frac{E_R}{E} \right) - \frac{\pi^2}{6} \right],
\end{aligned} \tag{9.78}$$

where Li_2 is the dilogarithm and we have defined the variables $E_L \equiv k_{12} + k_I$ (the energy flowing into the left vertex), $E_R \equiv k_{34} + k_I$ (same for the right vertex) and the total energy $E \equiv k_{12} + k_{34}$.

Exercise 9.6 Derive the result (9.78).



In order to compute the disconnected piece, it is simplest to do the time integrals directly, rather than trying to write them as energy integrals of the flat-space result. This is because we have to be careful about the $i\epsilon$ prescriptions for the contours, much like we did for (9.59). In this case, we can rotate the contours in the same direction as (9.59), so we get

$$\begin{aligned}
I_{++}^{(d)} &\equiv -\frac{\lambda^2 H^2 \eta_*^4}{32k_1 k_2 k_3 k_4 k_I} \int_{-\infty}^{\eta_*} \frac{d\eta}{\eta} e^{i(k_{12}+s)\eta} \int_{-\infty}^{\eta_*} \frac{d\eta'}{\eta'} e^{i(k_{34}+s)\eta'} \\
&= -\frac{\lambda^2 H^2 \eta_*^4}{32k_1 k_2 k_3 k_4 k_I} \log(-iE_L \eta_*) \log(-iE_R \eta_*).
\end{aligned} \tag{9.79}$$

As advertised, this contribution is divergent as $\eta_* \rightarrow 0$, which is why we split it off from $I_{++}^{(c)}$ to begin with. These divergences cancel with the $I_{\mp\pm}$ pieces, which we now want to calculate.

Using the Feynman rules, we find

$$\begin{aligned}
I_{+-} &\equiv \text{Diagram: A horizontal line with two vertices. From each vertex, a line goes up to a central point, and another line goes down to a common point below the line, forming a diamond shape with a central vertex and two outer vertices on the horizontal line.} \\
&= \lambda^2 \int_{-\infty}^{\eta_*} D\eta D\eta' W_{k_1}(\eta_*, \eta) W_{k_2}(\eta_*, \eta) W_{k_I}(\eta', \eta) W_{k_3}(\eta', \eta_*) W_{k_4}(\eta', \eta_*) \\
&= \frac{\lambda^2 H^2 \eta_*^4}{32k_1 k_2 k_3 k_4 k_I} \int_{-\infty}^{\eta_*} \frac{d\eta}{\eta} e^{i(k_{12}+k_I)\eta} \int_{-\infty}^{\eta_*} \frac{d\eta'}{\eta'} e^{-i(k_{34}+k_I)\eta'} \\
&= \frac{\lambda^2 H^2 \eta_*^4}{32k_1 k_2 k_3 k_4 k_I} \log(-iE_L \eta_*) \log(iE_R \eta_*).
\end{aligned} \tag{9.80}$$

In the last line, we evaluated the integrals, noting that the different $i\epsilon$ prescriptions required for convergence of the integrals lead to different signs of the arguments of the log for the η' and η'' integrals. Adding (9.80) to the result (9.79) for $I_{++}^{(d)}$, we get

$$I_{++}^{(d)} + I_{+-} = \frac{\lambda^2 H^2 \eta_*^4}{32k_1 k_2 k_3 k_4 k_I} (-i\pi \log(-iE_L \eta_*)). \tag{9.81}$$

Notice that this is similar in structure to what we found for the three-point function in (9.59). The sum of the $I_{--}^{(d)}$ and I_{-+} contributions gives the complex conjugate of (9.81). Putting everything together, we therefore find

$$I_{++}^{(d)} + I_{+-} + I_{--}^{(d)} + I_{-+} = \frac{\lambda^2 H^2 \eta_*^4}{32 k_1 k_2 k_3 k_4 k_I} \pi^2. \quad (9.82)$$

Note that there is a similar effect as in (9.60) where the kinematic part of the log drops out and we are left with just factors of π .

Assembling all the pieces, the full correlator is

$$\begin{aligned} \langle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \phi_{\mathbf{k}_4} \rangle' &= I_{++}^{(c)} + (I_{++}^{(d)} + I_{+-}) + \text{c.c.} \\ &= \left[\frac{\lambda^2 H^2 \eta_*^4}{16 k_1 k_2 k_3 k_4 k_I} \left[\text{Li}_2 \left(\frac{E - E_L}{E} \right) + \text{Li}_2 \left(\frac{E - E_R}{E} \right) + \log \left(\frac{E_L}{E} \right) \log \left(\frac{E_R}{E} \right) + \frac{\pi^2}{3} \right] \right]. \end{aligned} \quad (9.83)$$

Notice that this correlator has transcendentality two, compared to its flat-space counterpart (9.54), which is a rational function. This is rather interesting, and is in some sense a signature of the time evolution of the spacetime that gave rise to it. Aside from its intrinsic interest, and serving as a useful testing ground, this correlator is closely related to many correlation functions of practical interest, including those involving exchanges of massless particles with spin, which can be generated by differentiating this correlator in a suitable way [44, 66].

As a last example, we consider the four-point function that arises from the exchange of a massive particle $m > \frac{3}{2}H$ between two pairs of conformally coupled scalar fields. In this case, the order ν of the Hankel function in (9.56) is purely imaginary, so it is useful to write the mode functions as

$$f_k(\eta) = e^{\frac{i\pi}{4}} H \sqrt{\frac{\pi}{4}} (-\eta)^{3/2} e^{\frac{\pi}{2}\mu} H_{i\mu}^{(1)}(-k\eta), \quad \text{where} \quad i\mu = \sqrt{\frac{9}{4} - \frac{m^2}{H^2}}. \quad (9.84)$$

The definition of μ is such that it is a real number for $m > \frac{3}{2}H$. From this mode function, we can build the Wightman and Feynman propagators:²⁶

$$W_k(\eta, \eta') = \frac{H^2 \pi}{4} (\eta \eta')^{3/2} H_{i\mu}^{(1)}(-k\eta) H_{i\mu}^{(2)}(-k\eta'), \quad (9.86)$$

$$G_F(k; \eta, \eta') = W_k(\eta, \eta') \theta(\eta - \eta') + W_k(\eta', \eta) \theta(\eta' - \eta). \quad (9.87)$$

With these building blocks, we can write down the various exchange contributions. For example, I_{++} is

$$I_{++} = -\frac{\lambda^2 \eta_*^4}{16 k_1 k_2 k_3 k_4} \int_{-\infty}^{\eta_*} \frac{d\eta}{\eta^2} \int_{-\infty}^{\eta_*} \frac{d\eta'}{\eta'^2} e^{ik_{12}\eta} G_F(k_I; \eta, \eta') e^{ik_{34}\eta'}, \quad (9.88)$$

²⁶Here, we have used the fact that

$$\left(H_{i\mu}^{(2)}(x) \right)^* = H_{-i\mu}^{(1)}(x) = e^{-\pi\mu} H_{i\mu}^{(1)}(x), \quad (9.85)$$

which causes the explicit exponential factors of μ to cancel in the Wightman function.

and I_{--} is just the complex conjugate of this expression. Similarly, I_{+-} is

$$I_{+-} = \frac{\lambda^2 \eta_*^4}{16k_1 k_2 k_3 k_4} \int_{-\infty}^{\eta_*} \frac{d\eta}{\eta^2} \int_{-\infty}^{\eta_*} \frac{d\eta'}{\eta'^2} e^{ik_{12}\eta} W_{k_I}(\eta', \eta) e^{-ik_{34}\eta'}, \quad (9.89)$$

whose complex conjugate is I_{-+} .

The integral in (9.89) can be done by the same method employed in Exercise 9.4, but doing the nested time integrals involved in (9.88) is a challenge that we will return to later. However, in the limit that the exchanged momentum is small $k_I \rightarrow 0$, the computation is quite tractable.

Exercise 9.7 In the long-wavelength limit $k_I \rightarrow 0$, there is no difference between the Wightman and Feynman propagators:

$$\begin{aligned} \lim_{k_I \rightarrow 0} G_F(k_I; \eta, \eta') &= \lim_{k_I \rightarrow 0} W_{k_I}(\eta, \eta') \\ &= \frac{H^2}{4\pi} (\eta\eta')^{3/2} \left[\Gamma[-i\mu]^2 \left(\frac{k_I^2 \eta \eta'}{4} \right)^{i\mu} + \Gamma[i\mu]^2 \left(\frac{k_I^2 \eta \eta'}{4} \right)^{-i\mu} \right]. \end{aligned} \quad (9.90)$$

This is basically because we are probing separations between the two vertices that are outside each others' lightcones, so that the time-ordering does not matter. (This still captures the interesting non-analytic signature of particle exchange.) Use this fact to perform the time integrals (9.88) and (9.89), and combine them together with I_{--} and I_{-+} to obtain the correlator in the $k_I \rightarrow 0$ limit. ➔

9.6 Slow-Roll Inflation

One of the most famous applications of the in-in formalism in cosmology is Maldacena's computation of the bispectrum in slow-roll inflation [41]. In this section, we will present the full calculation. Although the computation is conceptually straightforward, it is algebraically quite involved because we need to compute the gravitational action to cubic order. Most explicit computations will be relegated to exercises, whose solutions can be found in Appendix B.

To compute the bispectrum, we need to expand the inflationary action (8.38) to third order in perturbations. As in Section 8.3, we write the metric in the ADM form (8.34):

$$ds^2 = -N^2 dt^2 + h_{ij}(N^i dt + dx^i)(N^j dt + dx^j), \quad (9.91)$$

and work in the comoving gauge, where the inflaton is unperturbed and the metric perturbations are

$$h_{ij} = a^2 e^{2\zeta(t, \mathbf{x})} \delta_{ij}, \quad N \equiv 1 + \alpha(t, \mathbf{x}), \quad N_i \equiv \partial_i \beta(t, \mathbf{x}). \quad (9.92)$$

Fortunately, to derive the cubic action for the curvature perturbation ζ , we only need N and N_i to first order since the second-order and third-order perturbations in N and N^i multiply the first-order and zeroth-order constraint equations, respectively. We found the first-order lapse and shift in (8.44):

$$\alpha = \frac{\dot{\zeta}}{H}, \quad \partial^2 \beta = -\frac{\partial^2 \zeta}{H} + a^2 \frac{\dot{\phi}^2}{2H^2} \dot{\zeta}. \quad (9.93)$$

We substitute these results into the action (8.38) and expand it to third order in ζ . After first, the result is rather nasty [69, 70]. It can be simplified using a lot of integration by parts, as well as the linearized equations of motion. One has to be careful with boundary terms: The spatial boundary terms of the form $\partial_i[\dots]$ can be dropped, but the temporal boundary terms $\partial_t[\dots]$ have to be kept. In Exercise 9.8, you are asked to show that this leads to [70]²⁷

$$S_3 = \int d\eta d^3x \left\{ a^2 \varepsilon^2 \left[\zeta(\zeta')^2 + \zeta(\partial_i \zeta)^2 - 2\zeta' \partial_i \zeta \partial_i^{-1} \zeta' \right] - \frac{1}{2} \partial_\eta (a^2 \varepsilon \kappa \zeta^2 \zeta') \right\} + \dots, \quad (9.94)$$

where the ellipses represent higher orders in slow-roll parameters, $\mathcal{O}(\varepsilon^3)$.

Exercise 9.8 Derive the cubic action (9.94).

Hint: Do not do this by hand, but use computer algebra!



Given the cubic action (9.94), we can now compute the bispectrum. Although the action has derivative couplings, the cubic Hamiltonian is still just given by $H_{\text{int}} = -L_3$; at quartic order the relation is more complicated. We substitute this into the *in-in* formula (9.22):

$$\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle = 2 \text{Im} \left(\int_{-\infty}^0 d\eta \langle 0 | \hat{\zeta}_{\mathbf{k}_1} \hat{\zeta}_{\mathbf{k}_2} \hat{\zeta}_{\mathbf{k}_3}(0) \hat{H}_{\text{int}}(\eta) | 0 \rangle \right), \quad (9.95)$$

where we have taken the late-time limit $\eta_* \rightarrow 0$. We compute the integral in (9.95) using the free-field mode functions for ζ . The result is [41]

$$\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle' = \frac{1}{32\varepsilon^2} \frac{H^4}{(k_1 k_2 k_3)^3} \left[\varepsilon \left(\sum_{i \neq j} k_i k_j^2 + \frac{8}{K} \sum_{i > j} k_i^2 k_j^2 \right) + (\kappa - \varepsilon) \sum_i k_i^3 \right], \quad (9.96)$$

where $K \equiv k_1 + k_2 + k_3$.

Exercise 9.9 Derive the result (9.96). Show that the amplitude of the bispectrum is

$$f_{\text{NL}} = \frac{5}{12} (9\varepsilon + \kappa). \quad (9.97)$$

Note that the slow-roll expansion $\{\varepsilon, |\kappa|\} \ll 1$ implies that $f_{\text{NL}} \ll 1$.



It is interesting to take the *squeezed limit*, $k_3 \ll k_1 \approx k_2$, of the bispectrum (9.96):

$$\begin{aligned} \lim_{k_3 \rightarrow 0} \langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle' &= \frac{1}{32\varepsilon^2} \frac{H^4}{k_1^6 k_3^3} \left[\varepsilon \left(2k_1^3 + \frac{8}{2k_1} k_1^4 \right) + (\kappa - \varepsilon) 2k_1^3 \right] \\ &= (2\varepsilon + \kappa) \left(\frac{H^2}{4\varepsilon} \frac{1}{k_1^3} \right) \left(\frac{H^2}{4\varepsilon} \frac{1}{k_3^3} \right) \\ &= (2\varepsilon + \kappa) P_\zeta(k_1) P_\zeta(k_3), \end{aligned} \quad (9.98)$$

where $P_\zeta(k)$ is the power spectrum. Note that the coefficient in the squeezed limit equals the deviation from scale invariance of the power spectrum, $1 - n_s = 2\varepsilon + \kappa$. In fact, this is a general

²⁷Maldacena removed the boundary term by a field redefinition [41].

result that applies to all models of single-field inflation, not necessarily just to slow-roll models. The *single-field consistency relation* states that [41, 71]

$$\lim_{k_3 \rightarrow 0} \langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle' = (1 - n_s) P_\zeta(k_1) P_\zeta(k_3), \quad (9.99)$$

i.e. the squeezed limit of the three-point function is suppressed by $(1 - n_s)$ and vanishes for perfectly scale-invariant perturbations.

Exercise 9.10 Prove the consistency relation (9.99) by considering the correlation between one long-wavelength mode, $k_L = k_3$, and two short-wavelength modes, $k_S = k_1 \approx k_2$:

$$\langle (\zeta_{\mathbf{k}_S})^2 \zeta_{\mathbf{k}_L} \rangle. \quad (9.100)$$

You will have to use that the long mode $\zeta_{\mathbf{k}_L}$ will be frozen and act as a classical background wave when the two short modes $\zeta_{\mathbf{k}_S}$ exit the horizon. ➔

The fact that the squeezed limit of the bispectrum for single-field inflation vanishes implies that it can be used as a clean diagnostic for extra fields during inflation. In Section 9.8, we will expand on this view of the squeezed limit as a particle detector.

9.7 EFT of Inflation

The gravitational nonlinearities in slow-roll inflation produce a minimal amount of non-Gaussianity which we will call the “gravitational floor”. Unfortunately, the amplitude of the signal is too small to be detectable in the foreseeable future. However, larger non-Gaussianities can be generated from additional inflaton self-interactions and we will briefly discuss this possibility.

Polynomial self-interactions like ϕ^n are part of the inflaton potential and are therefore constrained by the slow-roll conditions. In particular, it can be shown that the contribution of inflaton self-couplings to the cubic Lagrangian is

$$\mathcal{L}_3 \subset V''' \delta\phi^3 = \mathcal{O}(\epsilon^3) \zeta^3, \quad (9.101)$$

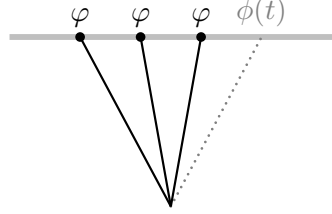
which is smaller than the leading gravitational interactions shown in (9.94).

In contrast, the inflaton’s derivative interactions, which naturally arise in an effective field theory (EFT) description of inflation, are not constrained in the same way. Consider a Lagrangian of the form

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_0(\phi) + \frac{(\partial_\mu \phi)^4}{8\Lambda^4} + \dots, \quad (9.102)$$

where $\mathcal{L}_0$ is the canonical slow-roll Lagrangian and Λ is a high-energy scale. Following [72], we will compute the bispectrum induced by the interaction $(\partial_\mu \phi)^4 = (g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi)^2$. This time it is useful to work in spatially flat gauge. At leading order in the slow-roll parameters, the action for the inflaton perturbation $\varphi \equiv \phi - \bar{\phi}(t)$ can then be calculated from (9.102) without including the contributions from $\sqrt{-g}R$. The intuitive reason for this simplification is that in the limit of a flat potential the perturbation φ does not induce a perturbation in the spacetime curvature.

Evaluating one of the legs of the interaction $(\partial_\mu \phi)^4$ on the background solution, we get



$$\Leftrightarrow \quad \mathcal{L}_{\text{eff},3} = -\frac{\dot{\phi}}{\Lambda^4} \dot{\varphi} (\partial_\mu \varphi)^2. \quad (9.103)$$

Using this interaction, we can compute the bispectrum $\langle \varphi \varphi \varphi \rangle$. We will relate the result to $\langle \zeta \zeta \zeta \rangle$ at horizon crossing to avoid the error that would be induced by the superhorizon evolution of φ . To perform this matching, we, in principle, need to know the relation between φ and ζ up to second order. However, the quadratic term is slow-roll suppressed, so it suffices to use the linear relationship $\zeta = -(H/\dot{\phi}) \varphi$. Equation (9.103) therefore implies

$$S = \int d^4x a^3 \left[\varepsilon \left(\dot{\zeta}^2 - \frac{1}{a^2} (\partial_i \zeta)^2 \right) - \frac{\dot{\phi}^4}{2\Lambda^4 H^3} \left(\dot{\zeta}^3 - \frac{1}{a^2} \dot{\zeta} (\partial_i \zeta)^2 \right) \right], \quad (9.104)$$

from which can read off the interaction Hamiltonian $H_{\text{int}} = -L_3$. Substituting this into (9.95), we can compute the bispectrum associated to the two cubic interactions:

$$B_{\zeta^3} = -\Delta_\zeta^4 \left(\frac{\dot{\phi}^2}{\Lambda^4} \right) \frac{6k_1 k_2 k_3}{K^3}, \quad (9.105)$$

$$B_{\dot{\zeta}(\partial_i \zeta)^2} = \Delta_\zeta^4 \left(\frac{\dot{\phi}^2}{\Lambda^4} \right) \frac{1}{k_1 k_2 k_3} \left[\frac{4}{K} \sum_{i>j} k_i^2 k_j^3 - \frac{2}{K^2} \sum_{i \neq j} k_i^2 k_j^3 + 6 \frac{(k_1 k_2 k_3)^2}{K^3} - \frac{1}{2} \sum_{i=1}^3 k_i^3 \right], \quad (9.106)$$

where $K \equiv k_1 + k_2 + k_3$.

Exercise 9.11 Derive the results (9.105) and (9.106). Show that the amplitude of the bispectrum is

$$f_{\text{NL}} = \frac{35}{108} \frac{\dot{\phi}^2}{\Lambda^4}. \quad (9.107)$$

Note that consistency of the truncation used in (9.102) requires $\dot{\phi} < \Lambda^2$, so that $f_{\text{NL}} < 1$. $\rightarrow$

In [42], an alternative “EFT of inflation” was introduced not for the inflaton field ϕ , but directly for the inflationary *fluctuations*. In this EFT, larger self-interactions of the field fluctuations are possible, leading to enhanced non-Gaussian signals.

The basic idea is to think of inflation as a symmetry breaking phenomenon and parameterize the interactions of the associated Goldstone mode. Recall that de Sitter space has the following line element $ds^2 = -dt^2 + e^{2Ht} d\mathbf{x}^2$. Although this line element looks time dependent, it has a time translation symmetry

$$t \mapsto t + \lambda, \quad x^i \mapsto e^{-H\lambda} x^i, \quad (9.108)$$

where λ is a constant parameter. In order to have true cosmological evolution, this time translation symmetry must be broken. For example, the Universe may contain some matter fields

ψ_m , with time-dependent expectation values. During inflation, this might be the inflaton field ϕ or the inflationary vacuum energy density ρ . The expectation value $\psi_m(t)$ defines a preferred time slicing, with the different time slices defined and labeled by distinct values of ψ_m . Time translation invariance is therefore broken. Although the spacetime-independent transformation $t \mapsto t + \xi$, with ξ a constant parameter, remains a redundancy of the system even in the presence of a preferred time slicing, the spacetime-dependent transformation

$$t \mapsto t + \pi(t, \mathbf{x}) \quad (9.109)$$

does not leave the action invariant unless π is constant. An immediate consequence of the broken symmetry is the existence of a Goldstone excitation, corresponding to a spacetime-dependent transformation along the broken generator, that is, $U(t, \mathbf{x}) \equiv t + \pi(t, \mathbf{x})$.

A key feature of the Goldstone fluctuations π is that they are closely connected to cosmological observables. In particular, recall that adiabatic fluctuations are defined as (see Section 7.4)

$$\delta\rho_i = \rho_i(t + \pi(t, \mathbf{x})) - \rho_i. \quad (9.110)$$

At linear order, adiabatic fluctuations are therefore proportional to the Goldstone mode, $\delta\rho_i = \dot{\rho}_i \pi$. Since observations do not show any signs of deviation from purely adiabatic initial conditions, the Goldstone boson is the most efficient way to describe the data. In spatially flat gauge, we have $g_{ij} = a^2(t)\delta_{ij}$ and all metric perturbations are related to the Goldstone mode by the Einstein constraint equations. However, for purely adiabatic fluctuations, we can also perform a time shift, $t \mapsto t - \pi(t, \mathbf{x})$, to remove all matter fluctuations, $\delta\rho_i \mapsto \delta\rho_i \equiv 0$. This transformation induces an isotropic perturbation to the spatial part of the metric,

$$\delta g_{ij} = a^2(t) e^{2\zeta(t, \mathbf{x})} \delta_{ij}, \quad \text{where} \quad \zeta = -H\pi. \quad (9.111)$$

Thus, the curvature perturbation ζ in comoving gauge is proportional to the Goldstone boson π in spatially flat gauge. For nearly constant H , we can therefore think of ζ and π interchangeably.

One way to construct the action for the Goldstone boson is by writing down the most general Lorentz-invariant action for the field $U \equiv t + \pi$ and then evaluate the time derivatives. Alternatively, we first go to unitary gauge ($\pi \equiv 0$), where the field π is “eaten” by the metric $g_{\mu\nu}$ and then write all fluctuations allowed by the residual symmetries. After gauge fixing, the action does not have to respect the full diffeomorphism invariance, but only has to be invariant under time-dependent spatial diffeomorphisms, $x^i \mapsto x^i + \xi^i(t, x^j)$. Besides terms that are invariant under all diffeomorphisms (such as curvature invariants like R and $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$), the reduced symmetry of the system now allows many new terms in the action. For example, arbitrary functions of g^{00} , as well as of the extrinsic and intrinsic curvatures of the spatial slices, are allowed.

To be concrete, we will restrict our discussion to the leading universal action, as well as corrections made from δg^{00} :

$$S = \int d^4x \sqrt{-g} \left[\left(\frac{M_{\text{Pl}}^2}{2} R - \Lambda(t) - c(t)g^{00} \right) + \sum_{n=2}^{\infty} \frac{M_n^4(t)}{n!} (\delta g^{00})^n + \dots \right]. \quad (9.112)$$

The functions $\Lambda(t)$ and $c(t)$ are fixed by the FRW background above which we are expanding. Terms linear in fluctuations correspond to tadpoles and must vanish if we expand around a

consistent background solution. Given an arbitrary cosmological background with expansion rate $H(t)$, this requires that $\Lambda(t) = M_{\text{Pl}}^2(3H^2 + \dot{H})$ and $c(t) = -M_{\text{Pl}}^2\dot{H}$.

To make the dynamics of the theory defined by (9.112) more transparent, we perform a spacetime-dependent time reparameterization,

$$t \mapsto t' = t + \pi(t, x^i), \quad (9.113)$$

$$x_i \mapsto x'_i = x_i. \quad (9.114)$$

The metric component g^{00} transforms as

$$g^{00} \mapsto g^{\mu\nu} \partial_\mu(t + \pi) \partial_\nu(t + \pi), \quad (9.115)$$

so that (9.112) becomes

$$\begin{aligned} S = \int d^4x \sqrt{-g} & \left[\frac{M_{\text{Pl}}^2}{2} R - M_{\text{Pl}}^2 \left(3H^2(t + \pi) + \dot{H}(t + \pi) \right) \right. \\ & + M_{\text{Pl}}^2 \dot{H} (g^{00} + 2\partial_\mu \pi g^{0\mu} + \partial_\mu \pi \partial_\nu \pi g^{\mu\nu}) \\ & \left. + \sum_{n=2}^{\infty} \frac{M_n^4(t + \pi)}{n!} (1 + g^{00} + 2\partial_\mu \pi g^{0\mu} + \partial_\mu \pi \partial_\nu \pi g^{\mu\nu})^n \right]. \end{aligned} \quad (9.116)$$

We see that the Goldstone mode π mixes with the metric perturbations in a nontrivial way. However, at frequencies above $\omega_{\text{mix}} = \sqrt{\varepsilon}H$, the Goldstone mode decouples from the metric perturbations $\delta g^{\mu\nu}$, and we can evaluate the action in the *unperturbed* spacetime $\bar{g}^{\mu\nu}$:

$$\begin{aligned} S = \int d^4x \sqrt{-g} & \left[\frac{M_{\text{Pl}}^2}{2} R - M_{\text{Pl}}^2 \left(3H^2(t + \pi) + \dot{H}(t + \pi) \right) + M_{\text{Pl}}^2 \dot{H} (-1 - 2\dot{\pi} + (\partial_\mu \pi)^2) \right. \\ & \left. + \sum_{n=2}^{\infty} \frac{M_n^4(t + \pi)}{n!} (-2\dot{\pi} + (\partial_\mu \pi)^2)^n \right]. \end{aligned} \quad (9.117)$$

Note that $-2M_{\text{Pl}}^2\dot{H}\dot{\pi}$ cancels against $6M_{\text{Pl}}^2H\dot{H}\pi$ after integration by parts.

So far, this is valid for arbitrary FRW backgrounds. During inflation, the spacetime is quasi-de Sitter, for which the fractional change in H and $\dot{H}$ per Hubble time is small. We will further assume that the same holds for the time dependence of all coefficients in the effective action,²⁸ e.g. $|\dot{M}_2| \ll HM_2$. We will therefore drop the explicit time dependence of the coefficient functions in (9.117) and expand the action in powers of π . In Exercise 9.12, you will show that, up to cubic order, the Goldstone action then is

$$S_\pi = \int d^4x a^3 \frac{f_\pi^4}{c_s^3} \left[\frac{1}{2} \left(\dot{\pi}^2 - \frac{c_s^2}{a^2} (\partial_i \pi)^2 \right) + \frac{1 - c_s^2}{2c_s^2} \left(A \dot{\pi}^3 - \frac{c_s^2}{a^2} \dot{\pi} (\partial_i \pi)^2 \right) \right], \quad (9.118)$$

where c_s is the sound speed of the fluctuations, A is a dimensionless constant and $f_\pi^4 \equiv 2c_s M_{\text{Pl}}^2 |\dot{H}|$ is the symmetry breaking scale. The fact that the interaction $\dot{\pi} (\partial_i \pi)^2$ is fixed in terms of the speed of sound is a consequence of the nonlinearly realized symmetries of the Goldstone mode.

²⁸Assuming that all coefficients vary slowly implies that the action for the *fluctuations* is approximately time translation invariant. This additional global symmetry should not be confused with the broken time translation symmetry of the *background*.

Exercise 9.12 Derive the action (9.118) starting from (9.112). →

Exercise 9.13 Compute the three-point function arising from (9.118). Show that the bispectrum amplitude is

$$f_{\text{NL}} = \frac{5}{18} \left(\frac{25}{18} - \frac{2}{9} A \right) \left(\frac{1}{c_s^2} - 1 \right). \quad (9.119)$$

Note that this can be large for $c_s \ll 1$. →

9.8 Cosmological Collider

As we have seen in the previous section, computing the quantum correlations in single-field slow-roll inflation is a well-defined problem leading to clean predictions for the cosmological correlation functions. In this final section, we will look for deviations from those predictions associated with the existence of extra particles during inflation.

Massive particles during inflation can have two types of effects:

- Particles with masses $M \gg H$ can be integrated out, leading to an EFT of contact interactions of the remaining light fields (including the inflaton field ϕ). These *local* interactions create characteristic imprints in the non-Gaussianity of the inflaton fluctuations. In particular, the soft limits of the resulting correlation functions must satisfy the single-field consistency relation discussed in the previous section.
- Particles with masses $M \lesssim \text{few} \times H$ cannot be integrated out completely, but are produced nonperturbatively by the expanding spacetime. When these particles decay, they produce characteristic *non-local* signatures in cosmological correlators. The single-field consistency relation will be violated.

To illustrate this cosmological collider physics, we consider the following two-field example:

$$\mathcal{L} = -\frac{1}{2}(\partial_\mu \phi)^2 - V(\phi) - \frac{1}{2}(\partial_\mu \sigma)^2 - \frac{1}{2}M^2\sigma^2 + \frac{\sigma(\partial_\mu \phi)^2}{\tilde{\Lambda}}. \quad (9.120)$$

For $M \gg H$, we can integrate out the massive field σ , which gives the effective action shown in (9.102), with $\Lambda^2 \equiv \tilde{\Lambda}M$. The bispectrum of this effective theory was computed in Exercise 9.11. It was shown that the signal peaks in the equilateral configuration, $k_1 \sim k_2 \sim k_3$, and satisfies the single-field consistency condition (9.99).

For $M \lesssim \text{few} \times H$, there are effects of massive particles that cannot be captured by local inflaton interactions, i.e. the field cannot be integrated out completely. The non-locality of the effective inflaton interactions will be reflected in a characteristic non-analyticity in its correlation functions. In

Recall that the two-point function of a massive scalar field in de Sitter space is

$$\langle \sigma_{\mathbf{k}}(\eta) \sigma_{\mathbf{k}'}(\eta') \rangle' = \frac{\pi}{4} H^2 (\eta \eta')^{3/2} e^{-\pi \mu} H_{i\mu}(-k\eta) H_{i\mu}^*(-k\eta'), \quad (9.121)$$

where $H_{i\mu} \equiv H_{i\mu}^{(1)}$ is the Hankel function of the first kind and $\mu \equiv \sqrt{M^2/H^2 - 9/4}$. For now, we assume that the field is massive enough for μ to be real. The local part of the two-point function has support only at coincident points in position space, while the non-local part describes correlations over long distances. In Fourier space, the local and non-local parts of the two-point function are analytic and non-analytic in the momentum k , respectively. In the late-time limit, we can split (9.121) into its local and non-local parts

$$\lim_{\eta, \eta' \rightarrow 0} \langle \sigma_{\mathbf{k}}(\eta) \sigma_{\mathbf{k}'}(\eta') \rangle'_{\text{local}} = \frac{H^2(\eta\eta')^{3/2}}{4\pi} \Gamma(-i\mu) \Gamma(i\mu) \left[e^{\pi\mu} \left(\frac{\eta}{\eta'} \right)^{i\mu} + e^{-\pi\mu} \left(\frac{\eta}{\eta'} \right)^{-i\mu} \right], \quad (9.122)$$

$$\lim_{\eta, \eta' \rightarrow 0} \langle \sigma_{\mathbf{k}}(\eta) \sigma_{\mathbf{k}'}(\eta') \rangle'_{\text{non-local}} = \frac{H^2(\eta\eta')^{3/2}}{4\pi} \left[\Gamma(-i\mu)^2 \left(\frac{k^2 \eta \eta'}{4} \right)^{i\mu} + \Gamma(i\mu)^2 \left(\frac{k^2 \eta \eta'}{4} \right)^{-i\mu} \right]. \quad (9.123)$$

Note that $\Gamma(\pm i\mu) \rightarrow e^{-\pi\mu/2}$ for large μ , resulting in an overall suppression of $e^{-\pi\mu}$ of the non-local contribution (9.123). In the following, we will see how the non-analyticity of the non-local part of the two-point function of σ is reflected in the inflationary correlators.

The correlations in the massive field σ become observable when they get converted to inflaton fluctuations through interactions such as the mixing term in (9.120):

$$\mathcal{L} \supset \frac{\sigma(\partial_\mu \phi)^2}{\Lambda}. \quad (9.124)$$

Evaluating one of the legs of the inflaton field on the background solution, $\bar{\phi}(t)$, leads to a bispectrum for the inflaton fluctuations φ (and hence of the curvature perturbations ζ); cf. Fig. 26. Computing this in the squeezed limit $k_3 \ll k_1$, we find (see Exercise 9.14):

$$\begin{aligned} \lim_{k_3 \rightarrow 0} \frac{\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle'}{P_\zeta(k_1) P_\zeta(k_3)} &= \frac{\varepsilon}{8\pi} \frac{M_{\text{Pl}}^2}{\Lambda^2} \frac{\pi^2}{\cosh^2(\pi\mu)} \left(\frac{k_3}{2k_1} \right)^{3/2} \\ &\times \left[(1 + i \sinh(\pi\mu)) \frac{(\frac{3}{2} + i\mu)(\frac{5}{2} + i\mu) \Gamma[-i\mu]^2}{\Gamma[\frac{1}{2} - i\mu]} \left(\frac{k_3}{4k_1} \right)^{i\mu} + (\mu \rightarrow -\mu) \right]. \end{aligned} \quad (9.125)$$

Let us comment on the main features of the result:

- The non-local effect associated with the massive particle exchange shows up as a non-analyticity in the squeezed limit. Between the horizon crossing times of the long mode and the short modes the amplitude of the massive field oscillates with a frequency set by the mass of the field (see Fig. 26). This leads to distinct oscillations in the bispectrum of curvature perturbations:

$$\lim_{k_3 \rightarrow 0} \langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle' \propto \left(\frac{k_3}{k_1} \right)^{3/2} \cos \left[\frac{M}{H} \log \left(\frac{k_3}{k_1} \right) + \delta \right], \quad (9.126)$$

where the mass M of the field is encoded in the frequency of the oscillations.

- Particles with spin J lead to a unique angular dependence:

$$\lim_{k_3 \rightarrow 0} \langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle' \propto P_J(\cos \theta), \quad (9.127)$$

where $\cos \theta \equiv \mathbf{k}_1 \cdot \mathbf{k}_3$ and P_J is a Legendre polynomial of degree J .

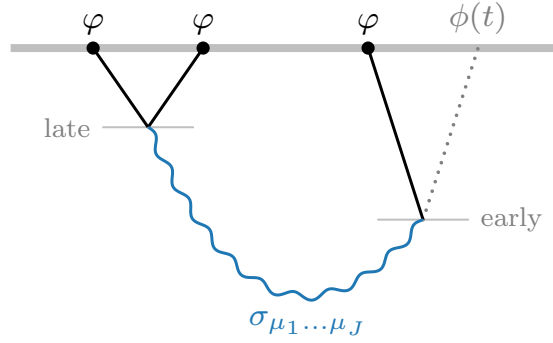


Figure 26: The spontaneous creation and decay of a massive particle σ produces a four-point function for the inflaton field ϕ . Evaluating one of the legs on the background solution, $\phi(t)$, leads to a three-point function for the inflaton fluctuations φ .

There are several effects that determine the amplitude of the signal:

- If we make the conservative assumption that the interaction in (9.124) is only of gravitational strength [43], then the scale determining the mixing between σ and ϕ is $\Lambda = M_{\text{Pl}}$. In that case, the amplitude of the bispectrum is at most of order the slow-roll signal, $f_{\text{NL}} \lesssim \mathcal{O}(\varepsilon)$. However, in principle, the mixing can be much larger. In particular, the mixing interaction in (9.124) remains perturbative as long as $\Lambda \geq \dot{\phi}^{1/2}$, allowing for non-Gaussianity of order $f_{\text{NL}} \lesssim \mathcal{O}(1)$. Finally, in the EFT for the inflationary fluctuations [42] the cutoff Λ can even be below $\dot{\phi}^{1/2}$ and the non-Gaussianity can be larger.
- In addition, the spontaneous production of massive particles with $M > \frac{3}{2}H$ is exponentially suppressed. This Boltzmann suppression is inherited by the amplitude of the bispectrum, $f_{\text{NL}} \propto e^{-\pi M/H}$. The creation of particles with $M < \frac{3}{2}H$, on the other hand, does not receive the same suppression and the signal can be larger [73, 74]. Instead of oscillations, the squeezed limit in that case has a monotonic scaling $(k_3/k_1)^\Delta$, where the scaling dimension Δ carries the information about the mass of the particle.
- For particles with $M > \frac{3}{2}H$, the squeezed limit is suppressed by a factor of $(k_3/k_1)^{3/2}$. This is to be compared to the squeezed limit for equilateral non-Gaussianity which scales as $(k_3/k_1)^2$. The signal from particles with $M < \frac{3}{2}H$ can be larger in the squeezed limit, $(k_3/k_1)^\Delta$, where $0 < \Delta < 3/2$.

Exercise 9.14 Derive the bispectrum (9.125) associated to the exchange of a massive field in the squeezed limit $k_3 \ll k_1$. ➔

10 Wavefunction Approach

Cosmological correlations can also be computed by means of the so-called “wavefunction of the universe.” This wavefunction is a slightly more primitive object than correlators themselves—the relation between them is similar to that between the S -matrix and scattering cross-sections. As a result, the wavefunction is somewhat simpler than correlators in certain ways that we will see. In this chapter, we will describe the wavefunction approach and show, in specific examples, how it reproduces the in-in results of the previous chapter. We will also discover that the wavefunction has an interesting analytic structure (i.e. a pattern of singularities) that will inform the bootstrap approach discussed in the next chapter.

10.1 Wavefunction of the Universe

Consider a set of bulk fields, $\Phi(t, \mathbf{x})$, which can include both matter fields (such as the inflaton) and metric fluctuations. We are interested in their spatial correlations. The information about these correlations is contained in the quantum-mechanical state of the system at a given time. It is convenient to describe states in the basis of field eigenstates, which (in the Heisenberg picture) satisfy

$$\hat{\Phi}(t, \mathbf{x})|\phi(\mathbf{x})\rangle = \phi(\mathbf{x})|\phi(\mathbf{x})\rangle, \quad (10.1)$$

where $\hat{\Phi}$ is a field operator and $\phi(\mathbf{x}) \equiv \phi(t, \mathbf{x})$ is the spatial profile of the field at a fixed time t . Any state of the system can be projected onto this basis, but we will primarily be interested in the (interacting) vacuum state of the system. Writing this state in the basis of field eigenstates, the overlap coefficients are

$$\Psi[\phi(\mathbf{x})] \equiv \langle \phi(\mathbf{x}) | 0 \rangle. \quad (10.2)$$

This *wavefunctional* expresses the overlap between the vacuum state and a given state of the late-time fluctuations, and therefore provides a probability density for spatial field configurations. Equal-time correlation functions of the field are then defined as

$$\langle \phi(\mathbf{x}_1) \cdots \phi(\mathbf{x}_n) \rangle = \frac{\int \mathcal{D}\phi \phi(\mathbf{x}_1) \cdots \phi(\mathbf{x}_n) |\Psi[\phi]|^2}{\int \mathcal{D}\phi |\Psi[\phi]|^2}, \quad (10.3)$$

which is a familiar formula from quantum mechanics applied to this field theory setup. This formula makes it clear that the wavefunctional and (equal-time) correlation functions contain the same information, though the translation between the two can be somewhat complicated in general. The fact that the wavefunction is in a sense a “square root” of correlation functions will lead to some simplifications in its structure, as we will see.

Since the wavefunctional is just a way of representing the state of a quantum field theory, it can be employed in a variety of situations, including in flat space. What is special about cosmology is that there is a distinguished time slice on which to consider the wavefunction—the future boundary of de Sitter space. In this $\eta_* \rightarrow 0$ limit, fields take on time-independent spatial profiles and the fact that the symmetries of de Sitter space preserve this spatial slice means that the invariance of the vacuum wavefunctional under de Sitter symmetry is easy to implement. In

many cases, this makes it possible to reconstruct the wavefunctional directly on the boundary from these symmetry principles (see Chapter 11).

The wavefunction at a given time (which we can set to $t_* = 0$ without loss of generality) is formally computed by the following path integral

$$\Psi[\phi] = \int_{\substack{\Phi(0)=\phi \\ \Phi(-\infty^-)=0}} \mathcal{D}\Phi e^{iS[\Phi]}, \quad (10.4)$$

which defines a sum over field configurations that interpolate between the free theory vacuum in the far past (selected via the usual $i\epsilon$ prescription, $-\infty^- \equiv -\infty(1-i\epsilon)$) and a Dirichlet boundary condition for the field at $t_* = 0$, $\phi(\mathbf{x}) \equiv \phi(t_*, \mathbf{x})$. Clearly, Ψ depends on this choice of late-time boundary condition for the field, making it a functional.

In practice, we compute the path integral (10.4) in perturbation theory. At tree level, the path integral can then be approximated by its saddle point

$$\Psi[\phi] \approx \exp(iS[\Phi_{\text{cl}}]), \quad (10.5)$$

where Φ_{cl} is a solution to the classical equations of motion with the future boundary condition $\Phi_{\text{cl}}(t_*) = \phi$. We will almost exclusively be interested in tree-level processes, so that this lowest-order approximation of the wavefunction will be sufficient for our purposes, but it can be systematically corrected by including quantum fluctuations around the classical saddle, as is typical in path-integral computations.

Perturbative expansion

In perturbation theory, we expect fluctuations to be small, and so it is convenient to expand the wavefunction (10.2) in powers of the field fluctuations:

$$\begin{aligned} \Psi[\phi] = \exp \bigg(& -\frac{1}{2} \int d^3x_1 d^3x_2 \Psi_2(\mathbf{x}_1, \mathbf{x}_2) \phi(\mathbf{x}_1) \phi(\mathbf{x}_2) \\ & + \sum_{n=3}^{\infty} \frac{1}{n!} \int d^3x_1 \cdots d^3x_n \Psi_n(\underline{\mathbf{x}}) \phi(\mathbf{x}_1) \cdots \phi(\mathbf{x}_n) \bigg), \end{aligned} \quad (10.6)$$

where $\underline{\mathbf{x}} \equiv \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ denotes the set of all spatial coordinates and the kernels $\Psi_n(\underline{\mathbf{x}})$ are called *wavefunction coefficients*. These functions capture the statistics of the fluctuations ϕ . It is useful to write the wavefunction coefficients in momentum (or Fourier) space,

$$\int d^3x_1 \cdots d^3x_n \Psi_n(\underline{\mathbf{x}}) \phi(\mathbf{x}_1) \cdots \phi(\mathbf{x}_n) = \int \frac{d^3k_1 \cdots d^3k_n}{(2\pi)^{3n}} \Psi_n(\underline{\mathbf{k}}) \phi_{\mathbf{k}_1} \cdots \phi_{\mathbf{k}_n}, \quad (10.7)$$

which is the natural habitat of cosmological correlations. As a consequence of translation invariance, the momentum-space wavefunction coefficients contain a momentum-conserving delta function. It is often useful to extract this delta function and write

$$\Psi_n(\underline{\mathbf{k}}) \equiv (2\pi)^3 \delta(\mathbf{k}_1 + \cdots + \mathbf{k}_n) \psi_n(\underline{\mathbf{k}}). \quad (10.8)$$

In de Sitter space, these wavefunction coefficients can also be interpreted as correlation functions of operators, O , which are dual to the bulk fields, $\Psi_n = \langle O_{\mathbf{k}_1} \cdots O_{\mathbf{k}_n} \rangle$, because they have the same kinematic properties [41]. We will just use $\langle O_{\mathbf{k}_1} \cdots O_{\mathbf{k}_n} \rangle$ as an alternative notation for the wavefunction coefficients Ψ_n that will be particularly useful when we consider non-identical fields.

Correlators

The wavefunction coefficients $\psi_n(\mathbf{k})$ then contain all the information about correlations in the quantum state. To see this, we parameterize the wavefunction as

$$\Psi[\phi] = \exp \left(-\frac{1}{2} \int \frac{d^3 k_1 d^3 k_2}{(2\pi)^6} \langle O_{\mathbf{k}_1} O_{\mathbf{k}_2} \rangle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} + \frac{1}{3!} \int \frac{d^3 k_1 d^3 k_2 d^3 k_3}{(2\pi)^9} \langle O_{\mathbf{k}_1} O_{\mathbf{k}_2} O_{\mathbf{k}_3} \rangle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \right. \\ \left. + \frac{1}{4!} \int \frac{d^3 k_1 d^3 k_2 d^3 k_3 d^3 k_4}{(2\pi)^{12}} \langle O_{\mathbf{k}_1} O_{\mathbf{k}_2} O_{\mathbf{k}_3} O_{\mathbf{k}_4} \rangle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \phi_{\mathbf{k}_4} + \cdots \right). \quad (10.9)$$

We want to see in perturbation theory how these wavefunction coefficients are related to correlators. To do this, we expand the wavefunction around a Gaussian:

$$\Psi[\phi] = \exp \left(-\frac{1}{2} \int \frac{d^3 k_1 d^3 k_2}{(2\pi)^6} \langle O_{\mathbf{k}_1} O_{\mathbf{k}_2} \rangle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \right) \\ \times \left[1 + \frac{1}{3!} \int \frac{d^3 k_1 d^3 k_2 d^3 k_3}{(2\pi)^9} \langle O_{\mathbf{k}_1} O_{\mathbf{k}_2} O_{\mathbf{k}_3} \rangle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \right. \\ \left. + \frac{1}{2(3!)^2} \int \frac{d^3 k_1}{(2\pi)^3} \cdots \frac{d^3 k_6}{(2\pi)^3} \langle O_{\mathbf{k}_1} O_{\mathbf{k}_2} O_{\mathbf{k}_3} \rangle \langle O_{\mathbf{k}_4} O_{\mathbf{k}_5} O_{\mathbf{k}_6} \rangle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \phi_{\mathbf{k}_4} \phi_{\mathbf{k}_5} \phi_{\mathbf{k}_6} \right. \\ \left. + \frac{1}{4!} \int \frac{d^3 k_1 d^3 k_2 d^3 k_3 d^3 k_4}{(2\pi)^{12}} \langle O_{\mathbf{k}_1} O_{\mathbf{k}_2} O_{\mathbf{k}_3} O_{\mathbf{k}_4} \rangle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \phi_{\mathbf{k}_4} + \cdots \right], \quad (10.10)$$

where we are displaying the terms that will contribute up to the four-point function at tree level. We can then use the quantum-mechanics formula (10.3) to compute correlation functions by performing Gaussian integrals over the fields ϕ .

For example, the simplest correlation function we could compute is the two-point function, which is given by

$$\langle \phi_1 \phi_2 \rangle' = \frac{1}{2 \text{Re} \langle O_1 O_2 \rangle'}, \quad (10.11)$$

where we have introduced the condensed notation $\phi_a \equiv \phi_{\mathbf{k}_a}$ (and similar for O), and where the prime on the wavefunction coefficient $\langle \cdots \rangle'$ indicates that we have removed the momentum-conserving delta function. Similarly, the three-point function is

$$\langle \phi_1 \phi_2 \phi_3 \rangle' = \frac{2 \text{Re} \langle O_1 O_2 O_3 \rangle'}{2 \text{Re} \langle O_1 O_1 \rangle' 2 \text{Re} \langle O_2 O_2 \rangle' 2 \text{Re} \langle O_3 O_3 \rangle'}, \quad (10.12)$$

and the four-point function becomes

$$\langle \phi_1 \phi_2 \phi_3 \phi_4 \rangle = \frac{2 \text{Re} \langle O_1 O_2 O_3 O_4 \rangle}{2 \text{Re} \langle O_1 O_1 \rangle' 2 \text{Re} \langle O_2 O_2 \rangle' 2 \text{Re} \langle O_3 O_3 \rangle' 2 \text{Re} \langle O_4 O_4 \rangle'} \\ + \frac{2 \text{Re} \langle O_1 O_2 O_I \rangle 2 \text{Re} \langle O_I O_3 O_4 \rangle}{2 \text{Re} \langle O_1 O_1 \rangle' 2 \text{Re} \langle O_2 O_2 \rangle' 2 \text{Re} \langle O_I O_I \rangle' 2 \text{Re} \langle O_3 O_3 \rangle' 2 \text{Re} \langle O_4 O_4 \rangle'} + \text{perms.}, \quad (10.13)$$

where “perms.” denotes the sum over other permutations of the external operators (i.e. other exchange channels) because we have taken all the operators to be identical for simplicity. Notice that the correlation functions of the fields (10.11), (10.12), and (10.13) only depend on the *real* part of the wavefunction coefficients. This is analogous to ordinary quantum mechanics where one must study expectation values of momenta in order to be sensitive to the phase of the wavefunction.

Exercise 10.1 Derive the formulas (10.11)–(10.13).



We see that in perturbation theory there is a direct correspondence between correlators and wavefunction coefficients. In the following, we will learn how to compute the wavefunction coefficients and then use the formulas of this section to maps them to correlators (which can then be compared to results derived directly in Chapter 9 using the in-in formalism).

10.2 Warmup in Quantum Mechanics

To get an intuitive understanding of the wavefunction approach, it is instructive to first apply it to a few simple examples in quantum mechanics.

Harmonic oscillator

Consider a *simple harmonic oscillator*, whose action is

$$S[\Phi] = \int dt \left(\frac{1}{2} \dot{\Phi}^2 - \frac{1}{2} \omega^2 \Phi^2 \right), \quad (10.14)$$

where Φ is the deviation from equilibrium (which we called q in Section 1.1) and ω is the constant frequency of the oscillator. On-shell, the action can equivalently be written as a pure boundary term

$$\begin{aligned} S[\Phi_{\text{cl}}] &= \int_{t_i}^{t_*} dt \left[\frac{1}{2} \partial_t (\dot{\Phi}_{\text{cl}} \Phi_{\text{cl}}) - \frac{1}{2} \Phi_{\text{cl}} \underbrace{(\ddot{\Phi}_{\text{cl}} + \omega^2 \Phi_{\text{cl}})}_{=0} \right] \\ &= \frac{1}{2} \dot{\Phi}_{\text{cl}} \Phi_{\text{cl}} \Big|_{t=t_*}, \end{aligned} \quad (10.15)$$

where we have integrated by parts and used the fact that the classical solution satisfies the equation of motion $\ddot{\Phi}_{\text{cl}} + \omega^2 \Phi_{\text{cl}} = 0$.

To determine the classical solution $\Phi_{\text{cl}}(t)$, we have to specify two boundary conditions: We require that 1) the late-time value of the oscillator position is $\Phi_{\text{cl}}(t_* \equiv 0) = \phi$, and 2) the early-time limit is $\Phi_{\text{cl}}(t \rightarrow -\infty) \sim e^{i\omega t}$. The latter condition ensures that the ground state wavefunction is normalizable. The unique solution satisfying these boundary conditions is $\Phi_{\text{cl}}(t) = \phi e^{i\omega t}$. Substituting this into (10.15), we get

$$S = \frac{i\omega}{2} \phi^2 \quad \implies \quad \exp(iS) = \exp\left(-\frac{\omega}{2} \phi^2\right) \quad \implies \quad |\Psi[\phi]|^2 = e^{-\omega \phi^2}. \quad (10.16)$$

We have recovered the familiar fact that the ground state wavefunction of the simple harmonic oscillator is a Gaussian.²⁹ The width of this Gaussian determines the size of the zero-point fluctuations of the oscillator:

$$\langle \phi^2 \rangle = \frac{1}{2\omega} . \quad (10.17)$$

In Exercise 1.1, we derived the same result using canonical quantization.

Free quantum field theory in Fourier space is essentially equivalent to a set of harmonic oscillators indexed by the continuous wavenumber $\mathbf{k}$. To see this, we write the action of a free scalar in Fourier space:

$$\begin{aligned} S &= \int dt d^3x \left(\frac{1}{2} \dot{\Phi}^2 - \frac{1}{2} (\nabla \Phi)^2 - \frac{1}{2} m^2 \Phi^2 \right) \\ &= \int dt \frac{d^3k}{(2\pi)^3} \left(\frac{1}{2} \dot{\Phi}_{\mathbf{k}} \dot{\Phi}_{-\mathbf{k}} - \frac{1}{2} (k^2 + m^2) \Phi_{\mathbf{k}} \Phi_{-\mathbf{k}} \right) , \end{aligned} \quad (10.18)$$

where $\Phi_{\mathbf{k}}(t)$ are the Fourier modes of the field. We see that these Fourier modes each satisfy the equation of a simple harmonic oscillator with frequencies $\omega_k \equiv \sqrt{k^2 + m^2}$. The wavefunction, and the associated zero-point fluctuations, can then be written as a sum over the corresponding results for each harmonic oscillator:

$$\Psi[\phi] = \exp \left(-\frac{1}{2} \int \frac{d^3k}{(2\pi)^3} \omega_k \phi_{\mathbf{k}} \phi_{-\mathbf{k}} \right) \implies \langle \phi_{\mathbf{k}} \phi_{-\mathbf{k}} \rangle' = \frac{1}{2\omega_k} . \quad (10.19)$$

The treatment of free fields in quantum field theory is therefore as simple as that of an infinite collection of uncoupled harmonic oscillators in quantum mechanics.

Time-dependent oscillator

In Section 8.2, we showed that a free field in de Sitter space (and also the fluctuations during inflation) satisfy the equation of motion of a harmonic oscillator with a time-dependent frequency (for each Fourier mode). It is therefore useful to consider the following *time-dependent oscillator*:

$$S[\Phi] = \int dt \left(\frac{1}{2} A(t) \dot{\Phi}^2 - \frac{1}{2} B(t) \Phi^2 \right) , \quad (10.20)$$

where $A(t)$ and $B(t)$ are arbitrary functions of time. On-shell, this action can again be written as a boundary term

$$\begin{aligned} S[\Phi_{\text{cl}}] &= \int_{t_i}^{t_*} dt \left[\frac{1}{2} \partial_t (A \dot{\Phi}_{\text{cl}} \Phi_{\text{cl}}) - \frac{1}{2} \Phi_{\text{cl}} \underbrace{(\partial_t (A \dot{\Phi}_{\text{cl}}) + B \Phi_{\text{cl}})}_{=0} \right] \\ &= \frac{1}{2} A \dot{\Phi}_{\text{cl}} \Phi_{\text{cl}} \Big|_{t=t_*} , \end{aligned} \quad (10.21)$$

where we have used that the classical solution satisfies $\partial_t (A \dot{\Phi}_{\text{cl}}) + B \Phi_{\text{cl}} = 0$.

²⁹If we wanted to reproduce the correct phase of the wavefunction $\sim e^{i\omega t/2}$, we would have to compute the Gaussian integral for fluctuations around this classical trajectory. See Appendix A of [75] for a discussion. Since this phase is independent of ϕ , it is not important for our purposes.

We again need to find the classical solution with the correct boundary conditions to describe the ground state. We can write the desired solution abstractly as $\Phi_{\text{cl}}(t) = \phi K(t)$. Substituting this into the equation of motion, we obtain a differential equation for $K(t)$:

$$\partial_t \left(A(t) \dot{K}(t) \right) + B(t) K(t) = 0. \quad (10.22)$$

The boundary conditions that we impose on the solution are $K(t_*) = 1$ and $K(-\infty) \sim e^{i\omega t}$. Of course, the precise form of $K(t)$ depends on $A(t)$ and $B(t)$, but given those functions we can then solve the equation of motion subject to the prescribed boundary conditions to obtain K .

Substituting this solution into (10.21), we have the following expression for the ground state wavefunction

$$\Psi[\phi] = \exp(iS) = \exp \left(\frac{i}{2} (A \partial_t \log K) \Big|_{t=t_*} \phi^2 \right), \quad (10.23)$$

and hence we can compute the two-point correlator using

$$|\Psi[\phi]|^2 = \exp \left(-\text{Im}(A \partial_t \log K) \Big|_{t=t_*} \phi^2 \right) \implies \langle \phi^2 \rangle = \frac{1}{2 \text{Im}(A \partial_t \log K) \Big|_{t=t_*}}. \quad (10.24)$$

As we will see in the next section, this result will be directly applicable to fields in de Sitter space.

10.3 Free Fields in De Sitter

Having gained some intuition for the wavefunction in the quantum-mechanical setting, we are now ready to use it to compute de Sitter correlators.

We start with the action of a free massive scalar field in a curved background:

$$S = \int d^4x \sqrt{-g} \left(-\frac{1}{2} g^{\mu\nu} \partial_\mu \Phi \partial_\nu \Phi - \frac{1}{2} m^2 \Phi^2 \right). \quad (10.25)$$

Substituting the de Sitter metric (in conformal time), this becomes

$$\begin{aligned} S &= \int d\eta d^3x a^4(\eta) \left[\frac{1}{2a^2(\eta)} ((\Phi')^2 - (\nabla \Phi)^2) - \frac{1}{2} m^2 \Phi^2 \right] \\ &= \frac{1}{2} \int d\eta \frac{d^3k}{(2\pi)^3} \left[\frac{1}{(H\eta)^2} \Phi_{\mathbf{k}}' \Phi_{-\mathbf{k}}' - \frac{1}{(H\eta)^2} \left(k^2 + \frac{m^2}{(H\eta)^2} \right) \Phi_{\mathbf{k}} \Phi_{-\mathbf{k}} \right], \end{aligned} \quad (10.26)$$

where, in the second line, we have written the field in terms of its Fourier components and used $a(\eta) = -1/(H\eta)$. We see that the action for each Fourier mode is the same as that of the time-dependent oscillator in (10.20), with the identifications

$$A(\eta) \equiv \frac{1}{(H\eta)^2}, \quad (10.27)$$

$$B(\eta) \equiv \frac{1}{(H\eta)^2} \left(k^2 + \frac{m^2}{(H\eta)^2} \right). \quad (10.28)$$

We can therefore apply the result (10.24) to compute the wavefunction and the variance of the field.

The classical equation of motion is

$$(A\Phi'_{\text{cl}})' + B\Phi_{\text{cl}} = 0 \quad \implies \quad \Phi''_{\text{cl}} + \frac{2}{\eta}\Phi'_{\text{cl}} + \left(k^2 + \frac{(m/H)^2}{\eta^2}\right)\Phi_{\text{cl}} = 0. \quad (10.29)$$

On small scales, $|k\eta| \gg 1$, the equation reduces to that of simple harmonic oscillator with frequency k . On large scales, $|k\eta| \rightarrow 0$, the harmonic oscillator becomes over-damped and the fluctuations freeze out (for $m = 0$) or decay slowly (for $m > 0$).

We write the solution to (10.29) as

$$\Phi_{\text{cl}}(\eta) \equiv \phi K(\eta), \quad (10.30)$$

where the function $K(\eta)$ approaches unity at late times and oscillates like $e^{ik\eta}$ at early times. For massless fields, we have

$$\begin{aligned} K(\eta) &= (1 - ik\eta)e^{ik\eta}, \\ \log K(\eta) &= \log(1 - ik\eta) + ik\eta. \end{aligned} \quad (10.31)$$

Substituting this into (10.24) gives

$$\begin{aligned} \text{Im}(A\partial_\eta \log K)|_{\eta=\eta_*} &= \frac{1}{(H\eta_*)^2} \text{Im} \left(\frac{-ik}{1 - ik\eta_*} + ik \right) \\ &= \frac{1}{(H\eta_*)^2} \text{Im} \left(\frac{k^2\eta_* + ik^3\eta_*^2}{1 + k^2\eta_*^2} \right) \xrightarrow{\eta_* \rightarrow 0} \frac{k^3}{H^2}, \end{aligned} \quad (10.32)$$

where we have taken $\eta_* \rightarrow 0$ at the end. The late-time two-point function of the field therefore is

$$\langle \phi_{\mathbf{k}} \phi_{-\mathbf{k}} \rangle' = \frac{H^2}{2k^3}, \quad (10.33)$$

which is the same as the result (8.32) derived by canonical quantization in Section 8.2.

For massive fields, the relevant solution to (10.29) has

$$\begin{aligned} K(\eta) &= \left(\frac{\eta}{\eta_*} \right)^{3/2} \frac{H_\nu^{(2)}(-k\eta)}{H_\nu^{(2)}(-k\eta_*)}, \\ \log K(\eta) &= \frac{3}{2} \log \eta + \log [H_\nu^{(2)}(-k\eta)], \end{aligned} \quad (10.34)$$

and hence the width of the wavefunction is given by

$$\begin{aligned} \text{Im}(A\partial_\eta \log K)|_{\eta=\eta_*} &= \frac{1}{(H\eta_*)^2} \text{Im} \left(\frac{\partial_\eta H_\nu^{(2)}(-k\eta)}{H_\nu^{(2)}(-k\eta)} \right) \Big|_{\eta=\eta_*} \\ &= \frac{1}{(H\eta_*)^2} \frac{1}{|H_\nu^{(2)}(-k\eta_*)|^2} \text{Im} \left(H_\nu^{(2)*}(-k\eta) \partial_\eta H_\nu^{(2)}(-k\eta) \right) \Big|_{\eta=\eta_*}. \end{aligned} \quad (10.35)$$

Consider first the case $m < \frac{3}{2}H$, where the order ν of the Hankel function is real. We then have $H_\nu^{(2)*} = H_\nu^{(1)}$ and hence

$$\text{Im} \left(H_\nu^{(2)*} \partial_\eta H_\nu^{(2)} \right) \Big|_{\eta=\eta_*} = \frac{1}{2i} \left(H_\nu^{(1)} \partial_\eta H_\nu^{(2)} - H_\nu^{(2)} \partial_\eta H_\nu^{(1)} \right). \quad (10.36)$$

We recognize the term in the brackets as the Wronskian of the Hankel function:

$$H_\nu^{(1)} \partial_\eta H_\nu^{(2)} - H_\nu^{(2)} \partial_\eta H_\nu^{(1)} = -\frac{4i}{\pi} \frac{1}{\eta}. \quad (10.37)$$

Substituting this into (10.35), we get

$$\text{Im}(A \partial_\eta \log K)|_{\eta=\eta_*} = \frac{1}{(H\eta_*)^2} \frac{1}{|H_\nu^{(2)}(-k\eta_*)|^2} \left(-\frac{2}{\pi} \frac{1}{\eta_*} \right). \quad (10.38)$$

The two-point function of the field is therefore

$$\langle \phi_{\mathbf{k}} \phi_{-\mathbf{k}} \rangle' = \frac{\pi}{4} H^2(-\eta_*)^3 H_\nu^{(1)}(-k\eta_*) H_\nu^{(2)}(-k\eta_*), \quad (10.39)$$

which is consistent with the result obtained in Exercise 8.6.

Exercise 10.2 Prove equation (10.37).

Hint: Take a derivative of the Wronskian W and use the Bessel equation to obtain a first-order differential equation for W . Find the solution of this equation, fixing the integration constant by considering the large argument limit of the Hankel functions. →

The two-point function (10.39) can actually be analytically continued to $m > \frac{3}{2}H$ by sending $\nu \rightarrow i\mu$, though showing this requires a little bit of work.

Exercise 10.3 Show that for $m > \frac{3}{2}H$, where $\nu \equiv i\mu$ is pure imaginary, the power spectrum becomes

$$\begin{aligned} \langle \phi_{\mathbf{k}} \phi_{-\mathbf{k}} \rangle' &= \frac{\pi}{4} H^2(-\eta_*)^3 e^{\pi\mu} |H_{i\mu}^{(2)}(-k\eta_*)|^2 \\ &= \frac{\pi}{4} H^2(-\eta_*)^3 H_{i\mu}^{(2)}(-k\eta_*) H_{i\mu}^{(1)}(-k\eta_*). \end{aligned} \quad (10.40)$$

The last expression holds both for real and imaginary orders of the Hankel function. →

10.4 Anharmonic Oscillator

Our main interest in these lectures is not in free fields, but in their interactions and the associated higher-point correlations. In this section, we will use the wavefunction approach to study higher-point correlations and show how this reproduces the in-in results of Chapter 9.

Before we study interacting field theory, it is again useful to first study a simpler example within quantum mechanics—the *anharmonic oscillator*. Consider the following action

$$S[\Phi] = \int dt \left(\frac{1}{2} \dot{\Phi}^2 - \frac{1}{2} \omega^2 \Phi^2 - \frac{1}{3} \lambda \Phi^3 \right), \quad (10.41)$$

which is the ordinary harmonic oscillator perturbed by the addition of a cubic interaction.³⁰ As before, we want to determine its ground state wavefunction by computing the on-shell action,

³⁰This model is, of course, unstable. However, since we are only interested in perturbatively correcting the ground state of the ordinary harmonic oscillator, this instability will not be important. One could alternatively consider the ϕ^4 anharmonic oscillator if this instability is troubling.

$S[\Phi_{\text{cl}}]$. The classical solution must obey the same boundary conditions as for the unperturbed oscillator: $\Phi_{\text{cl}}(t=0) = \phi$ and $\Phi_{\text{cl}}(t \rightarrow -\infty) \sim e^{i\omega t}$. The complication in this case is that the equation of motion is nonlinear

$$\ddot{\Phi} + \omega^2 \Phi = -\lambda \Phi^2, \quad (10.42)$$

and the solutions cannot be written in closed form. We can, however, write a formal solution to this equation by treating the $\lambda \Phi^2$ term as a source:

$$\Phi_{\text{cl}}(t) = \phi e^{i\omega t} + i \int dt' G(t, t') (-\lambda \Phi_{\text{cl}}^2(t')), \quad (10.43)$$

where $\phi e^{i\omega t}$ is the solution to the free equation of motion and $G(t, t')$ is a Green's function for the harmonic oscillator, which satisfies $(\partial_t^2 + \omega^2)G(t, t') = -i\delta(t - t')$. The boundary conditions we specify for Φ_{cl} determine what Green's function we should use. In this case, the nontrivial boundary condition that we have to impose is $G(0, t') = G(t, 0) = 0$. This guarantees that only the $\phi e^{i\omega t}$ part of the solution contributes at $t = 0$, as this piece already satisfies the boundary condition $\Phi_{\text{cl}}(0) = \phi$ by itself. The relevant Green's function is then

$$G(t, t') = \frac{1}{2\omega} \left(e^{-i\omega(t-t')} \theta(t-t') + e^{i\omega(t-t')} \theta(t'-t) - e^{i\omega(t+t')} \right), \quad (10.44)$$

which can be derived in the usual way.

Exercise 10.4 Derive the Green's function (10.44) for the harmonic oscillator. →

To derive the wavefunction of the oscillator in the saddle-point approximation, we have to compute the on-shell action, $S[\Phi_{\text{cl}}]$. It is convenient to first write (10.41) in the following form

$$\begin{aligned} S[\Phi] &= \int dt \left[\frac{1}{2} \partial_t(\Phi \dot{\Phi}) - \frac{1}{2} \Phi(\ddot{\Phi} + \omega^2 \Phi) - \frac{\lambda}{3} \Phi^3 \right] \\ &= \frac{1}{2} \Phi \dot{\Phi} \Big|_{t=0} + \int dt \left[-\frac{1}{2} \Phi(\ddot{\Phi} + \omega^2 \Phi) - \frac{\lambda}{3} \Phi^3 \right]. \end{aligned} \quad (10.45)$$

We have to be somewhat careful when we evaluate the boundary term for the classical solution (10.43). Although the Green's function vanishes for $t = 0$ (so that $\Phi_{\text{cl}}(0) = \phi$), its derivative does not, but instead satisfies the identity

$$\lim_{t \rightarrow 0} \partial_t G(t, t') = -ie^{i\omega t'}. \quad (10.46)$$

Using this, the boundary term becomes

$$\begin{aligned} \frac{1}{2} \Phi_{\text{cl}} \dot{\Phi}_{\text{cl}} \Big|_{t=0} &= \frac{1}{2} \phi \left(i\omega \phi - i\lambda \int dt' (-ie^{i\omega t'}) \Phi_{\text{cl}}^2(t') \right) \\ &= \frac{i\omega}{2} \phi^2 - \frac{\lambda}{2} \phi \int dt' e^{i\omega t'} \Phi_{\text{cl}}^2(t'), \end{aligned} \quad (10.47)$$

and substituting (10.43) into (10.45) gives

$$\begin{aligned} S[\Phi_{\text{cl}}] &= \frac{i\omega}{2} \phi^2 - \frac{\lambda}{2} \phi \int dt e^{i\omega t} \Phi_{\text{cl}}^2 \\ &\quad + \int dt \left[-\frac{1}{2} \left(\phi e^{i\omega t} - i\lambda \int dt' G(t, t') \Phi_{\text{cl}}^2(t') \right) \left(-\lambda \Phi_{\text{cl}}^2(t) \right) - \frac{\lambda}{3} \Phi_{\text{cl}}^3 \right]. \end{aligned} \quad (10.48)$$

The terms linear in ϕ cancel and we get the following *on-shell action*:

$$S[\Phi_{\text{cl}}] = \frac{i\omega}{2}\phi^2 - \frac{\lambda}{3} \int dt \Phi_{\text{cl}}^3(t) - \frac{i\lambda^2}{2} \int dt dt' G(t, t') \Phi_{\text{cl}}^2(t') \Phi_{\text{cl}}^2(t). \quad (10.49)$$

This way of organizing the terms makes it clear that the first term contributes only to the two-point function. At higher orders, both of the last two terms in (10.49) can be important.

Exercise 10.5 Consider an anharmonic oscillator with a general interaction Lagrangian L_{int} . Show that the on-shell action can be written as

$$S[\Phi_{\text{cl}}] = \frac{i\omega}{2}\phi^2 + \int dt L_{\text{int}} - \frac{i}{2} \int dt dt' G(t, t') \frac{\delta S_{\text{int}}}{\delta \Phi_{\text{cl}}(t')} \frac{\delta S_{\text{int}}}{\delta \Phi_{\text{cl}}(t)}, \quad (10.50)$$

which for $L_{\text{int}} = -\frac{1}{3}\lambda\Phi^3$ reduces to (10.49). ➔

Given the classical solution to the equations of motion, we can use (10.49) to compute the wavefunction. We obtain such a solution in perturbation theory by solving (10.43) order-by-order in the parameter λ . To do so, we expand

$$\Phi_{\text{cl}}(t) = \Phi^{(0)}(t) + \lambda\Phi^{(1)}(t) + \lambda^2\Phi^{(2)}(t) + \dots. \quad (10.51)$$

The lowest order ($\lambda = 0$) solution is just the solution to the unperturbed harmonic oscillator

$$\Phi^{(0)}(t) = \phi e^{i\omega t}. \quad (10.52)$$

At first order in λ , we find

$$\begin{aligned} \Phi^{(1)}(t) &= i \int dt' G(t, t') \left(-(\Phi^{(0)}(t'))^2 \right) \\ &= i \int dt' G(t, t') \left(-\phi^2 e^{2i\omega t'} \right) = \frac{\phi^2}{3\omega^2} (e^{2i\omega t} - e^{i\omega t}), \end{aligned} \quad (10.53)$$

where we have used the lowest-order solution as a source and performed the integral using the Green's function (10.44). By plugging this solution back into (10.43) and iterating, we can perturbatively solve to any order in λ desired.

Substituting (10.51) into (10.49), we obtain a perturbative expansion of the on-shell action:

$$\begin{aligned} S[\Phi_{\text{cl}}] &= \frac{i\omega}{2}\phi^2 - \frac{\lambda}{3} \int dt \left[(\Phi^{(0)})^3 + 3\lambda (\Phi^{(0)})^2 \Phi^{(1)} + \dots \right] \\ &\quad + \frac{\lambda^2}{2} \int dt (\Phi^{(0)}(t))^2 \int dt' iG(t, t') \left(-(\Phi^{(0)}(t'))^2 \right) + \dots, \end{aligned} \quad (10.54)$$

where we have displayed all terms up to $\mathcal{O}(\lambda^2)$. Identifying the last integral in the second line with the solution $\Phi^{(1)}$, we can also write

$$S[\Phi_{\text{cl}}] = \frac{i\omega}{2}\phi^2 - \frac{\lambda}{3} \int dt (\Phi^{(0)})^3 - \frac{\lambda^2}{2} \int dt (\Phi^{(0)})^2 \Phi^{(1)} + \dots. \quad (10.55)$$

Using the explicit solutions for the zeroth and first-order solutions—equations (10.52) and (10.53)—we find that the wavefunction of the anharmonic oscillator up to this order is

$$\Psi[\phi] \approx e^{iS[\Phi_{\text{cl}}]} = \exp\left(-\frac{\omega}{2}\phi^2 - \frac{\lambda}{9\omega}\phi^3 + \frac{\lambda^2}{72\omega^3}\phi^4 + \dots\right). \quad (10.56)$$

We see that the anharmonic interaction induces a deviation from the Gaussian shape of the wavefunction. These terms will contribute to the moments of the distribution of ϕ defined by $|\Psi[\phi]|^2$. For example, we could compute $\langle\phi^3\rangle$ in this theory and would get a nonzero answer, in contrast to its Gaussian counterpart where $\lambda = 0$. Similarly, higher moments $\langle\phi^n\rangle$ receive contributions order-by-order in λ . It is the analogue of these moments that we want to compute in the field theory context.

Exercise 10.6 Derive the first-order solution (10.53) by explicitly integrating the Green's function for the harmonic oscillator. Note that $\Phi^{(1)}(0) = 0$ as expected. Using (10.52) and (10.53), confirm the result (10.56) for the wavefunction. →

10.5 Wavefunction in Field Theory

After this brief detour into quantum mechanics, we return to our main problem of interest, computing the wavefunctional in interacting field theories.

On-shell action

Now that we know how to use the wavefunction, we turn to calculating it. Like we did for the harmonic oscillator, we are going to use the fact that, at tree level, the wavefunction is simply related to the on-shell action

$$\Psi[\phi] \approx \exp(iS[\Phi_{\text{cl}}]). \quad (10.57)$$

For concreteness, we consider the action of a massive scalar field in an arbitrary spacetime background:

$$S[\Phi] = \int d^4x \sqrt{-g} \left(-\frac{1}{2} g^{\mu\nu} \partial_\mu \Phi \partial_\nu \Phi - \frac{1}{2} m^2 \Phi^2 + \mathcal{L}_{\text{int}} \right), \quad (10.58)$$

where we have isolated the free part of the action and grouped all of the interactions schematically into $\mathcal{L}_{\text{int}}$. The equation of motion following from this action is

$$(\square - m^2)\Phi = -\frac{1}{\sqrt{-g}} \frac{\delta S_{\text{int}}}{\delta \Phi}, \quad (10.59)$$

where $\square \equiv g^{\mu\nu} \nabla_\mu \nabla_\nu$ is the d'Alembertian operator. As before, we can write a formal solution to this equation as

$$\Phi_{\text{cl}}(t, \mathbf{x}) = \int d^3x' \sqrt{\gamma} K(t, \mathbf{x}, \mathbf{x}') \phi(\mathbf{x}') + i \int d^4x' G(x, x') \frac{\delta S_{\text{int}}}{\delta \Phi(x')} \Big|_{\Phi=\Phi_{\text{cl}}}, \quad (10.60)$$

where γ_{ij} is the spatial metric and $K(t, \mathbf{x}, \mathbf{x}')$ is a solution to the free equation of motion that satisfies the boundary condition

$$K(t_*, \mathbf{x}, \mathbf{x}') = \frac{1}{\sqrt{\gamma}} \delta(\mathbf{x} - \mathbf{x}'), \quad (10.61)$$

and reduces to the vacuum solution as $t \rightarrow -\infty$ (we will see that this second boundary condition is most simply implemented in Fourier space). The solution $K(t, \mathbf{x}, \mathbf{x}')$ is often called the *bulk-to-boundary propagator*.³¹ The function $G(x, x')$ is a Green's function satisfying the equation

$$(\square - m^2)G(x, x') = \frac{i}{\sqrt{-g}} \delta(t - t') \delta(\mathbf{x} - \mathbf{x}'). \quad (10.62)$$

The precise Green's function of interest reduces to the vacuum in the far past and vanishes if we take one of the points to the boundary $G(x, x')|_{t=t_*} = G(x, x')|_{t'=t_*} = 0$. The latter constraint ensures that the formal solution (10.60) will satisfy the boundary condition $\Phi_{\text{cl}}(t_*, \mathbf{x}) = \phi(\mathbf{x})$. The function $G(x, x')$ is often called the *bulk-to-bulk propagator*.³² The two propagators K and G are related by the identity

$$\lim_{t \rightarrow t_*} g^{tt} \partial_t G(x, x') = iK(t', \mathbf{x}, \mathbf{x}'), \quad (10.63)$$

which is the analog of (10.46) for the harmonic oscillator.

As in the case of the anharmonic oscillator, we can substitute the formal solution (10.60) into the action (10.58). This is again most simply done by first integrating the kinetic term by parts:

$$S[\Phi] = - \int d^3x \sqrt{\gamma} \frac{1}{2} g^{tt} \Phi \dot{\Phi} \Big|_{t=t_*} + \int d^4x \sqrt{-g} \left(\frac{1}{2} \Phi (\square - m^2) \Phi + \mathcal{L}_{\text{int}}[\Phi] \right), \quad (10.64)$$

which generates a boundary term on the $t = t_*$ slice (with induced metric γ). As in the case of the anharmonic oscillator, we have to be careful when evaluating the boundary term. In particular, although $\Phi_{\text{cl}}(t_*, \mathbf{x}) = \phi(\mathbf{x})$, the time derivative of the solution (10.60) is

$$\begin{aligned} \partial_t \Phi_{\text{cl}}(t_*, \mathbf{x}) &= \int d^3x' \sqrt{\gamma} \partial_t K(t, \mathbf{x}, \mathbf{x}') \phi(\mathbf{x}') \Big|_{t=t_*} + i \int d^4x' \partial_t G(x, x') \frac{\delta S_{\text{int}}}{\delta \Phi_{\text{cl}}(x')} \\ &= \int d^3x' \sqrt{\gamma} \partial_t K(t, \mathbf{x}, \mathbf{x}') \phi(\mathbf{x}') \Big|_{t=t_*} - \int d^4x' \frac{K(t', \mathbf{x}, \mathbf{x}')}{g^{tt}} \frac{\delta S_{\text{int}}}{\delta \Phi_{\text{cl}}(x')}, \end{aligned} \quad (10.65)$$

where we have used the identity (10.63). Evaluating the boundary term in (10.64) on the classical solution therefore gives

$$\begin{aligned} - \int d^3x \sqrt{\gamma} \frac{1}{2} g^{tt} \Phi_{\text{cl}} \dot{\Phi}_{\text{cl}} \Big|_{t=t_*} &= - \frac{1}{2} \int d^3x \sqrt{\gamma} d^3x' \sqrt{\gamma} \phi(\mathbf{x}) \phi(\mathbf{x}') \left(g^{tt} \partial_t K(t, \mathbf{x}, \mathbf{x}') \right) \Big|_{t=t_*} \\ &\quad + \frac{1}{2} \int d^4x' d^3x \sqrt{\gamma} K(t', \mathbf{x}, \mathbf{x}') \phi(\mathbf{x}) \frac{\delta S_{\text{int}}}{\delta \Phi_{\text{cl}}(x')}, \end{aligned} \quad (10.66)$$

which should be compared to equation (10.47) in the anharmonic oscillator example. Note that the second line in (10.66) involves a bulk integral over x' . Equation (10.64) can then be written as

$$\begin{aligned} S[\Phi_{\text{cl}}] &= - \int d^3x \sqrt{\gamma} d^3x' \sqrt{\gamma} \frac{1}{2} \phi(\mathbf{x}) \phi(\mathbf{x}') \left(g^{tt} \partial_t K(t, \mathbf{x}, \mathbf{x}') \right) \Big|_{t=t_*} \\ &\quad - \frac{1}{2} \int d^4x d^3x' \sqrt{\gamma} K(t, \mathbf{x}', \mathbf{x}) \phi(\mathbf{x}') \frac{\delta S_{\text{int}}}{\delta \Phi_{\text{cl}}(x)} \\ &\quad + \int d^4x \left(- \frac{1}{2} \Phi_{\text{cl}} \frac{\delta S_{\text{int}}}{\delta \Phi_{\text{cl}}} + \sqrt{-g} \mathcal{L}_{\text{int}}[\Phi_{\text{cl}}] \right). \end{aligned} \quad (10.67)$$

³¹As we will see, in the diagrammatic interpretation of the computation of the wavefunction, this object is associated to lines connecting bulk vertices to the $t = t_*$ boundary slice. The nomenclature comes from AdS/CFT, where one computes the bulk on-shell action using essentially the same procedure [76, 77].

³²We will see that it is associated to internal lines in diagrams that connect two spacetime interaction vertices.

If we substitute (10.60) into the first term in the last line, the term involving ϕK will cancel with the second line and we will be left with the following *on-shell action*:

$$S[\Phi_{\text{cl}}] = - \int d^3x \sqrt{\gamma} d^3x' \sqrt{\gamma} \frac{1}{2} \phi(\mathbf{x}) \phi(\mathbf{x}') \left(g^{tt} \partial_t K(t, \mathbf{x}, \mathbf{x}') \right) \Big|_{t=t_*} + \int d^4x \sqrt{-g} \mathcal{L}_{\text{int}}[\Phi_{\text{cl}}] \\ - \frac{i}{2} \int d^4x d^4x' G(x, x') \frac{\delta S_{\text{int}}}{\delta \Phi_{\text{cl}}(x)} \frac{\delta S_{\text{int}}}{\delta \Phi_{\text{cl}}(x')}, \quad (10.68)$$

which is the analog of (10.49) for the anharmonic oscillator. Given a formula for the classical solution Φ_{cl} , we can then plug it into (10.68) to obtain the tree-level wavefunction.

Perturbation theory

We can solve (10.60) iteratively in perturbation theory. Like for the anharmonic oscillator, we expand the solution in the coupling constant λ :

$$\Phi_{\text{cl}}(x) = \Phi^{(0)}(x) + \lambda \Phi^{(1)}(x) + \lambda^2 \Phi^{(2)}(x) + \dots, \quad (10.69)$$

where $\Phi^{(0)}(x)$ is the solution to the free equation of motion, $\Phi^{(1)}(x)$ is the first-order correction,

$$\lambda \Phi^{(1)}(x) = i \int d^4x' G(x, x') \frac{\delta S_{\text{int}}}{\delta \Phi(x')} \Big|_{\Phi=\Phi^{(0)}}, \quad (10.70)$$

and so on.

In order to systematize the calculation, we assign a diagrammatic set of rules to the entire procedure of computing the classical solution order-by-order in perturbation theory and substituting this solution back into the action (10.68). To make this slightly more intuitive, consider the special case where $\mathcal{L}_{\text{int}} = -\frac{1}{3!} \lambda \Phi^3$. Up to $\mathcal{O}(\lambda^2)$, equation (10.68) can then be written as

$$iS[\Phi_{\text{cl}}] = -i \int d^3x \sqrt{\gamma} d^3x' \sqrt{\gamma} \frac{1}{2} \phi(\mathbf{x}) \phi(\mathbf{x}') \left(g^{tt} \partial_t K(t, \mathbf{x}, \mathbf{x}') \right) \Big|_{t=t_*} - \frac{i\lambda}{3!} \int d^4x \sqrt{-g} (\Phi^0(x))^3 \\ - \frac{\lambda^2}{8} \int d^4x \sqrt{-g} \int d^4x' \sqrt{-g} (\Phi^0(x))^2 G(x, x') (\Phi^0(x'))^2 + \dots. \quad (10.71)$$

The first term is just the ordinary Gaussian part of the action. The other terms have the following diagrammatic interpretation:

$$-i\lambda \int d^4x \sqrt{-g} (\Phi^0(x))^3 = \text{Diagram: A horizontal line at the top with three vertical lines descending from it to a single vertex at the bottom.} \quad (10.72)$$

$$-\lambda^2 \int d^4x \sqrt{-g} \int d^4x' \sqrt{-g} (\Phi^0(x))^2 G(x, x') (\Phi^0(x'))^2 = \text{Diagram: A horizontal line at the top with two vertical lines descending from it to two vertices. These two vertices are connected by a horizontal line at the bottom.} \quad (10.73)$$

and this can be continued to higher order in the boundary fluctuations ϕ .

Finally, it is convenient to express the computation in Fourier space, since this trivializes the integrals over space. Given the solution $f_k(t)$ to the free equation of motion for the field Φ in

Fourier space, we can write the bulk-to-boundary and bulk-to-bulk propagators as

$$K_k(t) = \frac{f_k(t)}{f_k(t_*)}, \quad (10.74)$$

$$G(k; t, t') = f_k^*(t) f_k(t') \theta(t - t') + f_k^*(t') f_k(t) \theta(t' - t) - \frac{f_k^*(t_*)}{f_k(t_*)} f_k(t) f_k(t'). \quad (10.75)$$

One can check that the function $G(k; t, t')$ satisfies the equation

$$(\square_k - m^2) G(k; t, t') = \frac{i}{\sqrt{-g}} \delta(t - t'), \quad (10.76)$$

where $\square_k$ is the spatial Fourier transform of the wave operator, while $K_k(t)$ solves the homogeneous version of the same equation. The classical solution (10.60) is then expressed in Fourier space as

$$\Phi_{\text{cl}}(t, \mathbf{k}) = K_k(t) \phi_{\mathbf{k}} + i \int dt' G(k; t, t') \frac{\delta S_{\text{int}}}{\delta \Phi_{\mathbf{k}}(t')} \Big|_{\Phi=\Phi_{\text{cl}}}. \quad (10.77)$$

We can now state the rules for the computation of wavefunction coefficients.

Feynman rules

Similar to the computation of correlators in Chapter 9, wavefunctional calculations can be simplified by organizing them into a systematic expansion in Feynman diagrams. The diagrams have two different types of propagators, which makes them slightly more complicated than their flat-space counterparts. However, compared to the in-in diagrams in Chapter 9, the wavefunction is a bit simpler, which is a benefit. Concretely, the Feynman rules are:

1. Draw the fixed-time boundary surface ($t = t_*$) on which you want to compute the wavefunction.
2. To compute the wavefunction coefficient ψ_n multiplying ϕ^n , draw all diagrams with a fixed number of lines, n , ending on the boundary.
3. Assign a vertex factor, iV , to each bulk interaction, where V is obtained by functionally differentiating the action as usual. (Note that this can lead to factors of momenta or time derivatives acting on propagators if there are derivative interactions.)
4. Assign a bulk-to-bulk propagator, G , to each internal line.
5. Assign a bulk-to-boundary propagator, K , to each external line (lines that end on the boundary).
6. Integrate over the time insertions of all bulk vertices using the measure $Dt \equiv dt a^3(t)$. For diagrams involving loops, integrate over the loop momenta.

We see that all told, the Feynman rules are fairly similar to those in the in-in formalism, except that the propagators that we use are slightly different, and that there are about half as many diagrams.

A small warning: As we saw above, there are some normalization factors that appear in the on-shell action multiplying the outputs of these Feynman diagrams to give the true wavefunction coefficients. For most of our purposes, the normalizations will not end up being particularly important, but if one wants to match exactly the result of an in-in calculation, the normalizations do matter. Working out these relative normalizations in complete generality is somewhat tedious, but for the case where one is computing the wavefunction coefficient coming from a $\mathcal{L}_{\text{int}} = -\lambda\Phi^n$ interaction, with I internal lines (at tree level), one should multiply the result of the Feynman diagram calculation by

$$c_{n,I} = \frac{\Gamma[I(n-1) + n]}{\Gamma[I+2]\Gamma[I(n-2) + n+1]\Gamma[n]^{I+1}}, \quad (10.78)$$

in order to convert it into a true wavefunction coefficient. In the following, we will often ignore these normalization factors and refer to the output of the Feynman diagram calculation as wavefunction coefficients.

Exercise 10.7 Using these Feynman rules, show explicitly that the expressions (10.12) and (10.13) are equivalent to the correlators derived in Chapter 9 using the in-in formalism.

Hint: You should find that the second line in (10.13) is equal to $I_{+-} + I_{-+}$, and that the first line is equal to $I_{++} + I_{--} + I_{\text{extra}}$, where I_{extra} precisely cancels against the third line. $\rightarrow$

10.6 Examples in Flat Space

We are now ready to compute correlators using the wavefunction formalism. We begin with examples of flat-space correlators where all time integrals can be performed explicitly. All correlators can be evaluated at $t_* \equiv 0$ without loss of generality.

Consider a massless scalar with a cubic interaction:

$$S = \int d^4x \left(-\frac{1}{2}(\partial\Phi)^2 - \frac{\lambda}{3!}\Phi^3 \right). \quad (10.79)$$

The free equation of motion in Fourier space is $(\partial_t^2 + k^2)\Phi_{\mathbf{k}} = 0$, so that the properly-normalized mode functions are

$$f_k(t) = \frac{1}{\sqrt{2k}} e^{ikt}. \quad (10.80)$$

Using this, the relevant flat-space propagators are

$$K_k(t) = e^{ikt}, \quad (10.81)$$

$$G(k; t, t') = \frac{1}{2k} \left(e^{-ik(t-t')} \theta(t-t') + e^{ik(t-t')} \theta(t'-t) - e^{ik(t+t')} \right). \quad (10.82)$$

The two-point wavefunction coefficient was computed in (10.19) and is $\langle O_1 O_2 \rangle' = k_1$, which by (10.11) implies the correct two-point function $\langle \phi_1 \phi_2 \rangle' = 1/(2k_1)$. Let us now compute the simplest tree-level correlators in this theory.

Three-point function Applying the Feynman rules to the three-point wavefunction coefficient, we get

$$\begin{aligned}
\langle O_1 O_2 O_3 \rangle' &\equiv \text{Diagram: A horizontal line at the top with two vertical lines descending from it to a single vertex at the bottom.} \\
&= i\lambda \int_{-\infty}^0 dt K_{k_1}(t) K_{k_2}(t) K_{k_3}(t) \\
&= i\lambda \int_{-\infty}^0 dt e^{i(k_1+k_2+k_3)t} = \boxed{\frac{\lambda}{k_1 + k_2 + k_3}}. \tag{10.83}
\end{aligned}$$

Substituting this into (10.12), we obtain the following three-point function

$$\langle \phi_1 \phi_2 \phi_3 \rangle' = -\frac{\text{Re}\langle O_1 O_2 O_3 \rangle'}{4k_1 k_2 k_3} = -\frac{\lambda}{4k_1 k_2 k_3} \frac{1}{k_1 + k_2 + k_3}, \tag{10.84}$$

which agrees with (9.46), the result of the in-in computation.

Four-point function Applying the Feynman rules to the four-point wavefunction coefficient, we get

$$\begin{aligned}
\langle O_1 O_2 O_3 O_4 \rangle' &\equiv \text{Diagram: A horizontal line at the top with two vertical lines descending from it to two vertices. These two vertices are connected by a horizontal line.} \\
&= -\lambda^2 \int_{-\infty}^0 dt dt' e^{ik_{12}t} G(k_I; t, t') e^{ik_{34}t'} \\
&= -\frac{\lambda^2}{2k_I} \int_{-\infty}^0 dt dt' e^{ik_{12}t} \left(e^{-ik_I(t-t')} \theta(t-t') + e^{ik_I(t-t')} \theta(t'-t) - e^{ik_I(t+t')} \right) e^{ik_{34}t'} \\
&= -\frac{\lambda^2}{2k_I} \int_{-\infty}^0 dt \int_{-\infty}^t dt' e^{i(k_{12}-k_I)t} e^{i(k_{34}+k_I)t'} \\
&\quad - \frac{\lambda^2}{2k_I} \int_{-\infty}^0 dt' \int_{-\infty}^{t'} dt e^{i(k_{12}+k_I)t} e^{i(k_{34}-k_I)t'} \\
&\quad + \frac{\lambda^2}{2k_I} \int_{-\infty}^0 dt \int_{-\infty}^0 dt' e^{i(k_{12}+k_I)t} e^{i(k_{34}+k_I)t'} \\
&= \frac{\lambda^2}{2k_I} \left[\frac{1}{(k_{12}+k_{34})(k_{34}+k_I)} + \frac{1}{(k_{12}+k_{34})(k_{12}+k_I)} - \frac{1}{(k_{12}+k_I)(k_{34}+k_I)} \right] \\
&= \boxed{\frac{\lambda^2}{(k_{12}+k_{34})(k_{12}+k_I)(k_{34}+k_I)}}. \tag{10.85}
\end{aligned}$$

We see that the answer has an interesting singularity structure. We will have much more to say about this later (cf. Section 11.3).

In summary, we have obtained the following wavefunction coefficients

$$\langle O_1 O_2 \rangle' = k_1, \quad (10.86)$$

$$\langle O_1 O_2 O_3 \rangle' = \frac{\lambda}{k_1 + k_2 + k_3}, \quad (10.87)$$

$$\langle O_1 O_2 O_3 O_4 \rangle' = \frac{\lambda^2}{(k_{12} + k_{34})(k_{12} + k_I)(k_{34} + k_I)}. \quad (10.88)$$

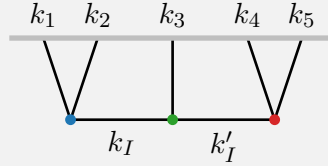
Note the simplicity of these expressions relative to the corresponding in-in correlation functions.

Finally, let us confirm that these results reproduce the four-point function that we derived in Section 9.4 using the in-in formalism. Since the wavefunction coefficients are real, equation (10.13) reduces to

$$\begin{aligned} \langle \phi_1 \phi_2 \phi_3 \phi_4 \rangle' &= \frac{1}{8 k_1 k_2 k_3 k_4} \left(\langle O_1 O_2 O_3 O_4 \rangle' + \frac{\langle O_1 O_2 O_I \rangle' \langle O_I O_3 O_4 \rangle'}{k_I} \right) \\ &= \frac{\lambda^2}{8 k_1 k_2 k_3 k_4} \left(\frac{1}{(k_{12} + k_{34})(k_{12} + k_I)(k_{34} + k_I)} + \frac{1}{k_I} \frac{1}{(k_{12} + k_I)(k_{34} + k_I)} \right), \end{aligned} \quad (10.89)$$

where we substituted (10.87) and (10.88) in the second line. It is not hard to confirm that (10.89) is the same as (9.54), so that we precisely recover the result of the in-in computation.

Exercise 10.8 Derive the wavefunction coefficient for the following five-point diagram:



You should find

$$\langle O_1 O_2 O_3 O_4 O_5 \rangle' = \frac{\lambda^3}{E_{\text{blue}} E_{\text{green}} E_{\text{red}}} \left[\frac{1}{k_{123} + k_I'} + \frac{1}{k_{345} + k_I} \right], \quad (10.90)$$

where $E_L \equiv k_{12} + k_I$, $E_M \equiv k_3 + k_I + k_I'$ and $E_R \equiv k_{45} + k_I'$ are the energies running into the three different vertices. ➔

10.7 Examples in De Sitter

It is straightforward to extend these computations to de Sitter space. The relevant propagators are given by (10.74) and (10.75), with

$$f_k(\eta) = H \sqrt{\frac{\pi}{4}} e^{-\frac{i\pi}{4}(1+2\nu)} (-\eta)^{3/2} H_\nu^{(2)}(-k\eta). \quad (10.91)$$

While the computations are conceptually exactly the same as in flat space, for generic fields the time integrals involving Hankel functions typically cannot be performed explicitly. As in

Section 9.5, we will therefore often specialize to situations involving scalars with conformal mass $m^2 = 2H^2$, whose mode functions are

$$f_k(\eta) = H(-\eta) \frac{e^{ik\eta}}{\sqrt{2k}}, \quad (10.92)$$

corresponding to $\nu = 1/2$ in (10.91). The simplicity of these functions will enable analytic results in many cases.

Contact diagrams We first consider contact diagrams coming from polynomial bulk interactions of the form Φ^n . For example, the three-point wavefunction coefficient is

$$\begin{aligned} \langle O_1 O_2 O_3 \rangle' &\equiv \text{Diagram: A horizontal line at the top with three vertical lines descending from it to a single vertex at the bottom.} \\ &= i\lambda \int_{-\infty}^{\eta_*} d\eta K_{k_1}(\eta) K_{k_2}(\eta) K_{k_3}(\eta) \\ &= \frac{i\lambda}{f_{k_1}(\eta_*) f_{k_2}(\eta_*) f_{k_3}(\eta_*)} \int_{-\infty}^{\eta_*} \frac{d\eta}{(H\eta)^4} f_{k_1}(\eta) f_{k_2}(\eta) f_{k_3}(\eta). \end{aligned} \quad (10.93)$$

In the generic case, this involves an integral over a product of three Hankel functions whose result in terms of an Appell F_4 function is not very illuminating.

Specializing to conformal scalars, and taking the limit $\eta_* \rightarrow 0$, we get

$$\langle O_1 O_2 O_3 \rangle' = -\frac{i\lambda}{H^4(-\eta_*)^3} \int_{-\infty}^{\eta_*} \frac{d\eta}{\eta} e^{iK\eta} = \boxed{\frac{i\lambda}{H^4\eta_*^3} \log(-iK\eta_*)}, \quad (10.94)$$

where $K = k_1 + k_2 + k_3$. As in (9.59) the factors of i come from the $i\epsilon$ prescription used to regulate the integral. Using $2\text{Re}\langle O_n O_n \rangle' = |f_{k_n}(\eta_*)|^{-2}$, the three-point correlator (10.12) becomes

$$\begin{aligned} \langle \phi_1 \phi_2 \phi_3 \rangle' &= -\frac{H^6 \eta_*^6}{8k_1 k_2 k_3} \left(\langle O_1 O_2 O_3 \rangle' + \langle O_1 O_2 O_3 \rangle'^* \right) \\ &= -\frac{\lambda}{8} \frac{H^2 \eta_*^3}{k_1 k_2 k_3} \left(i \log(-iK\eta_*) - i \log(iK\eta_*) \right) = \boxed{\frac{\pi}{8} \lambda \frac{H^2 \eta_*^3}{k_1 k_2 k_3}}, \end{aligned} \quad (10.95)$$

which is the same as the in-in result (9.60). Notice that the two contributions from the different branches of the in-in contour are reproduced from the wavefunction perspective by the contributions from Ψ and Ψ^* .

Exercise 10.9 Derive the three-point wavefunction coefficient of two conformally coupled scalars and one massive scalar with $m < \frac{3}{2}H$. You should find

$$\langle O_1 O_2 X_3 \rangle' = \frac{\sqrt{\pi}}{2^{\nu-\frac{1}{2}}} \frac{i\lambda}{H^4} \frac{\Gamma[1-\nu]}{\cot(\pi\nu)} \frac{(-ik_3\eta_*)^{\nu-\frac{1}{2}}}{(-\eta_*)^3} {}_2F_1 \left[\begin{matrix} \frac{1}{2} + \nu, & \frac{1}{2} - \nu \\ 1 \end{matrix} \middle| \frac{u-1}{2u} \right], \quad (10.96)$$

where $u \equiv k_3/(k_1 + k_2)$ and ${}_2F_1$ is the Gauss hypergeometric function.

Hint: You may use the same integral representations of the Hankel and hypergeometric functions as in Exercise 9.4. →

Exercise 10.10 Show that the contact n -point wavefunction coefficient and correlator for conformal scalars arising from a $-\lambda\Phi^n/n!$ interaction are ($n \geq 4$):

$$\langle O_1 \cdots O_n \rangle'_c = -i^n \frac{\lambda}{H^4 \eta_*^n} \frac{\Gamma[n-3]}{n!} \frac{1}{K^{n-3}}, \quad (10.97)$$

$$\langle \phi_1 \cdots \phi_n \rangle'_c = \begin{cases} (-1)^{\frac{n}{2}+1} \Gamma[n-3] \frac{\lambda H^{2n-4} \eta_*^n}{2^{n-1} k_1 \cdots k_4} \frac{1}{K^{n-3}}, & n \text{ even}, \\ 0, & n \text{ odd}, \end{cases} \quad (10.98)$$

where $K \equiv k_1 + k_2 + \cdots + k_n$. Since the Φ^4 interaction preserves the conformal symmetry of the free theory, the result for $n = 4$ is essentially the same as in flat space (except for a factor of $H\eta_*$ for each external field). →

Exchange diagrams Next, we look at exchange diagrams. For example, the four-point wavefunction coefficient arising from the exchange of a massive scalar is

$$\begin{aligned} \langle O_1 O_2 O_3 O_4 \rangle' &\equiv \text{Diagram: A trapezoid with a horizontal top line, a horizontal bottom line, and two vertical lines connecting them. The bottom line has two dots at its ends, representing external legs. The top line is a solid horizontal line. The two vertical lines are slanted, connecting the top line to the bottom line.} \\ &= -\lambda^2 \int_{-\infty}^{\eta_*} d\eta d\eta' K_{k_1}(\eta) K_{k_2}(\eta) G(k_I; \eta, \eta') K_{k_3}(\eta') K_{k_4}(\eta'). \end{aligned} \quad (10.99)$$

We first consider the case of a conformal scalar, for which the bulk-to-boundary and bulk-to-bulk propagators (for $\eta_* \rightarrow 0$) are

$$K_k(\eta) = \frac{\eta}{\eta_*} e^{ik\eta}, \quad (10.100)$$

$$G(k_I; \eta, \eta') = \frac{H^2 \eta \eta'}{2k_I} \left[e^{-ik|\eta-\eta'|} - e^{ik_I(\eta+\eta')} \right]. \quad (10.101)$$

Equation (10.99) then becomes

$$\begin{aligned} \langle O_1 O_2 O_3 O_4 \rangle' &= -\frac{\lambda^2}{H^6 \eta_*^4} \int_{-\infty}^0 \frac{d\eta}{\eta} \int_{-\infty}^0 \frac{d\eta'}{\eta'} e^{ik_{12}\eta} e^{ik_{34}\eta'} \frac{1}{2k_I} \left[e^{-ik_I|\eta-\eta'|} - e^{ik_I(\eta+\eta')} \right] \\ &= -\frac{\lambda^2}{H^6 \eta_*^4} \int_{k_{12}}^{\infty} dx \int_{k_{34}}^{\infty} dx' \int_{-\infty}^0 d\eta d\eta' e^{ix\eta} e^{ix'\eta'} G_{\text{(flat)}}(k_I; \eta, \eta') \\ &= \frac{1}{H^6 \eta_*^4} \int_{k_{12}}^{\infty} dx \int_{k_{34}}^{\infty} dx' \langle O_1 O_2 O_3 O_4 \rangle'_{\text{(flat)}}(x, x', k_I), \end{aligned} \quad (10.102)$$

where we have used the same trick as in Section 9.5 to write the expression as an integral over the corresponding flat-space result. Substituting (10.88) for $\langle O_1 O_2 O_3 O_4 \rangle'_{\text{(flat)}}$, gives the same

integral as for $I_{++}^{(c)}$ in (9.78):

$$\begin{aligned} \langle O_1 O_2 O_3 O_4 \rangle' &= -\frac{\lambda^2}{H^6 \eta_*^4} \int_0^\infty dx dx' \frac{1}{(E+x+x')(E_L+x)(E_R+x')} \\ &= \left[\frac{\lambda^2}{2H^6 \eta_*^4} \frac{1}{k_I} \left[\text{Li}_2 \left(\frac{k_{12} - k_I}{E} \right) + \text{Li}_2 \left(\frac{k_{34} - k_I}{E} \right) + \log \left(\frac{E_L}{E} \right) \log \left(\frac{E_R}{E} \right) - \frac{\pi^2}{6} \right] \right]. \end{aligned} \quad (10.103)$$

Notice that the wavefunction coefficient (10.103) is very similar to the full correlator (9.83). The primary difference is in the π^2 term (along with some overall normalization factors), which comes from the disconnected contributions to the correlator from $\langle O_1 O_2 O_3 \rangle$.

Exercise 10.11 Using the wavefunction coefficient that you just computed, along with the three-point result (10.94), check that the correlator derived from the wavefunction reproduces the in-in result (9.83). ➔

It is also straightforward to include the effects of derivative interactions in the vertices of diagrams. In many of the most phenomenologically interesting cases, the interactions of fields break the de Sitter symmetries of the background. Processes involving time derivatives of massless scalar fields are particularly simple [78–81].

Exercise 10.12 Consider the (de Sitter breaking) interaction

$$L_{\text{int}} = -\frac{\lambda}{2} a^3(t) \dot{\Phi}^2 \chi, \quad (10.104)$$

where Φ is a massless scalar and χ is a conformally coupled scalar. Compute the four-point wavefunction coefficient $\langle O_1 O_2 O_3 O_4 \rangle$ for massless scalars coming from the exchange of χ . You should find

$$\langle O_1 O_2 O_3 O_4 \rangle' = -\frac{\lambda^2}{H^2} k_1^2 k_2^2 k_3^2 k_4^2 \frac{2E_L E_R + E(E_L + E_R) + E^2}{E^3 E_L^2 E_R^2}, \quad (10.105)$$

where $E \equiv k_{12} + k_{34}$, $E_L \equiv k_{12} + k_I$, $E_R \equiv k_{34} + k_I$. ➔

10.8 Power-Law Cosmology

Our next example is the theory of a conformally coupled scalar field in a power-law FRW universe, with polynomial interactions:

$$S = \int d^4x \sqrt{-g} \left[-\frac{1}{2} (\partial\Phi)^2 - \frac{1}{12} R \Phi^2 - \sum_{p>2} \frac{\lambda_p}{p!} \Phi^p \right], \quad (10.106)$$

where R is the Ricci scalar. Although this is a toy model for cosmological correlations, it shares many features with more realistic models and is simple enough to study in great detail. In particular, if the universe evolves as a power law in conformal time, then the action (10.106) can be transformed into the action of a field in flat space with time-dependent couplings, which

greatly simplifies the computation of correlation functions (or wavefunction coefficients) in this theory.

A universe that is dominated by a single fluid with a constant equation of state will evolve as a power law in conformal time. In this case, the spacetime metric takes the form

$$ds^2 = a^2(\eta)(-d\eta^2 + d\mathbf{x}^2), \quad a(\eta) \equiv \left(\frac{\eta}{\eta_0}\right)^{-(1+\varepsilon)}, \quad (10.107)$$

where ε and η_0 are constant parameters. We will take the conformal time η to run from $-\infty$ to 0, and choose units such that $\eta_0 \equiv -1$. Many cosmologies of interest fit into this family, such as $\varepsilon = 0$ (de Sitter), $\varepsilon \approx 0$ (inflation), $\varepsilon = -1$ (Minkowski), $\varepsilon = -2$ (radiation) and $\varepsilon = -3$ (matter). With this parameterization, the universe experiences accelerated expansion for $\varepsilon > -1$ and decelerated collapse for $\varepsilon < -1$. We will be interested in correlations at a fixed time $\eta_* = 0$, which for an accelerating universe is the future boundary of the spacetime.

We now consider conformally coupled scalars in this background. Under a Weyl transformation $g_{\mu\nu} \mapsto a^2 g_{\mu\nu}$, $\Phi \mapsto a^{-1}\Phi$, the action (10.106) becomes that of a massless field in flat space with time-dependent couplings:

$$S = \int d^4x \left[-\frac{1}{2}(\partial\Phi)^2 - \sum_{p=3}^{\infty} \frac{\lambda_p(\eta)}{p!} \Phi^p \right], \quad (10.108)$$

where $\lambda_p(\eta) \equiv \lambda_p a(\eta)^{4-p}$. Specializing to the power-law scaling in (10.107), we have

$$\lambda_p(\eta) = \lambda_p(-\eta)^{-(1+\alpha_p)}, \quad \text{with} \quad \alpha_p \equiv (4-p)(1+\varepsilon) - 1. \quad (10.109)$$

For a cubic interaction, $\alpha_3 = \varepsilon$, while for a quartic interaction the theory is conformally invariant and $\alpha_4 = -1$, so that the coupling is time-independent as in flat space. All the time dependence has been absorbed into the couplings, $\lambda_p(\eta)$, so that the mode functions of the field are the same as in flat space, $f_k^{\text{flat}}(\eta) = e^{ik\eta}/\sqrt{2k}$, which simplifies computations in this theory.

The perturbative construction of the wavefunction for these power-law cosmologies closely parallels the flat-space case. In particular, since the quadratic action is the same, the bulk-to-boundary and bulk-to-bulk propagators are still given by (10.81) and (10.82), respectively (only written in terms of conformal time η rather than coordinate time t). The Feynman rules are also the same, the only difference being that now all vertex factors involve time-dependent couplings. This complicates the evaluation of the time integrals involved in the calculation of the wavefunction, and is the primary technical challenge involved in computing the cosmological wavefunction, as compared to flat space. However, an elegant way to deal with these time-dependent couplings is to use the following identity [82]:

$$\frac{1}{(-\eta)^{1+\alpha}} = \frac{ie^{\frac{i\pi\alpha}{2}}}{\Gamma(1+\alpha)} \int_0^\infty dx x^\alpha e^{ix\eta}. \quad (10.110)$$

The time integrals then become identical to their flat-space counterparts and can be performed explicitly.

Contact diagrams We first consider a three-point contact interaction. The associated wavefunction coefficient is³³

$$\begin{aligned}
\langle O_1 O_2 O_3 \rangle' &= \text{Diagram: A horizontal line with three points labeled } k_1, k_2, k_3. \text{ Three lines connect these points to a single vertex labeled } K. \\
&= i \int_{-\infty}^0 d\eta e^{iK\eta} \lambda_3(\eta) \\
&\propto \int_0^\infty dx x^\varepsilon \left(i \int d\eta e^{i(K+x)\eta} \right) \\
&\propto \int_0^\infty dx x^\varepsilon \langle O_1 O_2 O_3 \rangle'_{(\text{flat})}(K+x) = \int_0^\infty dx \frac{x^\varepsilon}{K+x}, \quad (10.111)
\end{aligned}$$

where we have used the identity (10.110) and then identified the integrand as the deformed flat-space wavefunction times a *twist factor*, x^ε . Performing the integral, we get

$$\boxed{\langle O_1 O_2 O_3 \rangle' = -\pi \csc(\pi\varepsilon) K^\varepsilon}. \quad (10.112)$$

We therefore see that the contact solution in a general FRW spacetime is a simple power law. In de Sitter space, with $\varepsilon = 0$, we instead found $\langle O_1 O_2 O_3 \rangle' \propto \log(iK)$.

Exchange diagram Next, consider the four-point wavefunction coefficient arising from a cubic interaction:

$$\begin{aligned}
\langle O_1 O_2 O_3 O_4 \rangle' &= \text{Diagram: A horizontal line with four points labeled } k_1, k_2, k_3, k_4. \text{ Two lines connect } k_1 \text{ and } k_2 \text{ to a vertex labeled } E_1. \text{ Two lines connect } k_3 \text{ and } k_4 \text{ to a vertex labeled } E_2. \text{ A horizontal line connects } E_1 \text{ and } E_2 \text{ labeled } k_I. \\
&= - \int_{-\infty}^0 d\eta d\eta' e^{iE_1\eta} \lambda_3(\eta) G(k_I; \eta, \eta') \lambda_3(\eta') e^{iE_2\eta'} \\
&\propto \int_0^\infty dx dx' (xx')^\varepsilon \int d\eta d\eta' e^{i(E_1+x)\eta} G(k_I; \eta, \eta') e^{i(E_2+x')\eta'} \\
&\propto \int_0^\infty dx dx' (xx')^\varepsilon \langle O_1 O_2 O_3 O_4 \rangle'_{(\text{flat})}(E_1+x, E_2+x', k_I). \quad (10.113)
\end{aligned}$$

We see that the FRW wavefunction can again be written as an integral over the deformed flat-space wavefunction with a twist factor $(xx')^\varepsilon$. Substituting (10.88), we get

$$\boxed{\langle O_1 O_2 O_3 O_4 \rangle' = \int_0^\infty dx dx' (xx')^\varepsilon \frac{1}{(E+x+x')(E_L+x)(E_R+x')}}. \quad (10.114)$$

Note that for $\varepsilon = 0$, this reduces to (10.103). This maps the challenge of computing the FRW wavefunction coefficient associated to the exchange to the task of evaluating this two-dimensional

³³Due to the Weyl rescaling $\Phi \mapsto a^{-1}\Phi$, the original FRW wavefunction coefficient evaluated at $\eta = \eta_*$ has an extra factor of $a^3(\eta_*)$. We will ignore this constant rescaling.

integral involving a rational integrand “twisted” by powers of the integration variables. There is a vast mathematical literature on these twisted integral and the specific integrals arising in the power-law cosmology were studied systematically in [83].

The integral shown in (10.114) can actually be computed in Mathematica. The result is

$$\begin{aligned} \langle O_1 O_2 O_3 O_4 \rangle' &= c_1(\varepsilon) (E_L E_R)^\varepsilon \\ &+ c_2(\varepsilon) E^{2\varepsilon} \left(1 - {}_2F_1 \left[\begin{matrix} 1, \varepsilon \\ 1 - \varepsilon \end{matrix} \middle| \frac{k_{34} - k_I}{E_L} \right] - {}_2F_1 \left[\begin{matrix} 1, \varepsilon \\ 1 - \varepsilon \end{matrix} \middle| \frac{k_{12} - k_I}{E_R} \right] \right), \end{aligned} \quad (10.115)$$

where

$$\begin{aligned} c_1(\varepsilon) &= \pi^2 \csc^2(\pi\varepsilon), \\ c_2(\varepsilon) &= 4^{-\varepsilon} \sqrt{\pi} \csc(\pi\varepsilon) \Gamma(\varepsilon) \Gamma(\tfrac{1}{2} - \varepsilon). \end{aligned} \quad (10.116)$$

We see that the solution is a combination of power laws and hypergeometric functions.

Exercise 10.13 Consider the time-integral representation of the function

$$F \equiv 2k_I \langle O_1 O_2 O_3 O_4 \rangle' = F_+ + F_- - F_0, \quad (10.117)$$

where we have separated the result according to the three parts of the bulk-to-bulk propagator:

$$F_+ \equiv -\lambda^2 \int \frac{d\eta d\eta'}{(\eta\eta')^{1+\varepsilon}} e^{i(E_1 - k_I)\eta} e^{i(E_2 + k_I)\eta'} \theta(\eta - \eta'), \quad (10.118)$$

$$F_- \equiv -\lambda^2 \int \frac{d\eta d\eta'}{(\eta\eta')^{1+\varepsilon}} e^{i(E_1 + k_I)\eta} e^{i(E_2 - k_I)\eta'} \theta(\eta' - \eta), \quad (10.119)$$

$$F_0 \equiv -\lambda^2 \int \frac{d\eta d\eta'}{(\eta\eta')^{1+\varepsilon}} e^{i(E_1 + k_I)\eta} e^{i(E_2 + k_I)\eta'}. \quad (10.120)$$

1. By taking derivatives with respect to E_1 and E_2 , show that these functions satisfy

$$dF_0 = \varepsilon (d \log(E_1 + k_I) + d \log(E_2 + k_I)) F_0, \quad (10.121)$$

$$dF_+ = (\varepsilon F_+ + F_c) d \log(E_1 - k_I) + (\varepsilon F_+ - F_c) d \log(E_2 + k_I), \quad (10.122)$$

$$dF_- = (\varepsilon F_- - F_c) d \log(E_1 + k_I) + (\varepsilon F_- + F_c) d \log(E_2 - k_I), \quad (10.123)$$

$$dF_c = 2\varepsilon F_c d \log(E_1 + E_2), \quad (10.124)$$

where $d \equiv \sum_i dE_i \partial_{E_i}$ and the extra source function F_c is to be determined.

2. Show that the solution of these differential equations is given by (10.115).

Hint: Use the boundary condition that none of the functions has a “folded singularity” at $E_1 - k_I = 0$ or $E_2 - k_I = 0$. Then impose the correct “factorization limit”:

$$\lim_{E_1 + k_I \rightarrow 0} F \propto (E_1 + k_I)^\varepsilon \left((E_2 - k_I)^\varepsilon - (E_2 + k_I)^\varepsilon \right). \quad (10.125)$$

The origin of these boundary conditions is explained in Section 11.3. The boundary condition in (10.125) can easily be obtained from the $E_1 + k_I \rightarrow 0$ limit of the integrals in (10.119) and (10.120). →

10.9 Cosmological Collider

So far, most of our explicit computations were for conformally coupled scalars. However, as soon as we allow for generic massive fields, the resulting integrals typically cannot be evaluated analytically. This problem is relevant for the exact predictions of the “cosmological collider” (see Section 9.8).

A simple example is the exchange of a generic massive scalar χ between two pairs of conformally coupled scalars Φ . The corresponding wavefunction coefficient is of the form (10.99) and can explicitly be written as

$$F \equiv \langle O_1 O_2 O_3 O_4 \rangle' = -\lambda^2 \int \frac{d\eta}{\eta^2} \int \frac{d\eta'}{\eta'^2} e^{ik_{12}\eta} e^{ik_{34}\eta'} G_\nu(k_I; \eta, \eta'), \quad (10.126)$$

where the propagator of the massive field is

$$G_\nu(k_I; \eta, \eta') = \frac{\pi}{4} (\eta\eta')^{3/2} \left[H_\nu^{(1)}(-k\eta) H_\nu^{(2)}(-k\eta') - \frac{H_\nu^{(1)}(-k\eta_*)}{H_\nu^{(2)}(-k\eta_*)} H_\nu^{(2)}(-k\eta) H_\nu^{(2)}(-k\eta') \right], \quad \text{for } \eta' < \eta. \quad (10.127)$$

The expression for $\eta < \eta'$ is the same but with η and η' interchanged. Here, ν can either be real or complex ($\nu = i\mu$), depending on the mass of the exchanged field.

The wavefunction coefficient (10.126) captures some of the physics of the on-shell production and decay of massive particles in de Sitter space, and can be used as a seed to compute the phenomenological signatures of these heavy particles [43, 44, 66]. Unfortunately, however, the time integrals involved cannot be evaluated by elementary means. Instead of trying to compute the integrals directly, we note that G_ν satisfies the differential equation

$$\left(\eta^2 \frac{\partial^2}{\partial \eta^2} - 2\eta \frac{\partial}{\partial \eta} + k_I^2 \eta^2 + \frac{m^2}{H^2} \right) G_\nu(k_I; \eta, \eta') = -iH^2 \eta^2 \eta'^2 \delta(\eta - \eta'), \quad (10.128)$$

which is just the explicit form of the Green’s function equation (10.76). The fact that G_ν satisfies this differential equation can be used to derive a differential equation for the function F . To do this, we consider the auxiliary object

$$\tilde{F} \equiv -\lambda^2 \int \frac{d\eta}{\eta^2} \frac{d\eta'}{\eta'^2} e^{ik_{12}\eta} e^{ik_{34}\eta'} \left(\eta^2 \frac{\partial^2}{\partial \eta^2} - 2\eta \frac{\partial}{\partial \eta} + k_I^2 \eta^2 + \frac{m^2}{H^2} \right) G_\nu(k_I; \eta, \eta'), \quad (10.129)$$

and evaluate it in two ways. First, we use (10.128) to write

$$\tilde{F} = i\lambda^2 H^2 \int_{-\infty}^0 d\eta e^{iE\eta} = \frac{\lambda^2 H^2}{E}, \quad (10.130)$$

with $E \equiv k_{12} + k_{34}$. Next, we integrate the time derivatives by parts in (10.129) and trade factors of η for $-i\partial_{k_{12}}$ to obtain³⁴

$$\begin{aligned}\tilde{F} &= -\lambda^2 \int_{-\infty}^0 \frac{d\eta}{\eta^2} \frac{d\eta'}{\eta'^2} \left((k_I^2 - k_{12}^2)\eta^2 + 2ik_{12}\eta + \frac{m^2}{H^2} - 2 \right) e^{ik_{12}\eta} e^{ik_{34}\eta'} G_\nu(k_I; \eta, \eta') \\ &= -\lambda^2 \int_{-\infty}^0 \frac{d\eta}{\eta^2} \frac{d\eta'}{\eta'^2} \left((k_{12}^2 - k_I^2)\partial_{k_{12}}^2 + 2k_{12}\partial_{k_{12}} + \frac{m^2}{H^2} - 2 \right) e^{ik_{12}\eta} e^{ik_{34}\eta'} G_\nu(k_I; \eta, \eta') \\ &= \left((k_{12}^2 - k_I^2)\partial_{k_{12}}^2 + 2k_{12}\partial_{k_{12}} + \frac{m^2}{H^2} - 2 \right) F, \end{aligned} \quad (10.131)$$

where, in the last line, we have pulled the differential operator outside the integral, which we then recognize as being precisely (10.126). Comparing (10.130) and (10.131), we find that F satisfies the following differential equation

$$\left((k_{12}^2 - k_I^2)\partial_{k_{12}}^2 + 2k_{12}\partial_{k_{12}} + \frac{m^2}{H^2} - 2 \right) F = \frac{\lambda^2 H^2}{E}. \quad (10.132)$$

This equation is rather interesting. Time evolution in the bulk spacetime has been traded for a differential operator in terms of the external energy k_{12} . When this operator acts on the exchange solution it produces a conformally coupled scalar contact solution; cf. (10.97). We can therefore think of the operator as collapsing the internal propagator, creating a contact interaction.

Exercise 10.14 Show that $\hat{F} \equiv k_I F$ satisfies

$$\left(u^2(1 - u^2)\partial_u^2 - 2u^3\partial_u + \frac{m^2}{H^2} - 2 \right) \hat{F} = \lambda^2 H^2 \frac{uv}{u + v}, \quad (10.133)$$

where $u \equiv k_I/k_{12}$ and $v \equiv k_I/k_{34}$.



This strategy of turning the integrals for the late-time wavefunction coefficients into differential equations is applicable in a wide range of situations. In the following exercise, you will derive an analogous differential equation for a three-point function.

Exercise 10.15 Derive a differential equation for the three-point wavefunction coefficient involving two conformally coupled scalars and one massive scalar, $G \equiv \langle OOX \rangle'$. Write the equation in terms of $u \equiv k_3/k_{12}$.



Notice that “time” has completely disappeared from the problem. The differential equation (10.132) is formulated purely in terms of the boundary momenta (or energies k_{12} , k_{34} and k_I). This suggests that there should be a more efficient way of understanding the correlations on the late-time boundary of de Sitter without ever explicitly referring to the (unobservable) bulk time evolution. In the next chapter, we will first show how the differential equation (10.132) can be derived directly on the boundary and then discuss its solutions.

³⁴We can drop the boundary term, because the bulk-to-bulk propagator vanishes on the boundary.

11 Cosmological Bootstrap

In the previous two chapters, we presented two methods for computing inflationary correlations by evolving the fluctuations through the bulk spacetime and evaluating the resulting correlators on the future boundary. This involved complex time integrals (one for each vertex in a diagram) that in general cannot be computed analytically. Numerical integration is possible, but comes at the cost of masking the physical structure of the answers.

In this chapter, we will describe a new approach to deriving the correlations directly on the boundary by imposing consistency with unitarity, locality and symmetry assumptions. These consistency conditions constrain the functional form of the boundary correlators without having to evaluate complex time integrals. Although “time” will not make any explicit appearance, the effects of the time-dependent bulk physics (such as the production and decay of massive particles) will be reflected in the momentum scalings of the resulting correlators (see Fig. 27). The cosmological bootstrap is a very active area of current research which we will *not* be able to fully review. Instead, we will try to give a pedagogical introduction to a few selected topics that hopefully will provide the reader with a useful entry point into the subject. More details and further references can found in [45, 84, 85].

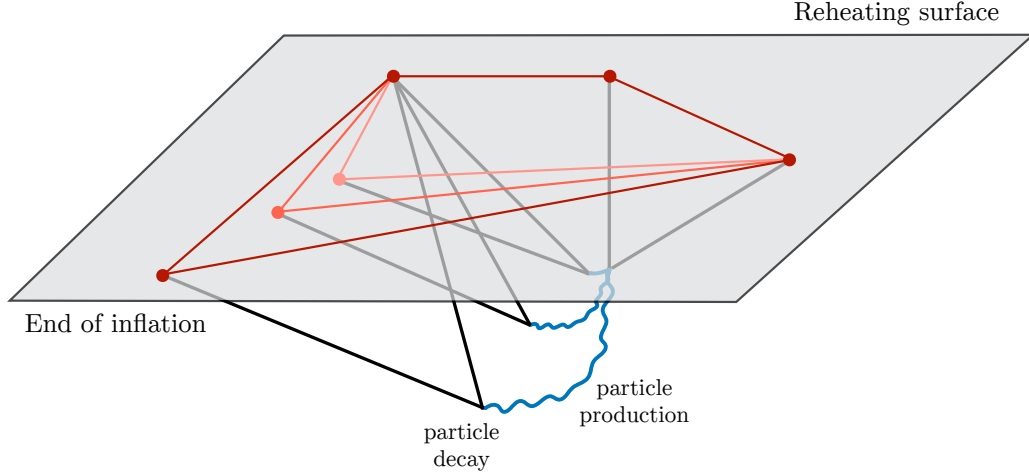


Figure 27: All cosmological correlations can be traced back to the spacelike boundary of the inflationary quasi-de Sitter spacetime. The time dependence of bulk interactions is encoded in the momentum dependence of these boundary correlators. The cosmological bootstrap is a way to derive these correlators purely from a boundary perspective.

11.1 Bootstrap Philosophy

You have probably learned that theories are defined by their Lagrangians. In particular, Lagrangians are a transparent way of encoding local and causal interactions between particles, which in relativistic quantum mechanics requires the introduction of fields. However, as illustrated in Fig. 28, Lagrangians are not really the fundamental starting point, but instead they are intermediate objects that take us from a set of fundamental principles to physical observables. To make locality and causality manifest, the Lagrangian formalism often requires the introduction

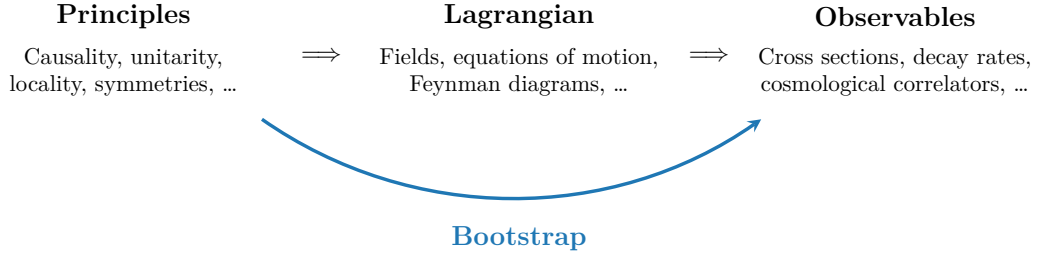


Figure 28: Graphical illustration of the bootstrap philosophy. Instead of encoding physical principles in a Lagrangian, the basic idea of the bootstrap is to take us directly from these principles to the final observables.

of additional, unphysical, degrees of freedom that are not observable, but nevertheless must be tracked in detail throughout the calculation.³⁵ This introduces a substantial complication, and so the basic bootstrap philosophy is to go directly from physical principles to final observables, without the intermediate step of introducing a Lagrangian and computing Feynman diagrams.

S-Matrix Bootstrap

The textbook approach to computing scattering amplitudes involves a perturbative expansion in terms of Feynman diagrams. Unfortunately, these calculations can quickly become very complicated, especially in gauge theories. For example, the five-point gluon amplitude involves 25 diagrams and the associated mathematical expressions span many pages. Similar, the four-point gravity amplitude involves thousands of diagrams, each involving hundreds of terms. In these cases, it required truly heroic efforts (by Parke and Taylor [86] and DeWitt [87], respectively) to compute these amplitudes explicitly. Yet, when restricted to physical (“on-shell”) configurations, the final answers are remarkably simple one-line expressions. This suggested that there should be a better way to obtain these results. Part of the problem is that individual Feynman diagrams are not completely physical, in the sense that they carry along unphysical gauge degrees of freedom that must cancel in the final answers.³⁶ In the modern S-matrix bootstrap, the focus is instead on the external on-shell data (like the momenta and helicities of the external particles) and the space of consistent amplitudes is reconstructed from consistency requirements alone, dispensing with the need for Lagrangians, Feynman diagrams and unphysical gauge degrees of freedom.

As a concrete example, consider the tree-level scattering of four identical scalars of mass m . All other particles are taken to be significantly heavier than m . We now review how basic physical requirements such as Lorentz invariance, locality and unitarity severely restrict the analytic structure of the four-particle scattering amplitude A_4 at tree level.

Firstly, *Lorentz symmetry* implies that the amplitude must be a function of the Mandelstam invariants $s \equiv -(p_1 + p_2)^2$, $t \equiv -(p_1 + p_4)^2$ and $u \equiv -(p_1 + p_3)^2$, which satisfy the constraint

³⁵For example, the two degrees of freedom of a photon are encoded in the four components of the electromagnetic vector potential A_μ . The additional (gauge) degrees of freedom are unphysical, but can appear at intermediate stages of the computations.

³⁶Even in scalar theories, Feynman diagrams are not invariant under field redefinitions. An infinite number of different Lagrangians and their corresponding Feynman diagrams can map to the same S-matrix. In that case, a poor choice of field basis can complicate the calculation and obscure important structures of the theory.

$s + t + u = 4m^2$ (because of momentum conservation). At low energies, the theory is described by contact interactions, and A_4 is an analytic function of s, t, u , which can be written as a power series for small s, t, u . At higher energies, we become sensitive to the exchange of massive particles and A_4 becomes non-analytic. For example, at tree level, the existence of a particle of mass M leads to poles as $s, t, u \rightarrow M^2$. As a consequence of *locality*, these poles must be simple poles. Moreover, the residues of the poles are fixed by kinematics. In particular, near the s -channel pole, the amplitude factorizes into a product of on-shell three-particle amplitudes

$$A_4(s, t) \xrightarrow{s \rightarrow M^2} \frac{1}{s - M^2} \sum_h A_3^h(p_1, p_2, p_I) A_3^{-h}(p_3, p_4, -p_I), \quad (11.1)$$

where $p_I \equiv p_1 + p_2$ is the four-momentum of the exchanged particles and we have summed over its helicities h . The relevant three-particle amplitudes A_3^h are simple because there are no kinematical invariants analogous to s, t, u for three-particle scattering, cf. $(p_1 + p_2)^2 = (-p_3)^2 = -m^2$, etc. Since $p^{\mu_i} \epsilon_{\mu_1 \dots \mu_i \dots \mu_J}^h = 0$, the most general three-particle amplitude is of the form

$$A_3^h(p_1, p_2, p_3) = \lambda (p_1 - p_2)^{\mu_1} \dots (p_1 - p_2)^{\mu_J} \epsilon_{\mu_1 \dots \mu_J}^h(p_3), \quad (11.2)$$

where λ is the coupling constant and $\epsilon_{\mu_1 \dots \mu_J}^h$ is the polarization tensor. Substituting this into (11.1), and performing the sum over helicities, one obtains the final form of the amplitude

$$A(s, t) \xrightarrow{s \rightarrow M^2} \lambda^2 \frac{(4m^2 - M^2)^J}{s - M^2} P_J \left(1 + \frac{2t}{M^2 - 4m^2} \right), \quad (11.3)$$

where P_J is a Legendre polynomial of order the spin J . Note that the coefficient of the Legendre polynomial is positive; this is a consequence of *unitarity*.

Exercise 11.1 Derive (11.3) for exchange particles with spin $J = 0$, $J = 1$ and $J \geq 1$.

Hint: Perform the sum over helicities in the rest frame of the massive particle, where only the spatial components of the polarization tensors are non-vanishing. ➔

The most general four-particle amplitude arising from tree-level exchange of a massive spin- J particle therefore takes the form

$$A_4(s, t, u) = \lambda^2 \frac{(4m^2 - M^2)^J}{s - M^2} P_J \left(1 + \frac{2t}{M^2 - 4m^2} \right) + t\text{- and } u\text{-channels} + \text{analytic}. \quad (11.4)$$

Symmetries and kinematics have fixed the structure of the amplitude near physical poles. The rest of the amplitude is analytic. Note that the separation of the amplitude into “pole” and “polynomial” parts is not unique; it is only the residues on the poles that have unambiguous meaning.

The above example of the bootstrap method may not have been that impressive, since the standard computation of this example is already rather simple. The full power of the bootstrap approach, however, becomes manifest when we consider the scattering of massless particles with spin where the computations in terms of Feynman diagrams are prohibitively complicated and the bootstrap method becomes a necessity. In this case, the requirement of consistent factorization

of four-particle amplitudes is highly restrictive, and makes almost all theories, other than the familiar gauge theories and gravity, inconsistent [88–91].

As we have seen in Section 10.7, the computation of cosmological correlators is complicated even for the tree-level exchange of massive particles. In that case, a bootstrap approach also becomes a powerful alternative to the standard Lagrangian method [44].

Cosmological Bootstrap

In this chapter, we will present attempts to develop a similar bootstrap method for cosmological correlators. This “cosmological bootstrap” is a vibrant and evolving subject, which we will not be able to review fully. We apologize for having to leave out many interesting recent developments.

- **Kinematics**—We are interested in the correlations defined on the spatial boundary of the cosmological spacetime. Since these correlations are statistically homogeneous, they are naturally described in Fourier space where the n -point wavefunction coefficients are functions of n wavevectors $\mathbf{k}_a$ that form closed n -gons. In a slight abuse of terminology, we refer to the side lengths of these n -gons, $k_a \equiv |\mathbf{k}_a|$, as “energies”. A natural question to ask is how the correlations change as we vary the kinematic data (i.e. deform the shapes of the n -gons). Often the answer to this question is a differential equation which describes a “flow” in the space of kinematic configurations. One of the main objectives of the cosmological bootstrap is to derive this “kinematic flow”.
- **Symmetries**—As we will describe in Section 11.2, consistent cosmological correlations are constrained by the isometries of the bulk spacetime and the symmetries of interactions. During inflation, the spacetime is approximately de Sitter space, and the isometries of this bulk spacetime become conformal symmetries on the late-time boundary. Assuming that the bulk interactions don’t strongly break this symmetry, the boundary correlators must then satisfy the Ward identities associated to this conformal symmetry. These Ward identities take the form of differential equations in terms of the external energies, which characterize how strength of the correlations depend on the energies. To solve these differential equations requires appropriate boundary conditions, which we impose on the singular limits of the correlators.
- **Singularities**—We have seen that singularities play an important role in determining the structure of scattering amplitudes. As we will show in Section 11.3, the same holds for cosmological correlators and the associated wavefunction coefficients. We will see that every wavefunction coefficient has a singularity when the sum of all external energies vanishes, $E \equiv |\mathbf{k}_1| + |\mathbf{k}_2| + \cdots + |\mathbf{k}_n| \rightarrow 0$. Moreover, the residue of this “total energy singularity” is the corresponding flat-space scattering amplitude. Rather remarkably, amplitudes therefore “live inside” the cosmological wavefunction.

More generally, the wavefunction coefficients have singularities whenever the energies entering any subgraph add up to zero. On these “partial energy singularities” the wavefunction coefficients factorize into an amplitude and a lower-point wavefunction coefficient. This is similar to the factorization of amplitudes that we discussed above. Like for amplitudes,

consistent factorization on all partial energy singularities is an important constraint on cosmological correlators. As in the S-matrix bootstrap, the challenge is to find a function with the correct singular limits, without introducing additional spurious singularities or other physical inconsistencies.

- **Unitarity**—An important feature of any quantum-mechanical theory is that probabilities are conserved. However, until recently, it wasn’t known how this unitarity of the bulk theory is encoded in cosmological boundary correlators. In [78, 92], it was pointed out that unitarity implies specific relations between wavefunction coefficients, quite similar to the optical theorem for scattering amplitudes in flat space. Moreover, these relations can be systematized into a set of cutting rules [81, 93–95] that are similar to the Cutkosky rules for the S-matrix [96]. These cutting rules have since become an important tool to bootstrap cosmological correlators.

The main difficulty faced by the standard way of computing cosmological correlators is the presence of complicated nested time integrals. We will introduce the concept of “cut wavefunctions” for which these nested integrals factorize. We will also show that, in the limit of vanishing partial energies, the cut wavefunction reduces to the original wavefunction. In this way, unitarity fixes the residues of all partial energy singularities (not just the leading singularity) and therefore put an important constraint on the wavefunction coefficients. Especially for rational correlators, the constraints imposed by unitarity cuts go a long way towards determining the full correlators.

In the rest of this chapter, we will first describe these bootstrap constraints in more detail and then apply them to a few concrete examples.

11.2 Symmetries

In this section, we review the role of symmetries to constrain the functional form of the boundary correlators in the inflationary quasi-de Sitter spacetime. We first derive the unitary representations associate to the de Sitter group and show how these representations are realized by bulk fields and boundary operators. We then demonstrate that the de Sitter isometries become a conformal symmetry on the late-time boundary and derive the associated Ward identities. This leads to differential equations describing the deformations of the correlators in momentum space.

11.2.1 De Sitter Isometries

In Section 5.1, we saw that D -dimensional de Sitter space can be described as a hypersurface in $(D + 1)$ -dimensional Minkowski space defined by the embedding constraint

$$\eta_{AB}X^AX^B = -X_0^2 + X_1^2 + \cdots + X_D^2 = H^{-2}. \quad (11.5)$$

The symmetries of de Sitter space then correspond to the higher-dimensional Lorentz transformations that leave the embedded hypersurface invariant. The generators of these Lorentz symmetries, J_{AB} , satisfy the $\mathfrak{so}(D, 1)$ algebra:

$$[J_{AB}, J_{CD}] = \eta_{BC}J_{AD} - \eta_{AC}J_{BD} + \eta_{AD}J_{BC} - \eta_{BD}J_{AC}. \quad (11.6)$$

It is natural, particularly in the inflationary slicing, to define

$$\begin{aligned}\hat{D} &\equiv J_{D0}, \\ P_i &\equiv J_{Di} - J_{0i}, \\ K_i &\equiv J_{Di} + J_{0i},\end{aligned}\tag{11.7}$$

so that the de Sitter algebra becomes

$$\begin{aligned}[J_{ij}, J_{kl}] &= \delta_{jk}J_{il} - \delta_{ik}J_{jl} + \delta_{il}J_{jk} - \delta_{jl}J_{ik}, & [\hat{D}, P_i] &= P_i, \\ [J_{ij}, K_l] &= \delta_{jl}K_i - \delta_{il}K_j, & [\hat{D}, K_i] &= -K_i, \\ [J_{ij}, P_l] &= \delta_{jl}P_i - \delta_{il}P_j, & [K_i, P_j] &= 2\delta_{ij}\hat{D} - 2J_{ij}.\end{aligned}\tag{11.8}$$

The generators P_i and J_{ij} create spatial translations and rotations, while $\hat{D}$ and K_i are associated with dilatations and special conformal transformations. The hat on the dilatation operator $\hat{D}$ was introduced to avoid confusion with the dimension D of the spacetime.

We will usually work with conformal coordinates, in which the de Sitter line element (in the flat slicing) takes the form

$$ds^2 = \frac{1}{H^2\eta^2} (-d\eta^2 + \delta_{ij}dx^i dx^j).\tag{11.9}$$

This is obviously invariant under spatial translations and rotations on the spatial slices $\mathbb{R}^d$. In addition, the line element has a *dilatation symmetry*:

$$\begin{aligned}\eta &\mapsto \lambda\eta, \\ x^i &\mapsto \lambda x^i,\end{aligned}\tag{11.10}$$

where λ is a constant parameter. Slightly harder to see (but easy to verify) is the symmetry under *special conformal transformations* (SCTs):

$$\begin{aligned}\eta &\mapsto \frac{\eta}{1 + 2b_i x^i + b^2(x^2 - \eta^2)}, \\ x^i &\mapsto \frac{x^i + (x^2 - \eta^2)b^i}{1 + 2b_i x^i + b^2(x^2 - \eta^2)},\end{aligned}\tag{11.11}$$

where b^i is a constant vector. As we will see below, the transformations (11.11) act precisely as special conformal transformations on the spacelike boundary of de Sitter, motivating the naming convention. This is not an accident, but a manifestation of the fact that the de Sitter group $\text{SO}(D, 1)$ is exactly the Euclidean conformal group in $D - 1$ dimensions.

The generators of infinitesimal transformations can be found by pulling back the ambient space Lorentz generators to find explicit expressions for the Killing vectors (11.7). This leads to

$$\begin{aligned}P_i &= \partial_i, \\ J_{ij} &= x_i \partial_j - x_j \partial_i, \\ \hat{D} &= -\eta \partial_\eta - x^i \partial_i, \\ K_i &= 2x_i \eta \partial_\eta + \left(2x^j x_i + (\eta^2 - x^2)\delta_i^j\right) \partial_j.\end{aligned}\tag{11.12}$$

It is easy to check that these differential operators satisfy the commutation relations in (11.8).

Exercise 11.2 The flat slicing (11.9) corresponds to the embedding

$$\begin{aligned} X^0 &= \frac{L}{2(-\eta)} (1 - \eta^2 + x^2), & X^i &= \frac{L}{(-\eta)} x^i, \\ X^D &= \frac{L}{2(-\eta)} (1 + \eta^2 - x^2), \end{aligned} \quad (11.13)$$

which foliates $\mathbb{R}^{D,1}$ by de Sitter hyperboloids of radius $L = H^{-1}$. Using these embedding functions, pull back the ambient space Lorentz transformations to derive the Killing vectors in (11.12). →

We want to understand how the symmetries of de Sitter space manifest themselves in the physics of cosmological correlations. In Section 11.2.2, we will present a brief summary of the unitary representations of the de Sitter group. In Section 11.2.3, we then show how the symmetries of de Sitter space are realized on bulk fields. We use these results, in Section 11.2.4, we determine how the symmetries act on fluctuations on the late-time boundary of the spacetime. We discover the famous fact that the isometries of the bulk spacetime become conformal symmetries on the spatial boundary. In Section 11.2.5, we present the associated conformal Ward identities, both for correlators and wavefunction coefficients. Finally, we solve these Ward identities for a few selected examples in Section 11.2.6.

11.2.2 Unitary Representations

The Hilbert space of a quantum field theory in de Sitter space arranges itself into representations of the de Sitter group. In this section, we will present the essential properties of these unitary irreducible representations (in the bosonic case). Further details can be found in [97–99].

The goal is to realize the generators J_{AB} as anti-Hermitian operators on a vector space with a positive-definite norm. It is natural to label representations by their eigenvalues of the quadratic and quartic Casimir operators [97, 99, 100]

$$\begin{aligned} \mathcal{C}_2 &\equiv \frac{1}{2} J^{AB} J_{AB} = \hat{D}(d - \hat{D}) + P^i K_i + \frac{1}{2} J^{ij} J_{ij}, \\ \mathcal{C}_4 &\equiv J_{AB} J^{BC} J_{CD} J^{DA}, \end{aligned} \quad (11.14)$$

since these commute with all the generators of the de Sitter symmetries. (The expression for $\mathcal{C}_4$ in terms of the generators in (11.7) is fairly long and unilluminating.)

Unitary representations of the de Sitter group can then be constructed by starting from a *primary state* $|\bar{\Delta}\rangle_{\mathbf{s}}$, which is annihilated by the SCT operator K_i , has dilatation eigenvalue $\bar{\Delta} = d - \Delta$, and $\text{so}(D)$ spin weight $\mathbf{s}$. Since Δ and $\mathbf{s}$ completely determine the Casimirs in (11.14), we can label representations by these quantum numbers.

Scalar representations: There are three families of unitary irreducible scalar representations on de Sitter space.³⁷

³⁷Note that this is already richer than in flat space, where there are only two families: the ordinary scalar representations with $m^2 \geq 0$ and tachyons with $m^2 < 0$.

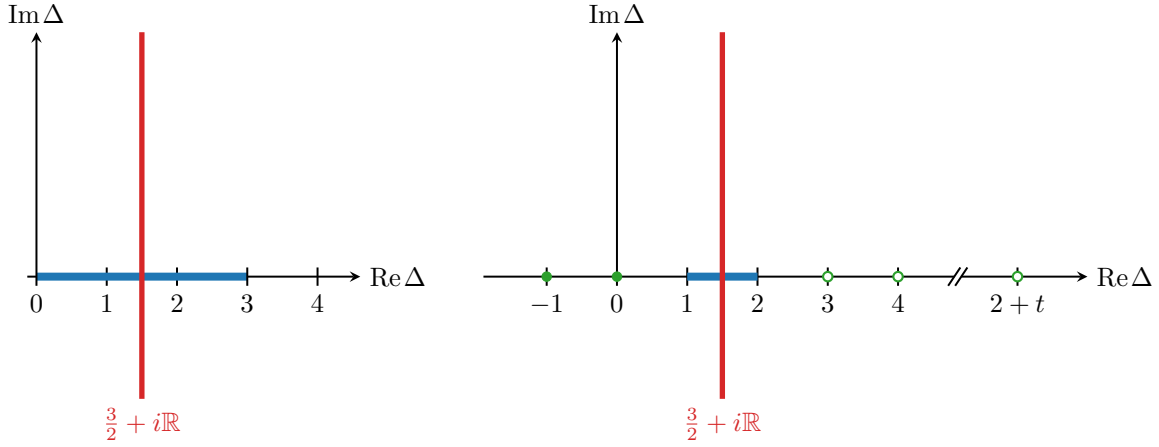


Figure 29: Illustration of scalar (*left*) and spin- J (*right*) representations of $\text{SO}(4,1)$ in the complex Δ -plane. The red lines corresponds to the principal series, while the blue lines mark the complementary series. The green circles for the spin- J case denote the representations of the exceptional series (which coincide with the discrete series in this case). Open circles are the shadows of the filled ones.

- **Principal series:** This family is generated by a primary with conformal dimension $\Delta = \frac{d}{2} + i\mu$, where $\mu \in \mathbb{R}$. We will see that these representations correspond to *heavy fields* in de Sitter with $m^2 > \frac{d^2}{4}H^2$.
- **Complementary series:** This family is generated by a primary with real conformal dimension in the range $0 < \Delta < d$. We will see that these representations are carried by *light fields* with $0 < m^2 < \frac{d^2}{4}H^2$.
- **Exceptional series:** These representations have $\Delta = -k$, with $k \in \mathbb{Z}_{\geq 0}$. The representation obtained from this primary state is reducible and so must be projected. It is unclear what field theory carries these representations.³⁸

It is interesting to compare these representations to the unitary scalar representations of the group $\text{SO}(D-1,2)$, which is the group of isometries of anti-de Sitter space. The representations are built in a similar manner, starting from a primary state, and in the AdS context, unitarity places a lower bound of the value of Δ for scalar representations: $\Delta \geq \frac{d}{2} - 1$.

Spinning representations: Fields with spin have a slightly different classification, which is somewhat richer than that of scalars. Along with the principal and complementary series, there are additional families of unitary representations, whose weights take on discrete values. The classification of general mixed-symmetry fields is rather intricate [97, 98], so we will focus on spin- J fields whose spin weight is $\mathbf{s} = (J, 0, \dots, 0)$:³⁹

³⁸A reasonable guess is that the shift-symmetric scalar fields considered in [101] carry these representations if they are quantized treating the shift symmetries like gauge redundancies. The mass values of these fields are $m^2 = -k(d+k)H^2$.

³⁹In odd d , general mixed-symmetry fields have another family of representations, called the *discrete series*. However, these representations require all spin weights to be activated [98], so they do not appear for spin- J representations, except in $d = 3$, where they happen to coincide with the exceptional series.

- **Principal series:** Spinning representations in the principal series have conformal dimensions satisfying $\Delta = \frac{d}{2} + i\mu$, with $\mu \in \mathbb{R}$. The corresponding bulk fields are heavy fields with

$$m^2 \geq (\tfrac{d}{2} + J - 2)^2 H^2. \quad (11.15)$$

- **Complementary series:** Fields in this family have conformal dimensions in the range $1 < \Delta < d - 1$. Note that this is bounded away from $\Delta = 0$. These representations correspond to bulk fields with masses in the range

$$(J - 1)(J + d - 3)H^2 < m^2 < (\tfrac{d}{2} + J - 2)^2 H^2. \quad (11.16)$$

A qualitative difference from the scalar case is that spinning representations with $J \geq 2$ have a lower bound on their mass in order to remain unitary. This lower bound is often called the *Higuchi bound* [102].

- **Exceptional series:** These representations have a primary state with $\Delta = 1 - t$, where $t \in \{0, 1, \dots, J - 1\}$ is the *depth*. The corresponding bulk gauge fields have spin J and mass

$$m^2 = (J - 1 - t)(J + t + d - 3)H^2. \quad (11.17)$$

The representation with $t = J - 1$ is a *massless* field, the other depths are known as *partially massless* fields, have gauge invariances with t -index gauge parameters, and have no flat-space analogues.

The unitary scalar and spin- J representations of $\text{SO}(4, 1)$ —which is the physically relevant case—are depicted in Fig. 29.

11.2.3 Realization on Bulk Fields

Next, we want to understand how the symmetries of de Sitter space are realized on bulk fields, and in particular what representations of $\text{SO}(D, 1)$ they carry. For simplicity, we will only present the scalar case.

Consider a real scalar field Φ , with mass m , in de Sitter space. It is straightforward to derive the transformation of Φ under the isometries generated by the Killing vectors (11.12); it just transforms by the corresponding *Lie derivative*⁴⁰

$$\begin{aligned} P_i \Phi &= \partial_i \Phi, \\ J_{ij} \Phi &= (x_i \partial_j - x_j \partial_i) \Phi, \\ \hat{D} \Phi &= -(\eta \partial_\eta + x^i \partial_i) \Phi, \\ K_i \Phi &= 2x_i \eta \partial_\eta \Phi + \left(2x^j x_i + (\eta^2 - x^2) \delta_i^j \right) \partial_j \Phi. \end{aligned} \quad (11.18)$$

From this, we can work out the action of the quadratic Casimir (11.14):

$$\mathcal{C}_2 \Phi = -\eta^2 \left(\frac{\partial^2}{\partial \eta^2} - \frac{d-1}{\eta} \frac{\partial}{\partial \eta} - \nabla^2 \right) \Phi. \quad (11.19)$$

⁴⁰Recall that the Lie derivative of a function is simply its directional derivative, $\mathcal{L}_\xi f = \xi(f) = \xi^a \partial_a f$.

This is exactly the wave operator acting on the scalar: $H^2 \mathcal{C}_2 \Phi = \square \Phi$, so that the equation of motion can be read as an eigenvalue equation

$$\mathcal{C}_2 \Phi = \frac{m^2}{H^2} \Phi. \quad (11.20)$$

Equating this to the Casimir of a scalar representation $\mathcal{C}_2 = \Delta(d - \Delta)$, we can solve for Δ :

$$\Delta_{\pm} = \frac{d}{2} \pm \sqrt{\frac{d^2}{4} - \frac{m^2}{H^2}}. \quad (11.21)$$

Notice first that the two branches Δ_+ and Δ_- are related by $\Delta_+ = d - \Delta_-$. Scaling weights with this relation are so-called *shadows* of each other, and correspond to equivalent representations, which makes sense because we anticipate that a scalar should carry a single representation that could be labeled by either of these two choices.

Notice that there are two qualitative regimes:

- **Complementary series:** For $0 \leq m^2/H^2 < d^2/4$, the values of $\Delta_{\pm}$ lie in the interval $[0, d]$, which is the range corresponding to the complementary series. We therefore see that “light” fields in de Sitter carry complementary series representations.
- **Principal series:** For $m^2/H^2 > d^2/4$, the values of $\Delta_{\pm}$ are of the form $\frac{d}{2} + i\mu$, with μ a real number. These are precisely the principal series values. Hence, “heavy” fields in de Sitter carry principal series representations.

We note in passing that fields with the (tachyonic) masses $m^2/H^2 = -k(D - 1 + k)$ have the correct weights to be the exceptional series scalars, but they have shift symmetries that are naively de Sitter symmetry-violating zero modes. It would therefore seem that they have to be quantized in some nonstandard way. Other tachyonic mass values do not correspond to any representations on the list in Section 11.2.2 and so are not unitary.

Cosmological correlations naturally live on the boundary of de Sitter space, so we now want to understand the action of these symmetries on the boundary.

11.2.4 Boundary Conformal Symmetry

There is a distinguished spatial slice in de Sitter space—the future boundary—which is invariant under the symmetries of de Sitter space. The points on this surface get mapped into each other, but the surface itself does not move. In the flat slicing, this is the $\eta = 0$ surface. In the idealized inflationary picture, this is where the universe reheats and the standard hot Big Bang evolution starts. The fluctuations on this surface serve as the initial conditions for the universe, and therefore the correlations at $\eta = 0$ have a distinguished status. We would like to understand how the de Sitter symmetries act on them.

Consider a scalar field, Φ , with mass m . Its equation of motion is

$$\Phi'' - \frac{(d-1)}{\eta} \Phi' - \nabla^2 \Phi + \frac{m^2}{H^2} \frac{1}{\eta^2} \Phi = 0. \quad (11.22)$$

At late times ($\eta \rightarrow 0$), we can neglect the spatial gradient term and the solutions become power laws in time:

$$\Phi(\mathbf{x}, \eta \rightarrow 0) = \phi(\mathbf{x}) \eta^\Delta + \bar{\phi}(\mathbf{x}) \eta^{\bar{\Delta}}, \quad (11.23)$$

where $\bar{\Delta} = d - \Delta$ and the *scaling dimension* (sometimes also called the scaling weight) is

$$\Delta \equiv \frac{d}{2} - \sqrt{\frac{d^2}{4} - \frac{m^2}{H^2}}. \quad (11.24)$$

Note that Δ is the same as Δ_- in (11.21). For “light fields”, with $m < \frac{d}{2}H$, the scaling dimension is purely real and the first term in (11.23) dominates at late times. Near the boundary, we can then replace time derivatives by $\eta \partial_\eta \mapsto \Delta$ and the de Sitter isometries (in the limit $\eta \rightarrow 0$) act as

$$\begin{aligned} P_i \phi &= \partial_i \phi, \\ J_{ij} \phi &= (x_i \partial_j - x_j \partial_i) \phi, \\ \hat{D} \phi &= -(\Delta + x^i \partial_i) \phi, \\ K_i \phi &= \left[2x_i \Delta + \left(2x^j x_i - x^2 \delta_i^j \right) \partial_j \right] \phi. \end{aligned} \quad (11.25)$$

These are precisely the transformation rules for a primary operator of conformal weight Δ in a conformal field theory (CFT). The late-time correlations of the field Φ can be thought of as being captured by the correlations of the spatial coefficients ϕ and $\bar{\phi}$. Since these coefficients transform like operators in a CFT, late-time correlators in dS are constrained by conformal symmetry.⁴¹ The constraints of conformal symmetry are well understood in position space, but, in cosmology, we need results in Fourier space. Unfortunately, the Fourier transforms of CFT correlators are highly nontrivial [67] and in practice is easier to redo the analysis directly in Fourier space.

11.2.5 Conformal Ward Identities

Our primary interest is in late-time cosmological correlators, and so we want to understand the constraints of de Sitter symmetry for these objects. In general, symmetries imply that correlation functions satisfy differential equations called *Ward identities*. We will now derive these Ward identities for both cosmological correlation functions and wavefunction coefficients.

Correlators: We can think of correlation functions as directly correlating the late-time field profiles of the bulk fields Φ . As such, they are essentially correlation functions of the coefficients of the field fall-offs in (11.23), which transform as in (11.25) under a de Sitter transformation. For light fields (in the complementary series), only the leading fall-off ϕ survives at late times, because the other branch decays more quickly. In this case, the requirements of de Sitter invariance can be phrased as

$$\sum_{a=1}^N \langle \phi_1 \cdots \delta \phi_a \cdots \phi_N \rangle = 0, \quad (11.26)$$

where $\delta \phi_a$ can stand for any of the field transformations in (11.25). The constraints of translation and rotation invariance are very easy to implement and simply require that the correlator is a

⁴¹This assumes that the bulk interactions don’t strongly break the symmetry. Especially in a time-dependent inflationary background this may not be the case. We will return to this issue below.

function only of the separations $|\mathbf{x}_a - \mathbf{x}_b|$. The constraints of dilatation and SCT invariance are more nontrivial, and lead to the equations

$$\text{D: } 0 = \sum_{a=1}^N - \left(\Delta_a + x_a^j \frac{\partial}{\partial x_a^j} \right) \langle \phi_1 \cdots \phi_a \cdots \phi_N \rangle, \quad (11.27)$$

$$\text{SCT: } 0 = \sum_{a=1}^N \left(2\Delta_a x_a^i + 2x_a^i x_a^j \frac{\partial}{\partial x_a^j} - x_a^2 \frac{\partial}{\partial x_{a,i}} \right) \langle \phi_1 \cdots \phi_a \cdots \phi_N \rangle. \quad (11.28)$$

Correlation functions computed directly from the bulk will automatically satisfy these equations. The idea of the bootstrap approach is to shortcut this procedure and solve these equations directly. However, in general this is rather difficult, and there are many solutions, so we require additional input in order to know which solutions are physically relevant.

Wavefunction coefficients: Next, we will determine how the conformal transformations act on the wavefunction coefficients. Consider the N -th order term in the perturbative expansion of the late-time wavefunction (10.6):

$$\Psi[\phi] = \exp \left(\cdots + \int d^d x_1 \cdots d^d x_N \Psi_N(\underline{\mathbf{x}}) \phi(\mathbf{x}_1) \cdots \phi(\mathbf{x}_N) + \cdots \right). \quad (11.29)$$

Under a de Sitter transformation, we have

$$\delta \Psi \supset \sum_{a=1}^N \int d^d x_1 \cdots d^d x_N \Psi_N(\underline{\mathbf{x}}) \phi(\mathbf{x}_1) \cdots \delta \phi(\mathbf{x}_a) \cdots \phi(\mathbf{x}_N), \quad (11.30)$$

where the transformations of the late-time field profiles $\delta \phi_a$ are given by (11.25). (We are again assuming a field in the complementary series, so that it only has a single fall-off, otherwise we should also keep track of the second mode $\bar{\phi}$.) We now integrate this expression by parts, so that the generators of the transformation acts on the wavefunction coefficient $\Psi_N(\underline{\mathbf{x}})$. For example, for the dilatation symmetry, we get

$$\begin{aligned} \delta \Psi &\supset \sum_{a=1}^N \int d^d x_1 \cdots d^d x_N \Psi_N(\underline{\mathbf{x}}) \phi(\mathbf{x}_1) \cdots \left(-\Delta_a - x_a^j \partial_{x_a^j} \right) \phi(\mathbf{x}_a) \cdots \phi(\mathbf{x}_N) \\ &= \sum_{a=1}^N \int d^d x_1 \cdots d^d x_N \left[\left((d - \Delta_a) + x_a^j \partial_{x_a^j} \right) \Psi_N(\underline{\mathbf{x}}) \right] \phi(\mathbf{x}_1) \cdots \phi(\mathbf{x}_N), \end{aligned} \quad (11.31)$$

and hence the wavefunction coefficients transform as

$$\delta \Psi_N(\underline{\mathbf{x}}) = \sum_{a=1}^N \left((d - \Delta_a) + x_a^j \partial_{x_a^j} \right) \Psi_N(\underline{\mathbf{x}}). \quad (11.32)$$

If we think of the wavefunction coefficients as being correlation functions in some putative dual theory, $\Psi_N(\underline{\mathbf{x}}) = \langle O(\mathbf{x}_1) \cdots O(\mathbf{x}_N) \rangle$, then the operators, O_a , transform as fields of weight $\bar{\Delta}_a = d - \Delta_a$. (There is also a flip of sign of the parameter of the transformation, but this is not important.) The same is true for the special conformal transformations, so that O_a behaves like

a primary operator of weight $d - \Delta_a$. Explicitly, the action of the de Sitter symmetry on the late-time wavefunction coefficients is captured by the following transformation rules for the dual operators

$$\hat{D} O = (\bar{\Delta} + x^j \partial_{x^j}) O, \quad (11.33)$$

$$K_i O = -(2\bar{\Delta} x^i + 2x^i x^j \partial_{x^j} - x^2 \partial_{x^i}) O. \quad (11.34)$$

Assuming that the wavefunction Ψ is *invariant* under the conformal symmetry, the wavefunction coefficients must then be annihilated by the action of the symmetry generators:

$$\text{D} : \sum_{a=1}^N \hat{D}_a \langle O_1 \cdots O_a \cdots O_N \rangle = 0, \quad (11.35)$$

$$\text{SCT} : \sum_{a=1}^N K_a^i \langle O_1 \cdots O_a \cdots O_N \rangle = 0, \quad (11.36)$$

where the operators $\hat{D}_a$ and K_a^i are given by (11.33) and (11.34), respectively. These equations are the analogue for the wavefunction coefficients of equations (11.27) and (11.28).

For the most part, we will be interested in solving these equations for the wavefunction because, as we have seen, wavefunction coefficients are somewhat simpler than the boundary correlators and contain the same information.

Fourier space: In cosmology, we typically work in Fourier space, so it will be useful to have expressions for the action of the dilatation (11.33) and special conformal (11.34) symmetries in Fourier space.⁴² On a scalar operator, they act as [43, 67, 105]

$$\hat{D} O_{\mathbf{k}} = -\left[(d - \bar{\Delta}) + k^j \partial_{k^j}\right] O_{\mathbf{k}}, \quad (11.37)$$

$$K_i O_{\mathbf{k}} = -i\left[2(d - \bar{\Delta}) \partial_{k^i} + 2k^j \partial_{k^j} \partial_{k^i} - k_i \partial_{k^j} \partial_{k^j}\right] O_{\mathbf{k}}. \quad (11.38)$$

One further subtlety arises when writing the Ward identities (11.35) and (11.36) in Fourier space. The derivatives with respect to the momenta can hit the momentum-conserving delta function which appears in front of a correlator:

$$\langle O_1 \cdots O_N \rangle = (2\pi)^d \delta^{(d)}(\mathbf{k}_1 + \cdots + \mathbf{k}_N) \langle O_1 \cdots O_N \rangle'. \quad (11.39)$$

We deal with this subtlety by treating the delta function as a distribution and integrating by parts the terms where the derivatives hit the delta function. The SCT operator (11.38) ends up passing through the delta function unchanged, but the dilatation operator becomes

$$\sum_{a=1}^N \hat{D}_a \langle O_1 \cdots O_N \rangle' = -\left[d(N-1) - \sum_{a=1}^N \bar{\Delta}_a + \sum_{a=1}^{N-1} k_a^j \partial_{k_a^j}\right] \langle O_1 \cdots O_N \rangle', \quad (11.40)$$

⁴²Reference [103] explains how to derive these transformations in Fourier space, though beware of some sign errors. See also [104] for details.

where we have imposed the constraint that the delta function implements by setting $\mathbf{k}_N = -\sum_{a=1}^{N-1} \mathbf{k}_a$. Concretely, this means that wavefunction coefficients (in Fourier space) satisfy

$$\text{D} : \left[d + \sum_{a=1}^N \hat{D}_a \right] \langle O_1 \cdots O_a \cdots O_N \rangle'(\underline{\mathbf{k}}) = 0, \quad (11.41)$$

$$\text{SCT} : \sum_{a=1}^N K_a^i \langle O_1 \cdots O - a \cdots O_N \rangle'(\underline{\mathbf{k}}) = 0, \quad (11.42)$$

where $\hat{D}_a$ and K_a^i are given by (11.37) and (11.38), respectively.

Exercise 11.3 Derive the Fourier-space action of the dilatation (11.37) and SCT (11.38) operators. Then, verify that their action on the wavefunction coefficients (with the momentum-conserving delta function removed) is given by (11.41) and (11.42). →

Dilatation symmetry is reflected simply in the overall scaling dimension of the correlator: a general N -point function has scaling dimension $\bar{\Delta}_t \equiv \sum_a \bar{\Delta}_a$ in position space. This implies that the momentum-space correlator has dimension $\bar{\Delta}_t - d \times N$. After stripping off the momentum-conserving delta function, the dimension of the remaining function is $\bar{\Delta}_t - d(N-1)$. This fixes the overall momentum scaling of the correlator, allowing us to make an ansatz for $\langle O_1 \cdots O_N \rangle'$ that automatically solves the dilatation Ward identity (11.41). All the juice is therefore in the SCT Ward identity (11.42).

11.2.6 Selected Examples

We now want to see how solving this Ward identity directly reproduces the de Sitter physics that we previously derived directly from the bulk.

Two-point functions

We first derive the constraints of conformal symmetry on the two-point functions of scalar operators. Following the above argument, it is straightforward to see that an ansatz that solves the dilatation Ward identity (11.41) is

$$\langle O_1 O_2 \rangle' \propto k_1^\alpha, \quad (11.43)$$

with $\alpha \equiv \bar{\Delta}_1 + \bar{\Delta}_2 - d$. Since the two-point function depends only on the magnitude of momenta, the SCT operator takes the simplified form:

$$b_i K^i O_{\mathbf{k}} = \mathbf{b} \cdot \mathbf{k} \left[\frac{\partial^2}{\partial k^2} + \frac{(d+1-2\bar{\Delta})}{k} \frac{\partial}{\partial k} \right] O_{\mathbf{k}}, \quad (11.44)$$

where b^i is an arbitrary vector. Demanding SCT invariance of the two-point function then implies

$$\alpha + d - 2\bar{\Delta}_1 = \bar{\Delta}_2 - \bar{\Delta}_1 = 0, \quad (11.45)$$

which forces the scaling dimensions of the two operators to be equal, $\bar{\Delta}_1 = \bar{\Delta}_2$. We therefore get

$$\langle O_1 O_2 \rangle' = A k_1^{2\bar{\Delta}_1 - d} \delta_{\bar{\Delta}_1, \bar{\Delta}_2}. \quad (11.46)$$

We have learned that the two-point function of two scalars vanishes unless their scaling dimensions are equal and that their momentum scaling is fixed in terms of that scaling dimension.

Exercise 11.4 Show that (11.46) is consistent with the explicit result for the power spectrum of a massive field derived in Exercise 8.6, which consists of one term with scaling weight Δ_+ and one with weight Δ_- . ➔

Three-point functions

As a more nontrivial example—which is still completely fixed by symmetry—we consider the three-point function of scalar fields in de Sitter space. The constraints of conformal invariance on the corresponding momentum space three-point function were treated comprehensively in [67]. An ansatz that solves the dilatation Ward identity (11.41) is

$$\langle O_1 O_2 O_3 \rangle' = k_3^{\bar{\Delta}_1 + \bar{\Delta}_2 + \bar{\Delta}_3 - 2d} \hat{G}(p, q), \quad (11.47)$$

where $p \equiv k_1/k_3$ and $q \equiv k_2/k_3$. Although it looks like k_3 is special here, it is not because of the presence of an arbitrary function, $\hat{G}$. The SCT Ward identity (11.42) then implies two differential equations for the dimensionless function $\hat{G}(p, q)$, whose solution can be expressed in terms of the Appell F_4 function.⁴³

For our purpose, it will be useful to work out more explicitly the special case of two conformally coupled fields Φ (with $\bar{\Delta} = 2$ for the dual operator O) and a generic field χ (with arbitrary $\bar{\Delta}$ for the dual operator X), in $d = 3$ dimensions. A dilatation-invariant ansatz for the wavefunction coefficient is

$$\langle O_1 O_2 X_3 \rangle' = k_3^{\bar{\Delta} - 2} \hat{G}(u, w), \quad (11.49)$$

where $u \equiv k_3/(k_1 + k_2)$ and $w \equiv k_3/(k_1 - k_2)$. This is a slightly different ansatz from (11.47), but is convenient for this situation because the variables u, w have definite parity under $1 \leftrightarrow 2$. We can now act with the SCT operator in the form (11.44) and choose $\mathbf{b} \cdot \mathbf{k}_3 = 0$. After using momentum conservation, we then find

$$\left(\frac{\partial^2}{\partial k_1^2} - \frac{\partial^2}{\partial k_2^2} \right) \hat{G}(u, w) = 0 \implies \frac{4u^2 w^2}{k_3^2} \partial_u \partial_w \hat{G}(u, w) = 0. \quad (11.50)$$

We can therefore take $\hat{G}(u, w) = \hat{G}(u)$.⁴⁴ Next, we consider the SCT Ward identity with $\mathbf{b} \cdot \mathbf{k}_2 = 0$, which leads to

$$\left(\frac{\partial^2}{\partial k_1^2} - \frac{\partial^2}{\partial k_3^2} - \frac{4 - 2\bar{\Delta}}{k_3} \frac{\partial}{\partial k_3} \right) \left(k_3^{\bar{\Delta} - 2} \hat{G}(u) \right) = 0. \quad (11.51)$$

Changing the derivatives to ones with respect to u , this implies the equation

$$\left[\Delta_u + \left(\mu^2 + \frac{1}{4} \right) \right] \hat{G}(u) = 0, \quad (11.52)$$

⁴³This special function is not so familiar, but there is a relatively nice representation in terms of an integral over products of modified Bessel functions. Using this integral representation, one can derive the Fourier space expression for the correlation function of three scalar operators [67]:

$$\langle O_1 O_2 O_3 \rangle' \propto k_1^{\bar{\Delta}_1 - \frac{d}{2}} k_2^{\bar{\Delta}_2 - \frac{d}{2}} k_3^{\bar{\Delta}_3 - \frac{d}{2}} \int_0^\infty dx x^{\frac{d}{2} - 1} K_{\nu_1}(k_1 x) K_{\nu_2}(k_2 x) K_{\nu_3}(k_3 x), \quad (11.48)$$

where K_ν , with $\nu = \bar{\Delta} - \frac{d}{2}$, is the modified Bessel function of the second kind. See [106, 107], for more details of the evaluation of integrals of this type for special mass values.

⁴⁴There is another solution that depends only on w , but we can obtain it from this one by sending $k_2 \rightarrow -k_2$.

where $\Delta_u \equiv u^2(1-u^2)\partial_u^2 - 2u^3\partial_u$ is the same differential operator as in (10.133) and $\mu^2 + \frac{1}{4} \equiv -(\bar{\Delta}-2)(\bar{\Delta}-1)$. In terms of the mass parameter of the bulk field χ , we have $\mu^2 + \frac{1}{4} = M^2/H^2 - 2$. Notice that this differential equation is precisely the one obtained for this wavefunction coefficient in Exercise 10.15 from the bulk perspective.

Equation (11.52) is of the hypergeometric type and is solved by

$$\hat{G}(u) \propto {}_2F_1 \left[\begin{matrix} \frac{1}{2} + i\mu, \frac{1}{2} - i\mu \\ 1 \end{matrix} \middle| \frac{u-1}{2u} \right], \quad (11.53)$$

where ${}_2F_1$ is the Gauss hypergeometric function. A second linearly-independent solution is given by the complex conjugate function $G^*(u)$. We normalize the solution, so that its Wronskian is $1/(1-u^2)$. The three-point wavefunction coefficient (11.49) then is

$$\langle O_1 O_2 X_3 \rangle' = \left(\frac{\pi}{2 \cosh(\pi\mu)} \right)^{\frac{1}{2}} k_3^{i\mu - \frac{1}{2}} {}_2F_1 \left[\begin{matrix} \frac{1}{2} + i\mu, \frac{1}{2} - i\mu \\ 1 \end{matrix} \middle| \frac{u-1}{2u} \right], \quad (11.54)$$

which is exactly the same as the wavefunction coefficient obtained in Exercise 10.9, and is simply related to the correlator from Exercise 9.4.

The solution (11.54) has a more complicated analytic structure than many of the examples we have encountered so far. In particular, it has branch points at $u = 0, 1$ (it is natural to choose the cut to not lie in the physical region, which is the interval between $u = 0$ and $u = 1$). This branch cut can be thought of as a signature of the production of massive particles with a continuum of possible relative momenta in the bulk spacetime. Note also that for special values of μ , equation (11.52) is not satisfied exactly in momentum space, due to the presence of anomalies.⁴⁵

We have seen that two- and three-point functions are fixed by conformal symmetry alone. To bootstrap the form of higher-point functions, however, we need additional input.

11.3 Singularities

An important feature of cosmology is that energy is not conserved—indeed, this is a direct consequence of the time dependence of the dynamics. As we will see, this leads to cosmological correlators possessing characteristic singularities at special points in the space of kinematic variables. We will see how these singularities arise and discuss how they are an important input for the bootstrap construction of wavefunctions and correlators.

⁴⁵For example, this happens when the third field also has $\bar{\Delta} = 2$. In Section 10.7, we derived the three-point wavefunction coefficient of three conformally coupled fields:

$$\langle O_1 O_2 O_3 \rangle' \propto \log(K/\bar{\mu}). \quad (11.55)$$

This expression solves the Ward identities “anomalously.” The scale variation of the logarithm does not vanish, but is instead a function that is analytic in the momenta (in this case it is just a constant), indicating that it is a contact term in position space. This correlation function therefore satisfies the conformal Ward identities at separated points, which is all that is required. (The failure of the conformal Ward identities at coincident points is a reflection of an anomaly [106].) Note that we can freely add an arbitrary constant to this correlation function by shifting the (arbitrary) scale $\bar{\mu}$.

Roughly speaking, wavefunction coefficients are singular when the sum of the energies flowing into a subgraph vanishes, and the coefficients of these singularities are fixed in terms of simpler building blocks (lower-point amplitudes and wavefunction coefficient). It is important to point out that these singularities cannot be accessed for physical kinematics, since for correlators all energies have to be positive. (For amplitudes, in contrast, energies can be positive or negative corresponding to incoming or outgoing particles, respectively.) Nevertheless, the fact that the analytic continuation of correlation functions must have these singularities will be an important constraint. Using this as an input for the bootstrap procedure will allow us to reconstruct the full correlator and extract answers in the physical domain.

11.3.1 Flat-Space Examples

Even the flat-space wavefunction has these energy singularities. In that case, the presence of a spatial boundary leads to a breaking of time translations which implies that energy is not conserved. It is useful to start with a few concrete examples and then try to draw some general lessons from them. In Section 10.6, we computed wavefunction coefficients in flat space. Let us look back at some of the results.

One-site graph

The three-point wavefunction coefficient is

$$\langle O_1 O_2 O_3 \rangle' \equiv \text{---}\bigvee = i\lambda \int_{-\infty}^0 dt e^{i(k_1+k_2+k_3)t} = \frac{\lambda}{k_1 + k_2 + k_3}. \quad (11.56)$$

We see that this has a singularity when the total energy of the process vanishes, $K \equiv k_1 + k_2 + k_3 \rightarrow 0$. For a general N -point contact interaction, we would get a similar result with a singularity at $E \equiv k_1 + k_2 + \dots + k_N \rightarrow 0$. We call these *total energy singularities* [105, 108].

We can understand this singularity as an avatar of the energy-conserving delta functions that appear in the computations of scattering amplitudes:

$$\int_{-\infty}^{\infty} dt e^{i(E_1 + \dots + E_N)t} = 2\pi\delta(E_1 + \dots + E_N). \quad (11.57)$$

(Recall that the time integral factorizes from the rest of the computation because of time-translation invariance.) In contrast, the flat-space wavefunction involves integration from $t = -\infty$ to $t = 0$, so instead of a delta function, we schematically find

$$i \int_{-\infty}^0 dt e^{i(k_1 + \dots + k_N)t} = \frac{1}{k_1 + \dots + k_N}, \quad (11.58)$$

which is indeed singular as $k_1 + \dots + k_N \rightarrow 0$. Physically, this singularity comes from the lower limit of the integration. This domain is normally regulated by the oscillatory factors in the integrand, but these vanish exactly when the sum of the energies vanishes, so the wavefunction diverges.

The integrals that we have to evaluate in realistic situations are typically more complicated than the one appearing in (11.58), but they will still (almost always) have singularities when the

total energy vanishes. More remarkably, the residues of these singularities are the corresponding scattering amplitudes. For the above contact examples, this is rather trivial since the residue of the total energy singularity is the coupling constant λ , which is the amplitude for a polynomial contact interaction. However, the fact that amplitudes live at the loci of energy singularities will be true more generally.⁴⁶

Two-site graph

A more nontrivial example is the four-point wavefunction coefficient arising from an exchange diagram:

$$\begin{aligned} \langle O_1 O_2 O_3 O_4 \rangle' \equiv & \text{Diagram: A trapezoid with a horizontal top line and a horizontal bottom line. Two vertical lines connect the top and bottom lines, forming a rectangle. The left vertical line is slightly to the left of the right vertical line. The top line is longer than the bottom line. The left and right sides are slanted outwards from bottom to top.} \\ & = -\frac{\lambda^2}{2k_I} \int_{-\infty}^0 dt \int_{-\infty}^t dt' e^{i(k_{12}-k_I)t} e^{i(k_{34}+k_I)t'} \\ & \quad - \frac{\lambda^2}{2k_I} \int_{-\infty}^0 dt \int_{-\infty}^t dt' e^{i(k_{12}+k_I)t'} e^{i(k_{34}-k_I)t} \\ & \quad + \frac{\lambda^2}{2k_I} \int_{-\infty}^0 dt \int_{-\infty}^0 dt' e^{i(k_{12}+k_I)t} e^{i(k_{34}+k_I)t'}. \end{aligned} \quad (11.59)$$

This combination of time integrals also has a singularity as $E \equiv k_{12} + k_{34} \rightarrow 0$. To make this manifest, it is useful to define $T \equiv t' - t$ in the first two integrals. We can then combine these integrals as

$$\begin{aligned} \langle O_1 O_2 O_3 O_4 \rangle' = & -\frac{\lambda^2}{2k_I} \int_{-\infty}^0 dt e^{iEt} \int_{-\infty}^0 dT (e^{iE_L T} + e^{iE_R T}) \\ & + \frac{\lambda^2}{2k_I} \int_{-\infty}^0 dt e^{iE_L t} \int_{-\infty}^0 dt' e^{iE_R t'}, \end{aligned} \quad (11.60)$$

where $E_L \equiv k_{12} + k_I$ and $E_R \equiv k_{34} + k_I$. Since we have decoupled the integration over T , it is clear that there will be a singularity when $E \rightarrow 0$. (It is also easy to see that only the time-ordered parts of the Feynman diagram contribute to the total energy singularity, which is both necessary and expected in order for it to be related to a scattering amplitude.) Indeed, performing the integrals in (11.60), we find

$$\langle O_1 O_2 O_3 O_4 \rangle' = \frac{\lambda^2}{2k_I} \left[\frac{1}{E} \left(\frac{1}{E_L} + \frac{1}{E_R} \right) - \frac{1}{E_L E_R} \right] = \frac{\lambda^2}{E E_L E_R}. \quad (11.61)$$

Taking the limit $E \rightarrow 0$, it is easy to confirm that the residue of the total energy singularity is indeed the corresponding exchange amplitude in the s -channel:

$$\lim_{E \rightarrow 0} \frac{\lambda^2}{E E_L E_R} = \frac{1}{E} \frac{\lambda^2}{(k_{12} + k_I)(-k_{12} + k_I)} = -\frac{1}{E} \frac{\lambda^2}{s} \equiv \frac{A_4}{E}, \quad (11.62)$$

where $s \equiv k_{12}^2 - k_I^2$ is the Mandelstam invariant.

⁴⁶In flat space, this is not so mysterious. The Feynman rules that we use to compute both the wavefunction coefficients and scattering amplitudes are nearly identical, the only material difference being the domain of integration over time; we integrate over all time to compute the S-matrix, while for the wavefunction we only integrate from $t = -\infty$ to $t = 0$. (The Green's functions for internal lines also differ, but this difference is regular as $E \rightarrow 0$.)

In addition to the total energy singularity at $E = 0$, the result in (11.61) also has *partial energy singularities* at $E_L = 0$ and $E_R = 0$. In the limit $E_L \rightarrow 0$, the wavefunction becomes

$$\lim_{E_L \rightarrow 0} \frac{\lambda^2}{E E_L E_R} = \frac{1}{E_L} \frac{\lambda^2}{(k_{34} + k_I)(k_{34} - k_I)} = \frac{A_3 \times \tilde{\psi}_3}{E_L}. \quad (11.63)$$

We see that the residue of this partial energy singularity is the product of a three-point scattering amplitude $A_3 = \lambda$ and a new object which we call a *shifted wavefunction coefficient*

$$\tilde{\psi}_3 \equiv \frac{1}{2k_I} \left[\psi_3(k_{34}, -k_I) - \psi_3(k_{34}, k_I) \right], \quad (11.64)$$

where ψ_3 is the three-point wavefunction coefficient given in (11.56). (The singularity at $E_R \rightarrow 0$ behaves similarly.) In fact, these simple flat-space correlators are completely fixed by their singularities. In particular, the result (11.61) is the unique function that has the correct factorization singularities. This is rather interesting, because it implies that the answer can be assembled from simpler building blocks (amplitudes and lower-point shifted wavefunctions).

By inspection of (11.60), we see that these energy singularities again come from the early-time regions of the integration. It is also easy to see why the residues of the partial energy singularities take the form shown in (11.63). In the early-time limit, the bulk-to-bulk propagator factorizes and the time-ordering disappears [82, 109]:

$$G(k_I; t, t') \xrightarrow{t' \rightarrow -\infty} \frac{e^{ik_I t}}{2k_I} \left(e^{-ik_I t'} - e^{+ik_I t'} \right). \quad (11.65)$$

On the partial energy poles, the wavefunction coefficient therefore factorizes into a product of an amplitude and a shifted wavefunction coefficient.

Finally, let us recall that the corresponding four-point correlator—computed in (9.54)—is

$$\langle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \phi_{\mathbf{k}_4} \rangle' = \frac{\lambda^2}{8k_1 k_2 k_3 k_4} \left[\frac{1}{E E_L E_R} + \frac{1}{k_I E_L E_R} \right]. \quad (11.66)$$

We note that this result has additional singularities at $k_a = 0$ (for $a = 1, 2, 3, 4$) and $k_I = 0$. These singularities imply that the analytic structure of correlators is more complicated than that of wavefunction coefficients. In (10.13), we see these singularities arising from factors of $\langle O_a O_a \rangle'$ and $\langle O_I O_I \rangle'$ in the map between the wavefunction function coefficients and the correlator.

Three-site graph

In Exercise 10.8, you were asked to derive the five-point wavefunction coefficient in flat space. You should have found the following result:

$$\begin{aligned} \langle O_1 O_2 O_3 O_4 O_5 \rangle' &\equiv \text{Diagram} \\ &= \frac{1}{E} \frac{\lambda^3}{E_L E_M E_R} \left[\frac{1}{k_{123} + k'_I} + \frac{1}{k_{345} + k_I} \right], \end{aligned} \quad (11.67)$$

where $E_L \equiv k_{12} + k_I$, $E_M \equiv k_3 + k_I + k'_I$ and $E_R \equiv k_{45} + k'_I$ are the energies running into the three different vertices. We again see that the result has singularities when the energies entering a subgraph add up to zero. It is also easy to check that the answer has the correct factorization properties on each pole.

This pattern generalizes to arbitrary graphs. Flat-space wavefunction coefficients are singular whenever the energy flowing into a connected subgraph vanishes [82]. At tree level, the singularities are just (simple) poles, but for loops the poles can become branch points.

11.3.2 Generalization to de Sitter

The singularity structure of the wavefunction in de Sitter space is very similar to that of its flat-space analog. It is still true that the wavefunction is singular at loci where the energy flowing into a subgraph vanishes, though now the singularities can be more complicated: They can be poles of higher order or branch points (even at tree level).

The reason that the singularities of the de Sitter wavefunction have the same structure is that the divergences in the time integrals are localized at early times, where the fluctuations evolve in the same way as in Minkowski space. Concretely, the bulk-to-bulk and bulk-to-boundary propagators are built from the mode functions (10.91). In the early-time limit, these can be expanded as⁴⁷

$$f_k(\eta) \xrightarrow{k\eta \rightarrow -\infty} \frac{(-H\eta)}{\sqrt{2k}} e^{ik\eta} \left((4\nu^2 - 1) + \frac{i}{8}(4\nu^2 - 1)(4\nu^2 - 9) \frac{1}{k\eta} + \dots \right), \quad (11.69)$$

where as usual $\nu \equiv \sqrt{9/4 - m^2/H^2}$. We see that in the region where the divergences are localized, the integrand takes the form of powers of η times the corresponding flat-space integrand. These additional factors of η can change the nature of the divergences, but not their locations.

Perhaps the simplest example is the three-point wavefunction of conformally coupled scalars that we computed in (10.94):

$$\langle O_1 O_2 O_3 \rangle' \equiv \text{[Diagram: A triangle with a horizontal line above it and a vertex at the bottom]} = \frac{i\lambda}{H^4 \eta_*^3} \log(iK\eta_*), \quad (11.70)$$

which has a branch point singularity when its total energy vanishes, $K \rightarrow 0$. Similarly, the four-point wavefunction in (10.103) has branch points both when its total energy and partial energies vanish:

$$\begin{aligned} \langle O_1 O_2 O_3 O_4 \rangle' &\equiv \text{[Diagram: A parallelogram with a horizontal line above it and vertices at the bottom]} \\ &= \frac{\lambda^2}{2H^6 \eta_*^4} \frac{1}{k_I} \left[\text{Li}_2 \left(\frac{k_{12} - k_I}{E} \right) + \text{Li}_2 \left(\frac{k_{34} - k_I}{E} \right) + \log \left(\frac{E_L}{E} \right) \log \left(\frac{E_R}{E} \right) - \frac{\pi^2}{6} \right]. \end{aligned} \quad (11.71)$$

⁴⁷Here, we are using the following asymptotic expansion for the Hankel function:

$$H_\nu^{(2)}(x) \xrightarrow{x \rightarrow \infty} \left(\frac{2}{\pi x} \right)^{\frac{1}{2}} e^{-ix + \frac{i\pi}{4}(1+2\nu)} \left((4\nu^2 - 1) - \frac{i}{8}(4\nu^2 - 1)(4\nu^2 - 9)x^{-1} + \dots \right). \quad (11.68)$$

See, for example, the [DLMF](#).

Note that despite appearance there are no singularities at $k_I = 0$, $k_{12} = k_I$ and $k_{34} = k_I$.

The general characterization of singularities in de Sitter space is similar to that in flat space: Wavefunctions can be singular whenever the energy flowing into a subgraph Γ vanishes, $E_\Gamma \rightarrow 0$. At these singularities, the wavefunction factorizes

$$\lim_{E_\Gamma \rightarrow 0} \psi_n = \frac{1}{E_\Gamma^p} \tilde{A}_\Gamma^{(m)} \times \tilde{\psi}_{\Gamma'}^{(n-m)}, \quad (11.72)$$

where Γ denotes a subgraph with m external lines and Γ' is the complementary subgraph with $n - m$ external lines. We have define a rescaled amplitude and a shifted wavefunction coefficient:

$$\tilde{A}_\Gamma \equiv k_1^{\nu_1 - \frac{1}{2}} \cdots k_m^{\nu_m - \frac{1}{2}} A_\Gamma, \quad (11.73)$$

$$\tilde{\psi}_{\Gamma'} \equiv P(k_I) \left(\psi_{\Gamma'}(k_a, -k_I) - \psi_{\Gamma'}(k_a, k_I) \right), \quad (11.74)$$

where $P(k_I)$ is the power spectrum of the field connecting the two graphs Γ and Γ' . The order of the singularity, p , in (11.72) is set by the number of derivatives involved in the interactions that gave rise to it. It is not difficult to check that all the examples of wavefunction coefficients that we have computed in de Sitter space only have singularities at these locations.

11.3.3 Folded Singularities

Let us emphasize again that, although the above singularities provide important constraints on the functional form of the wavefunction coefficients, they cannot be accessed for physical kinematics. However, in principle, there could also be additional singularities for physical momentum configurations such as the “folded limit” where two (or more) external momenta become collinear. It is called “folded” because some of the external edges of the kinematic polygon are folded onto an internal line. For example, in the case of a four-point function, the folded limit is

$$k_{12} \rightarrow k_I : \quad \begin{array}{c} \text{Diagram of a four-point function kinematic polygon. The left edge is blue and labeled } k_1 \text{ and } k_2. The right edge is red and labeled } k_3 \text{ and } k_4. The internal diagonal is red and labeled } k_I. \end{array} \quad (11.75)$$

A “folded singularity” would then be a singularity in the limit $k_{12} - k_I \rightarrow 0$. This requires that some of the modes have negative frequency components which is forbidden for Bunch–Davies initial conditions. An important additional constraint will therefore be that the wavefunction coefficients do not have such folded singularities and enforcing the absence of these singularities in the solutions will be an important boundary condition.

11.4 Unitarity

One of the most basic consequences of unitarity in quantum mechanics is that probability is conserved under time evolution. In this section, we will show that this imposes important constraints on the analytic structure of the wavefunction.

For concreteness, we consider the theory of a real scalar field Φ . Given the wavefunction $\Psi[\phi, t]$, for a complete set of field eigenstates ϕ , we can express the conservation of probability as

$$\frac{\partial}{\partial t} \int \mathcal{D}\phi |\Psi[\phi, t]|^2 = 0, \quad (11.76)$$

which follows from the unitary evolution of the wavefunction,

$$i\partial_t \Psi[\phi, t] = \hat{H} \Psi[\phi, t], \quad (11.77)$$

where $\hat{H}$ is a Hermitian operator. We will see that (11.76) implies an infinite number of constraints on the wavefunctions coefficients.

11.4.1 Flat-Space Examples

We start by deriving the consequences of unitarity in a simple model in flat space:

$$S = \int d^4x \left(-\frac{1}{2}(\partial\Phi)^2 - \frac{1}{2}m^2\Phi^2 - \lambda\Phi^3 \right). \quad (11.78)$$

After we have gained intuition from this example, we will see that the results are easily generalized to cosmological backgrounds.

We first note that the wavefunction of the theory defined in (11.78) can be expanded as in (10.6):

$$\Psi[\phi] = \exp \left(-\frac{1}{2} \int_{\underline{\mathbf{k}}} \psi_2(\underline{\mathbf{k}}) \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} + \frac{1}{n!} \sum_{n=3}^{\infty} \int_{\underline{\mathbf{k}}} \psi_n(\underline{\mathbf{k}}, t) \phi_{\mathbf{k}_1} \cdots \phi_{\mathbf{k}_n} \right), \quad (11.79)$$

where we have introduced the shorthand notation

$$\int_{\underline{\mathbf{k}}} \equiv \int \frac{d^3k_1 \cdots d^3k_n}{(2\pi)^{3n}} (2\pi)^3 \delta(\mathbf{k}_1 + \cdots + \mathbf{k}_n). \quad (11.80)$$

While the two-point wavefunction coefficient $\psi_2(\underline{\mathbf{k}})$ is a constant, the higher-point wavefunction coefficients $\psi_{n \geq 3}(\underline{\mathbf{k}}, t)$ depend on time. This time dependence is governed by the Schrödinger equation (11.77), with

$$\hat{H} = \frac{1}{2} \int_{\underline{\mathbf{k}}} \left(-\frac{\delta}{\delta\phi_{\mathbf{k}_1}} \frac{\delta}{\delta\phi_{\mathbf{k}_2}} + k_1^2 \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \right) + \lambda \int_{\underline{\mathbf{k}}} \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3}. \quad (11.81)$$

We assume that each wavefunction coefficient $\psi_{n \geq 3}$ scales as λ^{n-2} , which is the appropriate scaling at tree level. Acting with (11.81) on (11.79), and expanding in ϕ , we then obtain

$$\begin{aligned} & \frac{1}{3!} \int_{\underline{\mathbf{k}}} \left(i\partial_t \psi_3(\underline{\mathbf{k}}, t) - [\psi_2(k_1) + \psi_2(k_2) + \psi_2(k_3)] \psi_3(\underline{\mathbf{k}}, t) - 6\lambda \right) \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \\ & + \frac{1}{4!} \int_{\underline{\mathbf{k}}} \left(i\partial_t \psi_4(\underline{\mathbf{k}}, t) - [\psi_2(k_1) + \cdots + \psi_2(k_4)] \psi_4(\underline{\mathbf{k}}, t) + 3\psi_3(\underline{\mathbf{k}}, t) \psi_3(\underline{\mathbf{k}}, t) \right) \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \phi_{\mathbf{k}_4} \\ & + \cdots = 0. \end{aligned} \quad (11.82)$$

Note that the coefficients must vanish at each order separately, so the Schrödinger equation has an infinite number of consequences for wavefunction coefficients. At lowest order, we have

$$i\partial_t \psi_3(\underline{\mathbf{k}}, t) = [\psi_2(k_1) + \psi_2(k_2) + \psi_2(k_3)] \psi_3(\underline{\mathbf{k}}, t) + 6\lambda, \quad (11.83)$$

$$i\partial_t \psi_4(\underline{\mathbf{k}}, t) = [\psi_2(k_1) + \cdots + \psi_2(k_4)] \psi_4(\underline{\mathbf{k}}, t) - 3\psi_3(\underline{\mathbf{k}}, t) \psi_3(\underline{\mathbf{k}}, t). \quad (11.84)$$

These equations have a simple interpretation: The time derivative of ψ_n is related to all possible ways of gluing together lower-point wavefunctions to produce an n -point wavefunction coefficient (for example, the two-point wavefunction ψ_2 can be attached to an external line of a three-point wavefunction coefficient ψ_3 to produce a three-point wavefunction coefficient).⁴⁸

Let us now return to equation (11.76), which we write as

$$i \frac{\partial}{\partial t} \int \mathcal{D}\phi |\Psi[\phi, t]|^2 = \int \mathcal{D}\phi \left[\Psi^*[\phi, t] i \partial_t \Psi[\phi, t] + i \partial_t \Psi^*[\phi, t] \Psi[\phi, t] \right] = 0. \quad (11.85)$$

Substituting (11.79) into this expression, we get

$$\int \mathcal{D}\phi |\Psi[\phi, t]|^2 \left(\sum_{n=3}^{\infty} \frac{1}{n!} \int_{\underline{\mathbf{k}}} i \partial_t \psi_n(\underline{\mathbf{k}}, t) \phi_{\mathbf{k}_1} \cdots \phi_{\mathbf{k}_n} + \sum_{n=3}^{\infty} \frac{1}{n!} \int_{\underline{\mathbf{k}}} i \partial_t \psi_n^*(\underline{\mathbf{k}}, t) \phi_{\mathbf{k}_1}^* \cdots \phi_{\mathbf{k}_n}^* \right) = 0. \quad (11.86)$$

This expression is a nonperturbative sum rule on wavefunction coefficients. It is, however, not very practically useful because we usually don't know the wavefunction nonperturbatively. However, in perturbation theory, we can expand in the small parameter λ to obtain sum rules that are valid order-by-order in this small parameter.

- At first order, we have

$$\int \mathcal{D}\phi |\Psi_0[\phi, t]|^2 \left(\int_{\underline{\mathbf{k}}} i \partial_t \psi_3(\underline{\mathbf{k}}, t) \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} + \int_{\underline{\mathbf{k}}} i \partial_t \psi_3^*(\underline{\mathbf{k}}, t) \phi_{\mathbf{k}_1}^* \phi_{\mathbf{k}_2}^* \phi_{\mathbf{k}_3}^* \right) = 0, \quad (11.87)$$

where Ψ_0 is the Gaussian part of the wavefunction. This expression consists of a sum of two integrals, but we would like to extract identities that hold for the un-integrated wavefunction coefficients. In order to do so, it is convenient to combine the integrals. Notice that they are essentially the same, we just have to relate $\phi_{\mathbf{k}}$ to $\phi_{\mathbf{k}}^*$. Since the original field Φ was real, we have $\phi_{\mathbf{k}}^* = \phi_{-\mathbf{k}}$, and by relabeling $\mathbf{k} \rightarrow -\mathbf{k}$ in the second integral, we can write things as a single integral. *However*, it will pay off to be careful about the precise way that we deform the integration contour to send $\mathbf{k} \rightarrow -\mathbf{k}$. In particular, we want to make this analytic continuation such that it flips the sign of the magnitude of $\mathbf{k}$ as well: $k \rightarrow -k$ [78, 92].⁴⁹ This then leads to the relation

$$\int \mathcal{D}\phi |\Psi_0[\phi, t]|^2 \left(\int_{\underline{\mathbf{k}}} i \partial_t \left(\psi_3(\underline{\mathbf{k}}, t) + \psi_3^*(-\underline{\mathbf{k}}, t) \right) \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \right) = 0. \quad (11.88)$$

The simplest way to satisfy this constraint is to set the integrand to zero. Then, in order to remove the time derivatives we can use the Schrödinger equation in the form (11.83) and hence get

$$\begin{aligned} 6(\lambda^* - \lambda) &= \left[\psi_2(k_1) + \psi_2(k_2) + \psi_2(k_3) \right] \psi_3(\underline{\mathbf{k}}, t) \\ &\quad - \left[\psi_2^*(-k_1) + \psi_2^*(-k_2) + \psi_2^*(-k_3) \right] \psi_3^*(-\underline{\mathbf{k}}, t). \end{aligned} \quad (11.89)$$

⁴⁸This also makes the generalization to include loops fairly straightforward, we just additionally allow contributions that fuse two external legs of an $(n+2)$ -point function to produce an n -point function.

⁴⁹One way to understand why this is convenient is to notice that with this analytic continuation, the mode functions (which depend only on the magnitudes of momenta) are analytically continued as $f_k \rightarrow f_{-k} = f_k^*$.

In the theory (11.78), we have $\psi_2(k) = k$ and λ is real, so this reduces to

$$\boxed{\psi_3(\underline{\mathbf{k}}, t) + \psi_3^*(-\underline{\mathbf{k}}, t) = 0}, \quad (11.90)$$

where we evaluate the complex conjugate of the wavefunction coefficient at negative energies. It is straightforward to check that the explicit result in (10.83) satisfies this identity.

- At the next order, the computation is slightly more intricate. Expanding out (11.86), we obtain

$$\begin{aligned} 0 = \int \mathcal{D}\phi \, |\Psi_0[\phi, t]|^2 & \left(\int_{\underline{\mathbf{k}}} i\partial_t \psi_4 \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \phi_{\mathbf{k}_4} + \int_{\underline{\mathbf{k}}} i\partial_t \psi_4^* \phi_{\mathbf{k}_1}^* \phi_{\mathbf{k}_2}^* \phi_{\mathbf{k}_3}^* \phi_{\mathbf{k}_4}^* \right. \\ & \left. + \frac{2}{3} \int_{\underline{\mathbf{k}}} \left[\psi_3 \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_I} + \psi_3^* \phi_{\mathbf{k}_1}^* \phi_{\mathbf{k}_2}^* \phi_{\mathbf{k}_I}^* \right] \int_{\underline{\mathbf{k}}} \left[i\partial_t \psi_3 \phi_{\mathbf{k}_I} \phi_{\mathbf{k}_3} \phi_{\mathbf{k}_4} + i\partial_t \psi_3^* \phi_{\mathbf{k}_I}^* \phi_{\mathbf{k}_3}^* \phi_{\mathbf{k}_4}^* \right] \right), \end{aligned} \quad (11.91)$$

where at the moment $\mathbf{k}_I$ is just a dummy momentum variable. As before, we want to put everything under the same integral. This means that in the second line, we want to integrate over $\phi_{\mathbf{k}_I}$, which leads to

$$\begin{aligned} 0 = \int \mathcal{D}\phi \, |\Psi_0[\phi, t]|^2 & \left(\int_{\underline{\mathbf{k}}} i\partial_t \psi_4 \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \phi_{\mathbf{k}_4} + \int_{\underline{\mathbf{k}}} i\partial_t \psi_4^* \phi_{\mathbf{k}_1}^* \phi_{\mathbf{k}_2}^* \phi_{\mathbf{k}_3}^* \phi_{\mathbf{k}_4}^* \right. \\ & \left. + \int_{\underline{\mathbf{k}}} \left[\psi_3 \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} + \psi_3^* \phi_{\mathbf{k}_1}^* \phi_{\mathbf{k}_2}^* \right] \frac{1}{2\text{Re}[\psi_2(k_I)]} \int_{\underline{\mathbf{k}}} \left[i\partial_t \psi_3 \phi_{\mathbf{k}_3} \phi_{\mathbf{k}_4} + i\partial_t \psi_3^* \phi_{\mathbf{k}_3}^* \phi_{\mathbf{k}_4}^* \right] + \text{perms.} \right), \end{aligned} \quad (11.92)$$

where now $\mathbf{k}_I = \mathbf{k}_1 + \mathbf{k}_2$ and “perms.” denotes momentum permutations to other exchange channels. We analytically continue $\mathbf{k}$ in the terms involving $\phi_{\mathbf{k}}^*$, so that the wavefunction coefficients appearing there are functions of $-k_1, \dots, -k_4$. We will denote this by $\psi^*(-\underline{k})$, but it is important to keep in mind that they still depend on $+k_I$. We then have the identity

$$\begin{aligned} 0 = \int \mathcal{D}\phi \, |\Psi_0[\phi, t]|^2 & \int_{\underline{\mathbf{k}}} i\partial_t \left(\psi_4(\underline{k}) + \psi_4^*(-\underline{k}) \right) \\ & + \frac{1}{2\text{Re}[\psi_2(k_I)]} \left[\psi_3(\underline{k}) + \psi_3^*(-\underline{k}, k_I) \right] \left[\psi_3(\underline{k}) + \psi_3^*(k_I, -\underline{k}) \right] + \text{perms.} \Big) \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \phi_{\mathbf{k}_4}. \end{aligned} \quad (11.93)$$

Setting the time derivative in parentheses to zero and using the Schrödinger equation, we find that the following identity: **[DB: Introduce shifted wavefunction coefficients. Compare to (11.117).]**

$$\boxed{\psi_4(\underline{k}) + \psi_4^*(-\underline{k}) = -\frac{1}{2\text{Re}[\psi_2(k_I)]} \left(\psi_3(\underline{k}, k_I) + \psi_3^*(-\underline{k}) \right) \left(\psi_3(\underline{k}) + \psi_3^*(k_I, -\underline{k}) \right) + \text{perms.}}. \quad (11.94)$$

Conceptually, we interpret the identity (11.86) as telling us that order-by-order certain combinations of wavefunction coefficients are constants of motion [92]. Upon using the Schrödinger equation, these identities imply that combinations of n -point wavefunction coefficients are related to lower-point wavefunction coefficients. Since these relations split wavefunction coefficients coming from exchanges into smaller pieces, they can be interpreted as “cutting” the diagrams that produce the wavefunction in perturbation theory.

11.4.2 Cutting Rules

We have just seen that the perturbative expansion of the conservation of probability (11.76) implies constraints on the wavefunction coefficients that comprise the wavefunction Ψ . In principle, we could continue the procedure to higher orders. However, as we have already seen at next-to-leading order, this becomes algebraically fairly involved. We therefore would like to find a systematic way of implementing this procedure.

Optical theorem

We can draw inspiration from in-out quantum field theory, where unitarity of time evolution implies that the S -matrix is unitary, $S^\dagger S = 1$. If we write $S = 1 + iT$, then the transfer matrix T captures the nontrivial part of the S -matrix. Given an initial state $|i\rangle$ and a final state $|f\rangle$, we have

$$\langle f|T|i\rangle \equiv (2\pi)^4 \delta(p_i - p_f) i A(i \rightarrow f), \quad (11.95)$$

where $A(i \rightarrow f)$ is the scattering amplitude. The condition $S^\dagger S = 1$ then implies $T - T^\dagger = iT^\dagger T$. Sandwiching this between the states $\langle f|$ and $|i\rangle$, inserting a resolution of the identity, which sums over a complete set of multi-particle states $|X\rangle$ and using (11.95), this becomes the *generalized optical theorem*⁵⁰

$$2\text{Im} A(i \rightarrow f) = \sum_X \int d\Pi_X (2\pi)^4 \delta(p_i - p_X) A(i \rightarrow X) A^*(f \rightarrow X), \quad (11.97)$$

where $d\Pi_X$ is the phase space integral for the $|X\rangle$ states.

Cutkosky rules

The expression in (11.97) is a nonperturbative statement. Expanding the result in powers of the couplings in the theory leads to a constraint on the sum of Feynman diagrams at each order in perturbation theory. These identities relate the imaginary part of an amplitude to the possible ways of putting intermediate particles on-shell. Since this momentum configuration splits the diagram in two, these contributions are called *cuts*, and the corresponding identities are often called *Cutkosky rules*. Conceptually, we can think of these constraints as arising from the fact that the sum of the Feynman propagator $G_F(p^2)$ and its complex conjugate $\overline{G}_F(p^2)$ satisfy

$$G_F(p^2) + \overline{G}_F(p^2) = \frac{-i}{p^2 + m^2 - i\epsilon} + \frac{i}{p^2 + m^2 + i\epsilon} = 2\pi\delta(p^2 + m^2), \quad (11.98)$$

which is a version of the Sokhotski–Plemelj distributional identity. This formula relates the imaginary part of the Feynman propagator to a delta function that places the associated particle on shell.

⁵⁰The special case $|i\rangle = |f\rangle = |S\rangle$, for some state $|S\rangle$, is of interest in many applications. Equation (11.97) then reads

$$2\text{Im} A(S \rightarrow S) = \sum_X \int d\Pi_X (2\pi)^4 \delta(p_S - p_X) |A(S \rightarrow X)|^2. \quad (11.96)$$

When $|S\rangle$ is a single-particle state, the right-hand side can be written in term of the total decay rate of the particle. If $|S\rangle$ is a two-particle state, then the right-hand side can be expressed as the total cross section (in the center-of-mass frame). This latter case is often called the *optical theorem*.

The optical theorem (11.97) includes all the ways that internal lines can be put on shell via (11.98) in the sum over Feynman graphs that contribute to a given process. However, it is possible to derive constraints diagram-by-diagram by applying the identity (11.98) directly to a given graph (see e.g. [110]). In a sense these identities are stronger than the optical theorem, since they hold for individual diagrams, but the price to pay is that the relationship to unitarity is somewhat obscured, because similar identities also hold in a generic theory (albeit they relate physical quantities to unphysical ones). We now want to derive the analogs of these identities satisfied by the wavefunctional.

Cutting the wavefunction

Our starting point is the version of the identity (11.98) satisfied by the bulk-to-bulk propagator. Adding the bulk-to-bulk propagator (10.82) and its complex conjugate $\bar{G}(k; t, t') = -G(-k; t, t')$ (called the anti-bulk-to-bulk propagator), we find

$$\begin{aligned} G(k; t, t') + \bar{G}(k; t, t') &= -\frac{1}{2k} \left(e^{-ikt} - e^{ikt} \right) \left(e^{-ikt'} - e^{ikt'} \right) \\ &= -\frac{1}{2k} \text{Im}[K_k(t)] \text{Im}[K_k(t')], \end{aligned} \quad (11.99)$$

where we have written the right-hand side in terms of the bulk-to-boundary propagator. It will be useful to define the sum $\tilde{G}(k; t, t') \equiv G(k; t, t') + \bar{G}(k; t, t')$ as a new object, which we will call the *cut propagator*.

Similar to (11.98), the right-hand side of (11.99) has no time ordering. However, it importantly does not simply involve a delta function putting the particle on-shell. This is one indication that cutting rules for the wavefunction will be more complicated than those for the S -matrix. A further complication is that when computing the wavefunction, we assign bulk-to-boundary propagators to external lines, whereas we typically consider amputated S -matrix elements. To deal with these issues, it is convenient to define a *conjugate wavefunction*, $\bar{\Psi}$, whose wavefunction coefficients $\bar{\psi}$ are computed with slightly modified Feynman rules compared to those introduced in Section 10.5. We still use the normal bulk-to-boundary propagator for external lines, but we use the anti-bulk-to-bulk propagator $\bar{G}$ for internal lines, and assign a vertex factor $-iV$ (rather than $+iV$), to each interaction vertex. Obviously, this conjugate wavefunction is closely related to the original wavefunction, and in fact in a unitary theory, we have [78, 92, 111]

$$\bar{\psi}_n(k_1, \dots, k_n) = \psi_n^*(-k_1, \dots, -k_n). \quad (11.100)$$

This relation is clear for internal lines (where complex conjugation sends $\bar{G} \rightarrow G$), and for vertices (where unitary theories satisfy $V^\dagger = V$). In addition to this, we require the fact that the bulk-to-boundary propagator satisfies

$$K_{-k}^*(t) = K_k(t), \quad (11.101)$$

which is obviously true for the exponential bulk-to-boundary propagator in the situation at hand.

The strategy is now easy to state: We want to combine ψ and $\bar{\psi}$ in such a way that we can replace the internal bulk-to-bulk propagators using the identity (11.99).⁵¹ This can be done in any theory. In a unitary theory, we can additionally relate $\bar{\psi}$ to ψ^* using (11.100) in order to obtain identities satisfied by wavefunction coefficients themselves.

Let us begin by considering the simplest example of a single exchange. In this case, the wavefunction coefficient in Φ^3 theory in flat space is given by

$$\psi_4(k_{12}, k_{34}) = -\lambda^2 \int_{-\infty}^0 dt dt' e^{ik_{12}t} G(k_I; t, t') e^{ik_{34}t'}. \quad (11.103)$$

By adding this expression with its conjugate version, we obtain the identity

$$\psi_4(k_{12}, k_{34}) + \bar{\psi}_4(k_{12}, k_{34}) = -\lambda^2 \int_{-\infty}^0 dt dt' e^{ik_{12}t} \tilde{G}(k_I; t, t') e^{ik_{34}t'}, \quad (11.104)$$

where $\tilde{G}(k_I; t, t')$ is the cut propagator given by (11.99). The appearance of $\tilde{G}$ factorizes the time integrals, each of which takes the form

$$\tilde{\psi}_3(k_{12} \mp k_I) \equiv \frac{i}{2k_I} \int_{-\infty}^0 dt e^{ik_{12}t} \left(e^{-ik_I t} - e^{ik_I t} \right) = \frac{1}{2k_I} \left(\psi_3(k_{12} - k_I) - \psi_3(k_{12} + k_I) \right). \quad (11.105)$$

After relating $\bar{\psi}$ to ψ^* we obtain the identity

$$\psi_4(k_{12}, k_{34}) + \psi_4^*(-k_{12}, -k_{34}) = -2k_I \tilde{\psi}_3(k_{12} \mp k_I) \tilde{\psi}_3(-k_{34} \mp k_I), \quad (11.106)$$

which is valid in a unitary theory. It is straightforward to verify that this identity is satisfied by the wavefunction coefficients in the theory (11.78):

$$\begin{aligned} \psi_4 &= \frac{\lambda^2}{(k_{12} + k_{34})(k_{12} + k_I)(k_{34} + k_I)}, \\ \psi_3 &= \frac{\lambda}{k_1 + k_2 + k_3}. \end{aligned} \quad (11.107)$$

Though we have derived this cutting rule in the specific example of a Φ^3 theory, expressed in this way, it is valid more generally. Notice also that this formula exactly reproduces the unitarity constraint (11.94) derived from the Schrödinger equation.

Exercise 11.5 Consider the (non-unitary) action

$$S = \int d^4x \left(-\frac{1}{2}(\partial\phi)^2 + \frac{1}{2}(\partial\chi)^2 - \lambda\phi^2\chi \right), \quad (11.108)$$

which couples together a healthy field ϕ , with a ghostly field χ . Compute the four-point wavefunction coefficient for ϕ arising from exchanging a χ field, and verify that it does *not*

⁵¹Roughly speaking, we can think of the cutting rules for the wavefunction as implementing the following identity [110]

$$A_1 \cdots A_n - B_1 \cdots B_n = C_1 B_2 \cdots B_n + A_1 C_2 B_3 \cdots B_n + \cdots + A_1 \cdots A_{n-1} C_n, \quad (11.102)$$

for A, B and C which satisfy $A_j - B_j = C_j$. Here $A = G$, $B = -\bar{G}$, and $C = \tilde{G}$ and the cutting rules systematize the application of the identity (11.102), giving a relation between the wavefunction and conjugate wavefunction.

Cutting general graphs

We now wish to briefly describe the generalization of the formula (11.106) to general graphs. In order to do this, it is convenient to introduce some notation. We define the *cut wavefunction* as

$$\text{Cut}[\psi(k_a, k_I)] \equiv \psi(k_a, k_I) + \psi^*(-k_a, k_I), \quad (11.109)$$

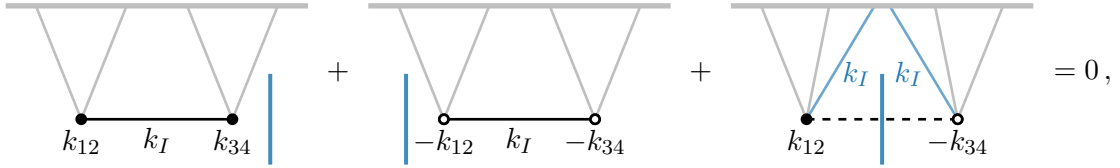
which adds together the wavefunction and its complex conjugate with its external energies flipped. In general, we can specify which of the energies are flipped under this operation. For example, we can write the shifted wavefunction coefficient appearing in (11.105) as

$$\tilde{\psi}_3(k_{12}, k_I) = \frac{i}{2k_I} \text{Cut}_{k_I}[\psi(k_{12}, k_I)], \quad (11.110)$$

where the subscript k_I indicates that we should flip the internal energy. With these definitions, we can write the formula (11.106) in a uniform way as

$$\text{Cut}_{k_a}[\psi_4(k_{12}, k_{34}, k_I)] = \frac{1}{2k_I} \text{Cut}_{k_I}[\psi_3(k_{12} + k_I)] \text{Cut}_{k_I}[\psi_3(-k_{34} + k_I)], \quad (11.111)$$

where on the left-hand side we flip the sign of the external energies, while on the right-hand side the flip the sign of the internal energy. This equation can be given a diagrammatic interpretation:



where the first two terms correspond to the two terms inside the “Cut” of the four-point wavefunction, and the remaining term is the product of the cuts of the two three-point functions obtained by dividing the diagram in two with an internal line, times (minus) the propagator of the cut line. We color vertices to the right of the dividing line white, and ones to the left black, and flip the energies of the white vertices. The diagrammatic version of (11.111) is then just the statement that the sum of all cuts of the diagram vanishes.

We can generalize this logic to higher-point graphs. By carefully applying the identity (11.99), we find that the following schematic formula holds:

$$\psi_n(X) + \psi_n^*(-X) = - \sum_{\text{cuts}} \psi_n. \quad (11.112)$$

On the left-hand side, we have separated out the cuts that pass to one side of the graph (where X stands for all external energies). The right-hand side stands for all cuts that pass through the graph, where we first flip the signs of all vertices to the right of the cut, and then act with the “Cut” operation (11.109) on i times each of the wavefunctions that the cut splits the graph into, and then multiply by $-P(k_I)$, with P the power spectrum of the cut lines. In the following, we will only make explicit use of the simplest formula (11.110), but see [81, 95] for proofs of the general statements, and examples.

11.4.3 Generalization to de Sitter

Our discussion of cutting correlators in flat space has only relied on very general properties of the Green's functions used to compute the wavefunction, so the essential features generalize almost immediately to other spacetimes. Our particular interest will be in de Sitter backgrounds because the relevant propagators take a simplified form, but the extension to even more general situations is also straightforward.

Recall that the line element of de Sitter space in the flat slicing is given by (11.9). The bulk dynamics of fields in de Sitter space give rise to correlations between fluctuations on the late-time ($\eta = 0$) surface. In the conventional inflationary picture, this is the reheating surface where these perturbations serve as the initial conditions for the subsequent evolution of the universe. The goal is to understand the structure of these fluctuations without tracking their detailed time evolution through the de Sitter spacetime. Cuts of correlators give us some partial information about this time evolution. An interesting feature of the cutting rules in de Sitter space is that they apply equally well in cases where the symmetries of the background spacetime are broken by the interaction vertices. These symmetry-breaking interactions are phenomenologically relevant in inflationary models with large non-Gaussianities.

Only two generalizations of the previous discussion are needed to import our results to de Sitter space. First, the bulk-to-bulk propagators are slightly different, so we have to re-derive the cut propagators. Second, the bulk-to-boundary propagators are no longer just exponentials, so we have to be careful about the precise analytic continuation that relates $\bar{\psi}$ to ψ .

We begin by collecting the necessary ingredients for cutting scalar lines in de Sitter. The bulk-to-boundary propagator for a general massive scalar is

$$G(k; \eta, \eta') \equiv \frac{\pi}{4} (\eta \eta')^{3/2} \left[H_\nu^{(2)}(-k\eta') H_\nu^{(1)}(-k\eta) \theta(\eta - \eta') + H_\nu^{(2)}(-k\eta) H_\nu^{(1)}(-k\eta') \theta(\eta' - \eta) - \frac{H_\nu^{(1)}(-k\eta_*)}{H_\nu^{(2)}(-k\eta_*)} H_\nu^{(2)}(-k\eta) H_\nu^{(2)}(-k\eta') \right]. \quad (11.113)$$

In flat space, the conjugate wavefunction coefficients $\bar{\psi}$ are related to the complex conjugate of the ordinary wavefunction coefficients, but with the external energies analytically continued to negative values. To apply a similar analytic continuation in de Sitter space, we note that the complex conjugate of the bulk-to-boundary propagator satisfies [78]

$$K^*(k, \eta) = K(e^{-i\pi} k, \eta). \quad (11.114)$$

This relation instructs us to analytically continue by rotating the energies clockwise in the complex plane to negative arguments. In flat space, the details of this analytic continuation did not matter because none of the answers had branch cuts, but some correlators in de Sitter space do, so we should navigate the branch cuts according to this prescription. In the following, we will typically leave this implicit, but negative energies should be understood in this sense.

Using bulk-to-bulk propagator (11.113), it is easy to show that the cut propagator $\tilde{G} \equiv G + \overline{G}$ can be written in terms of the bulk-to-boundary propagator as⁵²

$$\tilde{G}(k; \eta, \eta') = -P(k) \left(K(-k, \eta) - K(k, \eta) \right) \left(K(-k, \eta') - K(k, \eta') \right), \quad (11.115)$$

where we have introduced the power spectrum of the exchanged field

$$P(k) \equiv \frac{\pi}{4} \eta_*^3 H_\nu^{(1)}(-k\eta_*) H_\nu^{(2)}(-k\eta_*). \quad (11.116)$$

The cut propagator (11.115) is structurally the same as the formula (11.99) for the flat-space cut propagator. In particular, it also factorizes. As a result, the cutting formulas in de Sitter are essentially identical—we just have to replace all flat-space objects with their de Sitter counterparts, to obtain

$$\boxed{\psi_4(k_{12}, k_{34}) + \psi_4^*(-k_{12}, -k_{34}) = -P(k_I) \tilde{\psi}_3(k_{12} \mp k_I) \tilde{\psi}_3(-k_{34} \mp k_I)}, \quad (11.117)$$

where the shifted wavefunctions are defined as in (11.105) and the analytic continuation to negative arguments is clockwise in the complex plane as in (11.114).

Exercise 11.6 Verify the cutting rule (11.117) for the four-point function of conformally coupled scalars in de Sitter space, where

$$\psi_4 = \frac{1}{2k_I} \left[\text{Li}_2 \left(\frac{k_{12} - k_I}{E} \right) + \text{Li}_2 \left(\frac{k_{34} - k_I}{E} \right) + \log \left(\frac{E_L}{E} \right) \log \left(\frac{E_R}{E} \right) - \frac{\pi^2}{6} \right], \quad (11.118)$$

$$\psi_3 = i \log(K/\mu) - \frac{\pi}{2}. \quad (11.119)$$

Show that the real part in (11.119) is required in order for the three-point function to have vanishing cut. ➔

11.4.4 Singularities Revisited

The cosmological cutting rules is a powerful tool to bootstrap tree-level exchange diagrams without any assumption about dS boosts [79]. The general idea is to use the knowledge of the analytical structure of wavefunction coefficients to compute them via Cauchy's integral formula. Recall that tree-level wavefunction coefficients have a very simple analytic structure: the only singularities are poles where partial energies vanish. In Minkowski, these are all simple poles so the amplitude limit is sufficient to fix all relevant residues [82]. Conversely, in general FRW spacetimes and in dS poles can be of higher orders. The crucial insight is that the cosmological cutting rules fix the residues of all partial energy singularities of a given diagram in terms of lower-order diagrams [79], in a way analogous to factorization theorems for amplitudes. What is left is fixing the terms that are regular in all partial energies, which appear as boundary terms in Cauchy's integral formula.

⁵²In this expression, we have pulled out an overall factor of -1 on the right-hand side for uniformity with the other cut propagator expressions that we have presented. When evaluating this expression for explicit mass values, for example for $\nu = 3/2$ (appropriate for massless fields), this sign can be compensated by choosing the branch of the square root that appears appropriately. The result then recovers the previously shown expressions.

Exercise 11.7 Consider a theory with boost-breaking $\dot{\Phi}^3$ interaction in de Sitter space. The three-point wavefunction coefficient is

$$\psi_3 = 2\lambda \frac{(k_1 k_2 k_3)^2}{(k_1 + k_2 + k_3)^3}. \quad (11.120)$$

Using the cutting rule to derive the exchange four-point function. You should find

$$\begin{aligned} \psi_4 = 4\lambda^2 \frac{(k_1 k_2 k_3 k_4)^2}{E^5 E_L^3 E_R^3} & \left(k_I^2 \left(6E_L^2 E_R^2 + 3EE_L E_R(E_L + E_R) + E^2(E_L + E_R)^2 \right. \right. \\ & \left. \left. + E^3(E_L + E_R) + E^4 \right) - 6E_L^3 E_R^3 \right), \end{aligned} \quad (11.121)$$

where E , E_L , E_R are defined as before.



11.5 Cosmological Collider

As a concrete example of the bootstrap approach, we now revisit the challenge posed at the end of Section 10.7: We would like to derive the four-point function of conformally coupled scalars, mediated by the tree-level exchange of a generic massive scalar.⁵³ This is the fundamental seed function from which the signals of the cosmological collider can be derived [44]:

1. Acting with a set of differential operators—called *weight-shifting operators*—on the seed solution gives the result for massless scalars.
2. Another type of weight-shifting operators transforms the result for the exchange of a massive scalar to that of particles with spin.
3. Putting one of the external legs on the background (plus perturbing the scaling dimension of the associated field) leads to the results for the inflationary bispectrum.

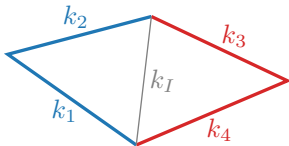
This shows that the four-point function of conformally coupled scalars is the fundamental seed from which a larger class of phenomenologically interesting inflationary correlators can be derived.

Ward identity

We first derive the Ward identity for the relevant wavefunction coefficient $\langle O_1 O_2 O_3 O_4 \rangle'$. A priori, this four-point function depends on four three-momenta, $\mathbf{k}_{a=1,2,3,4}$, and hence 12 scalar variables. However, translational and rotational invariance reduce this to six independent variables before imposing conformal symmetry. It is convenient to take these variables to be $k_{a=1,2,3,4} \equiv |\mathbf{k}_a|$, $k_I \equiv |\mathbf{k}_1 + \mathbf{k}_2|$ and $k'_I \equiv |\mathbf{k}_2 + \mathbf{k}_3|$. Constraints from dilatation symmetry and SCTs reduce these six independent variables to just two, which in position space are the conformally-invariant cross ratios. In momentum space, we will use the dimensionless variables

$$u \equiv \frac{k_I}{k_1 + k_2}, \quad v \equiv \frac{k_I}{k_3 + k_4}. \quad (11.122)$$

In Section 11.2.5, we explained that dilatation invariance forces an N -point function in d dimensional momentum space to have scaling dimension $\bar{\Delta}_t - d(N - 1)$. For the four-point function of conformally coupled scalars ($\bar{\Delta} = 2$) in $d = 3$ dimensions, this implies that $\langle O_1 O_2 O_3 O_4 \rangle'$ has dimension $4 \times 2 - 3 \times (4 - 1) = -1$. This motivates us to define

$$\langle O_1 O_2 O_3 O_4 \rangle' = \text{Diagram} = \frac{1}{k_I} \hat{F}(u, v), \quad (11.123)$$


which automatically solves the dilatation Ward identity.

At this point, the dimensionless function $\hat{F}(u, v)$ is still completely undetermined. Its functional form will be constrained by the special conformal Ward identity and the allowed singularities of a given perturbative process. After quite some work (see Appendix A of [44]), the

⁵³We will focus on the s -channel contribution, from which the t and u -channels can be obtained as simple permutations.

conformal Ward identity (11.42) can be written as

$$(\Delta_u - \Delta_v)\hat{F} = 0, \quad (11.124)$$

where $\Delta_u \equiv u^2(1 - u^2)\partial_u^2 - 2u^3\partial_u$ is our old friend the hypergeometric differential operator.

Exercise 11.8 Derive the result (11.124).

Warning: This is algebraically very involved and tedious.



Our task is to classify the solutions to (11.124). We will do this by making a tree-level approximation and organizing the solutions by their singularities.

Contact solutions

The simplest solutions to equation (11.124) correspond to the four-point functions arising from contact interactions:

$$\hat{F}_c = \begin{array}{c} \text{---} \\ \diagdown \quad \diagup \\ \text{---} \end{array}$$

These solutions are characterized by the simplest possible singularity structure with only total energy singularities. Consider first the case of a bulk Φ^4 interaction. As we have seen in Exercise 10.10, the four-point wavefunction coefficient coming from a Φ^4 interaction in both flat space and de Sitter space is proportional to $1/E$, where $E \equiv k_1 + k_2 + k_3 + k_4$ is the total energy. Up to an overall constant factor, we therefore have

$$\hat{F}_{c,0} = \frac{k_I}{E} = \frac{uv}{u+v}. \quad (11.125)$$

It is easy to confirm that this is indeed a solution of (11.124). Solutions corresponding to higher-derivative contact interactions in the bulk are created by the repeated application of the operator Δ_u .⁵⁴

$$\hat{F}_{c,n} \equiv \Delta_u^n \hat{F}_{c,0}, \quad (11.126)$$

It is easy to see that these are indeed solutions to (11.124). For example,

$$\begin{aligned} (\Delta_u - \Delta_v)\hat{F}_{c,1} &= (\Delta_u - \Delta_v)\Delta_u \hat{F}_{c,0} \\ &= (\Delta_u^2 - \Delta_v \Delta_u)\hat{F}_{c,0} \\ &= (\Delta_u^2 - \Delta_u \Delta_v)\hat{F}_{c,0} \\ &= \Delta_u(\Delta_u - \Delta_v)\hat{F}_{c,0} = 0, \end{aligned} \quad (11.127)$$

where we used that $\Delta_v \Delta_u \hat{F}_{c,0} = \Delta_u \Delta_v \hat{F}_{c,0}$ because the seed solution $\hat{F}_{c,0}$ is symmetric in u and v .

⁵⁴This only captures the contact solutions arising from integrating out massive scalars. Integrating out particles with spin can produce contact solutions with dependence on additional kinematic invariants.

Exercise 11.9 By acting with Δ_u on the seed solution (11.125), show that

$$\hat{F}_{c,1} = -2 \left(\frac{uv}{u+v} \right)^3 \frac{1+uv}{uv}, \quad (11.128)$$

$$\hat{F}_{c,2} = -4 \left(\frac{uv}{u+v} \right)^5 \frac{u^2 + v^2 + uv(3u^2 + 3v^2 - 4) - 6(uv)^2 - 6(uv)^3}{(uv)^3}. \quad (11.129)$$

We see that even these contact solutions are relatively intricate functions of the momenta, but are fully controlled by symmetries and singularities without any reference to a Lagrangian. →

The sum of contact solutions can be written as

$$\hat{F}_c = \sum_{n=0}^{\infty} \left(\frac{k_I}{E} \right)^{2n+1} \hat{f}_n(u, v) + \text{perms}, \quad (11.130)$$

where the functional form of the dimensionless functions $\hat{f}_n(u, v)$ is fixed by conformal invariance. The expansion in powers of E^{-1} corresponds to the derivative expansion of the bulk EFT, with higher-order poles corresponding to higher-derivative interactions. The limit $E \rightarrow 0$ of the individual contact solutions is

$$\lim_{E \rightarrow 0} \hat{F}_{c,n} = (2n)! \frac{s^{2n}}{E^{2n+1}}, \quad (11.131)$$

where s is the Mandelstam invariant. This confirms the expected relation between the singularity of the wavefunction coefficient at $E = 0$ and the scattering amplitude for contact interactions.

Exchange solutions

Next, we look for solutions of the Ward identity corresponding to the tree-level exchange of a scalar particle:

$$\hat{F}_e = \text{[Diagram: A trapezoidal shape representing a scalar particle exchange. It has a horizontal top line, a horizontal bottom line, and two slanted lines connecting them. The top line is longer than the bottom line. The left slanted line is on the left, and the right slanted line is on the right.]}$$

The solution will have additional partial-energy singularities.

To derive this solution, we will proceed by inspired guesswork. In Section 11.2.5, we have seen that the three-point function of two conformally coupled scalars and a generic massive scalar, $\langle O_1 O_2 X_3 \rangle' = k_3^{\bar{\Delta}-2} \hat{G}(u)$, with $u \equiv k_3/k_{12}$, satisfies

$$\left[\Delta_u + \left(\mu^2 + \frac{1}{4} \right) \right] \hat{G} = 0, \quad (11.132)$$

where Δ_u is the same operator as that appearing in the four-point Ward identity (11.124). It is therefore easy to see that a product of two three-point functions, $\hat{F}(u, v) = \hat{G}(u) \hat{H}(v)$, is also a

solution of (11.124):

$$\begin{aligned}
(\Delta_u - \Delta_v)\hat{G}(u)\hat{H}(v) &= (\Delta_u\hat{G})\hat{H} - \hat{G}(\Delta_v\hat{H}) \\
&= -\left(\mu^2 + \frac{1}{4}\right)\hat{G}\hat{H} - \hat{G}\left[-\left(\mu^2 + \frac{1}{4}\right)\hat{H}\right] \\
&= -\left(\mu^2 + \frac{1}{4}\right)(\hat{G}\hat{H} - \hat{G}\hat{H}) \\
&= 0.
\end{aligned} \tag{11.133}$$

This isn't surprising since a four-point function factorizes into a product of three-point functions in the OPE limit.

This motivates us to find a more general solution of (11.124), not of the factorized form, but obeying

$$\begin{aligned}
\left[\Delta_u + \left(\mu^2 + \frac{1}{4}\right)\right]\hat{F}_e &= \hat{C}(u, v), \\
\left[\Delta_v + \left(\mu^2 + \frac{1}{4}\right)\right]\hat{F}_e &= \hat{C}(u, v).
\end{aligned} \tag{11.134}$$

Note that these are two *ordinary* differential equations in u and v separately. The solution corresponding to the product of three-point functions are the homogeneous solutions to these equations, which explains the introduction of the μ -dependent terms, to dictate the mass of the exchanged particle.

Exercise 11.10 Show that for consistency the function $\hat{C}(u, v)$ appearing on the right-hand side of (11.134) must satisfy $(\Delta_u - \Delta_v)\hat{C} = 0$ and is thus itself conformally invariant. ➔

What is the physical interpretation of the source function $\hat{C}(u, v)$? Consider the limit of large μ , for which we expect the interaction to effectively reduce to a contact interaction, $\hat{F}_e \rightarrow \hat{F}_c = \mu^{-2}\hat{C}$. The source function on the right-hand side of (11.134) is therefore just one of our previously found contact solutions. Given our classification of these contact solutions, we can therefore also classify the corresponding exchange solutions. In Lagrangian language, this classification corresponds to choosing the bulk interactions of the theory.

Another way of understanding (11.134) is by considering the example of tree-exchange amplitudes. Multiplying an s -channel amplitude by the inverse propagator, we get

$$(s - M^2)A_e = C(s, t), \tag{11.135}$$

where the right-hand side is a rational function of the Mandelstam variables, i.e. a sum of contact amplitudes. The structure of (11.135) is similar to (11.134) except that it is an algebraic equation rather than a differential equation.

Exercise 11.11 In Section 10.7, we derived the four-point wavefunction coefficient of conformally coupled scalars:

$$\langle O_1 O_2 O_3 O_4 \rangle'_c = \frac{\lambda_4}{H^4 \eta_*^4} \frac{1}{E}, \quad (11.136)$$

$$\begin{aligned} \langle O_1 O_2 O_3 O_4 \rangle'_e = \frac{\lambda_3^2}{2H^6 \eta_*^4} \frac{1}{k_I} & \left[\text{Li}_2 \left(\frac{k_{12} - k_I}{E} \right) + \text{Li}_2 \left(\frac{k_{34} - k_I}{E} \right) \right. \\ & \left. + \log \left(\frac{k_{12} + k_I}{E} \right) \log \left(\frac{k_{34} + k_I}{E} \right) - \frac{\pi^2}{6} \right], \end{aligned} \quad (11.137)$$

where the subscripts denote the contact (c) and exchange (e) contributions. Show that these solutions solve the differential equations in (11.134) for $\lambda_4 = \lambda_3^2/H^2$. →

A formal solution of (11.134) can be obtained by inverting the differential operator:

$$\begin{aligned} \hat{F}_e = \frac{\hat{C}}{\Delta_u + M^2} &= \sum_n \frac{1}{n!} \left(\frac{\Delta_u}{M^2} \right)^n \frac{\hat{C}}{M^2} \\ &= \hat{F}_{c,0} + \frac{\hat{F}_{c,1}}{M^2} + \frac{1}{2} \frac{\hat{F}_{c,2}}{M^4} + \dots, \end{aligned} \quad (11.138)$$

where $M^2 \equiv \mu^2 + \frac{1}{4}$. We see that the solution has been written as an infinite expansion of contact solutions, which corresponds to the *EFT expansion* of the correlator as expected from integrating out a massive particle. However, as we will see, this EFT expansion misses nonperturbative terms associated with particle production in the expanding spacetime.

In the rest of this section, we will find an explicit solution to (11.134) corresponding to the exchange of a generic massive particle, thereby addressing the challenge posed in Section 10.7. For concreteness, we consider the simplest contact interaction, $\hat{F}_{c,0}$, as a source. The solutions for other sources, $\hat{F}_{c,n}$, can then simply be found by acting with Δ_u on the exchange solution $\hat{F}_{e,0}$. The equation we would like to solve then is

$$\left[u^2(1 - u^2)\partial_u^2 - 2u^3\partial_u + \left(\mu^2 + \frac{1}{4} \right) \right] \hat{F} = \frac{uv}{u + v}, \quad (11.139)$$

where we have rescaled $\hat{F}$ to set the normalization of the source function to unity. Note that this is the same as (10.133) if we recall that $\mu^2 + 1/4 = m^2/H^2 - 2$.

Singularities

To find the full solution of (11.139), we must impose boundary conditions for specific values of u and v . We will do this by looking at the singular points of the differential equation.⁵⁵ First, we

⁵⁵Alternatively, one can also use the correct normalization of the unitarity cut as a boundary condition. This was described in [94].

note that the differential operator on the left-hand side of (11.139) becomes singular at $u \rightarrow 0$ and $u \rightarrow \pm 1$. Let us discuss the physical meaning of these singularities:

- *Factorization limit:* In the limit $u \rightarrow -1$, the equation has a logarithmic singularity

$$\lim_{u \rightarrow -1} \hat{F} \propto A_3 \log(1+u). \quad (11.140)$$

This is precisely the partial energy singularity at $E_L = 0$ that we discussed in Section 11.3. Correctly normalizing this singularity (together with the corresponding singularity at $E_R = 0$ or $v \rightarrow -1$) will be one of the boundary conditions for our differential equation.

- *Folded limit:* Similarly, in the limit $u \rightarrow +1$, we have

$$\lim_{u \rightarrow +1} \hat{F} \propto \log(1-u). \quad (11.141)$$

This is a singularity in the folded limit, $k_1 + k_2 \rightarrow k_I$, where two of the momenta become collinear. As we discussed in Section 11.3, this singularity should be absent for Bunch–Davies initial conditions. We will take this as a second boundary condition for our differential equation.

- *Collapsed limit:* Imposing the boundary conditions at $u = \pm 1$ will completely fix the solution and the behavior at the remaining singularities becomes an output. For example, in the limit $u \rightarrow 0$, the solution becomes

$$\lim_{u \rightarrow 0} \hat{F} \propto u^{\frac{1}{2} + i\mu}. \quad (11.142)$$

This is the so-called collapsed limit of the correlator where the internal momentum becomes soft, $k_I \rightarrow 0$. As we will see, the boundary conditions of the problem will force us add a fixed amount of the homogeneous solutions which has this oscillatory behavior in the collapsed limit. Physically, the non-analytic scaling for $u \rightarrow 0$ is the signature of particle production in the bulk spacetime. We will have more to say about this below.

- *Flat-space limit:* Finally, the source term on the right-hand side of (11.139) has a singularity for $u \rightarrow -v$. Of course, this is simply the total energy singularity of the contact solution. We therefore expect the full exchange solution also to have this singularity:

$$\lim_{u \rightarrow -v} \hat{F} = A_4 (u+v) \log(u+v). \quad (11.143)$$

The coefficient of this singularity must be the correct four-point exchange amplitude (in the high-energy limit), $A_4 = 1/s$, where $s = k_{12}^2 - k_I^2$ is the Mandelstam variable. Reproducing this singularity will be an important consistency check for our solution.

Harmonic oscillator

Before discussing the general solution of (11.139), it is instructive to consider the limit $v \rightarrow 0$, where the source term is small. Writing $u \equiv \xi v$ and $\hat{F} \equiv v \tilde{F}$, the differential equation (11.139) then takes the form

$$\left[\xi^2 \partial_\xi^2 + \left(\mu^2 + \frac{1}{4} \right) \right] \tilde{F} = \frac{\xi}{1+\xi}, \quad (11.144)$$

and defining $t \equiv \log \xi$ and $f \equiv \tilde{F}/\sqrt{\xi}$, it becomes

$$\left[\frac{d^2}{dt^2} + \mu^2 \right] f = \frac{1}{2 \cosh(\frac{1}{2}t)}, \quad (11.145)$$

which is the equation of a *forced harmonic oscillator*.

- The *homogeneous solutions* of this equation are those of the simple harmonic oscillator:

$$f_{\pm} = e^{\pm i\mu t} = \xi^{\pm i\mu}, \quad (11.146)$$

where we have used $\xi = e^t$ in the final equality.

- We would now like to construct a *particular solution* of the inhomogeneous equation. Our strategy will be to find two series solutions around $\xi = 0$ and $\xi = \infty$, and then match them at $\xi = 1$ (where both solutions are convergent):



The matching will force us to add a fixed amount of the homogeneous solution.

Exercise 11.12 Show that, around $\xi = 0$, the inhomogeneous equation has the following series solution

$$f_{<}(\xi) = \sqrt{\xi} \sum_{n=0}^{\infty} (-1)^n \frac{\xi^n}{(n + \frac{1}{2})^2 + \mu^2}, \quad (11.147)$$

which is convergent for $\xi \leq 1$ and divergent for $\xi > 1$. ➔

Modulo the overall factor of $\sqrt{\xi}$, the solution (11.147) is analytic around $\xi = 0$. We would like to extend this solution into the regime $\xi > 1$. In this simple case, the solution can be written as a hypergeometric function and we could rely on the known analytic structure of the hypergeometric functions to find the solution for general ξ . However, as a warmup for the general case, it will be useful to proceed by finding a series solution to the differential equation ourselves.

Exercise 11.13 Show that, around $\xi = \infty$, the solution can be written as a series in ξ^{-1} :

$$f_{>}(\xi) = \frac{1}{\sqrt{\xi}} \sum_{n=0}^{\infty} (-1)^n \frac{\xi^{-n}}{(n + \frac{1}{2})^2 + \mu^2}, \quad (11.148)$$

which is convergent for $\xi \geq 1$ and divergent for $\xi < 1$. ➔

Modulo the overall factor of $1/\sqrt{\xi}$, the solution (11.148) is analytic around $\xi = \infty$. Importantly, the two solutions $f_{<}(\xi)$ and $f_{>}(\xi)$ are both convergent at $\xi = 1$. We can therefore find the full solution by matching the two series solutions at $\xi = 1$. Consider first the difference $f_{<}(\xi) - f_{>}(\xi)$, which must be a solution of the homogeneous equation:

$$f_{<}(\xi) - f_{>}(\xi) = \sum_{\pm} A_{\pm} \xi^{\pm i\mu}, \quad (11.149)$$

where continuity at $\xi = 1$ requires that $A_+ = -A_- \equiv A$ and

$$2i\mu A = \sum_{n=0}^{\infty} (-1)^n \frac{(2n+1)}{(n+\frac{1}{2})^2 + \mu^2} = \frac{\pi}{\cosh \pi\mu}. \quad (11.150)$$

We have therefore obtained an explicit solution of the differential equation which is analytic around the origin:

$$\tilde{F}_{<}(\xi) = \begin{cases} \sum_{n=0}^{\infty} (-1)^n \frac{\xi^{n+1}}{(n+\frac{1}{2})^2 + \mu^2} & \xi \leq 1, \\ \sum_{n=0}^{\infty} (-1)^n \frac{\xi^{-n}}{(n+\frac{1}{2})^2 + \mu^2} + \frac{\pi}{\cosh \pi\mu} \frac{\xi^{\frac{1}{2}-i\mu} - \xi^{\frac{1}{2}+i\mu}}{2i\mu} & \xi \geq 1. \end{cases} \quad (11.151)$$

We see that it is impossible for the solution which is analytic around $\xi = 0$ to be analytic around $\xi = \infty$. This important feature will persist in the general solution.

General solution

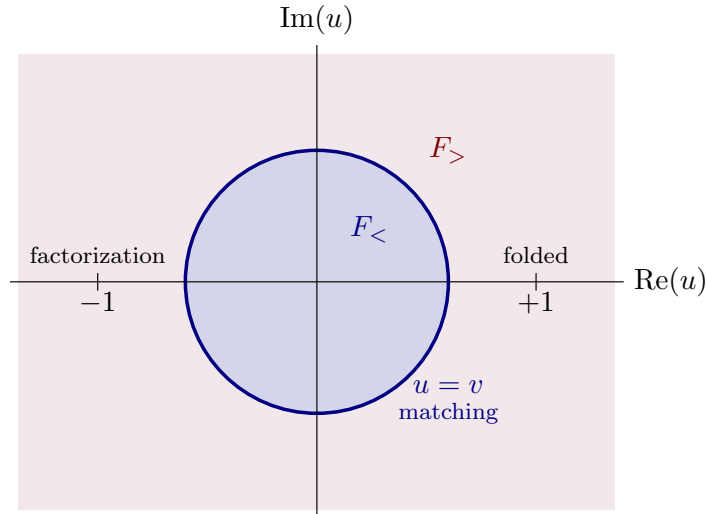
The general solution can be derived in a similar way. To avoid losing the forest for the trees, we will delegate some of the algebraic details to the exercises. The solutions to these exercises can be found in Appendix B.

We first note that the homogeneous solutions of (11.139), which we will denote by $\hat{F}_{\pm}(u)$, can be expressed as hypergeometric functions. The Wronksian of these solutions, $W \equiv \hat{F}_+ \partial_u \hat{F}_- - \hat{F}_- \partial_u \hat{F}_+$, satisfies $\partial_u W = 2u/(1-u^2)W$, so it is natural to normalized the solutions, so that $W = 1/(1-u^2)$. The correctly normalized homogeneous solutions are then

$$\hat{F}_{\pm}(u) = \left(\frac{iu}{2\mu}\right)^{\frac{1}{2} \pm i\mu} {}_2F_1 \left[\begin{matrix} \frac{1}{4} \pm \frac{i\mu}{2}, \frac{3}{4} \pm \frac{i\mu}{2} \\ 1 \pm i\mu \end{matrix} \middle| u^2 \right], \quad (11.152)$$

and equivalent solutions $\hat{F}_{\pm}(v)$ for the differential equation in terms of v .

Next, we want to find the particular solution of the inhomogeneous equation. The strategy will again be to find two series solutions around $u = 0$ and $u = \infty$, and then match these solutions at $u = v$:



After this, we still have the freedom to add linear combinations of the homogenous solutions (11.152). We fix the complete solution by:

1. Imposing the absence of folded singularities at $u = 1$ and $v = 1$.
2. Demanding the correct factorization at $u = v = -1$:

$$\lim_{u,v \rightarrow -1} \hat{F} = A_3 \log(1+u) \times A_3 \log(1+v), \quad (11.153)$$

where $A_3 \equiv \lambda$ (which we set to unity).

In the following, we describe this procedure in more detail.

Exercise 11.14 Show that, around $u = 0$, the inhomogeneous equation has the following series solution [44]

$$\hat{F}_{<}(u, v) = \sum_{m,n=0}^{\infty} c_{mn}(\mu) u^{2m+1} (u/v)^n, \quad (11.154)$$

where the series coefficients are

$$c_{mn}(\mu) \equiv \frac{(-1)^n}{n} \prod_{i=0}^{2m} \frac{(n+i)}{(n+\frac{1}{2}+i)^2 + \mu^2}. \quad (11.155)$$

This solution is convergent for $u \leq v$.



Note that (11.154) is the unique particular solution that is regular at the origin, since both homogeneous solutions are non-analytic at $u = 0$. In contrast, regularity around $u = \infty$ does not uniquely fix $\hat{F}_{>}(u, v)$. Instead, we will demand that the full particular solution is symmetric under the exchange $u \leftrightarrow v$. For $u > v$, the solution therefore is $\hat{F}_{>}(u, v) = \hat{F}_{<}(v, u)$, or

$$\hat{F}_{>}(u, v) = \sum_{m,n=0}^{\infty} c_{mn}(\mu) v^{2m+1} (v/u)^n, \quad (11.156)$$

where $c_{mn}(\mu)$ are same coefficients as in (11.155). It is straightforward to check that (11.156) solves (11.139) and that it is convergent for $u = v$.

Exercise 11.15 By matching the two solutions (11.154) and (11.156) at $u = v$, show that the particular solution which is regular around $u = 0$ is

$$\hat{F}_{<}(u, v) = \begin{cases} \sum_{m,n=0}^{\infty} c_{mn}(\mu) u^{2m+1} (u/v)^n & u \leq v, \\ \sum_{m,n=0}^{\infty} c_{mn}(\mu) v^{2m+1} (v/u)^n + \frac{\pi}{\cosh \pi \mu} \hat{F}_h(u, v) & u \geq v, \end{cases} \quad (11.157)$$

where $\hat{F}_h(u, v) \equiv \hat{F}_{+}(v) \hat{F}_{-}(u) - \hat{F}_{-}(v) \hat{F}_{+}(u)$.

Hint: You may use the identity

$$\sum_{n=0}^{\infty} (2m+2n+1) c_{mn}(\mu) = \frac{\pi}{\cosh \pi \mu}, \quad (11.158)$$

which can be proven by matching the residues on both sides. ➔

We still have the freedom to add homogeneous solutions to (11.157). In fact, we *must* add them to satisfy the boundary conditions [44, 94]

$$\lim_{u \rightarrow +1} \hat{F} = \text{regular}, \quad (11.159)$$

$$\lim_{u, v \rightarrow -1} \hat{F} = \log(1+u) \log(1+v). \quad (11.160)$$

The coefficient of the singularities in (11.160) is the product of two three-point amplitudes, which we normalized to unity, $A_3 \equiv 1$. It is easy to check that the homogeneous solution (11.152) has the following singular behavior in the limit $u \rightarrow 1$:

$$\lim_{u \rightarrow 1} \hat{F}_{\pm}(u) = \alpha_{\pm} \log(1-u), \quad \text{where} \quad \alpha_{\pm} \equiv - \left(\frac{i}{2\mu} \right)^{\frac{1}{2} \pm i\mu} \frac{\Gamma(1 \pm i\mu)}{\Gamma(\frac{1}{4} \pm \frac{i\mu}{2}) \Gamma(\frac{3}{4} \pm \frac{i\mu}{2})}. \quad (11.161)$$

By adding an appropriate amount of the homogeneous solution, we can cancel off any potential singularity in this limit. Finally, the solution in (11.157) isn't symmetric in $u \leftrightarrow v$, as required by consistency of the bulk evolution (and the symmetry of the conformally-invariant contact terms). In particular, the nonperturbative correction is absent as $u \rightarrow 0$, and as a result the solution is analytic in this limit. In the following exercise, you will fix these deficiencies.

Exercise 11.16 Show that the final result is [44, 94]

$$\hat{F}(u, v) = \begin{cases} \sum_{m,n=0}^{\infty} c_{mn} u^{2m+1} (u/v)^n + \frac{\pi}{2 \cosh \pi \mu} \hat{g}(u, v) & u \leq v, \\ \sum_{m,n=0}^{\infty} c_{mn} v^{2m+1} (v/u)^n + \frac{\pi}{2 \cosh \pi \mu} \hat{g}(v, u) & u \geq v, \end{cases} \quad (11.162)$$

where we have defined

$$\begin{aligned} \hat{g}(u, v) \equiv & (\beta - 1) \left[\hat{F}_+(u) \hat{F}_-(v) - \alpha \hat{F}_-(u) \hat{F}_-(v) \right] \\ & + (\beta + 1) \left[\hat{F}_-(u) \hat{F}_+(v) - \alpha^* \hat{F}_+(u) \hat{F}_+(v) \right], \end{aligned} \quad (11.163)$$

with $\beta \equiv 2/(i \sinh \pi \mu)$ and $\alpha = \alpha_+/\alpha_-$. ➔

It is interesting that the only solutions which are symmetric and free of any folded singularities are necessarily non-analytic around $u \rightarrow 0$ and $v \rightarrow 0$. This is how particle production in the time-dependent bulk spacetime is encoded in the boundary correlators.

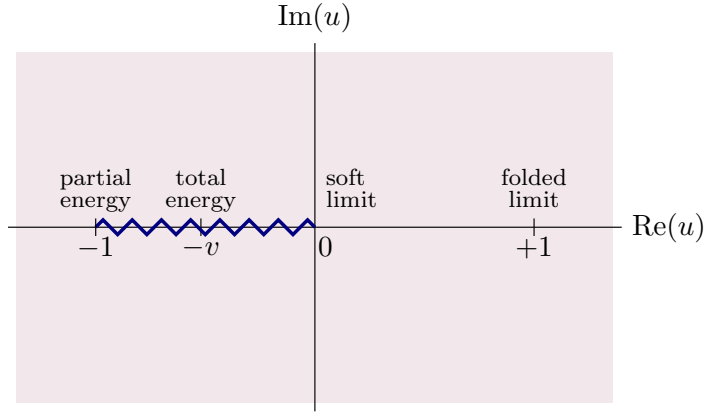


Figure 30: Analytic structure of the result (11.162). Note that the physical regime corresponds to $0 < u < 1$, and the branch cut for negative u is a signal of particle production.

The solution (11.162) has an interesting analytic structure, illustrated in Fig. 30. The function is analytic in the physical domain, $0 < u < 1$, but has a branch cut for $-1 < u \leq 0$. The existence of this branch cut is to be contrasted with the case of scattering amplitude which at tree level can only have poles. As we alluded to above, the discontinuity at $E = 0$ (or $u \rightarrow -v$) contains the flat-space scattering amplitude. To study the $E \rightarrow 0$ limit, we go back to the differential equation (11.139), which, in the limit $u \rightarrow -v$, becomes

$$\lim_{u \rightarrow -v} \frac{\partial^2 \hat{F}}{\partial u^2} = -\frac{1}{1-v^2} \frac{1}{u+v}. \quad (11.164)$$

This integrates to

$$\lim_{u \rightarrow -v} \frac{1}{k_I} \hat{F} = -\frac{1}{k_I(1-v^2)} (u+v) \log(u+v) = \frac{1}{s} (-E \log E), \quad (11.165)$$

where $s = k_{12}^2 - k_I^2$ is the Mandelstam variable. We see that the coefficient of the $E = 0$ singularity is indeed given by the high-energy limit of the flat-space amplitude, $A_4 = 1/s$.

Particle production

In Section 9.8, we found that massive particles leave a characteristic oscillatory signature in the soft limit of the inflationary bispectrum (see Exercise 9.14). In the following, we will show that the same feature arises in the collapsed limit $u \rightarrow 0$ of our results for the four-point function.⁵⁶

In the collapsed limit $u \rightarrow 0$, the particle production piece (homogeneous solution) dominates and the function (11.162) becomes

$$\lim_{u,v \rightarrow 0} \hat{F}(u,v) = \left(\frac{uv}{4}\right)^{\frac{1}{2}+i\mu} (1+i \sinh \pi\mu) \frac{\Gamma(\frac{1}{2}+i\mu)^2 \Gamma(-i\mu)^2}{2\pi} + \text{c.c.} \quad (11.166)$$

We see that the signal indeed oscillates with a frequency given by the mass of the new particles.

⁵⁶In [44], it was shown that this can be related to the squeezed limit of the inflationary bispectrum by putting one of the external legs on the background (plus perturbing the scaling dimension of the associated field).

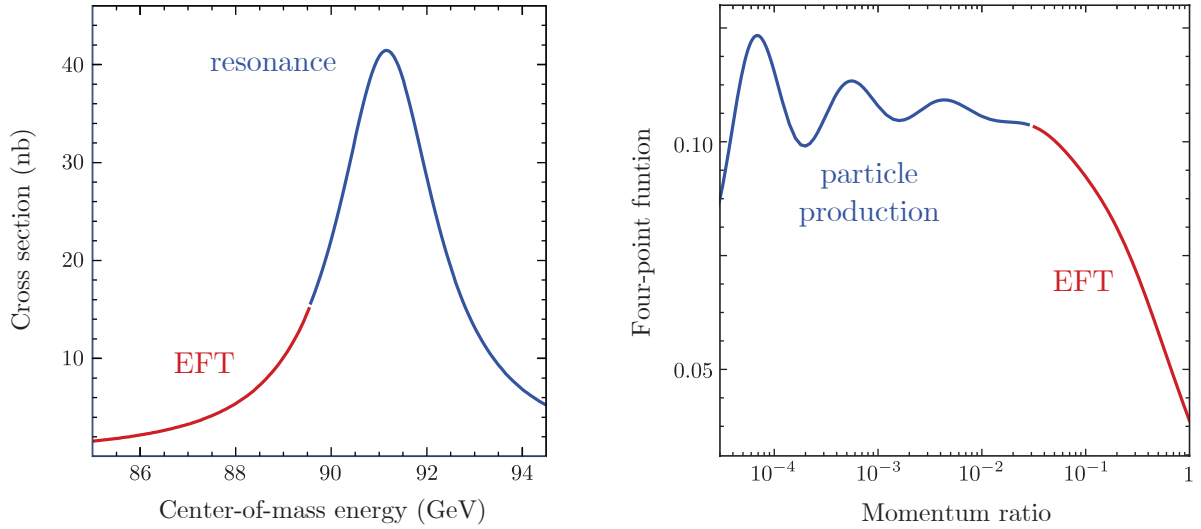


Figure 31: *Left panel:* Shape of the Z resonance as measured by LEP. *Right panel:* Example of scalar exchange, $u^{-1}\hat{F}(u, 0.5)$, for external particles with $\Delta = 1$ (conformally coupled) and an internal particle with $\mu = 3$. Note that the four-point function has been rescaled by u^{-1} which visually enhances the effect of the oscillations. In practice, the particle production effect will be harder to observe than the EFT contribution.

The right panel in Figure 31 displays our solution for the exchange of a massive scalar particle, $\hat{F}(u, v)$, for fixed $v = 0.5$. We see that the signal in the collapsed limit, $u \rightarrow 0$, oscillates with a frequency that is set by the mass of the exchange particle. Measuring these oscillations is the analog of measuring the position of a resonance peak in collider physics. It would prove the existence of new particles and determine their masses. Going away from the squeezed limit, the particular solution will start to dominate over the homogeneous solution. This provides a smooth contribution to the four-point function, whose shape will also be determined by the mass of the exchange particle. This is the analog of going off resonance and measuring the shape of the resonance peak in collider physics. Measuring both the oscillations and the smooth shape provides an important consistency check for the signal.

In colliders, we begin with low-energy measurements where all interactions are pure contact interactions. For example, at low energies the electroweak theory is approximated by the four-point interaction of Fermi theory. In the latter case, the energy dependence of the interaction hints at a violation of perturbative unitarity at a higher scale. This suggests the existence of new particles (in the case of the electroweak theory, W bosons) to improve the UV behavior of the effective theory. Going to higher energies, colliders may start producing these particles as resonances. Predicting the shape of the resonance is essential for extracting the detailed properties of the new particles. It also provides the opportunity to identify additional new physics. For example, any unexplained excess in the cross section may be due to additional particle exchanges.

In cosmology, we first expect to observe signals in the limit of relatively large momenta (small wavelengths). This is where the signals are strongest and the observations are most sensitive. Initially, we would see the shape of a pure contact interaction. With increased sensitivity we may

then be able to observe a small deviation from the pure contact shape.⁵⁷ Using the hypothesis of the exchange of a single massive particle to fit the smooth part of the signal would then allow us to predict the amplitude and frequency of the characteristic oscillations in the soft limit. Finding consistency between the smooth and oscillating parts of the signal would help to establish the underlying physics.

⁵⁷In practice, it will be hard to reliably extract the precise shape of the smooth part of the signal from large-scale structure observations because late-time nonlinearities produce non-Gaussianities of a similar form. Although the oscillatory part of the signal is smaller, it is more distinctive and cannot be mimicked by late-time effects.

Part III

Appendices

A Special Functions

In this appendix, we collect properties of a few selected special functions that will be useful for the computations and the exercises presented in the main text.

A.1 Legendre Functions

Legendre functions are solutions of

$$\left[(1-x^2) \frac{d^2}{dx^2} - 2x \frac{d}{dx} \right] P_l = -l(l+1)P_l, \quad (\text{A.1})$$

where l is a real number. In physics, these equations arise when we solve the Laplace equation in spherical coordinates. The variable x is then the cosine of the polar angle and its value is constrained to the range $-1 \leq x \leq 1$.

When the parameter l is an integer, the solutions of (A.1) are polynomials in x . The lowest-order *Legendre polynomials* are

$$P_0(x) = 1, \quad P_1(x) = x, \quad P_2(x) = \frac{1}{2}(3x^2 - 1), \quad (\text{A.2})$$

while higher-order solutions can be obtained from the recursion relation

$$(l+1)P_{l+1} = (2l+1)xP_l - lP_{l-1}. \quad (\text{A.3})$$

These solutions can also be obtained from Rodrigues' formula

$$P_l(x) = \frac{1}{2^l l!} \frac{d^l}{dx^l} (x^2 - 1)^l. \quad (\text{A.4})$$

Depending on the order, the Legendre polynomials have odd or even parity: $P_l(-x) = (-1)^l P_l(x)$.

The Legendre polynomials are orthogonal on the interval $[-1, 1]$:

$$\int_{-1}^1 dx P_l(x) P_{l'}(x) = \frac{2}{2+l} \delta_{ll'}. \quad (\text{A.5})$$

Any function defined on the interval can therefore be written as

$$f(x) = \sum_l a_l P_l(x), \quad \text{where} \quad a_l = \frac{2l+1}{2} \int_{-1}^1 dx f(x) P_l(x). \quad (\text{A.6})$$

Orthogonality of the Legendre polynomials guarantees that this expansion is unique.

A second class of solutions to (A.1) are the *Legendre functions of the second kind*. Even for integer l , these solutions are not polynomials in x . The first few lowest-order solutions are

$$\begin{aligned} Q_0(x) &= \frac{1}{2} \log \left(\frac{1+x}{1-x} \right), \\ Q_1(x) &= \frac{x}{2} \log \left(\frac{1+x}{1-x} \right) - 1, \\ Q_2(x) &= \frac{3x^2-1}{4} \log \left(\frac{1+x}{1-x} \right) - \frac{3x}{2}. \end{aligned} \quad (\text{A.7})$$

The functions $Q_l(x)$ satisfy the same recurrence relation (A.3) as the Legendre polynomials $P_l(x)$. This allows us to find higher-order solutions. Alternatively, we can use the Wronskian method to find

$$Q_l(x) = P_l(x) \int^x \frac{du}{(1-u^2)(P_l(u))^2}, \quad (\text{A.8})$$

where $P_l(x)$ are the Legendre polynomials (of the first kind).

An important generalization are the *associated Legendre functions* which are solutions of

$$\left[(1-x^2) \frac{d^2}{dx^2} - 2x \frac{d}{dx} \right] P_l^m(x) + \left[l(l+1) - \frac{m^2}{1-x^2} \right] P_l^m(x) = 0, \quad (\text{A.9})$$

where l and m are real parameters. For $m = 0$, this equation reduces to the ordinary Legendre equation (A.1).

In physical applications, the parameters l and m are often integers, with $|m| \leq l$. When m is even, the solution is a polynomial. For non-negative integer parameters l and m , we have

$$P_l^m(x) = (1-x^2)^{m/2} \frac{d^m P_l}{dx^m}. \quad (\text{A.10})$$

Sometimes a factor of $(-1)^m$ is included in this expression. Since $P_l(x)$ is a polynomial of order l , we have $P_l^m(x) = 0$ for $|m| > l$. Moreover, $P_l^m(x)$ is then also a polynomial of order l if m is even. Using Rodrigues' formula (A.4), this can be written as

$$P_l^m(x) = \frac{(-1)^m}{2^l l!} (1-x^2)^{m/2} \frac{d^{l+m}}{dx^{l+m}} (x^2-1)^l. \quad (\text{A.11})$$

The first few associated Legendre functions (of the first kind) then are

$$\begin{aligned} P_1^1(x) &= (1-x^2)^{1/2}, \\ P_2^1(x) &= 3x(1-x^2)^{1/2}, \quad P_2^2(x) = 3(1-x^2), \\ P_3^1(x) &= \frac{3}{2}(5x^2-1)(1-x^2)^{1/2}, \quad P_3^2(x) = 15x(1-x^2), \quad P_3^3(x) = 15(1-x^2)^{3/2}. \end{aligned} \quad (\text{A.12})$$

From (A.11), one can also show that

$$P_l^{-m}(x) = (-1)^m \frac{(l-m)!}{(l+m)!} P_l^m(x), \quad (\text{A.13})$$

which relates functions with positive and negative m .

For complex parameters and complex argument, the associated Legendre functions can be generalized via the hypergeometric function (see below).

A.2 Bessel and Hankel Functions

Ordinary Bessel functions satisfy

$$\frac{d^2 J_\nu}{dz^2} + \frac{1}{z} \frac{dJ_\nu}{dz} + \left(1 - \frac{\nu^2}{z^2} \right) J_\nu = 0, \quad (\text{A.14})$$

where ν is a constant parameter. A series solution of this equation is

$$J_\nu(z) = \left(\frac{z}{2}\right)^{-\nu} \sum_{n=0}^{\infty} \frac{(-1)^n}{n! \Gamma(\nu + n + 1)} \left(\frac{z}{2}\right)^{2n}. \quad (\text{A.15})$$

As long as ν is not an integer, then a second linearly independent solution is $J_{-\nu}(z)$. The functions $J_\nu(z)$ and $J_{-\nu}(z)$ are called *Bessel functions of the first kind*. If ν is an integer n , then $J_n(z)$ and $J_{-n}(z)$ are not independent, but related by

$$J_{-n}(z) = (-1)^n J_n(z). \quad (\text{A.16})$$

A function that remains linearly independent of $J_\nu(z)$, even for integer ν , is the *Bessel function of the second kind* (sometimes called the *Neumann function*):

$$Y_\nu(z) = \frac{J_\nu(z) \cos(\nu\pi) - J_{-\nu}(z)}{\sin(\nu\pi)}. \quad (\text{A.17})$$

For integer ν , this definition must be understood as the limit $\nu \rightarrow n$. The Bessel functions have the following limiting behavior for $z \rightarrow 0$:

$$\begin{aligned} J_\nu(z) &\xrightarrow{z \rightarrow 0} \frac{1}{\Gamma(\nu + 1)} \left(\frac{z}{2}\right)^\nu, \\ Y_\nu(z) &\xrightarrow{z \rightarrow 0} -\frac{1}{\pi} \Gamma(\nu) \left(\frac{z}{2}\right)^{-\nu}. \end{aligned} \quad (\text{A.18})$$

These limits make manifest that the Bessel equation (A.14) has a scaling symmetry for $z \rightarrow 0$. We see that $J_\nu(z)$ is regular for small argument, while $Y_\nu(z)$ diverges.

In the limit $z \rightarrow \infty$, the Bessel equation exhibits a translation symmetry and should have solutions that scale as $e^{\pm iz}$. This is made more manifest by defining the *Hankel functions* (or *Bessel functions of the third kind*):

$$\begin{aligned} H_\nu^{(1)}(z) &\equiv J_\nu(z) + iY_\nu(z), \\ H_\nu^{(2)}(z) &\equiv J_\nu(z) - iY_\nu(z). \end{aligned} \quad (\text{A.19})$$

We encountered Hankel functions when we derived the mode functions for inflationary fluctuations. The asymptotic limits of the Hankel functions for large arguments are

$$\begin{aligned} H_\nu^{(1)}(z) &\xrightarrow{z \rightarrow \infty} \sqrt{\frac{2}{\pi z}} e^{i(z - \frac{1}{2}\nu\pi - \frac{1}{4}\pi)}, \\ H_\nu^{(2)}(z) &\xrightarrow{z \rightarrow \infty} \sqrt{\frac{2}{\pi z}} e^{-i(z - \frac{1}{2}\nu\pi - \frac{1}{4}\pi)}. \end{aligned} \quad (\text{A.20})$$

These limits were used to define the initial conditions of the inflationary fluctuations. The scaling for small arguments follows from (A.18):

$$\begin{aligned} H_\nu^{(1)}(z) &\xrightarrow{z \rightarrow 0} -\frac{i}{\pi} \Gamma(\nu) \left(\frac{z}{2}\right)^{-\nu}, \\ H_\nu^{(2)}(z) &\xrightarrow{z \rightarrow 0} +\frac{i}{\pi} \Gamma(\nu) \left(\frac{z}{2}\right)^{-\nu}. \end{aligned} \quad (\text{A.21})$$

This limit is used to determine the superhorizon behavior of the inflationary fluctuations.

A.3 Polylogarithms

When computing the tree-level correlators of conformally coupled scalars in de Sitter space, we encountered polylogarithms.

A series definition of the polylogarithm is

$$\text{Li}_s(z) = \sum_{k=1}^{\infty} \frac{z^k}{k^s} = z + \frac{z^2}{2^s} + \frac{z^3}{3^s} + \dots \quad (\text{A.22})$$

which is valid for arbitrary complex order s and for all complex arguments z with $|z| < 1$. The special case $s = 1$ is the ordinary natural logarithm

$$\text{Li}_1(z) = -\log(1 - z), \quad (\text{A.23})$$

while $s = 2$ and $s = 3$ are called the dilogarithm and the trilogarithm, respectively. For non-positive integers s , polylogarithms are rational functions:

$$\text{Li}_0(z) = \frac{z}{1 - z}, \quad \text{Li}_{-1}(z) = \frac{z}{(1 - z)^2}, \quad \text{Li}_{-2}(z) = \frac{z(1 + z)}{(1 - z)^3}, \quad \dots \quad (\text{A.24})$$

Evaluated at $z = 1$, the polylogarithm becomes the Riemann zeta function: $\text{Li}_s(1) = \zeta(s)$.

Polylogarithms satisfy the following iterative integral

$$\text{Li}_{s+1}(z) = \int_0^z dt \frac{\text{Li}_s(t)}{t}. \quad (\text{A.25})$$

The dilogarithm is therefore an integral of the natural logarithm, the trilogarithm is an integral of the dilogarithm, and so on. The derivatives of the polylogarithm satisfy

$$z \frac{\partial \text{Li}_s(z)}{\partial z} = \text{Li}_{s-1}(z), \quad (\text{A.26})$$

which follows directly from (A.25).

The polylogarithm function can also be expressed in terms of integrals of the Bose–Einstein and Fermi–Dirac distributions

$$\text{Li}_s(z) = \frac{1}{\Gamma(s)} \int_0^\infty dt \frac{t^{s-1}}{e^t/z - 1}, \quad (\text{A.27})$$

$$-\text{Li}_s(-z) = \frac{1}{\Gamma(s)} \int_0^\infty dt \frac{t^{s-1}}{e^t/z + 1}. \quad (\text{A.28})$$

It therefore also makes an appearance in statistical mechanics.

A.4 Hypergeometric Functions

The hypergeometric function is a solution of

$$\boxed{z(1 - z) \frac{d^2 F}{dz^2} + [c - (1 + a + b)z] \frac{dF}{dz} - abF = 0}, \quad (\text{A.29})$$

where a, b and c are real numbers. The equation has regular singular points at $z = 0, 1, \infty$. In fact, it can be shown that any second-order differential equation with three regular singularities can be transformed into the hypergeometric equation by a change of variables. This allows other special functions—like the (associated) Legendre functions—to be written as special cases of hypergeometric functions. Often the (Gauss) hypergeometric function is denoted by ${}_2F_1$, while solutions of the ‘generalized hypergeometric equation’ are written as ${}_pF_q$, for integers p and q .

A series solution of (A.29) is

$${}_2F_1 \left[\begin{matrix} a, b \\ c \end{matrix} \middle| z \right] = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n} \frac{z^n}{n!} = 1 + \frac{ab}{c} \frac{z}{1!} + \frac{a(a+1)b(b+1)}{c(c+1)} \frac{z^2}{2!} + \cdots, \quad (\text{A.30})$$

where the Pochhammer symbol is defined by

$$(a)_{n>0} = a(a+1) \cdots (a+n-1), \quad (a)_0 = 1. \quad (\text{A.31})$$

Equation (A.30) can also be written as

$${}_2F_1 \left[\begin{matrix} a, b \\ c \end{matrix} \middle| z \right] = \frac{\Gamma(c)}{\Gamma(a)\Gamma(b)} \sum_{n=0}^{\infty} \frac{\Gamma(a+n)\Gamma(b+n)}{\Gamma(c+n)} \frac{z^n}{n!}, \quad (\text{A.32})$$

where $\Gamma(a)$ is the Euler gamma function.

A very useful integral representation of the hypergeometric function is

$${}_2F_1 \left[\begin{matrix} a, b \\ c \end{matrix} \middle| z \right] = \frac{\Gamma[c]}{\Gamma[b]\Gamma[c-b]} \int_0^{\infty} dt t^{-b+c-1} (t+1)^{a-c} (t-z+1)^{-a}. \quad (\text{A.33})$$

which converges for $c > b > 0$. This integral representation can be used to prove many general properties of the hypergeometric functions.

There are many relationships between hypergeometric functions with different arguments. Two particularly useful ones are

$${}_2F_1 \left[\begin{matrix} a, b \\ c \end{matrix} \middle| z \right] = (1-z)^{c-a-b} {}_2F_1 \left[\begin{matrix} c-a, c-b \\ c \end{matrix} \middle| z \right], \quad (\text{A.34})$$

$$\frac{d}{dz} {}_2F_1 \left[\begin{matrix} a, b \\ c \end{matrix} \middle| z \right] = \frac{ab}{c} {}_2F_1 \left[\begin{matrix} a+1, b+1 \\ c+1 \end{matrix} \middle| z \right]. \quad (\text{A.35})$$

In addition, there are relations between hypergeometric functions with shifted parameters. The six functions

$${}_2F_1 \left[\begin{matrix} a \pm 1, b \\ c \end{matrix} \middle| z \right], \quad {}_2F_1 \left[\begin{matrix} a, b \pm 1 \\ c \end{matrix} \middle| z \right], \quad {}_2F_1 \left[\begin{matrix} a, b \\ c \pm 1 \end{matrix} \middle| z \right] \quad (\text{A.36})$$

are called ‘contiguous’ to ${}_2F_1(a, b, c; z)$. As shown by Gauss, ${}_2F_1(a, b, c; z)$ can be written as a linear combination of any two of its contiguous functions, with rational coefficients depending on a, b, c and z . In total, there are 15 such relations. These ‘contiguous relations’ provide recurrence relations for the hypergeometric functions.

Many of the above special functions can also be written in terms of hypergeometric functions. Some examples are:

$$\begin{aligned}
{}_2F_1\left[\begin{matrix} a, & b \\ & b \end{matrix} \middle| x\right] &= (1-x)^{-a}, & {}_2F_1\left[\begin{matrix} \frac{1}{2}, & \frac{1}{2} \\ & \frac{3}{2} \end{matrix} \middle| x\right] &= x^{-1} \sin^{-1} x, \\
{}_2F_1\left[\begin{matrix} 1, & 1 \\ & 2 \end{matrix} \middle| -x\right] &= x^{-1} \log(1+x), & {}_2F_1\left[\begin{matrix} \frac{1}{2}, & 1 \\ & \frac{3}{2} \end{matrix} \middle| -x^2\right] &= x^{-1} \tan^{-1} x, \\
{}_2F_1\left[\begin{matrix} \frac{1}{2}, & -\frac{1}{2} \\ & \frac{1}{2} \end{matrix} \middle| \sin^2 x\right] &= \cos x, & {}_2F_1\left[\begin{matrix} l+1, & -l \\ & 1 \end{matrix} \middle| \frac{1-x}{2}\right] &= P_l(x),
\end{aligned} \tag{A.37}$$

where $P_l(x)$ is the Legendre polynomial. The associated Legendre functions are

$$P_l^m(z) = \frac{1}{\Gamma(1-m)} \left(\frac{1+z}{1-z} \right)^{m/2} {}_2F_1\left[\begin{matrix} -l, & l+1 \\ & 1-m \end{matrix} \middle| \frac{1-z}{2}\right], \tag{A.38}$$

$$Q_l^m(z) = \frac{\sqrt{\pi} \Gamma(l+m+1)}{2^{l+1} \Gamma(l+3/2)} \frac{1}{z^{l+m+1}} (1-z^2)^{m/2} {}_2F_1\left[\begin{matrix} \frac{1}{2}(l+m+1), & \frac{1}{2}(l+m+2) \\ & l+\frac{3}{2} \end{matrix} \middle| \frac{1}{z^2}\right], \tag{A.39}$$

which holds for complex parameters l, m and complex arguments z .

A.5 Gamma Functions

The Euler gamma function is

$$\Gamma(z) \equiv \int_0^\infty dt e^{-t} t^{z-1}, \tag{A.40}$$

where $\text{Re}(z) > 0$. It is an analytic function with simple poles at $z = -n$, for integer n . A useful relation is

$$\Gamma(z+1) = z\Gamma(z), \tag{A.41}$$

which implies that for integer arguments the gamma function reduces to factorials, $\Gamma(n+1) = n!$.

B Solutions to the Exercises

B.1 Exercises in Chapter 1

Solution to Exercise 1.1

The Hamiltonian of the simple harmonic oscillator (with unit mass) is

$$H = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 q^2, \quad (\text{B.1})$$

where $p = \dot{q}$ is the momentum. The position operator is written as

$$\hat{q}(t) = f(t) \hat{a} + f^*(t) \hat{a}^\dagger, \quad (\text{B.2})$$

where $f(t)$ is a solution to the classical equation of motion, $\ddot{f} + \omega^2 f = 0$. The most general such solution is

$$f(t) = \frac{1}{\sqrt{2\omega}} \left(c_1 e^{-i\omega t} + c_2 e^{i\omega t} \right), \quad (\text{B.3})$$

$$\dot{f}(t) = -i\sqrt{\frac{\omega}{2}} \left(c_1 e^{-i\omega t} - c_2 e^{i\omega t} \right). \quad (\text{B.4})$$

The correct normalization of the solution is

$$\begin{aligned} 1 &\equiv \langle f, f \rangle \\ &= i(f^* \dot{f} - \dot{f}^* f) \\ &= \frac{1}{2} \left(c_1 e^{-i\omega t} - c_2 e^{i\omega t} \right) \left(c_1^* e^{i\omega t} + c_2^* e^{-i\omega t} \right) + \frac{1}{2} \left(c_1^* e^{i\omega t} - c_2^* e^{-i\omega t} \right) \left(c_1 e^{-i\omega t} + c_2 e^{i\omega t} \right) \\ &= \frac{1}{2} \left(|c_1|^2 - |c_2|^2 + c_1 c_2^* e^{-2i\omega t} - c_1^* c_2 e^{2i\omega t} \right) + \frac{1}{2} \left(|c_1|^2 - |c_2|^2 + c_1^* c_2 e^{2i\omega t} - c_1 c_2^* e^{-2i\omega t} \right) \\ &= \left(|c_1|^2 - |c_2|^2 \right), \end{aligned} \quad (\text{B.5})$$

and hence

$$\boxed{|c_1|^2 - |c_2|^2 = 1}. \quad (\text{B.6})$$

Next, we consider the Hamiltonian operator

$$\hat{H} = \frac{1}{2}\hat{p}^2 + \frac{1}{2}\omega^2 \hat{q}^2, \quad (\text{B.7})$$

and the vacuum expectation value $\langle 0 | \hat{H} | 0 \rangle$. Substituting (B.2) into $\langle 0 | \hat{q}^2 | 0 \rangle$, we get

$$\begin{aligned} \langle 0 | \hat{q}^2 | 0 \rangle &= \langle 0 | (f(t) \hat{a} + f^*(t) \hat{a}^\dagger) (f(t) \hat{a} + f^*(t) \hat{a}^\dagger) | 0 \rangle \\ &= |f(t)|^2 \langle 0 | \hat{a} \hat{a}^\dagger | 0 \rangle \\ &= |f(t)|^2 \langle 0 | [\hat{a}, \hat{a}^\dagger] | 0 \rangle \\ &= |f(t)|^2 \langle 0 | 0 \rangle \\ &= |f(t)|^2, \end{aligned} \quad (\text{B.8})$$

where we have used that $\hat{a}$ annihilates $|0\rangle$ to the right and $\hat{a}^\dagger$ annihilates $\langle 0|$ to the left. Similarly, we get $\langle 0|\hat{p}^2|0\rangle = |\dot{f}(t)|^2$ and the expectation value of the Hamiltonian is

$$\begin{aligned}\langle 0|\hat{H}|0\rangle &= \frac{1}{2}(|\dot{f}|^2 + \omega^2|f|^2) \\ &= \frac{\omega}{4}(c_1 e^{-i\omega t} - c_2 e^{i\omega t})(c_1^* e^{i\omega t} - c_2^* e^{-i\omega t}) + \frac{\omega}{4}(c_1 e^{-i\omega t} + c_2 e^{i\omega t})(c_1^* e^{i\omega t} + c_2^* e^{-i\omega t}) \\ &= \frac{\omega}{4}(|c_1|^2 + |c_2|^2 - c_1 c_2^* e^{-2i\omega t} - c_1^* c_2 e^{2i\omega t}) + \frac{\omega}{4}(|c_1|^2 + |c_2|^2 + c_1 c_2^* e^{2i\omega t} + c_1^* c_2 e^{-2i\omega t}) \\ &= \frac{\omega}{2}(|c_1|^2 + |c_2|^2).\end{aligned}\tag{B.9}$$

Using (B.6), we then get

$$\langle 0|\hat{H}|0\rangle = \frac{\omega}{2}(1 + 2|c_2|^2).\tag{B.10}$$

Since $|c_2|^2$ is positive semi-definite, the expectation value of the Hamiltonian is minimized for $c_2 = 0$ (and hence $|c_1| = 1$). Up to an irrelevant phase, the mode function then is

$$f(t) = \frac{1}{\sqrt{2\omega}} e^{-i\omega t},\tag{B.11}$$

which is the unique ground state solution of the simple harmonic oscillator.

Solution to Exercise 1.2

Using (1.42), the annihilation operator can be written as

$$\hat{a}_{\mathbf{k}} \equiv \langle f_{\mathbf{k}}, \hat{\phi} \rangle = i \int d^3x (f_{\mathbf{k}}^* \hat{\pi} - \partial_t f_{\mathbf{k}}^* \hat{\phi}),\tag{B.12}$$

where $\hat{\pi} = \partial_t \hat{\phi}$ is the conjugate momentum. Taking the Hermitian conjugate, the creation operator is

$$\hat{a}_{\mathbf{k}}^\dagger \equiv -\langle f_{\mathbf{k}}^*, \hat{\phi} \rangle = -i \int d^3x (f_{\mathbf{k}} \hat{\pi} - \partial_t f_{\mathbf{k}} \hat{\phi}).\tag{B.13}$$

Substituting this into $[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{k}'}^\dagger]$, we get

$$\begin{aligned}[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{k}'}^\dagger] &= \int d^3x \int d^3x' [(f_{\mathbf{k}}^* \hat{\pi} - \partial_t f_{\mathbf{k}}^* \hat{\phi})(t, \mathbf{x}), (f_{\mathbf{k}'} \hat{\pi} - \partial_t f_{\mathbf{k}'} \hat{\phi})(t, \mathbf{x}')] \\ &= \int d^3x \int d^3x' (-f_{\mathbf{k}}^* \partial_t f_{\mathbf{k}'} [\hat{\pi}(t, \mathbf{x}), \hat{\phi}(t, \mathbf{x}')] - f_{\mathbf{k}'} \partial_t f_{\mathbf{k}} [\hat{\phi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{x}')]) \\ &= i \int d^3x (f_{\mathbf{k}}^* \partial_t f_{\mathbf{k}'} - \partial_t f_{\mathbf{k}}^* f_{\mathbf{k}'}) \\ &= \langle f_{\mathbf{k}}, f_{\mathbf{k}'} \rangle \\ &= \delta(\mathbf{k} - \mathbf{k}'),\end{aligned}\tag{B.14}$$

as required. In the same way, it also easy to show that

$$[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{k}'}] = -\langle f_{\mathbf{k}}, f_{\mathbf{k}'}^* \rangle = 0,\tag{B.15}$$

$$[\hat{a}_{\mathbf{k}}^\dagger, \hat{a}_{\mathbf{k}'}^\dagger] = -\langle f_{\mathbf{k}}^*, f_{\mathbf{k}'} \rangle = 0.\tag{B.16}$$

as required.

Solution to Exercise 1.3

The Hamiltonian operator is

$$\hat{H} = \int d^3x \left[\frac{1}{2}(\partial_t \hat{\phi})^2 + \frac{1}{2}(\nabla \hat{\phi})^2 + \frac{1}{2}m^2 \hat{\phi}^2 \right]. \quad (\text{B.17})$$

To write this in terms of creation and annihilation operators, we substitute the mode expansion of the field operator:

$$\hat{\phi}(t, \mathbf{x}) = \int d^3k \left[\hat{a}_{\mathbf{k}} f_{\mathbf{k}}(t, \mathbf{x}) + \hat{a}_{\mathbf{k}}^\dagger f_{\mathbf{k}}^*(t, \mathbf{x}) \right]. \quad (\text{B.18})$$

Consider first the mass term:

$$\begin{aligned} \frac{m^2}{2} \int d^3x \hat{\phi}^2 &= \frac{m^2}{2} \int d^3x d^3k d^3k' \left(\hat{a}_{\mathbf{k}} f_{\mathbf{k}} + \hat{a}_{\mathbf{k}}^\dagger f_{\mathbf{k}}^* \right) \left(\hat{a}_{\mathbf{k}'} f_{\mathbf{k}'} + \hat{a}_{\mathbf{k}'}^\dagger f_{\mathbf{k}'}^* \right) \\ &= \frac{m^2}{2} \int d^3x d^3k d^3k' \left(\hat{a}_{\mathbf{k}} \hat{a}_{\mathbf{k}'} f_{\mathbf{k}} f_{\mathbf{k}'} + \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}'} f_{\mathbf{k}}^* f_{\mathbf{k}'} + \hat{a}_{\mathbf{k}} \hat{a}_{\mathbf{k}'}^\dagger f_{\mathbf{k}} f_{\mathbf{k}'}^* + \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}'}^\dagger f_{\mathbf{k}}^* f_{\mathbf{k}'}^* \right). \end{aligned} \quad (\text{B.19})$$

Looks at the first term in the parenthesis:

$$\begin{aligned} \int d^3x d^3k d^3k' \hat{a}_{\mathbf{k}} \hat{a}_{\mathbf{k}'} f_{\mathbf{k}} f_{\mathbf{k}'} &= \int d^3k d^3k' \hat{a}_{\mathbf{k}} \hat{a}_{\mathbf{k}'} \frac{e^{-i(\omega+\omega')t}}{(2\pi)^3 \sqrt{2\omega} \sqrt{2\omega'}} \int d^3x e^{i(\mathbf{k}+\mathbf{k}') \cdot \mathbf{x}} \\ &= \int d^3k d^3k' \hat{a}_{\mathbf{k}} \hat{a}_{\mathbf{k}'} \frac{e^{-i(\omega+\omega')t}}{2\sqrt{\omega\omega'}} \delta(\mathbf{k} + \mathbf{k}') \\ &= \int d^3k \hat{a}_{\mathbf{k}} \hat{a}_{-\mathbf{k}} \frac{e^{-2i\omega t}}{2\omega}. \end{aligned} \quad (\text{B.20})$$

The other terms can be compute in the same way and we get

$$\frac{m^2}{2} \int d^3x \hat{\phi}^2 = \frac{m^2}{2} \int d^3k \frac{1}{2\omega} \left[\hat{a}_{\mathbf{k}} \hat{a}_{-\mathbf{k}} e^{-2i\omega t} + \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} + \hat{a}_{\mathbf{k}} \hat{a}_{\mathbf{k}}^\dagger + \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{-\mathbf{k}}^\dagger e^{2i\omega t} \right]. \quad (\text{B.21})$$

For the kinetic-energy term, the time-derivative pulls down as factor of $-i\omega$ when acting on $f_{\mathbf{k}}$ and a factor of $+i\omega$ when acting on $f_{\mathbf{k}}^*$. Similarly, for the gradient-energy term, the derivative pulls down as factor of $i\mathbf{k}$ when acting on $f_{\mathbf{k}}$ and a factor of $-i\mathbf{k}$ when acting on $f_{\mathbf{k}}^*$. Using these facts, we get

$$\frac{1}{2} \int d^3x (\partial_t \hat{\phi})^2 = \frac{1}{2} \int d^3k \frac{\omega^2}{2\omega} \left[-\hat{a}_{\mathbf{k}} \hat{a}_{-\mathbf{k}} e^{-2i\omega t} + \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} + \hat{a}_{\mathbf{k}} \hat{a}_{\mathbf{k}}^\dagger - \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{-\mathbf{k}}^\dagger e^{2i\omega t} \right], \quad (\text{B.22})$$

$$\frac{1}{2} \int d^3x (\nabla \hat{\phi})^2 = \frac{1}{2} \int d^3k \frac{\mathbf{k}^2}{2\omega} \left[\hat{a}_{\mathbf{k}} \hat{a}_{-\mathbf{k}} e^{-2i\omega t} + \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} + \hat{a}_{\mathbf{k}} \hat{a}_{\mathbf{k}}^\dagger + \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{-\mathbf{k}}^\dagger e^{2i\omega t} \right]. \quad (\text{B.23})$$

For example, the first term in (B.23) is computed as follows:

$$\begin{aligned} \int d^3x d^3k d^3k' \hat{a}_{\mathbf{k}} \hat{a}_{\mathbf{k}'} (-\mathbf{k} \cdot \mathbf{k}') f_{\mathbf{k}} f_{\mathbf{k}'} &= \int d^3k d^3k' \hat{a}_{\mathbf{k}} \hat{a}_{\mathbf{k}'} (-\mathbf{k} \cdot \mathbf{k}') \frac{e^{-i(\omega+\omega')t}}{(2\pi)^3 \sqrt{2\omega} \sqrt{2\omega'}} \int d^3x e^{i(\mathbf{k}+\mathbf{k}') \cdot \mathbf{x}} \\ &= \int d^3k d^3k' \hat{a}_{\mathbf{k}} \hat{a}_{\mathbf{k}'} (-\mathbf{k} \cdot \mathbf{k}') \frac{e^{-i(\omega+\omega')t}}{2\sqrt{\omega\omega'}} \delta(\mathbf{k} + \mathbf{k}') \\ &= \int d^3k \hat{a}_{\mathbf{k}} \hat{a}_{-\mathbf{k}} \frac{\mathbf{k}^2}{2\omega} e^{-2i\omega t}. \end{aligned} \quad (\text{B.24})$$

Combing (B.21), (B.22) and (B.23), and using $\omega^2 = \mathbf{k}^2 + m^2$, we finally get

$$\hat{H} = \frac{1}{2} \int d^3k \left[\hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} + \hat{a}_{\mathbf{k}} \hat{a}_{\mathbf{k}}^\dagger \right] \omega, \quad (\text{B.25})$$

as required. Notice that the terms proportional to $e^{\pm 2i\omega t}$ have cancelled. This had to happen since the Hamiltonian in flat space is time-independent.

B.2 Exercises in Chapter 2

Solution to Exercise 2.1

We consider the action

$$S = \int d^4x \sqrt{-g} \left(-\frac{1}{2} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{2} \xi \phi^2 R \right). \quad (\text{B.26})$$

Let us first determine how the kinetic term transforms under

$$g_{\mu\nu} \rightarrow \Omega^2(x) g_{\mu\nu} \quad \text{and} \quad \phi \rightarrow \Omega^{-\Delta}(x) \phi. \quad (\text{B.27})$$

Ignoring the common factor of $(-1/2)$, we get

$$\begin{aligned} \sqrt{-g} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi &\rightarrow \Omega^D \sqrt{-g} \Omega^{-2} g^{\mu\nu} \nabla_\mu (\Omega^{-\Delta} \phi) \nabla_\nu (\Omega^{-\Delta} \phi) \\ &= \Omega^{D-2-2\Delta} \sqrt{-g} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi + F, \end{aligned} \quad (\text{B.28})$$

where we defined

$$F \equiv \Omega^{D-2-2\Delta} \sqrt{-g} g^{\mu\nu} (-2\Delta \Omega^{-1} (\nabla_\mu \Omega) \phi \nabla_\nu \phi + \Delta^2 \Omega^{-2} \phi^2 \nabla_\mu \Omega \nabla_\nu \Omega). \quad (\text{B.29})$$

Looking at the first term in (B.28), we see that we must choose

$$D - 2 - 2\Delta = 0 \quad \implies \quad \boxed{\Delta = \frac{D-2}{2}}. \quad (\text{B.30})$$

Substituting this into (B.29), we get

$$F = \sqrt{-g} g^{\mu\nu} \left((2-D) \Omega^{-1} (\nabla_\mu \Omega) \phi \nabla_\nu \phi + \left(\frac{D-2}{2} \right)^2 \Omega^{-2} \phi^2 \nabla_\mu \Omega \nabla_\nu \Omega \right). \quad (\text{B.31})$$

Using $\phi \nabla_\nu \phi = \frac{1}{2} \nabla_\nu (\phi^2)$, and integrating by parts, this can also be written as

$$F = \sqrt{-g} g^{\mu\nu} \phi^2 \left(\frac{D-2}{2} \Omega^{-1} \nabla_\mu \nabla_\nu \Omega + \frac{(D-2)(D-4)}{4} \Omega^{-2} \nabla_\mu \Omega \nabla_\nu \Omega \right). \quad (\text{B.32})$$

This term will have to cancel against a contribution arising in the transformation of the curvature coupling.

Using

$$R \rightarrow \Omega^{-2} R - 2(D-1) \Omega^{-3} g^{\mu\nu} \nabla_\mu \nabla_\nu \Omega - (D-1)(D-4) \Omega^{-4} g^{\mu\nu} \nabla_\mu \Omega \nabla_\nu \Omega, \quad (\text{B.33})$$

the transformation of the curvature coupling is

$$\xi \sqrt{-g} \phi^2 R \rightarrow \xi \Omega^{D-2-2\Delta} \sqrt{-g} \phi^2 R + G, \quad (\text{B.34})$$

where we defined

$$G \equiv -\xi \Omega^{D-2-2\Delta} \sqrt{-g} g^{\mu\nu} \phi^2 \left(2(D-1) \Omega^{-1} \nabla_\mu \nabla_\nu \Omega + (D-1)(D-4) \Omega^{-2} \nabla_\mu \Omega \nabla_\nu \Omega \right). \quad (\text{B.35})$$

Using (B.30), the first term in (B.34) becomes $\xi \sqrt{-g} \phi^2 R$ and the second term reads

$$G = -\xi \sqrt{-g} g^{\mu\nu} \phi^2 \left(2(D-1) \Omega^{-1} \nabla_\mu \nabla_\nu \Omega + (D-1)(D-4) \Omega^{-2} \nabla_\mu \Omega \nabla_\nu \Omega \right). \quad (\text{B.36})$$

We would like this to cancel the contribution in (B.32), i.e. we require that $G = -F$. This is indeed the case if

$$\boxed{\xi = \frac{D-2}{4(D-1)}}. \quad (\text{B.37})$$

Conformal invariance has therefore fixed the parameters Δ and ξ in terms of the spacetime dimension D . For $D = 4$, we get $\Delta = 1$ and $\xi = 1/6$, as required. Another special case is $D = 2$, where the theory is conformal even without the curvature coupling, *i.e.* $\xi = 0$.

Solution to Exercise 2.2

Using (2.30), we get

$$\begin{aligned} \delta_{ij} &= [\hat{b}_i, \hat{b}_j^\dagger] = \left[\sum_k \left(\alpha_{ik}^* \hat{a}_k - \beta_{ik}^* \hat{a}_k^\dagger \right), \sum_l \left(\alpha_{jl} \hat{a}_l^\dagger - \beta_{jl} \hat{a}_l \right) \right] \\ &= \sum_{k,l} \left(\alpha_{ik}^* \alpha_{jl} [\hat{a}_k, \hat{a}_l^\dagger] + \beta_{ik}^* \beta_{jl} [\hat{a}_k^\dagger, \hat{a}_l] \right) \\ &= \sum_{k,l} (\alpha_{ik}^* \alpha_{jl} \delta_{kl} - \beta_{ik}^* \beta_{jl} \delta_{kl}) \\ &= \sum_k (\alpha_{ik}^* \alpha_{jk} - \beta_{ik}^* \beta_{jk}), \end{aligned} \quad (\text{B.38})$$

as required.

Solution to Exercise 2.3

In this exercise, we study an example of cosmological particle production that is exactly solvable [7].

1. We first consider the limit $\eta \rightarrow -\infty$ of the function $f_k^{(\text{in})}(\eta)$ in (2.58). In this limit, the hypergeometric function goes to 1 and $\log(2 \cosh(\rho\eta)) \rightarrow -\rho\eta$. We then get

$$f_k^{(\text{in})}(\eta) \rightarrow \frac{1}{\sqrt{2\omega_{\text{in}}}} e^{-i\omega_{\text{in}}\eta}, \quad (\text{B.39})$$

which is the correct positive-frequency solution at early times.

Similarly, for $\eta \rightarrow +\infty$, the hypergeometric function in (2.59) goes to 1 and $\log(2 \cosh(\rho\eta)) \rightarrow \rho\eta$. The function $f_k^{(\text{out})}(\eta)$ in (2.59) therefore becomes

$$f_k^{(\text{out})}(\eta) \rightarrow \frac{1}{\sqrt{2\omega_{\text{out}}}} e^{-i\omega_{\text{out}}\eta}, \quad (\text{B.40})$$

which is the correct positive-frequency solution at late times.

2. To apply the identity (2.64),

$$\begin{aligned} {}_2F_1 \left[\begin{matrix} a, b \\ c \end{matrix} \middle| z \right] &= \frac{\Gamma(c)\Gamma(c-a-b)}{\Gamma(c-a)\Gamma(c-b)} {}_2F_1 \left[\begin{matrix} a, b \\ 1+a+b-c \end{matrix} \middle| 1-z \right] \\ &+ \frac{\Gamma(c)\Gamma(a+b-c)}{\Gamma(a)\Gamma(b)} (1-z)^{c-a-b} z^{1-c} {}_2F_1 \left[\begin{matrix} 1-b, 1-a \\ 1+c-a-b \end{matrix} \middle| 1-z \right]. \end{aligned} \quad (\text{B.41})$$

to the hypergeometric function in (2.58), we write

$$a \equiv 1 + \frac{i\omega_-}{\rho}, \quad b \equiv \frac{i\omega_-}{\rho}, \quad c \equiv 1 - \frac{i\omega_{\text{in}}}{\rho}, \quad z \equiv \frac{1}{2}(1 + \tanh(\rho\eta)). \quad (\text{B.42})$$

This implies

$$(1-z)^{c-a-b} z^{1-c} = \left[\frac{1-z}{z} \right]^{-i\omega_+/\rho} \left[z(1-z) \right]^{-i\omega_-/\rho}, \quad (\text{B.43})$$

where

$$\frac{1-z}{z} = e^{-2\rho\eta}, \quad z(1-z) = \frac{1}{4 \cosh^2(\rho\eta)}. \quad (\text{B.44})$$

The hypergeometric function in (2.58) can then be written as

$$\begin{aligned} &{}_2F_1 \left[\begin{matrix} 1 + \frac{i\omega_-}{\rho}, \frac{i\omega_-}{\rho} \\ 1 - \frac{i\omega_{\text{in}}}{\rho} \end{matrix} \middle| \frac{1}{2}(1 + \tanh(\rho\eta)) \right] \\ &= \frac{\Gamma(1 - i\omega_{\text{in}}/\rho)\Gamma(-i\omega_{\text{out}}\rho)}{\Gamma(-i\omega_+/\rho)\Gamma(1 - i\omega_+/\rho)} {}_2F_1 \left[\begin{matrix} 1 + \frac{i\omega_-}{\rho}, \frac{i\omega_-}{\rho} \\ 1 + \frac{i\omega_{\text{out}}}{\rho} \end{matrix} \middle| \frac{1}{2}(1 + \tanh(\rho\eta)) \right] \\ &+ \frac{\Gamma(1 - i\omega_{\text{in}}/\rho)\Gamma(i\omega_{\text{out}}/\rho)}{\Gamma(1 + i\omega_-/\rho)\Gamma(i\omega_-/\rho)} {}_2F_1 \left[\begin{matrix} 1 - \frac{i\omega_-}{\rho}, -\frac{i\omega_-}{\rho} \\ 1 - \frac{i\omega_{\text{out}}}{\rho} \end{matrix} \middle| \frac{1}{2}(1 - \tanh(\rho\eta)) \right] \\ &\times e^{2i\omega_+\eta} (2 \cosh^2(\rho\eta))^{2i\omega_-/\rho}. \end{aligned} \quad (\text{B.45})$$

Using this in $f_k^{(\text{in})} = \alpha_k f_k^{(\text{out})} + \beta_k f_k^{(\text{out})*}$, we find

$$\alpha_k = \left(\frac{\omega_{\text{out}}}{\omega_{\text{in}}} \right)^{1/2} \frac{\Gamma(1 - i\omega_{\text{in}}/\rho)\Gamma(-i\omega_{\text{out}}/\rho)}{\Gamma(-i\omega_+/\rho)\Gamma(1 - i\omega_+/\rho)}, \quad (\text{B.46})$$

$$\beta_k = \left(\frac{\omega_{\text{out}}}{\omega_{\text{in}}} \right)^{1/2} \frac{\Gamma(1 - i\omega_{\text{in}}/\rho)\Gamma(i\omega_{\text{out}}/\rho)}{\Gamma(1 + i\omega_-/\rho)\Gamma(i\omega_-/\rho)}. \quad (\text{B.47})$$

Using

$$\Gamma(1-z)\Gamma(z) = \frac{\pi}{\sinh(\pi z)}, \quad \Gamma(1+z) = z\Gamma(z), \quad (\text{B.48})$$

this gives

$$|\alpha_k|^2 = \frac{\sinh^2(\pi\omega_+/\rho)}{\sinh(\pi\omega_{\text{in}}/\rho) \sinh(\pi\omega_{\text{out}}/\rho)}, \quad (\text{B.49})$$

$$|\beta_k|^2 = \frac{\sinh^2(\pi\omega_-/\rho)}{\sinh(\pi\omega_{\text{in}}/\rho) \sinh(\pi\omega_{\text{out}}/\rho)}, \quad (\text{B.50})$$

as required.

3. Massless particles in two dimensions are conformally invariant (see Exercise 2.1), so they don't feel the expansion of the Universe. We therefore expect no particle production in this limit. Indeed, in the limit $m \rightarrow 0$, we have $\omega_- \rightarrow 0$ and hence $n_k = |\beta_k|^2 \propto \sinh^2(\pi\omega_-) \rightarrow 0$.
4. In the limit of large k or large m , we have $\omega_{\text{in}}/\rho \sim \omega_{\text{out}}/\rho \gg 1$ and the density of produced particles is exponentially suppressed

$$n_k = |\beta_k|^2 \rightarrow e^{-2\pi\omega_{\text{in}}/\rho} \rightarrow 0. \quad (\text{B.51})$$

For both large k and large m the frequency ω is much larger than the expansion rate ρ and the efficiency of particle production due to the expansion is small.

Solution to Exercise 2.4

Given the scale factor (2.65), the mode functions satisfy the following equation of motion (in Fourier space):

$$f_k'' + k^2 f_k + m^2 (A + B \operatorname{sech}^2(\rho\eta)) f_k = 0. \quad (\text{B.52})$$

The positive-frequency solutions the in-region ($\eta \rightarrow -\infty$) and the out-region ($\eta \rightarrow \infty$) are

$$f_k^{(\text{in})} = \frac{1}{\sqrt{2\omega}} e^{-i\omega\eta} (1 + e^{2\rho\eta})^\gamma {}_2F_1 \left[\begin{matrix} \gamma, \gamma - \frac{i\omega}{\rho} \\ 1 - \frac{i\omega}{\rho} \end{matrix} \middle| e^{2\rho\eta} \right], \quad (\text{B.53})$$

$$f_k^{(\text{out})} = \frac{1}{\sqrt{2\omega}} e^{-i\omega\eta} (1 + e^{-2\rho\eta})^\gamma {}_2F_1 \left[\begin{matrix} \gamma, \gamma + \frac{i\omega}{\rho} \\ 1 + \frac{i\omega}{\rho} \end{matrix} \middle| e^{-2\rho\eta} \right], \quad (\text{B.54})$$

where we have defined the parameter

$$\gamma \equiv \frac{1}{2} + \sqrt{\frac{1}{4} + \frac{m^2 B}{\rho^2}}. \quad (\text{B.55})$$

Since $a(-\infty) = a(\infty) = A$, the frequencies in the two regions are the same, given by

$$\omega \equiv \sqrt{k^2 + m^2 A^2}. \quad (\text{B.56})$$

The negative-frequency solutions in the two regions are just complex conjugates of the positive-frequency solutions.

We want to write

$$f_k^{(\text{in})} = \alpha_k f_k^{(\text{out})} + \beta_k f_k^{(\text{out})*}. \quad (\text{B.57})$$

To do this, we use the hypergeometric connection formula

$${}_2F_1 \left[\begin{matrix} a, b \\ c \end{matrix} \middle| z \right] = \frac{\Gamma(c)\Gamma(b-a)}{\Gamma(b)\Gamma(c-a)} e^{i\pi a} z^{-a} {}_2F_1 \left[\begin{matrix} a, a-c+1 \\ a-b+1 \end{matrix} \middle| \frac{1}{z} \right] + \frac{\Gamma(c)\Gamma(a-b)}{\Gamma(a)\Gamma(c-b)} e^{i\pi b} z^{-b} {}_2F_1 \left[\begin{matrix} b, b-c+1 \\ b-a+1 \end{matrix} \middle| \frac{1}{z} \right], \quad (\text{B.58})$$

which then implies

$$\alpha_k = \frac{\Gamma(1-i\omega/\rho)\Gamma(-i\omega/\rho)}{\Gamma(\gamma-i\omega/\rho)\Gamma(1-\gamma-i\omega/\rho)}, \quad \beta_k = \frac{\Gamma(1-i\omega/\rho)\Gamma(i\omega/\rho)}{\Gamma(\gamma)\Gamma(1-\gamma)}. \quad (\text{B.59})$$

From the β coefficient, we can then read off the number density of particles

$$n_k = |\beta_k|^2 = \frac{\cos^2(\pi[\gamma - \frac{1}{2}])}{\sinh^2(\pi\omega/\rho)}, \quad (\text{B.60})$$

which is of the form (2.66) with $\omega_{\text{in}} = \omega$ and $\kappa = \gamma - \frac{1}{2}$.

Notice that

$$m^2 = \frac{\rho^2}{B} \ell(\ell+1) \implies \cos^2(\pi[\gamma - \frac{1}{2}]) = \sin^2(\pi\gamma) = \sin^2(\pi\ell), \quad (\text{B.61})$$

which vanishes for $\ell \in \mathbb{Z}$. There is therefore no particle production for these special mass values. We can understand this by looking at equation (B.52) for this mass value (and rescaling $\hat{\eta} \equiv \rho\eta$):

$$\frac{\partial^2 f_k}{\partial \hat{\eta}^2} + \ell(\ell+1) \text{sech}^2 \hat{\eta} f_k = -E_k f_k, \quad \text{with} \quad E_k \equiv k^2 + \frac{A}{B} \ell(\ell+1). \quad (\text{B.62})$$

Notice that this is of the form of a Schrödinger equation with the potential $V_\ell = -\ell(\ell+1)\text{sech}^2 \hat{\eta}$. This is a *Pöschl–Teller* potential [112]. For integer values of ℓ , the potentials V_ℓ and $V_{\ell+1}$ are supersymmetric partner potentials, in the sense of supersymmetric quantum mechanics. The fact that the potential does not change form under such a supersymmetry transformation, but just increments its parameter further means that this family of potentials is what is called shape-invariant. Such shape-invariant supersymmetric potentials have a number of interesting properties [113]. For our present purposes, the most important property is that they can be related to the potential with $\ell = 0$, which is reflection-less. By lifting this solution using supersymmetry transformations, one can obtain a solution at integer ℓ that is pure positive frequency in both asymptotic regimes [114, 115]. Consequently, there is no particle production for these mass values.

B.3 Exercises in Chapter 3

Solution to Exercise 3.1

The Fourier transform of the right-moving modes in the right Rindler wedge is

$$\begin{aligned} \tilde{g}_\omega(\omega') &\equiv \int_{-\infty}^{\infty} du e^{i\omega' u} g_\omega^{\text{R}}(U(u)) \\ &= \frac{1}{\sqrt{4\pi\omega}} \int_{-\infty}^0 du e^{i\omega' u} e^{i(\omega/a) \log(-au)}. \end{aligned} \quad (\text{B.63})$$

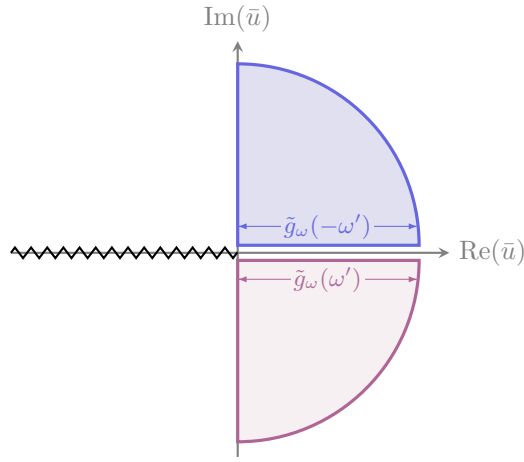


Figure 32: Integration contours chosen for the functions $\tilde{g}_\omega(\omega')$ and $\tilde{g}_\omega(-\omega')$.

Assuming $\omega' > 0$, we wish to relate this to

$$\tilde{g}_\omega(-\omega') = \frac{1}{\sqrt{4\pi\omega}} \int_{-\infty}^0 du e^{-i\omega' u} e^{i(\omega/a) \log(-au)}. \quad (\text{B.64})$$

Defining $\bar{u} = -u$, we can write these integrals as

$$\tilde{g}_\omega(\omega') = \frac{1}{\sqrt{4\pi\omega}} \int_0^\infty d\bar{u} e^{-i\omega' \bar{u}} e^{i(\omega/a) \log(a\bar{u})}, \quad (\text{B.65})$$

$$\tilde{g}_\omega(-\omega') = \frac{1}{\sqrt{4\pi\omega}} \int_0^\infty d\bar{u} e^{i\omega' \bar{u}} e^{i(\omega/a) \log(a\bar{u})}. \quad (\text{B.66})$$

We will analyze these integrals by complexifying the integration variable $\bar{u}$.

The logarithm has a branch cut that we chose to put on the negative real axis. We study the function $\tilde{g}_\omega(\omega')$ by closing the contour in the lower half-plane. Since the function is analytic in the lower right quadrant, the integral along the closed contour shown in Fig. 32 vanishes. Moreover, the contribution along the large arc vanishes, so the integral along the real axis can be related to an integral along the imaginary axis. We see this explicitly by defining $\bar{u} = -iy$, so that

$$\begin{aligned} \tilde{g}_\omega(\omega') &= -i \int_0^\infty dy e^{-\omega' y} e^{i(\omega/a) \log(-iay)} \\ &= -i e^{\pi\omega/(2a)} \int_0^\infty dy e^{-\omega' y} e^{i(\omega/a) \log(ay)} \equiv -i e^{\pi\omega/(2a)} I_{\omega, \omega'}, \end{aligned} \quad (\text{B.67})$$

where we have used

$$\log(-iay) = \log(ay) - i\frac{\pi}{2}. \quad (\text{B.68})$$

The latter identity is dictated by the location of the branch cut.

We can then repeat the same exercise for the function $\tilde{g}_\omega(-\omega')$. We now have to close the contour in the upper half-plane. Defining $\bar{u} = iy$, we get

$$\begin{aligned} \tilde{g}_\omega(-\omega') &= i \int_0^\infty dy e^{-\omega' y} e^{i(\omega/a) \log(iay)} \\ &= i e^{-\pi\omega/(2a)} \int_0^\infty dy e^{-\omega' y} e^{i(\omega/a) \log(ay)} \equiv i e^{-\pi\omega/(2a)} I_{\omega, \omega'}, \end{aligned} \quad (\text{B.69})$$

where we have used

$$\log(iay) = \log(ay) + i\frac{\pi}{2}. \quad (\text{B.70})$$

Comparing (B.69) to (B.67), we infer that

$$\boxed{\tilde{g}_\omega(-\omega') = -e^{-\pi\omega/a} \tilde{g}_\omega(\omega')}, \quad (\text{B.71})$$

as required.

Solution to Exercise 3.2

Restoring all fundamental constants, we have

$$k_B T = \frac{\hbar}{c} \frac{a}{2\pi}. \quad (\text{B.72})$$

Using

$$\begin{aligned} \hbar &= 1.05457182 \times 10^{-34} \text{ m}^2\text{kg/s}, \\ c &= 299\,792\,458 \text{ m/s}, \\ k_B &= 1.380649 \times 10^{-23} \text{ J/K}, \end{aligned} \quad (\text{B.73})$$

and $T_0 = 373.15 \text{ K}$, we get

$$\boxed{a_0 = \frac{2\pi c}{\hbar} k_B T_0 = 9.2 \times 10^{22} \text{ m/s}^2}. \quad (\text{B.74})$$

This explains why we do not experience this in everyday life.

Solution to Exercise 3.3

We start from the definition of the Wightman function

$$G(x, x') \equiv \langle 0 | \hat{\phi}(x) \hat{\phi}(x') | 0 \rangle. \quad (\text{B.75})$$

Writing the field operator as

$$\hat{\phi}(x) = \int d^3k \left[\hat{a}_{\mathbf{k}} f_{\mathbf{k}}(x) + \hat{a}_{\mathbf{k}}^\dagger f_{\mathbf{k}}^*(x) \right], \quad (\text{B.76})$$

this becomes

$$G(x, x') = \int d^3k f_{\mathbf{k}}(x) f_{\mathbf{k}}^*(x'). \quad (\text{B.77})$$

It is convenient to use translation invariance to set $x' = 0$, so that

$$G(x) = G(x, 0) = \int \frac{d^3k}{(2\pi)^3 2k} e^{-ikt + i\mathbf{k} \cdot \mathbf{x}}, \quad (\text{B.78})$$

where we have substituted the positive-frequency mode function for $f_{\mathbf{k}}(x)$. To evaluate this integral, we use polar coordinates

$$\begin{aligned} G(x) &= \int_0^\infty \frac{k dk}{2(2\pi)^3} \int_{-1}^1 d\cos\theta \int_0^{2\pi} d\phi e^{ik(r\cos\theta - t)} \\ &= -\frac{i}{2(2\pi)^2} \frac{1}{r} \int_0^\infty dk \left(e^{-ik(t-r)} - e^{-ik(t+r)} \right). \end{aligned} \quad (\text{B.79})$$

Note that this integral is divergent and has to be regularized. Defining $t \rightarrow t - i\epsilon$, with $\epsilon > 0$, we get

$$\int_0^\infty dk e^{-ik(t-r-i\epsilon)} = -\frac{i}{t-r-i\epsilon}, \quad (\text{B.80})$$

and hence

$$G(x) = \frac{1}{2(2\pi)^2} \frac{1}{r} \left[\frac{1}{t+r-i\epsilon} - \frac{1}{t-r-i\epsilon} \right] = \boxed{-\frac{1}{4\pi^2} \frac{1}{(t-i\epsilon)^2 - r^2}}, \quad (\text{B.81})$$

as required.

B.4 Exercises in Chapter 4

Solution to Exercise 4.1

We are asked to substitute $r = 2M + \rho$ into the metric (4.1) and expand the result for small $\rho \ll 2M$. At leading order, we have

$$\begin{aligned} 1 - \frac{2M}{r} &= 1 - \frac{2M}{2M + \rho} = 1 - \left(1 + \frac{\rho}{2M}\right)^{-1} \approx \frac{\rho}{2M} + O(\rho^2), \\ r^2 &= (2M + \rho)^2 \approx (2M)^2 + O(\rho), \end{aligned} \quad (\text{B.82})$$

so that the Schwarzschild metric becomes

$$ds^2 = \underbrace{-\frac{\rho}{2M} dt^2 + \frac{2M}{\rho} d\rho^2}_{\text{Rindler space}} + \underbrace{(2M)^2 d\Omega^2}_{S^2}. \quad (\text{B.83})$$

We see that the metric has separated into a two-sphere of fixed radius $2M$ and a 1+1 dimensional Lorentzian geometry. The claim is that the latter is the same Rindler space discussed in Section 3; cf. (3.18):

$$ds^2 = e^{2a\xi} (-d\eta^2 + d\xi^2). \quad (\text{B.84})$$

To see this, we consider the coordinate transformation $t = \eta$ and $\rho/2M = e^{2a\xi}$. The two-dimensional metric inside (B.83) then becomes

$$\begin{aligned} ds^2 &\equiv -\frac{\rho}{2M} dt^2 + \frac{2M}{\rho} d\rho^2 \\ &= -e^{2a\xi} d\eta^2 + e^{-2a\xi} \times (2M)^2 (2a)^2 e^{4a\xi} d\xi^2 \\ &= e^{2a\xi} (-d\eta^2 + (2M)^2 (2a)^2 d\xi^2), \end{aligned} \quad (\text{B.85})$$

which matches (B.84) if

$$a = \frac{1}{4M}. \quad (\text{B.86})$$

The effective acceleration is the *surface gravity* of the black hole.

Solution to Exercise 4.2

Restoring all fundamental constants, the Hawking temperature of a black hole of mass M is

$$T_{\text{BH}} = \frac{\hbar c^3}{k_B} \frac{1}{8\pi G M}. \quad (\text{B.87})$$

Using

$$\begin{aligned} \hbar &= 1.05457182 \times 10^{-34} \text{ m}^2\text{kg/s}, \\ c &= 299\,792\,458 \text{ m/s}, \\ G &= 6.67430 \times 10^{-11} \text{ Nm}^2/\text{kg}^2, \\ k_B &= 1.380649 \times 10^{-23} \text{ J/K}, \end{aligned} \quad (\text{B.88})$$

this gives

$$\boxed{T_{\text{BH}} = 61.7 \text{ nK} \frac{M_{\odot}}{M}}. \quad (\text{B.89})$$

For $M_{\odot} \approx 2 \times 10^{30} \text{ kg}$ (Solar mass), $M_{\oplus} \approx 6 \times 10^{24} \text{ kg}$ (Earth mass) and $M_{\text{Pl}} \approx 2 \times 10^{-8} \text{ kg}$ (Planck mass), we therefore have $T_{\text{BH}} \sim 60 \text{ nK}$, $T_{\text{BH}} \sim 0.02 \text{ K}$ and $T_{\text{BH}} \sim 10^{40} \text{ K}$, respectively.

Solution to Exercise 4.3

Restoring all fundamental constant, the lifetime of a black hole is

$$t_{\text{BH}} \equiv 5120\pi \frac{G^2 M_0^3}{\hbar c^4}. \quad (\text{B.90})$$

Using

$$\begin{aligned} \hbar &= 1.05457182 \times 10^{-34} \text{ m}^2\text{kg/s}, \\ c &= 299\,792\,458 \text{ m/s}, \\ G &= 6.67430 \times 10^{-11} \text{ Nm}^2/\text{kg}^2, \end{aligned} \quad (\text{B.91})$$

this gives

$$\boxed{t_{\text{BH}} = 6 \times 10^{74} \text{ s} \left(\frac{M_0}{M_{\odot}} \right)^3} = \begin{cases} 6 \times 10^{74} \text{ s} & \text{for } M_0 = M_{\odot}, \\ 2 \times 10^{58} \text{ s} & \text{for } M_0 = M_{\oplus}, \\ 7 \times 10^{-40} \text{ s} & \text{for } M_0 = M_{\text{Pl}}. \end{cases} \quad (\text{B.92})$$

For comparison, the age of the Universe is $t = 13.8 \text{ billion years} = 4 \times 10^{17} \text{ s}$.

Solution to Exercise 4.4

Let the two black holes have masses M_1 and M_2 . After the merger, we have a black hole with a mass $M_3 < M_1 + M_2$. The difference in the rest masses before and after the merger, $\Delta E \equiv M_1 + M_2 - M_3$, is radiated as gravitational wave. By the area theorem, we have

$$A_3 \geq A_1 + A_2 \quad \implies \quad M_3 \geq \sqrt{M_1^2 + M_2^2}. \quad (\text{B.93})$$

This implies

$$\frac{\Delta E}{E} \leq \frac{M_1 + M_2 - \sqrt{M_1^2 + M_2^2}}{M_1 + M_2} = 1 - \frac{\sqrt{1 + (M_2/M_1)^2}}{1 + (M_2/M_1)}. \quad (\text{B.94})$$

It is easy to check that the right-hand side is maximized for $M_2 = M_1$. Since the two black holes are equivalent, this had to be the case by symmetry. We therefore have

$$\boxed{\frac{\Delta E}{E} \leq 1 - \frac{1}{\sqrt{2}}} \approx 29\%, \quad (\text{B.95})$$

as required.

Solution to Exercise 4.5

1. The event horizon is defined by the condition $g^{rr}(r) = f(r) = 0$. This gives

$$1 - \frac{2M}{r} + \frac{Q^2}{r^2} = 0 \quad \implies \quad \boxed{r_{\pm} = M \pm \sqrt{M^2 - Q^2}}. \quad (\text{B.96})$$

We have two real solutions when $|Q| < M$.

2. The area of the outer horizon is

$$A = 4\pi r_+^2 = 4\pi \left(M + \sqrt{M^2 - Q^2} \right)^2. \quad (\text{B.97})$$

Letting $S = A/4$, this implies

$$\begin{aligned} dS &= \frac{dA}{4} \\ &= \frac{8\pi}{4} \left(M + \sqrt{M^2 - Q^2} \right) \left[\left(1 + \frac{M}{\sqrt{M^2 - Q^2}} \right) dM - \frac{Q}{\sqrt{M^2 - Q^2}} dQ \right] \\ &= 2\pi \frac{(M + \sqrt{M^2 - Q^2})^2}{\sqrt{M^2 - Q^2}} \left[dM - \frac{Q}{M + \sqrt{M^2 - Q^2}} dQ \right], \end{aligned} \quad (\text{B.98})$$

which can be put into the form

$$T dS = dM - \Phi dQ, \quad (\text{B.99})$$

if we identify the temperature and the potential with

$$T \equiv \frac{\sqrt{M^2 - Q^2}}{2\pi(M + \sqrt{M^2 - Q^2})^2} \equiv \frac{r_+ - r_-}{4\pi r_+^2}, \quad (\text{B.100})$$

$$\Phi \equiv \frac{Q}{M + \sqrt{M^2 - Q^2}} \equiv \frac{Q}{r_+}. \quad (\text{B.101})$$

Notice that the temperature vanishes in the extremal limit $|Q| \rightarrow M$ where $r_+ = r_-$.

3. We are given that the conserved energy of the particle is $e = -(p_\mu + qA_\mu)K^\mu$, where $K^\mu = (1, 0, 0, 0)$ is the Killing vector associated to the time-independent metric.

- (a) From the definition, we get

$$e = -(p_\mu + qA_\mu)K^\mu = -(p_t + qA_t) = -(g_{tt}p^t + qA_t), \quad (\text{B.102})$$

and hence

$$\boxed{e = f(r)p^t + \frac{qQ}{r}}. \quad (\text{B.103})$$

(b) Note that $f(r)p^t > 0$, since $f(r) > 0$ (outside the horizon) and $p^t = dt/d\tau > 0$. For $Q > 0$, the energy e can therefore only be negative if $q < 0$.

(c) The normalization $g_{\mu\nu}p^\mu p^\nu = -1$ implies

$$-f(r)(p^t)^2 + f^{-1}(r)(p^r)^2 = -1 \implies f(r)p^t = \sqrt{(p^r)^2 + f(r)}. \quad (\text{B.104})$$

The result (B.103) can then be written as

$$e = \sqrt{(p^r)^2 + f(r)} + \frac{qQ}{r}. \quad (\text{B.105})$$

(d) Evaluating (B.105) on the horizon, where $f(r_+) = 0$, we find

$$e = |p^r(r_+)| + \frac{qQ}{r_+}. \quad (\text{B.106})$$

Since $|p^r| \equiv |dr/d\tau| > 0$, we have

$$e > e_{\min} = \frac{qQ}{r_+}. \quad (\text{B.107})$$

The change in the entropy is

$$dS = \frac{1}{T} \left(e - \frac{qQ}{r_+} \right), \quad (\text{B.108})$$

which is positive since $e > qQ/r_+$.

Solution to Exercise 4.6

1. To determine r_+ and r_- , we impose that $g^{rr} = 0$, i.e. that $\Delta(r) = 0$. This implies

$$r_{\pm} = M \pm \sqrt{M^2 - a^2}. \quad (\text{B.109})$$

We have two real solutions when $M > a$.

2. The area of the outer horizon is given by

$$A = \int_0^\pi d\theta \int_0^{2\pi} d\phi \sqrt{g_{\theta\theta}(r_+) g_{\phi\phi}(r_+)}. \quad (\text{B.110})$$

To evaluate this, we consider

$$\begin{aligned} g_{\phi\phi}(r_+) &\equiv \left(r_+^2 + a^2 + \frac{2Mr_+ a^2 \sin^2 \theta}{\rho_+^2} \right) \sin^2 \theta \\ &= 2Mr_+ \left(1 + \frac{a^2 \sin^2 \theta}{\rho_+^2} \right) \sin^2 \theta \\ &= 2Mr_+ \frac{r_+^2 + a^2 \cos^2 \theta + a^2 \sin^2 \theta}{\rho_+^2} \sin^2 \theta \\ &= 2Mr_+ \frac{r_+^2 + a^2}{\rho_+^2} \sin^2 \theta \\ &= \frac{(2Mr_+)^2}{\rho_+^2} \sin^2 \theta, \end{aligned} \quad (\text{B.111})$$

where we used $r_+^2 + a^2 = 2Mr_+$ (implied by $\Delta(r_+) = 0$). Substituting this into (B.110), we get

$$A = 2\pi \int_0^\pi d\theta \sqrt{\rho_+^2 \frac{(2Mr_+)^2}{\rho_+^2} \sin^2 \theta} = 2\pi \int_{-1}^1 d(\cos \theta) (2Mr_+) = \boxed{8\pi Mr_+}. \quad (\text{B.112})$$

3. The entropy is

$$\begin{aligned} S &= \frac{A}{4} = 2\pi Mr_+ \\ &= 2\pi \left(M^2 + M\sqrt{M^2 - a^2} \right) \\ &= 2\pi \left(M^2 + \sqrt{M^4 - J^2} \right) \end{aligned} \quad (\text{B.113})$$

and hence

$$\begin{aligned} dS &= 2\pi \left[2M dM + \frac{1}{\sqrt{M^4 - J^2}} (2M^3 dM - J dJ) \right] \\ &= 4\pi \left[\frac{M(M^2 + \sqrt{M^4 - J^2})}{\sqrt{M^4 - J^2}} dM - \frac{1}{2} \frac{J}{\sqrt{M^4 - J^2}} dJ \right] \\ &= 4\pi \frac{M(M^2 + \sqrt{M^4 - J^2})}{\sqrt{M^4 - J^2}} \left[dM - \frac{1}{2} \frac{J}{M(M^2 + \sqrt{M^4 - J^2})} dJ \right], \end{aligned} \quad (\text{B.114})$$

which can be put into the form

$$T dS = dM - \Omega dJ, \quad (\text{B.115})$$

if we identify the temperature and the potential with

$$T \equiv \frac{\sqrt{M^4 - J^2}}{4\pi M(M^2 + \sqrt{M^4 - J^2})} \equiv \frac{r_+ - r_-}{8\pi Mr_+}, \quad (\text{B.116})$$

$$\Omega \equiv \frac{1}{2} \frac{J}{M(M^2 + \sqrt{M^4 - J^2})} \equiv \frac{a}{2Mr_+}. \quad (\text{B.117})$$

4. We consider the two Killing vectors $K = \partial_t$ and $R = \partial_\phi$.

(a) The norm of K is

$$g_{\mu\nu} K^\mu K^\nu = g_{tt} = -\frac{1}{\rho^2} (r^2 - 2Mr + a^2 \cos^2 \theta). \quad (\text{B.118})$$

We find that it is null when

$$r = M \pm \sqrt{M^2 - a^2 \cos^2 \theta}. \quad (\text{B.119})$$

It is clear that the larger root is bigger than r_+ for every θ (except for $\theta = 0$ or π , where they are the same). K is therefore timelike outside the ergoregion, and spacelike inside it.

- (b) Since K^μ is spacelike inside the ergoregion, and p^μ is timelike, their scalar product can be positive, thus yielding a negative energy particle. For a timelike K^μ , on the other hand, the scalar product would always be negative and the particle's energy positive.

5. The norm of ξ is null when:

$$\begin{aligned} 0 &= g_{\mu\nu} \xi^\mu \xi^\nu = g_{\mu\nu} (K^\mu + \Omega R^\mu) (K^\nu + \Omega R^\nu) \\ &= g_{\mu\nu} K^\mu K^\nu + 2\Omega g_{\mu\nu} K^\mu R^\nu + \Omega^2 g_{\mu\nu} R^\mu R^\nu \\ &= g_{tt} + 2g_{t\phi}\Omega + g_{\phi\phi}\Omega^2, \end{aligned} \quad (\text{B.120})$$

so that

$$\Omega(r, \theta) = \frac{-g_{t\phi} \pm \sqrt{g_{t\phi}^2 - g_{tt}g_{\phi\phi}}}{g_{\phi\phi}}. \quad (\text{B.121})$$

For $r = r_+$, we have $g_{t\phi}^2 - g_{tt}g_{\phi\phi} = \Delta(r_+) \sin^2 \theta = 0$ and the expression above simplifies to

$$\Omega_H \equiv \Omega(r_+, \theta) = -\frac{g_{t\phi}}{g_{\phi\phi}}. \quad (\text{B.122})$$

Using

$$\begin{aligned} g_{t\phi}(r_+, \theta) &= -\frac{2Mr_+a}{\rho_+^2} \sin^2 \theta, \\ g_{\phi\phi}(r_+, \theta) &= \frac{(r_+^2 + a^2)^2}{\rho_+^2} \sin^2 \theta = \frac{(2Mr_+)^2}{\rho_+^2} \sin^2 \theta, \end{aligned} \quad (\text{B.123})$$

this becomes

$$\boxed{\Omega_H = \frac{a}{2Mr_+}}. \quad (\text{B.124})$$

Note that this is the same as (B.117).

6. At the horizon, ξ^μ is lightlike everywhere and future pointing, so $\xi_\mu p^\mu \leq 0$ for any timelike P^μ . This implies

$$\xi_\mu p^\mu \Big|_{r=r_+} = (K_\mu + \Omega R_\mu) p^\mu \Big|_{r=r_+} = -E + \Omega_H L \leq 0 \quad \implies \quad \boxed{E \geq \Omega_H L}, \quad (\text{B.125})$$

so that $\boxed{\delta M \geq \Omega_H \delta J}$.

We write the horizon area as

$$A = 4\pi(r_+^2 + a^2) = 8\pi Mr_+ = 8\pi(M^2 + \sqrt{M^4 - J^2}), \quad (\text{B.126})$$

where we used again that $r_+^2 + a^2 = 2Mr_+$. The change of the horizon area after the absorption then is

$$\begin{aligned} \frac{\delta A}{8\pi} &= 2M\delta M + \frac{2M^3\delta M - J\delta J}{\sqrt{M^4 - J^2}} \\ &= \frac{1}{\sqrt{M^4 - J^2}} \left(2M(\sqrt{M^4 - J^2} + M^2) \delta M - J\delta J \right) \\ &= \frac{1}{\sqrt{M^4 - J^2}} (2M^2 r_+ \delta M - aM\delta J) \\ &= \frac{2M^2 r_+}{\sqrt{M^4 - J^2}} (\delta M - \Omega_H \delta J) \geq 0. \end{aligned} \quad (\text{B.127})$$

This proves the “area theorem”: Classically, the horizon area of a black hole is always increasing.

Solution to Exercise 4.7

The total energy and entropy of the system are

$$E = M + \sigma VT^4, \quad (\text{B.128})$$

$$S = 4\pi M^2 + \frac{4}{3}\sigma VT^3. \quad (\text{B.129})$$

Since E is fixed, it is useful to write the entropy as

$$S(T) = 4\pi(E - \sigma VT^4)^2 + \frac{4}{3}\sigma VT^3. \quad (\text{B.130})$$

We would like to find the extremum of this function $S(T)$, and determine for which range of V it is a maximum.

1. To derive the extremum of the entropy, we compute

$$\begin{aligned} \frac{dS}{dT} &= 8\pi(E - \sigma VT^4)(-4\sigma VT^3) + 4\sigma VT^2 \\ &= 4\sigma VT^2(1 - 8\pi MT), \end{aligned} \quad (\text{B.131})$$

which vanishes for

$$\boxed{T = \frac{1}{8\pi M} = T_{\text{BH}}}, \quad (\text{B.132})$$

as required.

2. The second derivative of the entropy is

$$\frac{d^2S}{dT^2} = 8\sigma VT [28\pi\sigma VT^5 - 12\pi ET + 1]. \quad (\text{B.133})$$

The extremum is a maximum if this is negative at $T = T_{\text{BH}}$, which is the case for

$$V < \frac{12\pi ET_{\text{BH}} - 1}{28\pi\sigma T_{\text{BH}}^5}. \quad (\text{B.134})$$

Using

$$\begin{aligned} E = M + \sigma VT_{\text{BH}}^4 &\Rightarrow 8\pi ET_{\text{BH}} = 1 + 8\pi\sigma VT_{\text{BH}}^5, \\ 12\pi ET_{\text{BH}} &= \frac{3}{2} + 12\pi\sigma VT_{\text{BH}}^5, \end{aligned} \quad (\text{B.135})$$

this can be written as

$$V < \frac{1}{56\pi\sigma T_{\text{BH}}^5} + \frac{3}{7}V \Rightarrow V < \frac{1}{32\pi\sigma T_{\text{BH}}^5}. \quad (\text{B.136})$$

However, we would like to obtain the upper limit purely in terms of the energy E . Note that (B.136) implies

$$4 \times 8\pi\sigma VT_{\text{BH}}^5 = 4(8\pi ET_{\text{BH}} - 1) < 1 \Rightarrow T_{\text{BH}} < \frac{5}{32\pi E}, \quad (\text{B.137})$$

which is saturated for

$$T_{\text{BH}} = \frac{5}{32\pi E}. \quad (\text{B.138})$$

Substituting this into (B.136), we find

$$V < \frac{2^{20}\pi^4 E^5}{5^5 \sigma}, \quad (\text{B.139})$$

as required.

For $V > V_c$, the black hole is in an unstable equilibrium with the thermal bath. It either absorbs all the radiation until $T = 0$ or it completely evaporates, leaving radiation with a temperature $T_{\text{max}} = (E/4\sigma V)^{1/4}$. The entropies of these two endpoints are

$$S(0) = 4\pi E^2, \quad (\text{B.140})$$

$$S(T_{\text{max}}) = \frac{4}{3}\sigma V \left(\frac{E}{\sigma V} \right)^{3/4}. \quad (\text{B.141})$$

Taking the ratio, we find

$$\frac{S(T_{\text{max}})}{S(0)} = \frac{2^{20}}{3^4 5^5} \frac{V}{V_c} \approx 4.14 \frac{V}{V_c} > 1. \quad (\text{B.142})$$

This shows that, for $V > V_c$, the black hole will evaporate.

Solution to Exercise 4.8

1. In Exercise 4.5, we showed that the outer horizon of the Reissner–Nordström black hole is

$$r_+ = M + \sqrt{M^2 - Q^2} \iff f(r_+) = 1 - \frac{2M}{r_+} + \frac{Q^2}{r_+^2} = 0. \quad (\text{B.143})$$

Its temperature and entropy therefore are

$$T_{\text{BH}} \equiv \frac{f'(r_+)}{4\pi} = \frac{1}{4\pi} \frac{r_+^2 - Q^2}{r_+^3}, \quad (\text{B.144})$$

$$S_{\text{BH}} \equiv \pi r_+^2 = \pi \left(2M^2 - Q^2 + 2M\sqrt{M^2 - Q^2} \right). \quad (\text{B.145})$$

The specific heat then is

$$C \equiv T_{\text{BH}} \left. \frac{\partial S_{\text{BH}}}{\partial T_{\text{BH}}} \right|_Q = T_{\text{BH}} \frac{dS_{\text{BH}}}{dr_+} \left(\frac{dT_{\text{BH}}}{dr_+} \right)_Q^{-1}. \quad (\text{B.146})$$

Using

$$\left(\frac{dT_{\text{BH}}}{dr_+} \right)_Q = \frac{1}{4\pi} \frac{3Q^2 - r_+^2}{r_+^4}, \quad (\text{B.147})$$

we get

$$\begin{aligned} C &= T_{\text{BH}} 2\pi r_+ \frac{4\pi r_+^4}{3Q^2 - r_+^2} \\ &= (2\pi r_+^2)(4\pi T_{\text{BH}} r_+^3) \frac{1}{3Q^2 - r_+^2} \\ &= 2S_{\text{BH}} \frac{r_+^2 - Q^2}{3Q^2 - r_+^2}. \end{aligned} \quad (\text{B.148})$$

Substituting (B.143) for r_+ gives

$$C = 2S_{\text{BH}} \frac{\sqrt{M^2 - Q^2}}{M - 2\sqrt{M^2 - Q^2}}, \quad (\text{B.149})$$

as required.

It is easy to see that C^{-1} vanishes for

$$M_c - 2\sqrt{M_c^2 - Q^2} = 0 \quad \implies \quad M_c = \frac{2}{\sqrt{3}}|Q|, \quad (\text{B.150})$$

and hence changes sign when M passes through $2|Q|/\sqrt{3}$.

2. Next, we put the black hole into a thermal reservoir (as in Exercise 4.7). The total energy and entropy of the system are

$$E = M + \sigma VT^4, \quad (\text{B.151})$$

$$S = \pi \left(2M^2 - Q^2 + 2M\sqrt{M^2 - Q^2} \right) + \frac{4}{3}\sigma VT^3. \quad (\text{B.152})$$

(a) To determine the extremum, we compute

$$\begin{aligned} \frac{dS}{dT} &= \frac{dS_{\text{BH}}}{dM} \frac{dM}{dT} + 4\sigma VT^2 \\ &= \pi \left(4M + 2\sqrt{M^2 - Q^2} + \frac{2M^2}{\sqrt{M^2 - Q^2}} \right) \frac{d}{dT} (E - \sigma VT^4) + 4\sigma VT^2 \\ &= 4\sigma VT^2 \left[1 - 2\pi T \left(2M + \sqrt{M^2 - Q^2} + \frac{M^2}{\sqrt{M^2 - Q^2}} \right) \right] \\ &= 4\sigma VT^2 \left[1 - 2\pi T \left(r_+ + M + \frac{M^2}{r_+ - M} \right) \right] \\ &= 4\sigma VT^2 \left[1 - 4\pi T \frac{r_+^2}{2(r_+ - M)} \right] \\ &= 4\sigma VT^2 \left[1 - 4\pi T \frac{r_+^3}{2r_+^2 - 2Mr_+} \right] \\ &= 4\sigma VT^2 \left[1 - 4\pi T \frac{r_+^3}{2r_+^2 - (r_+^2 + Q^2)} \right] \\ &= 4\sigma VT^2 \left[1 - 4\pi T \frac{r_+^3}{r_+^2 - Q^2} \right] \\ &= 4\sigma VT^2 \left[1 - \frac{T}{T_{\text{BH}}} \right], \end{aligned} \quad (\text{B.153})$$

which vanishes for

$$T = T_{\text{BH}} = \frac{1}{4\pi} \frac{r_+^2 - Q^2}{r_+^3}, \quad (\text{B.154})$$

as required.

(b) The second derivative of the entropy is

$$\frac{d^2 S}{dT^2} = 8\sigma VT \left[1 - \frac{T}{T_{\text{BH}}} \right] + 4\sigma VT^2 \left[-\frac{1}{T_{\text{BH}}} - T \frac{d}{dr_+} \left(\frac{1}{T_{\text{BH}}} \right) \frac{dr_+}{dT} \right]. \quad (\text{B.155})$$

Using

$$\frac{d}{dr_+} \left(\frac{1}{T_{\text{BH}}} \right) = -\frac{1}{T_{\text{BH}}^2} \frac{d}{dr_+} \left(\frac{1}{4\pi} \frac{r_+^2 - Q^2}{r_+^3} \right) = \frac{1}{4\pi T_{\text{BH}}^2} \frac{r_+^2 - 3Q^2}{r_+^4}, \quad (\text{B.156})$$

$$\frac{dr_+}{dT} = \frac{dr_+}{dM} \frac{dM}{dT} = -4\sigma VT^3 \frac{r_+}{r_+ - M}, \quad (\text{B.157})$$

this becomes

$$\frac{d^2 S}{dT^2} = 8\sigma VT \left[1 - \frac{T}{T_{\text{BH}}} \right] + 4\sigma VT \left[-\frac{T}{T_{\text{BH}}} - \frac{4\sigma VT^3}{r_+ - M} \frac{T^2}{T_{\text{BH}}^2} \frac{3Q^2 - r_+^2}{4\pi r_+^3} \right]. \quad (\text{B.158})$$

At $T = T_{\text{BH}}$, this reduces to

$$\frac{d^2 S}{dT^2} = -4\sigma VT_{\text{BH}} \left[1 + \gamma V \right], \quad \text{where} \quad \gamma \equiv \frac{4\sigma T_{\text{BH}}^4}{r_+ - M} \frac{3Q^2 - r_+^2}{r_+^2 - Q^2}. \quad (\text{B.159})$$

Substituting (B.143) for r_+ , we get

$$\boxed{\gamma \equiv 4\sigma T_{\text{BH}}^4 \frac{M - 2\sqrt{M^2 - Q^2}}{M^2 - Q^2}}, \quad (\text{B.160})$$

which is positive for $|Q| < M < \frac{2}{\sqrt{3}}|Q|$. For M in this range, $d^2 S/dT^2 < 0$ for any finite V . This shows that $V_c = \infty$, and the black hole is stable for all V .

B.5 Exercises in Chapter 5

Solution to Exercise 5.1

1. Comparing (5.3) and (5.10), we have

$$X_0 = L \sinh(t/L) = L \sinh(\tilde{t}/L) + \frac{\tilde{r}^2}{2L} e^{\tilde{t}/L}, \quad (\text{B.161})$$

$$X_1 = L \cosh(t/L) \cos \psi = L \cosh(\tilde{t}/L) - \frac{\tilde{r}^2}{2L} e^{\tilde{t}/L}. \quad (\text{B.162})$$

Computing $X_0 + X_1$, we find

$$L \left[\sinh(t/L) + \cosh(t/L) \cos \psi \right] = L \underbrace{\left[\sinh(\tilde{t}/L) + \cosh(\tilde{t}/L) \right]}_{e^{\tilde{t}/L}}, \quad (\text{B.163})$$

and hence

$$\boxed{e^{\tilde{t}/L} = \cos \psi \cosh(t/L) + \sinh(t/L)}, \quad (\text{B.164})$$

as required.

Computing $-X_0^2 + X_1^2$, we get

$$\begin{aligned} -\sinh^2(t/L) + \cosh^2(t/L) \cos^2 \psi &= -\left[\sinh(\tilde{t}/L) + \frac{\tilde{r}^2}{2L^2} e^{\tilde{t}/L} \right]^2 + \left[\cosh(\tilde{t}/L) - \frac{\tilde{r}^2}{2L^2} e^{\tilde{t}/L} \right]^2 \\ 1 - \cosh^2(t/L) \sin^2 \psi &= 1 - \underbrace{\left[\sinh(\tilde{t}/L) + \cosh(\tilde{t}/L) \right]}_{e^{\tilde{t}/L}} \frac{\tilde{r}^2}{L^2} e^{\tilde{t}/L}, \end{aligned} \quad (\text{B.165})$$

so that

$$\boxed{\frac{\tilde{r}}{L} e^{\tilde{t}/L} = \sin \psi \cosh(t/L)}, \quad (\text{B.166})$$

as required.

2. The differentials of the embedding coordinates are

$$dX_0 = \left[\cosh(\tilde{t}/L) + \frac{\tilde{r}^2}{2L^2} e^{\tilde{t}/L} \right] d\tilde{t} + \frac{\tilde{r}}{L} e^{\tilde{t}/L} d\tilde{r}, \quad (\text{B.167})$$

$$dX_1 = \left[\sinh(\tilde{t}/L) - \frac{\tilde{r}^2}{2L^2} e^{\tilde{t}/L} \right] d\tilde{t} - \frac{\tilde{r}}{L} e^{\tilde{t}/L} d\tilde{r}, \quad (\text{B.168})$$

$$dX_i = e^{\tilde{t}/L} \left[\frac{\tilde{x}_i}{L} d\tilde{t} + d\tilde{x}_i \right]. \quad (\text{B.169})$$

Substituting these into

$$ds^2 = -dX_0^2 + dX_1^2 + \sum_{i=2}^4 dX_i^2, \quad (\text{B.170})$$

we get the induced metric $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$.

- The temporal component $g_{\tilde{t}\tilde{t}}$ is

$$\begin{aligned} g_{\tilde{t}\tilde{t}} &= -\left[\cosh(\tilde{t}/L) + \frac{\tilde{r}^2}{2L^2} e^{\tilde{t}/L} \right]^2 + \left[\sinh(\tilde{t}/L) - \frac{\tilde{r}^2}{2L^2} e^{\tilde{t}/L} \right]^2 + e^{2\tilde{t}/L} \frac{\tilde{x}_i^2}{L^2} \\ &= -\cosh^2(\tilde{t}/L) - \cosh(\tilde{t}/L) \frac{\tilde{r}^2}{L^2} e^{\tilde{t}/L} + \sinh^2(\tilde{t}/L) - \sinh(\tilde{t}/L) \frac{\tilde{r}^2}{L^2} e^{\tilde{t}/L} + e^{2\tilde{t}/L} \frac{\tilde{r}^2}{L^2} \\ &= -1 - [\cosh(\tilde{t}/L) + \sinh(\tilde{t}/L)] e^{\tilde{t}/L} \frac{\tilde{r}^2}{L^2} + e^{2\tilde{t}/L} \frac{\tilde{r}^2}{L^2} \\ &= -1. \end{aligned} \quad (\text{B.171})$$

- The mixed component $g_{\tilde{t}\tilde{r}}$ vanishes:

$$\begin{aligned} g_{\tilde{t}\tilde{r}} &= -\left[\cosh(\tilde{t}/L) + \frac{\tilde{r}^2}{2L^2} e^{\tilde{t}/L} \right] \frac{\tilde{r}}{L} e^{\tilde{t}/L} - \left[\sinh(\tilde{t}/L) - \frac{\tilde{r}^2}{2L^2} e^{\tilde{t}/L} \right] \frac{\tilde{r}}{L} e^{\tilde{t}/L} + \frac{\tilde{r}}{L} e^{2\tilde{t}/L} \\ &= -[\cosh(\tilde{t}/L) + \sinh(\tilde{t}/L)] \frac{\tilde{r}}{L} e^{\tilde{t}/L} + \frac{\tilde{r}}{L} e^{2\tilde{t}/L} \\ &= -\frac{\tilde{r}}{L} e^{2\tilde{t}/L} + \frac{\tilde{r}}{L} e^{2\tilde{t}/L} \\ &= 0, \end{aligned} \quad (\text{B.172})$$

where we used that $\tilde{x}_i d\tilde{x}_i = \tilde{r} d\tilde{r}$.

- Finally, the spatial components g_{ij} are

$$g_{ij} = e^{2\tilde{t}/L} \delta_{ij}. \quad (\text{B.173})$$

The line element in planar coordinates then is

$$\boxed{ds^2 = -d\tilde{t}^2 + e^{2\tilde{t}/L} \delta_{ij} d\tilde{x}^i d\tilde{x}^j}, \quad (\text{B.174})$$

as required.

Solution to Exercise 5.2

The vacuum Einstein equations with a cosmological constant are

$$\left(R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R \right) + \Lambda g_{\mu\nu} = 0. \quad (\text{B.175})$$

Contracting both sides with the metric, we get $R = 4\Lambda$ and hence

$$R_{\mu\nu} = \Lambda g_{\mu\nu}. \quad (\text{B.176})$$

We first note that

$$\begin{aligned} e^{2(\beta-\alpha)} R_{tt} + R_{rr} &= e^{2(\beta-\alpha)} \Lambda g_{tt} + \Lambda g_{rr} \\ \frac{2}{r} (\partial_r \alpha + \partial_r \beta) &= \Lambda \left(e^{2(\beta-\alpha)} (-e^{2\alpha}) + e^{2\beta} \right) \\ &= 0, \end{aligned} \quad (\text{B.177})$$

so that $\alpha = -\beta + c$, where c is an arbitrary constant. This constant can be absorbed by a rescaling of the time coordinate, $t \rightarrow e^{-c} t$, after which we have

$$\alpha = -\beta. \quad (\text{B.178})$$

We have therefore reduced the number of free functions from two to just one. Next, we consider $R_{\theta\theta} = \Lambda g_{\theta\theta}$, which now becomes

$$e^{2\alpha} (2r \partial_r \alpha + 1) = 1 - r^2 \Lambda \quad \Rightarrow \quad \partial_r (r e^{2\alpha}) = 1 - r^2 \Lambda. \quad (\text{B.179})$$

Integrating the last expression, we find

$$e^{2\alpha} = 1 - \frac{\Lambda}{3} r^2, \quad (\text{B.180})$$

so that

$$\boxed{ds^2 = - \left(1 - \frac{r^2}{L^2} \right) dt^2 + \left(1 - \frac{r^2}{L^2} \right)^{-1} dr^2 + r^2 d\Omega}, \quad \text{with } L \equiv \sqrt{3/\Lambda}, \quad (\text{B.181})$$

as required.

Solution to Exercise 5.3

Comparing (5.10) and (5.16), we have

$$X_0 = L \sinh(\tilde{t}/L) + \frac{\tilde{r}^2}{2L} e^{\tilde{t}/L} = \sqrt{L^2 - r^2} \sinh(t/L), \quad (\text{B.182})$$

$$X_1 = L \cosh(\tilde{t}/L) - \frac{\tilde{r}^2}{2L} e^{\tilde{t}/L} = \sqrt{L^2 - r^2} \cosh(t/L), \quad (\text{B.183})$$

$$X_i = e^{\tilde{t}/L} \tilde{x}_i = r \omega_i. \quad (\text{B.184})$$

Using $\tilde{x}_i = \tilde{r} \omega_i$, the last expression implies

$$\boxed{r = \tilde{r} e^{\tilde{t}/L}}, \quad (\text{B.185})$$

as required. Computing $X_0 + X_1$, we find

$$L \underbrace{[\sinh(\tilde{t}/L) + \cosh(\tilde{t}/L)]}_{e^{\tilde{t}/L}} = \sqrt{L^2 - r^2} \underbrace{[\sinh(t/L) + \cosh(t/L)]}_{e^{t/L}}. \quad (\text{B.186})$$

Squaring both sides gives

$$L^2 e^{2\tilde{t}/L} = (L^2 - r^2) e^{2t/L}, \quad (\text{B.187})$$

and substituting (B.185) leads to

$$\boxed{e^{-2t/L} = e^{-2\tilde{t}/L} - \frac{\tilde{r}^2}{L^2}}, \quad (\text{B.188})$$

as required.

Solution to Exercise 5.4

The geodesics of de Sitter can be obtained by minimizing the distance in the embedding space, subject to the embedding constraint $X_A X^A \equiv \eta_{AB} X^A X^B = L^2$. The easiest way to derive this is from the Lagrangian

$$\mathcal{L} = -\frac{1}{2} \dot{X}_A \dot{X}^A + \lambda (X_A X^A - L^2). \quad (\text{B.189})$$

The associated Euler–Lagrange equation is

$$\ddot{X}_A = -2\lambda X_A. \quad (\text{B.190})$$

We therefore have

$$\ddot{X}_A X^A = -2\lambda X_A X^A = -2\lambda L^2, \quad (\text{B.191})$$

$$X_A X^A = L^2. \quad (\text{B.192})$$

Differentiating the second equation twice, we get

$$\ddot{X}_A X^A + \dot{X}_A \dot{X}^A = 0 \quad \Rightarrow \quad \ddot{X}_A X^A = -\delta, \quad (\text{B.193})$$

where $\delta \equiv \dot{X}_A \dot{X}^A = \mp 1$ for timelike/spacelike geodesics, respectively. Equation (B.191) then becomes

$$\lambda = \delta / (2L^2), \quad (\text{B.194})$$

and the equation of motion (B.190) reads

$$\ddot{X}^A = -\frac{\delta}{L^2} X^A. \quad (\text{B.195})$$

Consider first the case of timelike separated points, so that $\delta = -1$. The general solution of (B.195) then is

$$X^A = L(a^A e^{\tau/L} + b^A e^{-\tau/L}), \quad (\text{B.196})$$

where a^A and b^A are constant vectors. Differentiating this solution, we also have

$$\delta = \dot{X}_A \dot{X}^A = a^2 e^{2\tau/L} + b^2 e^{-2\tau/L} - 2a \cdot b = -1, \quad (\text{B.197})$$

so that

$$\begin{aligned} a^2 &= b^2 = 0, \\ 2a \cdot b &= 1. \end{aligned} \quad (\text{B.198})$$

Finally, we can take two points on the geodesic at τ and τ' . We then have

$$\begin{aligned} P(X, X') &= \frac{X \cdot X'}{L^2} = (a \cdot b) \left(e^{(\tau-\tau')/L} + e^{-(\tau-\tau')/L} \right) \\ &= \cosh(d(\tau, \tau')/L), \end{aligned} \quad (\text{B.199})$$

where $d(\tau, \tau') \equiv \tau - \tau'$ is the geodesic distance. Hence, it follows that

$$\boxed{d(x, x') = L \cosh^{-1}(P)}. \quad (\text{B.200})$$

A similar analysis for spacelike-separated points, with $\delta = +1$, gives

$$\boxed{d(x, x') = L \cos^{-1}(P)}. \quad (\text{B.201})$$

Finally, lightlike-separated points trivially have $d(x, x') = 0$.

Solution to Exercise 5.5

Evaluating the action (5.29) in the spacetime (5.15), we get

$$S = \frac{1}{2} \int d\eta d^{D-1}x a^{D-2} \left[(\phi')^2 - (\nabla\phi)^2 - (m^2 + \xi R) a^2 \phi^2 \right]. \quad (\text{B.202})$$

It is convenient to define $f \equiv a^{(D-2)/2}(\eta)\phi$, so that

$$\begin{aligned}
a^{D-2}(\phi')^2 &= a^{D-2}[(a^{-(D-2)/2}f)']^2 \\
&= \left[f' - \frac{D-2}{2} \frac{a'}{a} f \right]^2 \\
&= (f')^2 - (D-2) \frac{a'}{a} f' f + \frac{(D-2)^2}{4} \left(\frac{a'}{a} \right)^2 f^2 \\
&= (f')^2 - \frac{D-2}{2} \frac{a'}{a} (f^2)' + \frac{(D-2)^2}{4} \left(\frac{a'}{a} \right)^2 f^2 \\
&= (f')^2 - \frac{D-2}{2} \left[\frac{a'}{a} f^2 \right]' + \frac{D-2}{2} \left(\frac{a'}{a} \right)' f^2 + \frac{(D-2)^2}{4} \left(\frac{a'}{a} \right)^2 f^2 \\
&= (f')^2 + \frac{D-2}{2} \frac{a''}{a} f^2 + \frac{(D-2)(D-4)}{4} \left(\frac{a'}{a} \right)^2 f^2 - \frac{D-2}{2} \left[\frac{a'}{a} f^2 \right]' \\
&= (f')^2 + \frac{D-2}{2} \left[\frac{a''}{a} + \frac{D-4}{2} \left(\frac{a'}{a} \right)^2 \right] f^2 - \frac{D-2}{2} \left[\frac{a'}{a} f^2 \right]' \\
&= (f')^2 + \frac{D-2}{4(D-1)} a^2 R - \frac{D-2}{2} \left[\frac{a'}{a} f^2 \right]', \tag{B.203}
\end{aligned}$$

where we used that

$$a^2 R = 2(D-1) \left[\frac{a''}{a} + \frac{D-4}{2} \left(\frac{a'}{a} \right)^2 \right] = \frac{D(D-1)}{\eta^2}. \tag{B.204}$$

Dropping the total-derivative term, the action (B.202) becomes

$$S = \frac{1}{2} \int d\eta d^{D-1}x \left[(f')^2 - (\nabla f)^2 - (a^2 m^2 + (\xi - \bar{\xi}) a^2 R) f^2 \right], \tag{B.205}$$

and the equation of motion for the field is

$$\boxed{f'' - \nabla^2 f + \frac{m^2 L^2 + D(D-1)(\xi - \bar{\xi})}{\eta^2} f = 0}, \tag{B.206}$$

as required.

Solution to Exercise 5.6

We consider the Klein–Gordon inner product on the hypersurface $\eta = \text{const}$:

$$\langle \phi_{\mathbf{k}}, \phi_{\mathbf{k}'} \rangle = i \int d^{D-1}x a^{D-2}(\eta) \left[\phi_{\mathbf{k}}^* \partial_{\eta} \phi_{\mathbf{k}'} - \partial_{\eta} \phi_{\mathbf{k}}^* \phi_{\mathbf{k}'} \right], \tag{B.207}$$

where the factor of a^{D-2} comes from $\sqrt{\gamma} = a^{D-1}$ and $\partial_t = a^{-1} \partial_{\eta}$. Using

$$\phi_{\mathbf{k}}(\eta, \mathbf{x}) = \frac{e^{i\mathbf{k} \cdot \mathbf{x}}}{\sqrt{(2\pi)^{D-1}}} a^{(D-2)/2}(\eta) f_{\mathbf{k}}(\eta), \tag{B.208}$$

we have

$$\begin{aligned}\partial_\eta \phi_{\mathbf{k}}(\eta, \mathbf{x}) &= \frac{e^{i\mathbf{k}\cdot\mathbf{x}}}{\sqrt{(2\pi)^{D-1}}} \partial_\eta \left(a^{-(D-2)/2}(\eta) f_k(\eta) \right) \\ &= \frac{e^{i\mathbf{k}\cdot\mathbf{x}}}{\sqrt{(2\pi)^{D-1}}} \left[-\frac{D-2}{2} a^{-D/2} f_k + a^{-(D-2)/2} \partial_\eta f_k \right].\end{aligned}\quad (\text{B.209})$$

The first term in the sum—proportional to f_k —will cancel in the inner product, so we drop it now. Substituting (B.209) into (B.207), we then get

$$\begin{aligned}\langle \phi_{\mathbf{k}}, \phi_{\mathbf{k}'} \rangle &= i \int \frac{d^{D-1}x}{(2\pi)^{D-1}} e^{-i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{x}} \left[f_k^* \partial_\eta f_{k'} - \partial_\eta f_k^* f_{k'} \right] \\ &= i \left[f_k^* \partial_\eta f_k - \partial_\eta f_k^* f_k \right] \delta(\mathbf{k} - \mathbf{k}').\end{aligned}\quad (\text{B.210})$$

The normalization $\langle \phi_{\mathbf{k}}, \phi_{\mathbf{k}'} \rangle = \delta(\mathbf{k} - \mathbf{k}')$ then requires that

$$\boxed{i \left[f_k^* \partial_\eta f_k - \partial_\eta f_k^* f_k \right] = 1}.\quad (\text{B.211})$$

We will substitute

$$\begin{aligned}f_k(\eta) &= \sqrt{k|\eta|} \left[A_k J_\nu(k|\eta|) + B_k Y_\nu(k|\eta|) \right], \\ \partial_\eta f_k(\eta) &= \sqrt{k|\eta|} \left[A_k \partial_\eta J_\nu(k|\eta|) + B_k \partial_\eta Y_\nu(k|\eta|) \right] + \frac{1}{2\eta} f_k(\eta),\end{aligned}\quad (\text{B.212})$$

and their complex conjugates. We will use that $(J_\nu(z))^* = J_{\nu^*}(z^*)$ and $(Y_\nu(z))^* = Y_{\nu^*}(z^*)$. Assuming that both the order of the Bessel function and its argument are real, we have $(J_\nu(k|\eta|))^* = J_\nu(k|\eta|)$ and $(Y_\nu(k|\eta|))^* = Y_\nu(k|\eta|)$. Equation (B.211) then becomes

$$-ik^2\eta (A_k B_k^* - A_k^* B_k) W[J_\nu(k|\eta|), Y_\nu(k|\eta|)] = 1,\quad (\text{B.213})$$

where $W[f, g] = fg' - f'g$ is the Wronskian, and the derivative is with respect to $s = k|\eta|$. Using that

$$W[J_\nu(s), Y_\nu(s)] = \frac{2}{\pi s},\quad (\text{B.214})$$

we find

$$\boxed{A_k B_k^* - A_k^* B_k = -\frac{i\pi}{2k}},\quad (\text{B.215})$$

as required.

Solution to Exercise 5.7

First, we consider

$$\begin{aligned}\nabla^\mu P \nabla_\mu P &= g^{\mu\nu} \nabla_\mu P \nabla_\nu P \\ &= g^{\eta\eta} (\partial_\eta P)^2 + g^{ij} \partial_i P \partial_j P \\ &= -\frac{\eta^2}{L^2} \left[(\partial_\eta P)^2 - (\partial_i P)^2 \right].\end{aligned}\quad (\text{B.216})$$

Substituting

$$P = 1 + \frac{(\eta - \eta')^2 - (\mathbf{x} - \mathbf{x}')^2}{2\eta\eta'}, \quad (\text{B.217})$$

one finds

$$\boxed{\nabla^\mu P \nabla_\mu P = \frac{1 - P^2}{L^2}}, \quad (\text{B.218})$$

as required.

Next, we look at

$$\nabla_\mu \nabla_\nu P = \partial_\mu \partial_\nu P - \Gamma_{\mu\nu}^\lambda \partial_\lambda P. \quad (\text{B.219})$$

We look at the individual components:

- For $\mu = \nu = \eta$, this becomes

$$\begin{aligned} \nabla_\eta \nabla_\eta P &= \partial_\eta^2 P - \Gamma_{\eta\eta}^\lambda \partial_\lambda P \\ &= \partial_\eta^2 P - \Gamma_{\eta\eta}^\eta \partial_\eta P \\ &= \partial_\eta^2 P + \frac{1}{\eta} \partial_\eta P. \end{aligned} \quad (\text{B.220})$$

Substituting (B.217), one finds

$$\nabla_\eta \nabla_\eta P = \frac{P}{\eta^2} = -g_{\eta\eta} \frac{P}{L^2}. \quad (\text{B.221})$$

- Similarly, for $\mu = \eta$ and $\nu = i$, we get

$$\begin{aligned} \nabla_\eta \nabla_i P &= \partial_\eta \partial_i P - \Gamma_{\eta i}^\lambda \partial_\lambda P \\ &= \partial_\eta \partial_i P - \Gamma_{\eta i}^j \partial_j P \\ &= \partial_\eta \partial_i P + \frac{1}{\eta} \partial_i P. \end{aligned} \quad (\text{B.222})$$

Substituting (B.217), one finds

$$\nabla_\eta \nabla_i P = 0 = -g_{\eta i} \frac{P}{L^2}. \quad (\text{B.223})$$

- Similarly, for $\mu = \eta$ and $\nu = i$, we find

$$\begin{aligned} \nabla_i \nabla_j P &= \partial_i \partial_j P - \Gamma_{ij}^\lambda \partial_\lambda P \\ &= \partial_i \partial_j P - \Gamma_{ij}^\eta \partial_\eta P \\ &= \partial_i \partial_j P + \frac{1}{\eta} \partial_\eta P. \end{aligned} \quad (\text{B.224})$$

Substituting (B.217), one finds

$$\nabla_i \nabla_j P = -g_{ij} \frac{P}{L^2} = \begin{cases} 0 & \text{for } i \neq j \\ -P/\eta^2 & \text{for } i = j \end{cases}. \quad (\text{B.225})$$

Combining these results, we get

$$\boxed{\nabla_\mu \nabla_\nu P = -g_{\mu\nu} \frac{P^2}{L^2}}, \quad (\text{B.226})$$

as required.

Solution to Exercise 5.8

In (5.26), we found

$$P = 1 + \delta, \quad \delta \equiv \frac{(\eta - \eta')^2 - (\mathbf{x} - \mathbf{x}')^2}{2\eta\eta'}. \quad (\text{B.227})$$

If two points are close, then $|\delta| \ll 1$, and the geodesic distance between two points separated by a timelike interval is

$$d(x, x') = L \cosh^{-1}(P) = L \cosh^{-1}(1 + \delta) \xrightarrow{\delta \rightarrow 0} L\sqrt{2\delta}. \quad (\text{B.228})$$

We are given that the short-distance limit of the Wightman function in flat space is

$$G(x, x') \approx (-1)^{(D-2)/2} \frac{\Gamma((D-2)/2)}{4\pi^{D/2}} \frac{1}{d(x, x')^{D-2}}, \quad d(x, x') = \sqrt{(x - x')^2}. \quad (\text{B.229})$$

On the other hand, the hypergeometric function satisfies

$${}_2F_1 \left[\begin{matrix} h_+, h_- \\ D/2 \end{matrix} \middle| \frac{1+P}{2} \right] \xrightarrow{\delta \rightarrow 0} \frac{\Gamma(D/2)\Gamma(h_+ + h_- - D/2)}{\Gamma(h_+)\Gamma(h_-)} \left(-\frac{\delta}{2} \right)^{D/2 - h_+ - h_-}. \quad (\text{B.230})$$

Using that $h_+ + h_- = D - 1$, the solution (5.53) can then be written as

$$G(P) \xrightarrow{\delta \rightarrow 0} c_{m,D} (-1)^{(D-2)/2} \frac{\Gamma(D/2)\Gamma((D-2)/2)}{\Gamma(h_+)\Gamma(h_-)} \left(\frac{2L}{d(x, x')} \right)^{D-2}. \quad (\text{B.231})$$

This matches (B.229) if

$$c_{m,D} = \frac{L^{2-D}}{(4\pi)^{D/2}} \frac{\Gamma(h_+)\Gamma(h_-)}{\Gamma(D/2)}, \quad (\text{B.232})$$

as required.

B.6 Exercises in Chapter 6

Solution to Exercise 6.1

We start with a few mathematical preliminaries. Consider a space(time) with coordinates x^μ . A hypersurface Σ is defined the constraint $f(x^\mu) = 0$. The gradient of the constraint function defines a vector orthogonal to Σ . The unit normal vector then is

$$n_\mu = \frac{\epsilon f_{,\mu}}{\sqrt{\epsilon g^{\alpha\beta} f_{,\alpha} f_{,\beta}}}, \quad (\text{B.233})$$

where ϵ is $+1$ if the hypersurface Σ is spacelike and -1 if it is timelike. We will focus on the latter and hence set $\epsilon = -1$. Letting $\tilde{x}^i$ (with $i = 1, 2, 3$) be coordinates on Σ , we can define vectors that are tangential to the hypersurface:

$$t_i^\mu = \frac{\partial x^\mu}{\partial \tilde{x}^i}. \quad (\text{B.234})$$

Note that $n_\mu t_i^\mu = 0$. The *induced metric* (or first fundamental form) on Σ then is

$$h_{ij} = g_{\mu\nu} t_i^\mu t_j^\nu. \quad (\text{B.235})$$

We also define the so-called transverse metric as $h_{\mu\nu} = g_{\mu\nu} - n_\mu n_\nu$, which is related to the induced metric by $h_{ij} = h_{\mu\nu} t_i^\mu t_j^\nu$. The *extrinsic curvature* (or second fundamental form) is

$$K_{ij} = \nabla_\mu n_\nu t_i^\mu t_j^\nu, \quad (\text{B.236})$$

and its trace is

$$\begin{aligned} K &= h^{ij} K_{ij} = h^{ij} t_i^\mu t_j^\nu \nabla_\mu n_\nu \\ &= h^{\mu\nu} \nabla_\mu n_\nu \\ &= (n^\mu n^\nu + g^{\mu\nu}) \nabla_\mu n_\nu = \nabla_\mu n^\mu, \end{aligned} \quad (\text{B.237})$$

where we used $n^\nu \nabla_\mu n_\nu = \frac{1}{2} \nabla_\mu (n^\nu n_\nu) = 0$.

We are now ready to perform the actual computation. Varying the Einstein–Hilbert action, we get

$$\begin{aligned} (16\pi G) \delta I_{\text{EH}} &= - \int_M \delta (g^{\mu\nu} R_{\mu\nu} \sqrt{g}) d^4x \\ &= - \int_M (R_{\mu\nu} \sqrt{g} \delta g^{\mu\nu} + g^{\mu\nu} R_{\mu\nu} \delta \sqrt{g} + \sqrt{g} g^{\mu\nu} \delta R_{\mu\nu}) d^4x \\ &= - \int_M \left(R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R \right) \delta g^{\mu\nu} \sqrt{g} d^4x + \int_M g^{\mu\nu} \delta R_{\mu\nu} \sqrt{g} d^4x. \end{aligned} \quad (\text{B.238})$$

Using the Palatini identity, we have

$$g^{\mu\nu} \delta R_{\mu\nu} = \nabla_\mu \delta V^\mu, \quad \text{where} \quad \delta V^\mu \equiv g^{\alpha\beta} \delta \Gamma_{\alpha\beta}^\mu - g^{\alpha\mu} \delta \Gamma_{\alpha\beta}^\beta. \quad (\text{B.239})$$

To integrate this divergence, we use Stokes theorem:

$$\int_M \nabla_\mu \delta V^\mu \sqrt{g} d^4x = \int_{\partial M} \delta V^\mu d\Sigma_\mu = \int_{\partial M} \delta V^\mu n_\mu \sqrt{h} d^3x. \quad (\text{B.240})$$

We therefore need to evaluate $\delta V^\mu n_\mu$ on δM , keeping in mind that $\delta g_{\mu\nu}|_{\partial M} = \delta g^{\mu\nu}|_{\partial M} = 0$. The variation of the Christoffel symbol is

$$\delta \Gamma_{\alpha\beta}^\mu|_{\partial M} = \frac{1}{2} g^{\mu\nu} (\delta g_{\nu\alpha,\beta} + \delta g_{\nu\beta,\alpha} - \delta g_{\alpha\beta,\nu}). \quad (\text{B.241})$$

Substituting this into (B.239) gives

$$\delta V^\mu|_{\partial M} = g^{\mu\nu} g^{\alpha\beta} (\delta g_{\nu\beta,\alpha} - \delta g_{\alpha\beta,\nu}) \quad (\text{B.242})$$

and hence

$$\begin{aligned} \delta V^\mu n_\mu|_{\partial M} &= n^\mu g^{\alpha\beta} (\delta g_{\mu\beta,\alpha} - \delta g_{\alpha\beta,\mu}) \\ &= n^\mu (h^{\alpha\beta} - n^\alpha n^\beta) (\delta g_{\mu\beta,\alpha} - \delta g_{\alpha\beta,\mu}) \\ &= n^\mu h^{\alpha\beta} (\delta g_{\mu\beta,\alpha} - \delta g_{\alpha\beta,\mu}) - n^\beta [n^\mu n^\alpha (\delta g_{\mu\beta,\alpha} - \delta g_{\alpha\beta,\mu})] \\ &= n^\mu h^{\alpha\beta} (\delta g_{\mu\beta,\alpha} - \delta g_{\alpha\beta,\mu}). \end{aligned} \quad (\text{B.243})$$

Finally, we note that $\delta g_{\mu\nu}|_{\partial M} = 0$ implies $\delta g_{\mu\nu,\alpha} t_i^\alpha|_{\partial M} = 0$, i.e. the tangential derivative on the boundary vanishes. This leads to

$$h^{\alpha\beta} \delta g_{\mu\beta,\alpha}|_{\partial M} = h^{ij} t_i^\alpha t_j^\beta \delta g_{\mu\beta,\alpha}|_{\partial M} = 0 \quad (\text{B.244})$$

and hence

$$\delta V^\mu n_\mu|_{\partial M} = -h^{\alpha\beta} \delta g_{\alpha\beta,\mu} n^\mu. \quad (\text{B.245})$$

The variation of the Einstein–Hilbert action therefore is

$$(16\pi G) \delta I_{\text{EH}} = - \int_M G_{\mu\nu} \delta g^{\mu\nu} \sqrt{g} d^4x + \int_{\partial M} h^{\alpha\beta} \delta g_{\alpha\beta,\mu} n^\mu \sqrt{h} d^3x, \quad (\text{B.246})$$

as we were asked to show.

Solution to Exercise 6.2

The induced metric is fixed on ∂M , so the only quantity that is varied is the trace of the extrinsic curvature, $K = \nabla_\mu n^\mu$. Let us first write this as

$$\begin{aligned} K &= g^{\mu\nu} \nabla_\mu n_\nu \\ &= (h^{\mu\nu} + n^\mu n^\nu) \nabla_\mu n_\nu \\ &= h^{\mu\nu} \nabla_\mu n_\nu \\ &= h^{\mu\nu} (n_{\nu,\mu} - \Gamma_{\nu\mu}^\alpha n_\alpha), \end{aligned} \quad (\text{B.247})$$

where we used $n^\nu \nabla_\mu n_\nu = \frac{1}{2} \nabla_\mu (n^\nu n_\nu) = 0$. The variation of K then is

$$\begin{aligned} \delta K &= -h^{\mu\nu} \delta \Gamma_{\nu\mu}^\alpha n_\alpha \\ &= -\frac{1}{2} h^{\mu\nu} g^{\alpha\beta} (\delta g_{\beta\nu,\mu} + \delta g_{\beta\mu,\nu} - \delta g_{\nu\mu,\beta}) n_\alpha \\ &= -\frac{1}{2} h^{\mu\nu} (\delta g_{\beta\nu,\mu} + \delta g_{\beta\mu,\nu} - \delta g_{\nu\mu,\beta}) n^\beta \\ &= \frac{1}{2} h^{\mu\nu} \delta g_{\nu\mu,\beta} n^\beta, \end{aligned} \quad (\text{B.248})$$

where we have used that the tangential derivatives vanish on ∂M , i.e. $\delta g_{\beta\nu,\mu}|_{\partial M} = \delta g_{\beta\mu,\nu}|_{\partial M} = 0$. The variation of the Gibbons–Hawking–York term then is

$$(16\pi G) \delta I_{\text{GHY}} = - \int_{\partial M} h^{\alpha\beta} \delta g_{\alpha\beta,\mu} n^\mu \sqrt{h} d^3x. \quad (\text{B.249})$$

As required, this precisely cancels the boundary contribution in the variation of the Einstein–Hilbert action found in Exercise 6.1.

Solution to Exercise 6.3

1. Going to imaginary time $\tau = -it$, the Euclidean AdS–Schwarzschild metric is

$$ds^2 = f(r) d\tau^2 + \frac{1}{f(r)} dr^2 + r^2 d\Omega^2. \quad (\text{B.250})$$

The black hole horizon r_+ is given by the largest zero of the function $f(r)$:

$$1 - \frac{2GM}{r_+} + \frac{r_+^2}{L^2} = 0. \quad (\text{B.251})$$

Let us write $r = r_+ + \rho^2$, for $\rho^2 \ll r_+$. Expanding $f(r)$ near $r = r_+$ then gives

$$f(r) = f'(r_+)\rho^2 + \dots, \quad (\text{B.252})$$

where

$$\begin{aligned} f'(r_+) &= \frac{2GM}{r_+^2} + \frac{2r_+}{L^2} = \frac{1}{r_+} \left(\frac{2GM}{r_+} + \frac{2r_+^2}{L^2} \right) \\ &= \frac{1}{r_+} \left(1 + \frac{r_+^2}{L^2} + \frac{2r_+^2}{L^2} \right) \\ &= \frac{1}{r_+ L^2} (L^2 + 3r_+^2), \end{aligned} \quad (\text{B.253})$$

where we used (B.251) to replace $2GM/r_+$ in the second line. Using $dr = 2\rho d\rho$, we then have

$$\begin{aligned} ds^2 &= \frac{L^2 + 3r_+^2}{r_+ L^2} \rho^2 d\tau^2 + \frac{4r_+ L^2}{L^2 + 3r_+^2} d\rho^2 + r_+^2 d\Omega^2 \\ &= \underbrace{d\bar{\rho}^2 + \bar{\rho}^2 \left(\frac{L^2 + 3r_+^2}{2r_+ L^2} \right)^2 d\tau^2}_{\mathbb{R}^2} + \underbrace{r_+^2 d\Omega^2}_{S^2}. \end{aligned} \quad (\text{B.254})$$

We see that this is a smooth solution if τ is a periodic variable with period

$$\boxed{\beta = \frac{4\pi r_+ L^2}{L^2 + 3r_+^2}}, \quad (\text{B.255})$$

as required.

2. In the presence of a cosmological constant, the Einstein–Hilbert action is

$$I_{\text{EH}} = -\frac{1}{16\pi G} \int d^4x \sqrt{g} (R - 2\Lambda). \quad (\text{B.256})$$

Using that $R = 4\Lambda$ (from the Einstein equation), we have

$$I_{\text{EH}} = -\frac{\Lambda}{8\pi G} V = \frac{3}{8\pi G L^2} V, \quad (\text{B.257})$$

where V is the volume of the spacetime. To obtain a finite volume, we truncate the spacetime at $r = \bar{r}$ for some large $\bar{r}$. We then compute the difference of the volume of the truncated black hole solution and the AdS space with the same cutoff and a suitably adjusted period of τ , so that the two solutions have the same boundary metric. In the two

cases, the circumference of the circle parameterized by τ , determined using the metric at $\bar{r}$, is

$$\text{BH : } \beta \sqrt{1 - \frac{2GM}{\bar{r}} + \frac{\bar{r}^2}{L^2}}, \quad (\text{B.258})$$

$$\text{AdS : } \beta' \sqrt{1 + \frac{\bar{r}^2}{L^2}}. \quad (\text{B.259})$$

These are equal if

$$\beta' = \beta \frac{\sqrt{1 - \frac{2GM}{\bar{r}} + \frac{\bar{r}^2}{L^2}}}{\sqrt{1 + \frac{\bar{r}^2}{L^2}}} = \beta \left(1 - \frac{GML^2}{\bar{r}^3} + \dots \right). \quad (\text{B.260})$$

The volumes of the truncated solutions are

$$V(\text{BH}) = \frac{4\pi}{3} \beta (\bar{r}^3 - r_+^3), \quad (\text{B.261})$$

$$V(\text{AdS}) = \frac{4\pi}{3} \beta' \bar{r}^3 = \frac{4\pi}{3} \beta (\bar{r}^3 - GML^2) = \frac{4\pi}{3} \beta \left(\bar{r}^3 - \frac{L^2 r_+}{2} - \frac{r_+^3}{2} \right). \quad (\text{B.262})$$

The regularized action then is

$$\begin{aligned} I_E &= \frac{3}{8\pi GL^2} \Delta V \\ &= \frac{3}{8\pi GL^2} \times \frac{4\pi}{3} \beta \left[\left(\bar{r}^3 - r_+^3 \right) - \left(\bar{r}^3 - \frac{L^2 r_+}{2} - \frac{r_+^3}{2} \right) \right] \\ &= \frac{\beta r_+}{4GL^2} (L^2 - r_+^2). \end{aligned} \quad (\text{B.263})$$

Substituting (B.255), we get

$$I_E = \frac{\pi r_+^2 (L^2 - r_+^2)}{G(L^2 + 3r_+^2)}, \quad (\text{B.264})$$

as required.

3. Using

$$\frac{\partial \beta}{\partial r_+} = \frac{4\pi L^2 (L^2 - 3r_+^2)}{(L^2 + 3r_+^2)^2}, \quad (\text{B.265})$$

we find that β has a maximum at $r_0 = L/\sqrt{3}$. The value at the maximum is

$$\beta_0 \equiv \beta(r_0) = \frac{4\pi r_0 L^2}{L^2 + 3r_0^2} = \frac{2\pi}{\sqrt{3}} L. \quad (\text{B.266})$$

The minimum temperature is

$$T_0 = \frac{1}{\beta_0} = \frac{\sqrt{3}}{2\pi L}. \quad (\text{B.267})$$

For $T > T_0$, there are two black hole solutions, corresponding to the two solutions of

$$r_+^2 - \frac{4\pi}{3} L^2 T r_+ + \frac{L^2}{3} = 0 \quad \Leftrightarrow \quad \beta = \frac{4\pi r_+ L^2}{L^2 + 3r_+^2}. \quad (\text{B.268})$$

Thermodynamic stability requires that the specific heat is positive:

$$\frac{\partial M}{\partial T} > 0 \quad \Rightarrow \quad \frac{\partial M}{\partial \beta} < 0. \quad (\text{B.269})$$

Using (B.251), we have

$$M = \frac{r_+}{2G} \left(1 + \frac{r_+^2}{L^2} \right) \quad \Rightarrow \quad \frac{\partial M}{\partial r_+} > 0. \quad (\text{B.270})$$

We therefore require

$$\frac{\partial M}{\partial \beta} < 0 \quad \Rightarrow \quad \frac{\partial \beta}{\partial r_+} < 0. \quad (\text{B.271})$$

This is satisfied if

$$\boxed{r_+ > r_0 = \frac{L}{\sqrt{3}}}. \quad (\text{B.272})$$

Small black holes (with $r_+ < r_0$) are unstable, while large black holes (with $r_+ > r_0$) are stable.

B.7 Exercises in Chapter 7

Solution to Exercise 7.1

[DB: include]

Solution to Exercise 7.2

To compute the transformation (7.44), we require the Jacobian matrix

$$\frac{\partial \tilde{x}^\alpha}{\partial x^\mu} = \begin{pmatrix} \partial \tilde{\eta} / \partial \eta & \partial \tilde{\eta} / \partial x^j \\ \partial \tilde{x}^i / \partial \eta & \partial \tilde{x}^i / \partial x^j \end{pmatrix} = \begin{pmatrix} 1 + T' & \partial_j T \\ L'^i & \delta_j^i + \partial_j L^i \end{pmatrix}. \quad (\text{B.273})$$

We will apply this to the individual components of the metric (7.35).

- Consider $\mu = \nu = 0$ in (7.44):

$$g_{00}(x) = \frac{\partial \tilde{x}^\alpha}{\partial \eta} \frac{\partial \tilde{x}^\beta}{\partial \eta} \tilde{g}_{\alpha\beta}(\tilde{x}). \quad (\text{B.274})$$

The only term that contributes to the right-hand side is the one with $\alpha = \beta = 0$. Consider for example $\alpha = 0$ and $\beta = i$. The off-diagonal component of the metric $\tilde{g}_{0i}$ is proportional to $\tilde{B}_i$, so it is a first-order perturbation. However, $\partial \tilde{x}^i / \partial \eta$ is proportional to the first-order variable ξ^i , so the product is second order and can be neglected. A similar argument holds for $\alpha = i$ and $\beta = j$. Equation (B.274) therefore reduces to

$$g_{00}(x) = \left(\frac{\partial \tilde{\eta}}{\partial \eta} \right)^2 \tilde{g}_{00}(\tilde{x}). \quad (\text{B.275})$$

Substituting (7.35) on both sides, we get

$$\begin{aligned}
a^2(\eta)(1+2A) &= (1+T')^2 a^2(\eta+T)(1+2\tilde{A}) \\
&= (1+2T'+\dots)(a(\eta)+a'T+\dots)^2(1+2\tilde{A}) \\
&= a^2(\eta)(1+2\mathcal{H}T+2T'+2\tilde{A}+\dots).
\end{aligned} \tag{B.276}$$

Hence, we find that, at first order, the metric perturbation A transforms as

$$\boxed{A \mapsto \tilde{A} = A - T' - \mathcal{H}T}, \tag{B.277}$$

which is indeed the result quoted in (7.45).

- The transformation of the $0i$ -component of the metric is

$$\begin{aligned}
g_{0i}(x) &= \frac{\partial \tilde{x}^\alpha}{\partial \eta} \frac{\partial \tilde{x}^\beta}{\partial x^i} \tilde{g}_{\alpha\beta}(\tilde{x}) \\
&= \frac{\partial \tilde{\eta}}{\partial \eta} \left(\frac{\partial \tilde{\eta}}{\partial x^i} \tilde{g}_{00}(\tilde{x}) + \frac{\partial \tilde{x}^j}{\partial x^i} \tilde{g}_{0j}(\tilde{x}) \right) + \frac{\partial \tilde{x}^j}{\partial \eta} \left(\frac{\partial \tilde{\eta}}{\partial x^i} \tilde{g}_{j0}(\tilde{x}) + \frac{\partial \tilde{x}^k}{\partial x^i} \tilde{g}_{jk}(\tilde{x}) \right) \\
&= (1+T') \left(\partial_i T \tilde{g}_{00}(\tilde{x}) + (\delta_i^j + \partial_i L^j) \tilde{g}_{0j}(\tilde{x}) \right) + L^{j'} \left(\partial_i T \tilde{g}_{j0}(\tilde{x}) + (\delta_i^k + \partial_i L^k) \tilde{g}_{jk}(\tilde{x}) \right) \\
&= \tilde{g}_{0i}(x) + \partial_i T \tilde{g}_{00}(x) + L^{j'} \tilde{g}_{ji}(x) + O(2).
\end{aligned} \tag{B.278}$$

Substituting the perturbed metric components on both sides, we get

$$B_i = \tilde{B}_i - \partial_i T + L'_i, \tag{B.279}$$

and hence find

$$\boxed{B_i \mapsto \tilde{B}_i = B_i + \partial_i T - L'_i}. \tag{B.280}$$

The scalar and vector parts of B_i then transform as

$$\begin{aligned}
\tilde{B} &\mapsto B + T - L', \\
\hat{B}_i &\mapsto \hat{B}_i - \hat{L}'_i,
\end{aligned} \tag{B.281}$$

as required.

- Similarly, the transformation of the ij -component of the metric is

$$\begin{aligned}
g_{ij}(x) &= \frac{\partial \tilde{x}^\alpha}{\partial x^i} \frac{\partial \tilde{x}^\beta}{\partial x^j} \tilde{g}_{\alpha\beta}(\tilde{x}) \\
&= \frac{\partial \tilde{x}^k}{\partial x^i} \frac{\partial \tilde{x}^l}{\partial x^j} \tilde{g}_{kl}(\tilde{x}) + O(2) \\
&= (\delta_i^k + \partial_i L^k)(\delta_j^l + \partial_j L^l) \tilde{g}_{kl}(\tilde{x}) \\
&= \tilde{g}_{ij}(\tilde{x}) + \partial_i L^k \tilde{g}_{kj}(\tilde{x}) + \partial_j L^l \tilde{g}_{il}(\tilde{x}).
\end{aligned} \tag{B.282}$$

Substituting the perturbed metric components on both sides, this gives

$$\begin{aligned}
a^2(\delta_{ij} + 2E_{ij}) &= a^2(\eta + T)(\delta_{ij} + 2\tilde{E}_{ij}) + a^2(\eta)(\partial_i L^k \delta_{kj} + \partial_j L^i \delta_{il}) \\
&= a^2(1 + 2\mathcal{H}T)(\delta_{ij} + 2\tilde{E}_{ij}) + 2a^2\partial_{(i} L_{j)} \\
&= a^2(\delta_{ij} + 2\tilde{E}_{ij}) + 2a^2\mathcal{H}T \delta_{ij} + 2a^2\partial_{(i} L_{j)}.
\end{aligned} \tag{B.283}$$

Hence, we find

$$\boxed{E_{ij} \mapsto \tilde{E}_{ij} = E_{ij} - \mathcal{H}T\delta_{ij} - \partial_{(i} L_{j)}}. \tag{B.284}$$

In terms of the SVT decomposition, this implies

$$\begin{aligned}
C &\mapsto C - \mathcal{H}T - \frac{1}{3}\nabla^2 L, \\
E &\mapsto E - L, \\
\hat{E}_i &\mapsto \hat{E}_i - \hat{L}_i, \\
\hat{E}_{ij} &\mapsto \hat{E}_{ij},
\end{aligned} \tag{B.285}$$

as required.

Solution to Exercise 7.3

Under a coordinate transformation, scalar metric perturbations change as follows

$$\begin{aligned}
A &\mapsto A - T' - \mathcal{H}T, & B &\mapsto B + T - L', \\
C &\mapsto C - \mathcal{H}T - \frac{1}{3}\nabla^2 L, & E &\mapsto E - L.
\end{aligned} \tag{B.286}$$

This implies that

$$\begin{aligned}
B - E' &\mapsto B + T - L' - (E' - L') \\
&= B - E' + T.
\end{aligned} \tag{B.287}$$

Hence, we have

$$\begin{aligned}
\Psi &\equiv A + \mathcal{H}(B - E') + (B - E')' \\
&\mapsto A - T' - \mathcal{H}T + \mathcal{H}(B - E' + T) + (B - E' + T)' \\
&= A + \mathcal{H}(B - E') + (B - E')' = \Psi.
\end{aligned} \tag{B.288}$$

The Bardeen variable Ψ therefore doesn't change under a coordinate transformation.

Similarly, the second Bardeen variable Φ transforms as

$$\begin{aligned}
\Phi &\equiv -C + \frac{1}{3}\nabla^2 E - \mathcal{H}(B - E') \\
&\mapsto -C + \mathcal{H}T + \frac{1}{3}\nabla^2 L + \frac{1}{3}\nabla^2(E - L) - \mathcal{H}(B - E' + T) \\
&= -C + \frac{1}{3}\nabla^2 E - \mathcal{H}(B - E') = \Phi.
\end{aligned} \tag{B.289}$$

The second Bardeen variable Ψ therefore also doesn't change.

Solution to Exercise 7.4

To evaluate (7.57), we require $\partial x^\mu / \partial \tilde{x}^\alpha$ which, considered as a matrix, is the inverse of $\partial \tilde{x}^\alpha / \partial x^\mu$, since

$$\frac{\partial x^\mu}{\partial \tilde{x}^\alpha} \frac{\partial \tilde{x}^\alpha}{\partial x^\nu} = \delta_\nu^\mu. \quad (\text{B.290})$$

For the transformations in (7.42), the matrix $\partial \tilde{x}^\alpha / \partial x^\mu$ was given by (B.273):

$$\frac{\partial \tilde{x}^\alpha}{\partial x^\mu} = \begin{pmatrix} \partial \tilde{\eta} / \partial \eta & \partial \tilde{\eta} / \partial x^j \\ \partial \tilde{x}^i / \partial \eta & \partial \tilde{x}^i / \partial x^j \end{pmatrix} = \begin{pmatrix} 1 + T' & \partial_j T \\ L^{i'} & \delta_j^i + \partial_j L^i \end{pmatrix}. \quad (\text{B.291})$$

To determine the corresponding inverse matrix, we make use of the fact that the inverse of a matrix of the form $\mathbf{1} + \varepsilon$, where $\mathbf{1}$ is the identity and ε is a small perturbation, is $\mathbf{1} - \varepsilon$ to first order in ε . It follows that

$$\frac{\partial x^\mu}{\partial \tilde{x}^\alpha} = \begin{pmatrix} \partial \eta / \partial \tilde{\eta} & \partial \eta / \partial \tilde{x}^j \\ \partial x^i / \partial \tilde{\eta} & \partial x^i / \partial \tilde{x}^j \end{pmatrix} = \begin{pmatrix} 1 - T' & -\partial_j T \\ -L^{i'} & \delta_j^i - \partial_j L^i \end{pmatrix}. \quad (\text{B.292})$$

Substituting these results into (7.57), we can then determine how the perturbations of the energy-momentum tensor transform.

- Applying (7.57) to the temporal component of the energy-momentum tensor, we get

$$\begin{aligned} T^0_0(x) &= \frac{\partial \eta}{\partial \tilde{x}^\alpha} \frac{\partial \tilde{x}^\beta}{\partial \eta} \tilde{T}^\alpha_\beta(\tilde{x}) \\ &= \frac{\partial \eta}{\partial \tilde{\eta}} \frac{\partial \tilde{\eta}}{\partial \eta} \tilde{T}^0_0(\tilde{x}) + O(2). \end{aligned} \quad (\text{B.293})$$

Substituting $T^0_0 = -(\bar{\rho} + \delta\rho)$ on both sides, we get

$$\begin{aligned} \bar{\rho} + \delta\rho &= (1 - T')(1 + T')(\bar{\rho}(\eta + T) + \delta\tilde{\rho}(x)) \\ &= \bar{\rho}(\eta) + \bar{\rho}'T + \delta\tilde{\rho}(x) + O(2). \end{aligned} \quad (\text{B.294})$$

Solving this for $\delta\tilde{\rho}$ gives

$$\boxed{\delta\rho \mapsto \delta\tilde{\rho} = \delta\rho - \bar{\rho}'T}, \quad (\text{B.295})$$

as expected.

- Applying this to the ij -components, we get

$$\begin{aligned} T^i_j(x) &= \frac{\partial x^i}{\partial \tilde{x}^\alpha} \frac{\partial \tilde{x}^\beta}{\partial x^j} \tilde{T}^\alpha_\beta(\tilde{x}) \\ &= \frac{\partial x^i}{\partial \tilde{x}^l} \frac{\partial \tilde{x}^m}{\partial x^j} \tilde{T}^l_m(\tilde{x}) \\ &= (\delta_l^i - \partial_l L^i) (\delta_j^m + \partial_j L^m) \tilde{T}^l_m(\tilde{x}), \end{aligned} \quad (\text{B.296})$$

and substituting $T^i_j = (\bar{P} + \delta P) \delta_j^i$ on both sides gives

$$\begin{aligned} (\bar{P} + \delta P) \delta_j^i &= (\delta_l^i - \partial_l L^i) (\delta_j^m + \partial_j L^m) (\bar{P}(\eta + T) + \delta\tilde{P}) \delta_m^l \\ &= (\bar{P}(\eta) + \bar{P}'T + \delta\tilde{P}) \delta_j^i + O(2). \end{aligned} \quad (\text{B.297})$$

Solving this for $\delta\tilde{P}$, we find the transformation of the pressure perturbation

$$\boxed{\delta P \mapsto \delta\tilde{P} = \delta P - \bar{P}'T} , \quad (\text{B.298})$$

which agrees with the result quoted in the text.

Substituting $T^i_j = \Pi^i_j$ on both sides of (B.296), we get

$$\begin{aligned} \Pi^i_j &= (\delta^i_l - \partial_l L^i) (\delta^m_j + \partial_j L^m) \tilde{\Pi}^l_m \\ &= \tilde{\Pi}^i_j + O(2) , \end{aligned} \quad (\text{B.299})$$

i.e. the anisotropic stress does not transform at first order.

- Finally, we consider the $i0$ -component:

$$\begin{aligned} T^i_0(x) &= \frac{\partial x^i}{\partial \tilde{x}^\alpha} \frac{\partial \tilde{x}^\beta}{\partial x^0} \tilde{T}^\alpha_\beta(\tilde{x}) \\ &= \frac{\partial x^i}{\partial \tilde{\eta}} \left(\frac{\partial \tilde{\eta}}{\partial \eta} \tilde{T}^0_0 + \frac{\partial \tilde{x}^k}{\partial \eta} \tilde{T}^0_k \right) + \frac{\partial x^i}{\partial \tilde{x}^j} \left(\frac{\partial \tilde{\eta}}{\partial \eta} \tilde{T}^j_0 + \frac{\partial \tilde{x}^k}{\partial \eta} \tilde{T}^j_k \right) \\ &= L^{i'} \bar{\rho} + \tilde{T}^i_0 + L^{j'} T^i_j + O(2) \\ &= \tilde{T}^i_0(x) + (\bar{\rho} + \bar{P}) L^{i'} . \end{aligned} \quad (\text{B.300})$$

Solving this for $\tilde{T}^i_0 \equiv -(\bar{\rho} + \bar{P})\tilde{v}^i$ gives

$$\boxed{v^i \mapsto \tilde{v}^i = v^i + L^{i'}} , \quad (\text{B.301})$$

which agrees with the result quoted in the text.

Solution to Exercise 7.5

The induced metric on surfaces of constant time is

$$\gamma_{ij} = a^2 \left[(1 + 2C) \delta_{ij} + 2E_{ij} \right] . \quad (\text{B.302})$$

The associated connection is

$$\Gamma^i_{jk} = \frac{1}{2} \gamma^{il} (\partial_j \gamma_{kl} + \partial_k \gamma_{jl} - \partial_l \gamma_{jk}) , \quad (\text{B.303})$$

where γ^{ij} is the inverse of the induced metric. We only need the inverse metric at zeroth order, $\gamma^{ij} = a^{-2} \delta^{ij}$, since it is contracted with spatial derivatives of the metric (which are first order in perturbations). We therefore have

$$\begin{aligned} \Gamma^i_{jk} &= \delta^{il} \partial_j (C \delta_{kl} + E_{kl}) + \delta^{il} \partial_k (C \delta_{jl} + E_{jl}) - \delta^{il} \partial_l (C \delta_{jk} + E_{jk}) \\ &= (2\delta^i_{(j} \partial_{k)} C - \delta_{jk} \partial^i C) + (2\partial_{(j} E_{k)}^i - \partial^i E_{jk}) . \end{aligned} \quad (\text{B.304})$$

The Ricci scalar of the spatial slice is

$$\begin{aligned} R_{(3)} &= \gamma^{ik} R_{ik} \\ &= \gamma^{ik} \left(\partial_l \Gamma^l_{ik} - \partial_k \Gamma^l_{il} + \Gamma^l_{ik} \Gamma^m_{lm} - \Gamma^m_{il} \Gamma^l_{km} \right) . \end{aligned} \quad (\text{B.305})$$

At first order, this reduces to

$$a^2 R_{(3)} = \delta^{ik} \partial_l \Gamma_{ik}^l - \delta^{ik} \partial_k \Gamma_{il}^l. \quad (\text{B.306})$$

The first term on the right-hand side is

$$\begin{aligned} \delta^{ik} \partial_l \Gamma_{ik}^l &= \delta^{ik} \partial_l (2\delta_{(i}^l \partial_{k)} C - \delta_{ik} \partial^l C) + \delta^{ik} \partial_l (2\partial_{(i} E_{k)}^l - \partial^l E_{ik}) \\ &= 2\delta^{ik} \partial_i \partial_k C - 3\partial_l \partial^l C + 2\partial_l \partial_k E^{kl} - \partial_l \partial^l \underbrace{(\delta^{ik} E_{ik})}_0 \\ &= -\nabla^2 C + 2\partial_l \partial_k E^{kl}. \end{aligned} \quad (\text{B.307})$$

To evaluate the second term, we need

$$\begin{aligned} \Gamma_{il}^l &= \delta_i^l \partial_l C + \delta_i^l \partial_l C - \partial_i C + \underbrace{\partial_l E_i^l + \partial_i E_l^l - \partial_l E_i^l}_0 \\ &= 3\partial_i C. \end{aligned} \quad (\text{B.308})$$

Substituting these expressions into (B.306), we get

$$\begin{aligned} a^2 R_{(3)} &= -\nabla^2 C + 2\partial_i \partial_j E^{ij} - 3\nabla^2 C \\ &= -4\nabla^2 C + 2\partial_i \partial_j E^{ij}. \end{aligned} \quad (\text{B.309})$$

Vector and tensor perturbations don't contribute to $\partial_i \partial_j E^{ij}$. Substituting the scalar part of E^{ij} , we get

$$\begin{aligned} \partial_i \partial_j E^{ij} &= \partial_i \partial_j \left(\partial^i \partial^j E - \frac{1}{3} \delta^{ij} \nabla^2 E \right) \\ &= \nabla^2 \nabla^2 E - \frac{1}{3} \nabla^2 \nabla^2 E = \frac{2}{3} \nabla^4 E. \end{aligned} \quad (\text{B.310})$$

Hence, we find

$$\boxed{a^2 R_{(3)} = 4\nabla^2 \left(-C + \frac{1}{3} \nabla^2 E \right)}, \quad (\text{B.311})$$

which agrees with the quoted result.

Solution to Exercise 7.6

The given scalar field action is

$$S = \int d^4x \sqrt{-g} \left(-\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) \right), \quad (\text{B.312})$$

where $V(\phi)$ is the potential.

1. Varying the action with respect to the (inverse) metric, we get

$$\begin{aligned} \delta S &= \int d^4x \left[\delta \sqrt{-g} \left(-\frac{1}{2} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi - V(\phi) \right) - \sqrt{-g} \frac{1}{2} \delta g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi \right] \\ &= \frac{1}{2} \int d^4x \sqrt{-g} \left[g_{\mu\nu} \left(\frac{1}{2} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi + V(\phi) \right) - \partial_\mu \phi \partial_\nu \phi \right] \delta g^{\mu\nu}, \end{aligned} \quad (\text{B.313})$$

where we have used $\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}g_{\mu\nu}\delta g^{\mu\nu}$. Comparing this to the defining expression for the energy-momentum tensor

$$\delta S \equiv -\frac{1}{2} \int d^4x \sqrt{-g} T_{\mu\nu} \delta g^{\mu\nu}, \quad (\text{B.314})$$

we get

$$T_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi - g_{\mu\nu} \left(\frac{1}{2} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi + V(\phi) \right). \quad (\text{B.315})$$

2. For a homogeneous field configuration $\phi = \bar{\phi}(t)$, this indeed takes the form of a perfect fluid

$$\bar{T}^\mu{}_\nu = \begin{pmatrix} -\bar{\rho}_\phi & 0 & 0 & 0 \\ 0 & \bar{P}_\phi & 0 & 0 \\ 0 & 0 & \bar{P}_\phi & 0 \\ 0 & 0 & 0 & \bar{P}_\phi \end{pmatrix}, \quad (\text{B.316})$$

with

$$\bar{\rho}_\phi = -\bar{T}^0{}_0 = -\bar{g}^{00} \partial_0 \bar{\phi} \partial_0 \bar{\phi} + \delta_0^0 \left(\frac{1}{2} \bar{g}^{00} \partial_0 \bar{\phi} \partial_0 \bar{\phi} + V(\bar{\phi}) \right) = \boxed{\frac{1}{2a^2} (\bar{\phi}')^2 + V(\bar{\phi})}, \quad (\text{B.317})$$

$$\bar{P}_\phi = \frac{1}{3} \bar{T}^i{}_i = -\frac{1}{3} \delta_i^i \left(\frac{1}{2} \bar{g}^{00} \partial_0 \bar{\phi} \partial_0 \bar{\phi} + V(\bar{\phi}) \right) = \boxed{\frac{1}{2a^2} (\bar{\phi}')^2 - V(\bar{\phi})}. \quad (\text{B.318})$$

3. From its definition in (7.63), we infer that the comoving curvature perturbation in the spatially flat gauge is

$$\zeta_c = \mathcal{H}(v + B). \quad (\text{B.319})$$

Using (7.56), we have

$$T^0{}_i = (\bar{\rho} + \bar{P}) \partial_i(v + B) = \frac{(\bar{\phi}')^2}{a^2} \partial_i(v + B), \quad (\text{B.320})$$

which we can compute for the scalar field and then use in (B.319).

Given (B.315), we have

$$\begin{aligned} T^0{}_i &= g^{0\mu} T_{\mu i} \\ &= g^{0\mu} \left[\partial_\mu \phi \partial_\nu \phi - g_{\mu\nu} \left(\frac{1}{2} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi + V(\phi) \right) \right] \\ &= \bar{g}^{00} \partial_0 \bar{\phi} \partial_i \delta\phi + \delta_i^0 \left(\frac{1}{2} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi + V(\phi) \right) \\ &= -\frac{1}{a^2} \bar{\phi}' \partial_i \delta\phi, \end{aligned} \quad (\text{B.321})$$

so that (B.320) implies

$$v + B = -\frac{\delta\phi}{\bar{\phi}'}. \quad (\text{B.322})$$

Substituting this into (B.319), we then get

$$\boxed{\zeta_c = -\frac{\mathcal{H}}{\bar{\phi}'} \delta\phi}, \quad (\text{B.323})$$

as desired.

B.8 Exercises in Chapter 8

Solution to Exercise 8.1

The covariant derivative of the energy-momentum tensor (8.2) is

$$\begin{aligned}\nabla_\mu T^{\mu\nu} &= g^{\mu\alpha} g^{\nu\beta} \nabla_\mu (\partial_\alpha \phi \partial_\beta \phi) - g^{\mu\nu} \nabla_\mu \left(\frac{1}{2} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi + V(\phi) \right) \\ &= g^{\mu\alpha} g^{\nu\beta} \left[\partial_\mu (\partial_\alpha \phi \partial_\beta \phi) - \left(\Gamma_{\mu\alpha}^\rho \partial_\beta \phi + \Gamma_{\mu\beta}^\rho \partial_\alpha \phi \right) \partial_\rho \phi \right] \\ &\quad - g^{\mu\nu} \partial_\mu \left(\frac{1}{2} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi + V(\phi) \right).\end{aligned}\tag{B.324}$$

Using that $g^{0i} = 0$ and $\partial_i \phi = 0$, this expression simplifies to

$$\nabla_\mu T^{\mu\nu} = \delta^{\nu 0} \partial_0 \dot{\phi}^2 + \Gamma_{\mu\alpha}^0 g^{\mu\alpha} \delta^{\nu 0} \dot{\phi}^2 + \Gamma_{0\beta}^0 g^{\nu\beta} \dot{\phi}^2 + \delta^{0\nu} \partial_0 \left(-\frac{1}{2} \dot{\phi}^2 + V(\phi) \right).\tag{B.325}$$

To get all the signs right, you have to remember that $g^{00} = -1$. The relevant Christoffel symbols for the FRW metric are $\Gamma_{0\nu}^0 = 0$ and $\Gamma_{ij}^0 = H g_{ij}$, so that $\Gamma_{\mu\alpha}^0 g^{\mu\alpha} = \Gamma_{ij}^0 g^{ij} = H g_{ij} g^{ij} = 3H$. We therefore get

$$\begin{aligned}\nabla_\mu T^{\mu\nu} &= \delta^{\nu 0} \left(2\dot{\phi}\ddot{\phi} + 3H\dot{\phi}^2 + \left(-\dot{\phi}\ddot{\phi} + \dot{\phi} \frac{dV}{d\phi} \right) \right) \\ &= \delta^{\nu 0} \dot{\phi} \left(\ddot{\phi} + 3H\dot{\phi} + \frac{dV}{d\phi} \right).\end{aligned}\tag{B.326}$$

The conservation of the energy-momentum tensor, $\nabla_\mu T^{\mu\nu} = 0$, then implies the Klein–Gordon equation

$$\boxed{\ddot{\phi} + 3H\dot{\phi} + \frac{dV}{d\phi} = 0},\tag{B.327}$$

as expected.

Solution to Exercise 8.2

Taking the time derivative of the Friedmann equation (8.4) gives

$$2H\dot{H} = \frac{1}{3M_{\text{Pl}}^2} \dot{\phi} \left(\ddot{\phi} + \frac{dV}{d\phi} \right) = \frac{1}{3M_{\text{Pl}}^2} \dot{\phi} (-3H\dot{\phi}) = -\frac{H\dot{\phi}^2}{M_{\text{Pl}}^2},\tag{B.328}$$

where we have used the Klein–Gordon equation (8.5). Substituting this into (8.8), we find

$$\varepsilon = -\frac{\dot{H}}{H^2} = \frac{\frac{1}{2}\dot{\phi}^2}{M_{\text{Pl}}^2 H^2} = \boxed{\frac{\frac{3}{2}\dot{\phi}^2}{\frac{1}{2}\dot{\phi}^2 + V}},\tag{B.329}$$

as required. Inflation ($\varepsilon < 1$) occurs when $\dot{\phi}^2 < V$.

Solution to Exercise 8.3

In the slow-roll approximation, the equations of motion are

$$\begin{aligned} H^2 &= \frac{1}{3M_{\text{Pl}}^2} \left(\frac{1}{2} \dot{\phi}^2 + V \right) & H^2 &\approx \frac{V}{3M_{\text{Pl}}^2}, \\ \ddot{\phi} + 3H\dot{\phi} &= -\frac{dV}{d\phi} & 3H\dot{\phi} &\approx -\frac{dV}{d\phi}. \end{aligned} \quad (\text{B.330})$$

The slow-roll parameter ε then becomes

$$\varepsilon = \frac{\frac{1}{2} \dot{\phi}^2}{M_{\text{Pl}}^2 H^2} = \frac{\frac{1}{2} (V_{,\phi})^2}{M_{\text{Pl}}^2 (3H^2)^2} = \boxed{\frac{M_{\text{Pl}}^2}{2} \left(\frac{V_{,\phi}}{V} \right)^2}, \quad (\text{B.331})$$

as required.

Taking the time derivative of $3H\dot{\phi} = -V_{,\phi}$, we get

$$3\dot{H}\dot{\phi} + 3H\ddot{\phi} = -V_{,\phi\phi}\dot{\phi}, \quad (\text{B.332})$$

and dividing this by $(-3H^2\dot{\phi})$ leads to

$$\varepsilon + \delta = \frac{V_{,\phi\phi}}{3H^2} \approx \boxed{M_{\text{Pl}}^2 \frac{V_{,\phi\phi}}{V}}, \quad (\text{B.333})$$

as required.

Solution to Exercise 8.4

We begin with the ansatz

$$v_k(\eta) = \alpha_k u_k(\eta) + \beta_k u_k^*(\eta). \quad (\text{B.334})$$

The Wronskian of this function is

$$\begin{aligned} &v_k(\eta) \partial_\eta v_k^*(\eta) - v_k^*(\eta) \partial_\eta v_k(\eta) \\ &= [\alpha_k u_k(\eta) + \beta_k u_k^*(\eta)] \partial_\eta [\alpha_k^* u_k^*(\eta) + \beta_k^* u_k(\eta)] - [\alpha_k^* u_k^*(\eta) + \beta_k^* u_k(\eta)] \partial_\eta [\alpha_k u_k(\eta) + \beta_k u_k^*(\eta)] \\ &= [|\alpha_k|^2 - |\beta_k|^2] [u_k(\eta) \partial_\eta u_k^*(\eta) - u_k^*(\eta) \partial_\eta u_k(\eta)] \\ &= i[|\alpha_k|^2 - |\beta_k|^2]. \end{aligned} \quad (\text{B.335})$$

This satisfies (8.24) if

$$\boxed{|\alpha_k|^2 - |\beta_k|^2 = 1}, \quad (\text{B.336})$$

as required.

Solution to Exercise 8.5

The de Sitter metric is invariant under the following dilatation symmetry:

$$\begin{aligned}\eta &\rightarrow \lambda\eta, \\ \mathbf{x} &\rightarrow \lambda\mathbf{x},\end{aligned}\tag{B.337}$$

where λ is a constant parameter. It is easy to see that the action of a free massive scalar field is invariant under this transformation,

$$S = \frac{1}{2} \int \frac{d\eta d^3x}{H^2 \eta^2} \left[(\phi')^2 - (\nabla\phi)^2 - \frac{m^2}{H^2 \eta^2} \phi^2 \right],\tag{B.338}$$

if the field $\phi(\eta, \mathbf{x})$ (in real space) doesn't transform. Since

$$\phi(\eta, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \phi_{\mathbf{k}}(\eta) e^{i\mathbf{k}\cdot\mathbf{x}},\tag{B.339}$$

this implies that

$$\phi_{\mathbf{k}} \rightarrow \lambda^3 \phi_{\mathbf{k}} \quad \text{and} \quad \mathbf{k} \rightarrow \lambda^{-1} \mathbf{k}.\tag{B.340}$$

The metric is also invariant under spatial translations and so momentum ($\mathbf{k}$) is conserved. The most general two-point correlation function that respects both of these symmetries is

$$\langle \phi_{\mathbf{k}}(\eta) \phi_{\mathbf{k}'}(\eta) \rangle = (2\pi)^3 \delta(\mathbf{k} + \mathbf{k}') \frac{F(k\eta)}{k^3},\tag{B.341}$$

where we used that $\delta(\lambda^{-1}\mathbf{k}) = \lambda^3 \delta(\mathbf{k})$. A massless field freezes outside the horizon and hence the late-time power spectrum must be time independent. The function $F(k\eta)$ must therefore be a constant and we get

$$\boxed{P_\phi(k) \propto \frac{1}{k^3}},\tag{B.342}$$

as required.

A massive field satisfies the equation of motion

$$\phi_{\mathbf{k}}'' - \frac{2}{\eta} \phi_{\mathbf{k}}' + \left(k^2 + \frac{m^2}{H^2 \eta^2} \right) \phi_{\mathbf{k}} = 0.\tag{B.343}$$

At late times, the mode is outside the horizon, $|k\eta| \ll 1$, and we get

$$\phi_{\mathbf{k}}'' - \frac{2}{\eta} \phi_{\mathbf{k}}' + \frac{m^2}{H^2 \eta^2} \phi_{\mathbf{k}} = 0.\tag{B.344}$$

This is solved by a power law, $\phi_{\mathbf{k}}(\eta) \sim \eta^\alpha$, with

$$\alpha_{\pm} = \frac{3}{2} \pm \sqrt{\frac{9}{4} - \frac{m^2}{H^2}}.\tag{B.345}$$

This implies that (B.341) takes the form

$$\langle \phi_{\mathbf{k}}(\eta) \phi_{\mathbf{k}'}(\eta) \rangle = (2\pi)^3 \delta(\mathbf{k} + \mathbf{k}') \frac{(-k\eta)^{2\alpha_-}}{k^3},\tag{B.346}$$

where we used that $(-k\eta)^{\alpha_+} \ll (-k\eta)^{\alpha_-}$ for small $k\eta$. For $m^2 \ll H^2$, we have $\alpha_- \approx m^2/(3H^2)$ and the power spectrum scales as

$$P_\phi(k) \propto k^{n-4}, \quad \text{with} \quad \boxed{n = 1 + \frac{2m^2}{3H^2}}.\tag{B.347}$$

Solution to Exercise 8.6

The action for a massive scalar field in an FRW background is

$$S = \int d^4x \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} m^2 \phi^2 \right] \quad (\text{B.348})$$

$$= \frac{1}{2} \int d\eta d^3x \left[a^2(\eta) ((\phi')^2 - (\nabla \phi)^2) - m^2 a^4(\eta) \phi^2 \right]. \quad (\text{B.349})$$

Introducing the canonically-normalized field $f \equiv a\phi$, and substituting $a(\eta) = -(H\eta)^{-1}$, this becomes

$$S = \frac{1}{2} \int d\eta d^3x \left[(f')^2 - (\nabla f)^2 - \left(m^2 a^2 - \frac{a''}{a} \right) f^2 \right] \quad (\text{B.350})$$

$$= \frac{1}{2} \int d\eta d^3x \left[(f')^2 - (\nabla f)^2 - \frac{m^2/H^2 - 2}{\eta^2} f^2 \right]. \quad (\text{B.351})$$

The equation of motion (in Fourier space) then is

$$f_k'' + \left(k^2 + \frac{m^2/H^2 - 2}{\eta^2} \right) f_k = 0. \quad (\text{B.352})$$

Defining $x \equiv -k\eta$, this becomes

$$x^2 \frac{d^2 f_k}{dx^2} + \left(x^2 - \nu^2 + \frac{1}{4} \right) f_k = 0, \quad \text{where} \quad \nu \equiv \sqrt{\frac{9}{4} - \frac{m^2}{H^2}}. \quad (\text{B.353})$$

Writing $f_k(\eta) \equiv \sqrt{x} g(x)$, this takes the form of a *Bessel equation* (see Appendix A)

$$x^2 \frac{d^2 g}{dx^2} + x \frac{dg}{dx} + (x^2 - \nu^2) g(x) = 0, \quad (\text{B.354})$$

which has the following solution in terms of Hankel functions:

$$g(x) = c_1 H_\nu^{(1)}(x) + c_2 H_\nu^{(2)}(x). \quad (\text{B.355})$$

Using

$$H_\nu^{(1)}(x) \xrightarrow{x \rightarrow \infty} \sqrt{\frac{2}{\pi x}} e^{i(x - \frac{1}{2}\nu\pi - \frac{1}{4}\pi)}, \quad (\text{B.356})$$

$$H_\nu^{(2)}(x) \xrightarrow{x \rightarrow \infty} \sqrt{\frac{2}{\pi x}} e^{-i(x - \frac{1}{2}\nu\pi - \frac{1}{4}\pi)}, \quad (\text{B.357})$$

the early-time limit of the canonically-normalized field becomes

$$\lim_{-k\eta \rightarrow \infty} f_k(\eta) = \sqrt{\frac{2}{\pi}} \left(c_1 e^{-\frac{i}{4}\pi(1+2\nu)} e^{-ik\eta} + c_2 e^{\frac{i}{4}\pi(1+2\nu)} e^{ik\eta} \right). \quad (\text{B.358})$$

This matches the Bunch–Davies initial condition,

$$\lim_{-k\eta \rightarrow \infty} f_k(\eta) = \frac{1}{\sqrt{2k}} e^{-ik\eta}, \quad (\text{B.359})$$

if $c_2 = 0$ and $c_1 = e^{\frac{i}{4}\pi(1+2\nu)} \sqrt{\pi/(4k)}$, which fixes the solution to be

$$f_k(\eta) = e^{\frac{i}{4}\pi(1+2\nu)} \sqrt{\frac{\pi}{4}} \sqrt{-\eta} H_\nu^{(1)}(-k\eta). \quad (\text{B.360})$$

The de Sitter mode function for the massive field, $\phi_k = f_k/a$, then is

$$\boxed{\phi_k(\eta) = H \sqrt{\frac{\pi}{4}} e^{\frac{i}{4}\pi(1+2\nu)} (-\eta)^{3/2} H_\nu^{(1)}(-k\eta)}. \quad (\text{B.361})$$

For small argument, the Hankel functions have the following expansions:

$$H_\nu^{(1)}(x) \xrightarrow{x \rightarrow 0} -\frac{i}{\pi} \left(\Gamma(\nu) \left(\frac{x}{2}\right)^{-\nu} + e^{-i\pi\nu} \Gamma(-\nu) \left(\frac{x}{2}\right)^\nu \right), \quad (\text{B.362})$$

$$H_\nu^{(2)}(x) \xrightarrow{x \rightarrow 0} \frac{i}{\pi} \left(\Gamma(\nu) \left(\frac{x}{2}\right)^{-\nu} + e^{i\pi\nu} \Gamma(-\nu) \left(\frac{x}{2}\right)^\nu \right). \quad (\text{B.363})$$

Note that these expressions hold also for complex ν . If ν has a positive real part, we can drop the x^ν contribution.

Consider first the case $m < \frac{3}{2}H$, for which ν is purely real (and positive). The complex conjugate of the Hankel function is $(H_\nu^{(1)}(x))^* = H_\nu^{(2)}(x)$, so that

$$\begin{aligned} \lim_{k\eta \rightarrow 0} |H_\nu^{(1)}(-k\eta)|^2 &= \lim_{k\eta \rightarrow 0} H_\nu^{(1)}(-k\eta) H_\nu^{(2)}(-k\eta) \\ &= \frac{1}{\pi^2} \left(\Gamma^2(\nu) \left(\frac{-k\eta}{2}\right)^{-2\nu} + \Gamma^2(-\nu) \left(\frac{-k\eta}{2}\right)^{2\nu} + 2\Gamma(\nu)\Gamma(-\nu) \cos(\pi\nu) \right) \\ &= \frac{1}{\pi^2} \left(\Gamma^2(\nu) \left(\frac{-k\eta}{2}\right)^{-2\nu} + \Gamma^2(-\nu) \left(\frac{-k\eta}{2}\right)^{2\nu} - \frac{2\pi \cot(\pi\nu)}{\nu} \right), \end{aligned} \quad (\text{B.364})$$

where we used Euler's reflection formula $\Gamma(\nu)\Gamma(-\nu) = -\pi/(\nu \sin(\pi\nu))$. The late-time limit of the power spectrum then becomes

$$\begin{aligned} \lim_{k\eta \rightarrow 0} P_\phi(k, \eta) &= \lim_{k\eta \rightarrow 0} |\phi_k(\eta)|^2 \\ &= \frac{\pi}{4} H^2(-\eta)^3 \lim_{k\eta \rightarrow 0} |H_\nu^{(1)}(-k\eta)|^2 = \boxed{\frac{4\Gamma^2(\nu)}{\pi} \frac{H^2}{2k^3} \left(\frac{-k\eta}{2}\right)^{3-2\nu}}. \end{aligned} \quad (\text{B.365})$$

As a consistency check, we take the limit of a massless field, $m^2 = 0$ or $\nu = 3/2$. Using $\Gamma(3/2) = \sqrt{\pi}/2$, we find

$$\lim_{k\eta \rightarrow 0} P_\phi(k, \eta) = \frac{H^2}{2k^3}, \quad (\text{B.366})$$

which matches the result (8.32).

Next, we take $m > \frac{3}{2}H$, for which ν is purely imaginary. It is convenient to write $\nu = i\mu$, where μ is purely real (and positive). The complex conjugate of the Hankel function then is $(H_{i\mu}^{(1)}(x))^* = H_{-i\mu}^{(2)}(x) = e^{-\pi\mu} H_{i\mu}^{(2)}(x)$, so that

$$\begin{aligned} \lim_{k\eta \rightarrow 0} |H_{i\mu}^{(1)}(-k\eta)|^2 &= e^{-\pi\mu} \lim_{k\eta \rightarrow 0} H_{i\mu}^{(1)}(-k\eta) H_{i\mu}^{(2)}(-k\eta) \\ &= \frac{e^{-\pi\mu}}{\pi^2} \left(\Gamma^2(i\mu) \left(\frac{-k\eta}{2}\right)^{-2i\mu} + \Gamma^2(-i\mu) \left(\frac{-k\eta}{2}\right)^{2i\mu} - \frac{2\pi \cot(i\pi\mu)}{i\mu} \right), \end{aligned} \quad (\text{B.367})$$

and the late-time limit of the power spectrum becomes

$$\begin{aligned}
\lim_{k\eta \rightarrow 0} P_\phi(k, \eta) &= \lim_{k\eta \rightarrow 0} |\phi_k(\eta)|^2 \\
&= \frac{\pi}{4} e^{\pi\mu} H^2(-\eta)^3 \lim_{k\eta \rightarrow 0} |H_\nu^{(1)}(-k\eta)|^2 \\
&= \frac{1}{4\pi} H^2(-\eta)^3 \left(\Gamma^2(i\mu) \left(\frac{-k\eta}{2} \right)^{-2i\mu} + \Gamma^2(-i\mu) \left(\frac{-k\eta}{2} \right)^{2i\mu} + \frac{2\pi \coth(\pi\mu)}{\mu} \right). \quad (\text{B.368})
\end{aligned}$$

Writing $\Gamma(\pm i\mu) \equiv |\Gamma(\pm i\mu)|e^{\pm i\delta}$, and using

$$|\Gamma(\pm i\mu)|^2 = \frac{\pi}{\mu \sinh(\pi\mu)}, \quad (\text{B.369})$$

we can write this as

$$\begin{aligned}
\lim_{k\eta \rightarrow 0} P_\phi(k, \eta) &= \frac{H^2(-\eta)^3}{4\mu \sinh(\pi\mu)} \left[e^{2i\delta(\mu)} \left(\frac{-k\eta}{2} \right)^{-2i\mu} + e^{-2i\delta(\mu)} \left(\frac{-k\eta}{2} \right)^{2i\mu} + 2 \cosh(\pi\mu) \right] \\
&= \frac{H^2}{2k^3} \frac{(-k\eta)^3}{\mu \sinh(\pi\mu)} \left[\cos \left(2\mu \log \left(\frac{-k\eta}{2} \right) - 2\delta \right) + \cosh(\pi\mu) \right]. \quad (\text{B.370})
\end{aligned}$$

For large masses, $\mu \gg 1$, the power spectrum scales as $1/\sinh(\pi\mu) \rightarrow e^{-\pi\mu}$, corresponding to the Boltzmann suppression of the spontaneous production of massive particles. We also see that the signal oscillates (as a function of $-k\eta$), with a frequency that depends on the mass of the particle.

Solution to Exercise 8.7

The induced metric on surfaces of constant time is

$$h_{ij} = a^2 \left[e^{2\zeta} \delta_{ij} \right]. \quad (\text{B.371})$$

The associated connection is

$$\begin{aligned}
\Gamma_{jk}^i &= \frac{1}{2} h^{il} (\partial_j h_{kl} + \partial_k h_{jl} - \partial_l h_{jk}) \\
&= \frac{1}{2} a^{-2} e^{-2\zeta} \delta^{il} \left(\partial_j (a^2 e^{2\zeta} \delta_{kl}) + \partial_k (a^2 e^{2\zeta} \delta_{jl}) - \partial_l (a^2 e^{2\zeta} \delta_{jk}) \right) \\
&= \delta_k^i \partial_j \zeta + \delta_j^i \partial_k \zeta - \delta_{jk} \delta^{il} \partial_l \zeta. \quad (\text{B.372})
\end{aligned}$$

The Ricci scalar of the spatial slice is

$$\begin{aligned}
R_{(3)} &= h^{ik} R_{ik} \\
&= a^{-2} e^{-2\zeta} \delta^{ik} \left(\partial_l \Gamma_{ik}^l - \partial_k \Gamma_{il}^l + \Gamma_{ik}^l \Gamma_{lm}^m - \Gamma_{il}^m \Gamma_{km}^l \right). \quad (\text{B.373})
\end{aligned}$$

We now evaluate the four terms on the right-hand side:

- The first term is

$$\begin{aligned}
\delta^{ik} \partial_l \Gamma_{ik}^l &= \delta^{ik} \partial_l (\delta_k^l \partial_i \zeta + \delta_i^l \partial_k \zeta - \delta_{ik} \delta^{lm} \partial_m \zeta) \\
&= 2\delta^{ik} \partial_i \partial_k \zeta - 3\partial_l \partial^l \zeta \\
&= -\partial^2 \zeta, \quad (\text{B.374})
\end{aligned}$$

where $\partial^2 \zeta \equiv \delta^{ik} \partial_i \partial_k \zeta$.

- Similarly, the second term is

$$\begin{aligned}
-\delta^{ik}\partial_k\Gamma_{il}^l &= -\delta^{ik}\partial_k(\delta_i^l\partial_l\zeta + \delta_i^l\partial_l\zeta - \delta_{il}\delta^{lm}\partial_m\zeta) \\
&= -3\delta^{ik}\partial_k\partial_i\zeta + \delta^{lk}\partial_k\partial_l\zeta - \delta^{mk}\partial_k\partial_m\zeta \\
&= -3\partial^2\zeta.
\end{aligned} \tag{B.375}$$

- To evaluate the third term, we first note that $\Gamma_{lm}^m = 3\partial_l\zeta$, and hence get

$$\begin{aligned}
\delta^{ik}\Gamma_{ik}^l\Gamma_{lm}^m &= 3\delta^{ik}(\delta_k^l\partial_i\zeta + \delta_i^l\partial_k\zeta - \delta_{ik}\delta^{lm}\partial_m\zeta)\partial_l\zeta \\
&= 3\delta^{ik}(2\partial_i\zeta\partial_k\zeta - \delta_{ik}\delta^{lm}\partial_l\zeta\partial_m\zeta) \\
&= -3(\partial\zeta)^2,
\end{aligned} \tag{B.376}$$

where $(\partial\zeta)^2 \equiv \delta^{ij}\partial_i\zeta\partial_j\zeta$.

- Finally, the fourth term is

$$\begin{aligned}
-\delta^{ik}\Gamma_{il}^m\Gamma_{km}^l &= -\delta^{ik}(\delta_l^m\partial_i\zeta + \delta_i^m\partial_l\zeta - \delta_{il}\delta^{mn}\partial_n\zeta)(\delta_m^l\partial_k\zeta + \delta_k^l\partial_m\zeta - \delta_{km}\delta^{lo}\partial_o\zeta) \\
&= -\delta^{ik}(5\partial_i\zeta\partial_k\zeta - 2\delta_{ik}\delta^{mn}\partial_m\zeta\partial_n\zeta) \\
&= -5(\partial\zeta)^2 + 6(\partial\zeta)^2 \\
&= (\partial\zeta)^2.
\end{aligned} \tag{B.377}$$

Substituting these expressions into (B.373), we get

$$\boxed{R_{(3)} = -2a^{-2}e^{-2\zeta}(2\partial^2\zeta + (\partial\zeta)^2)}, \tag{B.378}$$

as required.

Next, we compute the extrinsic curvature

$$\begin{aligned}
E_{ij} &\equiv \frac{1}{2}(\dot{h}_{ij} - \nabla_i N_j - \nabla_j N_i) \\
&= \frac{1}{2}(\dot{h}_{ij} - \partial_i N_j - \partial_j N_i + 2\Gamma_{ij}^k N_k).
\end{aligned} \tag{B.379}$$

Substituting (B.371) and (B.372), we get

$$E_{ij} = \frac{1}{2}\left[\partial_t(a^2 e^{2\zeta}\delta_{ij}) - 2\partial_{(i}N_{j)} + 2\left(\delta_j^k\partial_i\zeta + \delta_i^k\partial_j\zeta - \delta_{ij}\delta^{kl}\partial_l\zeta\right)N_k\right] \tag{B.380}$$

Using $\partial_t(a^2 e^{2\zeta}) = 2a^2 e^{2\zeta}(H + \dot{\zeta})$, where $H = \dot{a}/a$ is the Hubble parameter, we find

$$\boxed{E_{ij} = a^2 e^{2\zeta}(H + \dot{\zeta})\delta_{ij} - \partial_{(i}N_{j)} + 2N_{(i}\partial_{j)}\zeta - N_k\partial_k\zeta\delta_{ij}}. \tag{B.381}$$

as required.

Solution to Exercise 8.8

The momentum and Hamiltonian constraint equations (in the comoving gauge) are

$$\nabla_i [N^{-1}(E_j^i - E\delta_j^i)] = 0, \quad (\text{B.382})$$

$$R^{(3)} - 2V - N^{-2}(E_{ij}E^{ij} - E^2) - N^{-2}\dot{\phi}^2 = 0. \quad (\text{B.383})$$

We will evaluate them in turn:

- To evaluate (B.382), we need to compute $E_j^i - E\delta_j^i$. Using $E_j^i = h^{ik}E_{kj}$, we have

$$\begin{aligned} E_j^i &= a^{-2}e^{-2\zeta}\delta^{ik}\left(a^2e^{2\zeta}(H + \dot{\zeta})\delta_{kj} - \partial_{(k}N_{j)} + 2N_{(k}\partial_{j)}\zeta - N_m\partial_m\zeta\delta_{kj}\right) \\ &= (H + \dot{\zeta})\delta_j^i - a^{-2}\delta^{ik}\partial_{(k}N_{j)} + \dots, \end{aligned} \quad (\text{B.384})$$

where we have only kept terms at first order in the fluctuations ζ and $N_i = \partial_i\beta$. This implies $E = 3(H + \dot{\zeta}) - a^{-2}\partial_i N_i$ and hence

$$E_j^i - E\delta_j^i = -2(H + \dot{\zeta})\delta_j^i + a^{-2}(\partial^2\beta\delta_j^i - \delta^{ik}\partial_{(k}\partial_{j)}\beta). \quad (\text{B.385})$$

Using $N = 1 + \alpha$, we can then write (B.382) as

$$\begin{aligned} 0 &= \nabla_i [N^{-1}(E_j^i - E\delta_j^i)] = \nabla_i [(1 - \alpha)(-2(H + \dot{\zeta})\delta_j^i + a^{-2}(\partial^2\beta\delta_j^i - \delta^{ik}\partial_{(k}\partial_{j)}\beta))] \\ &= 2\nabla_i(H\alpha - \dot{\zeta}) + a^{-2}\nabla_i(\partial^2\beta\delta_j^i - \delta^{ik}\partial_{(k}\partial_{j)}\beta) \\ &= 2\nabla_i(H\alpha - \dot{\zeta}) + a^{-2}(\partial_j\partial^2\beta - \partial_i\partial_{(i}\partial_{j)}\beta) \\ &= 2\nabla_i(H\alpha - \dot{\zeta}). \end{aligned} \quad (\text{B.386})$$

Hence, we have

$$\boxed{\alpha = \frac{\dot{\zeta}}{H}}, \quad (\text{B.387})$$

as required.

- To evaluate (B.383), we need to compute $E_{ij}E^{ij} - E^2$. Using $E^{ij} = h^{ik}h^{jl}E_{kl}$ and $E = h^{ij}E_{ij}$, we have

$$E_{ij}E^{ij} - E^2 = a^{-4}e^{-4\zeta}\left(\delta^{ik}\delta^{jl} - \delta^{ij}\delta^{kl}\right)E_{ij}E_{kl}. \quad (\text{B.388})$$

Substituting (B.381) for E_{ij} , we then find

$$\begin{aligned} E_{ij}E^{ij} - E^2 &= -6(H + \dot{\zeta})^2 + 4a^{-2}e^{-2\zeta}(H + \dot{\zeta})(\partial_i N_i + N_i\partial_i\zeta) \\ &\quad - a^{-4}e^{-4\zeta}\left[(\partial_i N_i)^2 + 2(N_i\partial_i\zeta)^2 - (\partial_{(i}N_{j)} - 2N_{(i}\partial_{j)}\zeta)^2\right]. \end{aligned} \quad (\text{B.389})$$

Assuming that $N_i = \partial_i\beta$ vanishes for the background metric, we expand (B.389) to first order in fluctuations:

$$E_{ij}E^{ij} - E^2 = -6H^2 - 12H\dot{\zeta} + \frac{4}{a^2}H\partial^2\beta. \quad (\text{B.390})$$

Similarly, the first-order expansion of (B.378) gives

$$R^{(3)} = -\frac{4}{a^2}\partial^2\zeta. \quad (\text{B.391})$$

At first order, the constraint equation (B.383) then reads

$$\begin{aligned} 0 &= R^{(3)} - 2V - N^{-2}(E_{ij}E^{ij} - E^2 + \dot{\phi}^2) \\ &= -\frac{4}{a^2}\partial^2\zeta - 2V - (1 - 2\alpha) \left[-6H^2 - 12H\dot{\zeta} + \frac{4}{a^2}H\partial^2\beta + \dot{\phi}^2 \right] \\ &= 2 \left[3H^2 - \left(\frac{1}{2}\dot{\phi}^2 + V \right) \right] + 12H\dot{\zeta} - \frac{4}{a^2}\partial^2\zeta - 12H^2\alpha + 2\alpha\dot{\phi}^2 - \frac{4}{a^2}H\partial^2\beta \\ &= -\frac{4}{a^2}H \left[\partial^2\beta + \frac{\partial^2\zeta}{H} - a^2\frac{\dot{\phi}^2}{2H^2}\dot{\zeta} \right], \end{aligned} \quad (\text{B.392})$$

where we have used the Friedmann equation in the zeroth-order term and substituted $\alpha = \dot{\zeta}/H$. Hence, we found that

$$\partial^2\beta = -\frac{\partial^2\zeta}{H} + a^2\frac{\dot{\phi}^2}{2H^2}\dot{\zeta}, \quad (\text{B.393})$$

as required.

Solution to Exercise 8.9

The inflaton action in the comoving gauge is

$$\begin{aligned} S &= \frac{1}{2} \int d^4x \sqrt{-g} \left[R - g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - 2V \right] \\ &= \frac{1}{2} \int d^4x \sqrt{h} N \left[R^{(3)} - 2V + N^{-2}(E^{ij}E_{ij} - E^2 + \dot{\phi}^2) \right], \end{aligned} \quad (\text{B.394})$$

where

$$R_{(3)} = a^{-2}e^{-2\zeta} (-4\partial^2\zeta - 2(\partial\zeta)^2), \quad (\text{B.395})$$

$$\begin{aligned} E_{ij}E^{ij} - E^2 &= -6(H + \dot{\zeta})^2 + 4a^{-2}e^{-2\zeta}(H + \dot{\zeta})(\partial_i N_i + N_i \partial_i \zeta) \\ &\quad - a^{-4}e^{-4\zeta} \left[(\partial_i N_i)^2 + 2(N_i \partial_i \zeta)^2 - (\partial_{(i} N_{j)})^2 \right]. \end{aligned} \quad (\text{B.396})$$

Substituting this into the action (together with $N = 1 + \alpha$ and $N_i = \partial_i \beta$) and expanding to second order, we get

$$\begin{aligned} S &= \frac{1}{2} \int d^4x \left[(1 + \alpha) \left(ae^\zeta (-4\partial^2\zeta - 2(\partial\zeta)^2) - 2a^3e^{3\zeta}V \right) \right. \\ &\quad \left. + (1 - \alpha + \alpha^2) \left(-6a^3e^{3\zeta}(H + \dot{\zeta})^2 + 4ae^\zeta(H + \dot{\zeta})(\partial^2\beta + \partial_i \beta \partial_i \zeta) + a^3e^{3\zeta}\dot{\phi}^2 \right) \right], \end{aligned} \quad (\text{B.397})$$

where we have used integration by parts and dropped third orders terms to get rid of the contributions in the second line of the expression for $E_{ij}E^{ij} - E^2$. After some integrations by parts and using the background equations of motion $2V + \dot{\phi}^2 = 6H^2$ and $\ddot{a}/a + 2H^2 = V$, we get

$$S = \int dt d^3x a^3 \varepsilon \left(\dot{\zeta}^2 - \frac{1}{a^2}(\partial\zeta)^2 \right), \quad (\text{B.398})$$

where $\varepsilon \equiv \dot{\phi}^2/(2H^2)$.

Solution to Exercise 8.10

We start with a few preliminaries. The comoving Hubble parameter is

$$\mathcal{H} \equiv \frac{a'}{a} = \dot{a} = aH, \quad (\text{B.399})$$

where $a' \equiv \partial_\eta a$ and $\dot{a} \equiv \partial_t a$. Moreover, the first Hubble slow-variation parameter is

$$\varepsilon \equiv -\frac{\dot{H}}{H^2} = 1 - \frac{\mathcal{H}'}{\mathcal{H}^2}, \quad (\text{B.400})$$

so that $\mathcal{H}' = \mathcal{H}^2(1 - \varepsilon)$. For $\varepsilon \approx \text{const}$, this can be integrated to give

$$\mathcal{H} = -\frac{1}{(1 - \varepsilon)\eta} \approx -\frac{1 + \varepsilon}{\eta}. \quad (\text{B.401})$$

Starting from the definition $z \equiv a\sqrt{2\varepsilon}$, we then have

$$z' = z\mathcal{H} \left(1 + \frac{1}{2}\kappa\right), \quad \text{where} \quad \kappa \equiv \frac{\varepsilon'}{\mathcal{H}\varepsilon}. \quad (\text{B.402})$$

Taking a further derivative, we find

$$\begin{aligned} z'' &= z'\mathcal{H} \left(1 + \frac{1}{2}\kappa\right) + z\mathcal{H}' \left(1 + \frac{1}{2}\kappa\right) + \frac{1}{2}z\mathcal{H}\kappa' \\ &\approx z\mathcal{H}^2 \left(1 + \frac{1}{2}\kappa\right) \left(1 + \frac{1}{2}\kappa\right) + z\mathcal{H}^2(1 - \varepsilon) \left(1 + \frac{1}{2}\kappa\right) \\ &\approx z\mathcal{H}^2 \left(2 - \varepsilon + \frac{3}{2}\kappa\right), \end{aligned} \quad (\text{B.403})$$

where we have dropped terms beyond linear order in the slow-variation parameters. Substituting (B.401), this becomes

$$\boxed{\frac{z''}{z} = \frac{1}{\eta^2} \left[2 + 3 \left(\varepsilon + \frac{1}{2}\kappa\right)\right]}, \quad (\text{B.404})$$

as required.

Solution to Exercise 8.11

In the previous exercise, we determined

$$\frac{z'}{z} = \mathcal{H} \left(1 + \frac{1}{2}\kappa\right), \quad (\text{B.405})$$

$$\mathcal{H} = -\frac{1 + \varepsilon}{\eta}. \quad (\text{B.406})$$

Combining these expressions, we get

$$\frac{z'}{z} = -\frac{1 + \varepsilon}{\eta} \left(1 + \frac{1}{2}\kappa\right) \approx -\frac{1 + \varepsilon + \frac{1}{2}\kappa}{\eta} = \frac{\frac{1}{2} - \nu}{\eta}, \quad \text{where} \quad \nu \equiv \frac{3}{2} + \varepsilon + \frac{1}{2}\kappa. \quad (\text{B.407})$$

Integrating this, we find

$$\boxed{z(\eta) = z_*(\eta/\eta_*)^{\frac{1}{2} - \nu}}, \quad (\text{B.408})$$

as required.

B.9 Exercises in Chapter 9

Solution to Exercise 9.1

We start from the master formula (9.21):

$$\langle \hat{\mathcal{O}}(t) \rangle = \langle 0 | \overline{T} e^{i \int_{-\infty}^t dt' \hat{H}_{\text{int}}(t')} \hat{\mathcal{O}}(t) T e^{-i \int_{-\infty}^t dt' \hat{H}_{\text{int}}(t')} | 0 \rangle, \quad (\text{B.409})$$

which to linear order in $\hat{H}_{\text{int}}$ becomes

$$\langle \hat{\mathcal{O}}(t) \rangle = -i \int_{-\infty}^t dt' \langle 0 | \hat{\mathcal{O}}(t) \hat{H}_{\text{int}}(t') | 0 \rangle + i \int_{-\infty}^t dt' \langle 0 | \hat{H}_{\text{int}}(t') \hat{\mathcal{O}}(t) | 0 \rangle. \quad (\text{B.410})$$

We notice that

$$\left(\int_{-\infty}^t dt' \langle 0 | \hat{\mathcal{O}}(t) \hat{H}_{\text{int}}(t') | 0 \rangle \right)^\dagger = \int_{-\infty}^t dt' \langle 0 | \hat{H}_{\text{int}}(t') \hat{\mathcal{O}}(t) | 0 \rangle, \quad (\text{B.411})$$

so that (B.410) is of the form $-iA + iA^\dagger = 2\text{Im} A$, or

$$\langle \hat{\mathcal{O}}(t) \rangle = 2\text{Im} \left(\int_{-\infty}^t dt' \langle 0 | \hat{\mathcal{O}}(t) \hat{H}_{\text{int}}(t') | 0 \rangle \right), \quad (\text{B.412})$$

which is precisely (9.22). Returning to (B.410), we see that this is exactly of the form of the commutator

$$\boxed{\langle \hat{\mathcal{O}}(t) \rangle = i \int_{-\infty}^t dt' \langle 0 | [\hat{H}_{\text{int}}(t'), \hat{\mathcal{O}}(t)] | 0 \rangle}, \quad (\text{B.413})$$

as desired.⁵⁸

Next, we expand (B.409) to second order in $\hat{H}_{\text{int}}$:

$$\begin{aligned} \langle \hat{\mathcal{O}}(t) \rangle &= \int_{-\infty}^t dt' \int_{-\infty}^t dt'' \langle 0 | \hat{H}_{\text{int}}(t') \hat{\mathcal{O}}(t) \hat{H}_{\text{int}}(t'') | 0 \rangle \\ &\quad - \int_{-\infty}^t dt' \int_{-\infty}^{t'} dt'' \left(\langle 0 | \hat{\mathcal{O}}(t) \hat{H}_{\text{int}}(t') \hat{H}_{\text{int}}(t'') | 0 \rangle + \langle 0 | \hat{H}_{\text{int}}(t'') \hat{H}_{\text{int}}(t') \hat{\mathcal{O}}(t) | 0 \rangle \right), \end{aligned} \quad (\text{B.414})$$

where the time integrals in the second line have the same limits because one term is time-ordered and the other is anti-time-ordered. As before, the terms in the second line can be combined to give twice the real part, reproducing (9.31). In order to write this expression in commutator form, we note that

$$\begin{aligned} &\int_{-\infty}^t dt' \int_{-\infty}^{\textcolor{red}{t}} dt'' \langle 0 | \hat{H}_{\text{int}}(t') \hat{\mathcal{O}}(t) \hat{H}_{\text{int}}(t'') | 0 \rangle \\ &= 2 \int_{-\infty}^t dt' \int_{-\infty}^{\textcolor{red}{t}'} dt'' \langle 0 | \hat{H}_{\text{int}}(t') \hat{\mathcal{O}}(t) \hat{H}_{\text{int}}(t'') | 0 \rangle, \end{aligned} \quad (\text{B.415})$$

because it is symmetric under the interchange of $t' \leftrightarrow t''$. Now that all the terms are under the same integrals, it is straightforward to rearrange (B.414) into the form (9.33).

⁵⁸In writing this commutator form, we have glossed over a small technical subtlety associated to the $i\epsilon$ prescription: looking back at (B.410), we see that the two terms have different $i\epsilon$ prescriptions. In order to combine them into (B.413), we are implicitly integrating both terms along the same contour, which seems a bit suspect [116, 117]. In practice, this is not much of a problem because we typically break up the commutator into its constituent pieces and perform the integrals so that they converge. More formally, we can choose a different regularization scheme that is compatible with the commutator grouping. See [118, 119], for detailed discussions.

Solution to Exercise 9.2

Starting from the action

$$S = \int d^4x \left(-\frac{1}{2}(\partial\phi)^2 - \frac{1}{6\Lambda^2}\dot{\phi}^3 \right), \quad (\text{B.416})$$

we can derive the canonical momentum conjugate to ϕ :

$$\pi \equiv \frac{\delta\mathcal{L}}{\delta\dot{\phi}} = \dot{\phi} - \frac{1}{2\Lambda^2}\dot{\phi}^2. \quad (\text{B.417})$$

The Hamiltonian is then given by

$$\begin{aligned} H &= \int d^3x \left(\pi\dot{\phi} - \mathcal{L} \right) \\ &= \int d^3x \left(\frac{1}{2}\dot{\phi}^2 + (\nabla\phi)^2 - \frac{1}{3\Lambda^2}\dot{\phi}^3 \right) \\ &= \int d^3x \left(\frac{1}{2}\pi^2 + (\nabla\phi)^2 + \frac{1}{6\Lambda^2}\pi^3 + \frac{1}{8\Lambda^4}\pi^4 + \dots \right), \end{aligned} \quad (\text{B.418})$$

where, in the last line, we have used the relation between $\dot{\phi}$ and the canonical momentum

$$\frac{1}{\Lambda^2}\dot{\phi} = 1 - \sqrt{1 - \frac{2\pi}{\Lambda^2}}, \quad (\text{B.419})$$

expanded out perturbatively for $\pi \ll \Lambda^2$. We therefore see that the interaction Hamiltonian up to quartic order is

$$H_{\text{int}} = \int d^3x \left(\frac{1}{6\Lambda^2}\pi^3 + \frac{1}{8\Lambda^4}\pi^4 \right), \quad (\text{B.420})$$

which, importantly, is not just $-L_{\text{int}}$ appearing in (B.416). To quantize the theory, we treat these interactions perturbatively in the interaction picture, which amounts to setting $\pi = \dot{\phi}$ and expanding ϕ in modes as usual, so that

$$H_{\text{int}} = \int d^3x \left(\frac{1}{6\Lambda^2}\dot{\phi}^3 + \frac{1}{8\Lambda^4}\dot{\phi}^4 \right), \quad (\text{B.421})$$

where we are suppressing interaction-picture labels.

In order to compute the four-point function in this theory, we can employ the Feynman rules from Section 9.3. There will now be both exchange and contact contributions. Moreover, we have to keep track of one additional subtlety: since there are derivatives acting on the fields in the interaction vertices, lines connected to interactions should now be associated with factors of

$$\dot{f}_k(t) = -ik f_k(t), \quad (\text{B.422})$$

the time derivative of the mode function. For lines that are not time-ordered (i.e. external lines or lines that cross the $t = 0$ surface) we can proceed in the naive way by differentiating the Wightman propagator. For bulk internal lines, however, we have to keep in mind that they should be associated to

$$\langle T \dot{\phi}_{\mathbf{k}}(t) \dot{\phi}_{\mathbf{k}'}(t') \rangle' = \partial_t \partial_{t'} W_k(t, t') \theta(t - t') + \partial_t \partial_{t'} W_k(t', t) \theta(t' - t), \quad (\text{B.423})$$

which is the time-ordered two-point function of $\dot{\phi}$. Importantly, this is *not* the time derivative of the Feynman propagator:

$$\partial_t \partial_{t'} G_F(k; t, t') = \partial_t \partial_{t'} \langle T \phi_{\mathbf{k}}(t) \phi_{\mathbf{k}}(t') \rangle' = \langle T \dot{\phi}_{\mathbf{k}}(t) \dot{\phi}_{\mathbf{k}}(t') \rangle' + i\delta(t - t'). \quad (\text{B.424})$$

With this in mind, we can compute

$$\begin{aligned} I_{++} &\equiv \text{Diagram: A horizontal line with two vertices. From each vertex, two lines extend upwards and outwards, meeting at a point above the line.} \\ &= \left(\frac{-i}{\Lambda^2} \right)^2 \int_{-\infty}^0 dt dt' \partial_t W_{k_1}(0, t) \partial_t W_{k_2}(0, t) \langle T \dot{\phi}_{\mathbf{k}_I}(t) \dot{\phi}_{\mathbf{k}_I}(t') \rangle' \partial_{t'} W_{k_3}(0, t') \partial_{t'} W_{k_4}(0, t') \\ &= -\frac{1}{\Lambda^4} k_1 k_2 k_3 k_4 k_I^2 \int_{-\infty}^0 dt dt' W_{k_1}(0, t) W_{k_2}(0, t) G_F(k_I; t, t') W_{k_3}(0, t') W_{k_4}(0, t') \\ &= \frac{1}{32\Lambda^4} \frac{k_I}{k_{12} + k_{34}} \left(\frac{1}{k_{12} + k_I} + \frac{1}{k_{34} + k_I} \right), \end{aligned} \quad (\text{B.425})$$

where we evaluated the derivatives to put the time integral in the form (9.48). We can similarly evaluate

$$\begin{aligned} I_{+-} &\equiv \text{Diagram: A horizontal line with two vertices. From each vertex, two lines extend upwards and outwards, meeting at a point above the line. The lines are connected in a different way than in I_{++}.} \\ &= \frac{1}{\Lambda^4} \int_{-\infty}^0 dt dt' \partial_t W_{k_1}(0, t) \partial_t W_{k_2}(0, t) \partial_t \partial_{t'} W_{k_I}(t', t) \partial_{t'} W_{k_3}(t', 0) \partial_{t'} W_{k_4}(t', 0) \\ &= \frac{1}{\Lambda^4} k_1 k_2 k_3 k_4 k_I^2 \int_{-\infty}^0 dt dt' W_{k_1}(0, t) W_{k_2}(0, t) W_{k_I}(t', t) W_{k_3}(t', 0) W_{k_4}(t', 0) \\ &= \frac{1}{32\Lambda^4} \frac{k_I}{(k_{12} + k_I)(k_{34} + k_I)}. \end{aligned} \quad (\text{B.426})$$

Finally, we have to compute the contact contribution

$$\begin{aligned} I_+ &= \text{Diagram: A horizontal line with two vertices. From each vertex, two lines extend upwards and outwards, meeting at a point above the line. The lines are connected in a different way than in I_{++}.} \\ &= -\frac{3i}{\Lambda^4} \int_{-\infty}^0 dt \partial_t W_{k_1}(0, t) \partial_t W_{k_2}(0, t) \partial_t W_{k_3}(0, t) \partial_t W_{k_4}(0, t) \\ &= -\frac{3}{16\Lambda^4} \frac{1}{k_1 + k_2 + k_3 + k_4}. \end{aligned} \quad (\text{B.427})$$

All of these contributions are purely real, so the other time-orderings will just lead to a factor of two. Adding everything up, we obtain

$$\boxed{\langle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \phi_{\mathbf{k}_4} \rangle' = -\frac{2(k_{12} + k_I)(k_{34} + k_I) + k_{12} k_{34}}{8(k_{12} + k_{34})(k_{12} + k_I)(k_{34} + k_I)}}. \quad (\text{B.428})$$

There were two important subtleties in this computation involving (time) derivative interactions. First, we saw that starting at quartic order, the interaction Hamiltonian was no longer just

minus the interaction Lagrangian. Additionally, we saw that we have to be careful about how time derivatives act on internal lines; we cannot just replace the time-ordered two-point function of $\dot{\phi}$ with the time derivative of the Feynman propagator. Here, we were careful about both of these subtleties. Interestingly enough, most often one can disregard both of these things and the mistakes will cancel out. Roughly speaking, the way that this happens is that the time derivatives acting on the theta functions in the Feynman propagator will produce a delta function term as in (B.424), which will reproduce the contribution from the quartic interaction Hamiltonian. If you are motivated, you should check that this happens for this simple example. See [120–122], for a more detailed discussion.

Solution to Exercise 9.3

We start from equation (9.30):

$$\langle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \rangle'(\eta_*) = 2\lambda \operatorname{Im} \left[f_{k_1}(\eta_*) f_{k_2}(\eta_*) f_{k_3}(\eta_*) \int_{-\infty}^{\eta_*} d\eta a^4(\eta) f_{k_1}^*(\eta) f_{k_2}^*(\eta) f_{k_3}^*(\eta) \right]. \quad (\text{B.429})$$

Substituting the Bunch–Davies mode function,

$$f_k(\eta) = \frac{u_k(\eta)}{a(\eta)} = \frac{iH}{\sqrt{2k^3}} (1 + ik\eta) e^{-ik\eta}, \quad (\text{B.430})$$

the integral in (B.429) can be expressed as

$$\begin{aligned} I(\eta_*) &\equiv \int_{-\infty}^{\eta_*} d\eta a^4(\eta) f_{k_1}^*(\eta) f_{k_2}^*(\eta) f_{k_3}^*(\eta) \\ &= \frac{i}{H} \frac{1}{\sqrt{8k_1^3 k_2^3 k_3^3}} \int_{-\infty}^{\eta_*} \frac{d\eta}{\eta^4} (1 - ik_1\eta)(1 - ik_2\eta)(1 - ik_3\eta) e^{iK\eta} \\ &= \frac{i}{H} \frac{1}{\sqrt{8k_1^3 k_2^3 k_3^3}} \left[\mathcal{I}_4(\eta_*) - iK \mathcal{I}_3(\eta_*) - (k_1 k_2 + k_2 k_3 + k_3 k_1) \mathcal{I}_2(\eta_*) + ik_1 k_2 k_3 \mathcal{I}_1(\eta_*) \right], \end{aligned} \quad (\text{B.431})$$

where $K \equiv k_1 + k_2 + k_3$ and

$$\mathcal{I}_N(\eta_*) \equiv \int_{-\infty}^{\eta_*} d\eta \frac{e^{iK\eta}}{\eta^N}. \quad (\text{B.432})$$

Substituting $iK e^{iK\eta} = d(e^{iK\eta})/d\eta$ into this expression, and integrating by parts, we get the following recursion relation

$$\mathcal{I}_{N+1}(\eta_*) = \frac{iK}{N} \mathcal{I}_N(\eta_*) - \frac{1}{N} \frac{e^{iK\eta_*}}{\eta_*^N}. \quad (\text{B.433})$$

The integrals $\mathcal{I}_2, \mathcal{I}_3, \mathcal{I}_4$ can therefore all be expressed in terms of $\mathcal{I}_1$:

$$\mathcal{I}_2(\eta_*) = iK \mathcal{I}_1(\eta_*) - \frac{e^{iK\eta_*}}{\eta_*}, \quad (\text{B.434})$$

$$\mathcal{I}_3(\eta_*) = -\frac{K^2}{2} \mathcal{I}_1(\eta_*) - \frac{iK}{2} \frac{e^{iK\eta_*}}{\eta_*} - \frac{1}{2} \frac{e^{iK\eta_*}}{\eta_*^2}, \quad (\text{B.435})$$

$$\mathcal{I}_4(\eta_*) = -\frac{iK^3}{6} \mathcal{I}_1(\eta_*) + \frac{K^2}{6} \frac{e^{iK\eta_*}}{\eta_*} - \frac{iK}{6} \frac{e^{iK\eta_*}}{\eta_*^2} - \frac{1}{3} \frac{e^{iK\eta_*}}{\eta_*^3}. \quad (\text{B.436})$$

Equation (B.431) can then be written as

$$I(\eta_*) = \frac{i}{H} \frac{1}{\sqrt{8k_1^3 k_2^3 k_3^3}} \left[i \left(\frac{K^3}{3} - (k_1 k_2 + k_2 k_3 + k_3 k_1) K + k_1 k_2 k_3 \right) \mathcal{I}_1(\eta_*) \right. \\ \left. - \left(\frac{K^2}{3} - (k_1 k_2 + k_2 k_3 + k_3 k_1) \right) \frac{e^{iK\eta_*}}{\eta_*} + \frac{iK}{3} \frac{e^{iK\eta_*}}{\eta_*^2} - \frac{1}{3} \frac{e^{iK\eta_*}}{\eta_*^3} \right]. \quad (\text{B.437})$$

In addition, we have

$$f_{k_1} f_{k_2} f_{k_3}(\eta_*) = \frac{iH^3}{\sqrt{8k_1^3 k_2^3 k_3^3}} \left(1 - iK\eta_* - (k_1 k_2 + k_2 k_3 + k_3 k_1) \eta_*^2 + ik_1 k_2 k_3 \eta_*^3 \right) e^{-iK\eta_*}. \quad (\text{B.438})$$

We now multiply (B.438) and (B.437), and extract the imaginary part. Dropping terms that vanish in the limit $\eta_* \rightarrow 0$, we find

$$\langle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \rangle' = -\frac{\lambda H^2}{12} \frac{(\gamma + \log(|K\eta_*|)) \sum_i k_i^3 - 3K \sum_{i<j} k_i k_j + K^3}{(k_1 k_2 k_3)^3}, \quad (\text{B.439})$$

where we used that $\mathcal{I}_1(\eta_*) \rightarrow \gamma + \log(|K\eta_*|)$.

Solution to Exercise 9.4

We consider an interaction of the form $\mathcal{L}_{\text{int}} = -\frac{1}{2}\lambda\phi^2\chi$, where ϕ is a conformally coupled scalar and χ is a scalar of general mass. Formula (9.58) still applies, so that

$$I_+ = -i\lambda f_{k_1}(\eta_*) f_{k_2}(\eta_*) \chi_{k_3}(\eta_*) \int_{-\infty}^{\eta_*} \frac{d\eta}{(H\eta)^4} f_{k_1}^*(\eta) f_{k_2}^*(\eta) \chi_{k_3}^*(\eta), \quad (\text{B.440})$$

where $f_k(\eta)$ are the mode functions of a conformally coupled scalar (9.57) and $\chi_k(\eta)$ is the mode function of a general mass scalar (9.56):

$$f_k(\eta) = H(-\eta) \frac{e^{-ik\eta}}{\sqrt{2k}}, \quad (\text{B.441})$$

$$\chi_k(\eta) = H \sqrt{\frac{\pi}{4}} e^{\frac{i\pi}{4}(1+2\nu)} (-\eta)^{3/2} H_\nu^{(1)}(-k\eta). \quad (\text{B.442})$$

Their complex conjugates are

$$f_k^*(\eta) = H(-\eta) \frac{e^{ik\eta}}{\sqrt{2k}}, \quad (\text{B.443})$$

$$\chi_k^*(\eta) = H \sqrt{\frac{\pi}{4}} e^{\frac{-i\pi}{4}(1+2\nu)} (-\eta)^{3/2} H_\nu^{(2)}(-k\eta). \quad (\text{B.444})$$

Using the integral representation of the Hankel function (9.64), we can write $\chi_k^*(\eta)$ as

$$\chi_k^*(\eta) = \frac{2^\nu e^{\frac{i\pi}{4}(1+2\nu)}}{\Gamma[\nu + \frac{1}{2}]} \frac{H}{k^{3/2}} (-k\eta)^{3/2+\nu} e^{ik\eta} \int_0^\infty ds e^{2ik\eta s} s^{\nu-\frac{1}{2}} (1+s)^{\nu-\frac{1}{2}}. \quad (\text{B.445})$$

Equation (B.440) then becomes

$$I_+ = \frac{e^{-\frac{i\pi}{4}(1-2\nu)}}{\Gamma[\nu + \frac{1}{2}]} \lambda H \eta_*^2 \chi_{k_3}(\eta_*) \frac{(2k_3)^\nu}{4k_1 k_2} \times \int_0^\infty ds s^{\nu-\frac{1}{2}} (1+s)^{\nu-\frac{1}{2}} \int_{-\infty}^{\eta_*} d\eta e^{i[k_{12}+(1+2s)k_3]\eta} (-\eta)^{\nu-\frac{1}{2}}, \quad (\text{B.446})$$

where we have dropped the factor of $e^{-ik_{12}\eta_*}$ (which becomes unity in the limit $\eta_* \rightarrow 0$). To perform the time integral in this expression, we use the following identity

$$\int_{-\infty}^0 d\eta e^{i\alpha\eta} (-\eta)^{\Delta-2} = (-i)^{\Delta-1} \Gamma[\Delta-1] \alpha^{1-\Delta}. \quad (\text{B.447})$$

Setting $\Delta = \frac{3}{2} + \nu$ and $\alpha = k_{12} + (1+2s)k_3$, this gives

$$\begin{aligned} \int_{-\infty}^0 d\eta e^{i[k_{12}+(1+2s)k_3]\eta} (-\eta)^{\nu-\frac{1}{2}} &= (-i)^{\frac{1}{2}+\nu} \Gamma[\nu + \frac{1}{2}] [k_{12} + k_3(2s+1)]^{-\frac{1}{2}-\nu} \\ &= (-i)^{\frac{1}{2}+\nu} \Gamma[\nu + \frac{1}{2}] (2k_3)^{-\frac{1}{2}-\nu} \left(s + \frac{1+u}{2u}\right)^{-\frac{1}{2}-\nu}, \end{aligned} \quad (\text{B.448})$$

where we have defined $u \equiv k_3/k_{12}$. Equation (B.446) then becomes

$$I_+ = -i\lambda H \eta_*^2 \chi_{k_3}(\eta_*) \frac{(2k_3)^{-1/2}}{4k_1 k_2} \int_0^\infty ds s^{\nu-\frac{1}{2}} (1+s)^{\nu-\frac{1}{2}} \left(s + \frac{1+u}{2u}\right)^{-\frac{1}{2}-\nu}. \quad (\text{B.449})$$

Recalling the integral formula for the hypergeometric function (valid for $\text{Re } c > \text{Re } b > 0$ and $|\arg(1-z)| < \pi$),

$${}_2F_1 \left[\begin{matrix} a, & b \\ c \end{matrix} \middle| z \right] = \frac{\Gamma[c]}{\Gamma[b]\Gamma[c-b]} \int_0^\infty dt t^{-b+c-1} (t+1)^{a-c} (t-z+1)^{-a}, \quad (\text{B.450})$$

we see that (B.449) can be written as

$$I_+ = -i\lambda H \eta_*^2 \chi_{k_3}(\eta_*) \Gamma[\frac{1}{2}-\nu] \Gamma[\frac{1}{2}+\nu] \frac{(2k_3)^{-1/2}}{4k_1 k_2} {}_2F_1 \left[\begin{matrix} \frac{1}{2} + \nu, & \frac{1}{2} - \nu \\ 1 \end{matrix} \middle| \frac{u-1}{2u} \right]. \quad (\text{B.451})$$

We can simplify this expression by noting that this particular hypergeometric function is essentially how the associated Legendre function is defined:

$$P_\nu^\mu(x) = \left(\frac{x+1}{x-1}\right)^{\mu/2} \frac{1}{\Gamma[1-\mu]} {}_2F_1 \left[\begin{matrix} \nu+1, & -\nu \\ 1-\mu \end{matrix} \middle| \frac{1}{2} - \frac{x}{2} \right]. \quad (\text{B.452})$$

The Legendre function has a branch cut running from $x = -\infty$ to $x = 1$, and is real along the positive real interval $x \in (1, \infty)$. We can then write (B.451) as

$$I_+ = -i\lambda H \eta_*^2 \chi_{k_3}(\eta_*) \Gamma[\frac{1}{2}-\nu] \Gamma[\frac{1}{2}+\nu] \frac{(2k_3)^{-1/2}}{4k_1 k_2} P_{\nu-\frac{1}{2}}^0(u^{-1}), \quad (\text{B.453})$$

where the physical region is $u \in (0, 1)$, so that there is no branch cut of the Legendre function in the physical region. The full correlator is given by $I_+ + I_-$, where I_- is the complex conjugate:

$$\langle \phi \phi \chi \rangle' = -\left(i\chi_{k_3}(\eta_*) - i\chi_{k_3}^*(\eta_*)\right) \lambda H \eta_*^2 \Gamma[\frac{1}{2}-\nu] \Gamma[\frac{1}{2}+\nu] \frac{(2k_3)^{-1/2}}{4k_1 k_2} P_{\nu-\frac{1}{2}}^0(u^{-1}). \quad (\text{B.454})$$

Note that this formula holds for both real and complex ν . For real ν it is obvious, for complex $\nu = i\mu$, one has to use the fact that (see p. 140 of [123])

$$P_\nu^\mu(x) = P_{-\nu-1}^\mu(x). \quad (\text{B.455})$$

To finally extract the late-time correlator, we use (B.362) and (B.363) to write

$$\begin{aligned} i\chi_{k_3}(\eta_*) - i\chi_{k_3}^*(\eta_*) &= \\ &= H(-\eta_*)^{3/2} \sqrt{\frac{\pi}{4}} \left(e^{\frac{i\pi}{4}(1+2\nu)} iH_\nu^{(1)}(-k\eta_*) - e^{-\frac{i\pi}{4}(1+2\nu)} iH_\nu^{(2)}(-k\eta_*) \right) \\ &\rightarrow \frac{H(-\eta_*)^{3/2}}{\sqrt{\pi}} \left[\cos\left(\frac{\pi}{4}(1+2\nu)\right) \Gamma[\nu] \left(\frac{-k_3\eta_*}{2}\right)^{-\nu} + \cos\left(\frac{\pi}{4}(1-2\nu)\right) \Gamma[-\nu] \left(\frac{-k_3\eta_*}{2}\right)^\nu \right]. \end{aligned} \quad (\text{B.456})$$

Substituting this into (B.454), we find

$$\begin{aligned} \langle \phi\phi\chi \rangle' &= -\frac{\lambda(H\eta_*)^2}{2\sqrt{\pi}} \frac{1}{k_1 k_2 k_3^2} \Gamma[\tfrac{1}{2} - \nu] \Gamma[\tfrac{1}{2} + \nu] P_{\nu-\frac{1}{2}}^0(u^{-1}) \\ &\quad \left[\cos\left(\frac{\pi}{4}(1+2\nu)\right) \Gamma[\nu] \left(\frac{-k_3\eta_*}{2}\right)^{\frac{3}{2}-\nu} + \cos\left(\frac{\pi}{4}(1-2\nu)\right) \Gamma[-\nu] \left(\frac{-k_3\eta_*}{2}\right)^{\frac{3}{2}+\nu} \right], \end{aligned} \quad (\text{B.457})$$

which is valid for both real and imaginary ν . When ν is real, the branch with $\eta_*^{-\nu}$ is more important as $\eta_* \rightarrow 0$.

Let us check that (B.457) reproduces (9.60). A conformally coupled scalar has $\nu = 1/2$. In this case, the first term in brackets decays the most slowly at late times. Using the fact that $P_0^0(x) = 1$, we see that the divergence in $\Gamma[\frac{1}{2} - \nu]$ cancels against the zero in $\cos(\frac{\pi}{4}(1+2\nu))$, leaving

$$\boxed{\langle \phi\phi\chi \rangle' \xrightarrow{\nu \rightarrow \frac{1}{2}} \frac{\pi}{8} \lambda \frac{H^2 \eta_*^3}{k_1 k_2 k_3}}, \quad (\text{B.458})$$

as required.

Solution to Exercise 9.5

The D -dimensional equation of motion for a generic massive scalar is

$$\phi_k'' - \frac{(D-2)}{\eta} \phi_k' + \left(k^2 + \frac{m^2}{H^2 \eta^2} \right) \phi_k = 0. \quad (\text{B.459})$$

Choosing the conformally coupled mass,

$$\frac{m^2}{H^2} = \frac{D(D-2)}{4}, \quad (\text{B.460})$$

the relevant mode function is

$$f_k(\eta) = -iH^{(D-2)/2} (-\eta)^{(D-2)/2} \frac{e^{-ik\eta}}{\sqrt{2k}}. \quad (\text{B.461})$$

The Wightman function is then given by

$$W_k(\eta, \eta') = \frac{H^{D-2}}{2k} (\eta\eta')^{\frac{D-2}{2}} e^{-ik(\eta-\eta')}. \quad (\text{B.462})$$

The two diagrams we then want to calculate are

$$\begin{aligned} I_+ &= i\lambda \int_{-\infty}^{\eta_*} d\eta a^D(\eta) W_{k_1}(\eta, \eta_*) W_{k_2}(\eta, \eta_*) W_{k_3}(\eta, \eta_*) W_{k_4}(\eta, \eta_*) \\ &= i \frac{\lambda H^{3D-8} (-\eta_*)^{2(D-2)}}{16k_1 k_2 k_3 k_4} e^{iK\eta_*} \int_{-\infty}^{\eta_*} d\eta (-\eta)^{D-4} e^{-iK\eta} \\ &= i^{D-2} \frac{\lambda H^{3D-8} (-\eta_*)^{2(D-2)}}{16k_1 k_2 k_3 k_4 K^{D-3}} \Gamma[D-3], \end{aligned} \quad (\text{B.463})$$

$$\begin{aligned} I_- &= -i\lambda \int_{-\infty}^{\eta_*} d\eta a^D(\eta) W_{k_1}(\eta_*, \eta) W_{k_2}(\eta_*, \eta) W_{k_3}(\eta_*, \eta) W_{k_4}(\eta_*, \eta) \\ &= -i \frac{\lambda H^{3D-8} (-\eta_*)^{2(D-2)}}{16k_1 k_2 k_3 k_4} e^{-iK\eta_*} \int_{-\infty}^{\eta_*} d\eta (-\eta)^{D-4} e^{iK\eta} \\ &= (-i)^{D-2} \frac{\lambda H^{3D-8} (-\eta_*)^{2(D-2)}}{16k_1 k_2 k_3 k_4 K^{D-3}} \Gamma[D-3]. \end{aligned} \quad (\text{B.464})$$

Adding these two contributions together, we obtain the result for the four-point function:

$$\langle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \phi_{\mathbf{k}_4} \rangle' = \cos\left(\frac{(D-2)\pi}{2}\right) \Gamma[D-3] \frac{\lambda H^{3D-8} \eta_*^{2(D-2)}}{8k_1 k_2 k_3 k_4} \frac{1}{K^{D-3}}. \quad (\text{B.465})$$

This has the interesting feature that it vanishes for odd D due to interference between the two branches of the in-in contour. This effect was pointed out in [124], which has more examples.

Solution to Exercise 9.6

Effectively, we just need to do the integral

$$\tilde{I} \equiv (-2k_I)I = \int_{k_{12}}^{\infty} dx \int_{k_{34}}^{\infty} dx' \frac{2k_I}{(x+x')(x+k_I)(x'+k_I)}, \quad (\text{B.466})$$

where the factor of $(-2k_I)$ is for convenience. While this particular integral can be done via computer algebra with sufficient care, it is an opportunity to introduce some mathematical technology to do integrals like this.

First, we note that this integral can be written as

$$I(k_{12}, k_{34}, k_I) = - \int_{k_{12}}^{\infty} dx \int_{k_{34}}^{\infty} dx' d \log \left(\frac{x+k_I}{x+x'} \right) \wedge d \log \left(\frac{x'+k_I}{x+x'} \right), \quad (\text{B.467})$$

which is an integral of a two-form with logarithmic singularities on the boundaries of the triangle formed by the lines $x+k_I=0$, $x'+k_I=0$ and $x+x'=0$ in the (x, x') plane. This two-form is integrated over the projective simplex bounded by $x=k_{12}$, $x'=k_{34}$ and the line at infinity. The function obtained by this integration is an example of an *Aomoto polylogarithm* [125, 126].

(In fact, we will see that this particular function is even simpler and is related to classical polylogarithms.)

There is a useful way to organize the properties of a polylogarithmic function by noting that it can be thought of as an iterated integral of the form

$$F^{(n)} = \int d \log R_1 \circ \cdots \circ d \log R_n, \quad (\text{B.468})$$

where there are n nested integrations and R_a are rational functions of the variables that the function depends on. For functions of this type, it is useful to define the concept of *transcendentality*, which is the number n . The structure of (B.468) implies that its total differential is also quite simple:

$$dF^{(n)} = \sum_a F_a^{(n-1)} d \log R_a, \quad (\text{B.469})$$

which implies that the derivative of a function of transcendentality n can be written as a sum of functions with transcendentality $n - 1$. From this, it is clear that the important information about the function (the rational functions R_a) can be obtained by repeatedly differentiating and reading off the arguments of the $d \log$ terms. This motivates defining the *symbol* [126–128] of a polylogarithmic function recursively as

$$\mathcal{S}(F^{(n)}) = \sum_a \mathcal{S}(F_a^{(n-1)}) \otimes R_a, \quad (\text{B.470})$$

where we are using the tensor product notation $\otimes$ because the symbol is an element of the tensor product space of rational functions R_a . By convention, we have dropped the $d \log$, which is present in (B.468), but it is important that the product $\otimes$ has the distributional properties of the argument of a logarithm. This means that when performing algebraic manipulations, we should implicitly think of each of the R_a as being inside a log; e.g.

$$a_1 \otimes \frac{b_1}{b_2} = a_1 \otimes b_1 - a_1 \otimes b_2. \quad (\text{B.471})$$

Many interesting properties and applications of the symbol are described in [126–129]. The only class of functions whose symbol we will need to consider are classical polylogs, which are defined recursively via

$$\text{Li}_n(z) = \int_0^z \text{Li}_{n-1}(t) d \log t, \quad (\text{B.472})$$

with $\text{Li}_1 \equiv -\log(1 - z)$. This implies that the differential is

$$d\text{Li}_n(z) = \text{Li}_{n-1}(z) d \log z, \quad (\text{B.473})$$

and we can read off the symbol as

$$\mathcal{S}(\text{Li}_n(z)) = -(1 - z) \otimes z \otimes \cdots \otimes z. \quad (\text{B.474})$$

With these preliminaries out of the way, we can now “do” the integral (B.466) by deriving its symbol and then matching to the appropriate polylogarithmic function.

Following [82], we first differentiate with respect to the endpoints of integration

$$\begin{aligned}\frac{\partial \tilde{I}}{\partial k_{12}} &= \int_{k_{34}}^{\infty} dx' \frac{2k_I}{(k_{12} + x')(k_{12} + k_I)(x' + k_I)} \\ &= \left[\int_{k_{34}}^{\infty} dx' \left(\frac{1}{x' + k_I} - \frac{1}{k_{12} + x'} \right) \right] d \log \left(\frac{k_{12} + k_I}{k_{12} - k_I} \right),\end{aligned}\tag{B.475}$$

and similarly for k_{34} . Note that this is of the form (B.469) with

$$R_1 = \frac{k_{12} + k_I}{k_{12} - k_I}, \quad F_1^{(1)} = \int_{k_{34}}^{\infty} dx' \left(\frac{1}{x' + k_I} - \frac{1}{k_{12} + x'} \right),\tag{B.476}$$

and $F_2^{(1)} = F_1^{(1)}(k_{12} \leftrightarrow k_{34})$. Repeating the procedure for these integrals, we find

$$\frac{\partial F_1^{(1)}}{\partial k_{34}} = d \log \left(\frac{k_{34} + k_I}{k_{12} + k_{34}} \right),\tag{B.477}$$

and the same with $k_{12} \leftrightarrow k_{34}$. Putting this all together, we see that the symbol of $\tilde{I}$ is

$$\mathcal{S}(\tilde{I}) = \frac{k_{34} + k_I}{k_{12} + k_{34}} \otimes \frac{k_{12} + k_I}{k_{12} - k_I} + \frac{k_{12} + k_I}{k_{12} + k_{34}} \otimes \frac{k_{34} + k_I}{k_{34} - k_I}.\tag{B.478}$$

We can write this in a more transparent form by defining the variables

$$u \equiv \frac{k_{12} - k_I}{k_{12} + k_{34}}, \quad v \equiv \frac{k_{34} - k_I}{k_{12} + k_{34}},\tag{B.479}$$

and then expanding everything to obtain [130]

$$\mathcal{S}(\tilde{I}) = (1 - u) \otimes u + (1 - v) \otimes v - (1 - v) \otimes (1 - u) - (1 - u) \otimes (1 - v).\tag{B.480}$$

From (B.474), we see that the first two terms are symbols for $\text{Li}_2(u)$ and $\text{Li}_2(v)$, while the last two terms are the symbols of products of ordinary logarithms. This implies that

$$\tilde{I} = -\text{Li}_2(u) - \text{Li}_2(v) - \log(1 - u) \log(1 - v) + C,\tag{B.481}$$

where the integration constant C accounts for the fact that the symbol can only determine the function up to a constant. However, we can fix this constant from physics. Looking at (B.466), it is regular when $k_I \rightarrow 0$. In fact, it vanishes. In terms of u, v , this corresponds to $u + v = 1$. We can use this to fix C by requiring [130]

$$-\text{Li}_2(u) - \text{Li}_2(1 - u) - \log(1 - u) \log(u) + C = 0.\tag{B.482}$$

Famously, the dilogarithm obeys a reflection identity of this type with $C = \pi^2/6$, which fixes this value. Substituting (B.481) into (9.78) then gives

$$\begin{aligned}I_{++}^{(c)} &\equiv -\frac{\lambda^2 H^2 \eta_*^4}{32k_1 k_2 k_3 k_4 k_I} \int_{k_{12}}^{\infty} dx \int_{k_{34}}^{\infty} dx' \tilde{I}(x, x') \\ &= \frac{\lambda^2 H^2 \eta_*^4}{32k_1 k_2 k_3 k_4 k_I} \left[\text{Li}_2 \left(\frac{E - E_L}{E} \right) + \text{Li}_2 \left(\frac{E - E_R}{E} \right) + \log \left(\frac{E_L}{E} \right) \log \left(\frac{E_R}{E} \right) - \frac{\pi^2}{6} \right],\end{aligned}\tag{B.483}$$

as expected.

Solution to Exercise 9.7

We start from (9.88) and (9.89):

$$I_{++} = -\frac{\lambda^2 \eta_*^4}{16k_1 k_2 k_3 k_4} \int_{-\infty}^{\eta_*} \frac{d\eta}{\eta^2} \int_{-\infty}^{\eta_*} \frac{d\eta'}{\eta'^2} e^{ik_{12}\eta} G_F(k_I; \eta, \eta') e^{ik_{34}\eta'}, \quad (\text{B.484})$$

$$I_{+-} = \frac{\lambda^2 \eta_*^4}{16k_1 k_2 k_3 k_4} \int_{-\infty}^{\eta_*} \frac{d\eta}{\eta^2} \int_{-\infty}^{\eta_*} \frac{d\eta'}{\eta'^2} e^{ik_{12}\eta} W_{k_I}(\eta', \eta) e^{-ik_{34}\eta'}. \quad (\text{B.485})$$

In the $k_I \rightarrow 0$ limit, we can use that both the Feynman and Wightman propagators are given by

$$\begin{aligned} \lim_{k_I \rightarrow 0} G_F(k_I; \eta, \eta') &= \lim_{k_I \rightarrow 0} W_{k_I}(\eta, \eta') \\ &= \frac{H^2}{4\pi} (\eta\eta')^{3/2} \left[\Gamma[-i\mu]^2 \left(\frac{k_I^2 \eta \eta'}{4} \right)^{i\mu} + \Gamma[i\mu]^2 \left(\frac{k_I^2 \eta \eta'}{4} \right)^{-i\mu} \right]. \end{aligned} \quad (\text{B.486})$$

With this simplification, all integrals will be of the form

$$\int_{-\infty}^0 d\eta e^{\pm i\alpha\eta} (-\eta)^{\Delta-2} = (\mp i)^{\Delta-1} \alpha^{1-\Delta} \Gamma[\Delta-1]. \quad (\text{B.487})$$

Using this formula, we find

$$I_{++} = -\frac{\lambda^2 H^2 \eta_*^4}{64\pi k_1 k_2 k_3 k_4 (k_{12} k_{34})^{1/2}} \left[(-i)^{1+2i\mu} \Gamma[\tfrac{1}{2} + i\mu]^2 \Gamma[-i\mu]^2 \left(\frac{k_I^2}{4k_{12} k_{34}} \right)^{i\mu} + \text{c.c.} \right], \quad (\text{B.488})$$

$$I_{+-} = \frac{\lambda^2 H^2 \eta_*^4}{64\pi k_1 k_2 k_3 k_4 (k_{12} k_{34})^{1/2}} \left[\Gamma[\tfrac{1}{2} + i\mu]^2 \Gamma[-i\mu]^2 \left(\frac{k_I^2}{4k_{12} k_{34}} \right)^{i\mu} + \text{c.c.} \right]. \quad (\text{B.489})$$

Combining these contributions, along with $I_{--} = I_{++}^*$ and $I_{-+} = I_{+-}^*$, we obtain [\[AJ: check the sign here, pull out overall cosh factor\]](#)

$$\begin{aligned} \langle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \phi_{\mathbf{k}_4} \rangle' &= \frac{\lambda^2 H^2 \eta_*^4}{64\pi k_1 k_2 k_3 k_4 (k_{12} k_{34})^{1/2}} \left[(1 - i \sinh \pi\mu) \Gamma[\tfrac{1}{2} + i\mu]^2 \Gamma[-i\mu]^2 \left(\frac{k_I^2}{4k_{12} k_{34}} \right)^{i\mu} \right. \\ &\quad \left. + (1 + i \sinh \pi\mu) \Gamma[\tfrac{1}{2} - i\mu]^2 \Gamma[i\mu]^2 \left(\frac{k_I^2}{4k_{12} k_{34}} \right)^{-i\mu} \right]. \end{aligned} \quad (\text{B.490})$$

One of the interesting features of this expression is that it oscillates as a function of $k_I^2/(k_{12} k_{34})$, where the period of the oscillation is set by μ . Since μ is in turn fixed by the mass of the exchanged particle, this oscillatory behavior can be used to perform a kind of spectroscopy on the exchanged particles to extract their mass. See [\[43\]](#), for more details on this ‘‘cosmological collider physics.’’

Solution to Exercise 9.8

This solution follows the treatment in [\[70\]](#). We start with the inflaton action in the comoving gauge:

$$\begin{aligned} S &= \frac{1}{2} \int d^4x \sqrt{-g} \left[R - g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - 2V \right] \\ &= \frac{1}{2} \int d^4x \sqrt{h} N \left[R^{(3)} - 2V + N^{-2} (E^{ij} E_{ij} - E^2 + \dot{\phi}^2) \right], \end{aligned} \quad (\text{B.491})$$

where

$$R_{(3)} = a^{-2}e^{-2\zeta} (-4\partial^2\zeta - 2(\partial\zeta)^2), \quad (\text{B.492})$$

$$\begin{aligned} E_{ij}E^{ij} - E^2 = & -6(H + \dot{\zeta})^2 + 4a^{-2}e^{-2\zeta}(H + \dot{\zeta})(\partial_i N_i + N_i \partial_i \zeta) \\ & - a^{-4}e^{-4\zeta} \left[(\partial_i N_i)^2 + 2(N_i \partial_i \zeta)^2 - (\partial_{(i} N_{j)}) - 2N_{(i} \partial_{j)} \zeta \right]^2. \end{aligned} \quad (\text{B.493})$$

Substituting this into the action (together with $N = 1 + \alpha$ and $N_i = \partial_i \beta$) and expanding to third order, we get

$$\begin{aligned} S_3 = \int dt d^3x a^3 \Big\{ & 3H^2\alpha^3 - \varepsilon H^2\alpha^3 - 9H^2\alpha^2\dot{\zeta} + 3\varepsilon H^2\alpha^2\dot{\zeta} + \frac{27}{2}H^2\alpha\dot{\zeta}^2 - \frac{9}{2}\varepsilon H^2\alpha\dot{\zeta}^2 - \frac{27}{2}H^2\dot{\zeta}^3 \\ & + \frac{9}{2}\varepsilon H^2\dot{\zeta}^3 - 6H\alpha^2\dot{\zeta} + 18H\alpha\dot{\zeta}\dot{\zeta} - 27H\dot{\zeta}^2\dot{\zeta} + 3\alpha(\dot{\zeta})^2 - 9\dot{\zeta}(\dot{\zeta})^2 - \frac{2}{a^2}H\alpha\partial_i\beta\partial_i\zeta + \frac{2}{a^2}H\dot{\zeta}\partial_i\beta\partial_i\zeta \\ & + \frac{2}{a^2}\partial_i\beta\dot{\zeta}\partial_i\zeta - \frac{1}{a^2}\alpha(\partial_i\zeta)^2 - \frac{1}{a^2}\dot{\zeta}(\partial_i\zeta)^2 + \frac{2}{a^2}H\alpha^2\partial^2\beta - \frac{2}{a^2}H\alpha\dot{\zeta}\partial^2\beta + \frac{1}{a^2}H\dot{\zeta}^2\partial^2\beta \\ & - \frac{2}{a^2}\alpha\dot{\zeta}\partial^2\beta + \frac{2}{a^2}\dot{\zeta}\dot{\zeta}\partial^2\beta - \frac{2}{a^4}\partial_j\beta\partial_i\zeta\partial_i\partial_j\beta - \frac{1}{2a^4}\alpha(\partial_i\partial_j\beta)^2 - \frac{1}{2a^4}\dot{\zeta}(\partial_i\partial_j\beta)^2 + \frac{1}{2a^4}\alpha(\partial^2\beta)^2 \\ & + \frac{1}{2a^4}\dot{\zeta}(\partial^2\beta)^2 - \frac{2}{a^2}\alpha\dot{\zeta}\partial^2\zeta - \frac{1}{a^2}\dot{\zeta}^2\partial^2\zeta - \frac{9}{2}V\alpha\dot{\zeta}^2 - \frac{9}{2}V\dot{\zeta}^3 \Big\}. \end{aligned} \quad (\text{B.494})$$

Inserting the constrained variables

$$\alpha = \frac{\dot{\zeta}}{H}, \quad \partial^2\beta = -\frac{\partial^2\zeta}{H} + a^2\varepsilon\dot{\zeta}, \quad (\text{B.495})$$

and doing a lot of integrating by parts, we get

$$\begin{aligned} S_3 = \int dt d^3x \Big\{ & a^3\varepsilon^2\dot{\zeta}\dot{\zeta}^2 + a\varepsilon^2\dot{\zeta}(\partial\zeta)^2 - 2a\varepsilon\dot{\zeta}\partial\zeta\partial\chi + \mathcal{L}_b \\ & + \frac{a^3\varepsilon}{2}\dot{\kappa}\dot{\zeta}^2\dot{\zeta} + \frac{\varepsilon}{2a}\partial\zeta\partial\chi\partial^2\chi + \frac{\varepsilon}{4a}\partial^2\zeta(\partial\chi)^2 + 2f(\zeta)\frac{\delta\mathcal{L}_2}{\delta\zeta} \Big\}, \end{aligned} \quad (\text{B.496})$$

where we defined $\partial^2\chi \equiv a^2\varepsilon\dot{\zeta}$ and

$$\frac{\delta\mathcal{L}_2}{\delta\zeta} \equiv \partial_t(a^3\varepsilon\dot{\zeta}) - a\varepsilon\partial^2\zeta, \quad (\text{B.497})$$

$$f(\zeta) \equiv \frac{\kappa}{4}\dot{\zeta}^2 + \frac{\dot{\zeta}\dot{\zeta}}{H} + \frac{-(\partial\zeta)^2 + \partial^{-2}(\partial_i\partial_j(\partial_i\zeta\partial_j\zeta))}{4a^2H^2} + \frac{\partial\zeta\partial\chi - \partial^{-2}(\partial_i\partial_j(\partial_i\zeta\partial_j\chi))}{2a^2H^2}. \quad (\text{B.498})$$

Using the linearized equation of motion, the term proportional to $f(\zeta)$ vanishes. Alternatively, Maldacena showed that its contribution can be removed by the field redefinition $\zeta \rightarrow \tilde{\zeta} + f(\tilde{\zeta})$. The other terms in the second line of are higher order in the slow-roll parameter, $\mathcal{O}(\varepsilon^2)$, and will therefore be dropped.

The boundary terms in (B.9) are

$$\mathcal{L}_b = \partial_t T + \partial_i S_i, \quad (\text{B.499})$$

where

$$T \equiv -9a^3 H \zeta^3 + \frac{a}{H} \zeta (\partial \zeta)^2 - \frac{1}{4aH^3} (\partial \zeta)^2 \partial^2 \zeta - \frac{a\varepsilon}{H} \zeta (\partial \zeta)^2 - \frac{\varepsilon a^3}{H} \zeta \dot{\zeta}^2 + \frac{1}{2aH^2} \zeta (\partial_i \partial_j \zeta \partial_i \partial_j \chi - \partial^2 \zeta \partial^2 \chi) - \frac{a\kappa}{2} \zeta^2 \partial^2 \chi - \frac{1}{2aH} \zeta (\partial_i \partial_j \chi \partial_i \partial_j \chi - \partial^2 \chi \partial^2 \chi), \quad (\text{B.500})$$

$$S_i \equiv a^3 \varepsilon H \zeta^2 \partial_i \partial^{-2} \dot{\zeta} - 2a \zeta^2 \partial_i \dot{\zeta} - \frac{2a}{H} \zeta \dot{\zeta} \partial_i \zeta + \frac{2a\varepsilon}{H} \zeta \partial_j \zeta \partial_i \partial_j \partial^{-2} \dot{\zeta} + \frac{a\varepsilon}{H} \partial_j \zeta \partial_j \zeta \partial_i \partial^{-2} \dot{\zeta} + \frac{1}{4aH^2} (-3\partial_i \zeta \partial_j \zeta \partial_j \dot{\zeta} - 2\zeta \partial_j \zeta \partial_i \partial_j \dot{\zeta} + 2\zeta \partial_i \zeta \partial^2 \dot{\zeta}) + \frac{1}{4aH^3} (2\partial^2 \zeta \partial_i \dot{\zeta} - 2\partial_i \partial_j \zeta \partial_j \dot{\zeta} + \partial_j \zeta \partial_j \partial_i \dot{\zeta}) + \frac{\varepsilon}{4aH^2} (2\zeta \partial^2 \zeta \partial_i \dot{\zeta} - 2\zeta \partial_i \partial_j \zeta \partial_j \dot{\zeta} + \partial_i \zeta \partial_j \zeta \partial_j \dot{\zeta}) + \frac{a\varepsilon}{2H^2} (\dot{\zeta} \partial_j \zeta \partial_i \partial_j \zeta^{-2} \dot{\zeta} - \zeta \partial_j \dot{\zeta} \partial_i \partial_j \zeta^{-2} \dot{\zeta} + \zeta \dot{\zeta} \partial_i \dot{\zeta} - \dot{\zeta}^2 \partial_i \zeta). \quad (\text{B.501})$$

The spatial part $\partial_i S_i$ doesn't contribute to the three-point function and so can be dropped. The total time derivative $\partial_t T$ must be kept because we do have a time boundary. However, in the late-time limit, where $a \rightarrow \infty$, only one term survives in $\partial_t T$ and we get [131, 132]

$$\mathcal{L}_b = \partial_t \left(-\frac{\varepsilon \kappa}{2} a^3 \zeta^2 \dot{\zeta} \right). \quad (\text{B.502})$$

Hence, we are left with

$$S_3 = \int d\eta d^3x \left\{ a^2 \varepsilon^2 \left[\zeta (\zeta')^2 + \zeta (\partial_i \zeta)^2 - 2\zeta' \partial_i \zeta \partial_i^{-1} \zeta' \right] - \frac{1}{2} \partial_\eta (a^2 \varepsilon \kappa \zeta^2 \zeta') \right\}, \quad (\text{B.503})$$

as quoted in the text.

Solution to Exercise 9.9

The interaction Hamiltonian is

$$H_{\text{int}} = -L_3 = - \int d^3x \left\{ a^2 \varepsilon^2 \left[\zeta (\zeta')^2 + \zeta (\partial_i \zeta)^2 - 2\zeta' \partial_i \zeta \partial_i^{-2} \zeta' \right] - \frac{1}{2} \partial_\eta (a^2 \varepsilon \kappa \zeta^2 \zeta') \right\}. \quad (\text{B.504})$$

To compute the associated bispectrum, we feed this into

$$\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle = 2 \text{Im} \left(\int_{-\infty}^0 d\eta \langle 0 | \hat{\zeta}_{\mathbf{k}_1} \hat{\zeta}_{\mathbf{k}_2} \hat{\zeta}_{\mathbf{k}_3}(0) \hat{H}_{\text{int}}(\eta) | 0 \rangle \right), \quad (\text{B.505})$$

where the mode expansion of the field operator is

$$\hat{\zeta}_{\mathbf{k}}(\eta) = f_{\mathbf{k}}(\eta) \hat{a}_{\mathbf{k}} + \text{h.c.}, \quad f_{\mathbf{k}}(\eta) \equiv \frac{H}{\sqrt{4\varepsilon k^3}} (1 + ik\eta) e^{-ik\eta}. \quad (\text{B.506})$$

We consider each interaction in turn:

- For the interaction $\zeta (\zeta')^2$, we get

$$\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle'_{(1)} = -2\varepsilon^2 \text{Im} \left(\prod_{i=1}^3 f_{k_i}(0) \int_{-\infty}^0 \frac{d\eta}{(H\eta)^2} f_{k_1}^* \partial_\eta f_{k_2}^* \partial_\eta f_{k_3}^* \right) + \text{perms.}, \quad (\text{B.507})$$

where

$$\begin{aligned} f_k^*(\eta) &= \frac{H}{\sqrt{4\varepsilon k^3}}(1 - ik\eta)e^{ik\eta}, \\ \partial_\eta f_k^* &= \frac{H}{\sqrt{4\varepsilon k^3}}k^2\eta e^{ik\eta}. \end{aligned} \quad (\text{B.508})$$

Hence, we have

$$\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle'_{(1)} = -\frac{1}{32\varepsilon} \frac{H^4}{(k_1 k_2 k_3)^3} \left[\text{Im}(\mathcal{I}(k_1, k_2, k_3)) + \text{perms.} \right], \quad (\text{B.509})$$

where

$$\begin{aligned} \mathcal{I}(k_1, k_2, k_3) &\equiv \int_{-\infty}^0 d\eta (1 - ik_1\eta) k_2^2 k_3^2 e^{i(k_1+k_2+k_3)\eta} \\ &= k_2^2 k_3^2 (1 - k_1 \partial_{k_1}) \int_{-\infty}^0 d\eta e^{i(k_1+k_2+k_3)\eta} \\ &= k_2^2 k_3^2 (1 - k_1 \partial_{k_1}) \frac{1}{i(k_1 + k_2 + k_3)} \\ &= -ik_2^2 k_3^2 \left(\frac{1}{k_1 + k_2 + k_3} + \frac{k_1}{(k_1 + k_2 + k_3)^2} \right). \end{aligned} \quad (\text{B.510})$$

Equation (B.509) then becomes

$$\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle'_{(1)} = \frac{1}{16\varepsilon^2} \frac{H^4}{(k_1 k_2 k_3)^2} F_{(1)}, \quad (\text{B.511})$$

where

$$F_{(1)} = \frac{\varepsilon}{k_1 k_2 k_3} \left[\frac{k_1^2 k_2^2 + k_1^2 k_3^2 + k_2^2 k_3^2}{K} + \frac{k_1 k_2^2 k_3^2 + k_2 k_1^2 k_3^2 + k_3 k_1^2 k_2^2}{K^2} \right], \quad (\text{B.512})$$

with $K \equiv k_1 + k_2 + k_3$. The result in (B.512) can also be written as

$$F_{(1)} = \frac{\varepsilon}{k_1 k_2 k_3} \left[\frac{2}{K} \sum_{i>j} k_i^2 k_j^2 - \frac{1}{K^2} \sum_{i \neq j} k_i^2 k_j^3 \right]. \quad (\text{B.513})$$

Just write out (B.513) explicitly to see that it is equivalent to (B.512).

- Similarly, for the interaction $\zeta(\partial_i \zeta)^2$, we get

$$\begin{aligned} \langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle'_{(2)} &= 2\varepsilon^2 (\mathbf{k}_2 \cdot \mathbf{k}_3) \text{Im} \left(\prod_{i=1}^3 f_{k_i}(0) \int_{-\infty}^0 \frac{d\eta}{(H\eta)^2} f_{k_1}^* f_{k_2}^* f_{k_3}^* \right) + \text{perms.} \\ &= \frac{1}{32\varepsilon} \frac{H^4}{(k_1 k_2 k_3)^3} \left[(\mathbf{k}_2 \cdot \mathbf{k}_3) \text{Im}(\mathcal{J}(k_1, k_2, k_3)) + \text{perms.} \right], \end{aligned} \quad (\text{B.514})$$

where

$$\begin{aligned} \mathcal{J}(k_1, k_2, k_3) &\equiv \int_{-\infty}^0 \frac{d\eta}{\eta^2} (1 - ik_1\eta)(1 - ik_2\eta)(1 - ik_3\eta) e^{i(k_1+k_2+k_3)\eta} \\ &= \int_{-\infty}^0 d\eta \left(\frac{1}{\eta^2} - \frac{iK}{\eta} - (k_1 k_2 + k_1 k_3 + k_2 k_3) + ik_1 k_2 k_3 \eta \right) e^{iK\eta}. \end{aligned} \quad (\text{B.515})$$

In Exercise 9.3, we showed that

$$\int_{-\infty}^{\eta_*} d\eta \frac{e^{iK\eta}}{\eta^2} = iK \int_{-\infty}^{\eta_*} d\eta \frac{e^{iK\eta}}{\eta} - \frac{e^{iK\eta_*}}{\eta_*}. \quad (\text{B.516})$$

Using this identity, we can write (B.515) as

$$\begin{aligned} \mathcal{J}(k_1, k_2, k_3) &= -\frac{e^{iK\eta_*}}{\eta_*} - \left(k_1 k_2 + k_1 k_3 + k_2 k_3 - k_1 k_2 k_3 \partial_K \right) \int_{-\infty}^0 d\eta e^{iK\eta} \\ &= -\left(\frac{1}{\eta_*} + iK + \dots \right) + i \frac{k_1 k_2 + k_1 k_3 + k_2 k_3}{K} + i \frac{k_1 k_2 k_3}{K^2}. \end{aligned} \quad (\text{B.517})$$

Notice that this integral diverges for $\eta_* \rightarrow 0$. However, this divergent part is purely real, so that

$$\text{Im}(\mathcal{J}(k_1, k_2, k_3)) = -K + \frac{k_1 k_2 + k_1 k_3 + k_2 k_3}{K} + \frac{k_1 k_2 k_3}{K^2}. \quad (\text{B.518})$$

Using

$$\begin{aligned} \mathbf{k}_2 \cdot \mathbf{k}_3 + \text{perms.} &= \frac{1}{2}(k_1^2 - k_2^2 - k_3^2) + \text{perms.} \\ &= 2 \times \left[\frac{1}{2}(k_1^2 - k_2^2 - k_3^2) + \frac{1}{2}(k_2^2 - k_1^2 - k_3^2) + \frac{1}{2}(k_3^2 - k_1^2 - k_2^2) \right] \\ &= -(k_1^2 + k_2^2 + k_3^2), \end{aligned} \quad (\text{B.519})$$

the bispectrum contribution (B.514) can be written in the form (B.511), with

$$F_{(2)} = \frac{\varepsilon}{k_1 k_2 k_3} \frac{k_1^2 + k_2^2 + k_3^2}{2} \left(K - \frac{k_1 k_2 + k_1 k_3 + k_2 k_3}{K} - \frac{k_1 k_2 k_3}{K^2} \right), \quad (\text{B.520})$$

which is equivalent to

$$F_{(2)} = \frac{\varepsilon}{k_1 k_2 k_3} \left[\frac{1}{2} \sum_i k_i^3 + \frac{2}{K} \sum_{i>j} k_i^2 k_j^2 - \frac{1}{K^2} \sum_{i \neq j} k_i^2 k_j^3 \right]. \quad (\text{B.521})$$

- Next, we consider the non-local interaction $\zeta' \partial_i \zeta \partial_i (\partial^{-2} \zeta')$. The associated bispectrum is

$$\begin{aligned} \langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle'_{(3)} &= 4\varepsilon^2 \frac{(\mathbf{k}_2 \cdot \mathbf{k}_3)}{k_3^2} \text{Im} \left(\prod_{i=1}^3 f_{k_i}(0) \int_{-\infty}^0 \frac{d\eta}{(H\eta)^2} \partial_\eta f_{k_1}^* f_{k_2}^* \partial_\eta f_{k_3}^* \right) + \text{perms.} \\ &= \frac{1}{32\varepsilon} \frac{H^4}{(k_1 k_2 k_3)^3} \left[\frac{k_1^2 - k_2^2 - k_3^2}{k_3^2} \text{Im}(\mathcal{K}(k_1, k_2, k_3)) + \text{perms.} \right], \end{aligned} \quad (\text{B.522})$$

where

$$\begin{aligned} \mathcal{K}(k_1, k_2, k_3) &\equiv \int_{-\infty}^0 d\eta k_1^2 k_3^2 (1 - ik_2 \eta) e^{i(k_1 + k_2 + k_3)\eta} \\ &= -ik_1^2 k_3^2 \left(\frac{1}{k_1 + k_2 + k_3} + \frac{k_2}{(k_1 + k_2 + k_3)^2} \right). \end{aligned} \quad (\text{B.523})$$

The bispectrum contribution (B.522) then takes the form (B.511), with

$$F_{(3)} = \frac{\varepsilon}{k_1 k_2 k_3} \left[-\frac{k_1^2(k_1^2 - k_2^2 - k_3^2)}{2} \left(\frac{1}{K} + \frac{k_2}{K^2} \right) + \text{perms.} \right], \quad (\text{B.524})$$

which is equivalent to

$$F_{(3)} = \frac{\varepsilon}{k_1 k_2 k_3} \left[-\sum_i k_i^3 + \frac{1}{2} \sum_{i \neq j} k_i k_j^2 + \frac{2}{K^2} \sum_{i \neq j} k_i^2 k_j^3 \right]. \quad (\text{B.525})$$

- Finally, from the boundary term, we get

$$\begin{aligned} \langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle'_{(4)} &= -\varepsilon \kappa \text{Im} \left(a^2(\eta_*) |f_{k_1}(\eta_*)|^2 |f_{k_2}(\eta_*)|^2 f_{k_3}(\eta_*) \partial_\eta f_{k_3}^*(\eta_*) \right) + \text{perms.} \\ &= -\frac{\kappa}{64\varepsilon^2} \frac{H^4}{(k_1 k_2 k_3)^3} \text{Im} \left(\frac{1}{\eta_*^2} (1 - i k_3 \eta_*) k_3^2 \eta_* \right) + \text{perms.} \\ &= -\frac{\kappa}{64\varepsilon^2} \frac{H^4}{(k_1 k_2 k_3)^3} \left[\text{Im} \left(\frac{k_3^2}{\eta_*} - i k_3^3 \right) + \text{perms.} \right]. \end{aligned} \quad (\text{B.526})$$

Notice that the divergence for $\eta_* \rightarrow 0$ doesn't contribute when we take the imaginary part. We therefore find

$$\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle'_{(4)} = \frac{1}{16\varepsilon^2} \frac{H^4}{(k_1 k_2 k_3)^2} F_{(4)}, \quad \text{with} \quad F_{(4)} = \frac{\kappa}{k_1 k_2 k_3} \left(\frac{1}{2} \sum_i k_i^3 \right). \quad (\text{B.527})$$

Combining these four contributions, we find

$$F = \sum_{i=1}^4 F_{(i)} = \frac{1}{k_1 k_2 k_3} \left[\frac{\varepsilon}{2} \left(\sum_{i \neq j} k_i k_j^2 + \frac{8}{K} \sum_{i > j} k_i^2 k_j^2 \right) + \frac{\kappa - \varepsilon}{2} \sum_i k_i^3 \right], \quad (\text{B.528})$$

as required. Note that the K^{-2} poles have cancelled.

Recall from (8.72) that the dimensionless bispectrum is defined as

$$\begin{aligned} B_\zeta(k_1, k_2, k_3) &\equiv \frac{(k_1 k_2 k_3)^2}{(2\pi^2)^2} \langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle' = \left(\frac{H^2}{8\pi^2 \varepsilon} \right)^2 F(k_1, k_2, k_3) \\ &= \Delta_\zeta^4 \times F(k_1, k_2, k_3), \end{aligned} \quad (\text{B.529})$$

and its amplitude is

$$\begin{aligned} f_{\text{NL}} &\equiv \frac{5}{18} \frac{B_\zeta(k, k, k)}{\Delta_\zeta^4(k)} = \frac{5}{18} F(k, k, k) \\ &= \frac{5}{18} \left[\frac{\varepsilon}{2} (6 + 8 \times 3) + \frac{\kappa - \varepsilon}{2} \times 3 \right] \\ &= \frac{5}{12} (9\varepsilon + \kappa), \end{aligned} \quad (\text{B.530})$$

as desired.

Solution to Exercise 9.10

The squeezed triangle correlates one long-wavelength mode, $k_L = k_3$, to two short-wavelength modes, $k_S = k_1 \approx k_2$,

$$\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle \rightarrow \langle (\zeta_{\mathbf{k}_S})^2 \zeta_{\mathbf{k}_L} \rangle. \quad (\text{B.531})$$

Modes with longer wavelengths freeze earlier. The long mode $\zeta_{\mathbf{k}_L}$ will therefore already be frozen and act as a classical background wave when the two short modes $\zeta_{\mathbf{k}_S}$ exit the horizon. This provides an intuitive way to study the correlations between long and short modes.

Why should $(\zeta_{\mathbf{k}_S})^2$ be correlated with $\zeta_{\mathbf{k}_L}$? The theorem says that it isn't correlated if $\zeta_{\mathbf{k}}$ is scale-invariant, but that the short scale power does get modified by the long-wavelength mode if $n_s \neq 1$. Let's see why. We decompose the evaluation of (B.531) into two steps:

- i) we calculate the power spectrum of short fluctuations $\langle \zeta_S^2 \rangle_{\zeta_L}$ in the presence of ζ_L ;
- ii) we then calculate the correlation $\langle (\zeta_S)^2 \zeta_L \rangle$, i.e. average the short-scale power spectrum over realizations of the long modes.

The calculation of $\langle \zeta_S^2 \rangle_{\zeta_L}$ is simplest in real space: When the background mode is homogeneous, $\zeta_L(x) \equiv \bar{\zeta}_L$, it can be reabsorbed simply by a rescaling of the spatial coordinates, $\tilde{x}^i = e^{\bar{\zeta}_L} x^i$ (recall that $ds^2 = -dt^2 + a^2(t)e^{2\zeta(\mathbf{x},t)}d\mathbf{x}^2$). After this rescaling, $\bar{\zeta}_L$ no longer appears in the action, so that the two-point function in the new coordinates is the same as in the absence of $\bar{\zeta}_L$. In other words, in the limit of constant $\bar{\zeta}_L$, we have

$$\langle \zeta_S(\mathbf{x}_1) \zeta_S(\mathbf{x}_2) \rangle_{\bar{\zeta}_L} = \langle \zeta_S(\tilde{\mathbf{x}}_1) \zeta_S(\tilde{\mathbf{x}}_2) \rangle. \quad (\text{B.532})$$

When ζ_L is slowly varying, we can evaluate it at the middle point $\mathbf{x}_+ \equiv \frac{1}{2}(\mathbf{x}_1 + \mathbf{x}_2)$ to get

$$\tilde{\mathbf{x}}_- \simeq \mathbf{x}_- + \zeta_L(\mathbf{x}_+) \cdot \mathbf{x}_- + \dots, \quad (\text{B.533})$$

where we defined $\mathbf{x}_- \equiv \mathbf{x}_1 - \mathbf{x}_2$. The two-point function at linear order in ζ_L therefore is

$$\langle \zeta_S(\mathbf{x}_2) \zeta_S(\mathbf{x}_3) \rangle_{\zeta_L(x)} \approx \xi_S(|\mathbf{x}_-|) + \zeta_L(\mathbf{x}_+) [\mathbf{x}_- \cdot \nabla \xi_S(|\mathbf{x}_-|)], \quad (\text{B.534})$$

where

$$\xi_S(|\mathbf{x}_-|) \equiv \int \frac{d^3 k_S}{(2\pi)^3} P_\zeta(k_S) e^{i\mathbf{k}_S \cdot \mathbf{x}_-}. \quad (\text{B.535})$$

The three-point function then is

$$\begin{aligned} \langle \zeta_S(\mathbf{x}_1) \zeta_S(\mathbf{x}_2) \zeta_L(\mathbf{x}_3) \rangle &\approx \langle \zeta_L(\mathbf{x}_3) \zeta_L(\mathbf{x}_+) \rangle [\mathbf{x}_- \cdot \nabla \xi_S(|\mathbf{x}_-|)] \\ &= \int \frac{d^3 k_L}{(2\pi)^3} \int \frac{d^3 k_S}{(2\pi)^3} e^{i\mathbf{k}_L \cdot (\mathbf{x}_3 - \mathbf{x}_+)} P_\zeta(k_L) P_\zeta(k_S) \left[\mathbf{k}_S \cdot \frac{\partial}{\partial \mathbf{k}_S} \right] e^{i\mathbf{k}_S \cdot \mathbf{x}_-}. \end{aligned} \quad (\text{B.536})$$

Integrating by parts, inserting $1 = \int d^3 k_3 \delta(\mathbf{k}_3 + \mathbf{k}_L)$ and using

$$\frac{\partial}{\partial \mathbf{k}_S} \cdot [\mathbf{k}_S P_\zeta(k_S)] = P_\zeta(k_S) \frac{d \log(k_S^3 P_\zeta(k_S))}{d \log k_S}, \quad (\text{B.537})$$

we get

$$\begin{aligned} \langle \zeta_S(\mathbf{x}_1) \zeta_S(\mathbf{x}_2) \zeta_L(\mathbf{x}_3) \rangle &= - \int \frac{d^3 k_3}{(2\pi)^3} \int \frac{d^3 k_L}{(2\pi)^3} \int \frac{d^3 k_S}{(2\pi)^3} e^{-i\mathbf{k}_1 \cdot \mathbf{x}_1 - i\mathbf{k}_L \cdot \mathbf{x}_+ + i\mathbf{k}_S \cdot \mathbf{x}_-} \\ &\quad \times \left[(2\pi)^3 \delta(\mathbf{k}_1 + \mathbf{k}_L) P_\zeta(k_1) P_\zeta(k_S) \frac{d \log(k_S^3 P_\zeta(k_S))}{d \log k_S} \right]. \end{aligned} \quad (\text{B.538})$$

Letting $\mathbf{k}_L = \mathbf{k}_2 + \mathbf{k}_3$ and $\mathbf{k}_S = \frac{1}{2}(\mathbf{k}_2 - \mathbf{k}_3)$, we get $-i\mathbf{k}_L \cdot \mathbf{x}_+ + i\mathbf{k}_S \cdot \mathbf{x}_- = -i\mathbf{k}_2 \cdot \mathbf{x}_2 - i\mathbf{k}_3 \cdot \mathbf{x}_3$. Changing variables in the integration and Fourier transforming, we get

$$\begin{aligned} \lim_{k_3 \rightarrow 0} \langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle &= -(2\pi)^3 \delta(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3) P_\zeta(k_1) P_\zeta(k_3) \frac{d \log k_3^3 P_\zeta(k_3)}{d \log k_3}, \\ &= (2\pi)^3 \delta(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3) (1 - n_s) P_\zeta(k_1) P_\zeta(k_3). \end{aligned} \quad (\text{B.539})$$

This completes the proof.

Solution to Exercise 9.11

Our starting point in the cubic action (9.104), which in conformal time reads

$$S = \int d\eta d^3 x a^2(\eta) \left[\varepsilon \left((\zeta')^2 - (\partial_i \zeta)^2 \right) - \frac{\dot{\phi}^4}{2\Lambda^4 H^3} \frac{1}{a(\eta)} \left((\zeta')^3 - \zeta' (\partial_i \zeta)^2 \right) \right], \quad (\text{B.540})$$

and the interaction Hamiltonian is

$$H_{\text{int}} = \frac{\dot{\phi}^4}{2\Lambda^4 H^3} \int d^3 x a(\eta) \left[(\zeta')^3 - \zeta' (\partial_i \zeta)^2 \right]. \quad (\text{B.541})$$

To compute the associated bispectrum, we feed this into

$$\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle = 2 \text{Im} \left(\int_{-\infty}^0 d\eta \langle 0 | \hat{\zeta}_{\mathbf{k}_1} \hat{\zeta}_{\mathbf{k}_2} \hat{\zeta}_{\mathbf{k}_3}(0) \hat{H}_{\text{int}}(\eta) | 0 \rangle \right), \quad (\text{B.542})$$

where the mode expansion of the field operator is

$$\hat{\zeta}_{\mathbf{k}}(\eta) = f_k(\eta) \hat{a}_{\mathbf{k}} + \text{h.c.}, \quad f_k(\eta) \equiv \frac{H}{\sqrt{4\varepsilon k^3}} (1 + ik\eta) e^{-ik\eta}. \quad (\text{B.543})$$

We consider each interaction in turn:

- For the interaction ζ^3 , we get

$$\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle'_{\zeta^3} = \frac{6\dot{\phi}^4}{\Lambda^4 H^3} \text{Im} \left(\prod_{i=1}^3 f_{k_i}(0) \int_{-\infty}^0 \frac{d\eta}{(-H\eta)} \partial_\eta f_{k_1}^* \partial_\eta f_{k_2}^* \partial_\eta f_{k_3}^* \right), \quad (\text{B.544})$$

where we have summed over all the permutations of the external legs. Since the interaction is symmetric, this just produces a factor of $3! = 6$. In order to evaluate this integral, we note that the derivative of the mode function (B.543) is very simple

$$\partial_\eta f_k^* = \frac{H}{\sqrt{4\varepsilon k^3}} k^2 \eta e^{ik\eta}. \quad (\text{B.545})$$

Hence, we get

$$\begin{aligned}
\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle'_{\dot{\zeta}^3} &= -\frac{3}{16} \frac{\dot{\phi}^2}{\Lambda^4} \frac{H^4}{\varepsilon^2} \frac{1}{k_1 k_2 k_3} \operatorname{Im} \left(\int_{-\infty}^0 d\eta \eta^2 e^{i(k_1+k_2+k_3)\eta} \right) \\
&= -\frac{3}{16} \frac{\dot{\phi}^2}{\Lambda^4} \frac{H^4}{\varepsilon^2} \frac{1}{k_1 k_2 k_3} \operatorname{Im} \left(-\partial_K^2 \int_{-\infty}^0 d\eta e^{iK\eta} \right) \\
&= -\frac{3}{16} \frac{\dot{\phi}^2}{\Lambda^4} \frac{H^4}{\varepsilon^2} \frac{1}{k_1 k_2 k_3} \operatorname{Im} \left(-\partial_K^2 \frac{1}{iK} \right) \\
&= -\frac{3}{8} \frac{\dot{\phi}^2}{\Lambda^4} \frac{H^4}{\varepsilon^2} \frac{1}{k_1 k_2 k_3} \frac{1}{K^3}, \tag{B.546}
\end{aligned}$$

where $K \equiv k_1 + k_2 + k_3$.

Recall from (8.72) that the dimensionless bispectrum is

$$\begin{aligned}
B_{\dot{\zeta}^3}(k_1, k_2, k_3) &\equiv \frac{(k_1 k_2 k_3)^2}{(2\pi^2)^2} \langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle'_{\dot{\zeta}^3} = -\left(\frac{H^2}{8\pi^2 \varepsilon} \right)^2 \left(\frac{\dot{\phi}^2}{\Lambda^4} \right) \frac{6 k_1 k_2 k_3}{K^3} \\
&= \boxed{-\Delta_{\dot{\zeta}}^4 \left(\frac{\dot{\phi}^2}{\Lambda^4} \right) \frac{6 k_1 k_2 k_3}{K^3}}, \tag{B.547}
\end{aligned}$$

as required.

- Next, we consider the $\dot{\zeta}(\partial_i \zeta)^2$ interaction. Following the same steps as for the $\dot{\zeta}^3$ interaction, we obtain

$$\begin{aligned}
\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle'_{\dot{\zeta}(\partial_i \zeta)^2} &= -\frac{\dot{\phi}^4}{\Lambda^4 H^3} (\mathbf{k}_2 \cdot \mathbf{k}_3) \operatorname{Im} \left(\prod_{i=1}^3 f_{k_i}(0) \int_{-\infty}^0 \frac{d\eta}{(-H\eta)} \partial_\eta f_{k_1}^* f_{k_2}^* f_{k_3}^* \right) \\
&\quad + 2 \text{ perms.} \tag{B.548}
\end{aligned}$$

The primary difference from the previous case is that the spatial derivatives turn into factors of momenta, $\nabla \mapsto -i\mathbf{k}$, and that we now have to sum explicitly over all permutations of the external fields in order to make the final correlator symmetric. Using

$$\mathbf{k}_2 \cdot \mathbf{k}_3 = \frac{1}{2}(k_1^2 - k_2^2 - k_3^2), \tag{B.549}$$

we can write (B.548) as

$$\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle'_{\dot{\zeta}(\partial_i \zeta)^2} = \frac{\dot{\phi}^2}{\Lambda^4} \frac{H^4}{32\varepsilon^2} \left[\frac{k_1^2 - k_2^2 - k_3^2}{(k_1 k_2 k_3)^3} \operatorname{Im}(\mathcal{I}(k_1, k_2, k_3)) + 2 \text{ perms.} \right], \tag{B.550}$$

where

$$\begin{aligned}
\mathcal{I}(k_1, k_2, k_3) &\equiv \int_{-\infty}^0 d\eta k_1^2 (1 - ik_2\eta)(1 - ik_3\eta) e^{i(k_1+k_2+k_3)\eta} \\
&= k_1^2 (1 - k_2\partial_{k_2})(1 - k_3\partial_{k_3}) \int_{-\infty}^0 d\eta e^{i(k_1+k_2+k_3)\eta} \\
&= k_1^2 (1 - k_2\partial_{k_2})(1 - k_3\partial_{k_3}) \frac{1}{i(k_1 + k_2 + k_3)} \\
&= -i k_1^2 \frac{k_1^2 + 3k_1(k_2 + k_3) + 2(k_2^2 + k_3^2) + 6k_2k_3}{(k_1 + k_2 + k_3)^3}. \tag{B.551}
\end{aligned}$$

Summing over permutations in (B.550), we then get

$$\begin{aligned}
\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle'_{\zeta(\partial_i \zeta)^2} &= \frac{\dot{\phi}^2}{\Lambda^4} \frac{H^4}{32\varepsilon^2} \frac{1}{(k_1 k_2 k_3)^3} \left[\frac{8}{K} \sum_{i>j} k_i^2 k_j^3 - \frac{4}{K^2} \sum_{i \neq j} k_i^2 k_j^3 \right. \\
&\quad \left. + 12 \frac{(k_1 k_2 k_3)^2}{K^3} - \sum_{i=1}^3 k_i^3 \right], \tag{B.552}
\end{aligned}$$

and hence find

$$\begin{aligned}
B_{\zeta(\partial_i \zeta)^2}(k_1, k_2, k_3) &\equiv \frac{(k_1 k_2 k_3)^2}{(2\pi^2)^2} \langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle'_{\zeta(\partial_i \zeta)^2} = \\
&= \left[\Delta_\zeta^4 \left(\frac{\dot{\phi}^2}{\Lambda^4} \right) \frac{1}{k_1 k_2 k_3} \left[\frac{4}{K} \sum_{i>j} k_i^2 k_j^3 - \frac{2}{K^2} \sum_{i \neq j} k_i^2 k_j^3 + 6 \frac{(k_1 k_2 k_3)^2}{K^3} - \frac{1}{2} \sum_{i=1}^3 k_i^3 \right] \right], \tag{B.553}
\end{aligned}$$

as required.

The total bispectrum therefore is

$$\begin{aligned}
B_\zeta &\equiv B_{\zeta^3} + B_{\zeta(\partial_i \zeta)^2} \\
&= \Delta_\zeta^4 \left(\frac{\dot{\phi}^2}{\Lambda^4} \right) \frac{1}{k_1 k_2 k_3} \left[\frac{4}{K} \sum_{i>j} k_i^2 k_j^3 - \frac{2}{K^2} \sum_{i \neq j} k_i^2 k_j^3 - \frac{1}{2} \sum_{i=1}^3 k_i^3 \right], \tag{B.554}
\end{aligned}$$

and its amplitude is

$$f_{\text{NL}} \equiv \frac{5}{18} \frac{B_\zeta(k, k, k)}{\Delta_\zeta^4(k)} = \boxed{\frac{35}{108} \frac{\dot{\phi}^2}{\Lambda^4}}, \tag{B.555}$$

as required.

Solution to Exercise 9.12

In the following, we derive the Goldstone action starting from action (9.112) for metric fluctuations in the unitary gauge.

- We first consider the universal part of the action, which in unitary gauge reads

$$S_0 = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R - M_{\text{Pl}}^2 (3H^2 + \dot{H}) + M_{\text{Pl}}^2 \dot{H} g^{00} \right]. \quad (\text{B.556})$$

For $-\dot{H} \ll H^2$, this is just single-field slow-roll inflation in disguise. To see this, consider the canonical inflaton Lagrangian

$$\mathcal{L}_0 = -\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi). \quad (\text{B.557})$$

In unitary gauge, $\phi = \bar{\phi}(t)$, this becomes

$$\begin{aligned} \mathcal{L}_0 &= -\frac{1}{2} \dot{\bar{\phi}}^2 g^{00} - V(\bar{\phi}) \\ &= M_{\text{Pl}}^2 \dot{H} g^{00} - M_{\text{Pl}}^2 (3H^2 + \dot{H}), \end{aligned} \quad (\text{B.558})$$

where in the second line we have used the Friedmann equations. The slow-roll action in unitary gauge therefore matches (B.556).

We introduce the Goldstone boson into (B.556) as in (9.116):

$$\begin{aligned} \mathcal{L}_\pi &= -M_{\text{Pl}}^2 \left(3H^2(t + \pi) + \dot{H}(t + \pi) \right) \\ &\quad + M_{\text{Pl}}^2 \dot{H} (g^{00} + 2\partial_\mu \pi g^{0\mu} + \partial_\mu \pi \partial_\nu \pi g^{\mu\nu}). \end{aligned} \quad (\text{B.559})$$

The leading mixing of π with gravitational fluctuations⁵⁹ is $2M_{\text{Pl}}^2 \dot{H} \dot{\pi} \delta g^{00}$. This should be compared with the Goldstone kinetic term $-M_{\text{Pl}}^2 \dot{H} \dot{\pi}^2$. Using the Einstein constraint equation [133],

$$\delta g^{00} = 2\varepsilon H \pi, \quad (\text{B.560})$$

the mixing term becomes $4M_{\text{Pl}}^2 \dot{H} (\varepsilon H \dot{\pi} \pi)$. Notice that $\dot{\pi} \pi$ is a total derivative in flat space, so we should integrate the mixing term by parts before we compare it with the kinetic term. This gives

$$4M_{\text{Pl}}^2 \dot{H} (\varepsilon H \dot{\pi} \pi) \rightarrow -6M_{\text{Pl}}^2 \dot{H} (\varepsilon H^2 \pi^2) + \dots. \quad (\text{B.561})$$

A similar term arises from the expansion

$$-3M_{\text{Pl}}^2 H^2(t + \pi) = 3M_{\text{Pl}}^2 \dot{H} (\varepsilon H^2 \pi^2) + \dots. \quad (\text{B.562})$$

Combining (B.561) and (B.562), the ratio of the mixing term and the kinetic term is therefore

$$\frac{\text{mixing}}{\text{kinetic}} = \frac{3\varepsilon H^2 \pi^2}{\dot{\pi}^2} = \frac{3\varepsilon H^2}{\omega^2}, \quad (\text{B.563})$$

and the mixing is negligible in the decoupling limit:

$$\omega \gg \omega_{\text{mix}} \equiv \sqrt{\varepsilon} H. \quad (\text{B.564})$$

⁵⁹The contribution to the quadratic Goldstone action of the fluctuations δg^{0i} in (B.559) is canceled by a corresponding term from the Einstein–Hilbert action, and so can be neglected at this stage.

It is a special feature of quasi-de Sitter backgrounds, where $\varepsilon \ll 1$, that decoupling happens at relatively low frequencies. In particular, horizon crossing, $\omega = H$, falls in the decoupling regime. Evaluating (B.559) in the unperturbed spacetime, we find

$$\mathcal{L}_\pi^{(2)} = M_{\text{Pl}}^2 |\dot{H}| \left(\dot{\pi}^2 - \frac{(\partial_i \pi)^2}{a^2} \right), \quad (\text{B.565})$$

where $(\partial_i \pi)^2 \equiv \delta^{ij} \partial_i \pi \partial_j \pi$. From the normalization of the kinetic term, we read off the symmetry breaking scale $f_\pi^4 \equiv 2M_{\text{Pl}}^2 |\dot{H}|$

Had we included the mixing with gravity (B.560) in (B.559), we would have found a small mass term for the Goldstone boson,

$$\mathcal{L}_\pi^{(2)} = M_{\text{Pl}}^2 |\dot{H}| \left(\dot{\pi}^2 - \frac{(\partial_i \pi)^2}{a^2} + 3\varepsilon H^2 \pi^2 \right), \quad (\text{B.566})$$

where we have kept terms at next-to-leading order in slow-roll parameters. Expressed in terms of the comoving curvature perturbation, $\zeta = -H\pi$, the action (B.566) takes the form

$$\mathcal{L}_\zeta^{(2)} = \frac{1}{2} \frac{f_\pi^4}{H^2} \left(\dot{\zeta}^2 - \frac{(\partial_i \zeta)^2}{a^2} \right). \quad (\text{B.567})$$

Although (B.566) involves a slow-roll approximation, (B.567) is actually an exact result. The action (B.567) has precisely the form that is required in order for ζ to be massless, and therefore conserved outside the horizon. The power spectrum of vacuum fluctuations then is

$$\Delta_\zeta^2 = \frac{1}{4\pi^2} \left(\frac{H}{f_\pi} \right)^4. \quad (\text{B.568})$$

We see that the size of the fluctuations is determined by the ratio of the Hubble rate H to the symmetry-breaking scale f_π . This is a universal result that will continue to hold even for more complicated models of inflation.

- Next, we consider the following class of corrections:

$$\Delta \mathcal{L} = \sum_{n=2}^{\infty} \frac{M_n^4(t)}{n!} (\delta g^{00})^n. \quad (\text{B.569})$$

This is equivalent to studying $P(X)$ theories, in which the inflaton Lagrangian is a functional of the inflaton kinetic term, $X \equiv -\frac{1}{2}(\partial_\mu \phi)^2$,

$$\mathcal{L}_{P(X)} = P(X, \phi). \quad (\text{B.570})$$

Evaluating (B.570) in unitary gauge, $\phi = \bar{\phi}(t)$, we obtain

$$\begin{aligned} \mathcal{L}_{P(X)} &= P\left(-\frac{1}{2}\dot{\bar{\phi}}^2 g^{00}, \bar{\phi}\right) \\ &= M_{\text{Pl}}^2 \dot{H} g^{00} - M_{\text{Pl}}^2 (3H^2 + \dot{H}) + \sum_{n=2}^{\infty} \frac{M_n^4(t)}{n!} (\delta g^{00})^n, \end{aligned} \quad (\text{B.571})$$

where

$$M_n^4(t) \equiv \left(-\frac{1}{2}\dot{\phi}^2\right)^n \frac{\partial^n P}{\partial X^n}. \quad (\text{B.572})$$

We will now analyze the dynamics of these theories in terms of the Goldstone fluctuations. We will build up the action in stages, starting with the operator $(\delta g^{00})^2$ and then adding higher-order terms.

– Introducing the Goldstone boson in the operator $(\delta g^{00})^2$, we get

$$\frac{M_2^4}{2}(\delta g^{00})^2 \mapsto \frac{M_2^4}{2} \left(1 + g^{00} + 2\partial_\mu \pi g^{0\mu} + \partial_\mu \pi \partial_\nu \pi g^{\mu\nu}\right)^2. \quad (\text{B.573})$$

For $M_2^4 \gg M_{\text{Pl}}^2 |\dot{H}|$, the dominant mixing with gravity now comes from the interaction $M_2^4 \dot{\pi} \delta g^{00}$, while the kinetic term becomes $M_2^4 \dot{\pi}^2$. The Einstein constraint equation still implies [133]

$$\delta g^{00} = 2\varepsilon H \pi. \quad (\text{B.574})$$

The condition for being able to ignore the coupling to the metric perturbations is therefore given, as before, by (B.564). Evaluating (B.573) in the unperturbed spacetime, we find

$$\frac{M_2^4}{2}(\delta g^{00})^2 \mapsto 2M_2^4 \left(\dot{\pi}^2 + \dot{\pi}^3 - \frac{\dot{\pi}(\partial_i \pi)^2}{a^2}\right) + \mathcal{O}(\pi^4). \quad (\text{B.575})$$

We see that the operator $(\delta g^{00})^2$ changes the Goldstone kinetic term, $\dot{\pi}^2$, but not the spatial gradient term, $(\partial_i \pi)^2$, whose coefficient is fixed by the symmetries of the FRW background. Adding $(\delta g^{00})^2$ to the universal part of the Goldstone action therefore induces a nontrivial speed of sound for the fluctuations. In particular, adding (B.575) to (B.565), we get the quadratic action

$$\mathcal{L}_\pi^{(2)} = \frac{M_{\text{Pl}}^2 |\dot{H}|}{c_s^2} \left(\dot{\pi}^2 - c_s^2 \frac{(\partial_i \pi)^2}{a^2}\right), \quad (\text{B.576})$$

where we have defined the *speed of sound*

$$c_s^2 \equiv \frac{M_{\text{Pl}}^2 \dot{H}}{M_{\text{Pl}}^2 \dot{H} - 2M_2^4}. \quad (\text{B.577})$$

In order to avoid superluminal propagation for the fluctuations, we must impose $M_2^4 > 0$, while for $M_2^4 \gg M_{\text{Pl}}^2 |\dot{H}|$, we have $c_s \ll 1$.

It is convenient to absorb the sound speed into a redefinition of the spatial coordinates, $x^i \mapsto \tilde{x}^i \equiv c_s^{-1} x^i$, so that fake Lorentz invariance is restored:

$$\tilde{\mathcal{L}}_\pi^{(2)} \equiv c_s^3 \mathcal{L}_\pi^{(2)} = M_{\text{Pl}}^2 |\dot{H}| c_s \left(\dot{\pi}^2 - \frac{(\tilde{\partial}_i \pi)^2}{a^2}\right). \quad (\text{B.578})$$

This allows us to read off the symmetry-breaking scale directly from the normalization of the kinetic term $f_\pi^4 \equiv 2M_{\text{Pl}}^2 |\dot{H}| c_s$.

Had we included the mixing with gravity (B.574), we would have found

$$\tilde{\mathcal{L}}_\pi^{(2)} = \frac{1}{2} f_\pi^4 \left(\dot{\pi}^2 - \frac{(\tilde{\partial}_i \pi)^2}{a^2} + 3\varepsilon H^2 \pi^2 \right), \quad (\text{B.579})$$

or, in terms of $\zeta = -H\pi$,

$$\tilde{\mathcal{L}}_\zeta^{(2)} = \frac{1}{2} \frac{f_\pi^4}{H^2} \left(\dot{\zeta}^2 - \frac{(\tilde{\partial}_i \zeta)^2}{a^2} \right). \quad (\text{B.580})$$

Notice that the action (B.580) is identical to (B.567), but with rescaled spatial coordinates and a redefined f_π . The power spectrum of curvature perturbations is therefore still given by (B.568). Horizon crossing corresponds to $\tilde{k} = aH$ in the rescaled coordinates, and $c_s k = aH$ in the original coordinates.

- The nonlinearly-realized time-translation symmetry relates a small sound speed, $c_s \ll 1$, to large interactions. This is manifest in (B.575), where several interaction terms are proportional to M_2^4 , and therefore related to c_s . Consider the cubic part of the Goldstone action,

$$\mathcal{L}_\pi^{(3)} = \frac{M_{\text{Pl}}^2 |\dot{H}|}{c_s^2} (1 - c_s^2) \left(\dot{\pi}^3 - \frac{\dot{\pi}(\partial_i \pi)^2}{a^2} \right), \quad (\text{B.581})$$

or, in terms of the rescaled coordinates,

$$\tilde{\mathcal{L}}_\pi^{(3)} = \frac{1}{2} f_\pi^4 (1 - c_s^2) \left(\dot{\pi}^3 - \frac{1}{c_s^2} \frac{\dot{\pi}(\tilde{\partial}_i \pi)^2}{a^2} \right). \quad (\text{B.582})$$

For $c_s \ll 1$, the operator $\dot{\pi}(\tilde{\partial}_i \pi)^2$ has a boosted coefficient.

To complete the cubic Goldstone action at single-derivative level, we add the operator $(\delta g^{00})^3$, whose representation in terms of π is

$$\frac{M_3^4}{3!} (\delta g^{00})^3 \mapsto \frac{M_3^4}{3!} (-2\dot{\pi} + (\partial_\mu \pi)^2)^3 = -\frac{4}{3} M_3^4 \dot{\pi}^3 + \dots, \quad (\text{B.583})$$

where the Goldstone interactions have been evaluated in the decoupling limit. Adding this to (B.581), we get

$$\boxed{\mathcal{L}_\pi^{(3)} = \frac{M_{\text{Pl}}^2 |\dot{H}|}{c_s^2} (1 - c_s^2) \left[\frac{A}{c_s^2} \dot{\pi}^3 - \frac{\dot{\pi}(\partial_i \pi)^2}{a^2} \right]}, \quad (\text{B.584})$$

where we have defined

$$\frac{A}{c_s^2} \equiv 1 - \frac{2}{3} \frac{M_3^4}{M_2^4}. \quad (\text{B.585})$$

The cubic Goldstone Lagrangian therefore has two independent operators, $\dot{\pi}^3$ and $\dot{\pi}(\partial_i \pi)^2$. We will compute the two different shapes of non-Gaussianity associated with these operators in the next exercise.

Solution to Exercise 9.13

In this problem, we are computing the simplest non-Gaussianities in the effective field theory of inflation [42].

We start from the Goldstone action:

$$S_\pi = \int d^4x a^3 \frac{f_\pi^4}{c_s^3} \left[\frac{1}{2} \left(\dot{\pi}^2 - \frac{c_s^2}{a^2} (\partial_i \pi)^2 \right) + \frac{1 - c_s^2}{2c_s^2} \left(A \dot{\pi}^3 - \frac{c_s^2}{a^2} \dot{\pi} (\partial_i \pi)^2 \right) \right], \quad (\text{B.586})$$

where c_s is the sound speed of the fluctuations, A is a dimensionless constant and $f_\pi^4 \equiv 2M_{\text{Pl}}^2 |\dot{H}| c_s$ is the symmetry breaking scale. It is convenient to absorb the (constant) sound speed into a rescaling of the spatial coordinates, $\mathbf{x} \rightarrow c_s \mathbf{x}$, which restores a fake Lorentz invariance of the free theory. The action for the curvature perturbation $\zeta = -H\pi$ then is

$$S_\pi = \int d\eta d^3\tilde{x} a^2(\eta) \left[\frac{1}{2} \frac{f_\pi^4}{H^2} \left((\zeta')^2 - (\tilde{\partial}_i \zeta)^2 \right) - \frac{1 - c_s^2}{2c_s^2} \frac{f_\pi^4}{H^3} \frac{1}{a(\eta)} \left(A (\zeta')^3 - \zeta' (\tilde{\partial}_i \zeta)^2 \right) \right], \quad (\text{B.587})$$

which is similar to the action in (B.540). To write down the associated bispectra, we therefore just have to rescale the amplitudes of the results from Exercise 9.11:

$$B_{\zeta^3} = -\Delta_\zeta^4 \times A \left(\frac{1}{c_s^2} - 1 \right) \frac{6k_1 k_2 k_3}{K^3}, \quad (\text{B.588})$$

$$B_{\zeta(\partial_i \zeta)^2} = \Delta_\zeta^4 \left(\frac{1}{c_s^2} - 1 \right) \frac{1}{k_1 k_2 k_3} \left[\frac{4}{K} \sum_{i>j} k_i^2 k_j^3 - \frac{2}{K^2} \sum_{i \neq j} k_i^2 k_j^3 + 6 \frac{(k_1 k_2 k_3)^2}{K^3} - \frac{1}{2} \sum_{i=1}^3 k_i^3 \right]. \quad (\text{B.589})$$

Using $B_\zeta \equiv B_{\zeta^3} + B_{\zeta(\partial_i \zeta)^2}$, the amplitude of the total bispectrum then is

$$f_{\text{NL}} = \frac{5}{18} \frac{B_\zeta(k, k, k)}{\Delta_\zeta^4(k)} = \boxed{\frac{5}{18} \left(\frac{25}{18} - \frac{2}{9} A \right) \left(\frac{1}{c_s^2} - 1 \right)}, \quad (\text{B.590})$$

as required.

Solution to Exercise 9.14

Consider an action of the form

$$S = \int d^4x \sqrt{-g} \left(\frac{M_{\text{Pl}}^2}{2} R - \frac{1}{2} (\partial\phi)^2 - V(\phi) - \frac{1}{2} (\partial_\mu \sigma)^2 - \frac{1}{2} M^2 \sigma^2 + \frac{1}{2\Lambda} \sigma (\partial_\mu \phi)^2 \right), \quad (\text{B.591})$$

where we imagine that the potential $V(\phi)$ is chosen in order to be able to support inflation. We will compute the contribution to the inflaton's three-point function from the presence of the heavy field following [72]. First, we perturb the action around the inflationary solution:

$$\phi = \bar{\phi}(t) + \varphi(t, \mathbf{x}), \quad (\text{B.592})$$

so that the effective action for fluctuations contains the interactions

$$S_{\text{int}} = \int d^4x a^3(t) \left(-\frac{\dot{\bar{\phi}}}{\Lambda} \sigma \dot{\varphi} + \frac{1}{2\Lambda} \sigma (\partial_\mu \varphi)^2 \right) \quad (\text{B.593})$$

$$= \int d\eta d^3x a^2(\eta) \left(-a(\eta) \frac{\dot{\bar{\phi}}}{\Lambda} \sigma \varphi' - \frac{1}{2\Lambda} \sigma \varphi'^2 + \frac{1}{2\Lambda} \sigma (\partial_i \varphi)^2 \right), \quad (\text{B.594})$$

where in the second line we have written things in terms of conformal time, with $a(\eta) = (-H\eta)^{-1}$. The fluctuation of the inflaton, φ , is massless in this approximation, and so has mode function

$$f_k(\eta) = \frac{1}{\sqrt{2k^3}}(1 + ik\eta)e^{-ik\eta}, \quad (\text{B.595})$$

while the mode function for the massive field σ is

$$g_k(\eta) = e^{\frac{i\pi}{4}} H \sqrt{\frac{\pi}{4}} (-\eta)^{3/2} e^{\frac{\pi}{2}\mu} H_{i\mu}^{(1)}(-k\eta), \quad \text{with} \quad i\mu = \sqrt{\frac{9}{4} - \frac{M^2}{H^2}}. \quad (\text{B.596})$$

Using these ingredients, we now can compute the three-point function associated to the interactions in (B.594).

Since we are interested in the limit where $k_3 \ll k_1, k_2$ the following diagram topology gives the leading contribution

$$I_{++} \equiv \text{Diagram} \quad (\text{B.597})$$

The two cubic interactions in (B.594) imply the following two contributions

$$I_{++}^{(1)} = -\frac{\dot{\phi}}{4\Lambda^2} \frac{1}{\sqrt{8k_1^3 k_2^3 k_3^3}} \int d\eta d\eta' a^2(\eta) a^3(\eta') \left[\partial_\eta f_{k_1} \partial_\eta f_{k_2} G_F(k_I, \eta, \eta') \partial_{\eta'} f_{k_3} \right], \quad (\text{B.598})$$

$$I_{++}^{(2)} = -\frac{\dot{\phi}}{4\Lambda^2} \frac{1}{\sqrt{8k_1^3 k_2^3 k_3^3}} \int d\eta d\eta' a^2(\eta) a^3(\eta') \left[(\mathbf{k}_1 \cdot \mathbf{k}_2) f_{k_1}(\eta) f_{k_2}(\eta) G_F(k_I, \eta, \eta') \partial_{\eta'} f_{k_3} \right], \quad (\text{B.599})$$

where $G_F(k_I, \eta, \eta')$ is the Feynman propagator made from the mode functions in (B.596). Using the following identities

$$\partial_\eta f_k(\eta) = \frac{k^2 \eta}{\sqrt{2k^3}} e^{-ik\eta}, \quad (\text{B.600})$$

$$(1 + ik\eta) e^{-ik\eta} = (1 - k\partial_k) e^{-ik\eta}, \quad (\text{B.601})$$

$$k^2 \eta^2 e^{-ik\eta} = -k^2 \partial_k^2 e^{-ik\eta}, \quad (\text{B.602})$$

$$\mathbf{k}_1 \cdot \mathbf{k}_2 = \frac{1}{2}(k_3^2 - k_1^2 - k_2^2), \quad (\text{B.603})$$

we get

$$\begin{aligned} I_{++} &= I_{++}^{(1)} + I_{++}^{(2)} \\ &= \frac{\dot{\phi}}{32\Lambda^2 H^5} \frac{1}{k_1^3 k_2^3 k_3^3} O_{12} \int \frac{d\eta}{\eta^2} \frac{d\eta'}{\eta'^2} e^{-ik_{12}\eta} G_F(k_I; \eta, \eta') e^{-ik_3\eta'}, \end{aligned} \quad (\text{B.604})$$

where we have defined the differential operator

$$O_{12} \equiv -k_1^2 k_2^2 \partial_{k_{12}}^2 + \frac{1}{2}(k_3^2 - k_1^2 - k_2^2)(1 - k_1 \partial_{k_1})(1 - k_2 \partial_{k_2}). \quad (\text{B.605})$$

For general momentum configurations, the integral in (B.604) is rather difficult to compute because the massive propagator involves products of Hankel functions. However, in the squeezed

limit $k_3 \ll k_1, k_2$, we can take advantage of the fact that the Feynman propagator greatly simplifies and becomes effectively the same as the Wightman function [43]

$$\begin{aligned} \lim_{k_I \rightarrow 0} G_F(k_I; \eta, \eta') &= \lim_{k_I \rightarrow 0} W_{k_I}(\eta, \eta') \\ &= \frac{H^2}{4\pi} (\eta\eta')^{3/2} \left[\Gamma[-i\mu]^2 \left(\frac{k_I^2 \eta\eta'}{4} \right)^{i\mu} + \Gamma[i\mu]^2 \left(\frac{k_I^2 \eta\eta'}{4} \right)^{-i\mu} \right]. \end{aligned} \quad (\text{B.606})$$

In this limit, the integral (B.604) reduces to

$$I_{++} = A \frac{\Gamma[-i\mu]^2}{k_1^3 k_2^3 k_3} \left(\frac{k_3}{2} \right)^{2i\mu} O_{12} \int d\eta d\eta' (\eta\eta')^{-\frac{1}{2}+i\mu} e^{-ik_{12}\eta} e^{-ik_3\eta'} + (\mu \rightarrow -\mu), \quad (\text{B.607})$$

where $A \equiv \dot{\phi}/(128\pi\Lambda^2 H^3)$. The integrals over η and η' can be done using the formula

$$\int_{-\infty}^0 d\eta e^{\pm iK\eta} (-\eta)^\Delta = (\mp i)^{\Delta+1} K^{-(1+\Delta)} \Gamma[\Delta+1], \quad (\text{B.608})$$

which leads to

$$\int d\eta d\eta' (\eta\eta')^{-\frac{1}{2}+i\mu} e^{-ik_{12}\eta} e^{-ik_3\eta'} = i^{1+2i\mu} \Gamma[\frac{1}{2} + i\mu]^2 k_{12}^{-(\frac{1}{2}+i\mu)} k_3^{-(\frac{1}{2}+i\mu)}. \quad (\text{B.609})$$

Using

$$O_{12} k_{12}^{-(\frac{1}{2}+i\mu)} \xrightarrow{k_3 \rightarrow 0} -2^{-\frac{3}{2}-i\mu} (\frac{3}{2} + i\mu) (\frac{5}{2} + i\mu) k_1^{\frac{3}{2}-i\mu}, \quad (\text{B.610})$$

we then obtain

$$I_{++} = -A i^{1+2i\mu} \frac{2^{-\frac{3}{2}-2i\mu}}{k_1^3 k_3^3} (\frac{3}{2} + i\mu) (\frac{5}{2} + i\mu) \Gamma[-i\mu]^2 \Gamma[\frac{1}{2} + i\mu]^2 \left(\frac{k_3}{k_1} \right)^{\frac{3}{2}+i\mu} + (\mu \rightarrow -\mu), \quad (\text{B.611})$$

which implies

$$\begin{aligned} I_{++} + I_{--} &= I_{++} + I_{++}^* \\ &= 2iA \sinh(\pi\mu) \frac{2^{-\frac{3}{2}-2i\mu}}{k_1^3 k_3^3} (\frac{3}{2} + i\mu) (\frac{5}{2} + i\mu) \Gamma[-i\mu]^2 \Gamma[\frac{1}{2} + i\mu]^2 \left(\frac{k_3}{k_1} \right)^{\frac{3}{2}+i\mu} + (\mu \rightarrow -\mu), \end{aligned} \quad (\text{B.612})$$

where we used that

$$i^{1+2i\mu} + (-i)^{1+2i\mu} = e^{\frac{i\pi}{2}(1+2i\mu)} + e^{-\frac{i\pi}{2}(1+2i\mu)} = -2i \sinh(\pi\mu). \quad (\text{B.613})$$

In the soft limit, the disconnected contribution is given by almost the same integral as (B.607), we just complex conjugate one of the vertices:

$$I_{-+} = A \frac{\Gamma[-i\mu]^2}{k_1^3 k_2^3 k_3} \left(\frac{k_3}{2} \right)^{2i\mu} O_{12} \int d\eta d\eta' (\eta\eta')^{-\frac{1}{2}+i\mu} e^{+ik_{12}\eta} e^{-ik_3\eta'} + (\mu \rightarrow -\mu). \quad (\text{B.614})$$

The only thing that this changes is that when we use the formula (B.608) the factors of i cancel, so we have

$$\begin{aligned} I_{-+} &= I_{+-} \\ &= A \frac{2^{-\frac{3}{2}-2i\mu}}{k_1^3 k_3^3} (\tfrac{3}{2} + i\mu)(\tfrac{5}{2} + i\mu) \Gamma[-i\mu]^2 \Gamma[\tfrac{1}{2} + i\mu]^2 \left(\frac{k_3}{k_1}\right)^{\frac{3}{2}+i\mu} + (\mu \rightarrow -\mu). \end{aligned} \quad (\text{B.615})$$

It is also convenient to use the identity

$$\Gamma[\tfrac{1}{2} + i\mu] = \frac{\pi \text{sech}(\pi\mu)}{\Gamma[\tfrac{1}{2} - i\mu]}. \quad (\text{B.616})$$

Putting everything together, we get

$$\begin{aligned} \lim_{k_3 \rightarrow 0} \langle \varphi_{\mathbf{k}_1} \varphi_{\mathbf{k}_2} \varphi_{\mathbf{k}_3} \rangle' &= \frac{\dot{\phi}}{64\pi\Lambda^2 H^3} \frac{\pi^2}{\cosh^2(\pi\mu)} \frac{1}{k_1^3 k_3^3} \left(\frac{k_3}{2k_1}\right)^{\frac{3}{2}} \\ &\times \left[(1 + i \sinh(\pi\mu)) \frac{(\frac{3}{2} + i\mu)(\frac{5}{2} + i\mu) \Gamma[-i\mu]^2}{\Gamma[\frac{1}{2} - i\mu]^2} \left(\frac{k_3}{4k_1}\right)^{i\mu} + (\mu \rightarrow -\mu) \right]. \end{aligned} \quad (\text{B.617})$$

The reason that it was useful to pull out a factor of $1/\cosh^2(\pi\mu)$ is that as $\mu \rightarrow \infty$ this factor scales like $\sim e^{-2\pi\mu}$. Recalling that $\mu \sim M/H$, this is precisely the expected Boltzmann suppression $\sim e^{-M/T_{\text{dS}}}$, with $T_{\text{dS}} = H/(2\pi)$ the de Sitter temperature (5.64).

To convert this to the three-point function of the curvature perturbation ζ , we note that it is related to the inflaton perturbation φ (in spatially-flat gauge) via

$$\zeta = -\frac{H}{\dot{\phi}} \varphi = -\frac{1}{\sqrt{2\varepsilon} M_{\text{Pl}}} \varphi. \quad (\text{B.618})$$

The natural dimensionless object is the three-point function divided by two powers of the power spectrum

$$P_\zeta(k) = \frac{H^2}{4M_{\text{Pl}}^2 \varepsilon} \frac{1}{k^3}, \quad (\text{B.619})$$

so that

$$\boxed{\begin{aligned} \lim_{k_3 \rightarrow 0} \frac{\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle'}{P_\zeta(k_1) P_\zeta(k_3)} &= \frac{\varepsilon}{8\pi} \frac{M_{\text{Pl}}^2}{\Lambda^2} \frac{\pi^2}{\cosh^2(\pi\mu)} \left(\frac{k_3}{2k_1}\right)^{\frac{3}{2}} \\ &\times \left[(1 + i \sinh(\pi\mu)) \frac{(\frac{3}{2} + i\mu)(\frac{5}{2} + i\mu) \Gamma[-i\mu]^2}{\Gamma[\frac{1}{2} - i\mu]^2} \left(\frac{k_3}{4k_1}\right)^{i\mu} + (\mu \rightarrow -\mu) \right] \end{aligned}}, \quad (\text{B.620})$$

as required.

B.10 Exercises in Chapter 10

Solution to Exercise 10.1

These formulas are obtained from (10.3) by expanding the wavefunction perturbatively as in (10.10) and performing the resulting Gaussian integrals.

Recall the one-dimensional Gaussian integral

$$\int dx e^{-ax^2+bx} = \sqrt{\frac{\pi}{a}} e^{b^2/(4a)}. \quad (\text{B.621})$$

A field theory version of this is

$$\int \mathcal{D}\phi \exp\left(-\int_{\mathbf{k}} G(k_1, k_2) \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} + \int_{\mathbf{k}} \phi_{-\mathbf{k}} J_{\mathbf{k}}\right) = \mathcal{N} \exp\left(\frac{1}{4} \int_{\mathbf{k}} J_{\mathbf{k}_1} G^{-1}(k_1, k_2) J_{\mathbf{k}_2}\right), \quad (\text{B.622})$$

where $\int_{\mathbf{k}} \equiv \int \prod d^3k/(2\pi)^3$ indicates integration over all the $\mathbf{k}$ arguments, and $\mathcal{N}$ is a (formally infinite) normalization factor. By functionally differentiating this expression with respect to J and then setting $J = 0$, we can obtain formulas for arbitrary moments of the Gaussian. We will also need the formula

$$\frac{\delta J_{\mathbf{k}}}{\delta J_{\mathbf{q}}} = (2\pi)^3 \delta(\mathbf{k} - \mathbf{q}). \quad (\text{B.623})$$

First, consider the Gaussian part of the wavefunction

$$\Psi_0[\phi] \equiv \exp\left(-\frac{1}{2} \int_{\mathbf{k}_1} \int_{\mathbf{k}_2} \langle O_1 O_2 \rangle \phi_1 \phi_2\right). \quad (\text{B.624})$$

We can then express the two-point function of ϕ as⁶⁰

$$\langle \phi_1 \phi_2 \rangle = \frac{\int \mathcal{D}\phi \phi_1 \phi_2 |\Psi_0[\phi]|^2}{\int \mathcal{D}\phi |\Psi_0[\phi]|^2} = \frac{\int \mathcal{D}\phi \phi_1 \phi_2 \exp\left(-\int_{\mathbf{k}_1} \int_{\mathbf{k}_2} \text{Re}\langle O_1 O_2 \rangle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2}\right)}{\int \mathcal{D}\phi |\Psi_0[\phi]|^2}. \quad (\text{B.625})$$

We can perform the integral in the numerator by differentiating (B.622) twice. The normalization factor cancels out with the denominator, and we are left with

$$\langle \phi_1 \phi_2 \rangle = \frac{1}{2\text{Re}\langle O_1 O_2 \rangle'} (2\pi)^3 \delta(\mathbf{k}_1 + \mathbf{k}_2), \quad (\text{B.626})$$

where the $'$ indicates that we have removed the momentum-conserving delta function from the kernel $\text{Re}\langle O_1 O_2 \rangle'$.

The calculation at higher orders is similar. The wavefunction takes the schematic form (10.9), which we then expand around a Gaussian as in (10.10):

$$\begin{aligned} \Psi[\phi] = \Psi_0[\phi] & \left[1 + \frac{1}{3!} \int_{\mathbf{k}_1} \int_{\mathbf{k}_2} \int_{\mathbf{k}_3} \langle O_{\mathbf{k}_1} O_{\mathbf{k}_2} O_{\mathbf{k}_3} \rangle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \right. \\ & + \frac{1}{2(3!)^2} \int_{\mathbf{k}_1} \cdots \int_{\mathbf{k}_6} \langle O_{\mathbf{k}_1} O_{\mathbf{k}_2} O_{\mathbf{k}_3} \rangle \langle O_{\mathbf{k}_4} O_{\mathbf{k}_5} O_{\mathbf{k}_6} \rangle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \phi_{\mathbf{k}_4} \phi_{\mathbf{k}_5} \phi_{\mathbf{k}_6} \\ & \left. + \frac{1}{4!} \int_{\mathbf{k}_1} \cdots \int_{\mathbf{k}_4} \langle O_{\mathbf{k}_1} O_{\mathbf{k}_2} O_{\mathbf{k}_3} O_{\mathbf{k}_4} \rangle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \phi_{\mathbf{k}_4} + \cdots \right]. \end{aligned} \quad (\text{B.627})$$

⁶⁰We have implicitly used rotation invariance to relate $\phi_{\mathbf{k}}$ to its complex conjugate. If ϕ is real, then $\phi_{\mathbf{k}}^* = \phi_{-\mathbf{k}}$. In a rotationally-invariant theory none of the kernels will depend on individual $\mathbf{k}$'s, so we can rotate them all to make them positive.

After squaring this expression and multiplying by factors of ϕ , we can perform the Gaussian integrals using (B.622) to obtain a final formula for the correlators.

Three points: The three-point function is given by

$$\langle \phi_1 \phi_2 \phi_3 \rangle = \int \mathcal{D}\phi \phi_1 \phi_2 \phi_3 |\Psi_0[\phi]|^2 \left(1 + \frac{1}{3!} \int_{\mathbf{k}_3} \int_{\mathbf{k}_4} \int_{\mathbf{k}_5} 2\text{Re}\langle O_4 O_5 O_6 \rangle \phi_4 \phi_5 \phi_6 + \dots \right), \quad (\text{B.628})$$

where we are dropping the explicit normalization factor for simplicity. The $\dots$ denotes terms with more powers of ϕ , which would contribute at higher orders in perturbation theory. The Gaussian integral can again be done using (B.622). Repeated differentiation with respect to the source brings down factors of the Gaussian kernel $\text{Re}\langle O_1 O_2 \rangle$. Since the source appears as J^2 in the exponent of (B.622), when we differentiate with respect to J , we end up with a factor of J multiplying the Gaussian. At the end when we set $J = 0$, only terms where the additional derivatives remove these sources will contribute. This means that effectively we pair up the field operators when we functionally differentiate. This is the wavefunctional manifestation of Wick contractions, and so the momentum dependence of the integrated correlator is most simply understood by Wick contraction. We pair up the fields $\phi_1 \dots \phi_3$ that appear in the correlator with the ones $\phi_4 \dots \phi_6$ that appear in the wavefunction. Much like in canonical quantization, the connected correlator only gets contributions from pairings across these sets, so for example we cannot contract ϕ_1 with ϕ_2 . Each contraction yields a factor of $(2\text{Re}\langle OO \rangle')^{-1}$, with the momentum dependence of the field appearing in the correlator. There are $3!$ ways of pairing up the fields, so we get

$$\langle \phi_1 \phi_2 \phi_3 \rangle = \frac{2\text{Re}\langle O_1 O_2 O_3 \rangle'}{2\text{Re}\langle O_1 O_1 \rangle' 2\text{Re}\langle O_2 O_2 \rangle' 2\text{Re}\langle O_3 O_3 \rangle'} (2\pi)^3 \delta(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3). \quad (\text{B.629})$$

We see that the three-point correlator is simply related to the real part of the three-point wavefunction coefficient.

Four points: At the next order, we consider

$$\begin{aligned} |\Psi[\phi]|^2 = |\Psi_0[\phi]|^2 & \left[1 + \dots + \frac{1}{(3!)^2} \int_{\mathbf{k}_1} \dots \int_{\mathbf{k}_6} \langle O_{\mathbf{k}_1} O_{\mathbf{k}_2} O_{\mathbf{k}_3} \rangle \langle O_{\mathbf{k}_4} O_{\mathbf{k}_5} O_{\mathbf{k}_6} \rangle^* \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \phi_{\mathbf{k}_4} \phi_{\mathbf{k}_5} \phi_{\mathbf{k}_6} \right. \\ & + \frac{1}{2(3!)^2} \int_{\mathbf{k}_1} \dots \int_{\mathbf{k}_6} \langle O_{\mathbf{k}_1} O_{\mathbf{k}_2} O_{\mathbf{k}_3} \rangle \langle O_{\mathbf{k}_4} O_{\mathbf{k}_5} O_{\mathbf{k}_6} \rangle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \phi_{\mathbf{k}_4} \phi_{\mathbf{k}_5} \phi_{\mathbf{k}_6} + \text{c.c.} \\ & \left. + \frac{1}{4!} \int_{\mathbf{k}_1} \dots \int_{\mathbf{k}_4} 2\text{Re}\langle O_{\mathbf{k}_1} O_{\mathbf{k}_2} O_{\mathbf{k}_3} O_{\mathbf{k}_4} \rangle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \phi_{\mathbf{k}_4} + \dots \right], \quad (\text{B.630}) \end{aligned}$$

where we have only shown the part of the squared wavefunction that will contribute to the four-point correlator. Note that we can combine the first two lines to write

$$\begin{aligned} |\Psi[\phi]|^2 = |\Psi_0[\phi]|^2 & \left[1 + \frac{1}{2(3!)^2} \int_{\mathbf{k}_1} \dots \int_{\mathbf{k}_6} 2\text{Re}\langle O_{\mathbf{k}_1} O_{\mathbf{k}_2} O_{\mathbf{k}_3} \rangle 2\text{Re}\langle O_{\mathbf{k}_4} O_{\mathbf{k}_5} O_{\mathbf{k}_6} \rangle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \phi_{\mathbf{k}_4} \phi_{\mathbf{k}_5} \phi_{\mathbf{k}_6} \right. \\ & \left. + \frac{1}{4!} \int_{\mathbf{k}_1} \dots \int_{\mathbf{k}_4} 2\text{Re}\langle O_{\mathbf{k}_1} O_{\mathbf{k}_2} O_{\mathbf{k}_3} O_{\mathbf{k}_4} \rangle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \phi_{\mathbf{k}_4} + \dots \right], \quad (\text{B.631}) \end{aligned}$$

Substituting this into

$$\langle \phi_1 \phi_2 \phi_3 \phi_4 \rangle = \int \mathcal{D}\phi \phi_1 \phi_2 \phi_3 \phi_4 |\Psi[\phi]|^2, \quad (\text{B.632})$$

and performing the Gaussian integrals, we find

$$\begin{aligned} \langle \phi_1 \phi_2 \phi_3 \phi_4 \rangle = (2\pi)^3 \delta(\mathbf{k}_1 + \dots + \mathbf{k}_4) & \left[\frac{2\text{Re}\langle O_1 O_2 O_3 O_4 \rangle}{2\text{Re}\langle O_1 O_1 \rangle' 2\text{Re}\langle O_2 O_2 \rangle' 2\text{Re}\langle O_3 O_3 \rangle' 2\text{Re}\langle O_4 O_4 \rangle'} \right. \\ & \left. + \frac{2\text{Re}\langle O_1 O_2 O_I \rangle 2\text{Re}\langle O_I O_3 O_4 \rangle}{2\text{Re}\langle O_1 O_1 \rangle' 2\text{Re}\langle O_2 O_2 \rangle' 2\text{Re}\langle O_I O_I \rangle' 2\text{Re}\langle O_3 O_3 \rangle' 2\text{Re}\langle O_4 O_4 \rangle'} \right]. \end{aligned} \quad (\text{B.633})$$

Note that there is a new phenomenon at this order. In addition to the piece proportional to the real part of the four-point wavefunction coefficient $\text{Re}\langle OOOO \rangle$, there is also a *disconnected* contribution, that is proportional to the square of the three-point wavefunction coefficient $\text{Re}\langle OOO \rangle^2$, and depends on an internal momenta, which connects these two wavefunction coefficients.

Solution to Exercise 10.2

Recall the Bessel equation

$$x^2 \frac{d^2 g(x)}{dx^2} + x \frac{dg(x)}{dx} + (x^2 - \nu^2)g(x) = 0, \quad (\text{B.634})$$

which has the Hankel functions $H_\nu^{(1,2)}(x)$ as its solutions. The Wronskian of the two Hankel functions is defined as

$$W \equiv H_\nu^{(1)} \partial_x H_\nu^{(2)} - H_\nu^{(2)} \partial_x H_\nu^{(1)}. \quad (\text{B.635})$$

Taking a derivative of W gives

$$\begin{aligned} \frac{dW}{dx} &= H_\nu^{(1)} \partial_x^2 H_\nu^{(2)} - H_\nu^{(2)} \partial_x^2 H_\nu^{(1)} \\ &= \frac{H_\nu^{(1)}}{x^2} \left(-x \partial_x H_\nu^{(2)} - (x^2 - \nu) H_\nu^{(2)} \right) - \frac{H_\nu^{(2)}}{x^2} \left(-x \partial_x H_\nu^{(1)} - (x^2 - \nu) H_\nu^{(1)} \right) \\ &= -\frac{1}{x} \left(H_\nu^{(1)} \partial_x H_\nu^{(2)} - H_\nu^{(2)} \partial_x H_\nu^{(1)} \right) \\ &= -\frac{W}{x}. \end{aligned} \quad (\text{B.636})$$

We have therefore found a first-order differential equation for the Wronskian, which has the following general solution

$$W(x) = \frac{A}{x}. \quad (\text{B.637})$$

To fix the normalization A , we consider the large argument limit of the Hankel functions:

$$H_\nu^{(1)}(x) \sim \sqrt{\frac{2}{\pi x}} e^{i(x - \nu\pi/2 - \pi/4)} + O(x^{-3/2}), \quad (\text{B.638})$$

$$H_\nu^{(2)}(x) \sim \sqrt{\frac{2}{\pi x}} e^{-i(x - \nu\pi/2 - \pi/4)} + O(x^{-3/2}). \quad (\text{B.639})$$

The derivations of these limiting functions are

$$\partial_x H_\nu^{(1)} \sim i H_\nu^{(1)}, \quad (\text{B.640})$$

$$\partial_x H_\nu^{(2)} \sim -i H_\nu^{(2)}, \quad (\text{B.641})$$

and the Wronskian becomes

$$\begin{aligned} W &= H_\nu^{(1)} \partial_x H_\nu^{(2)} - H_\nu^{(2)} \partial_x H_\nu^{(1)} \\ &\sim \sqrt{\frac{2}{\pi x}} (-i) \sqrt{\frac{2}{\pi x}} - \sqrt{\frac{2}{\pi x}} i \sqrt{\frac{2}{\pi x}} = -\frac{4i}{\pi} \frac{1}{x}. \end{aligned} \quad (\text{B.642})$$

This fixed the constant A in (B.643), so that

$$\boxed{W(x) = -\frac{4i}{\pi} \frac{1}{x}}, \quad (\text{B.643})$$

which is the identity we used in the text.

Solution to Exercise 10.3

For purely imaginary order, $\nu = i\mu$, the complex conjugate of the Hankel function satisfies $H_{i\mu}^{(2)*} = e^{-\pi\mu} H_{i\mu}^{(1)}$. We therefore have

$$\begin{aligned} \text{Im} \left(H_\nu^{(2)*} \partial_\eta H_\nu^{(2)} \right) \Big|_{\eta=\eta_*} &= \frac{1}{2i} e^{-\pi\mu} \left(H_\nu^{(1)} \partial_\eta H_\nu^{(2)} - H_\nu^{(2)} \partial_\eta H_\nu^{(1)} \right) \\ &= -\frac{2}{\pi} \frac{1}{\eta_*} e^{-\pi\mu}, \end{aligned} \quad (\text{B.644})$$

where we have used the Wronskian of the Hankel function to obtain the second line. Substituting this into (10.35), we get

$$\text{Im}(A \partial_\eta \log K) \Big|_{\eta=\eta_*} = \frac{1}{(H\eta_*)^2} \frac{1}{|H_\nu^{(2)}(-k\eta_*)|^2} \left(-\frac{2}{\pi} \frac{1}{\eta_*} e^{-\pi\mu} \right). \quad (\text{B.645})$$

The two-point function of the field therefore is

$$\boxed{\langle \phi_{\mathbf{k}} \phi_{-\mathbf{k}} \rangle' = \frac{\pi}{4} H^2 (-\eta_*)^3 e^{\pi\mu} |H_{i\mu}^{(2)}(-k\eta_*)|^2}. \quad (\text{B.646})$$

Using $H_{i\mu}^{(2)*} = e^{-\pi\mu} H_{i\mu}^{(1)}$, we may also write

$$\boxed{\langle \phi_{\mathbf{k}} \phi_{-\mathbf{k}} \rangle' = \frac{\pi}{4} H^2 (-\eta_*)^3 H_\nu^{(1)}(-k\eta_*) H_\nu^{(2)}(-k\eta_*)}. \quad (\text{B.647})$$

The latter form of writing the result holds for both real and imaginary ν .

Solution to Exercise 10.4

The Green's function solves the equation

$$(\partial_t^2 + \omega^2)G(t, t') = -i\delta(t - t'), \quad (\text{B.648})$$

as well as the same equation with $t \leftrightarrow t'$. This implies that the Green's function will be symmetric under interchanging t and t' , as it should be. Away from $t = t'$, the most general solution to (B.648) for t is

$$G(t) = c_1 e^{i\omega t} + c_2 e^{-i\omega t}, \quad (\text{B.649})$$

and similarly for t' . The general Green's function is then a product of these two solutions

$$G(t, t') = (c_1 e^{i\omega t} + c_2 e^{-i\omega t})(c_3 e^{i\omega t'} + c_4 e^{-i\omega t'}). \quad (\text{B.650})$$

We now impose boundary conditions to select the Green's function of interest:

- *Asymptotics:* When either $t \rightarrow -\infty$ or $t' \rightarrow -\infty$, we want the field to scale as $e^{i\omega t}$. The most general solution compatible with this boundary condition is

$$G(t, t') = \begin{cases} (c_1 e^{i\omega t} + c_2 e^{-i\omega t})e^{i\omega t'} & \text{when } t' < t, \\ (c_3 e^{i\omega t'} + c_4 e^{-i\omega t'})e^{i\omega t} & \text{when } t < t'. \end{cases} \quad (\text{B.651})$$

- *Continuity:* We want the Green's function to be continuous at $t = t'$. This requires that $c_1 = c_3$ and $c_2 = c_4$.
- *Jump condition:* Integrating (B.648) across the jump, we find that the discontinuity in the first derivative of G must be [AJ: I think should be $-i$, check sign]

$$\Delta \partial_t G(k; t, t') \Big|_{t=t'-\epsilon}^{t=t'+\epsilon} = i. \quad (\text{B.652})$$

From (B.651), with $c_1 = c_3$ and $c_2 = c_4$, we find

$$\Delta \partial_t G(t, t') \Big|_{t=t'-\epsilon}^{t=t'+\epsilon} = 2i\omega c_2, \quad (\text{B.653})$$

which implies that $c_2 = 1/(2\omega)$.

- *Vanishing on boundary:* Finally, the solution must go to zero as $t \rightarrow t_*$.⁶¹ This fixes $c_1 = -(2\omega)^{-1}e^{-2\omega t_*}$ for $t > t'$, so that

$$G(t, t') = \begin{cases} \frac{1}{2\omega} e^{-i\omega(t-t')} - \frac{1}{2\omega} e^{i\omega(t-t_*)} e^{i\omega k(t'-t_*)} & \text{when } t > t', \\ \frac{1}{2\omega} e^{i\omega(t-t')} - \frac{1}{2\omega} e^{i\omega(t-t_*)} e^{i\omega(t'-t_*)} & \text{when } t' > t. \end{cases} \quad (\text{B.654})$$

⁶¹Note that this is kind of a nonstandard boundary condition, but it is important. We are forced to impose this condition because we are trying to compute the wavefunctional as a function of some boundary conditions specified on some time slice at t_* . This Dirichlet problem requires this set of boundary condition.

This can be written as a single expression

$$G(t, t') = \frac{1}{2\omega} \left(e^{-i\omega(t-t')} \theta(t-t') + e^{i\omega(t-t')} \theta(t'-t) - e^{i\omega(t-t_*)} e^{i\omega(t'-t_*)} \right). \quad (\text{B.655})$$

For $t_* \rightarrow 0$, this becomes

$$\boxed{G(t, t') = \frac{1}{2\omega} \left(e^{-i\omega(t-t')} \theta(t-t') + e^{i\omega(t-t')} \theta(t'-t) - e^{i\omega(t+t')} \right)}, \quad (\text{B.656})$$

which is the answer quoted in the text.

Solution to Exercise 10.5

We consider the following anharmonic oscillator

$$S[\Phi] = \int dt \left(\frac{1}{2} \dot{\Phi}^2 - \frac{1}{2} \omega^2 \Phi^2 + L_{\text{int}} \right), \quad (\text{B.657})$$

for a general interaction Lagrangian $L_{\text{int}}[\Phi]$. The classical equation of motion is

$$\ddot{\Phi} + \omega^2 \Phi = \frac{\delta S_{\text{int}}}{\delta \Phi}. \quad (\text{B.658})$$

A formal solution of this equation is

$$\Phi_{\text{cl}}(t) = \phi e^{i\omega t} + i \int dt' G(t, t') \frac{\delta S_{\text{int}}}{\delta \Phi(t')} \bigg|_{\Phi=\Phi_{\text{cl}}}, \quad (\text{B.659})$$

where $G(t, t')$ is the Green's function of the harmonic oscillator. Since we have extracted the solution of the harmonic oscillator, $\phi e^{i\omega t}$, the boundary condition $\Phi_{\text{cl}}(t \rightarrow t_* = 0) = \phi$ implies $G(0, t') = G(t, 0) = 0$. The solution of $(\partial_t^2 + \omega^2)G(t, t') = -i\delta(t - t')$, with this boundary condition, is

$$G(t, t') = \frac{1}{2\omega} \left(e^{-i\omega(t-t')} \theta(t-t') + e^{i\omega(t-t')} \theta(t'-t) - e^{i\omega(t+t')} \right), \quad (\text{B.660})$$

which you were invited to derived in Exercise 10.4.

Although equation (B.659) is a bit formal, it is useful because we can easily substitute it back into the action. In order to do this, it is convenient to integrate (B.657) by parts:

$$S[\Phi] = \frac{1}{2} \Phi \dot{\Phi} \bigg|_{t=t_*} + \int dt \left(-\frac{1}{2} \Phi (\ddot{\Phi} + \omega^2 \Phi) + L_{\text{int}} \right). \quad (\text{B.661})$$

Using (B.658), the on-shell action then is

$$S[\Phi_{\text{cl}}] = \frac{1}{2} \Phi_{\text{cl}} \dot{\Phi}_{\text{cl}} \bigg|_{t=t_*} + \int dt \left(-\frac{1}{2} \Phi_{\text{cl}} \frac{\delta S_{\text{int}}}{\delta \Phi_{\text{cl}}} + L_{\text{int}} \right). \quad (\text{B.662})$$

For the moment, let us focus on the boundary term. When we substitute in (B.659), the first term produces the expected Gaussian part of the wavefunction. The second term in (B.659) is,

however, more subtle. In order to evaluate it, we have to note that the Green's function (B.660) satisfies the identity

$$\lim_{t \rightarrow t_*} \partial_t G(t, t') = -i e^{i\omega t'} , \quad (\text{B.663})$$

for $t_* = 0$. Using this, we find

$$S[\Phi_{\text{cl}}] = \frac{i}{2} \omega \phi^2 + \int dt \left(\frac{1}{2} \phi e^{i\omega t} \frac{\delta S_{\text{int}}}{\delta \Phi_{\text{cl}}} - \frac{1}{2} \Phi_{\text{cl}} \frac{\delta S_{\text{int}}}{\delta \Phi_{\text{cl}}} + L_{\text{int}} \right) , \quad (\text{B.664})$$

where we have relabeled $t' \rightarrow t$ in the first term in the integral. Plugging (B.659) into the second term inside the integral, the terms proportional to ϕ cancel and we are left with

$$S[\Phi_{\text{cl}}] = \frac{i}{2} \omega \phi^2 + \int dt L_{\text{int}} - \frac{i}{2} \int dt dt' G(t, t') \frac{\delta S_{\text{int}}}{\delta \Phi_{\text{cl}}(t')} \frac{\delta S_{\text{int}}}{\delta \Phi_{\text{cl}}(t)} , \quad (\text{B.665})$$

as required.

Solution to Exercise 10.6

We have seen that the solution of the anharmonic oscillator at order λ is

$$\Phi^{(1)}(t) = i \int_{-\infty}^0 dt' G(t, t') \left(-\phi^2 e^{2i\omega t'} \right) . \quad (\text{B.666})$$

Substituting the Green's function (B.656), this becomes

$$\begin{aligned} \Phi^{(1)}(t) &= -\frac{i\phi^2}{2\omega} \int_{-\infty}^0 dt' \left(e^{-i\omega(t-t')} \theta(t-t') + e^{i\omega(t-t')} \theta(t'-t) - e^{i\omega(t+t')} \right) e^{2i\omega t'} \\ &= -\frac{i\phi^2}{2\omega} \int_{-\infty}^0 dt' \left(e^{-i\omega(t-3t')} \theta(t-t') + e^{i\omega(t+t')} \theta(t'-t) - e^{i\omega(t+3t')} \right) \\ &= -\frac{i\phi^2}{2\omega} \left[e^{-i\omega t} \int_{-\infty}^t dt' e^{3i\omega t'} + e^{i\omega t} \int_t^0 dt' e^{i\omega t'} - e^{i\omega t} \int_{-\infty}^0 dt' e^{3i\omega t'} \right] \\ &= -\frac{i\phi^2}{2\omega} \left[\frac{1}{3i\omega} e^{2i\omega t} + \frac{e^{i\omega t}}{i\omega} (1 - e^{i\omega t}) - \frac{e^{i\omega t}}{3i\omega} \right] \\ &= \boxed{\frac{\phi^2}{3\omega^2} (e^{2i\omega t} - e^{i\omega t})} . \end{aligned} \quad (\text{B.667})$$

In equation (10.55), we wrote the on-shell action as

$$\begin{aligned} S[\Phi_{\text{cl}}] &= \frac{i\omega}{2} \phi^2 - \frac{\lambda}{3} \int dt (\Phi^{(0)})^3 - \frac{\lambda^2}{2} \int dt (\Phi^{(0)})^2 \Phi^{(1)} + \dots \\ &\equiv S^{(0)} + \lambda S^{(1)} + \lambda^2 S^{(2)} + \dots . \end{aligned} \quad (\text{B.668})$$

Substituting the explicit solutions for $\Phi^{(0)}$ and $\Phi^{(1)}$, we can compute this up to order λ^2 :

$$S^{(1)} \equiv -\frac{1}{3} \int dt (\Phi^{(0)})^3 = -\frac{1}{3} \int_{-\infty}^0 dt \phi^3 e^{3i\omega t} = -\frac{\phi^3}{3} \frac{1}{3i\omega} = \boxed{i \frac{\phi^3}{9\omega}}, \quad (\text{B.669})$$

$$\begin{aligned} S^{(2)} &\equiv -\frac{1}{2} \int dt (\Phi^{(0)})^2 \Phi^{(1)} = -\frac{1}{2} \int_{-\infty}^0 dt \phi^2 e^{2i\omega t} \frac{\phi^2}{3\omega^2} (e^{2i\omega t} - e^{i\omega t}) \\ &= -\frac{\phi^4}{6\omega^2} \int_{-\infty}^0 dt (e^{4i\omega t} - e^{3i\omega t}) \\ &= -\frac{\phi^4}{6\omega^2} \left(\frac{1}{4i\omega} - \frac{1}{3i\omega} \right) = \boxed{-i \frac{\phi^4}{72\omega^3}}. \end{aligned} \quad (\text{B.670})$$

Up to order λ^2 , the perturbative wavefunction therefore is

$$\Psi[\phi] \approx e^{iS[\Phi_{\text{cl}}]} = \exp \left(-\frac{\omega}{2} \phi^2 - \frac{\lambda}{9\omega} \phi^3 + \frac{\lambda^2}{72\omega^3} \phi^4 + \dots \right), \quad (\text{B.671})$$

which is the result we quoted in the text.

Solution to Exercise 10.7

We start by writing the bulk-to-boundary propagator (10.74) and the bulk-to-bulk propagator (10.75) in terms of the Wightman propagator (9.34) and Feynman propagator (9.35):

$$K_k(t) = \frac{W_k(t_*, t)}{|f_k(t_*)|^2}, \quad (\text{B.672})$$

$$G(k; t, t') = G_F(k; t, t') - \frac{W_k(t_*, t) W_k(t_*, t')}{|f_k(t_*)|^2}. \quad (\text{B.673})$$

We will now use these to express (10.12) and (10.13) in terms of the building blocks of the in-in formalism.

Three-point function The three-point wavefunction coefficient is

$$\begin{aligned} \langle O_1 O_2 O_3 \rangle' &= i\lambda \int_{-\infty}^{t_*} Dt K_{k_1}(t) K_{k_2}(t) K_{k_3}(t) \\ &= \frac{i\lambda}{|f_{k_1}(t_*)|^2 |f_{k_2}(t_*)|^2 |f_{k_3}(t_*)|^2} \int_{-\infty}^{t_*} Dt W_{k_1}(t_*, t) W_{k_2}(t_*, t) W_{k_3}(t_*, t), \end{aligned} \quad (\text{B.674})$$

where $Dt \equiv dt a^3(t)$. The integral is the same that appeared in the computation of I_+ in (9.37), so we have

$$\langle O_1 O_2 O_3 \rangle' = 8 \langle O_1 O_1 \rangle' \langle O_2 O_2 \rangle' \langle O_3 O_3 \rangle' \times I_+. \quad (\text{B.675})$$

where we have used that $2 \langle O_n O_n \rangle' = |f_{k_n}(t_*)|^{-2}$. Equation (10.12) then reads

$$\langle \phi_1 \phi_2 \phi_3 \rangle' = \frac{\langle O_1 O_2 O_3 \rangle' + \langle O_1 O_2 O_3 \rangle'^*}{8 \langle O_1 O_1 \rangle' \langle O_2 O_2 \rangle' \langle O_3 O_3 \rangle'} = I_+ + I_-, \quad (\text{B.676})$$

which is indeed the in-in expression for the three-point function.

Four-point function We first consider the second line of (10.13). Using (B.674), we have

$$\begin{aligned} \langle O_1 O_2 O_I \rangle' \langle O_I O_3 O_4 \rangle'^* &= (-i\lambda) \int Dt \frac{W_{k_1}(t_*, t)}{|f_{k_1}(t_*)|^2} \frac{W_{k_2}(t_*, t)}{|f_{k_1}(t_*)|^2} \frac{W_{k_I}(t_*, t)}{|f_{k_I}(t_*)|^2} \\ &\quad \times (i\lambda) \int Dt' \frac{W_{k_I}^*(t_*, t')}{|f_{k_I}(t_*)|^2} \frac{W_{k_3}^*(t_*, t')}{|f_{k_3}(t_*)|^2} \frac{W_{k_4}^*(t_*, t')}{|f_{k_4}(t_*)|^2}, \end{aligned} \quad (\text{B.677})$$

and hence get

$$\begin{aligned} I_2 &\equiv \frac{\langle O_1 O_2 O_I \rangle' \langle O_I O_3 O_4 \rangle'^* + c.c.}{32 \langle O_I O_I \rangle' \prod_{n=1}^4 \langle O_n O_n \rangle'} = \lambda^2 \int Dt Dt' W_{k_1}(t_*, t) W_{k_2}(t_*, t) \\ &\quad \times \frac{W_{k_I}(t_*, t) W_{k_I}^*(t_*, t')}{|f_{k_I}(t_*)|^2} W_{k_3}^*(t_*, t') W_{k_4}^*(t_*, t') + c.c. \end{aligned} \quad (\text{B.678})$$

Since

$$\begin{aligned} \frac{W_{k_I}(t_*, t) W_{k_I}^*(t_*, t')}{|f_{k_I}(t_*)|^2} &= \frac{f_{k_I}^*(t_*) f_{k_I}(t) (f_{k_I}^*(t_*) f_{k_I}(t'))^*}{f_{k_I}(t_*) f_{k_I}^*(t_*)} \\ &= f_{k_I}^*(t') f_{k_I}(t) \\ &= W_{k_I}(t', t), \end{aligned} \quad (\text{B.679})$$

we see that the integral in (B.678) is equal to $I_{+-}^* = I_{+-}$; cf. (9.52). Adding the complex conjugate, we find that the second line of (10.13) is

$$I_2 = I_{+-} + I_{-+}. \quad (\text{B.680})$$

Next, we consider the first line in (10.13). The four-point wavefunction coefficient can be written as

$$\begin{aligned} \langle O_1 O_2 O_3 O_4 \rangle' &= -\lambda^2 \int Dt Dt' K_{k_1}(t) K_{k_2}(t) G(k_I; t, t') K_{k_3}(t') K_{k_4}(t') \\ &= -\lambda^2 \int Dt Dt' \frac{W_{k_1}(t_*, t)}{|f_{k_1}(t_*)|^2} \frac{W_{k_2}(t_*, t)}{|f_{k_2}(t_*)|^2} \left[G_F(k_I; t, t') - \frac{W_{k_I}(t_*, t) W_{k_I}(t_*, t')}{|f_{k_I}(t_*)|^2} \right] \\ &\quad \times \frac{W_{k_3}(t_*, t')}{|f_{k_3}(t_*)|^2} \frac{W_{k_4}(t_*, t')}{|f_{k_2}(t_*)|^2}. \end{aligned} \quad (\text{B.681})$$

The term involving the Feynman propagator is equal to $16 \prod_{n=1}^4 \langle O_n O_n \rangle' \times I_{++}$. Similarly, $\langle O_1 O_2 O_3 O_4 \rangle'^*$ will contain a term equal to $16 \prod_{n=1}^4 \langle O_n O_n \rangle' \times I_{--}$. The first line in (10.13) therefore becomes

$$I_1 \equiv \frac{2 \text{Re} \langle O_1 O_2 O_3 O_4 \rangle'}{16 \prod_{n=1}^4 \langle O_n O_n \rangle'} = I_{++} + I_{--} + 2 \text{Re}(\Delta I_1), \quad (\text{B.682})$$

where

$$\Delta I_1 \equiv \lambda^2 \int Dt Dt' W_{k_1}(t_*, t) W_{k_2}(t_*, t) \frac{W_{k_I}(t_*, t) W_{k_I}(t_*, t')}{|f_{k_I}(t_*)|^2} W_{k_3}(t_*, t') W_{k_4}(t_*, t'). \quad (\text{B.683})$$

To reproduce the correct in-in result, the extra term ΔI_1 must cancel against the last line of (10.13).

Up to one complex conjugation, the derivation of the last line of (10.13) is the same as that of the second line; cf. (B.678). We find that

$$\begin{aligned}
I_3 &\equiv \frac{\langle O_1 O_2 O_I \rangle' \langle O_I O_3 O_4 \rangle' + c.c.}{32 \langle O_I O_I \rangle' \prod_{n=1}^4 \langle O_n O_n \rangle'} = -\lambda^2 \int Dt Dt' W_{k_1}(t_*, t) W_{k_2}(t_*, t) \\
&\quad \times \frac{W_{k_I}(t_*, t) W_{k_I}(t_*, t')}{|f_{k_I}(t_*)|^2} W_{k_3}(t_*, t') W_{k_4}(t_*, t') + c.c. \\
&= -2 \operatorname{Re}(\Delta I_1).
\end{aligned} \tag{B.684}$$

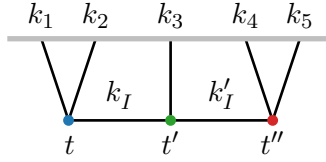
The sum of all contributions therefore indeed reproduces the correct in-in result:

$$\begin{aligned}
\langle \phi_1 \phi_2 \phi_3 \phi_4 \rangle' &= I_1 + I_2 + I_3 \\
&= (I_{++} + I_{--} + 2 \operatorname{Re}(\Delta I_1)) + (I_{+-} + I_{-+}) - 2 \operatorname{Re}(\Delta I_1) \\
&= I_{++} + I_{--} + I_{+-} + I_{-+}.
\end{aligned} \tag{B.685}$$

Note that the nontrivial cancellation between the terms required the correct form of the bulk-to-bulk propagator.

Solution to Exercise 10.8

We are interested in the following Feynman graph:



The associated wavefunction coefficient is

$$\psi = -i\lambda^3 \int_{-\infty}^0 dt \int_{-\infty}^0 dt' \int_{-\infty}^0 dt'' e^{ik_{12}t} G(k_I; t, t') e^{ik_3 t'} G(k'_I; t', t'') e^{ik_{45}t''}. \tag{B.686}$$

To organize the computation, we write the bulk-to-bulk propagator as

$$G = G_+ + G_- - G_0, \tag{B.687}$$

where G_+ is the time-ordered piece, G_- is the anti-time-ordered piece and G_0 is the non-time-ordered piece. The wavefunction coefficient can then be separated into nine parts:

$$\psi = \psi_{00} - (\psi_{0+} + \psi_{0-}) - (\psi_{+0} + \psi_{-0}) + (\psi_{++} + \psi_{--} + \psi_{+-} + \psi_{-+}). \tag{B.688}$$

We start with the fully non-time-ordered part ψ_{00} , where the three time integral become independent:

$$\begin{aligned}
\psi_{00} &= -\frac{i\lambda^3}{4k_I k'_I} \int_{-\infty}^0 dt dt' dt'' e^{ik_{12}t} e^{ik_I(t+t')} e^{ik_3 t'} e^{ik'_I(t'+t'')} e^{ik_{45}t''} \\
&= -\frac{i\lambda^3}{4k_I k'_I} \int_{-\infty}^0 dt e^{i(k_{12}+k_I)t} \int_{-\infty}^0 dt' e^{i(k_3+k_I+k'_I)t'} \int_{-\infty}^0 dt'' e^{i(k_{45}+k'_I)t''} \\
&= -\frac{i\lambda^3}{4k_I k'_I} \int_{-\infty}^0 dt e^{iE_L t} \int_{-\infty}^0 dt' e^{iE_M t'} \int_{-\infty}^0 dt'' e^{iE_R t''} \\
&= \boxed{\frac{\lambda^3}{4k_I k'_I} \frac{1}{E_L E_M E_R}}.
\end{aligned} \tag{B.689}$$

Next, we look at a contribution with a single non-time-ordered propagator:

$$\begin{aligned}
\psi_{0+} &= -\frac{i\lambda^3}{4k_I k'_I} \int_{-\infty}^0 dt \int_{-\infty}^0 dt' \int_{-\infty}^{t'} dt'' e^{ik_{12}t} e^{ik_I(t+t')} e^{ik_3 t'} e^{-ik'_I(t'-t'')} e^{ik_{45}t''} \\
&= -\frac{i\lambda^3}{4k_I k'_I} \int_{-\infty}^0 dt e^{i(k_{12}+k_I)t} \int_{-\infty}^0 dt' e^{i(k_3+k_I-k'_I)t'} \int_{-\infty}^{t'} dt'' e^{i(k_{45}+k'_I)t''} \\
&= -\frac{i\lambda^3}{4k_I k'_I} \frac{1}{i(k_{12}+k_I)} \int_{-\infty}^0 dt' e^{i(k_3+k_I-k'_I)t'} \frac{1}{i(k_{45}+k'_I)} e^{i(k_{45}+k'_I)t'} \\
&= \frac{i\lambda^3}{4k_I k'_I} \frac{1}{E_L E_R} \int_{-\infty}^0 dt' e^{i(k_{345}+k_I)t'} \\
&= \boxed{\frac{\lambda^3}{4k_I k'_I} \frac{1}{E_L E_R (k_{345}+k_I)}}.
\end{aligned} \tag{B.690}$$

Similarly, ψ_{0-} becomes

$$\begin{aligned}
\psi_{0-} &= -\frac{i\lambda^3}{4k_I k'_I} \int_{-\infty}^0 dt \int_{-\infty}^0 dt'' \int_{-\infty}^{t''} dt' e^{ik_{12}t} e^{ik_I(t+t')} e^{ik_3 t'} e^{ik'_I(t'-t'')} e^{ik_{45}t''} \\
&= -\frac{i\lambda^3}{4k_I k'_I} \int_{-\infty}^0 dt e^{i(k_{12}+k_I)t} \int_{-\infty}^0 dt'' e^{i(k_{45}-k'_I)t''} \int_{-\infty}^{t''} dt' e^{i(k_3+k_I+k'_I)t'} \\
&= -\frac{i\lambda^3}{4k_I k'_I} \frac{1}{i(k_{12}+k_I)} \int_{-\infty}^0 dt'' e^{i(k_{45}-k'_I)t''} \frac{1}{i(k_3+k_I+k'_I)} e^{i(k_3+k_I+k'_I)t'} \\
&= \frac{i\lambda^3}{4k_I k'_I} \frac{1}{E_L E_M} \int_{-\infty}^0 dt'' e^{i(k_{345}+k_I)t''} \\
&= \boxed{\frac{\lambda^3}{4k_I k'_I} \frac{1}{E_L E_M (k_{345}+k_I)}}.
\end{aligned} \tag{B.691}$$

By symmetry, ψ_{+0} and ψ_{-0} are

$$\psi_{+0} = \boxed{\frac{\lambda^3}{4k_I k'_I} \frac{1}{E_R E_M (k_{123}+k'_I)}}, \tag{B.692}$$

$$\psi_{-0} = \boxed{\frac{\lambda^3}{4k_I k'_I} \frac{1}{E_L E_R (k_{123}+k'_I)}}. \tag{B.693}$$

Finally, we compute the fully time-ordered contributions. For ψ_{++} and ψ_{--} , we get

$$\begin{aligned}
\psi_{++} &= -\frac{i\lambda^3}{4k_I k'_I} \int_{-\infty}^0 dt \int_{-\infty}^t dt' \int_{-\infty}^{t'} dt'' e^{ik_{12}t} e^{-ik_I(t-t')} e^{ik_3 t'} e^{-ik'_I(t'-t'')} e^{ik_{45}t''} \\
&= -\frac{i\lambda^3}{4k_I k'_I} \int_{-\infty}^0 dt e^{i(k_{12}-k_I)t} \int_{-\infty}^t dt' e^{i(k_3+k_I-k'_I)t'} \int_{-\infty}^{t'} dt'' e^{i(k_{45}+k'_I)t''} \\
&= -\frac{i\lambda^3}{4k_I k'_I} \int_{-\infty}^0 dt e^{i(k_{12}-k_I)t} \int_{-\infty}^t dt' e^{i(k_3+k_I-k'_I)t'} \frac{1}{i(k_{45}+k'_I)} e^{i(k_{45}+k'_I)t'} \\
&= -\frac{i\lambda^3}{4k_I k'_I} \frac{1}{iE_R} \int_{-\infty}^0 dt e^{i(k_{12}-k_I)t} \int_{-\infty}^0 dt' e^{i(k_{345}+k_I)t'} \\
&= \frac{i\lambda^3}{4k_I k'_I} \frac{1}{E_R(k_{345}+k_I)} \int_{-\infty}^0 dt e^{i(k_{12345})t} \\
&= \boxed{\frac{\lambda^3}{4k_I k'_I} \frac{1}{E E_R(k_{345}+k_I)}} , \tag{B.694}
\end{aligned}$$

$$\psi_{--} = \boxed{\frac{\lambda^3}{4k_I k'_I} \frac{1}{E E_L(k_{123}+k'_I)}} . \tag{B.695}$$

For ψ_{-+} , we have $t < t'$ and $t'' < t'$, but no relative time-ordering between t and t'' . The integral for the wavefunction coefficient therefore is

$$\begin{aligned}
\psi_{-+} &= -\frac{i\lambda^3}{4k_I k'_I} \int_{-\infty}^0 dt' \int_0^{t'} dt \int_0^{t'} dt'' e^{ik_{12}t} e^{ik_I(t-t')} e^{ik_3 t'} e^{-ik'_I(t'-t'')} e^{ik_{45}t''} \\
&= -\frac{i\lambda^3}{4k_I k'_I} \int_{-\infty}^0 dt' e^{i(k_3-k_I-k'_I)t'} \int_0^{t'} dt e^{i(k_{12}+k_I)t} \int_0^{t'} dt'' e^{i(k_{45}+k'_I)t''} \\
&= \frac{i\lambda^3}{4k_I k'_I} \frac{1}{E_L E_R} \int_{-\infty}^0 dt' e^{i(k_{12}+k_3+k_{45})t'} \\
&= \boxed{\frac{\lambda^3}{4k_I k'_I} \frac{1}{E E_L E_R}} . \tag{B.696}
\end{aligned}$$

Similarly, for ψ_{+-} , we have $t > t'$ and $t'' > t'$, but no relative time-ordering between t and t'' , and the integral for the wavefunction coefficient is

$$\begin{aligned}
\psi_{+-} &= -\frac{i\lambda^3}{4k_I k'_I} \int_{-\infty}^0 dt' \int_{t'}^0 dt \int_{t'}^0 dt'' e^{ik_{12}t} e^{-ik_I(t-t')} e^{ik_3 t'} e^{ik'_I(t'-t'')} e^{ik_{45}t''} \\
&= -\frac{i\lambda^3}{4k_I k'_I} \int_{-\infty}^0 dt' e^{i(k_3+k_I+k'_I)t'} \int_{t'}^0 dt e^{i(k_{12}-k_I)t} \int_{t'}^0 dt'' e^{i(k_{45}-k'_I)t''} \\
&= -\frac{i\lambda^3}{4k_I k'_I} \int_{-\infty}^0 dt' e^{i(k_3+k_I+k'_I)t'} \frac{1}{i(k_{12}-k_I)} \left[1 - e^{i(k_{12}-k_I)t'}\right] \frac{1}{i(k_{45}-k'_I)} \left[1 - e^{i(k_{45}-k'_I)t'}\right] \\
&= \frac{i\lambda^3}{4k_I k'_I} \frac{1}{(k_{12}-k_I)(k_{45}-k'_I)} \int_{-\infty}^0 dt' \left[e^{iE_M t'} - e^{i(k_{123}+k'_I)t'} - e^{i(k_{345}+k_I)t'} + e^{iE t'} \right] \\
&= \frac{\lambda^3}{4k_I k'_I} \frac{1}{(k_{12}-k_I)(k_{45}-k'_I)} \left[\frac{1}{E_M} - \frac{1}{k_{123}+k'_I} - \frac{1}{k_{345}+k_I} + \frac{1}{E} \right]
\end{aligned}$$

$$= \frac{\lambda^3}{4k_I k'_I} \frac{E + E_M}{EE_M} \frac{1}{(k_{123} + k'_I)(k_{345} + k_I)}. \quad (\text{B.697})$$

Substituting these results into (B.688), we get

$$\begin{aligned} \psi &= \frac{\lambda^3}{4k_I k'_I} \left[\frac{1}{E_L E_M E_R} - \left(\frac{1}{E_L E_R} + \frac{1}{E_L E_M} \right) \frac{1}{k_{345} + k_I} - \left(\frac{1}{E_L E_R} + \frac{1}{E_R E_M} \right) \frac{1}{k_{123} + k'_I} \right. \\ &\quad \left. + \frac{1}{EE_R(k_{345} + k_I)} + \frac{1}{EE_L(k_{123} + k'_I)} + \frac{1}{EE_L E_R} + \frac{E + E_M}{EE_M} \frac{1}{(k_{123} + k'_I)(k_{345} + k_I)} \right] \\ &= \frac{\lambda^3}{E E_L E_M E_R} \left[\frac{1}{k_{123} + k'_I} + \frac{1}{k_{345} + k_I} \right], \end{aligned} \quad (\text{B.698})$$

as required.

Solution to Exercise 10.9

Using the Feynman rules, the three-point wavefunction coefficient of two conformally coupled scalars and a generic massive scalar is

$$\langle O_1 O_2 X_3 \rangle' = \frac{i\lambda}{f_{k_1}(\eta_*) f_{k_2}(\eta_*) \chi_{k_3}(\eta_*)} \int_{-\infty}^{\eta_*} \frac{d\eta}{(H\eta)^4} f_{k_1}(\eta) f_{k_2}(\eta) \chi_{k_3}(\eta), \quad (\text{B.699})$$

where the relevant mode functions are

$$f_k(\eta) = H(-\eta) \frac{e^{ik\eta}}{\sqrt{2k}}, \quad (\text{B.700})$$

$$\chi_k(\eta) = H \sqrt{\frac{\pi}{4}} e^{-\frac{i\pi}{4}(1+2\nu)} (-\eta)^{3/2} H_\nu^{(2)}(-k\eta). \quad (\text{B.701})$$

Equation (B.699) can then be written as

$$\langle O_1 O_2 X_3 \rangle' = \frac{i\lambda}{(-\eta_*)^{7/2} H_\nu^{(2)}(-k_3 \eta_*)} \frac{1}{H^4} \int_{-\infty}^{\eta_*} d\eta (-\eta)^{-1/2} e^{ik_{12}\eta} H_\nu^{(2)}(-k_3 \eta). \quad (\text{B.702})$$

where we have dropped the factor of $e^{-ik_{12}\eta_*}$ (which becomes unity in the limit $\eta_* \rightarrow 0$). For the Hankel function, we then use the integral representation

$$H_\nu^{(2)}(x) = \frac{i}{\sqrt{\pi}} \frac{2^{\nu+1} e^{i\nu\pi}}{\Gamma[\nu + \frac{1}{2}]} x^\nu e^{-ix} \int_0^\infty ds e^{-2is} s^{\nu-\frac{1}{2}} (1+s)^{\nu-\frac{1}{2}}. \quad (\text{B.703})$$

Substituting this into (B.702), we get

$$\begin{aligned} \langle O_1 O_2 X_3 \rangle' &= -\frac{2\lambda}{\sqrt{\pi} H^4} \frac{e^{i\nu\pi}}{\Gamma[\nu + \frac{1}{2}]} \frac{(2k_3)^\nu}{(-\eta_*)^{7/2} H_\nu^{(2)}(-k_3 \eta_*)} \\ &\quad \times \int_0^\infty ds s^{\nu-\frac{1}{2}} (1+s)^{\nu-\frac{1}{2}} \int_{-\infty}^{\eta_*} d\eta e^{i[k_{12} + (1+2s)k_3]\eta} (-\eta)^{\nu-\frac{1}{2}}. \end{aligned} \quad (\text{B.704})$$

The integrals in the second line have already appeared in Exercise 9.4; cf. (B.446). Following the same steps, we find

$$\begin{aligned} \langle O_1 O_2 X_3 \rangle' &= -\sqrt{\frac{2}{\pi}} \frac{\lambda}{H^4} e^{i\frac{\pi}{2}\nu} \frac{(-ik_3\eta_*)^{-1/2}}{(-\eta_*)^3 H_\nu^{(2)}(-k_3\eta_*)} \\ &\quad \times \Gamma[\tfrac{1}{2} - \nu] \Gamma[\tfrac{1}{2} + \nu] {}_2F_1\left[\begin{matrix} \tfrac{1}{2} + \nu, & \tfrac{1}{2} - \nu \\ 1 \end{matrix} \middle| \frac{u-1}{2u} \right], \end{aligned} \quad (\text{B.705})$$

where $u \equiv k_3/k_{12}$. To evaluate this at late times, $\eta_* \rightarrow 0$, we use the following limit of the Hankel function

$$H_\nu^{(2)}(x) \xrightarrow{x \rightarrow 0} \frac{i}{\pi} \left(\Gamma(\nu) \left(\frac{x}{2}\right)^{-\nu} + e^{i\pi\nu} \Gamma(-\nu) \left(\frac{x}{2}\right)^\nu \right). \quad (\text{B.706})$$

Assuming a small mass, $m < \frac{3}{2}H$, the parameter ν is real and positive. The first term in (B.706) then dominates at late times. In addition, we use the identities

$$\Gamma[\tfrac{1}{2} - \nu] \Gamma[\tfrac{1}{2} + \nu] = \frac{\pi}{\cos(\pi\nu)}, \quad (\text{B.707})$$

$$\Gamma[\nu] \Gamma[1 - \nu] = \frac{\pi}{\sin(\pi\nu)}. \quad (\text{B.708})$$

Equation (B.705) then becomes

$$\langle O_1 O_2 X_3 \rangle' = \frac{\sqrt{\pi}}{2^{\nu-\frac{1}{2}}} \frac{i\lambda}{H^4} \frac{\Gamma[1 - \nu]}{\cot(\pi\nu)} \frac{(-ik_3\eta_*)^{\nu-\frac{1}{2}}}{(-\eta_*)^3} {}_2F_1\left[\begin{matrix} \tfrac{1}{2} + \nu, & \tfrac{1}{2} - \nu \\ 1 \end{matrix} \middle| \frac{u-1}{2u} \right], \quad (\text{B.709})$$

as expected.

Solution to Exercise 10.10

Start from the action of a conformally coupled scalar with a polynomial interaction

$$S = \int d^4x \sqrt{-g} \left(-\frac{1}{2}(\partial\Phi)^2 - H^2\Phi^2 - \frac{\lambda}{n!}\Phi^n \right). \quad (\text{B.710})$$

The bulk-to-boundary propagator for this field is given by

$$K_k(\eta) = \frac{\eta}{\eta_*} e^{ik\eta}. \quad (\text{B.711})$$

Using the Feynman rules, the wavefunction coefficient of interest is

$$\begin{aligned} \langle O_1 \cdots O_n \rangle' &\equiv \text{[Diagram: A vertex connected to } n \text{ external lines, with an ellipsis in the middle line]} \\ &= -i\lambda \int_{-\infty}^{\eta_*} \frac{d\eta}{(H\eta)^4} K_{k_1}(\eta) \cdots K_{k_n}(\eta) \\ &= -\frac{i\lambda}{H^4 \eta_*^n} \int_{-\infty}^{\eta_*} d\eta \eta^{n-4} e^{iK\eta}, \end{aligned} \quad (\text{B.712})$$

where we have defined $K \equiv k_1 + \dots + k_n$. Using (B.487), we can perform the integral and get

$$\langle O_1 \dots O_n \rangle'_c = -i^n \frac{\Gamma[n-3]}{n!} \frac{\lambda}{H^4 \eta_*^n} \frac{1}{K^{n-3}}, \quad (\text{B.713})$$

where we have multiplied by the normalization factor (10.78) (in this case $1/n!$) in order to turn the Feynman diagram into a true wavefunction coefficient. Notice that the Gamma function diverges as $n \rightarrow 3$, but being careful about this divergence, we can extract the log behavior of the conformally coupled scalar three-point function. Here, we will specialize to $n \geq 4$.

We can repeat the calculation using the in-in formalism. The same Feynman diagram corresponds to

$$\begin{aligned} I_+ &= -i\lambda f_{k_1}^*(\eta_*) \dots f_{k_n}^*(\eta_*) \int_{-\infty}^{\eta_*} \frac{d\eta}{(H\eta)^4} f_{k_1}(\eta) \dots f_{k_n}(\eta) \\ &= -i\lambda \frac{H^{2n-4} \eta_*^n}{2^n k_1 \dots k_n} \int_{-\infty}^{\eta_*} d\eta \eta^{n-4} e^{iK\eta} \\ &= -i^n \lambda \Gamma[n-3] \frac{H^{2n-4} \eta_*^n}{2^n k_1 \dots k_n} \frac{1}{K^{n-3}}. \end{aligned} \quad (\text{B.714})$$

Adding this together with $I_- = I_+^*$, we get

$$\langle \phi_1 \dots \phi_n \rangle'_c = -2 \cos\left(\frac{n\pi}{2}\right) \Gamma[n-3] \frac{\lambda H^{2n-4} \eta_*^n}{2^n k_1 \dots k_n} \frac{1}{K^{n-3}}, \quad (\text{B.715})$$

where for n odd, the relative phases of I_+ and I_- cause the in-in correlator to vanish. We can also see this from the wavefunction perspective where both $\langle O_1 \dots O_n \rangle'_c$ and $\langle O_1 \dots O_n \rangle_c'^*$ contribute to the correlator, but these two contributions cancel for $n \geq 4$ odd. This is the same type of cancellation seen in Exercise 9.5, but now as a function of n rather than D . (See [124] for more examples.)

In order to match the wavefunction and in-in calculations, we need the conformally coupled power spectrum

$$\langle \phi_{\mathbf{k}} \phi_{-\mathbf{k}} \rangle' = \frac{H^2 \eta_*^2}{2k}. \quad (\text{B.716})$$

The difference between the in-in correlator and $\langle O_1 \dots O_n \rangle'_c + \langle O_1 \dots O_n \rangle_c'^*$ is therefore just factors of the power spectrum and combinatoric factors from the Gaussian integrals.

Solution to Exercise 10.11

The relevant wavefunction coefficients are

$$\langle O_1 O_1 \rangle' = \frac{k_1}{H^2 \eta_*^2}, \quad (\text{B.717})$$

$$\langle O_1 O_2 O_3 \rangle' = \frac{i\lambda}{H^4 \eta_*^3} \log(iK\eta_*), \quad (\text{B.718})$$

$$\begin{aligned} \langle O_1 O_2 O_3 O_4 \rangle' &= \frac{\lambda^2}{2H^6 \eta_*^4} \frac{1}{k_I} \left[\text{Li}_2\left(\frac{E - E_L}{E}\right) + \text{Li}_2\left(\frac{E - E_R}{E}\right) \right. \\ &\quad \left. + \log\left(\frac{E_L}{E}\right) \log\left(\frac{E_R}{E}\right) - \frac{\pi^2}{6} \right], \end{aligned} \quad (\text{B.719})$$

and the four-point correlator (in the s -channel) is given by

$$\langle \phi_1 \phi_2 \phi_3 \phi_4 \rangle' = \frac{1}{8 \prod_{n=1}^4 \langle O_n O_n \rangle'} \left(\langle O_1 O_2 O_3 O_4 \rangle' + \frac{\langle O_1 O_2 O_I \rangle' \langle O_I O_3 O_4 \rangle'^* + \text{c.c.}}{4 \langle O_I O_I \rangle'} + \frac{\langle O_1 O_2 O_I \rangle' \langle O_I O_3 O_4 \rangle' + \text{c.c.}}{4 \langle O_I O_I \rangle'} \right). \quad (\text{B.720})$$

The connected component of the correlator simply is

$$\begin{aligned} \langle O_1 O_2 O_3 O_4 \rangle'_{(c)} &\equiv \frac{H^8 \eta_*^8}{8 k_1 k_2 k_3 k_4} \langle O_1 O_2 O_3 O_4 \rangle' \\ &= \frac{\lambda^2 H^2 \eta_*^4}{16 k_1 k_2 k_3 k_4 k_I} \left[\text{Li}_2 \left(\frac{E - E_L}{E} \right) + \text{Li}_2 \left(\frac{E - E_R}{E} \right) \right. \\ &\quad \left. + \log \left(\frac{E_L}{E} \right) \log \left(\frac{E_R}{E} \right) - \frac{\pi^2}{6} \right], \end{aligned} \quad (\text{B.721})$$

which is the same as (2 times) $I_{++}^{(c)}$ in equation (9.78). The disconnected component is

$$\begin{aligned} \langle O_1 O_2 O_3 O_4 \rangle'_{(d)} &\equiv \frac{H^{10} \eta_*^{10}}{32 k_1 k_2 k_3 k_4 k_I} \left(\langle O_1 O_2 O_I \rangle' \langle O_I O_3 O_4 \rangle'^* + \langle O_1 O_2 O_I \rangle' \langle O_I O_3 O_4 \rangle' + \text{c.c.} \right) \\ &= \boxed{\frac{\lambda^2 H^2 \eta_*^4}{32 k_1 k_2 k_3 k_4 k_I} \left(\log(i E_L \eta_*) \log(-i E_R \eta_*) - \log(i E_L \eta_*) \log(i E_R \eta_*) + \text{c.c.} \right)}. \end{aligned} \quad (\text{B.722})$$

We recognize the first term as I_{+-} in (9.80) and the second term as $I_{++}^{(d)}$ in (9.79). As we have seen in (9.82), these contributions combine with their complex conjugates into

$$\boxed{\langle O_1 O_2 O_3 O_4 \rangle'_{(d)} = \frac{\lambda^2 H^2 \eta_*^4}{32 k_1 k_2 k_3 k_4 k_I} \pi^2}. \quad (\text{B.723})$$

The sum of the connected and disconnected contributions then correctly reproduces the in-in correlator (9.83).

Solution to Exercise 10.12

Start from the action

$$S = \int d^4 x a^3(t) \left(\frac{1}{2} \dot{\Phi}^2 - \frac{1}{2a^2(t)} (\nabla \Phi)^2 + \frac{1}{2} \dot{\chi}^2 - \frac{1}{2a^2(t)} (\nabla \chi)^2 - H^2 \chi^2 - \frac{\lambda}{2} \Phi^2 \chi \right), \quad (\text{B.724})$$

where the interaction breaks the de Sitter symmetries. Writing this in terms of conformal time, we get

$$S = \int d\eta d^3 x a^2(\eta) \left(\frac{1}{2} \Phi'^2 - \frac{1}{2} (\nabla \Phi)^2 + \frac{1}{2} \chi'^2 - \frac{1}{2} (\nabla \chi)^2 - a^2 H^2 \chi^2 - \frac{\lambda}{2} \Phi'^2 \chi \right). \quad (\text{B.725})$$

The bulk-to-boundary propagator for a massless field and its derivative are

$$K_k(\eta) = (1 - ik\eta) e^{ik\eta}, \quad (\text{B.726})$$

$$K'_k(\eta) = k^2 \eta e^{ik\eta}. \quad (\text{B.727})$$

The bulk-to-bulk propagator for a conformally coupled scalar can be written simply in terms of the flat-space propagator:

$$G_{(\text{cc})}(k_I; \eta, \eta') = H^2 \eta \eta' G_{(\text{flat})}(k_I; \eta, \eta'), \quad (\text{B.728})$$

where $G_{(\text{flat})}$ is given by (10.82). Applying the Feynman rules, we then see that the four-point wavefunction coefficient for the massless scalar is

$$\begin{aligned} \langle O_1 O_2 O_3 O_4 \rangle' &= -\frac{\lambda^2}{H^2} k_1^2 k_2^2 k_3^2 k_4^2 \int d\eta d\eta' \eta \eta' e^{ik_{12}\eta} e^{ik_{34}\eta'} G_{(\text{flat})}(k_I; \eta, \eta') \\ &= \frac{\lambda^2}{H^2} k_1^2 k_2^2 k_3^2 k_4^2 \partial_{k_{12}} \partial_{k_{34}} \int d\eta d\eta' e^{ik_{12}\eta} e^{ik_{34}\eta'} G_{(\text{flat})}(k_I; \eta, \eta'). \end{aligned} \quad (\text{B.729})$$

Notice that this can be written as energy derivatives of the flat-space result (10.88), so that

$$\langle O_1 O_2 O_3 O_4 \rangle' = -\frac{\lambda^2}{H^2} k_1^2 k_2^2 k_3^2 k_4^2 \partial_{k_{12}} \partial_{k_{34}} \frac{1}{(k_{12} + k_{34})(k_{12} + k_I)(k_{34} + k_I)}. \quad (\text{B.730})$$

Explicitly taking the derivatives, we obtain

$$\boxed{\langle O_1 O_2 O_3 O_4 \rangle' = -\frac{\lambda^2}{H^2} k_1^2 k_2^2 k_3^2 k_4^2 \frac{2E_L E_R + E(E_L + E_R) + E^2}{E^3 E_L^2 E_R^2}}, \quad (\text{B.731})$$

where we have defined $E \equiv k_{12} + k_{34}$, $E_L \equiv k_{12} + k_I$, and $E_R \equiv k_{34} + k_I$. This strategy of relating de Sitter processes to simpler flat-space ones is quite useful for de Sitter-violating interactions involving massless fields. See [80, 81], for more examples.

Solution to Exercise 10.13

1. We start with the functions:

$$F_+ \equiv -\lambda^2 \int \frac{d\eta d\eta'}{(\eta\eta')^{1+\varepsilon}} e^{i(E_1 - k_I)\eta} e^{i(E_2 + k_I)\eta'} \theta(\eta - \eta'), \quad (\text{B.732})$$

$$F_- \equiv -\lambda^2 \int \frac{d\eta d\eta'}{(\eta\eta')^{1+\varepsilon}} e^{i(E_1 + k_I)\eta} e^{i(E_2 - k_I)\eta'} \theta(\eta' - \eta), \quad (\text{B.733})$$

$$F_0 \equiv -\lambda^2 \int \frac{d\eta d\eta'}{(\eta\eta')^{1+\varepsilon}} e^{i(E_1 + k_I)\eta} e^{i(E_2 + k_I)\eta'}, \quad (\text{B.734})$$

We will take derivatives of these functions to derive the coupled differential equations that these integrals satisfy.

- First, we consider the E_1 -derivative of the non-time-ordered contribution (B.734):

$$\begin{aligned} \partial_{E_1} F_0 &= -\lambda^2 \int \frac{d\eta d\eta'}{(\eta\eta')^{1+\varepsilon}} \left[\partial_{E_1} e^{i(E_1 + k_I)\eta} \right] e^{i(E_2 + k_I)\eta'} \\ &= -\lambda^2 \int \frac{d\eta d\eta'}{(\eta\eta')^{1+\varepsilon}} \left[\frac{\eta}{E_1 + k_I} \partial_\eta e^{i(E_1 + k_I)\eta} \right] e^{i(E_2 + k_I)\eta'} \\ &= \frac{\lambda^2}{E_1 + k_I} \int \frac{d\eta d\eta'}{(\eta\eta')^{1+\varepsilon}} \left[\partial_\eta \eta^{-\varepsilon} \right] e^{i(E_1 + k_I)\eta} e^{i(E_2 + k_I)\eta'} \\ &= \frac{\varepsilon}{E_1 + k_I} F_0. \end{aligned} \quad (\text{B.735})$$

In the second line, we traded the E_1 -derivative for an η -derivative using the relation between derivatives, $x\partial_x e^{ix\eta} = \eta\partial_\eta e^{ix\eta}$, and then integrated by parts in the third line. This generates a singularity at $E_1 + k_I = 0$, with a coefficient that is ε times the original function F_0 . Following the same steps, the X_2 -derivative is

$$\partial_{E_2} F_0 = \frac{\varepsilon}{E_2 + k_I} F_0. \quad (\text{B.736})$$

The derivative with respect to the internal energy, $\partial_{k_I} F_0$, can be inferred from the derivatives with respect to the external energies E_1 and E_2 , and therefore doesn't have to be considered separately. Hence, we conclude that the total differential is

$$\boxed{dF_0 = \varepsilon(d\log(E_1 + k_I) + d\log(E_2 + k_I)) F_0}, \quad (\text{B.737})$$

as required.

- Next, we look at the E_1 -derivative of the time-ordered contribution (B.732):

$$\begin{aligned} \partial_{E_1} F_+ &= -\lambda^2 \int \frac{d\eta d\eta'}{(\eta\eta')^{1+\varepsilon}} \left[\partial_{E_1} e^{i(E_1 - k_I)\eta} \right] e^{i(E_2 + k_I)\eta} \theta(\eta - \eta') \\ &= -\lambda^2 \int \frac{d\eta d\eta'}{(\eta\eta')^{1+\varepsilon}} \left[\frac{\eta}{E_1 - k_I} \partial_\eta e^{i(E_1 - k_I)\eta} \right] e^{i(E_2 + k_I)\eta'} \theta(\eta - \eta') \\ &= \frac{\lambda^2}{E_1 - k_I} \int \frac{d\eta d\eta'}{(\eta\eta')^{1+\varepsilon}} \partial_\eta \left[\eta^{-\varepsilon} \theta(\eta - \eta') \right] e^{i(E_1 - k_I)\eta} e^{i(E_2 + k_I)\eta'} \\ &= \frac{\varepsilon}{E_1 - k_I} F_+ + \frac{1}{E_1 - k_I} F_c, \end{aligned} \quad (\text{B.738})$$

where we got an extra function from the derivative of the theta-function:

$$F_c \equiv -\lambda^2 \int \frac{d\eta}{(-\eta)^{1+2\varepsilon}} e^{i(E_1 + E_2)\eta}. \quad (\text{B.739})$$

Note that this function is obtained by the replacement $\theta(\eta - \eta') \rightarrow \eta\delta(\eta - \eta')$ in (B.732). Similarly, the E_2 -derivative yields

$$\begin{aligned} \partial_{E_2} F_+ &= -\lambda^2 \int \frac{d\eta d\eta'}{(\eta\eta')^{1+\varepsilon}} e^{i(E_1 - k_I)\eta} \left[\partial_{E_2} e^{i(E_2 + k_I)\eta'} \right] \theta(\eta - \eta') \\ &= -\lambda^2 \int \frac{d\eta d\eta'}{(\eta\eta')^{1+\varepsilon}} e^{i(E_1 - k_I)\eta} \left[\frac{\eta'}{E_2 + k_I} \partial_{\eta'} e^{i(E_2 + k_I)\eta'} \right] \theta(\eta - \eta') \\ &= \frac{\lambda^2}{E_2 + k_I} \int \frac{d\eta d\eta'}{\eta^{1+\varepsilon}} \partial_{\eta'} \left[(\eta')^{-\varepsilon} \theta(\eta - \eta') \right] e^{i(E_1 - k_I)\eta} e^{i(E_2 + k_I)\eta'} \\ &= \frac{\varepsilon}{E_2 + k_I} F_+ - \frac{1}{E_2 + k_I} F_c. \end{aligned} \quad (\text{B.740})$$

Together, these imply that the differential of the time-ordered contribution is

$$\boxed{dF_+ = (\varepsilon F_+ + F_c) d\log(E_1 - k_I) + (\varepsilon F_+ - F_c) d\log(E_2 + k_I)}, \quad (\text{B.741})$$

as required.

- By symmetry, the differential of the anti-time-ordered contribution (B.733) is

$$\boxed{dF_- = (\varepsilon F_- - F_c) d\log(E_1 + k_I) + (\varepsilon F_- + F_c) d\log(E_2 - k_I)} , \quad (\text{B.742})$$

as required.

- Finally, the derivatives of the new function F_c can be computed as

$$\begin{aligned} \partial_{E_1} F_c &= -\lambda^2 \int \frac{d\eta}{(-\eta)^{1+2\varepsilon}} \partial_{E_1} e^{i(E_1+E_2)\eta} \\ &= -\lambda^2 \int \frac{d\eta}{(-\eta)^{1+2\varepsilon}} \frac{\eta}{E_1 + E_2} \partial_\eta e^{i(E_1+E_2)\eta} \\ &= -\frac{\lambda^2}{E_1 + E_2} \int d\eta e^{i(E_1+E_2)\eta} \partial_\eta (-\eta)^{-2\varepsilon} \\ &= \frac{2\varepsilon}{E_1 + E_2} F_c , \end{aligned} \quad (\text{B.743})$$

$$\partial_{E_2} F_c = \frac{2\varepsilon}{E_1 + E_2} F_c , \quad (\text{B.744})$$

and hence the differential is

$$\boxed{F_c = 2\varepsilon F_c d\log(E_1 + E_2)} , \quad (\text{B.745})$$

as required.

2. We would now like to solve the equations

$$dF_0 = \varepsilon(d\log(E_1 + k_I) + d\log(E_2 + k_I)) F_0 , \quad (\text{B.746})$$

$$dF_+ = (\varepsilon F_+ + F_c) d\log(E_1 - k_I) + (\varepsilon F_+ - F_c) d\log(E_2 + k_I) , \quad (\text{B.747})$$

$$dF_- = (\varepsilon F_- - F_c) d\log(E_1 + k_I) + (\varepsilon F_- + F_c) d\log(E_2 - k_I) , \quad (\text{B.748})$$

$$dF_c = 2\varepsilon F_c d\log(E_1 + E_2) , \quad (\text{B.749})$$

The solutions of (B.746) and (B.749) are simple power laws

$$F_0 = c_0 (E_1 + k_I)^\varepsilon (E_2 + k_I)^\varepsilon , \quad (\text{B.750})$$

$$F_c = c_c (E_1 + E_2)^{2\varepsilon} , \quad (\text{B.751})$$

where c_0 and c_c are unfixed integration constants. The relative size of the two constants will later be fixed by imposing the correct factorization limit.

The solution (B.751) then enters as a source in (B.747) and (B.748). For example, equation (B.747) can be expressed as two separate ODEs as

$$\left(\partial_{E_1} - \frac{\varepsilon}{E_1 - k_I} \right) F_+ = + \frac{c_c}{E_1 - k_I} (E_1 + E_2)^{2\varepsilon} , \quad (\text{B.752})$$

$$\left(\partial_{E_2} - \frac{\varepsilon}{E_2 + k_I} \right) F_+ = - \frac{c_c}{E_2 + k_I} (E_1 + E_2)^{2\varepsilon} , \quad (\text{B.753})$$

where we have substituted the source term for F_c . To solve these equations, it is useful to write

$$F_+ \equiv (E_1 + E_2)^{2\varepsilon} (f(\xi) + h(\xi)), \quad (\text{B.754})$$

where $\xi \equiv (E_1 - k_I)/(E_2 + k_I)$ and $h(\xi)$ is a solution to the homogeneous equation. The equations for the inhomogeneous solution become

$$\left(\xi \partial_\xi + \varepsilon - \frac{2\varepsilon}{1+\xi} \right) f = c_c, \quad (\text{B.755})$$

$$\left(\xi \partial_\xi - \varepsilon + \frac{2\varepsilon}{1+\xi^{-1}} \right) f = c_c. \quad (\text{B.756})$$

We solve this using a series expansion, in two regimes separately: $|\xi| \leq 1$ and $|\xi| \geq 1$.

- For $|\xi| \leq 1$, we start with the ansatz

$$f_{<}(\xi) = \sum_{m=0}^{\infty} a_m \xi^m, \quad (\text{B.757})$$

which is regular at $\xi = 0$. This is a solution of (B.755) and (B.756) if

$$\begin{aligned} a_0 &= -\frac{c_c}{\varepsilon}, \\ a_m &= -c_c \frac{2}{\varepsilon} \frac{(\varepsilon)_m}{(1-\varepsilon)_m} (-1)^m \quad (m \geq 1). \end{aligned} \quad (\text{B.758})$$

The solution can be expressed as

$$f_{<}(\xi) = c_c \left[\frac{1}{\varepsilon} - \frac{2}{\varepsilon} \sum_{m=0}^{\infty} \frac{(\varepsilon)_m}{(1-\varepsilon)_m} (-\xi)^m \right] \quad (|\xi| \leq 1). \quad (\text{B.759})$$

This series converges absolutely at $|\xi| = 1$ (or $E_1 + E_2 = 0$) for $2\varepsilon < 0$. For $2\varepsilon \geq 0$, the branch cut at $|\xi| = 1$ cancels with the prefactor $(E_1 + E_2)^{2\varepsilon}$ in (B.754), so that the limit remains well defined. In this specific example, the series solution in (B.759) could also be written in terms of the Gauss hypergeometric function ${}_2F_1$, which could then be analytically continued to the regime $|\xi| > 1$. In other examples, however, we won't have this luxury, so we will instead continue working with the explicit series solution.

- For $|\xi| \geq 1$, we use as the ansatz a series expansion around $\xi^{-1} = 0$:

$$f_{>}(\xi) = \sum_{m=0}^{\infty} \tilde{a}_m \frac{1}{\xi^m}. \quad (\text{B.760})$$

The solution can be found in a similar manner and is given by

$$f_{>}(\xi) = -c_c \left[\frac{1}{\varepsilon} - \frac{2}{\varepsilon} \sum_{m=0}^{\infty} \frac{(\varepsilon)_m}{(1-\varepsilon)_m} \frac{1}{(-\xi)^m} \right] \quad (|\xi| \geq 1), \quad (\text{B.761})$$

which differs from (B.759) by the exchange $\xi \leftrightarrow 1/\xi$, and an overall minus sign.

The homogeneous solution is

$$h(\xi) = c_+ \frac{\xi^\varepsilon}{(1 + \xi)^{2\varepsilon}}. \quad (\text{B.762})$$

We will fix the homogeneous solution by imposing the absence of a folded singularity at $\xi = 0$ (or $E_1 - k_I = 0$) and continuity at $|\xi| = 1$.

- For $|\xi| \leq 1$, the solution (B.762) has a branch point at $\xi = 0$. Since the inhomogeneous solution $f_{<}(\xi)$ is regular at $\xi = 0$, we must set $c_+ = 0$ to forbid the appearance of a folded singularity. The final solution, for $|\xi| \leq 1$, then is

$$F_+ = c_c (E_1 + E_2)^{2\varepsilon} \left[\frac{1}{\varepsilon} - \frac{2}{\varepsilon} \sum_{m=0}^{\infty} \frac{(\varepsilon)_m}{(1 - \varepsilon)_m} (-\xi)^m \right] \quad (|\xi| \leq 1), \quad (\text{B.763})$$

which is the same as $(E_1 + E_2)^{2\varepsilon} f_{<}(\xi)$.

- For $|\xi| \geq 1$, we avoid the branch point at $\xi = 0$, so we can, in principle, add a nonzero homogeneous piece, $f_{>}(\xi) + h(\xi)$. To fix the homogeneous solution, we impose continuity at $\xi = -1$. This gives

$$h(\xi) = c_c \frac{\Gamma(\varepsilon)\Gamma(\varepsilon)}{\Gamma(-2\varepsilon)} \frac{\xi^\varepsilon}{(1 + \xi)^{2\varepsilon}} \quad (|\xi| \geq 1). \quad (\text{B.764})$$

We see that the size of the homogeneous solution, for $|\xi| \geq 1$, is determined by the integration constant c_c .

Combining these results, we thus have

$$F_+ = (E_1 + E_2)^{2\varepsilon} \times \begin{cases} f_{<}(\xi) & |\xi| \leq 1, \\ f_{>}(\xi) + h(\xi) & |\xi| \geq 1. \end{cases} \quad (\text{B.765})$$

The solution for F_- can be obtained from this solution by the replacements $E_1 \leftrightarrow E_2$.

The solution still has two free coefficients c_0 and c_c , which appear in the total wavefunction

$$F = F_+ + F_- - F_0. \quad (\text{B.766})$$

As our final boundary condition, we impose the correct factorization behavior in the limit $E_1 + k_I \rightarrow 0$:

$$\lim_{E_1 + k_I \rightarrow 0} F \propto (E_1 + k_I)^\varepsilon \left((E_2 - k_I)^\varepsilon - (E_2 + k_I)^\varepsilon \right). \quad (\text{B.767})$$

Substituting (B.766) in the left-hand side, one finds

$$c_0 = -\frac{\Gamma(-\varepsilon)\Gamma(-\varepsilon)}{\Gamma(-2\varepsilon)} c_c. \quad (\text{B.768})$$

The final solution for the wavefunction then is

$$F = c_c \left[-\frac{\Gamma(-\varepsilon)\Gamma(-\varepsilon)}{\Gamma(-2\varepsilon)} (E_L E_R)^\varepsilon + \frac{2}{\varepsilon} E^{2\varepsilon} \left(1 - {}_2F_1 \left[\begin{matrix} 1, \varepsilon \\ 1 - \varepsilon \end{matrix} \middle| \frac{k_{34} - k_I}{E_L} \right] - {}_2F_1 \left[\begin{matrix} 1, \varepsilon \\ 1 - \varepsilon \end{matrix} \middle| \frac{k_{12} - k_I}{E_R} \right] \right) \right], \quad (\text{B.769})$$

where, for compactness, we have introduced the hypergeometric function and defined $E_L \equiv E_1 + k_I$, $E_R \equiv E_2 + k_I$ and $E \equiv E_1 + E_2$. Up to an overall normalization, this is the same as the solution (10.115).

Solution to Exercise 10.14

From the definition $u \equiv k_I/k_{12}$, we have

$$\partial_{k_{12}} = -\frac{u^2}{k_I} \partial_u, \quad (\text{B.770})$$

$$\partial_{k_{12}}^2 = -\frac{u^2}{k_I} \partial_u \left(-\frac{u^2}{k_I} \partial_u \right) = \frac{1}{k_I^2} (u^4 \partial_u^2 + 2u^3 \partial_u), \quad (\text{B.771})$$

and hence

$$(k_{12}^2 - k_I^2) \partial_{k_{12}}^2 = (1 - u^2) k_{12}^2 \partial_{k_{12}}^2 = (1 - u^2) (u^2 \partial_u^2 + 2u \partial_u), \quad (\text{B.772})$$

$$2k_{12} \partial_{k_{12}} = -2u \partial_u. \quad (\text{B.773})$$

Moreover, we have

$$\frac{1}{E} = \frac{1}{k_{12} + k_{34}} = \frac{1}{k_I(u^{-1} + v^{-1})} = \frac{1}{k_I} \frac{uv}{u + v}. \quad (\text{B.774})$$

Substituting these expressions in (10.132), we get

$$\boxed{\left(u^2(1 - u^2) \partial_u^2 - 2u^3 \partial_u + \frac{m^2}{H^2} - 2 \right) \hat{F} = \lambda^2 H^2 \frac{uv}{u + v}}, \quad (\text{B.775})$$

as required.

Solution to Exercise 10.15

The integral form of the three-point wavefunction coefficient of two conformally coupled scalars and a generic massive scalar is

$$G \equiv \langle O_1 O_2 X_3 \rangle' = \frac{i\lambda}{f_{k_1}(\eta_*) f_{k_2}(\eta_*)} \int_{-\infty}^{\eta_*} \frac{d\eta}{(H\eta)^4} f_{k_1}(\eta) f_{k_2}(\eta) \mathcal{K}_\nu(k_3\eta) \quad (\text{B.776})$$

$$= \frac{i\lambda}{H^4 \eta_*^2} \int_{-\infty}^{\eta_*} \frac{d\eta}{\eta^2} e^{i(k_1 + k_2)\eta} \mathcal{K}_\nu(k_3\eta), \quad (\text{B.777})$$

where $\mathcal{K}_\nu(k_3\eta)$ is the bulk-to-boundary propagator of the massive field. The propagator satisfies the following equation of motion

$$\left(\eta^2 \frac{\partial^2}{\partial \eta^2} - 2\eta \frac{\partial}{\partial \eta} + k_3^2 \eta^2 + \frac{m^2}{H^2} \right) \mathcal{K}_\nu(k_3\eta) = 0. \quad (\text{B.778})$$

This implies that

$$\frac{i\lambda}{H^4\eta_*^2} \int_{-\infty}^{\eta_*} \frac{d\eta}{\eta^2} e^{ik_{12}\eta} \left(\eta^2 \frac{\partial^2}{\partial \eta^2} - 2\eta \frac{\partial}{\partial \eta} + k_3^2 \eta^2 + \frac{m^2}{H^2} \right) \mathcal{K}_\nu(k_3\eta) = 0. \quad (\text{B.779})$$

As in the four-point example shown in the text, we integrate the time derivatives by parts and trade factors of η for $-i\partial_{k_{12}}$ to obtain

$$\begin{aligned} 0 &= \frac{i\lambda}{H^4\eta_*^2} \int_{-\infty}^0 \frac{d\eta}{\eta^2} \left((k_{12}^2 - k_3^2) \partial_{k_{12}}^2 + 2k_{12} \partial_{k_{12}} + \frac{m^2}{H^2} - 2 \right) e^{ik_{12}\eta} \mathcal{K}_\nu(k_3\eta) \\ &= \left((k_{12}^2 - k_3^2) \partial_{k_{12}}^2 + 2k_{12} \partial_{k_{12}} + \frac{m^2}{H^2} - 2 \right) G, \end{aligned} \quad (\text{B.780})$$

where, in the second line, we have pulled the differential operator outside the integral which we then recognize as (B.777). Defining $u \equiv k_3/k_{12}$, we have (see Exercise 10.14)

$$\partial_{k_{12}} = -\frac{u^2}{k_3} \partial_u, \quad \partial_{k_{12}}^2 = \frac{1}{k_3^2} (u^4 \partial_u^2 + 2u^3 \partial_u), \quad (\text{B.781})$$

so that

$$\boxed{\left[(1-u^2) u^2 \partial_u^2 - 2u^3 \partial_u + \left(\mu^2 + \frac{1}{4} \right) \right] G = 0}, \quad (\text{B.782})$$

where we have defined $\nu = i\mu$.

B.11 Exercises in Chapter 11

Solution to Exercise 11.1

We derive (11.3) for exchange particles with different spin.

- **$J = 0$:** If the internal state is a massive scalar, then the three-particle amplitude A_3 is just a constant,

$$A_3(p_1, p_2, p_3) = \lambda, \quad (\text{B.783})$$

where λ determines the interaction strength.

- **$J = 1$:** If the internal state is a massive vector, then A_3^h must depend on the polarization vector ϵ_μ^h . Since $p^\mu \epsilon_\mu^h = 0$, the most general amplitude is of the form

$$A_3^h(p_1, p_2, p_3) = \lambda (p_1^\mu - p_2^\mu) \epsilon_\mu^h(p_3). \quad (\text{B.784})$$

- **$J \geq 1$:** If the internal state is a massive particle of general spin J , then we have

$$A_3^h(p_1, p_2, p_3) = \lambda (p_1 - p_2)^{\mu_1} \cdots (p_1 - p_2)^{\mu_J} \epsilon_{\mu_1 \dots \mu_J}^h(p_3). \quad (\text{B.785})$$

The polarization sum in the four-point amplitude (11.1) then reads

$$f(k, q) \equiv \sum_h k^{\mu_1} \cdots k^{\mu_J} q^{\nu_1} \cdots q^{\nu_J} \epsilon_{\mu_1 \dots \mu_J}^h \epsilon_{\nu_1 \dots \nu_J}^{-h}, \quad (\text{B.786})$$

where $k \equiv p_1 - p_2$ and $q \equiv p_3 - p_4$. This sum is evaluated most easily in the rest frame of the massive particle, where only the spatial components of the polarization tensors are non-vanishing. Switching to real space variables for the sake of notational familiarity, we then have

$$f(\mathbf{x}, \mathbf{y}) = \sum_h x^{i_1} \cdots x^{i_J} y^{j_1} \cdots y^{j_J} \epsilon_{i_1 \dots i_J}^h \epsilon_{j_1 \dots j_J}^{-h} \equiv |\mathbf{x}|^J |\mathbf{y}|^J f(\cos \theta), \quad (\text{B.787})$$

where $\cos \theta \equiv \mathbf{x} \cdot \mathbf{y} / |\mathbf{x}| |\mathbf{y}|$. The sum over the polarization tensors must be made out of Kronecker delta's, be symmetric in all i 's and j 's and traceless in the i 's and j 's separately. The tracelessness condition implies

$$\nabla_{\mathbf{x}}^2 f(\mathbf{x}, \mathbf{y}) = 0. \quad (\text{B.788})$$

The solution to the Laplace equation is

$$\frac{1}{|\mathbf{x} - \mathbf{y}|} = \frac{1}{(|\mathbf{x}|^2 - 2|\mathbf{x}||\mathbf{y}| \cos \theta + |\mathbf{y}|^2)^{1/2}} = \frac{1}{|\mathbf{y}|} \sum_{J=0}^{\infty} \left(\frac{|\mathbf{x}|}{|\mathbf{y}|} \right)^J P_J(\cos \theta), \quad (\text{B.789})$$

where we have introduced the Legendre polynomials. From the term proportional to $|\mathbf{x}|^J$, we read off $f(\cos \theta) = P_J(\cos \theta)$. Writing the amplitude in a Lorentz-invariant form gives (11.3).

Solution to Exercise 11.2

The generators of the $(D+1)$ -dimensional Lorentz transformations are

$$J_{AB} = X_A \partial_B - X_B \partial_A. \quad (\text{B.790})$$

The corresponding Killing vector is

$$K_A \equiv \omega_{AB} X^B, \quad (\text{B.791})$$

with ω_{AB} an anti-symmetric matrix. It is straightforward to check that this satisfies the Killing equation $\partial_{(A} K_{B)} = 0$.

The flat slicing corresponds to the embedding

$$\begin{aligned} X^0 &= \frac{L}{2(-\eta)} (1 - \eta^2 + x^2), \\ X^i &= \frac{Lx^i}{(-\eta)}, \\ X^D &= \frac{L}{2(-\eta)} (1 + \eta^2 - x^2), \end{aligned} \quad (\text{B.792})$$

where $L \in (0, \infty)$, $\eta \in (-\infty, 0)$ and $x^i \in (-\infty, \infty)$. It is straightforward to check that the embedding coordinates satisfy $\eta_{AB} X^A X^B = H^{-2}$, if we define $L \equiv H^{-1}$; i.e. they parameterize the dS hyperboloid with Hubble parameter H . In fact, these coordinates foliate half the dS hyperboloid with flat slices, and in these coordinates the $(D+1)$ -dimensional Minkowski metric takes the form

$$ds^2 = d\rho^2 + (LH)^2 ds_{\text{dS}_D}^2, \quad (\text{B.793})$$

where the dS metric in the inflationary slicing is given by

$$ds_{\text{dS}_D}^2 = \frac{1}{H^2 \eta^2} (-d\eta^2 + \delta_{ij} dx^i dx^j). \quad (\text{B.794})$$

By inverting the embedding (B.792), we can express the ambient space derivatives in terms of intrinsic coordinates [134]

$$\begin{aligned} \frac{\partial}{\partial X^0} &= \frac{1}{2(-\eta)} (-1 + \eta^2 - x^2) \partial_L + \frac{1}{2L} (1 + \eta^2 + x^2) \partial_\eta + \frac{\eta}{L} x^i \partial_i, \\ \frac{\partial}{\partial X^i} &= \frac{x^i}{(-\eta)} \partial_L - \frac{x^i}{L} \partial_\eta - \frac{\eta}{L} \partial_i, \\ \frac{\partial}{\partial X^D} &= \frac{1}{2(-\eta)} (1 + \eta^2 - x^2) \partial_L + \frac{1}{2L} (-1 + \eta^2 + x^2) \partial_\eta + \frac{\eta}{L} x^i \partial_i. \end{aligned} \quad (\text{B.795})$$

Equations (B.792) and (B.795) can then be used to translate ambient space formulae into (A)dS formula in the flat slicing.

It is natural to define

$$\begin{aligned} \hat{D} &\equiv J_{D0}, \\ P_i &\equiv J_{Di} - J_{0i}, \\ K_i &\equiv J_{Di} + J_{0i}. \end{aligned} \quad (\text{B.796})$$

Using (B.792) and (B.795), we then obtain

$$\begin{aligned} J_{ij} &= x_i \partial_j - x_j \partial_i, \\ P_i &= \partial_i, \\ \hat{D} &= -\eta \partial_\eta - x^i \partial_i, \\ K_i &= 2x_i \eta \partial_\eta + \left(2x^j x_i + (\eta^2 - x^2) \delta_i^j \right) \partial_j. \end{aligned} \quad (\text{B.797})$$

Notice that all L dependence has dropped out, which is a manifestation of the fact that these transformations are parallel to the dS hyperboloid. One can check that these do indeed generate an $\text{so}(D, 1)$ algebra.

Solution to Exercise 11.3

In position space, the generators of dilatations and special conformal transformations are

$$\hat{D} O(\mathbf{x}) = (\bar{\Delta} + x^j \partial_{x^j}) O(\mathbf{x}), \quad (\text{B.798})$$

$$K_i O(\mathbf{x}) = -(2\bar{\Delta} x^i + 2x^i x^j \partial_{x^j} - x^2 \partial_{x^i}) O(\mathbf{x}), \quad (\text{B.799})$$

where $\bar{\Delta} = d - \Delta$ is the scaling dimension of the operator O , and d is the spatial dimension. We would like to express this generators in Fourier space.

First, we note that we can write the position-space operator in Fourier space as

$$O(\mathbf{x}) = \int \frac{d^d k}{(2\pi)^d} O_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{x}}. \quad (\text{B.800})$$

Acting on this with the generator of dilatations (B.798), we get

$$\hat{D}O(\mathbf{x}) = \int \frac{d^d k}{(2\pi)^d} O_{\mathbf{k}} (\bar{\Delta} + x^j \partial_{x^j}) e^{-i\mathbf{k}\cdot\mathbf{x}} \quad (\text{B.801})$$

$$= \int \frac{d^d k}{(2\pi)^d} O_{\mathbf{k}} (\bar{\Delta} + k^j \partial_{k^j}) e^{-i\mathbf{k}\cdot\mathbf{x}}, \quad (\text{B.802})$$

where we replaced $\partial_{\mathbf{x}}$ by $-i\mathbf{k}$ and $\mathbf{x}$ by $i\partial_{\mathbf{k}}$. Now, we integrate by parts, so that the derivative with respect to $\mathbf{k}$ acts on the Fourier-space operator $O_{\mathbf{k}}$. Crucially, in doing this, the derivative *also* acts on the vector k^j , giving a factor of the dimension d . Putting this together, we get

$$\hat{D}O(\mathbf{x}) = \int \frac{d^d k}{(2\pi)^d} \left[-(d - \bar{\Delta}) - k^j \partial_{k^j} \right] O_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{x}}, \quad (\text{B.803})$$

so that we can infer the action of the dilatation operator in Fourier space as

$$\boxed{\hat{D}O_{\mathbf{k}} = -\left[(d - \bar{\Delta}) + k^j \partial_{k^j}\right] O_{\mathbf{k}}}. \quad (\text{B.804})$$

Although the combination $d - \bar{\Delta}$ appears, which we previously called Δ , we write things solely in terms of $\bar{\Delta}$ throughout, for clarity.

We can do the same for the SCT generator. Acting with (B.799) on (B.800), we have

$$K_i O(\mathbf{x}) = - \int \frac{d^d k}{(2\pi)^d} O_{\mathbf{k}} \left(2\bar{\Delta} x^i + 2x^i x^j \partial_{x^j} - x^2 \partial_{x^i} \right) e^{-i\mathbf{k}\cdot\mathbf{x}} \quad (\text{B.805})$$

$$= -i \int \frac{d^d k}{(2\pi)^d} O_{\mathbf{k}} \left(-2\bar{\Delta} \partial_{k^i} + 2k^j \partial_{k^j} \partial_{k^i} - k_i \partial_{k^j} \partial_{k^j} \right) e^{-i\mathbf{k}\cdot\mathbf{x}}. \quad (\text{B.806})$$

Integrating the last expression by parts, we obtain

$$\boxed{K_i O_{\mathbf{k}} = -i \left[2(d - \bar{\Delta}) \partial_{k^i} + 2k^j \partial_{k^j} \partial_{k^i} - k^i \partial_{k^j} \partial_{k^j} \right] O_{\mathbf{k}}}. \quad (\text{B.807})$$

The transformations (B.804) and (B.806) tell us how single operators transform infinitesimally under conformal transformations in momentum space. We now want to understand how correlation functions consisting of many operators transform. The essential subtlety is that in Fourier space translation invariance implies the presence of a momentum-conserving delta function

$$\langle O_1 \cdots O_N \rangle = (2\pi)^d \delta^{(d)}(\mathbf{k}_1 + \cdots + \mathbf{k}_N) \langle O_1 \cdots O_N \rangle'. \quad (\text{B.808})$$

Since the infinitesimal conformal transformations are differential operators, these transformations can hit the delta function. We then integrate these pieces by parts in order to infer the action on the correlation function with the delta function removed (and momentum conservation enforced).

We begin by considering the dilatation operator. Conceptually, the result is clear: since the delta function has dimension d , in order for the full correlation function to be dilatation invariant, the primed correlation function (with the delta function removed) must have total

scaling dimension $-d$. We now want to verify this by explicit calculation. Dilatations act on the correlation function as

$$\begin{aligned} \sum_{a=1}^N \hat{D}_a \langle O_1 \cdots O_N \rangle &= \sum_{a=1}^N \left(-(d - \bar{\Delta}_a) - k_a^j \partial_{k_a^j} \right) \left[(2\pi)^d \delta^{(d)}(\mathbf{K}) \langle O_1 \cdots O_N \rangle' \right] \\ &= \left[-(2\pi)^d \mathbf{K} \cdot \partial_{\mathbf{K}} \delta^{(d)}(\mathbf{K}) \right] \langle O_1 \cdots O_N \rangle' + (2\pi)^d \delta^{(d)}(\mathbf{K}) \sum_{a=1}^N \hat{D}_a \langle O_1 \cdots O_N \rangle', \end{aligned} \quad (\text{B.809})$$

where we have defined the total momentum $\mathbf{K} \equiv \mathbf{k}_1 + \cdots + \mathbf{k}_N$. We treat the term involving the derivative of the delta function distributionally and integrate by parts to obtain a factor of d . The term where the derivative $\partial_{\mathbf{K}}$ acts on $\langle O_1 \cdots O_N \rangle'$ vanishes because it multiplies $\mathbf{K} \delta^{(d)}(\mathbf{K}) = 0$. We then have

$$\sum_{a=1}^N \hat{D}_a \langle O_1 \cdots O_N \rangle = (2\pi)^d \delta^{(d)}(\mathbf{K}) \left[d + \sum_{a=1}^N \left(-(d - \bar{\Delta}_a) - k_a^j \partial_{k_a^j} \right) \right] \langle O_1 \cdots O_N \rangle'. \quad (\text{B.810})$$

From this, we infer that the action of dilatations on Fourier space correlation functions with the delta function removed is

$$\begin{aligned} \sum_{a=1}^N \hat{D}_a \langle O_1 \cdots O_N \rangle' &= \left[d + \sum_{a=1}^N \left(-(d - \bar{\Delta}_a) - k_a^j \partial_{k_a^j} \right) \right] \langle O_1 \cdots O_N \rangle' \\ &= \left[d + \sum_{a=1}^N \hat{D}_a \right] \langle O_1 \cdots O_N \rangle'. \end{aligned} \quad (\text{B.811})$$

We can alternatively write this as

$$\boxed{\sum_{a=1}^N \hat{D}_a \langle O_1 \cdots O_N \rangle' = - \left[d(N-1) - \sum_{a=1}^N \bar{\Delta}_a + \sum_{a=1}^{N-1} k_a^j \partial_{k_a^j} \right] \langle O_1 \cdots O_N \rangle'}. \quad (\text{B.812})$$

Next, we repeat the exercise for special conformal transformations. In this case, the SCT generator actually commutes past the delta function, but let's check this explicitly. Acting with the SCT generator (B.807) on an N -point correlator, we get a number of contributions:

$$\begin{aligned} \sum_{a=1}^N K_a^i \langle O_1 \cdots O_N \rangle &= -i(2\pi)^d \delta^{(d)}(\mathbf{K}) \sum_{a=1}^N \left(2(d - \bar{\Delta}) \partial_{k_a^i} + 2k_a^j \partial_{k_a^j} \partial_{k_a^i} - k_i \partial_{k_a^j} \partial_{k_a^j} \right) \langle O_1 \cdots O_N \rangle' \\ &\quad + T_1 + T_2 + T_3, \end{aligned} \quad (\text{B.813})$$

where

$$T_1 \equiv 2i(2\pi)^d \sum_{a=1}^N \partial_{k_a^j} \delta^{(d)}(\mathbf{K}) \left(k_a^i \partial_{k_a^j} - k_a^j \partial_{k_a^i} \right) \langle O_1 \cdots O_N \rangle', \quad (\text{B.814})$$

$$T_2 \equiv 2i(2\pi)^d \sum_{a=1}^N \partial_{k_a^i} \delta^{(d)}(\mathbf{K}) \hat{D}_a \langle O_1 \cdots O_N \rangle', \quad (\text{B.815})$$

$$T_3 \equiv i(2\pi)^d \sum_{a=1}^N \left(k_a^i \partial_{k_a^j} \partial_{k_a^j} \delta^{(d)}(\mathbf{K}) - 2k_a^j \partial_{k_a^i} \partial_{k_a^j} \delta^{(d)}(\mathbf{K}) \right) \langle O_1 \cdots O_N \rangle'. \quad (\text{B.816})$$

In order to simplify this, we note that we can trade the derivatives acting on the delta function for derivatives with respect to the total momentum

$$\partial_{k_a^j} \delta^{(d)}(\mathbf{K}) = \partial_{K^j} \delta^{(d)}(\mathbf{K}). \quad (\text{B.817})$$

Using this, we can see, for example, that

$$T_1 = 2i(2\pi)^d \partial_{K^j} \delta^{(d)}(\mathbf{K}) \sum_{a=1}^N \left(k_a^i \partial_{k_a^j} - k_a^j \partial_{k_a^i} \right) \langle O_1 \cdots O_N \rangle', \quad (\text{B.818})$$

where (B.817) has allowed us to move the delta function outside the sum. We then now recognize the operator acting on the primed correlator in (B.818) as the generator of rotations in Fourier space. Assuming that the correlator is rotation invariant, this vanishes.⁶² We can similarly rewrite the term in (B.816) as

$$T_3 = i(2\pi)^d \left(K^i \partial_{K^j} \partial_{K^j} - 2K^j \partial_{K^i} \partial_{K^j} \right) \delta^{(d)}(\mathbf{K}) \langle O_1 \cdots O_N \rangle', \quad (\text{B.819})$$

where we were able to do the full sum in this case. We can now treat the derivatives of the delta function distributionally, and integrate by parts. We do this for a single derivative in each of the two terms. There will be a contribution where the derivative acts on the unprimed correlator, but this will be zero because it will multiply $\mathbf{K} \delta^{(d)}(\mathbf{K}) = 0$. So, we are left with

$$T_3 = i(2\pi)^d \times 2d \partial_{K^i} \delta^{(d)}(\mathbf{K}) \langle O_1 \cdots O_N \rangle'. \quad (\text{B.820})$$

We can then substitute this back into (B.813), along with the fact that (B.818) vanishes, to obtain

$$\begin{aligned} \sum_{a=1}^N K_a^i \langle O_1 \cdots O_N \rangle &= -i(2\pi)^d \delta^{(d)}(\mathbf{K}) \sum_{a=1}^N \left(2(d - \bar{\Delta}) \partial_{k_a^i} + 2k_a^j \partial_{k_a^j} \partial_{k_a^i} - k_i \partial_{k_a^j} \partial_{k_a^j} \right) \langle O_1 \cdots O_N \rangle' \\ &\quad + 2i(2\pi)^d \partial_{K^i} \delta^{(d)}(\mathbf{K}) \left[d + \sum_{a=1}^N \hat{D}_a \right] \langle O_1 \cdots O_N \rangle'. \end{aligned} \quad (\text{B.821})$$

We recognize the term in brackets on the second line as the action of the dilatation operator on the primed correlator (B.811). Assuming that the correlator is dilatation invariant, we see that indeed the SCT operator commutes through the delta function and we have

$$\boxed{\sum_{a=1}^N K_a^i \langle O_1 \cdots O_N \rangle' = -i \sum_{a=1}^N \left(2(d - \bar{\Delta}) \partial_{k_a^i} + 2k_a^j \partial_{k_a^j} \partial_{k_a^i} - k_i \partial_{k_a^j} \partial_{k_a^j} \right) \langle O_1 \cdots O_N \rangle'}, \quad (\text{B.822})$$

as desired.

⁶²Here, we are specializing to scalar operators; if the operators have spin, rotation invariance involves also acting on the indices carried by the operator. See [105], for an example involving spin-2 operators.

Solution to Exercise 11.4

The late-time two-point function of a massive scalar field, computed in Exercise 8.6, is

$$\lim_{k\eta \rightarrow 0} P_\phi(k, \eta) = \frac{H^2}{4\pi} (-\eta)^3 \left(\Gamma^2(i\mu) \left(\frac{-k\eta}{2} \right)^{-2i\mu} + \Gamma^2(-i\mu) \left(\frac{-k\eta}{2} \right)^{2i\mu} + \frac{2\pi \coth(\pi\mu)}{\mu} \right), \quad (\text{B.823})$$

where

$$i\mu = \sqrt{\frac{d^2}{4} - \frac{m^2}{H^2}}. \quad (\text{B.824})$$

Using the relation between the mass parameter and the scaling dimensions, $\pm 2i\mu = 2\Delta_\pm - d$, the two-point function (B.823) then becomes

$$\lim_{k\eta \rightarrow 0} P_\phi(k, \eta) \propto \left(\Gamma^2(i\mu) \left(\frac{-k\eta}{2} \right)^{2\Delta_- - d} + \Gamma^2(-i\mu) \left(\frac{-k\eta}{2} \right)^{2\Delta_+ - d} + \frac{2\pi \coth(\pi\mu)}{\mu} \right). \quad (\text{B.825})$$

The first two terms of this expression are manifestly of the form (11.46), and it is straightforward to directly check that each of these terms individually satisfy the Ward identities (11.41) and (11.42) with scaling dimensions Δ_- and Δ_+ , respectively. (For a light field, the contribution scaling with Δ_- will be the most important at late times.) We therefore see that the late-time two-point function of a massive field consists of two pieces with the scaling weights Δ_- and Δ_+ . Note that in addition to this, there is a position-space contact term contribution in (B.823) (which is k -independent in momentum space). Thus, if one wants to assign “dual” operators to the two fall-offs of the fields, they must be correlated in such a way to reproduce this momentum-independent piece.

Solution to Exercise 11.5

[DB: To be included]

Solution to Exercise 11.6

[DB: To be included]

Solution to Exercise 11.7

[DB: To be included]

Solution to Exercise 11.8

We begin by collecting some basic formulas. Differentiating a momentum vector gives

$$\frac{\partial k_n^i}{\partial k_n} = \frac{\partial k_n}{\partial k_n^i} = \hat{k}_n^i, \quad \frac{\partial k_n^i}{\partial k_n^j} = \delta^{ij}, \quad \frac{\partial \hat{k}_n^i}{\partial k_n^j} = \frac{\delta^{ij} - \hat{k}_n^i \hat{k}_n^j}{k_n}, \quad \frac{\partial \hat{k}_n^i}{\partial k_n} = 0. \quad (\text{B.826})$$

In addition, it will be useful to define the variable $s \equiv |\mathbf{k}_1 + \mathbf{k}_2|$ and $t \equiv |\mathbf{k}_2 + \mathbf{k}_3|$ (physically, these are the exchange energies in the s - and t -channels). The derivatives of the s, t variables are

$$\begin{aligned} \frac{\partial s}{\partial k_1^i} &= \frac{\partial s}{\partial k_2^i} = \frac{k_{12}^i}{s}, & \frac{\partial s}{\partial k_3^i} &= \frac{\partial s}{\partial k_4^i} = 0, \\ \frac{\partial t}{\partial k_2^i} &= \frac{\partial t}{\partial k_3^i} = \frac{k_{23}^i}{t}, & \frac{\partial t}{\partial k_1^i} &= \frac{\partial t}{\partial k_4^i} = 0, \end{aligned} \quad (\text{B.827})$$

where $k_{nm}^i \equiv k_n^i + k_m^i$. Using the chain rule, the first derivatives with respect to the momentum vectors can be replaced by

$$\begin{aligned} \frac{\partial}{\partial \hat{k}_1^i} &= \frac{k_1^i}{k_1} \frac{\partial}{\partial k_1} + \frac{k_{12}^i}{s} \frac{\partial}{\partial s}, & \frac{\partial}{\partial \hat{k}_3^i} &= \frac{k_3^i}{k_3} \frac{\partial}{\partial k_3} + \frac{k_{23}^i}{t} \frac{\partial}{\partial t}, \\ \frac{\partial}{\partial \hat{k}_2^i} &= \frac{k_2^i}{k_2} \frac{\partial}{\partial k_2} + \frac{k_{12}^i}{s} \frac{\partial}{\partial s} + \frac{k_{23}^i}{t} \frac{\partial}{\partial t}, & \frac{\partial}{\partial \hat{k}_4^i} &= \frac{k_4^i}{k_4} \frac{\partial}{\partial k_4}. \end{aligned} \quad (\text{B.828})$$

Moreover, the second derivatives are

$$\begin{aligned} \frac{\partial^2}{\partial \hat{k}_1^i \partial \hat{k}_1^j} &= \hat{k}_1^i \hat{k}_1^j \frac{\partial^2}{\partial k_1^2} + \frac{\delta^{ij} - \hat{k}_1^i \hat{k}_1^j}{k_1} \frac{\partial}{\partial k_1} + \frac{k_1^i k_{12}^j + k_{12}^i k_1^j}{s k_1} \frac{\partial^2}{\partial s \partial k_1} + \frac{\delta^{ij} - \hat{s}^{ij}}{s} \frac{\partial}{\partial s} + \hat{s}^{ij} \frac{\partial^2}{\partial s^2}, \\ \frac{\partial^2}{\partial \hat{k}_2^i \partial \hat{k}_2^j} &= \hat{k}_2^i \hat{k}_2^j \frac{\partial^2}{\partial k_2^2} + \frac{\delta^{ij} - \hat{k}_2^i \hat{k}_2^j}{k_2} \frac{\partial}{\partial k_2} + \frac{k_{23}^i k_{12}^j + k_{12}^i k_{23}^j}{s t} \frac{\partial^2}{\partial s \partial t} \\ &\quad + \frac{\delta^{ij} - \hat{s}^{ij}}{s} \frac{\partial}{\partial s} + \frac{k_2^i k_{12}^j + k_{12}^i k_2^j}{s k_2} \frac{\partial^2}{\partial s \partial k_2} + \hat{s}^{ij} \frac{\partial^2}{\partial s^2} \\ &\quad + \frac{\delta^{ij} - \hat{t}^{ij}}{t} \frac{\partial}{\partial t} + \frac{k_2^i k_{23}^j + k_{23}^i k_2^j}{t k_2} \frac{\partial^2}{\partial t \partial k_2} + \hat{t}^{ij} \frac{\partial^2}{\partial t^2}, \\ \frac{\partial^2}{\partial \hat{k}_3^i \partial \hat{k}_3^j} &= \hat{k}_3^i \hat{k}_3^j \frac{\partial^2}{\partial k_3^2} + \frac{\delta^{ij} - \hat{k}_3^i \hat{k}_3^j}{k_3} \frac{\partial}{\partial k_3} + \frac{k_3^i k_{23}^j + k_{23}^i k_3^j}{t k_3} \frac{\partial^2}{\partial t \partial k_3} + \frac{\delta^{ij} - \hat{t}^{ij}}{t} \frac{\partial}{\partial t} + \hat{t}^{ij} \frac{\partial^2}{\partial t^2}, \\ \frac{\partial^2}{\partial \hat{k}_4^i \partial \hat{k}_4^j} &= \hat{k}_4^i \hat{k}_4^j \frac{\partial^2}{\partial k_4^2} + \frac{\delta^{ij} - \hat{k}_4^i \hat{k}_4^j}{k_4} \frac{\partial}{\partial k_4}, \end{aligned} \quad (\text{B.829})$$

where we have defined

$$\hat{s}^{ij} \equiv \frac{(k_1^i + k_2^i)(k_1^j + k_2^j)}{s^2}, \quad \hat{t}^{ij} \equiv \frac{(k_2^i + k_3^i)(k_2^j + k_3^j)}{t^2}. \quad (\text{B.830})$$

Below, we write down the constraint equations arising from dilatations and SCTs.

In the previous exercise, we have shown that the conformal Ward identities in Fourier space are

$$\left[9 - \sum_{n=1}^4 \left(\Delta_n - k_n^j \frac{\partial}{\partial k_n^j} \right) \right] F = 0, \quad (\text{B.831})$$

$$\sum_{n=1}^4 \left[k_n^i \frac{\partial^2}{\partial k_n^j \partial k_n^j} - 2k_n^j \frac{\partial^2}{\partial k_n^j \partial k_n^i} + 2(\Delta_n - 3) \frac{\partial}{\partial k_n^i} \right] F = 0. \quad (\text{B.832})$$

We will now massage them into a nicer form.

Dilatation Expanding the dilatation Ward identity (B.831) gives

$$0 = (9 - \Delta_t)F + k_1 \frac{\partial F}{\partial k_1} + k_2 \frac{\partial F}{\partial k_2} + k_3 \frac{\partial F}{\partial k_3} + k_4 \frac{\partial F}{\partial k_4} + s \frac{\partial F}{\partial s} + t \frac{\partial F}{\partial t}. \quad (\text{B.833})$$

where $\Delta_t \equiv \sum_{n=1}^4 \Delta_n$ denotes the total conformal weight. Differentiating this with respect to s and t gives

$$-\frac{\partial^2 F}{\partial s^2} = \frac{10 - \Delta_t}{s} \frac{\partial F}{\partial s} + \frac{k_1}{s} \frac{\partial^2 F}{\partial s \partial k_1} + \frac{k_2}{s} \frac{\partial^2 F}{\partial s \partial k_2} + \frac{k_3}{s} \frac{\partial^2 F}{\partial s \partial k_3} + \frac{k_4}{s} \frac{\partial^2 F}{\partial s \partial k_4} + \frac{t}{s} \frac{\partial^2 F}{\partial s \partial t}, \quad (\text{B.834})$$

$$-\frac{\partial^2 F}{\partial t^2} = \frac{10 - \Delta_t}{t} \frac{\partial F}{\partial t} + \frac{k_1}{t} \frac{\partial^2 F}{\partial t \partial k_1} + \frac{k_2}{t} \frac{\partial^2 F}{\partial t \partial k_2} + \frac{k_3}{t} \frac{\partial^2 F}{\partial t \partial k_3} + \frac{k_4}{t} \frac{\partial^2 F}{\partial t \partial k_4} + \frac{s}{t} \frac{\partial^2 F}{\partial s \partial t}. \quad (\text{B.835})$$

We will substitute these derivatives in our derivation of the SCT constraints.

Linear term We first consider the first-order term in (B.832). We have

$$\begin{aligned} \sum_{n=1}^4 2(\Delta_n - 3) \frac{\partial F}{\partial k_n^i} &= k_1^i \left[2(\Delta_1 - 3) \frac{1}{k_1} \frac{\partial F}{\partial k_1} + 2(\Delta_{12} - 6) \frac{1}{s} \frac{\partial F}{\partial s} \right] \\ &\quad + k_2^i \left[2(\Delta_2 - 3) \frac{1}{k_2} \frac{\partial F}{\partial k_2} + 2(\Delta_{12} - 6) \frac{1}{s} \frac{\partial F}{\partial s} + 2(\Delta_{23} - 6) \frac{1}{t} \frac{\partial F}{\partial t} \right] \\ &\quad + k_3^i \left[2(\Delta_3 - 3) \frac{1}{k_3} \frac{\partial F}{\partial k_3} + 2(\Delta_{23} - 6) \frac{1}{t} \frac{\partial F}{\partial t} \right] \\ &\quad + k_4^i \left[2(\Delta_4 - 3) \frac{1}{k_4} \frac{\partial F}{\partial k_4} \right], \end{aligned} \quad (\text{B.836})$$

where $\Delta_{nm} \equiv \Delta_n + \Delta_m$.

Cross term There are two second-order terms in (B.832). The cross term is given by

$$\begin{aligned}
\sum_{n=1}^4 k_n^j \frac{\partial^2 F}{\partial k_n^i \partial k_n^j} &= \textcolor{red}{k}_1^i \left[\frac{\partial^2}{\partial k_1^2} + \frac{s_1^2 + k_1^2}{sk_1} \frac{\partial^2 F}{\partial s \partial k_1} + \frac{t_2^2}{st} \frac{\partial^2 F}{\partial s \partial t} + \frac{k_2}{s} \frac{\partial^2 F}{\partial s \partial k_2} + \frac{\partial^2 F}{\partial s^2} \right] \\
&+ \textcolor{orange}{k}_2^i \left[\frac{k_1^2}{sk_1} \frac{\partial^2 F}{\partial s \partial k_1} + \frac{\partial^2 F}{\partial s^2} + \frac{\partial^2 F}{\partial k_2^2} + \frac{s_2^2 + t_2^2}{st} \frac{\partial^2 F}{\partial s \partial t} + \frac{s_2^2 + k_2^2}{sk_2} \frac{\partial^2 F}{\partial s \partial k_2} + \frac{t_2^2 + k_2^2}{tk_2} \frac{\partial^2 F}{\partial t \partial k_2} \right. \\
&\quad \left. + \frac{k_3^3}{tk_3} \frac{\partial^2 F}{\partial t \partial k_3} + \frac{\partial^2 F}{\partial t^2} \right] \\
&+ \textcolor{green}{k}_3^i \left[\frac{\partial^2 F}{\partial k_3^2} + \frac{t_3^2 + k_3^2}{tk_3} \frac{\partial^2 F}{\partial t \partial k_3} + \frac{s_2^2}{st} \frac{\partial^2 F}{\partial s \partial t} + \frac{k_2}{t} \frac{\partial^2 F}{\partial t \partial k_2} + \frac{\partial^2 F}{\partial t^2} \right] \\
&+ \textcolor{blue}{k}_4^i \left[\frac{\partial^2 F}{\partial k_4^2} \right], \tag{B.837}
\end{aligned}$$

where we have defined

$$s_1^2 \equiv k_1^i(k_1^i + k_2^i), \quad s_2^2 \equiv k_2^i(k_1^i + k_2^i), \quad t_2^2 \equiv k_2^i(k_2^i + k_3^i), \quad t_3^2 \equiv k_3^i(k_2^i + k_3^i). \tag{B.838}$$

Quadratic term Lastly, we consider the ‘‘Laplacian’’ term in (B.832). This is given by

$$\begin{aligned}
\sum_{n=1}^4 k_n^i \frac{\partial^2 F}{\partial k_n^j \partial k_n^j} &= \textcolor{red}{k}_1^i \left[\frac{\partial^2 F}{\partial k_1^2} + \frac{2}{k_1} \frac{\partial F}{\partial k_1} + \frac{2s_1^2}{sk_1} \frac{\partial^2 F}{\partial s \partial k_1} + \frac{2}{s} \frac{\partial F}{\partial s} + \frac{\partial^2 F}{\partial s^2} \right] \\
&+ \textcolor{orange}{k}_2^i \left[\frac{\partial^2 F}{\partial k_2^2} + \frac{2}{k_2} \frac{\partial F}{\partial k_2} + \frac{2s_2^2 + 2t_2^2 + 2k_1^i k_3^i - 2k_2^2}{st} \frac{\partial^2 F}{\partial s \partial t} + \frac{2}{s} \frac{\partial F}{\partial s} + \frac{2s_2^2}{sk_2} \frac{\partial^2 F}{\partial s \partial k_2} \right. \\
&\quad \left. + \frac{\partial^2 F}{\partial s^2} + \frac{2}{t} \frac{\partial F}{\partial t} + \frac{2t_2^2}{tk_2} \frac{\partial^2 F}{\partial t \partial k_2} + \frac{\partial^2 F}{\partial t^2} \right] \\
&+ \textcolor{green}{k}_3^i \left[\frac{\partial^2 F}{\partial k_3^2} + \frac{2}{k_3} \frac{\partial F}{\partial k_3} + \frac{2t_3^2}{tk_3} \frac{\partial^2 F}{\partial t \partial k_3} + \frac{2}{t} \frac{\partial F}{\partial t} + \frac{\partial^2 F}{\partial t^2} \right] \\
&+ \textcolor{blue}{k}_4^i \left[\frac{\partial^2 F}{\partial k_4^2} + \frac{2}{k_4} \frac{\partial F}{\partial k_4} \right]. \tag{B.839}
\end{aligned}$$

Total We now sum over the contributions from the terms above. This gives

$$\begin{aligned}
\sum_{n=1}^4 K_n^i F &= \textcolor{red}{k}_1^i \left[\frac{2(\Delta_1 - 2)}{k_1} \frac{\partial F}{\partial k_1} + \frac{2(\Delta_{12} - 5)}{s} \frac{\partial F}{\partial s} - \frac{\partial^2 F}{\partial k_1^2} - \frac{2k_1}{s} \frac{\partial^2 F}{\partial s \partial k_1} - \frac{2t_2^2}{st} \frac{\partial^2 F}{\partial s \partial t} \right. \\
&\quad \left. - \frac{2k_2}{s} \frac{\partial^2 F}{\partial s \partial k_2} - \frac{\partial^2 F}{\partial s^2} \right] \\
&+ \textcolor{orange}{k}_2^i \left[\frac{2(\Delta_2 - 2)}{k_2} \frac{\partial F}{\partial k_2} + \frac{2(\Delta_{12} - 5)}{s} \frac{\partial F}{\partial s} + \frac{2(\Delta_{23} - 5)}{t} \frac{\partial F}{\partial t} - \frac{\partial^2 F}{\partial k_2^2} - \frac{\partial^2 F}{\partial s^2} - \frac{\partial^2 F}{\partial t^2} \right. \\
&\quad \left. - \frac{2k_1}{s} \frac{\partial^2 F}{\partial s \partial k_1} - \frac{2k_2}{s} \frac{\partial^2 F}{\partial s \partial k_2} - \frac{2k_2}{t} \frac{\partial^2 F}{\partial t \partial k_2} - \frac{2k_3}{t} \frac{\partial^2 F}{\partial t \partial k_3} + \frac{2(k_1^j k_3^j - k_2^2)}{st} \frac{\partial^2 F}{\partial s \partial t} \right] \\
&+ \textcolor{green}{k}_3^i \left[\frac{2(\Delta_3 - 2)}{k_3} \frac{\partial F}{\partial k_3} + \frac{2(\Delta_{23} - 5)}{t} \frac{\partial F}{\partial t} - \frac{\partial^2 F}{\partial k_3^2} - \frac{2k_3}{t} \frac{\partial^2 F}{\partial t \partial k_3} - \frac{2s_2^2}{st} \frac{\partial^2 F}{\partial s \partial t} \right]
\end{aligned}$$

$$\begin{aligned}
& -\frac{2k_2}{t} \frac{\partial^2 F}{\partial t \partial k_2} - \frac{\partial^2 F}{\partial t^2} \Big] \\
& + k_4^i \left[\frac{2(\Delta_4 - 2)}{k_4} \frac{\partial F}{\partial k_4} - \frac{\partial^2 F}{\partial k_4^2} \right].
\end{aligned} \tag{B.840}$$

Because of the asymmetric definitions of s and t in terms of the momenta k_n , the result is not manifestly symmetric under cyclic permutations. We can use momentum conservation to bring these into a more symmetric form. Moreover, to make the resulting expression symmetric between s and t , we use the constraints (B.834) and (B.835). As a consequence of momentum conservation, we can drop all terms that are equal in each square brackets. These include terms such as $\partial^2 F / \partial s^2$, $\partial^2 F / \partial t^2$ and $\partial F / \partial s$, $\partial F / \partial t$.

We see that the final result is of the form

$$\sum_{n=1}^4 k_n^i D_n F = 0, \tag{B.841}$$

where we have defined

$$\begin{aligned}
D_1 F \equiv & \left[\frac{\partial^2}{\partial k_1^2} + \frac{1}{s} \frac{\partial}{\partial s} \left(k_1 \frac{\partial}{\partial k_1} + k_2 \frac{\partial}{\partial k_2} \right) + \frac{1}{t} \frac{\partial}{\partial t} \left(k_1 \frac{\partial}{\partial k_1} + k_4 \frac{\partial}{\partial k_4} \right) - \frac{k_3^2}{st} \frac{\partial^2}{\partial s \partial t} \right. \\
& \left. - \frac{2(\Delta_1 - 2)}{k_1} \frac{\partial}{\partial k_1} + \frac{\Delta_1 + \Delta_2}{s} \frac{\partial}{\partial s} + \frac{\Delta_1 + \Delta_4}{t} \frac{\partial}{\partial t} \right] F,
\end{aligned} \tag{B.842}$$

and the rest are given by the cyclic permutation of the indices (remembering that $t \rightarrow s$ under a cyclic shift). The only nontrivial way to satisfy (B.841) is to demand that all $D_n F$ are equal to each other, so that $\sum_n k_n^i D_n F$ vanishes as a consequence of momentum conservation.⁶³ To satisfy the SCT constraint, we can therefore pick any three of the six conditions

$$(D_n - D_m) F = 0, \tag{B.843}$$

for $n, m = 1, \dots, 4$.

An object of particular interest is the four-point function of conformally coupled scalars, mediated by the tree-level exchange of massive scalars. In that case, the s -channel contribution⁶⁴ can be written as $F = s^{-1} \hat{F}(u, v)$, an ansatz which automatically satisfies the equations $D_{12} F = 0$ and $D_{34} F = 0$, where $D_{nm} \equiv D_n - D_m$. The remaining conformal invariance equation $D_{13} F = 0$ becomes

$$\boxed{(\Delta_u - \Delta_v) \hat{F} = 0}, \tag{B.844}$$

where we have introduced the differential operator

$$\Delta_u \equiv u^2(1 - u^2) \partial_u^2 - 2u^3 \partial_u. \tag{B.845}$$

This completes the exercise!

⁶³If all momenta are collinear, then perhaps other possibilities are allowed. However, in that case the four-point function would have to vanish in any non-collinear configuration, which we rule out on the basis of continuity.

⁶⁴It suffices to impose conformal invariance of a single channel to fix the whole four-point function. To see this, note that the correlator $\langle OOO'O' \rangle$ with $\Delta_O = \Delta_{O'}$, which is conformally invariant, coincides with the s -channel of $\langle OOOO \rangle$. The other channels can be included by cyclic permutations, e.g. by replacing u with $|\mathbf{k}_2 + \mathbf{k}_3|/(k_2 + k_3)$ and $|\mathbf{k}_2 + \mathbf{k}_4|/(k_2 + k_4)$ for the t - and u -channels, respectively. Finally, for contact interactions, “ s -channel” refers to a specific permutation of the external momenta.

Solution to Exercise 11.9

We start from the seed solution

$$\hat{F}_{c,0} = \frac{uv}{u+v}, \quad (\text{B.846})$$

and act on it with the operator

$$\Delta_u \equiv u^2(1-u^2)\partial_u^2 - 2u^3\partial_u. \quad (\text{B.847})$$

This gives

$$\hat{F}_{c,1} = \Delta_u \hat{F}_{c,0} = \boxed{-2 \left(\frac{uv}{u+v} \right)^3 \frac{1+uv}{uv}}, \quad (\text{B.848})$$

$$\hat{F}_{c,2} = \Delta_u^2 \hat{F}_{c,0} = \boxed{-4 \left(\frac{uv}{u+v} \right)^5 \frac{u^2 + v^2 + uv(3u^2 + 3v^2 - 4) - 6(uv)^2 - 6(uv)^3}{(uv)^3}}, \quad (\text{B.849})$$

as required.

Solution to Exercise 11.10

Consider equation (11.134):

$$\left[\Delta_u + \left(\mu^2 + \frac{1}{4} \right) \right] \hat{F}_e = \hat{C}, \quad (\text{B.850})$$

$$\left[\Delta_v + \left(\mu^2 + \frac{1}{4} \right) \right] \hat{F}_e = \hat{C}. \quad (\text{B.851})$$

Acting on (B.850) with Δ_v and on (B.851) with Δ_u , we get

$$\Delta_v \left[\Delta_u + \left(\mu^2 + \frac{1}{4} \right) \right] \hat{F}_e = \Delta_v \hat{C}, \quad (\text{B.852})$$

$$\Delta_u \left[\Delta_v + \left(\mu^2 + \frac{1}{4} \right) \right] \hat{F}_e = \Delta_u \hat{C}. \quad (\text{B.853})$$

Subtracting these equations from each other gives

$$(\Delta_u \Delta_v - \Delta_v \Delta_u) \hat{F}_e + \left(\mu^2 + \frac{1}{4} \right) (\Delta_u - \Delta_v) \hat{F}_e = (\Delta_u - \Delta_v) \hat{C}. \quad (\text{B.854})$$

The first term on the left-hand side vanishes, because $\hat{F}_e$ is symmetric under exchange of u and v , while the second term vanishes by conformal invariance. Hence, we find that

$$\boxed{(\Delta_u - \Delta_v) \hat{C} = 0}, \quad (\text{B.855})$$

i.e. the source term $\hat{C}$ has to be conformally invariant.

Solution to Exercise 11.11

The differential equation for the four-point exchange diagram of a conformally coupled scalar is

$$\left[u^2(1-u^2)\partial_u^2 - 2u^3\partial_u \right] \hat{F}_e = \hat{F}_c. \quad (\text{B.856})$$

In terms of $u \equiv k_I/k_{12}$ and $v \equiv k_I/k_{34}$, we have

$$\begin{aligned} \hat{F}_e \equiv k_I \langle O_1 O_2 O_3 O_4 \rangle'_e &= \frac{\lambda_3^2}{2H^6 \eta_*^4} \left[\text{Li}_2 \left(\frac{v(1-u)}{u+v} \right) + \text{Li}_2 \left(\frac{u(1-v)}{u+v} \right) \right. \\ &\quad \left. + \log \left(\frac{v(1+u)}{u+v} \right) \log \left(\frac{u(1+v)}{u+v} \right) - \frac{\pi^2}{6} \right]. \end{aligned} \quad (\text{B.857})$$

Substituting this into the left-hand side of (B.856) gives

$$\left[u^2(1-u^2)\partial_u^2 - 2u^3\partial_u \right] \hat{F}_e = \frac{\lambda_3^2}{H^6 \eta_*^4} \frac{uv}{u+v} = \frac{\lambda_3^2}{H^6 \eta_*^4} \frac{k_I}{E}, \quad (\text{B.858})$$

which equals the contact solution $\hat{F}_c \equiv k_I \langle O_1 O_2 O_3 O_4 \rangle'_c$ if we set $\lambda_4 = \lambda_3^2/H^2$. This confirms that our previous solutions for the contact and exchange four-point functions of a conformally coupled scalar satisfy the conformal Ward identity.

Solution to Exercise 11.12

Consider equation (11.144):

$$\left[\xi^2 \partial_\xi^2 + \left(\mu^2 + \frac{1}{4} \right) \right] \tilde{F} = \frac{\xi}{1+\xi}, \quad (\text{B.859})$$

where $\tilde{F} \equiv \sqrt{\xi} f$. Around $\xi = 0$, we write the particular solution of this equation as a series solution

$$\tilde{F}_<(\xi) = \sum_{n=1}^{\infty} c_n \xi^n. \quad (\text{B.860})$$

Substituting this into (B.859), we get

$$\sum_{n=1}^{\infty} c_n \left[n(n-1) + \left(\mu^2 + \frac{1}{4} \right) \right] \xi^n = \frac{\xi}{1+\xi}. \quad (\text{B.861})$$

Multiplying through by $(1+\xi)$ gives

$$c_1 \left(\mu^2 + \frac{1}{4} \right) \xi + \sum_{n=1}^{\infty} \left(c_{n+1} \left[\left(n + \frac{1}{2} \right)^2 + \mu^2 \right] + c_n \left[\left(n - \frac{1}{2} \right)^2 + \mu^2 \right] \right) \xi^{n+1} = \xi, \quad (\text{B.862})$$

which implies that

$$c_1 = \left(\mu^2 + \frac{1}{4} \right)^{-1} \quad \text{and} \quad c_{n+1} = -\frac{\left(n - \frac{1}{2} \right)^2 + \mu^2}{\left(n + \frac{1}{2} \right)^2 + \mu^2} c_n. \quad (\text{B.863})$$

The solution for $\tilde{F}_<$ then is

$$\tilde{F}_< = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{\xi^n}{(n - \frac{1}{2})^2 + \mu^2}, \quad (\text{B.864})$$

and $f_< = \tilde{F}_< / \sqrt{\xi}$ can be written as

$$f_<(\xi) = \sqrt{\xi} \sum_{n=0}^{\infty} (-1)^n \frac{\xi^n}{(n + \frac{1}{2})^2 + \mu^2}, \quad (\text{B.865})$$

as required.

Solution to Exercise 11.13

Let $x = 1/\xi$. Using

$$\partial_\xi = \frac{dx}{d\xi} \partial_x = -\frac{1}{\xi^2} \partial_x = -x^2 \partial_x, \quad (\text{B.866})$$

$$\partial_\xi^2 = -x^2 \partial_x (-x^2 \partial_x) = x^2 (x^2 \partial_x^2 + 2x \partial_x), \quad (\text{B.867})$$

we find that equation (B.859) becomes

$$\left[x^2 \partial_x^2 + 2x \partial_x + \left(\mu^2 + \frac{1}{4} \right) \right] \tilde{F} = \frac{1}{1+x}. \quad (\text{B.868})$$

Around $x = 0$, we write the particular solution of this equation as a series solution

$$\tilde{F}_>(x) = \sum_{n=0}^{\infty} c_n x^n. \quad (\text{B.869})$$

Substituting this into (B.868), we get

$$\sum_{n=0}^{\infty} c_n \left[n(n-1) + 2n + \left(\mu^2 + \frac{1}{4} \right) \right] x^n = \frac{1}{1+x}. \quad (\text{B.870})$$

Multiplying through by $(1+x)$, this implies that

$$c_0 \left(\mu^2 + \frac{1}{4} \right) + \sum_{n=1}^{\infty} \left(c_n \left[\left(n + \frac{1}{2} \right)^2 + \mu^2 \right] + c_{n-1} \left[\left(n - \frac{1}{2} \right)^2 + \mu^2 \right] \right) x^n = 1. \quad (\text{B.871})$$

Hence, we have

$$c_0 = \left(\mu^2 + \frac{1}{4} \right)^{-1} \quad \text{and} \quad c_n = -\frac{(n - \frac{1}{2})^2 + \mu^2}{(n + \frac{1}{2})^2 + \mu^2} c_{n-1}. \quad (\text{B.872})$$

The solution for $f_> = \tilde{F}_> / \sqrt{\xi}$ can then be written as

$$f_>(\xi) = \frac{1}{\sqrt{\xi}} \sum_{n=0}^{\infty} (-1)^n \frac{\xi^{-n}}{(n + \frac{1}{2})^2 + \mu^2}, \quad (\text{B.873})$$

as required.

Solution to Exercise 11.14

We start from the differential equation (11.139):

$$\left[u^2(1-u^2)\partial_u^2 - 2u^3\partial_u + \left(\mu^2 + \frac{1}{4} \right) \right] \hat{F} = \frac{uv}{u+v}. \quad (\text{B.874})$$

For $u < v$, we seek a power series solution of the form

$$\hat{F}_{<}(u, v) = \sum_{m,n=0}^{\infty} c_{mn} u^{m+1} (u/v)^n. \quad (\text{B.875})$$

This form of the series solution is motivated by the series expansion of the source term $uv/(u+v)$. Plugging this ansatz into the differential equation, we find that the series coefficients obey the following recursive relation:

$$c_{0n} = \frac{(-1)^n}{(n + \frac{1}{2})^2 + \mu^2}, \quad c_{1n} = 0, \quad c_{m+2,n} = \frac{(m+n+2)(m+n+1)}{(m+n+\frac{5}{2})^2 + \mu^2} c_{mn}, \quad (\text{B.876})$$

where the condition $c_{1n} = 0$ implies that c_{mn} vanishes for all odd m . Redefining $m \rightarrow 2m$ and solving the recursive relation, we find

$$\boxed{\hat{F}_{<}(u, v) = \sum_{m,n=0}^{\infty} c_{mn} u^{2m+1} (u/v)^n}, \quad (\text{B.877})$$

with

$$\boxed{c_{mn} = \frac{(-1)^n (n+1)(n+2) \cdots (n+2m)}{[(n+\frac{1}{2})^2 + \mu^2][(n+\frac{5}{2})^2 + \mu^2] \cdots [(n+\frac{1}{2}+2m)^2 + \mu^2]}}. \quad (\text{B.878})$$

Note that the series solution (11.154) is the unique particular solution that is regular at the origin, since both homogeneous solutions are non-analytic at $u = 0$.

Solution to Exercise 11.15

In Exercise 11.14, we found the solution $\hat{F}_{<}(u, v)$ for $u < v$; cf. equation (B.877). The full solution is symmetric under the exchange $u \leftrightarrow v$. For $u > v$, the solution therefore is

$$\hat{F}_{>}(u, v) = \hat{F}_{<}(v, u) = \sum_{m,n=0}^{\infty} c_{mn} v^{2m+1} (v/u)^n, \quad (\text{B.879})$$

where c_{mn} are same coefficients as in (B.878). It is straightforward to check that (B.879) solves (B.874).

As before, the difference between the two particular solutions must be a homogeneous solution, so we can write

$$\hat{F}_{<}(u, v) - \hat{F}_{>}(u, v) = \sum_{\pm} A_{\pm}(v; \mu) \hat{F}_{\pm}(u), \quad (\text{B.880})$$

where the functions $\hat{F}_\pm$ are given by (11.152). Evaluating this at $u = v$ gives $A_\pm(v; \mu) = \mp a(v; \mu) \hat{F}_\mp(v)$, for some function $a(v; \mu)$. Matching the u -derivative at $u = v$ fixes the function in terms of the Wronskian of the homogeneous solutions, namely

$$\begin{aligned} \left(\partial_u \hat{F}_< - \partial_u \hat{F}_> \right) \Big|_{u \rightarrow v} &= a(v; \mu) \left(\hat{F}_+(v) \hat{F}'_-(v) - \hat{F}'_+(v) \hat{F}_-(v) \right) \\ &= \frac{a(v; \mu)}{1 - v^2} = a(v; \mu) \times \sum_{m=0}^{\infty} v^{2m}. \end{aligned} \quad (\text{B.881})$$

It remains to evaluate the left-hand side

$$\left(\partial_u \hat{F}_< - \partial_u \hat{F}_> \right) \Big|_{u \rightarrow v} = \sum_{m,n=0}^{\infty} (2m + 2n + 1) c_{mn} v^{2m}. \quad (\text{B.882})$$

Somewhat remarkably, the sum over n above is m -independent; we have the identity

$$\sum_{n=0}^{\infty} (2m + 2n + 1) c_{mn} = \sum_{n=0}^{\infty} \frac{(-1)^n (2m + 2n + 1) (n + 1) \cdots (n + 2m)}{[(n + \frac{1}{2})^2 + \mu^2] \cdots [(n + \frac{1}{2} + 2m)^2 + \mu^2]} = \frac{\pi}{\cosh \pi \mu}. \quad (\text{B.883})$$

Once again, this identity can be established by matching the residues on both sides. For instance, for $m = 1$, only the terms in the sum with $n = k - 2, k$ have a residue as $\mu \rightarrow \pm i(k + \frac{1}{2})$; the residue is

$$-\frac{(-1)^k (1 + k)(2 + k)}{2(1 + 2k)} + \frac{(-1)^k (k - 1)k}{2(1 + 2k)} = (-1)^k, \quad (\text{B.884})$$

which matches that of the right-hand side. This allows us to fix $a(v; \mu) = \pi / \cosh \pi \mu$ independent of v .

The explicit solution for $\hat{F}_<$ is

$$\hat{F}_<(u, v) = \begin{cases} \sum_{m,n=0}^{\infty} c_{mn} u^{2m+1} (u/v)^n & u \leq v, \\ \sum_{m,n=0}^{\infty} c_{mn} v^{2m+1} (v/u)^n + \frac{\pi}{\cosh \pi \mu} \left(\hat{F}_+(v) \hat{F}_-(u) - \hat{F}_-(v) \hat{F}_+(u) \right) & u \geq v, \end{cases} \quad (\text{B.885})$$

as required.

Solution to Exercise 11.16

The most general solution of the differential equation that is symmetric in $u \leftrightarrow v$ is

$$\begin{aligned} \hat{F}(u, v) &= \hat{F}_<(u, v) + \frac{\pi}{2 \cosh \pi \mu} \left\{ [\hat{F}_+(u) \hat{F}_-(v) - \hat{F}_-(u) \hat{F}_+(v)] \right. \\ &\quad \left. + [\beta_+ \hat{F}_+(u) \hat{F}_+(v) + \beta_- \hat{F}_-(u) \hat{F}_-(v) + \beta [\hat{F}_-(u) \hat{F}_+(v) + \hat{F}_+(u) \hat{F}_-(v)] \right\}. \end{aligned} \quad (\text{B.886})$$

Demanding the absence of the spurious singularity at $u = 1$ implies $\beta_\pm = -(\beta \pm 1) \alpha_\mp / \alpha_\pm$, but still leaves β unfixed. To fix β , we take the limit $u = v \rightarrow -1$. Since the function $\hat{F}(u, v)$ has a

branch point at $u = v = -1$ (see Fig. 30), the limit will depend on how this point is approached. In particular, for $u = v = -1^+ \equiv e^{i\pi}$ and $u = v = -1^- \equiv e^{-i\pi}$, we get

$$\lim_{u,v \rightarrow -1^+} \hat{F}(u, v) = \frac{i}{2} (\beta \sinh \pi\mu - \cosh \pi\mu) \log(1+u) \log(1+v), \quad (\text{B.887})$$

$$\lim_{u,v \rightarrow -1^-} \hat{F}(u, v) = \frac{i}{2} (\beta \sinh \pi\mu + \cosh \pi\mu) \log(1+u) \log(1+v). \quad (\text{B.888})$$

We see that these limits have a universal $\beta \sinh \pi\mu$ contribution and ambiguous $\cosh \pi\mu$ terms. To isolate the physically relevant contribution, we define the $u = v \rightarrow -1$ limit as

$$\begin{aligned} \lim_{u,v \rightarrow -1} \hat{F}(u, v) &\equiv \frac{1}{2} \left(\lim_{u,v \rightarrow -1^+} + \lim_{u,v \rightarrow -1^-} \right) \hat{F}(u, v) \\ &= \frac{i\beta \sinh \pi\mu}{2} \log(1+u) \log(1+v). \end{aligned} \quad (\text{B.889})$$

Restoring the coupling constant λ^2 , the coefficient of this singularity takes the form of a product of three-particle amplitudes. Since we defined $\lambda \equiv 1$, we must then have

$$\beta = \frac{2}{i \sinh \pi\mu}. \quad (\text{B.890})$$

Another way to determine β is by comparing the $u \rightarrow -1$ limit, for general $0 < v < 1$, with the corresponding limit of the bulk calculation. In that limit, the four-point function factorizes into the product of a three-point correlation function and a three-particle amplitude. This limit has the advantage that it doesn't depend on the way the branch point is approached.

References

- [1] J. Donoghue, “General Relativity as an Effective Field Theory: The Leading Quantum Corrections,” *Phys. Rev. D* **50** (1994) 3874–3888, [arXiv:gr-qc/9405057](#).
- [2] W. Unruh, “Notes on Black Hole Evaporation,” *Phys. Rev. D* **14** (1976) 870.
- [3] S. W. Hawking, “Black Hole Explosions,” *Nature* **248** (1974) 30–31.
- [4] S. W. Hawking, “Particle Creation by Black Holes,” *Commun. Math. Phys.* **43** (1975) 199–220.
- [5] G. Gibbons and S. W. Hawking, “Cosmological Event Horizons, Thermodynamics, and Particle Creation,” *Phys. Rev. D* **15** (1977) 2738–2751.
- [6] T. Jacobson, Introduction to Quantum Fields in Curved Spacetime and the Hawking Effect, 8, 2003. [arXiv:gr-qc/0308048](#).
- [7] N. Birrell and P. Davies, *Quantum Fields in Curved Space*. Cambridge Monographs on Mathematical Physics. Cambridge University Press, Cambridge, UK, 1982.
- [8] I. Komissarov, A. Nicolis, and J. Staunton, “Cosmology as a Weak Gravitational Field and the Trans-Planckian Problem,” *JHEP* **05** (2023) 216, [arXiv:2210.11508 \[hep-th\]](#).
- [9] D. Tong, Lectures Notes on General Relativity.
- [10] H. Reall, Lectures Notes on Black Holes.
- [11] J. Bekenstein, “Black Holes and Entropy,” *Phys. Rev. D* **7** (1973) 2333–2346.
- [12] S. W. Hawking, “Black Holes in General Relativity,” *Commun. Math. Phys.* **25** (1972) 152–166.
- [13] A. Strominger and C. Vafa, “Microscopic Origin of the Bekenstein-Hawking Entropy,” *Phys. Lett. B* **379** (1996) 99–104, [arXiv:hep-th/9601029](#).
- [14] J. Bardeen, B. Carter, and S. W. Hawking, “The Four Laws of Black Hole Mechanics,” *Commun. Math. Phys.* **31** (1973) 161–170.
- [15] G. ’t Hooft, “Dimensional Reduction in Quantum Gravity,” *Conf. Proc. C* **930308** (1993) 284–296, [arXiv:gr-qc/9310026](#).
- [16] L. Susskind, “The World as a Hologram,” *J. Math. Phys.* **36** (1995) 6377–6396, [arXiv:hep-th/9409089](#).
- [17] J. Maldacena, “The Large N Limit of Superconformal Field Theories and Supergravity,” *Adv. Theor. Math. Phys.* **2** (1998) 231–252, [arXiv:hep-th/9711200](#).
- [18] S. W. Hawking, “Breakdown of Predictability in Gravitational Collapse,” *Phys. Rev. D* **14** (1976) 2460–2473.
- [19] D. Page, “Information in Black Hole Radiation,” *Phys. Rev. Lett.* **71** (1993) 3743–3746, [arXiv:hep-th/9306083](#).
- [20] G. Penington, “Entanglement Wedge Reconstruction and the Information Paradox,” *JHEP* **09** (2020) 002, [arXiv:1905.08255 \[hep-th\]](#).
- [21] A. Almheiri, N. Engelhardt, D. Marolf, and H. Maxfield, “The Entropy of Bulk Quantum Fields and the Entanglement Wedge of an Evaporating Black Hole,” *JHEP* **12** (2019) 063, [arXiv:1905.08762 \[hep-th\]](#).
- [22] D. Harlow, “Jerusalem Lectures on Black Holes and Quantum Information,” *Rev. Mod. Phys.* **88** (2016) 015002, [arXiv:1409.1231 \[hep-th\]](#).

- [23] A. Almheiri, T. Hartman, J. Maldacena, E. Shaghoulian, and A. Tajdini, “The Entropy of Hawking Radiation,” *Rev. Mod. Phys.* **93** no. 3, (2021) 035002, [arXiv:2006.06872 \[hep-th\]](#).
- [24] S. Raju, “Lessons from the Information Paradox,” *Phys. Rept.* **943** (2022) 1–80, [arXiv:2012.05770 \[hep-th\]](#).
- [25] T. Hartman, Lectures Notes on Quantum Gravity and Black Holes.
- [26] S. Perlmutter et al. [Supernova Cosmology Project Collaboration], “Measurements of Ω and Λ from 42 High Redshift Supernovae,” *Astrophys. J.* **517** (1999) 565–586, [arXiv:astro-ph/9812133](#).
- [27] A. Riess *et al.* [Supernova Search Team], “Observational Evidence from Supernovae for an Accelerating Universe and a Cosmological Constant,” *Astron. J.* **116** (1998) 1009–1038, [arXiv:astro-ph/9805201](#).
- [28] A. Guth, “The Inflationary Universe: A Possible Solution to the Horizon and Flatness Problems,” *Adv. Ser. Astrophys. Cosmol.* **3** (1987) 139–148.
- [29] A. Linde, “A New Inflationary Universe Scenario: A Possible Solution of the Horizon, Flatness, Homogeneity, Isotropy and Primordial Monopole Problems,” *Adv. Ser. Astrophys. Cosmol.* **3** (1987) 149–153.
- [30] A. Albrecht and P. Steinhardt, “Cosmology for Grand Unified Theories with Radiatively Induced Symmetry Breaking,” *Adv. Ser. Astrophys. Cosmol.* **3** (1987) 158–161.
- [31] D. Baumann, “Inflation,” in Theoretical Advanced Study Institute in Elementary Particle Physics: Physics of the Large and the Small, pp. 523–686. 2011. [arXiv:0907.5424 \[hep-th\]](#).
- [32] M. Spradlin, A. Strominger, and A. Volovich, Les Houches Lectures on de Sitter Space. [arXiv:hep-th/0110007](#).
- [33] M. Marino, Lectures Notes on Quantum Field Theory in Curved Space.
- [34] T. Bunch and P. Davies, “Quantum Field Theory in de Sitter Space: Renormalization by Point Splitting,” *Proc. Roy. Soc. Lond.* **A360** (1978) 117–134.
- [35] D. Anninos, “De Sitter Musings,” *Int. J. Mod. Phys. A* **27** (2012) 1230013, [arXiv:1205.3855 \[hep-th\]](#).
- [36] G. Gibbons and S. W. Hawking, eds., Euclidean Quantum Gravity. 1994.
- [37] G. Gibbons and S. W. Hawking, “Action Integrals and Partition Functions in Quantum Gravity,” *Phys. Rev. D* **15** (1977) 2752–2756.
- [38] R. Wald, “Black Hole Entropy is the Noether Charge,” *Phys. Rev. D* **48** no. 8, (1993) R3427–R3431, [arXiv:gr-qc/9307038](#).
- [39] S. W. Hawking and D. Page, “Thermodynamics of Black Holes in Anti-De Sitter Space,” *Commun. Math. Phys.* **87** (1983) 577.
- [40] C. Egan and C. Lineweaver, “A Larger Estimate of the Entropy of the Universe,” *The Astrophysical Journal* **710** no. 2, (2010) 1825–1834.
- [41] J. Maldacena, “Non-Gaussian Features of Primordial Fluctuations in Single-Field Inflationary Models,” *JHEP* **05** (2003) 013, [arXiv:astro-ph/0210603 \[astro-ph\]](#).
- [42] C. Cheung, P. Creminelli, A. L. Fitzpatrick, J. Kaplan, and L. Senatore, “The Effective Field Theory of Inflation,” *JHEP* **03** (2008) 014, [arXiv:0709.0293 \[hep-th\]](#).
- [43] N. Arkani-Hamed and J. Maldacena, “Cosmological Collider Physics,” [arXiv:1503.08043 \[hep-th\]](#).

- [44] N. Arkani-Hamed, D. Baumann, H. Lee, and G. L. Pimentel, “The Cosmological Bootstrap: Inflationary Correlators from Symmetries and Singularities,” *JHEP* **04** (2020) 105, [arXiv:1811.00024 \[hep-th\]](#).
- [45] D. Baumann, D. Green, A. Joyce, E. Pajer, G. L. Pimentel, C. Sleight, and M. Taronna, “Snowmass White Paper: The Cosmological Bootstrap,” in 2022 Snowmass Summer Study. 3, 2022. [arXiv:2203.08121 \[hep-th\]](#).
- [46] R. Scoccimarro, Lectures Notes on Cosmology.
- [47] D. Baumann, *Cosmology*. Cambridge University Press, 2022.
- [48] N. Aghanim et al. [Planck Collaboration], “Planck 2018 Results. VI. Cosmological Parameters,” *Astron. Astrophys.* **641** (2020) A6, [arXiv:1807.06209 \[astro-ph.CO\]](#).
- [49] S. Chabanier, M. Millea, and N. Palanque-Delabrouille, “Matter Power Spectrum: From Ly α Forest to CMB Scales,” *Mon. Not. Roy. Astron. Soc.* **489** no. 2, (2019) 2247–2253, [arXiv:1905.08103 \[astro-ph.CO\]](#).
- [50] S. Dodelson, Modern Cosmology. Academic Press, Amsterdam, 2003.
- [51] D. Lyth and A. Riotto, “Particle Physics Models of Inflation and the Cosmological Density Perturbation,” *Phys. Rept.* **314** (1999) 1–146, [arXiv:hep-ph/9807278](#).
- [52] D. Baumann and L. McAllister, *Inflation and String Theory*. Cambridge Monographs on Mathematical Physics. Cambridge University Press, 5, 2015. [arXiv:1404.2601 \[hep-th\]](#).
- [53] R. Arnowitt, S. Deser, and C. Misner, “The Dynamics of General Relativity,” *Gen. Rel. Grav.* **40** (2008) 1997–2027, [arXiv:gr-qc/0405109 \[gr-qc\]](#).
- [54] P. Creminelli, M. Luty, A. Nicolis, and L. Senatore, “Starting the Universe: Stable Violation of the Null Energy Condition and Non-standard Cosmologies,” *JHEP* **12** (2006) 080, [arXiv:hep-th/0606090](#).
- [55] D. Green and R. Porto, “Signals of a Quantum Universe,” *Phys. Rev. Lett.* **124** no. 25, (2020) 251302, [arXiv:2001.09149 \[hep-th\]](#).
- [56] J. Schwinger, “Brownian Motion of a Quantum Oscillator,” *J. Math. Phys.* **2** (1961) 407–432.
- [57] P. Bakshi and K. Mahanthappa, “Expectation Value Formalism in Quantum Field Theory,” *J. Math. Phys.* **4** (1963) 1–11.
- [58] L. Keldysh, “Diagram Technique for Nonequilibrium Processes,” *Zh. Eksp. Teor. Fiz.* **47** (1964) 1515–1527.
- [59] R. Jordan, “Effective Field Equations for Expectation Values,” *Phys. Rev. D* **33** (1986) 444–454.
- [60] E. Calzetta and B. Hu, “Closed Time Path Functional Formalism in Curved Space-Time: Application to Cosmological Back Reaction Problems,” *Phys. Rev. D* **35** (1987) 495.
- [61] S. Weinberg, “Quantum Contributions to Cosmological Correlations,” *Phys. Rev. D* **72** (2005) 043514, [arXiv:hep-th/0506236 \[hep-th\]](#).
- [62] X. Chen, “Primordial Non-Gaussianities from Inflation Models,” *Adv. Astron.* **2010** (2010) 638979, [arXiv:1002.1416 \[astro-ph.CO\]](#).
- [63] Y. Wang, “Inflation, Cosmic Perturbations and Non-Gaussianities,” *Commun. Theor. Phys.* **62** (2014) 109–166, [arXiv:1303.1523 \[hep-th\]](#).
- [64] S. Weinberg, The Quantum Theory of Fields. Vol. 1: Foundations. Cambridge University Press, 6, 2005.

- [65] S. Giddings and M. Sloth, “Cosmological Diagrammatic Rules,” *JCAP* **1007** (2010) 015, [arXiv:1005.3287 \[hep-th\]](#).
- [66] D. Baumann, C. Duaso Pueyo, A. Joyce, H. Lee, and G. L. Pimentel, “The Cosmological Bootstrap: Weight-Shifting Operators and Scalar Seeds,” *JHEP* **12** (2020) 204, [arXiv:1910.14051 \[hep-th\]](#).
- [67] A. Bzowski, P. McFadden, and K. Skenderis, “Implications of Conformal Invariance in Momentum Space,” *JHEP* **03** (2014) 111, [arXiv:1304.7760 \[hep-th\]](#).
- [68] M. Zaldarriaga, “Non-Gaussianities in Models with a Varying Inflaton Decay Rate,” *Phys. Rev. D* **69** (2004) 043508, [arXiv:astro-ph/0306006](#).
- [69] E. Lim, Part III Advanced Cosmology: Primordial Non-Gaussianities.
- [70] Y. Wang, “Inflation, Cosmic Perturbations and Non-Gaussianities,” *Commun. Theor. Phys.* **62** (2014) 109–166, [arXiv:1303.1523 \[hep-th\]](#).
- [71] P. Creminelli and M. Zaldarriaga, “Single-Field Consistency Relation for the Three-Point Function,” *JCAP* **10** (2004) 006, [arXiv:astro-ph/0407059](#).
- [72] P. Creminelli, “On Non-Gaussianities in Single-Field Inflation,” *JCAP* **10** (2003) 003, [arXiv:astro-ph/0306122](#).
- [73] X. Chen and Y. Wang, “Quasi-Single Field Inflation and Non-Gaussianities,” *JCAP* **04** (2010) 027, [arXiv:0911.3380 \[hep-th\]](#).
- [74] D. Baumann and D. Green, “Signatures of Supersymmetry from the Early Universe,” *Phys. Rev. D* **85** (2012) 103520, [arXiv:1109.0292 \[hep-th\]](#).
- [75] J. Polchinski, *String Theory. Vol. 1: An Introduction to the Bosonic String*. Cambridge Monographs on Mathematical Physics. Cambridge University Press, 12, 2007.
- [76] E. Witten, “Anti-de Sitter Space and Holography,” *Adv. Theor. Math. Phys.* **2** (1998) 253–291, [arXiv:hep-th/9802150](#).
- [77] I. Klebanov and E. Witten, “AdS / CFT Correspondence and Symmetry Breaking,” *Nucl. Phys. B* **556** (1999) 89–114, [arXiv:hep-th/9905104](#).
- [78] H. Goodhew, S. Jazayeri, and E. Pajer, “The Cosmological Optical Theorem,” *JCAP* **04** (2021) 021, [arXiv:2009.02898 \[hep-th\]](#).
- [79] S. Jazayeri, E. Pajer, and D. Stefanyszyn, “From Locality and Unitarity to Cosmological Correlators,” *JHEP* **10** (2021) 065, [arXiv:2103.08649 \[hep-th\]](#).
- [80] A. Hillman and E. Pajer, “A Differential Representation of Cosmological Wavefunctions,” *JHEP* **04** (2022) 012, [arXiv:2112.01619 \[hep-th\]](#).
- [81] D. Baumann, W.-M. Chen, C. Duaso Pueyo, A. Joyce, H. Lee, and G. L. Pimentel, “Linking the Singularities of Cosmological Correlators,” *JHEP* **09** (2022) 010, [arXiv:2106.05294 \[hep-th\]](#).
- [82] N. Arkani-Hamed, P. Benincasa, and A. Postnikov, “Cosmological Polytopes and the Wavefunction of the Universe,” [arXiv:1709.02813 \[hep-th\]](#).
- [83] N. Arkani-Hamed, D. Baumann, A. Hillman, A. Joyce, H. Lee, and G. L. Pimentel, “Differential Equations for Cosmological Correlators,” [arXiv:2312.05303 \[hep-th\]](#).
- [84] P. Benincasa, “Amplitudes Meet Cosmology: A (Scalar) Primer,” [arXiv:2203.15330 \[hep-th\]](#).
- [85] C. Duaso Pueyo, A Bootstrap Approach to Cosmology. PhD thesis, University of Amsterdam, 2022.

- [86] S. Parke and T. Taylor, “An Amplitude for n -Gluon Scattering,” *Phys. Rev. Lett.* **56** (1986) 2459.
- [87] B. DeWitt, “Quantum Theory of Gravity. III. Applications of the Covariant Theory,” *Physical Review* **162** no. 5, (1967) 1239.
- [88] S. Weinberg, “Photons and Gravitons in S -Matrix Theory: Derivation of Charge Conservation and Equality of Gravitational and Inertial Mass,” *Phys. Rev.* **135** (1964) B1049–B1056.
- [89] S. Weinberg and E. Witten, “Limits on Massless Particles,” *Phys. Lett. B* **96** (1980) 59–62.
- [90] P. Benincasa and F. Cachazo, “Consistency Conditions on the S -Matrix of Massless Particles,” [arXiv:0705.4305 \[hep-th\]](#).
- [91] D. McGady and L. Rodina, “Higher-Spin Massless S -Matrices in Four Dimensions,” *Phys. Rev. D* **90** no. 8, (2014) 084048, [arXiv:1311.2938 \[hep-th\]](#).
- [92] S. C  spedes, A.-C. Davis, and S. Melville, “On the Time Evolution of Cosmological Correlators,” *JHEP* **02** (2021) 012, [arXiv:2009.07874 \[hep-th\]](#).
- [93] D. Meltzer and A. Sivaramakrishnan, “CFT Unitarity and the AdS Cutkosky Rules,” *JHEP* **11** (2020) 073, [arXiv:2008.11730 \[hep-th\]](#).
- [94] H. Goodhew, S. Jazayeri, M. H. Gordon Lee, and E. Pajer, “Cutting Cosmological Correlators,” *JCAP* **08** (2021) 003, [arXiv:2104.06587 \[hep-th\]](#).
- [95] S. Melville and E. Pajer, “Cosmological Cutting Rules,” *JHEP* **05** (2021) 249, [arXiv:2103.09832 \[hep-th\]](#).
- [96] R. Cutkosky, “Singularities and Discontinuities of Feynman Amplitudes,” *J. Math. Phys.* **1** (1960) 429–433.
- [97] V. Dobrev, G. Mack, V. Petkova, S. Petrova, and I. Todorov, *Harmonic Analysis on the n -Dimensional Lorentz Group and Its Application to Conformal Quantum Field Theory*, vol. 63. 1977.
- [98] T. Basile, X. Bekaert, and N. Boulanger, “Mixed-Symmetry Fields in de Sitter Space: A Group Theoretical Glance,” *JHEP* **05** (2017) 081, [arXiv:1612.08166 \[hep-th\]](#).
- [99] Z. Sun, “A Note on the Representations of $SO(1, d + 1)$,” [arXiv:2111.04591 \[hep-th\]](#).
- [100] S. Deser and A. Waldron, “Arbitrary Spin Representations in De Sitter from dS/CFT with Applications to dS Supergravity,” *Nucl. Phys. B* **662** (2003) 379–392, [arXiv:hep-th/0301068 \[hep-th\]](#).
- [101] J. Bonifacio, K. Hinterbichler, A. Joyce, and R. Rosen, “Shift Symmetries in (Anti) de Sitter Space,” *JHEP* **02** (2019) 178, [arXiv:1812.08167 \[hep-th\]](#).
- [102] A. Higuchi, “Forbidden Mass Range for Spin-2 Field Theory in De Sitter Space-time,” *Nucl. Phys. B* **282** (1987) 397–436.
- [103] P. Creminelli, A. Joyce, J. Khoury, and M. Simonovic, “Consistency Relations for the Conformal Mechanism,” *JCAP* **1304** (2013) 020, [arXiv:1212.3329](#).
- [104] L. Berezhiani, J. Khoury, and J. Wang, “Non-Trivial Checks of Novel Consistency Relations,” *JCAP* **1406** (2014) 056, [arXiv:1401.7991 \[hep-th\]](#).
- [105] J. Maldacena and G. L. Pimentel, “On Graviton Non-Gaussianities During Inflation,” *JHEP* **09** (2011) 045, [arXiv:1104.2846 \[hep-th\]](#).
- [106] A. Bzowski, P. McFadden, and K. Skenderis, “Scalar Three-Point Functions in CFT: Renormalisation, Beta Functions and Anomalies,” *JHEP* **03** (2016) 066, [arXiv:1510.08442](#)

[hep-th].

- [107] A. Bzowski, P. McFadden, and K. Skenderis, “Evaluation of Conformal Integrals,” *JHEP* **02** (2016) 068, [arXiv:1511.02357 \[hep-th\]](#).
- [108] S. Raju, “New Recursion Relations and a Flat Space Limit for AdS/CFT Correlators,” *Phys. Rev. D* **85** (2012) 126009, [arXiv:1201.6449 \[hep-th\]](#).
- [109] D. Baumann, C. Duaso Pueyo, A. Joyce, H. Lee, and G. L. Pimentel, “The Cosmological Bootstrap: Spinning Correlators from Symmetries and Factorization,” *SciPost Phys.* **11** (2021) 071, [arXiv:2005.04234 \[hep-th\]](#).
- [110] J. Bourjaily, H. Hannesdottir, A. McLeod, M. Schwartz, and C. Vergu, “Sequential Discontinuities of Feynman Integrals and the Monodromy Group,” *JHEP* **01** (2021) 205, [arXiv:2007.13747 \[hep-th\]](#).
- [111] S. Albayrak, P. Benincasa, and C. Duaso Pueyo, “Perturbative Unitarity and the Wavefunction of the Universe,” *SciPost Phys.* **16** no. 6, (2024) 157, [arXiv:2305.19686 \[hep-th\]](#).
- [112] G. Poschl and E. Teller, “Bemerkungen zur Quantenmechanik des Anharmonischen Oszillators,” *Z. Phys.* **83** (1933) 143–151.
- [113] F. Cooper, A. Khare, and U. Sukhatme, “Supersymmetry and Quantum Mechanics,” *Phys. Rept.* **251** (1995) 267–385, [arXiv:hep-th/9405029](#).
- [114] P. Lagogiannis, A. Maloney, and Y. Wang, “Odd-dimensional de Sitter Space is Transparent,” [arXiv:1106.2846 \[hep-th\]](#).
- [115] L. Hui, A. Joyce, R. Penco, L. Santoni, and A. Solomon, “Ladder Symmetries of Black Holes. Implications for Love Numbers and No-Hair Theorems,” *JCAP* **01** no. 01, (2022) 032, [arXiv:2105.01069 \[hep-th\]](#).
- [116] P. Adshead, R. Easter, and E. Lim, “The ‘in-in’ Formalism and Cosmological Perturbations,” *Phys. Rev. D* **80** (2009) 083521, [arXiv:0904.4207 \[hep-th\]](#).
- [117] L. Senatore and M. Zaldarriaga, “On Loops in Inflation,” *JHEP* **12** (2010) 008, [arXiv:0912.2734 \[hep-th\]](#).
- [118] A. Kaya, “On $i\epsilon$ Prescription in Cosmology,” *JCAP* **04** (2019) 002, [arXiv:1810.12324 \[gr-qc\]](#).
- [119] M. Baumgart and R. Sundrum, “Manifestly Causal In-In Perturbation Theory about the Interacting Vacuum,” *JHEP* **03** (2021) 080, [arXiv:2010.10785 \[hep-th\]](#).
- [120] S. Coleman, “Notes from Sidney Coleman’s Physics 253a: Quantum Field Theory,” [arXiv:1110.5013 \[physics.ed-ph\]](#).
- [121] X. Chen, Y. Wang, and Z.-Z. Xianyu, “Schwinger-Keldysh Diagrammatics for Primordial Perturbations,” *JCAP* **12** (2017) 006, [arXiv:1703.10166 \[hep-th\]](#).
- [122] A. Abolhasani and H. Goodhew, “Derivative Interactions During Inflation: A Systematic Approach,” *JCAP* **06** no. 06, (2022) 032, [arXiv:2201.05117 \[hep-th\]](#).
- [123] H. Bateman, *Higher Transcendental Functions*. McGraw-Hill Book Company, 1953.
- [124] C. Sleight, “A Mellin Space Approach to Cosmological Correlators,” *JHEP* **01** (2020) 090, [arXiv:1906.12302 \[hep-th\]](#).
- [125] K. Aomoto, “Addition Theorem of Abel Type for Hyper-Logarithms,” *Nagoya Mathematical Journal* **88** no. none, (1982) 55 – 71.
- [126] A. Goncharov, “A simple construction of grassmannian polylogarithms,” *arXiv preprint*

arXiv:0908.2238 (2009) .

- [127] A. B. Goncharov, M. Spradlin, C. Vergu, and A. Volovich, “Classical Polylogarithms for Amplitudes and Wilson Loops,” *Phys. Rev. Lett.* **105** (2010) 151605, [arXiv:1006.5703 \[hep-th\]](#).
- [128] C. Duhr, H. Gangl, and J. Rhodes, “From Polygons and Symbols to Polylogarithmic Functions,” *JHEP* **10** (2012) 075, [arXiv:1110.0458 \[math-ph\]](#).
- [129] C. Duhr and F. Dulat, “PolyLogTools — Polylogs for the Masses,” *JHEP* **08** (2019) 135, [arXiv:1904.07279 \[hep-th\]](#).
- [130] A. Hillman, “Symbol Recursion for the dS Wave Function,” [arXiv:1912.09450 \[hep-th\]](#).
- [131] F. Arroja and T. Tanaka, “A Note on the Role of the Boundary Terms for the Non-Gaussianity in General k-Inflation,” *JCAP* **05** (2011) 005, [arXiv:1103.1102 \[astro-ph.CO\]](#).
- [132] S. Renaux-Petel, “On the Redundancy of Operators and the Bispectrum in the Most General Second-Order Scalar-Tensor Theory,” *JCAP* **02** (2012) 020, [arXiv:1107.5020 \[astro-ph.CO\]](#).
- [133] C. Cheung, A. L. Fitzpatrick, J. Kaplan, and L. Senatore, “On the Consistency Relation of the Three-Point Function in Single-Field Inflation,” *JCAP* **02** (2008) 021, [arXiv:0709.0295 \[hep-th\]](#).
- [134] G. Goon, K. Hinterbichler, and M. Trodden, “Symmetries for Galileons and DBI Scalars on Curved Space,” *JCAP* **07** (2011) 017, [arXiv:1103.5745 \[hep-th\]](#).